
MARROW ED8

Community Medicine

Comprehensive Question Bank

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History of Medicine

Question 1:

The origin of social and preventive medicine dates back to the:

- a) 17th century
- b) 18th century
- c) 16th century
- d) 19th century

Question 2:

Which medicine claims to be the world's first organized body of medical knowledge?

- a) Indian
- b) Chinese
- c) Egyptian
- d) Mesopotamian

Question 3:

All of the following are true about the ministry of AYUSH except:

- a) It was formed in 2014 in India for education and research on AYUSH
- b) It aims to mainstream AYUSH as part of the National Rural Health Mission
- c) It was previously known as department of Indian system of medicine and homeopathy
- d) It regulates the fields of Ayurveda, Yoga, Homeopathy and Siddha only

Question 4:

Samuel Hahnemann is referred to as the founding father of:

- a) Ayurveda
- b) Homeopathy
- c) Unani

d) Naturopathy

Question 5:

The Tridosha theory of disease explains disease as a disturbance in the equilibrium of the three humors. To which system of medicine does this theory belong to?

- a) Ayurveda**
- b) Homeopathy**
- c) Siddha**
- d) Unani – Tibb**

Question 6:

Which system of medicine originated in Tamil-speaking areas in south India?

- a) Ayurveda**
- b) Naturopathy**
- c) Unani – Tibb**
- d) Siddha**

Question 7:

Where did the Unani–Tibb system of medicine originate?

- a) Egypt**
- b) China**
- c) Greece**
- d) Mesopotamia**

Question 8:

Inoculating a person with a small amount of infective material to obtain protection against it was first pioneered by:

- a) Robert Koch**
- b) Edward Jenner**

- c) Louis Pasteur
- d) Chinese medicine

Question 9:

Who is referred to as the father of Indian medicine?

- a) Susruta
- b) Charaka
- c) Chanakya
- d) Vagbhata

Question 10:

Who was responsible for the first codification of medical practice?

- a) Aristotle
- b) Hippocrates
- c) Socrates
- d) Hammurabi

Question 11:

Who is known as the first true epidemiologist in the history of medicine?

- a) Hippocrates
- b) Susruta
- c) Louis Pasteur
- d) John Snow

Question 12:

Who compiled the 21 volume encyclopedia known as 'the Canon of Medicine'?

- a) Andreas Vesalius
- b) Avicenna
- c) Hippocrates

d) John Snow

Question 13:

Who was the mastermind behind the great medical work, Fabrica?

- a) Andreas Vesalius
- b) Ambroise Paré
- c) Joseph Lister
- d) Ferdinand Cohn

Question 14:

The father of public health is:

- a) Tuberculosis
- b) John Snow
- c) Cholera
- d) Hippocrates

Question 15:

Match the following scientists with their discoveries:

- a) 1-a, 2-b, 3-d, 4-c
- b) 1-b, 2-a, 3-c, 4-d
- c) 1-d, 2-c, 3-b, 4-a
- d) 1-c, 2-d, 3-a, 4-b

Question 16:

Who was the architect of the Public Health Act of 1848?

- a) Joseph Lister
- b) James Lind
- c) William Farr
- d) Edwin Chadwick

Question 17:

Identify the scientist who showed that anthrax was caused by a bacteria.

- a) Robert Koch
- b) Edward Jenner
- c) James Lind
- d) John Snow

Question 18:

In 1902, it was discovered that malaria was transmitted by the Anopheles mosquito. Who was behind this discovery?

- a) Ronald Ross
- b) Charles Alphonse Laveran
- c) Walter Reed
- d) David Bruce

Question 19:

The first bacterium discovered as a cause of the disease was:

- a) TB bacillus
- b) Anthrax bacillus
- c) Leprosy bacillus
- d) Plague bacillus

Question 20:

Max Joseph von Pettenkofer described the concept of:

- a) Multi-factorial causation of disease
- b) Supernatural theory of disease
- c) Spontaneous generation theory
- d) Theory of web of causation

Question 21:

In which country was the last case of naturally occurring smallpox in the world recorded?

- a) India
- b) Chile
- c) Somalia
- d) New Zealand

Question 22:

Who first coined the term social medicine?

- a) John Ryle
- b) Jules Guérin
- c) Alfred Grotjahn
- d) Francis Albert Eley Crew

Question 23:

How would you define socialized medicine?

- a) Subsidized medical care at government's expense
- b) Free medical care provided and regulated by government alone
- c) Free medical care at government expense, regulated by professional groups
- d) Integration of social medicine with health care

Question 24:

All of the following are true about socialized medicine except:

- a) Ensures social equity and universal coverage
- b) Reduces competition among physicians
- c) Use state funds for free medicine
- d) Promotes utilization of health facilities

Question 25:

Who said that the secret of national health lies in the homes of people?

- a) Florence Nightingale
- b) Indira Gandhi
- c) Abraham Lincoln
- d) Joseph Bore

Answer Key

Question No.	Correct Option
1	b
2	b
3	d
4	b
5	a
6	d
7	c
8	d
9	b
10	d
11	a
12	b
13	a
14	c
15	d
16	d
17	a
18	a
19	c
20	a
21	c
22	b
23	c
24	d

25	a
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Detailed Explanations

Solution to Question 1:

The origin of social and preventive medicine dates back to the 18th century. It began even before the germ theory had been proposed.

Solution to Question 2:

Chinese medicine claims to be the world's first organized body of medical knowledge. It dates back to 2700 B.C. They are also considered to be the pioneers of immunization (used variolation against smallpox).

Modern Chinese medicine includes barefoot doctors and acupuncture.

Solution to Question 3:

The Ministry of AYUSH was established to regulate the fields of Ayurveda, Yoga, and Naturopathy, Unani, Siddha, and Homeopathy.

It was formed in 2014 in India for the development of education and research in AYUSH. It was previously known as the Department of Indian System of Medicine and Homeopathy (ISM&H). It aims to mainstream AYUSH as part of the National Rural Health Mission (NRHM).

The Ayurveda, Siddha, and Yoga systems of medicine are of Indian origin. Ayurveda means knowledge of life. Its origin can be traced back to 5000 B.C (Vedic times). The medical knowledge in the Atharvaveda eventually developed into Ayurveda.

The Unani system is of Greek origin.

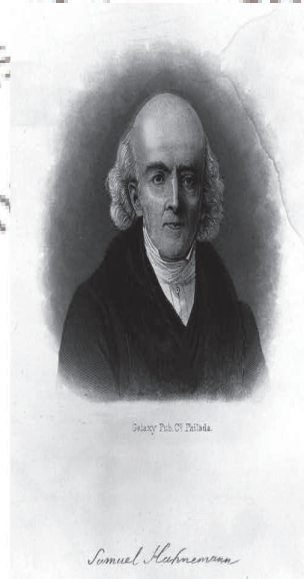
Solution to Question 4:

Samuel Hahnemann is the founding father of homeopathy. He was German.

There are 3 principles in homeopathy:

- Similia similibus curentur or human drug pathogenicity study: Treatment of disease using small amounts of drug that, in healthy people, would produce similar symptoms of the disease itself.
- A single medicine is used for the treatment.
- The minimum dose of the drug should be used.

Scientist/ Doctor	Achievements
Susruta	Father of Indian Surgery Father of Plastic and Cosmetic Surgery Author of "Susruta Samhita"
Hippocrates	First True Epidemiologist Father of Medicine Author of "Airs, Water and Place"
Charaka	Father of Indian Medicine Author of "Charaka Samhita"
Avicenna	Author of "Canon of Medicine" and "The Book on Healing"
Andreas Vesalius	Father of Modern Anatomy Author of "Fabrica"
Hammurabi	Author of "Babylonian Code of Hammurabi"
John Snow	Father of Epidemiology
Louis Pasteur Robert Koch / Ferdinand Cohn	Father of Bacteriology
Samuel Hahnemann	Father of Homeopathy



Solution to Question 5:

Tridosha theory of disease belongs to Ayurveda. It explains disease as a disturbance in the equilibrium of the three humors, namely:

- Vata (wind)

- Pitta (gall)
- Kapha (mucus).

A healthy person has a perfect balance of these humors.

This theory is similar to the Unani-tibb system of medicine which is based on four humors.

Solution to Question 6:

The Siddha system of medicine originated from Tamil culture. It means achievement.

This system believes that sociodemographic, psychological, and physiological factors related to a person must be taken into account while deciding the medical treatment. They believe that disease is a disturbance in the equilibrium of three humors called vaadham, pittam, and kapham.

Solution to Question 7:

The Unani-Tibb system of medicine originated in ancient Greece.

The Greeks believed that matter was made up of four elements. These elements had certain qualities that were represented in the body by the 4 humors and also determined the person's temperament.

Elements	Qualities	Humors	Temperament
Earth	Cold	Phlegm	Phlegmatic
Air	Dry	Yellow bile	Choleric
Fire	Hot	Blood	Sanguine
Water	Moist	Black bile	Melancholic

Solution to Question 8:

The procedure described is immunization. It was first pioneered in ancient Chinese medicine. They practiced variolation to prevent smallpox.

Note: Edward Jenner discovered the smallpox vaccine in 1796, the first vaccine to be discovered. He also coined the term vaccination.

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Solution to Question 9:

Charaka is the father of Indian medicine. He wrote the Charaka Samhita.

Note: Susruta is known as the father of Indian surgery. He is depicted in the following image.



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Solution to Question 10:

Hammurabi, a Mesopotamian king was responsible for the first codification of medical practice.

Known as the Babylonian Code, it contained laws for the conduct of physicians, health practices, costs, and penalties of harmful therapy.

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Solution to Question 11:

Hippocrates is known as the first true epidemiologist in the history of medicine. He is also known as the father of medicine.

He introduced the concept of the relationship between the environment and human health. He was the author of *Airs, Water, and Places*, a treatise on social medicine and hygiene as well as the *Corpus Hippocraticum*, a collection of 72 volumes of clinical case histories.

He also developed the Hippocratic Oath, which is considered the keystone of medical ethics.



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Samuel Hahnemann	Father of Homeopathy

Solution to Question 12:

Avicenna compiled the book, the Canon of Medicine which is a 21 volume encyclopedia.

He was a Persian who was also known as Ibn Sina. He was the leader of Arabic medicine. The Arabs invented the art of prescription writing.

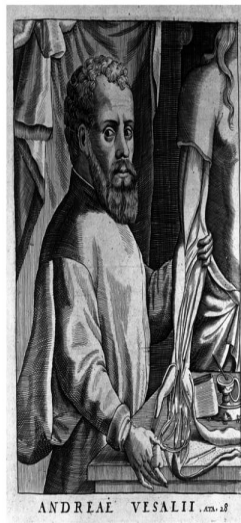


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Solution to Question 13:

Andreas Vesalius' work is called the de humani corporis Fabrica. He is regarded as the father of modern anatomy.



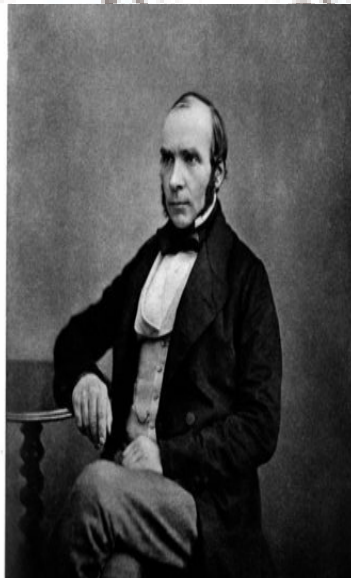
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Solution to Question 14:

The father of public health is cholera, the disease, not a person.

John Snow studied the epidemiology of cholera in London from 1848 to 1854 and established the role of polluted drinking water in the spread of cholera. He is called the father of modern epidemiology. He also calculated the dose of ether and chloroform as surgical anesthesia.



Note: Tuberculosis is called the barometer of social welfare.

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Samuel Hahnemann	Father of Homeopathy

Solution to Question 15:

The discoveries of the given scientists are as follows:

- John Snow - association of cholera with polluted water
- Edward Jenner - smallpox vaccination
- Louis Pasteur - the germ theory of disease
- James Lind - prevention of scurvy by intake of citrus fruits

Louis Pasteur was the first to advance the germ theory of disease, which stated that most diseases were caused by microorganisms. This theory was later expanded by Robert Koch, who put forth Koch's postulates.

Pasteur is also known for the following contributions:

- Disproved the theory of spontaneous generation
- Developed vaccines for rabies and fowl cholera
- Pioneer of techniques of pasteurization and sterilization

The following images show Pasteur and Lind respectively:



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John Snow	Father of Epidemiology
Louis Pasteur Robert Koch / Ferdinand Cohn	Father of Bacteriology
Samuel Hahnemann	Father of Homeopathy

Solution to Question 16:

Edwin Chadwick was the architect of the Public Health Act of 1848.

He was an English lawyer during the 1832 cholera epidemic. He compiled a report on the Sanitary Condition of the Labouring Population in Great Britain that eventually led to the passing of this act. It is considered a landmark act in the history of public health.

Solution to Question 17:

Robert Koch discovered that anthrax was caused by bacteria.

He also discovered the causative agents of tuberculosis and cholera. He won the Nobel prize in Physiology in 1905.

He also proposed Koch's postulates linking specific microorganisms to specific diseases. They state that the microorganism must be:

- Found in abundance in diseased organisms, but not in healthy organisms.
- Isolated from a diseased organism and grown in pure culture
- Able to cause disease when inoculated into a healthy organism
- Re-isolated from the diseased experimental host and identical to the original specific causative agent

The following images show Robert Koch and his drawing of the anthrax bacillus:



Solution to Question 18:

Ronald Ross showed that malaria was transmitted by Anopheles mosquito.

Option B: Charles Alphonse Laveran discovered the malarial parasite.

Option C: Walter Reed showed that yellow fever transmitted by Aedes mosquito.

Option D: David Bruce showed that African sleeping sickness/trypanosomiasis transmitted by the tsetse fly.

Solution to Question 19:

The first bacterium discovered as a cause of the disease is leprosy bacillus.

In 1873, Armauer Hansen demonstrated Mycobacterium leprae (rod-like bodies) from a leprosy nodule. This is the first time that chronic infectious disease in humans was shown to be related to a bacillus.

In 1876, Robert Koch validated the germ theory of disease after discovering the anthrax bacillus as the cause of anthrax.

Solution to Question 20:

Pettenkofer of Munich put forth the theory of multifactorial causation of disease.

Other theories of causation of disease include:

- Spontaneous generation theory - given by Aristotle
- Web of causation theory - given by MacMahon and Pugh
- Supernatural theory of disease

Solution to Question 21:

The last case of naturally occurring smallpox was in Somalia in 1977.

The last case of smallpox occurred in 1978 when a medical photographer was accidentally exposed to the smallpox virus in a research laboratory in Birmingham.

India was declared smallpox-free on 24th May 1975. WHO declared smallpox as eradicated globally on 8th May 1980.

Solution to Question 22:

Jules Guérin coined this term as editor of the Gazette médicale de Paris. He was a French orthopaedist in the mid-1800s.

Option A: John Ryle promoted social medicine in England in the early 1900s.

Option C: Alfred Grotjahn described social pathology, the social factors influencing diseases.

Option D: Francis Albert Eley Crew stated that social medicine stood on the pillars of medicine and sociology.

Solution to Question 23:

Socialized medicine is the provision of medical service at the government's expense but is operated and regulated by professional groups, and not the government. Russia was the first

country to completely socialize medicine.

Free medical care by a government alone is called state medicine.

Solution to Question 24:

Increasing the utilization of health services is not a component of socialized medicine.

It refers to the provision of medical service through the state funds; however, it is operated and regulated by professional groups. Russia was the first to completely socialize medicine.

It eliminates competition among physicians as well as ensures social equity and universal health coverage.

Solution to Question 25:

Florence Nightingale said that the secret of national health lies in the homes of people. She is considered the founder of modern nursing.



Health Determinants and Indicators

Question 1:

Which of the following is not included in the WHO definition of health?

- a) Social well-being
- b) Emotional well-being
- c) Mental well-being
- d) Physical well-being

Question 2:

Which of the following is the biomedical concept of disease based upon?

- a) Germ theory of disease
- b) Combination of social, economic and political influences
- c) Social and psychological factors
- d) Equilibrium between man and environment

Question 3:

Which of the following is a mortality indicator?

- a) Disability adjusted life years
- b) Sullivan's Index
- c) Life expectancy
- d) Health adjusted life expectancy

Question 4:

An epidemiologist is tasked with measuring the overall health status of a community using a morbidity indicator. Which of the following indicators will she not use?

- a) Duration of stay in hospital
- b) Attendance rates at OPD

- c) Number of days of restricted activity
- d) Notification rates

Question 5:

Which of the following indicators can be used to estimate the number of years a newborn would be expected to live in full health based on the current rates of ill health and mortality?

- a) DALY
- b) Sullivan's index
- c) HALE
- d) Life expectancy

Question 6:

While assessing the health needs of a community, an epidemiologist is tasked with estimating the burden of cholera in the area. Which of the following is the simplest measure he can use?

- a) Case fatality rate
- b) Incidence
- c) Crude death rate
- d) Proportional mortality rate

Question 7:

You are researching the overall burden of COVID on a given population. Which of the following disability indicators is the best measure of this?

- a) Sullivan's index
- b) DALY
- c) QALY
- d) YLL

Question 8:

Which of the following is not a health utilisation rate?

- a) Bed occupancy rate

- b) Bed turnover ratio
- c) Bed disability days
- d) Average length of stay in hospital

Question 9:

As the medical superintendent of a hospital, you wish to optimise the utilization of health services. What bed turnover interval should you aim for?

- a) Slightly positive
- b) Slightly negative
- c) Largely positive
- d) Equal to zero

Question 10:

Which of the following is the best indicator of availability, utilization, and effectiveness of health services?

- a) Infant mortality rate
- b) Maternal mortality rate
- c) Proportional mortality rate
- d) Disability adjusted life expectancy

Question 11:

While reviewing a national health programme, you are asked to assess the cost-effectiveness of the interventions provided under it. Which indicator will you use to measure this?

- a) Admission, readmission and discharge rates
- b) QALY
- c) HALE
- d) Bed turnover rates

Question 12:

Which of the following is not a socio-economic determinant of health?

- a) Lifestyle
- b) Political system
- c) Education
- d) GNP per capita

Question 13:

Which of the following is not a socio-economic indicator of health?

- a) Per capita GNP
- b) Family size
- c) Literacy rate
- d) Annual family income

Question 14:

Which of the following statements is false regarding positive health in today's world?

- a) It is dependent on social, economic, and cultural determinants
- b) It is a mirage because of changing environment
- c) It means that the body and mind are at peace
- d) It includes provision of free health camps by the government

Question 15:

As per the WHO, which of the following is not included in the standard of living?

- a) Income
- b) Occupation
- c) Sanitation
- d) Social security

Question 16:

Which of the following are components of the physical quality of life index (PQLI)?

- a) 1,3 and 5

- b) 2 and 5
- c) 1,2 and 4
- d) 1,3 and 4

Question 17:

Which of the following is the best measure of the standard of living?

- a) Under 5 mortality rate
- b) Sullivan index
- c) HDI
- d) PQLI

Question 18:

Which of the following is not a component of the human development index (HDI)?

- a) Infant Mortality Rate
- b) Income
- c) Longevity
- d) Knowledge

Question 19:

What are the minimum and maximum values established for the calculation of the life expectancy index in HDI?

- a) 0 years and 65 years
- b) 20 years and 85 years
- c) 25 years and 75 years
- d) 0 years and 100 years

Question 20:

Which of the following is not true regarding the education index used for calculating the HDI?

- a) It includes expected years of schooling
- b) Gross enrollment ratio is considered
- c) Adult literacy rate is not considered
- d) Minimum and maximum values are based on global data

Question 21:

Which of the following indices measures the deprivation in basic dimensions of the country in human development?

- a) HDI
- b) PQLI
- c) HPI
- d) DALY

Question 22:

Which of the following is not included in the Ottawa charter for health promotion?

- a) Develop personal skills
- b) Strengthen community action for health
- c) Build social security system
- d) Reorient health services

Answer Key

Question No.	Correct Option
1	b
2	a
3	c
4	c
5	c
6	d
7	b
8	c

9	a
10	a
11	b
12	a
13	d
14	d
15	d
16	d
17	c
18	a
19	b
20	b
21	c
22	c

Detailed Explanations

Solution to Question 1:

According to the WHO, health is “a state of complete physical, mental and social well-being and not merely the absence of disease or infirmity”.

In recent years, the definition has been modified to also include the ability to lead a “socially and economically productive life.”

Solution to Question 2:

The biomedical concept of disease is based on the germ theory of disease, which was postulated by Louis Pasteur.

The germ theory of disease states that microorganisms known as pathogens or germs can lead to disease. It was criticized that it did not include environmental, social, psychological, and cultural determinants of health.

The ecological concept of disease states that disease is the maladjustment of the human organism to the environment.

The psychosocial concept of disease hypothesised that health is also influenced by social, psychological, cultural and economic factors as well.

The holistic concept of disease includes all the above concepts to describe health.

Solution to Question 3:

Life expectancy is a positive mortality indicator.

A set of values that measure the rate of mortality or death is known as mortality indicators.

Disability-adjusted life years (DALY), health adjusted life expectancy (HALE), and Sullivan's Index are disability rates.

Solution to Question 4:

Number of days of restricted activity is a disability indicator.

Morbidity is defined as public health outcomes focusing on diseases, traumas, or injury. Morbidity indicators can be used to measure the overall health and extent of healthcare utilisation in a given population.

Disability is defined as public health outcomes related to health conditions that include limitations in personal activities and societal participation. There are two types of disability indicators as follows:

- Event type indicators-
- Number of days of restricted activity
- Bed disability days
- Work loss days (or school loss days) within a specified period
- Person type indicators-
- Limitation of mobility: e.g confined to bed
- Limitation of activity: e.g limitation to perform the basic activities of daily living such as eating and going to the toilet

Solution to Question 5:

HALE (Health Adjusted Life Expectancy) is an indicator that can be used to estimate the number of years a newborn is expected to live in full health based on the current rates of ill health and mortality.

HALE is calculated using the life expectancy at birth with an adjustment for time spent in poor health. It captures both fatal and non-fatal health outcomes of average levels of population health. It is also sensitive to changes over time or differences between countries in the distribution of health states.

Sullivan's index is also known as disability-free life expectancy (DFLE) or active life expectancy. It is the average number of years an individual is expected to live free of disability if the current patterns of mortality and disability continue to apply. It doesn't take into account premature death. As HALE includes not just disability but the larger definition of poor health, it is a better option with respect to the question.

Solution to Question 6:

The proportional mortality rate is the simplest measure of estimating the burden of disease in the community. It signifies the proportion of all deaths that can be attributed to the disease.

It is calculated as follows:

Proportional mortality rate of a disease = (No. of deaths due to that disease in the year/ Total no. of deaths in the year) *100.

Option A: Case fatality rate measures the risk of persons dying from a certain disease within a given time period. It measures the killing power of a disease. It is used to link mortality to morbidity.

Option B: Incidence measures the rate at which new cases are occurring in a population. It is not influenced by the duration of the disease. The use of incidence is generally restricted to acute diseases.

Option C: Crude death rate is considered a fair indicator of the comparative health of the people. It is the rate of total deaths in a year compared to the mid-year population.

Solution to Question 7:

Disability adjusted life years (DALY) is the best measure of the overall burden of a disease in a defined population. It is also the best measure of the effectiveness of interventions.

DALY is calculated as - Years of lost life prematurely (YLL) + Years lost to disability (YLD). It can measure both mortality and disability together.

As DALY is affected by premature loss of life, conditions that affect younger individuals like road traffic accidents have a high DALY. In comparison to this, Alzheimer's disease has a lower DALY as it is a disease that mostly develops during old age and so has a lower YLL.

Solution to Question 8:

Bed disability days is not a utilisation rate. It is an event-type disability indicator. It is defined as the number of days a person spent in a hospital bed due to illness or injury.

Health utilisation rate is defined as the number of people making use of the available health services to the total number of individuals in that population.

Health utilisation rates include:

- Proportion of infants immunized against 9 diseases in the expanded program on immunization (EPI)
- Proportion of women receiving antenatal care and supervised delivery
- Percentage of population using family planning methods

- Bed occupancy rate
- Average length of stay in the hospital
- Bed turnover ratio

Solution to Question 9:

For optimum utilization of health services in a hospital, bed turnover interval must always be slightly positive.

Bed turnover interval is defined as the time interval where a hospital bed is left unoccupied following a patient's discharge till the next patient occupies it after admission.

Solution to Question 10:

Infant mortality rate (IMR) is the best indicator of the availability, utilization, and effectiveness of health care. It is a very sensitive indicator.

Infant mortality rate (IMR)= $\frac{\text{Number of deaths in the first year of life}}{\text{Number of live births}} \times 1000$

It is also one of the best indicators of state failure. It is the most universally accepted health status indicator of the whole population and of the socio-economic status in which they live.

Note: The best indicator of socio-economic development is the under-5 mortality rate.

Solution to Question 11:

QALY(Quality Adjusted Life Years) is used in assessing the cost-effectiveness (value of money) of a medical intervention. It is a measure of both the quality and quantity of life lived. It indicates the number of years of life that would be added by the intervention.

Solution to Question 12:

Lifestyle is a behavioural and socio-cultural determinant of health. It is not a socio-economic determinant.

Socio-economic determinants of health are the following:

- Economic status (per capita GNP)
- Education
- Occupation
- Political system

Lifestyle is simply the way people live. It reflects our social values, attitudes, and activities. It is composed of:

- Cultural patterns (daily nutrition, superstitions)
- Behavioral patterns (personal hygiene, sedentary life)
- Lifelong personal habits (e.g. smoking, alcoholism)

Lifestyle is learned by social interaction with parents, siblings, friends, school, and mass media.

Our political system determines our health. It can help shape community health services with regards to resource allocation, manpower policy, choice of technology, and the degree to which these services are made available & accessible to all segments of the society.

The percentage of GDP spent on health is an indicator of political commitment. WHO recommends at least 5% of GDP to be spent on health care. India spends about 2.1 % of its GDP on health(budget 2022-23).

Solution to Question 13:

Annual family income is not a socioeconomic indicator of health.

Socio-economic indicators of health (mnemonic: H – FLAGGED):

- Housing (number of persons per room)
- Family size
- Literacy rate (especially female literacy rate)
- Availability of per capita calorie
- GNP per capita
- Growth rate of population
- UnEmployment level
- Dependency ratio

Solution to Question 14:

The provision of free health camps by the government is a concept of state medicine.

Health does not merely mean the absence of disease or provision of diagnostic, curative, and preventive services. It is a state of physical, mental, and social well-being.

The state of positive health is based on the idea of perfect functioning of the body and mind. According to this concept, health is:

- Biologically a state in which every cell and organ in the body is functioning at optimum capacity
- Psychologically a state in which the individual feels a sense of perfect well-being and mastery over his environment

- Socially a state in which the individual's capacities for participation in the social system are optimal

This concept is too idealistic and can never become a reality. Humans can never be so perfectly adapted to the environment because everything in our life is subject to change.

As the goal of positive health is unattainable, a broader concept of improving the quality of life has emerged. This highlights that positive health depends on medical action and also on economic, cultural, and social factors in the community.

Solution to Question 15:

Social security is a component of the level of living as per the United Nations and not standard of living.

According to the WHO, the standard of living includes:

- Income and occupation
- Standards of housing, sanitation, and nutrition
- Level of provision of health, educational, and recreational services

It depends on per capita GNP.

Solution to Question 16:

Infant mortality rate, literacy rate, and life expectancy at 1 year of age are the components of the physical quality of life index (PQLI).

Solution to Question 17:

Human development index (HDI) is the best measure of the standard of living of a population.

Physical quality of life index (PQLI) measures the quality of life.

Solution to Question 18:

Infant mortality rate (IMR) is not a component of HDI. It is a component of the physical quality of life index (PQLI).

The human development index (HDI) is used to compare living standards in different countries. It includes the following dimensions or components:

Each dimension index is calculated using the formula: $\text{Dimension Index} = \frac{(\text{Actual Value} - \text{Minimum Value})}{(\text{Maximum Value} - \text{Minimum Value})}$

HDI is the geometric mean of the three indices: $I_{Life}^{1/3} \times I_{Education}^{1/3} \times I_{Income}^{1/3}$.

As per the Human Development Report 2023-2024, India's Human Development Index (HDI) value is 0.644. This places India at 134th out of 193 countries.

Component	Indicator	Index
Longevity	Life expectancy at birth	Life expectancy index
Knowledge	Literacy and enrolment	Education index
Income	GNI per capita US\$ (purchasing power parity-PPP)	GNI index(income index)

Solution to Question 19:

The minimum value established for the calculation of the life expectancy index in HDI is 20 years and the maximum is 85 years.

These minimum and maximum values are used to calculate the dimension indices in HDI and are called goalposts.

Each dimension index is calculated using the formula.

Dimension Index = $(\text{Actual Value} - \text{Minimum Value}) / (\text{Maximum Value} - \text{Minimum Value})$.

Solution to Question 20:

The gross enrollment ratio is no longer used in obtaining the education index in HDI. It was used as an indicator earlier.

The education index is calculated by taking into account the mean years of schooling and expected years of schooling. Maximum and minimum values are based on global data.

Adult literacy rate and gross enrollment ratio were used as indicators before 2009.

Solution to Question 21:

Human poverty index (HPI) measures deprivation in basic dimensions of human development.

HPI-1 is used for developing countries and HPI-2 is used for developed countries. It is based upon the following indicators:

The range of HPI is from 0 to 100. The term HPI has now been replaced with multidimensional poverty index (MPI).

	Human Poverty Index(HPI-1) for developing countries	Human Poverty Index(HPI-2) for selected OECD countries
A long and healthy life	Probability at birth of not surviving to age 40	Probability at birth of not surviving to age 60
Knowledge	Adult illiteracy rate	Percentage of adults lacking functional literacy skills
A decent standard of living	Percentage of the population not using an improved water source Percentage of children underweight for age Deprivation of a decent standard of living	Percentage of people living below the poverty line
Social exclusion	Not included as a dimension	Long term unemployment rate

Solution to Question 22:

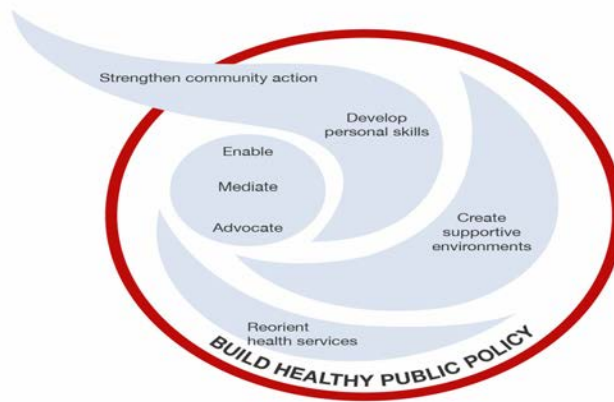
Building social security is not included in the Ottawa charter for health promotion.

The first international conference for health promotion was held in Ottawa on 21st Nov 1986. It produced a charter known as the Ottawa charter.

The charter identified five key action areas as given below:

- Build healthy public policy
- Create supportive environment for health
- Strengthen community action for health
- Develop personal skills
- Reorient health services

Ottawa charter



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Concepts of Disease and Prevention

Question 1:

Which of the following is not a component of the epidemiological triad for tuberculosis?

- a) A malnourished child.
- b) *Mycobacterium tuberculosis*
- c) Duration of illness
- d) Overcrowding

Question 2:

Which of the following is the most logical sequence?

- a) Disease → Impairment → Disability → Handicap
- b) Impairment → Disease → Handicap → Disability
- c) Disability → Disease → Impairment → Handicap
- d) Disease → Disability → Impairment → Handicap

Question 3:

A salesman met with an unfortunate accident that led to him eventually losing his left foot. He has been unable to walk after this incident and so been rendered unemployed. Match the following examples with the corresponding events explaining the concept of disability.

- a) 1-B, 2-C, 3-D, 4-A
- b) 1-D, 2-A, 3-B, 4-C
- c) 1-A, 2-B, 3-C, 4-D
- d) 1-C, 2-D, 3-A, 4-B

Question 4:

Which of the following is most appropriate about the web of disease causation?

- a) It implies that individual factors in the web have equal weightage
- b) It implies that all the chains of causation must be controlled to prevent disease

- c) It is mostly applicable for acute and rare diseases
- d) It suggests ways of interrupting the disease occurrence

Question 5:

Which of the following is not true regarding the pathogenesis period of a disease?

- a) Primary prevention is possible
- b) Recovery is seen in this period
- c) Entry of causative agent is seen
- d) Includes subclinical cases

Question 6:

In the iceberg phenomenon of disease, which of the following does not represent the submerged portion of the iceberg?

- a) Pre-symptomatic cases
- b) Latent cases
- c) Diagnosed cases
- d) Undiagnosed cases

Question 7:

Which of the following diseases shows the iceberg phenomenon?

- a) Tetanus
- b) Rubella
- c) Measles
- d) Rabies

Question 8:

What is the term used to describe the interruption in the transmission of a disease?

- a) Elimination
- b) Eradication

- c) Surveillance
- d) Control

Question 9:

Among the following options, which aids the most in disease elimination?

- a) Isolation
- b) Quarantine
- c) Herd immunity
- d) Complete treatment

Question 10:

While collecting surveillance data for disease X, you find that the incidence of the disease is zero in your region. Which of the following statements best describes this scenario?

- a) Disease is eliminated from the region.
- b) Disease is well controlled in the region.
- c) Disease is eradicated from the region.
- d) Disease shows iceberg phenomenon in the region.

Question 11:

Which of the following is a silent epidemic?

- a) Bipolar disorders
- b) Manic depressive disorder
- c) Alzheimer's disease
- d) Generalised anxiety disorder

Question 12:

What is the analysis of routine measurements aimed at detecting changes in the health status of a population called?

- a) Monitoring

- b) Surveillance
- c) Systematic review
- d) Sentinel surveillance

Question 13:

In which of the following does surveillance of disease not have a direct role?

- a) Health planning
- b) Control of disease
- c) Data on new and changing trends
- d) Breaking chain of transmission

Question 14:

Measures in sentinel surveillance do not aim at identifying:

- a) Missing cases of the disease
- b) Old cases of the disease
- c) New cases of the disease
- d) Cases free of disability

Question 15:

Which of the following is public health surveillance preferentially aimed at?

- a) Treatment of disease
- b) Disease eradication
- c) Disease monitoring
- d) Health planning

Question 16:

Which of the following is included under primordial prevention of coronary artery disease?

- a) Lifestyle change in adults
- b) Coronary bypass

- c) Regular BP estimation
- d) Lifestyle changes in childhood

Question 17:

What type of prevention is the prescriptive screening of a disease?

- a) Primordial
- b) Primary
- c) Secondary
- d) Tertiary

Question 18:

What form of prevention is prospective screening?

- a) Primary
- b) Secondary
- c) Tertiary
- d) Primordial

Question 19:

You advise a reproductive age woman to vaccinate against HPV. What is this type of prevention called?

- a) Primordial prevention
- b) Primary prevention
- c) Secondary prevention
- d) Tertiary prevention

Question 20:

Who among the following is not a target for primary prevention?

- a) Apparently healthy individuals
- b) Individuals with risk factor

- c) Children less than 5years
- d) Asymptomatic patients

Question 21:

As a part of the iodine deficiency control programme, you advocate the usage of iodized salt in hilly areas. What is this type of prevention an example of?

- a) Primordial prevention
- b) Primary prevention
- c) Secondary prevention
- d) Tertiary prevention

Question 22:

You are performing a routine pap smear on a 30-year-old woman in the gynecology OPD. What level of prevention does this fall under?

- a) Primordial
- b) Secondary
- c) Primary
- d) Tertiary

Question 23:

What mode of prevention is disability prevention by appropriate treatment an example of?

- a) Primordial prevention
- b) Primary prevention
- c) Secondary prevention
- d) Tertiary prevention

Question 24:

Which of the following is an example of disability limitation?

- a) Prescribing spectacles for refractive error

- b) Starting a patient on anti-tuberculous therapy
- c) Abstinence from smoking
- d) Early fixation of fractures

Question 25:

Which among the following diseases was eliminated in India in 2015?

- a) Leprosy
- b) Neonatal tetanus
- c) Tuberculosis
- d) Yellow fever

Question 26:

Which of the following statements is false about ICD- 11?

- a) It includes a new chapter on traditional medicine
- b) Gender incongruence is now listed as part of mental health
- c) The codes relating to antimicrobial resistance are more closely in line with the Global Antimicrobial Resistance Surveillance System
- d) It has come into full effect from January 2022

Answer Key

Question No.	Correct Option
1	c
2	a
3	a
4	d
5	a
6	c
7	b
8	a
9	c

10	a
11	c
12	a
13	d
14	d
15	d
16	d
17	c
18	a
19	b
20	d
21	b
22	b
23	c
24	a
25	b
26	b

Detailed Explanations

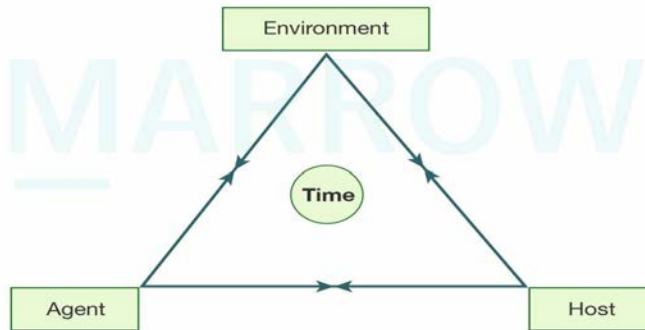
Solution to Question 1:

Duration of illness is not a part of the epidemiological triad for tuberculosis.

The epidemiological triad includes host, agent and environment:

- Host factors: Age, sex, religion, race, genetic profile, immune system
- Agent factors:
- Biological - bacteria, viruses
- Chemical - poisons, alcohol, smoke
- Physical - trauma, fire, radiation
- Nutrition - lack/excess
- Environmental factors:
- Temperature, humidity, altitude
- Water, milk, food
- Radiation, pollution, noise
- Crowding, housing, neighborhood

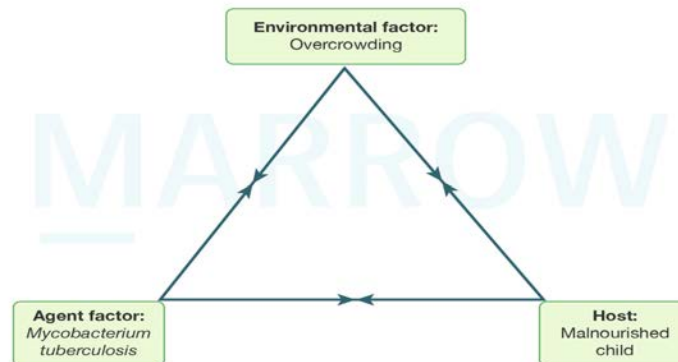
Triangle of epidemiology



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Hence, a malnourished child (host), *Mycobacterium tuberculosis* (agent) and overcrowding (environment) form the epidemiological triad for tuberculosis.

Epidemiological triad for tuberculosis



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Solution to Question 2:

Disease → Impairment → Disability → Handicap is the most logical sequence among the given options.

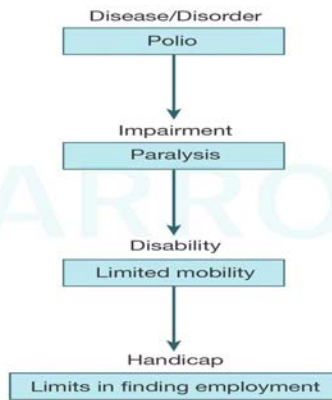
Disease is a state of physiological/psychological dysfunction.

Impairment is any loss or abnormality in normal structure or function.

Disability is the inability to carry out normal actions due to impairment.

Handicap is the disadvantage due to disability that limits fulfilment of a role that is normal.

Sequence of development of handicap



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Solution to Question 3:

The correctly matched events relating to the given example are as follows:

The WHO definition of impairment is "any loss or abnormality of psychological, physiological or anatomical structure or function". E.g., loss of a foot. An initial impairment may cause a secondary impairment as well.

The WHO defines disability as "any restriction or lack of ability to perform an activity in the manner or within the range considered normal for a human being". Because of impairment, the affected person may be unable to carry out certain activities and hence the person is considered disabled.

According to the WHO, a handicap is "a disadvantage for a given individual, resulting from an impairment or a disability, that limits or prevents the fulfilment of a role that is normal (depending on age, sex, and social and cultural factors) for that individual". The disadvantages experienced by a person as a result of disability makes them handicapped.

Event	Example
Disease/Disorder	Accident
Impairment	Loss of foot
Disability	Cannot Walk
Handicap	Unemployed

Solution to Question 4:

The web of causation for a disease suggests a number of ways to interrupt the disease occurrence/progression.

This model was suggested by MacMahon and Pugh in their book 'Epidemiologic Principles and Methods'.

Web of causation considers all the predisposing factors and their complex interrelationship with each other. This model is ideally suited in the study of chronic disease. The web of causation does not imply that the disease is controlled only when all the multiple chains of causation are removed. Sometimes removal of one significant link is sufficient to control the disease.

In a multifactorial event, individual factors are by no means of equal weightage. The importance of each factor is expressed in terms of relative risk.

Solution to Question 5:

Primary prevention is not possible in the pathogenesis period of a disease. It can only be done in the period of pre-pathogenesis.

The two phases in the natural history of a disease include:

- **Period of pre-pathogenesis:** It is before the onset of disease in humans. The disease agent has not entered the host, but the factors favouring the disease already exist. Primary prevention (health promotion and specific protection) is applicable in this period.
- **Period of pathogenesis:** It begins with the entry of the organism till recovery, death or disability. Secondary (halting of the disease process) and tertiary prevention (measures to limit disability) are applicable in this period. It isn't always characterised by the presence of disease symptoms as it also includes subclinical cases.

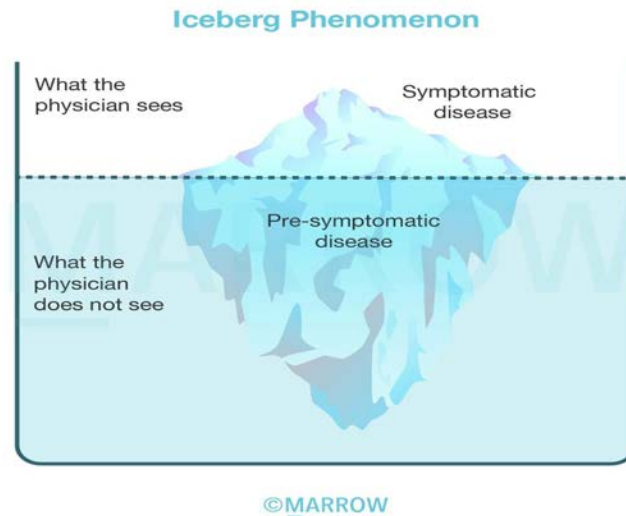
Solution to Question 6:

Epidemiologically, the iceberg phenomenon of disease, diagnosed cases are represented by the floating tip of the iceberg.

The iceberg phenomenon of disease is related to the spectrum of disease. The water surface is the line of demarcation between the apparent and inapparent infections.

The submerged portion of the iceberg represents the hidden mass of diseases like asymptomatic, latent, and undiagnosed cases. Screening is done to identify these cases.

The iceberg phenomenon of disease is depicted in the image below.



Solution to Question 7:

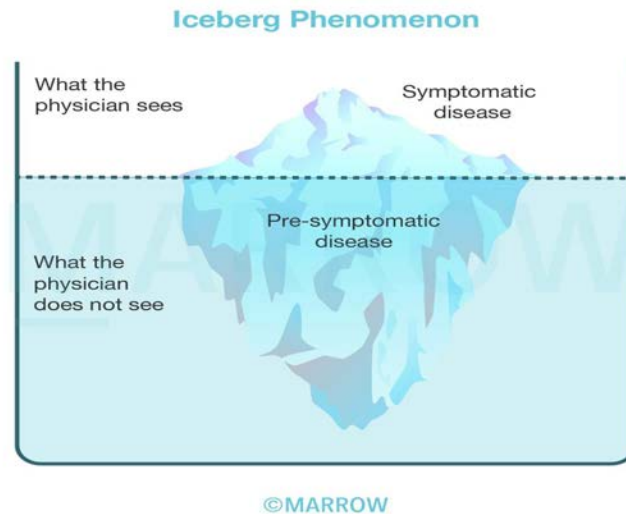
Rubella shows the iceberg phenomenon of disease.

The iceberg phenomenon of diseases is not shown by diseases such as tetanus, rabies, and measles.

Iceberg phenomenon is shown by:

- Polio
- Influenza
- Rubella
- Mumps
- Japanese Encephalitis
- Diphtheria
- Hepatitis A and B
- Tuberculosis
- Diabetes
- Malnutrition
- Hypertension

Note: If a disease has more number of subclinical infections, it shows the iceberg phenomenon of disease depicted in the image below.



Solution to Question 8:

Interruption in the transmission of a disease is called elimination. It is a step towards eradication.

Elimination is a reduction in the number of new cases of an infectious disease to zero (or a very low defined target rate) in a defined geographical area as a result of deliberate efforts. Continued measures to prevent re-establishment of transmission are required.

Eradication is a complete and permanent worldwide reduction to zero new cases of an infectious disease through deliberate efforts. No further control measures are required.

Surveillance is the continuous scrutiny of factors that determine the occurrence and distribution of disease and ill health.

Control aims at decreasing the incidence, duration and transmission of the disease. It also aims at decreasing the physical and psychological complications of a disease.

Solution to Question 9:

Herd immunity aids the most in disease elimination.

Herd immunity achieved, especially through immunization, goes a long way in helping the elimination of diseases. When a high level of herd immunity is maintained, the susceptible population is reduced to a small proportion, thereby eliminating the disease.

The purpose of isolation is to protect the community by preventing the transfer of infection from the reservoir to the possible susceptible hosts. Quarantine refers to restrictions on the healthy contacts of an infectious disease.

Solution to Question 10:

In the given scenario, as the disease has zero incidence, we can say the disease has been eliminated from the region.

Elimination is a reduction in the number of new cases of an infectious disease to zero (or a very low defined target rate) in a defined geographical area as a result of deliberate efforts. Continued measures to prevent re-establishment of transmission are required.

Eradication is a complete and permanent worldwide reduction to zero new cases of an infectious disease through deliberate efforts. No further control measures are required.

Control aims at decreasing the incidence, duration and transmission of the disease. It also aims at decreasing the physical and psychological complications of a disease.

The iceberg phenomenon of disease is related to the spectrum of disease. The submerged portion of the iceberg represents the hidden mass of diseases like asymptomatic, latent, and undiagnosed cases. The clinical or diagnosed cases are represented by the floating tip of the iceberg and not the submerged portion. As disease incidence is zero here, there are no cases to form the iceberg.

Solution to Question 11:

Alzheimer's disease is referred to as the silent epidemic of the century.

Nearly 50 million people globally are living with this disease. Since it strikes so often, so arbitrarily, and with such little warning, many have dubbed it the silent epidemic.

Solution to Question 12:

The analysis of routine measurements aimed at detecting changes in the health status or environment of a population is called monitoring.

Surveillance is the continuous scrutiny of factors that determine the occurrence and distribution of disease and ill health.

Sentinel surveillance is a method of identifying the missing cases and thereby supplementing the notified cases.

Systematic reviews involve a detailed plan with the goal of reducing bias by identifying, appraising, and synthesizing all relevant studies on a particular topic.

Solution to Question 13:

Surveillance of a disease does not have a direct role in breaking the chain of transmission.

Surveillance is the continuous scrutiny of factors that determine the occurrence and distribution of disease and ill health.

It helps in health planning and provides information about public health disasters including data on new and changing trends of disease.

Surveillance requires professional analysis and sophisticated judgment of data, leading to recommendations for control activities.

Solution to Question 14:

Sentinel surveillance does not include identifying cases free of disability.

Sentinel surveillance helps in identifying the missing cases, in addition to the already existing and new cases. It is used for supplementing routine notifications.

Solution to Question 15:

Public health surveillance is preferentially aimed at health planning.

Public health surveillance provides important feedback on the health status of the population, about planning control and prevention activities, and also for public health disasters.

Solution to Question 16:

Adoption of a healthy lifestyle from childhood would fall under the category of primordial prevention of coronary artery disease.

Primordial prevention is preventing the emergence of risk factors among population groups in which these risk factors have not yet appeared. Most adult health problems (obesity, hypertension) have their early origins in childhood. It is in childhood that lifestyles are formed i.e. eating patterns, physical exercise, smoking, etc.

In primordial prevention, efforts are made to discourage children from adopting harmful lifestyles.

These lifestyle changes when made by adults are known as primary prevention.

Solution to Question 17:

Prescriptive screening of a disease is an example of secondary prevention.

Prescriptive screening of disease can lead to early diagnosis of disease. By early diagnosis, intervention can be made to prevent disability. Hence, it comes under secondary prevention.

Solution to Question 18:

Prospective screening is a type of primary prevention.

Prospective screening is done on healthy individuals to screen for disease. It is done prior to the onset of the disease. E.g., screening donated blood for HIV before it is transfused. This type of prevention reduces the incidence of disease.

Note: Prescriptive screening is done for case detection. People are screened for their own benefits. It is an early diagnosis of diseases existing in them. This is a form of secondary prevention. E.g., neonatal screening for hypothyroidism, pap smear for detection of cervical cancer.

Solution to Question 19:

The given scenario of immunisation is an example of primary prevention.

Primary prevention is an action taken before the onset of disease. This reduces the possibility of disease occurrence. Hence, immunization is a form of primary prevention, through specific protection.

Examples of primary prevention include immunization, sanitation, solid and liquid waste management etc.

Solution to Question 20:

Asymptomatic patients are the target population for secondary prevention but not for primary prevention.

Primary prevention is implemented before the onset of the disease. Hence, all patients including asymptomatic ones are not included in the target group.

Solution to Question 21:

The usage of iodized salt in iodine-deficient areas like in hilly regions is an example of primary prevention as it provides specific protection.

Solution to Question 22:

Pap smear is a screening test for carcinoma cervix and is an example of secondary prevention.

Early diagnosis (including screening) and treatment are the components of secondary prevention. Some of the examples of secondary prevention are as follows:

- Papanicolaou (Pap) smear for early detection of cervical cancer
- Mammography for early detection of breast cancer
- Colonoscopies for early detection of colon cancer

- Blood Pressure Screening

Secondary prevention measures focus on the sub-clinical stage and the early clinical stage, such as dysplasia in this case.

These measures enable early detection and prompt effective intervention to correct departures from a state of health.

Solution to Question 23:

Disability prevention by appropriate treatment is an example of secondary prevention. This is because disability has not occurred yet so correct treatment can stop it from developing.

After the disability has set in, prevention of progression of disability from transitioning into a handicap is tertiary prevention. It is also called disability limitation.

Solution to Question 24:

Prescribing spectacles for refractive error is an example of disability limitation which is a component of tertiary prevention. This mode of intervention is applicable when the patient presents late in the pathogenesis phase.

The objective of this intervention is to prevent or halt the transition of the disease process from impairment to handicap.

Option B: Starting a patient on anti-tuberculous therapy is secondary prevention as it is a part of treatment.

Option C: Abstinence from smoking is a modification of the risk factor. Hence it is an example of primary prevention.

Option D: Early fixation of fractures will prevent disability. Hence, it is an example of secondary prevention.

Solution to Question 25:

Maternal and neonatal tetanus was eliminated from India in 2015, as certified by the WHO.

Important years for India (declared disease-free):

- Infective trachoma - 2017
- Yaws - 2016
- Maternal & neonatal tetanus - 2015
- Polio - 2014
- Guinea worm disease (dracunculiasis) - 2000

Worldwide eradication:

- Type-2 wild poliovirus - 2015
- Type-3 wild poliovirus- 2019
- Smallpox - 1980

Solution to Question 26:

The new ICD-11 has a new chapter on sexual health that contains gender incongruence, which was previously listed under mental health conditions in the ICD-10.

The ICD 11 was released by the WHO in June 2018 and was presented at the World Health Assembly in May 2019 for adoption by the Member States. It has come into effect from 1 January 2022.

Its designations for antibiotic resistance align with the Global Antimicrobial Resistance Surveillance System (GLASS). It is given by the WHO and is revised every 10 years.

New chapters added in ICD-11 are:

- Traditional medicine
- Sexual health: Gender incongruence has been moved from mental health to sexual health.
- Gaming disorder has been added under Addictive disorders.
- Immune system disorders, which will enable the better study of allergies.

It has more detail on cancer, specifically different types of skin cancer, as well as on heart health and weight disorders compared to ICD-10.

Principles of Epidemiology

Question 1:

Which of the following is not included in the format of the death certificate recommended by the WHO?

- a) Condition directly leading to death
- b) Antecedent causes
- c) Other significant conditions contributing to death
- d) Age of person at the time of death

Question 2:

You are filling the death certificate of a patient who died from rupture of the myocardium following an acute myocardial infarction. This was because of thrombosis in the coronary artery due to longstanding atherosclerosis. What will you record in line I (c) of the death certificate?

- a) Rupture of myocardium
- b) Atherosclerosis
- c) Coronary artery thrombosis
- d) Myocardial infarction

Question 3:

An HIV positive patient was suffering from multi-drug resistant tuberculosis. He died of respiratory failure due to lobar pneumonia. Which of the following options is the most appropriate for the line I (a) of the death certificate?

- a) Respiratory failure
- b) Lobar pneumonia
- c) HIV
- d) Multi drug resistant tuberculosis

Question 4:

Which of the following is not a morbidity indicator?

- a) Notification rates
- b) Years of potential life lost
- c) Spells of sickness
- d) Incidence

Question 5:

Which of the following statements about rate is false?

- a) It measures the occurrence of an event in a population.
- b) It has no specification of time.
- c) It has a multiplier.
- d) It's numerator is mostly part of the denominator.

Question 6:

To calculate the crude death rate of a country, at what time of the year is the population assessed?

- a) 1st July
- b) 31st March
- c) 31st December
- d) 1st May

Question 7:

For which of the following purposes is age-specific death rates not used?

- a) Identifying specific age groups at risk
- b) Comparison for different causes within the same population
- c) Comparison between different sexes in relation to accidents
- d) Comparison between different age groups in relation to accidents

Question 8:

An epidemiologist is assessing an outbreak of acute diarrhoeal disease in a region. He wants to predict the virulence of the causative agent. Which of the following parameters will he use for this purpose?

- a) Secondary attack rate
- b) Case fatality rate
- c) Survival rate
- d) Incidence rate

Question 9:

Case fatality rate is _____.

- a) Spreading power of a disease
- b) Killing power of a disease in time
- c) Resistance of a disease
- d) Killing power of a disease with no time interval

Question 10:

A village has a population of 30,000 people. 500 of them contracted a disease X, out of which 75 died. What is the case fatality rate?

- a) 15%
- b) 6%
- c) 13%
- d) 8%

Question 11:

In a village of 5000 people, an outbreak of cholera led to 200 cases with 50 deaths. What is the case fatality rate?

- a) 1%
- b) 2.5%
- c) 10%
- d) 25%

Question 12:

Which of the following is the simplest measure of the burden of a disease in a community?

- a) Case fatality rate
- b) Survival rate
- c) Proportional mortality rate
- d) Crude death rate

Question 13:

Among the following, which measurement can be used as a yardstick for the assessment of standards of therapy?

- a) Prevalence
- b) Case fatality rate
- c) Standardized mortality rate
- d) Survival rate

Question 14:

Standardisation was done before comparing the mortality rates of two countries. What is the rationale behind this?

- a) Due to difference in number of deaths.
- b) Due to difference in age distribution.
- c) Due to difference in the population density.
- d) Due to difference in cause of death.

Question 15:

Which of the following statements is false about the standardized mortality ratio?

- a) It is a form of direct standardization.
- b) Observed deaths and expected deaths are part of its formula.
- c) It permits adjustments for age.
- d) It can be used if the event of interest is occurrence of disease.

Question 16:

A party was attended by 110 people out of which, 40 did not eat fruit salad. Among the rest of those who ate fruit salad, 55 developed food poisoning. What is the attack rate?

- a) 46
- b) 56
- c) 78
- d) 50

Question 17:

Out of a family of 4 members, one of them developed COVID-19 infection. Four days later, 2 other members developed the same. What is the secondary attack rate?

- a) 50%
- b) 33%
- c) 66%
- d) 75%

Question 18:

Which of the following study designs would you prefer for calculating the incidence of a disease?

- a) Case control study
- b) Randomized control study
- c) Cohort studies
- d) Field trials

Question 19:

Which of the following statements about incidence are true?

- a) 3,5
- b) 1,3,4
- c) 2,4
- d) 1,2,5

Question 20:

A medical officer noticed that there were 300 new cases of cholera in an area in a year. The total population in that area is 30,000. What is the incidence of cholera?

- a) 0.1 per 1000
- b) 10 per 1000
- c) 100 per 1000
- d) 1 per 1000

Question 21:

Which of the following is a good indicator of efficacy of health programs?

- a) Case fatality rate
- b) Prevalence
- c) Secondary attack rate
- d) Incidence

Question 22:

Which of the following is false with regard to prevalence?

- a) It is a product of incidence and duration.
- b) It can be determined by a cross-sectional study.
- c) It is an example of rate.
- d) It is point prevalence unless specified.

Question 23:

The incidence of a disease is 4 per 1000 of the population with an average duration of 2 years. What is the prevalence?

- a) 8/100
- b) 2/1000
- c) 4/1000

d) 8/1000

Question 24:

A village has a total population of 100,000 with an under-16 population at 3%. The prevalence of goitre in the under-16 population is (0.3/100). What is the total number of under-16 population affected by goitre?

- a) 90
- b) 9
- c) 10
- d) 100

Question 25:

In a population of 5000, new cases of TB are 500 and old cases are 50. What is the prevalence of TB in this case?

- a) 11%
- b) 10%
- c) 13%
- d) 12%

Question 26:

In a village having a population of 5000, a diagnostic test for a particular disease showed the following results. What is the prevalence of the disease?

- a) 20%
- b) 15%
- c) 30%
- d) 10%

Question 27:

A randomized control trial showed an effective treatment against an infectious disease. It was then incorporated by the health system. Which of the following is subsequently expected to

happen?

- a) Incidence decreases.
- b) Prevalence increases.
- c) Prevalence remains same.
- d) Prevalence decreases.

Question 28:

There was an outbreak of hepatitis A infection in your town. Proper prevention strategies were enforced. What is expected to happen at first to the number of cases of hepatitis A?

- a) Decrease in prevalence.
- b) Decrease in incidence.
- c) No change in prevalence.
- d) No change in incidence.

Question 29:

A new drug was approved for the treatment of prostate cancer. It has been shown to prevent early deaths but not cure the cancer. What is expected to happen to the cases of prostate cancer over a period of time?

- a) Increase in incidence, no change in prevalence
- b) Decrease in incidence, increase in prevalence
- c) Increase in incidence, decrease in prevalence
- d) Increase in prevalence, no change in incidence

Question 30:

Cross product ratio can be arrived from:

- a) Ecological study
- b) Cohort study
- c) Cross sectional study
- d) Case control study

Question 31:

A person wants to study about the relationship between smoking and lung cancer, so he collects data about people diseased with lung cancer from government hospitals and the number of cigarette packets sold at the same time period. Which type of study is this?

- a) Cross sectional
- b) Ecological
- c) Experimental
- d) Quasi-experimental

Answer Key

Question No.	Correct Option
1	d
2	c
3	b
4	b
5	b
6	a
7	c
8	b
9	d
10	a
11	d
12	c
13	d
14	b
15	a
16	c
17	c
18	c
19	a
20	b
21	d
22	c

23	d
24	b
25	a
26	c
27	d
28	b
29	d
30	d
31	b

Detailed Explanations

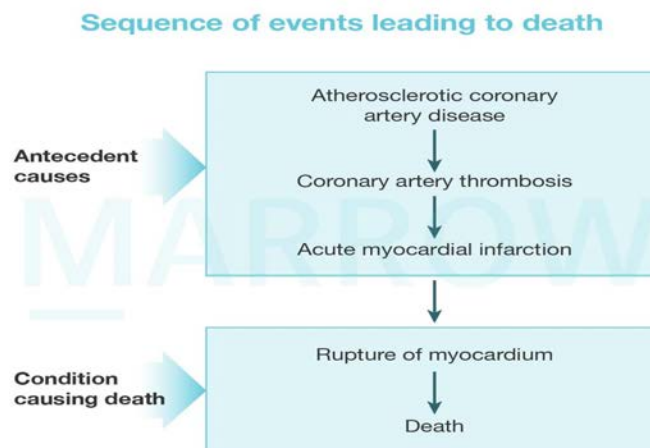
Solution to Question 1:

Age at the time of death is not included in the WHO format for a death certificate.

Solution to Question 2:

Coronary artery thrombosis is to be filled in line I (c) of the death certificate as one of the antecedent causes in the order of the events that took place.

The flow of events in this patient is as follows:



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CAUSE OF DEATH		Approximate interval between onset and death
I <i>Disease or condition directly leading to death*</i>	Rupture of myocardium (a) due to (or as a consequence of)	
<i>Antecedent causes</i> Morbidity conditions, if any, giving rise to above cause, stating the underlying condition last	Acute myocardial infarction (b) due to (or as a consequence of) Coronary artery thrombosis (c) due to (or as a consequence of) Atherosclerotic coronary artery disease (d) due to (or as a consequence of)	
II <i>Other significant conditions contributing to the death, but not related to the disease or condition causing it.</i>		

* This does not mean the mode of dying e.g. heart failure, respiratory failure. It means the disease, injury or complication that caused death.

Death Certificate: The International Standard Form

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As per the recommendations of Forty-Third World Health Assembly 1990, an additional line (d) was added below (c) in part I of the death certificate, as approved by the WHO.

Part II will only be filled up with a significant condition contributing to the death but not related to the disease or condition causing it.

Solution to Question 3:

Lobar pneumonia is appropriate for line I (a) of the death certificate. Line I (a) consists of the immediate cause of death such as the disease/condition. Immediate cause is the immediate/terminal event leading to death

In this patient:

- I(a) - Disease directly leading to death: Lobar pneumonia
- I(b) - Antecedent cause: MDR tuberculosis.
- II- HIV

Respiratory failure is a mode of dying in this patient. It is not the cause of death.

Terminal events like circulatory failure and respiratory failure, are modes of dying. They should not be written as the immediate cause of death in the death certificate because they are just signs of death.

Simply put, respiratory failure is 'how he died' whereas lobar pneumonia is 'why he died'. Therefore, lobar pneumonia (why he died) is the cause of the death.

CAUSE OF DEATH		Approximate interval between onset and death
I <i>Disease or condition directly leading to death*</i>	(a) Lobar pneumonia due to (or as a consequence of)	_____
<i>Antecedent causes</i> Morbidity conditions, if any, giving rise to above cause, stating the underlying condition last	(b) MDR tuberculosis due to (or as a consequence of)	_____
	(c) _____ due to (or as a consequence of)	_____
	(d) _____ due to (or as a consequence of)	_____
	_____	_____
II <i>Other significant conditions contributing to the death, but not related to the disease or condition causing it.</i>	HIV _____	_____
* This does not mean the mode of dying e.g. heart failure, respiratory failure. It means the disease, injury or complication that caused death.		_____

Death Certificate: The International Standard Form

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Solution to Question 4:

Years of potential life lost is not a morbidity indicator. It is a mortality indicator. It is the years of potential life lost through premature death.

Notification rates, spells of sickness and incidence are examples of morbidity indicators.

Solution to Question 5:

Rate is the measurement of the occurrence of a particular event in a given time. Hence, rates have a specification in time.

Examples include crude death rate, incidence rate etc.

Solution to Question 6:

The crude death rate is a mortality indicator, which uses the mid-year population as its denominator. The mid-year population is always attributed as the population on 1st July of any year.

Hence for calculating the crude death rate, the population is assessed on the 1st of July.

Solution to Question 7:

Sex-specific death rates are used for the comparison between sexes and not age-specific death rates.

Age-specific death rates are used for:

- Comparison for different causes within the same population.
- Comparison between different age groups in relation to a particular event.
- Identifying specific age groups at risk.

Solution to Question 8:

Case fatality rate is used in this scenario as it is the most useful parameter to predict the virulence of an acute illness. It represents the killing power of disease.

Note:

Case fatality rate is an incidence proportion and not a true rate as no time interval is used for its calculation.

Solution to Question 9:

Case fatality rate reflects the killing power of a disease with no specific time interval.

Solution to Question 10:

Case fatality rate is given by the formula:

$$\text{Case fatality rate} = \frac{\text{Total number of deaths due to a particular disease}}{\text{Total number of cases due to the same disease}} \times 100$$

The total number of cases of disease X= 500

The total number of deaths= 75

Hence the case fatality rate= $75/500 \times 100 = 15\%$

Solution to Question 11:

The case fatality rate is 25%.

Case fatality rate: (Total number of deaths due to a particular disease)/(Total number of cases due to the same disease) x 100

$$: 50/200 * 100 = 25\%$$

Solution to Question 12:

The proportional mortality rate is the simplest measure to estimate the burden of a disease in a community.

It is commonly used for broad disease groups and when population data is not available. It is of limited value in making comparisons between populations and in identifying risk groups.

Proportional mortality rate

Proportional mortality from a specific disease	=	$\frac{\text{Number of deaths from the specific disease in a year}}{\text{Total deaths from all causes in that year}} \times 100$
Under-5 proportionate mortality rate	=	$\frac{\text{Number of deaths under 5 years of age in the given year}}{\text{Total number of deaths during the same period}} \times 100$
Proportional mortality rate for aged 50 years and above	=	$\frac{\text{Number of deaths of persons aged 50 years and above}}{\text{Total deaths of all age groups in that year}} \times 100$

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Solution to Question 13:

The survival rate can be used as a yardstick for the assessment of standards of therapy.

It is used as a method of describing prognosis and is widely employed in cancer studies.

The survival rate is given by the following formula.

$$\text{Survival Rate} = \frac{\text{Total number of patients alive after 5 years}}{\text{Total number of patients diagnosed or treated}} \times 100$$

Solution to Question 14:

While comparing the mortality rates of two countries, the standardization of rates is done due to differences in age distribution.

The two types of standardisation include direct and indirect.

In both methods, a standard population is chosen where the numbers in each age and sex group are known and its choice is arbitrary.

The commonly used standard populations are the Segi world population and the European standard population.

Solution to Question 15:

Standardized mortality ratio is a form of indirect standardization.

Solution to Question 16:

The attack rate is calculated by the formula:

$$\text{Attack Rate} = \frac{\text{Number of new cases in a specified period of time}}{\text{Total population at risk during the given time}} \times 100$$

In this scenario, the number of new cases is 55 and the total number of those at risk is 70 as they ate the fruit salad.

Therefore attack rate = $55/70 \times 100 = 78.6\%$.

Attack rate is an incidence rate. It is used only when the population is exposed to risk for a limited period.

Solution to Question 17:

Secondary attack rate is calculated by the following:

$$\text{Secondary Attack Rate} = \frac{\text{Number of exposed persons developing the disease within the range of incubation period}}{\text{Total number of exposed or susceptible contacts}} \times 100$$

In this scenario, there is one primary case. The total exposed contacts are the other 3 members of the family. The number of persons developing the disease is 2.

So, secondary attack rate = $\frac{2}{3} \times 100 = 66\%$.

Note: The primary case is excluded from both the numerator and denominator.

Solution to Question 18:

The study designs which can be used to calculate incidence are cohort studies and longitudinal studies.

Solution to Question 19:

Incidence is not affected by the duration of the disease. Analysis of incidence rates may provide useful insights into the effectiveness of the health services provided.

Incidence is the number of new cases occurring in a defined population during a period of time.

Prevalence is more than incidence as it is the product of incidence and duration.

Incidence is a type of rate and not a proportion.

Solution to Question 20:

The incidence of cholera is 10 per 1000.

Incidence is given by the formula:

$$\text{Incidence} = \frac{\text{Number of new cases of a disease in a specified period of time}}{\text{Population at risk during that time}} \times 1000$$

In this scenario number of new cases = 300

Total population = 30,000

Hence the incidence rate = $(300/30,000) \times 1000 = 10$

Therefore it is 10 per 1000 per year.

Solution to Question 21:

Incidence is a good indicator of the efficacy of health services and programs.

Solution to Question 22:

Prevalence is a proportion, not a rate.

Prevalence is the proportion of all cases, old and new, of a particular disease at a given point in time or over a period of time in a given population. It can be determined from cross-sectional studies.

Types of prevalence:

- Point prevalence is the prevalence at a point in time. Usually, prevalence is point prevalence unless otherwise specified.
- Period prevalence is the prevalence over a period of time.

Solution to Question 23:

The prevalence in the given situation is 8/1000 population.

Prevalence is given by the formula :

Prevalence = Incidence x Mean Duration

Prevalence = $4/1000 \times 2 = 8/1000$ population

Solution to Question 24:

The total number of persons in the under-16 with goitre is 9.

Number of total under-16 population = 3% of 100,000 = 3000.

Thus, total number of goitre patients who are under-16 with the prevalence at the given value = $3000 \times 0.3/100 = 9$

Solution to Question 25:

Prevalence is given by the formula:

$$\text{Prevalence} = \frac{\text{Number of all cases of a specified disease at a given point in time}}{\text{Estimated population at the same point in time}} \times 100$$

In the given scenario total no. of cases : 50 (old) + 500 (new) = 550.

Hence prevalence = $(550/5000) \times 100 = 11\%$

Solution to Question 26:

Prevalence is given by the formula:

$$\text{Prevalence} = \frac{\text{Number of all cases of a specified disease at a given point in time}}{\text{Estimated population at the same point in time}} \times 100$$

In this scenario, the number of cases: $1000 + 500 = 1500$.

Total population: 5000

Prevalence = $1500/5000 \times 100 = 30\%$.

Solution to Question 27:

A new effective treatment for infectious disease would reduce the duration of the disease, thereby reducing prevalence.

Note: Both prevalence and incidence decrease but prevalence is the first to show a change in pattern. Hence, 'prevalence decreases' is the better answer.

Solution to Question 28:

In the given scenario, prevention of hepatitis A would lead to decrease in incidence at first.

Prevention of an acute disease would prevent new cases, thus first decreasing incidence.

Solution to Question 29:

A new drug that prevents death yet does not cure the disease will cause an increase in prevalence but the incidence remains the same.

In the given scenario, the drug reduces early deaths but does not cure prostate cancer. Hence, there is a paradoxical increase in the prevalence while the incidence remains the same.

Prevalence depends on two factors, the incidence and duration of illness. If the population is stable and incidence and duration are unchanging, the relationship between incidence and prevalence can be expressed as:

$$P = I \times D$$

(I - Incidence and D - Mean duration)

This equation shows longer the duration of the disease, the greater the prevalence.

Note:

- In chronic diseases like TB, prevalence will be higher than incidence.
- If a new treatment for a disease is effective, incidence does not change but prevalence decreases as the duration of disease decreases.
- If the treatment only prevents death but doesn't cure the disease, it may cause a paradoxical increase in prevalence.

Solution to Question 30:

Cross product ratio (Odds ratio) can be obtained from the case-control study.

Solution to Question 31:

The type of study described is an ecological study as the unit of study is the population. The number of cigarette packets sold at the same time is data about the population.

Option A: The unit of study in a cross-sectional study is an individual.

Option C: There is no experiment done here. Hence, it is not experimental study.

Option D: Quasi-experiments are nonrandomized, pre-post intervention studies done when it is not feasible or ethical to conduct a randomized controlled trial. The unit of study is therefore patients or individuals and not population.

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you may have been scammed.

Descriptive Epidemiology

Question 1:

A researcher is planning to study the etiology of certain painful idiopathic peripheral neuropathies. Which of the following study methods shall he choose?

- a) Cohort study
- b) Randomized clinical trial
- c) Ecological study
- d) Descriptive study

Question 2:

The preliminary observations made by Burkitt in Africa eventually found that the Epstein-Barr Virus is an etiological factor for Burkitt lymphoma. What is this study an example of?

- a) Descriptive study
- b) Analytical study
- c) Case-control study
- d) Retrospective cohort study

Question 3:

Which of the following are included in descriptive epidemiology?

- a) 1, 2 and 3
- b) 1, 3 and 4
- c) 2, 3 and 5
- d) 1, 4 and 5

Question 4:

Which of the following best describes the term hypothesis?

- a) Axiom

- b) Verified variable
- c) Variable to be tested
- d) Established theory

Question 5:

The National Institute of Allergy and Infectious Diseases is planning to study the natural history of COVID-19. Which of the following studies can be used?

- a) Longitudinal studies
- b) Cross-sectional studies
- c) Descriptive studies
- d) Case-control studies

Question 6:

What are prevalence studies also known as?

- a) Longitudinal studies
- b) Cohort studies
- c) Case-control studies
- d) Cross-sectional studies

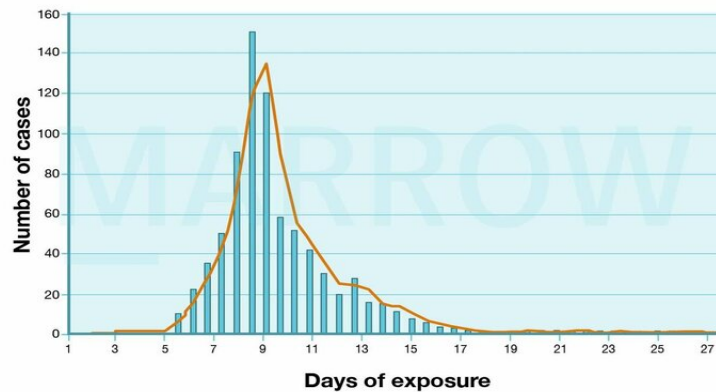
Question 7:

Which of the following statements about point source epidemic is false?

- a) Secondary waves do not occur
- b) Clustering of cases occur within a short period of time
- c) It occurs only due to an infectious agent
- d) All cases usually develop within one incubation period

Question 8:

Identify the type of epidemic depicted from the following epidemic curve.



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- a) Point-source epidemic
- b) Continuous epidemic
- c) Progressive epidemic
- d) Propagated epidemic

Question 9:

An outbreak of viral hepatitis A was reported from a town between June and September of a particular year. The peak of the outbreak occurred in July. What is this an example of?

- a) Common source, multiple exposures
- b) Common source, single exposure
- c) Propagated epidemic, single exposure
- d) Propagated epidemic, multiple exposure

Question 10:

The sales team of a corporate company presents to the hospital with complaints of sudden onset nausea, vomiting and loose stools. They give a history of having food from a nearby bakery three hours ago. What is this an example of?

- a) Common source, single exposure epidemic
- b) Common source, multiple exposure epidemic
- c) Propagated epidemic, single exposure

- d) Progressive epidemic, single exposure

Question 11:

Which of the following is an example of disease following a seasonal trend?

- a) Occurrence of rubella every 6-9 years in the pre-vaccination era
- b) Occurrence of influenza pandemic every 7-10 years
- c) Occurrence of bacterial gastroenteritis in summer
- d) Increase in the occurrence of diabetes in developed countries

Question 12:

The influenza pandemic is anticipated to occur every 7-10 years due to the antigenic variation of the virus. What is this an example of?

- a) Seasonal trend
- b) Secular trend
- c) Progressive trend
- d) Cyclical trend

Question 13:

How often did the cyclical trends of measles occurred in the pre-vaccination era?

- a) Every 2-3 years
- b) Every 5-7 years
- c) Every 6-9 years
- d) Every 3-5 years

Question 14:

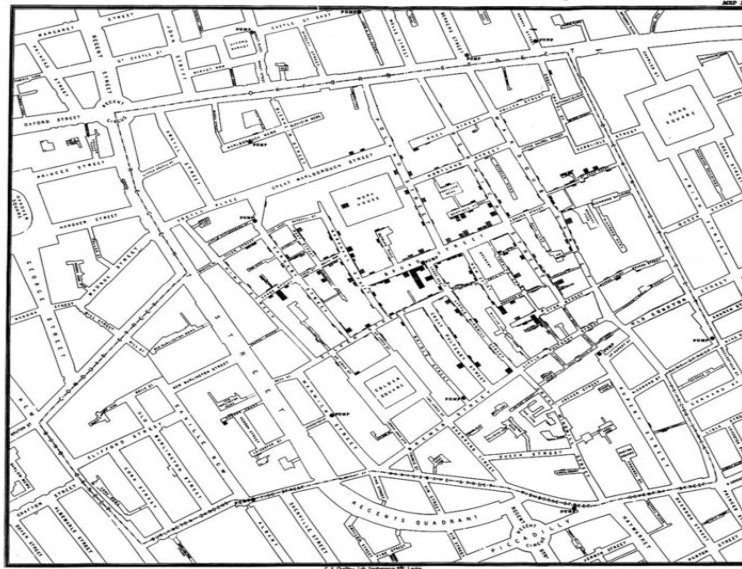
What is known as the change in the occurrence of a disease over very long periods of time?

- a) Propagated epidemic
- b) Secular trend
- c) Cyclical trend

d) Modern epidemic

Question 15:

Which of the following cannot be depicted by the image given below?



- a) Frequency of disease distribution
- b) Rural - urban distributions of a disease
- c) Pattern of disease distribution
- d) Boundaries of disease distribution

Question 16:

What are migration studies used to evaluate?

- a) Socio-demographic reasons for migration of a population
- b) Prevalence of diseases in a population
- c) Environmental and genetic factors of a disease in a population
- d) Incidence of disease in the immigrant population

Question 17:

What does bimodality in person distribution refer to?

- a) Description of two etiological hypotheses of a disease in one person

- b) Evaluation of two different environmental factors of a person in disease causation
- c) Two separate peaks in the age incidence curve of a disease
- d) Equal incidence of a disease in male and female population

Answer Key

Question No.	Correct Option
1	d
2	a
3	b
4	c
5	a
6	d
7	c
8	a
9	a
10	a
11	c
12	d
13	a
14	b
15	b
16	c
17	c

Detailed Explanations

Solution to Question 1:

Diseases of unknown etiology or new diseases can be studied with descriptive epidemiology.

Solution to Question 2:

The given scenario is an example of a descriptive study.

Descriptive studies are used to describe a disease of unknown etiology. These studies also led to the discovery of prions as the etiological agents of chronic degenerative neurological disorders (Kuru) in humans. This is also an example of a descriptive study.

Types of epidemiological studies:

- Observational studies - allow nature to take its own course, the investigator measures but does not intervene.
- Descriptive studies - limited to the description of the occurrence of a disease in a population.
- Analytical studies - go further by analyzing the relationship between health status and other variables.
- Ecological studies
- Cross-sectional studies
- Case-control studies
- Cohort studies
- Experimental studies or intervention studies involve an active attempt to change a disease determinant or the progress of a disease and are similar in design to experiments in other sciences.
- Randomized controlled trials
- Field trials
- Community trials

Solution to Question 3:

Descriptive epidemiology involves describing the disease in terms of time, place, and person distribution.

Solution to Question 4:

An epidemiological hypothesis is a supposition or variable. It is arrived at from observation and needs to be tested.

Descriptive studies primarily focus on the formulation of a hypothesis.

Analytical studies which include study designs like case-control and cohort are used to test the formulated hypothesis. The hypothesis can subsequently be rejected or accepted.

Experimental studies involve an intervention to provide scientific proof for etiological risk factors of diseases and to measure the efficiency of health services.

Solution to Question 5:

Longitudinal studies are useful to study the natural history of a disease.

It is also used to identify risk factors, and to find the incidence rate. In longitudinal studies, observations are repeated in the same population over a prolonged period of time by follow-up examinations. Their disadvantage is that they are difficult to organize and take a very long time.

Solution to Question 6:

Cross-sectional studies help in obtaining the prevalence of a disease. Hence, they are also known as prevalence studies.

Cross-sectional study is a type of observational study where a cross-section of a population is examined once at one point in time. It is more useful in studying chronic diseases than acute ones. They do not provide information on the natural history of disease or incidence and also do not establish the time sequence.

Solution to Question 7:

Point source epidemics are caused not only due to infectious agents but also by environmental pollution and contamination of food.

Point source epidemics belong to the common source type of epidemics. Exposure to disease agents is brief so they are also called single exposure epidemics.

All cases develop within one incubation period. In the epidemic curve, there is a sharp rise and fall as all cases are clustered within one incubation period. There are no secondary waves and these epidemics tend to be explosive.

Solution to Question 8:

Point-source epidemic is depicted in the given image.

In the epidemic curve for a point-source epidemic, there is a sharp rise and fall of the curve with no secondary waves. It tends to be explosive as all the cases are clustered in a narrow period of time. It occurs within one incubation period of the disease.

Solution to Question 9:

This outbreak of viral hepatitis A is an example of a common source, multiple exposure type of epidemic.

The incubation period of hepatitis A is approximately 28 days (range 15 to 50 days). In this scenario, the cases of viral hepatitis A have occurred over a period from June to September (120 days) which is more than one incubation period of the disease. This is suggestive of multiple

exposures from a common source.

If the period of the outbreak falls within one incubation period, then the type of epidemic is common source, single exposure or point source epidemic.

Difference between single exposure and multiple exposure epidemics:

Propagated epidemic results from person to person transmission of the infectious agent over a prolonged duration of time. For e.g. Polio epidemic, COVID-19.

Note: Hepatitis A epidemic is generally an example of a propagated epidemic. However, it does not always follow this pattern. In the given scenario, since the outbreak lasts for a short duration of 3 months (about 3 incubation periods for hepatitis A), it is considered to be a point source, multiple exposure outbreak.

	Common source, single exposure epidemic	Common source, multiple exposure epidemic
Nature of exposure	Single, brief exposure	Multiple, continuous or intermittent exposures for a prolonged time
Nature of epidemic	Explosive with clustering of cases within a short interval of time	Less chance of clustering of cases within short interval of time
Duration of epidemic	Falls < 1 incubation period of a particular disease	Extends > 1 incubation period of a particular disease
Example	An epidemic of food poisoning	Outbreak of cholera

Solution to Question 10:

The given scenario is suggestive of food poisoning. It is a type of common-source (point-source), single exposure epidemic.

Solution to Question 11:

Occurrence of bacterial gastroenteritis in summer is an example of a disease following seasonal trend.

Seasonal trend is a type of periodic fluctuation. It refers to the occurrence of a disease in seasonal variation influenced by environmental conditions like climate, rainfall, life cycle of vectors, heat, etc.

Examples are:

- Measles and varicella in early spring
- Bacterial gastroenteritis in summer
- Heatstroke in summer

- Dengue in July-November
- Respiratory infections in winter.

Solution to Question 12:

The influenza pandemic occurring every 7–10 years due to antigenic variations of the virus is an example of a disease showing cyclical trend.

Diseases occurring in cycles spread over a short period of time show cyclical trends of time distribution. Cyclical trend is a type of periodic fluctuation.

Examples include:

- Measles: every 2-3 years in the pre-vaccination era.
- Rubella: 6-9 years in the pre-vaccination era.

Non-infectious conditions can also show cyclical trends.

Solution to Question 13:

In the pre-vaccination era, the cyclical trends of measles occurred every 2-3 years, while that of rubella occurred every 6-9 years.

Solution to Question 14:

A secular trend refers to the changes in the occurrence of a disease over long periods of time (decade).

Usually, disease distributions tend to move in one particular direction over a period of time.

Examples: Decrease in diseases like polio, tuberculosis, and typhoid fever, but increase in diabetes and coronary artery disease in developed countries.

Solution to Question 15:

The given image is of a spot map. Spot maps are not used to depict rural-urban distributions of a disease. They help to understand the local distributions of a disease.

They are also called shaded maps. This method was used by John Snow to investigate the epidemic of cholera in London in 1854. They depict areas of high or low frequency, the boundaries, and patterns of disease distribution.

Solution to Question 16:

Migration studies are used to evaluate the role of genetic and environmental factors of a disease in a population.

The two methods used are:

- Comparing disease and death rates of migrants with kin who stayed at home
- Comparing disease and death rates of migrants with the local population of the host country

Limitations of these studies:

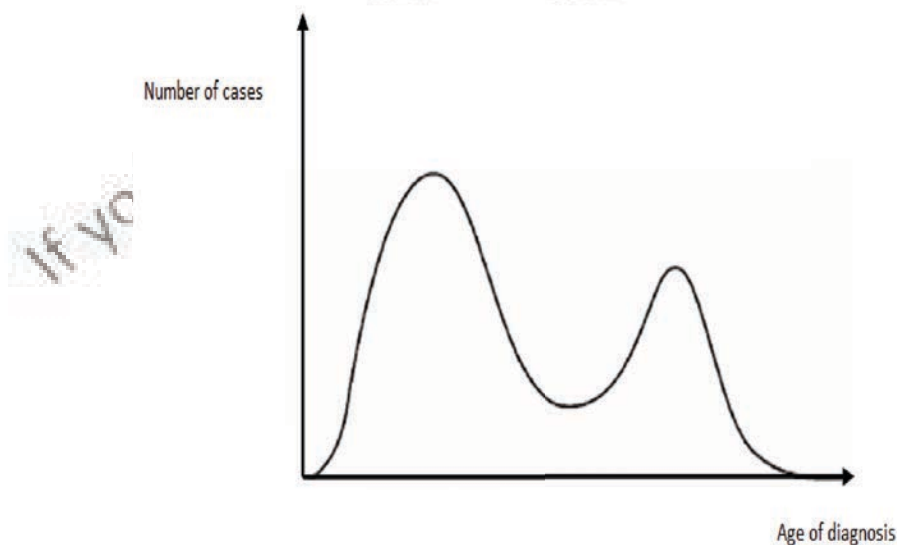
- Non-randomization of these groups
- Environmental factors may act only up to a certain age
- Not useful for diseases with long incubation periods

Solution to Question 17:

Bimodality refers to the occurrence of two separate peaks in the age incidence curve of a disease.

Example: Hodgkin's lymphoma shows a bimodal curve with peaks at 15–20 years and 70–80 years.

The image below shows bimodality in the age incidence curve:



Analytical Epidemiology

Question 1:

You are conducting a cohort study in a group of headload workers to identify the risk factors for developing low back pain in future. This type of study falls under which of the following?

- a) Experimental analytical studies
- b) Quasi-experimental analytical studies
- c) Non-experimental analytical studies
- d) Interventional analytical studies

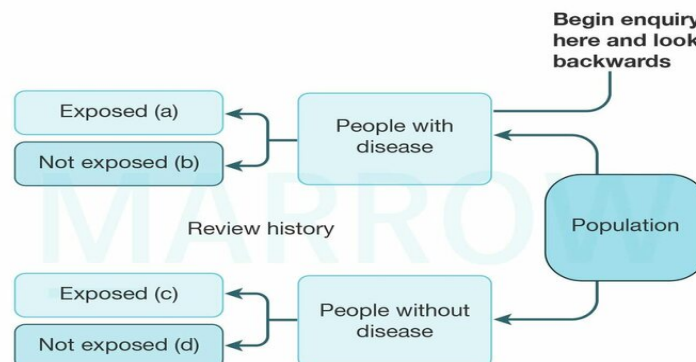
Question 2:

What is the unit of observation in ecological studies?

- a) Population
- b) Patient
- c) Healthy individuals
- d) Case

Question 3:

Identify the type of study from the following design:



- a) Prospective cohort study
- b) Retrospective cohort study
- c) Case-control study
- d) Follow-up study

Question 4:

A study was conducted among a group of 500 patients with lung cancer and another group of 300 people chosen from the general population. Both the groups were questioned about their exposure to smoking. What kind of study is this?

- a) Cohort study
- b) Cross-sectional study
- c) Correlational study
- d) Case-control study

Question 5:

A case-control study was conducted to evaluate the association between lifestyle risk factors and the prevalence of dyslipidemia. Which of the following is not an advantage of this type of study?

- a) Incidence can be calculated
- b) No risk to subjects
- c) Easy to carry out
- d) Suitable for rare diseases

Question 6:

A case-control study was conducted to see if symptoms of upper gastrointestinal dysfunction occurred more frequently in patients with Duchenne muscular dystrophy. Which of the following cannot be used as controls in this study?

- a) Hospital patients
- b) Siblings
- c) General public
- d) Healthcare workers

Question 7:

Which of the following is not a cause of bias in a case-control study?

- a) Recall bias
- b) Confounding bias
- c) Hawthorne bias
- d) Selection bias

Question 8:

Which of the following is not true about random errors?

- a) Sampling errors are random errors
- b) It cannot be completely eliminated
- c) It can cause bias
- d) Systematic errors are different from random errors

Question 9:

Which of the following factors in a study will cause Berksonian bias?

- a) Interviewer asking leading questions to the subjects
- b) Different rates of hospital admission of different diseases
- c) Researcher's belief in the treatment changing the outcome of the study
- d) Cases that are lost to follow-up

Question 10:

An epidemiologist plans to conduct a prospective study instead of a retrospective study. Which of the following disadvantages does he aim to eliminate by conducting this study?

- a) Recall bias
- b) Selection bias
- c) Confounding bias
- d) Berksonian bias

Question 11:

A study conducted in a group revealed that consumption of alcohol increases the risk of esophageal cancer, but it may be actually due to cigarette smoking. This happens due to:

- a) Confounding factor
- b) Misclassification bias
- c) Ecological fallacy
- d) Effect modifier

Question 12:

Which of the following is false regarding confounding factor?

- a) It can be reduced by matching
- b) It is associated individually with both cause and effect
- c) It is distributed equally in both study and control groups
- d) It is associated with the exposure of the study

Question 13:

A researcher is planning to conduct a study to investigate the relationship between oral contraceptives and breast cancer. He wants to remove age as a confounding factor. Which of the following methods will not help him reduce this bias?

- a) Stratification
- b) Randomization
- c) Deletion
- d) Regression analysis

Question 14:

Matching is done for which of the following?

- a) Known confounding
- b) Unknown confounding
- c) Known and unknown confounding

d) Sampling errors

Question 15:

Which among the following is not a type of matching?

- a) Frequency matching
- b) Collective matching
- c) Group matching
- d) Matching by pairs

Question 16:

An epidemiologist determines to evaluate the strength of association between asbestos exposure and mesothelioma in a case-control study. Which of the following variables would he use for it?

- a) Absolute risk ratio
- b) Relative risk
- c) Attributable risk
- d) Odds ratio

Question 17:

Which of the following statements is false about the odds ratio?

- a) $OR = 1$ indicates positive association
- b) It is obtained from case-control studies
- c) $OR < 1$ indicates possible protection
- d) It is a measure of strength of association

Question 18:

A case-control study was done among 500 lung cancer patients and 400 controls. The study groups had 400 and 50 smokers respectively. Calculate the odds ratio.

- a) 2.8
- b) 0.28

- c) 0.20
- d) 28

Question 19:

A research scholar is preparing to do a prospective cohort study in a hospital to evaluate the association of H. pylori with inflammatory bowel disease. Which of the following measures must she take, before starting the study?

- a) Enrol all patients taking proton pump inhibitors
- b) Enrol all patients on whom endoscopy was done
- c) Exclude all patients diagnosed with inflammatory bowel disease
- d) Exclude all patients with positive urea breath test for H. pylori

Question 20:

Which of the following is not true about cohort study?

- a) Incidence can be calculated
- b) It is always a prospective study design
- c) Attrition is very common
- d) Outcome is not present at the start of study

Question 21:

Hawthorne effect is seen in which of the following studies?

- a) Case-control studies
- b) Retrospective studies
- c) Cohort studies
- d) Cross-sectional studies

Question 22:

What is the ratio of the incidence of the disease in the exposed to that of the non-exposed?

- a) Attributable risk

- b) Odds ratio
- c) Relative risk
- d) Population attributable risk

Question 23:

Calculate the risk ratio when the incidence of the disease in exposed is 30 and that in non-exposed is 5.

- a) 6
- b) 5
- c) 0.6
- d) 0.5

Question 24:

A cohort study was conducted among 10,000 smokers and 10,000 non-smokers. They were followed up over a period of 20 years. It was found that 100 of the smokers and 10 of the non-smokers developed oral squamous cell carcinoma. Calculate the relative risk.

- a) 0.1
- b) 0.001
- c) 1
- d) 10

Question 25:

While testing the deleterious health hazards on a group of people after giving a new drug and a placebo, the following results were obtained as tabulated. Based on these results, which of the following is true?

- a) Absolute risk reduction (ARR) is 1% and relative risk reduction (RRR) is 0.1
- b) Absolute risk reduction (ARR) is 10% and relative risk reduction (RRR) is 0.9
- c) Absolute risk reduction (ARR) is 1% and relative risk reduction (RRR) is 10
- d) Absolute risk reduction (ARR) is 10% and relative risk reduction (RRR) is 9

Question 26:

The difference between the incidence of disease among exposed and incidence among non exposed is given by which of the following variables?

- a) Relative risk
- b) Attributable risk
- c) Population Attributable risk
- d) Odds Ratio

Question 27:

A study conducted among smokers shows that 80% of the cases of lung cancer are due to cigarette smoking. Which of the following variables indicates this?

- a) Risk ratio
- b) Relative risk
- c) Attributable risk
- d) Population attributable risk

Question 28:

A study found that the incidence of bladder carcinoma is 5 times higher in smokers than in non smokers. Calculate the attributable risk.

- a) 20%
- b) 40%
- c) 60%
- d) 80%

Question 29:

In a population of 9000 people, 2100 are alcoholics. 70 alcoholics and 23 non-alcoholics developed cirrhosis. What is the attributable risk?

- a) 40%
- b) 60%
- c) 70%
- d) 90%

Question 30:

Which of the following is the numerator in the formula for population-attributable risk?

- a) Incidence among exposed minus incidence among non-exposed
- b) Incidence among non-exposed minus incidence among exposed
- c) Incidence in the total population minus incidence among non-exposed
- d) Incidence in the total population minus incidence among exposed

Question 31:

A study was done in the Indian population to correlate the deaths due to myocardial infarction and dietary practices. What is this type of study known as?

- a) Cohort study
- b) Case reference study
- c) Ecological study
- d) Experimental study

Question 32:

Which among the following is the weakest study design to prove causation?

- a) Cohort study
- b) Case control study
- c) Randomized control trials
- d) Ecological study

Question 33:

What kind of bias is implied by assigning group characteristics to people which they may not possess as individuals?

- a) Ecological fallacy
- b) Hawthorne effect
- c) Berksonian bias
- d) Attrition

Answer Key

Question No.	Correct Option
1	c
2	a
3	c
4	d
5	a
6	b
7	c
8	c
9	b
10	a
11	a
12	c
13	c
14	a
15	b
16	d
17	a
18	d
19	c
20	b
21	c
22	c
23	a
24	d
25	a
26	b
27	c
28	d
29	d
30	c
31	c
32	d

Detailed Explanations

Solution to Question 1:

Cohort studies fall under non-experimental (observational) analytical studies.

Classification of epidemiological studies:

- Observational studies: allow nature to take its own course, the investigator measures but does not intervene. It is further classified into:
 - Descriptive studies: limited to describing the occurrence of a disease in a population.
 - Analytical studies: go further by analyzing the relationship between health status and other variables. It includes:
 - Ecological studies
 - Cross-sectional studies
 - Case-control studies
 - Cohort studies
- Experimental studies or interventional studies: it involves an active attempt to change a disease determinant or the progress of a disease, and is similar in design to experiments in other sciences.
 - Randomized controlled trials
 - Field trials
 - Community trials

Solution to Question 2:

In ecological studies, the unit of observation is the population or community.

Type of study	Unit
Ecological/ Correlational	Population
Cross-sectional/ Prevalence	Individual
Case-control/ Case reference	Individual
Cohort/ Follow up	Individual
Randomised Controlled Trials	Patients
Field Trial	Healthy people

Type of study	Unit
Community trial	Community

Solution to Question 3:

The design given is of a case-control study. It proceeds from effect to cause and is done after the exposure and outcome have occurred.

It tries to determine the possible exposure factors after a known disease incidence. This is the first approach to test causal hypotheses and uses controls to support or refute an inference.

Retrospective or historical cohort study, on the other hand, tries to compare the risk of developing a disease to some already known exposure factors. Ex: risk of lung cancer in uranium miners, risk of angiosarcoma in persons exposed to polyvinyl chloride etc.

Retrospective cohort studies are more economical and quicker than prospective cohort studies.

Solution to Question 4:

This is a case-control study. In this scenario, 500 patients constitute the cases and 300 people from the general population constitute the controls. The study is done from effect (lung cancer) to cause (smoking).

Solution to Question 5:

Incidence cannot be calculated from case-control studies. Incidence can be calculated from cohort studies, also known as longitudinal studies.

Solution to Question 6:

Siblings cannot be used as controls while studying genetic conditions because the outcome would be distorted due to their similarities.

Controls in a case-control study:

- Usually disease-free
- Can be up to 4 times the number of cases in small studies
- In large studies, a 1:1 ratio of cases and controls can be sought
- Sources: general population, hospitals, relatives (spouses and siblings), neighbors, or colleagues
- Although, siblings cannot be used as controls for studies on genetic conditions

Solution to Question 7:

Hawthorne bias is not seen in case-control studies. This bias occurs when the subjects change their attitude/ actions while under observation. This distorts the results of the cohort or experimental studies.

The bias that are seen in the case-control study are:

- Confounding
- Recall bias
- Selection bias
- Berkesonian bias
- Interviewer's bias

Solution to Question 8:

Random errors are not causes of bias.

Random error arises when a value of the sample measurement diverges, due to chance alone, from that of the true population value. They can never be completely eliminated.

Sources of random errors are individual biological variations, sampling errors, and measurement errors.

Systematic errors are also called biases. They are different from random errors.

Solution to Question 9:

Berkesonian bias occurs due to the different rates of hospital admission of people with different diseases. It is a type of selection bias termed after Joseph Berkeson who described it.

Solution to Question 10:

Recall bias can be eliminated in a prospective study.

Recall bias is also called memory bias. It is widely seen in retrospective (case-control) studies as extensive history is taken from the participants.

Solution to Question 11:

In this scenario, cigarette smoking is related to both the outcome of esophageal cancer and consumption of alcohol. Thus, this is an example of a confounding factor.

A confounding factor is defined as one which is associated with both exposure and outcome and is distributed unequally in study and control groups. In this study to determine the role of alcohol in the etiology of esophageal cancer, smoking is considered as a confounding factor because:

- It is associated with the consumption of alcohol (exposure).
- It is an independent risk factor for oesophageal cancer (outcome).

In these conditions, the effects of alcohol consumption can be determined only if the influence of smoking is neutralized by matching.

Solution to Question 12:

A confounding factor is a factor that is associated with both exposure and diseases and is distributed unequally in study and control groups. It can be reduced by matching.

For example, if the study is that a sedentary lifestyle can lead to coronary artery diseases, then age can act as a confounding factor as it is individually linked to both sedentary lifestyle and coronary artery diseases.

Solution to Question 13:

Deletion is not a method to reduce confounding.

Methods to reduce confounding are:

- At the time of designing the study:
 - Randomization
 - Matching
 - Restriction
- At the time of analysis:
 - Stratification
 - Statistical modelings like regression analysis, multivariate analysis, and the Cochran-Mantel-Haenszel method.

Solution to Question 14:

Matching helps only in removing the known confounding factors.

Matching is the process by which controls are selected in a way that they are similar to cases in certain pertinent variables (e.g., age) that are known to influence the outcome of the disease. If not adequately matched for comparability, these variables could confound the results.

Solution to Question 15:

There is no type of matching called collective matching.

Types of matching:

- Group matching: It is also called frequency matching. It is done when a large stratum is used for study and a 1:1 ratio of cases and controls can be achieved.
- Individual matching: It is also called matching by pairs. It can be done when the study is small and more controls can be selected for one case.

Solution to Question 16:

Odds ratio (OR), also known as cross-product ratio, is used to measure the strength of association between risk factor (asbestos exposure) and outcome (mesothelioma) in case-control studies.

A case-control study yields only odds ratio and not incidence or relative risk.

Solution to Question 17:

An odds ratio (OR) of 1 indicates that there is no association or that the event is likely in both the groups.

Solution to Question 18:

In the given scenario, the odds ratio (OR) is 28.

The given data is represented as:

Odds Ratio = ad/bc

$$= (400 \times 35) / (50 \times 100)$$

$$= (4 \times 35)/5$$

$$= 4 \times 7$$

$$= 28$$

	Disease +	Disease -
Exposure +	400 (a)	50 (b)

	Disease +	Disease -
Exposure +	100 (c)	350 (d)
Exposure -		

Solution to Question 19:

All patients diagnosed with inflammatory bowel disease (IBD) must be excluded from a prospective cohort study, as the study progresses from cause to effect. The study begins before the disease under study has occurred.

Solution to Question 20:

Although cohort studies are called prospective studies, there are subtypes of cohort studies that do not follow the prospective study design like historical (retrospective) cohort studies.

Framingham Heart Study is one of the most famous long-term cardiovascular prospective cohort studies.

Solution to Question 21:

Hawthorne effect is seen in cohort studies. It refers to the phenomenon when study subjects change or alter their behaviour during the course of the study, which results in different outcomes and underestimation of risks.

Solution to Question 22:

The ratio of incidence of disease or death among exposed to that among the non-exposed is called relative risk (RR).

Solution to Question 23:

In the given scenario, risk ratio or relative risk (RR) is 6.

Here, the incidence of disease in the exposed is 30 and that in the non-exposed is 5.

Thus, $RR = 30/5 = 6$.

Solution to Question 24:

The relative risk in the given scenario is 10.

Here, the incidence in exposed is 0.01 (100/10000). Incidence in non-exposed is 0.001 (10/10000).

Relative Risk = Incidence in exposed/ Incidence in non-exposed = $0.01/0.001 = 10$.

Solution to Question 25:

In the given scenario, absolute risk reduction (ARR) is 1% and relative risk reduction (RRR) is 0.1.

Let us reconstruct the design table in the following way:

Relative risk (RR) is given as $a/(a+b) \div c/(c+d)$.

$RR = 1603/15225 \div 1804/15225 = 0.106/0.118 = 0.89 \approx 0.90$.

Relative risk reduction (RRR) = $1 - \text{Relative risk}$. In this scenario, $RRR = 1 - 0.9 = 0.1$ or 10%.

Incidence of a disease is the absolute risk of contracting it.

Absolute risk rate of the group using new drug = $a/(a+b) = 1603/15225 = 0.106$ or 10.6% $\approx 11\%$.

Absolute risk rate of the group using placebo = $c/(c+d) = 1804/15225 = 0.118$ or 11.8% $\approx 12\%$.

Absolute risk reduction (ARR) is the difference in the absolute risk rates between the placebo and the drug. In this scenario, $ARR = 12\% - 11\% = 1\%$.

	Health hazard (+)	Health hazard (-)	Total
New drug	1603 (a)	13622 (b)	15225 (a + b)
Placebo	1804 (c)	13421 (d)	15225 (c + d)

Solution to Question 26:

The difference between the incidence of disease among exposed and incidence among non-exposed is given by attributable risk.

Attributable risk is often expressed as a percentage.

$$\text{Attributable risk (AR)} = \frac{\text{Incidence of disease rate among exposed} - \text{Incidence of disease rate among non-exposed}}{\text{Incidence of disease rate among exposed}} \times 100$$

Solution to Question 27:

Attributable risk indicates to what extent the disease under study can be attributed to the exposure. Here, cigarette smoking is attributed to be the cause of 80% of lung cancers in smokers.

Solution to Question 28:

In the given scenario, the attributable risk (AR) is 80%.

$$\text{Attributable risk (AR)} = \frac{\text{Incidence of disease rate among exposed} - \text{Incidence of disease rate among non-exposed}}{\text{Incidence of disease rate among exposed}} \times 100$$

Taking the incidence of disease in non-exposed as x, the incidence of disease in exposed would be 5x.

Attributable risk (AR) would be given as $[(5x - x)/5x] \times 100 = 80\%$.

Solution to Question 29:

In the given scenario, the attributable risk (AR) is 90%.

Number of people exposed (to alcohol) = 2100

Number of people not exposed (to alcohol) = 6900

Incidence of cirrhosis among exposed = $70/2100$ (I_{Exposed})

Incidence of cirrhosis among non-exposed = $23/6900$ (I_{Non-exposed})

Attributable risk (AR) = $[(I_{\text{Exposed}} - I_{\text{Non-exposed}}) / I_{\text{Exposed}}] \times 100$

= $[(70/2100 - 23/6900) / (70/2100)] \times 100$

= $[9 / 10] \times 100$

= 90

Thus, the attributable risk (AR) is 90%.

Solution to Question 30:

Incidence in the total population minus incidence among non-exposed is the numerator in population-attributable risk (PAR).

Population-attributable risk (PAR)

$$\text{Population-attributable risk (PAR)} = \frac{\text{Incidence of disease in total population} - \text{Incidence of disease in non-exposed}}{\text{Incidence of disease in total population}} \times 100$$

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Population attributable risk is obtained from cohort studies. It provides an estimate of the amount by which the disease could be reduced in that population if the suspected factor was eliminated or modified.

Solution to Question 31:

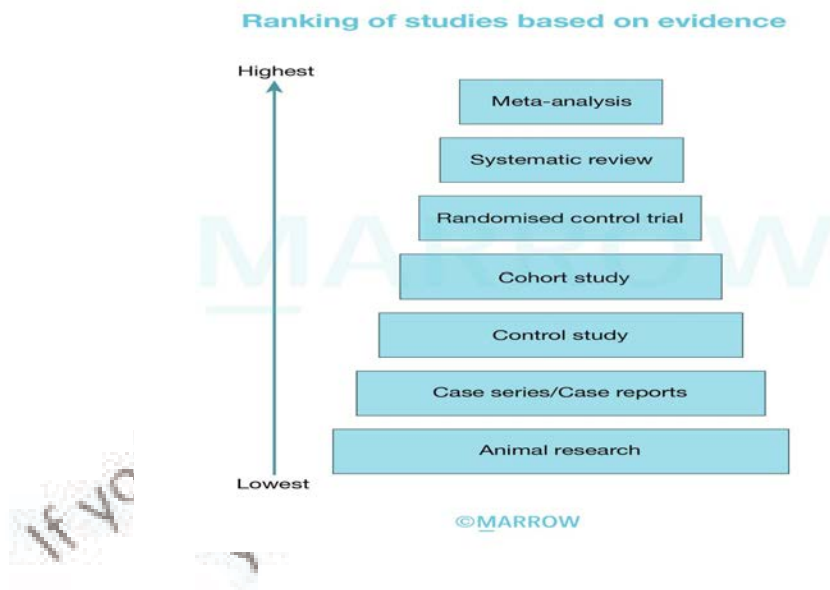
This is an example of an ecological study. As the study was done in the Indian population, it can be inferred that population was taken as the unit of the study.

The study was done to correlate the deaths and dietary practices, and hence ecological studies are also called correlational studies.

Solution to Question 32:

Ecological studies take a population as a unit. They provide the weakest evidence for causality because of extrapolation to individuals from regional data.

Randomized control trials (RCTs) provide the strongest design amongst the options given. Case-control and cohort studies have moderate ability to provide evidence for causality.



Solution to Question 33:

An ecological fallacy results when characteristics of the group are assigned inappropriately to individuals.

Experimental Epidemiology

Question 1:

Which of the following study designs is the best to prove causation?

- a) Case control studies
- b) Randomized controlled trials
- c) Cohort studies
- d) Ecological studies

Question 2:

In a clinical trial to evaluate a drug, randomisation was done before starting the trial. What is the purpose behind this?

- a) The two groups will be similar in prognostic factors
- b) The clinician has no control over the allocation of participants in each group
- c) The clinician can predict in advance which treatment the subjects will receive
- d) The sample may be referenced to a known population

Question 3:

Which of the following statements is false about randomised control trial?

- a) Baseline characteristics are similar in both arms
- b) Investigator's bias is minimized by double blinding
- c) The sample size required depends upon the hypothesis
- d) The dropouts from the trial are not included in the analysis

Question 4:

A researcher decided to use double-blinding in his trial evaluating two new therapies for cystic fibrosis. What is the characteristic feature of this concept?

- a) Study and control groups are blinded
- b) Observer and analyzer are blinded

- c) Observer and group being observed are blinded
- d) Group members and analyzer are blinded

Question 5:

In a clinical trial, what would be the best study design that ensures comparability between experimental and control groups?

- a) Cross-over study design
- b) Concurrent parallel study design
- c) Quasi trial
- d) Per protocol analysis

Question 6:

A new drug for multiple sclerosis is undergoing clinical trials where its therapeutic efficacy is being confirmed and adverse reactions are being monitored. Which of the following phases is the drug most likely in?

- a) Phase I
- b) Phase II
- c) Phase III
- d) Phase IV

Question 7:

In which of the following phases is the maximum drug failure rate seen?

- a) Phase I
- b) Phase II
- c) Phase IV
- d) Phase 0

Question 8:

The maximum tolerable dose of a new drug for Alzheimer's disease is to be evaluated. In which of the following phases is this done?

- a) Phase I
- b) Phase II
- c) Phase III
- d) Phase IV

Question 9:

A hypothesis stating that people who continue to smoke are at a higher risk of developing lung carcinoma than those who have quit is to be tested. Which of the following study designs would you choose for this purpose?

- a) Trial of etiological agents
- b) Clinical trial
- c) Cessation experiment
- d) Uncontrolled trial

Question 10:

Which of the following is not an example of a randomised controlled trial?

- a) Trial of aetiological agent
- b) Before and after comparison studies
- c) Preventive trials
- d) Evaluation of health services

Question 11:

What is the suspected cause preceding the observed effect an example of?

- a) Consistency
- b) Specificity
- c) Plausibility
- d) Temporality

Question 12:

While investigating a new disease, you are trying to establish the aetiological agent. Which of the following criterion is the most difficult to establish?

- a) Temporality
- b) Specificity
- c) Coherence
- d) Plausibility

Question 13:

What does evidence-based medicine refer to?

- a) Use of experiments to prove the cause of a disease
- b) Use of clinical trials to prove safety of drugs
- c) Use of research findings for taking decisions about patient care
- d) Use of behavioral modification to prevent disease

Question 14:

In the hierarchy of evidence, which of the following rankings is correct?

- a) 3 > 1 > 4 > 2 > 5
- b) 4 > 3 > 1 > 5 > 2
- c) 3 > 4 > 1 > 2 > 5
- d) 4 > 3 > 2 > 5 > 1

Question 15:

Which of the following is incorrect about systematic reviews?

- a) Meta-analysis is always performed
- b) Search for literature is compulsory
- c) Critical appraisal is criteria based
- d) Research question is always focused

Question 16:

A meta-analysis is being performed to evaluate the relationship between obesity and endometrial carcinoma. Which among the following is false regarding this technique?

- a) File drawer effect can be seen here
- b) It increases the statistical power by increasing sample size
- c) Apples and oranges effect can be seen here
- d) Its validity does not depend on the quality of the systematic review

Question 17:

Which of the following is not a use of epidemiology?

- a) Ensuring community participation
- b) Community diagnosis
- c) Syndrome identification
- d) Evaluation of individual risk

Question 18:

For what purpose were knowledge, attitude, and practice (KAP) studies first used in India?

- a) HIV
- b) Family planning
- c) Diabetes mellitus
- d) Malaria

Question 19:

Which of the following quantitative methods is the best to study previous research studies?

- a) Cohort study
- b) Systematic review
- c) Meta-analysis
- d) RCT

Answer Key

Question No.	Correct Option
1	b
2	b
3	d
4	c
5	a
6	c
7	b
8	a
9	c
10	b
11	d
12	b
13	c
14	c
15	a
16	d
17	a
18	b
19	c

Detailed Explanations

Solution to Question 1:

Randomized controlled trials (RCT) are the strongest and best study design to prove causation. They are the gold standard of clinical research.

Case-control and cohort studies have a moderate ability to provide evidence for causality. Ecological studies or correlational studies are considered as the weakest study design in this aspect.

The difference between analytical studies and randomised controlled trials is that in the former there is no randomisation because it requires the participants to be divided into diseased and non-diseased. This can lead to a number of biases that are avoided in RCTs.

Solution to Question 2:

The main purpose behind randomisation is to eliminate selection bias.

In an RCT, participants are randomly assigned to a study group or a control group. It is done only after obtaining consent. It removes both known and unknown confounding factors.

The heart of a randomized controlled trial (RCT) is randomisation. This allows comparability between the study and the control population.

Solution to Question 3:

The dropouts from an RCT are included in the final analysis. This type of analysis is called intention-to-treat (ITT) analysis.

It ignores noncompliance, protocol change, withdrawal, cross-over, and anything that happens after randomization during analysis. The exclusion of dropouts may lead to the introduction of bias.

Note: Per-protocol analysis is a comparison of treatment groups that include only those patients who completed the treatment originally allocated to them. Hence, comparability between the study and control group is limited and randomisation is lost.

Solution to Question 4:

In double-blind trials, both the group members (study and control group) and the observers are not aware of the group allocation.

Solution to Question 5:

A cross-over study would be the best method to ensure comparability between the experimental and control group.

In a cross over study design, patients are randomly assigned to the study and control group, and then after a period of time, their groups are switched. Here, each patient serves as his/her own control which allows for maximum comparability.

In a concurrent parallel study design, comparisons are made between two randomly assigned groups and there is no switching of groups. This allows only for limited comparability between experimental and control groups.

Quasi-experiments are studies that aim to evaluate interventions but that do not use randomization. The lack of randomisation is the major weakness of quasi trials. The comparability in a quasi trial is limited.

A per-protocol analysis is a comparison of treatment groups that include only those patients who completed the treatment originally allocated. Hence, comparability is limited.

Solution to Question 6:

The confirmation of therapeutic efficacy and monitoring of adverse reactions of a new drug happens in phase III of the clinical trial.

Purpose of phase III:

- Confirmation of the effectiveness of the drug in patients with the target disease
- Monitoring of adverse reactions

Phase II involves therapeutic exploration which is the evaluation of the efficacy of the drug and dose-ranging (determination of the optimum tolerated dose).

Solution to Question 7:

Maximum drug failure rates are seen in phase II followed by phase III of a clinical trial.

Solution to Question 8:

The maximum tolerated dose of a new drug is tested in phase I of a clinical trial.

Phase I trials are usually performed on healthy volunteers to check the safety and pharmacokinetic properties of a drug. The participants in this phase do not gain any direct health benefits from taking part. They are generally paid a fee for their time, with the payments being regulated and not related to the risk involved.

Solution to Question 9:

To test the given hypothesis that smokers are at a higher risk of developing lung carcinoma than those who have quit, the study design chosen is a cessation experiment.

The outcomes of termination of a habit or removal of an agent considered to be causally related to a disease can be evaluated in a cessation experiment.

It is a type of RCT and preventive trial. If the outcome is a significant reduction in disease then the association of that habit or agent to the disease is strengthened.

A trial of aetiological agents is an experimental epidemiological study to confirm or refute an aetiological hypothesis.

Clinical trials, for the most part, have been concerned with evaluating therapeutic agents, mainly drugs.

An uncontrolled trial is an example of a non-randomised trial where there is no control group for comparison.

Solution to Question 10:

Before and after comparison studies are a type of non-randomized trials.

These are community trials and centre around comparing the incidence of disease before and after the introduction of a preventive measure. They are of two distinct types:

- Before and after comparison studies without control.
- Before and after comparison studies with control.

Non-randomized trials include:

- Before and after comparison studies.
- Uncontrolled trials.
- Natural experiments.

Solution to Question 11:

Temporality refers to the condition when the suspected cause precedes the observed effect. It is a part of Hill's criteria for causal association.

Solution to Question 12:

Specificity is the most difficult criterion to establish in both chronic and acute diseases.

This is because a single factor or cause can give rise to more than one disease. Also, most diseases are multifactorial in origin with no possibility of demonstrating a one-to-one relationship which would indicate specificity.

Solution to Question 13:

Evidence-based medicine (EBM) refers to the use of modern research findings to guide decision-making about the treatment of individual patients.

EBM integrates clinical experience and patient values with the best available research information. Research and studies are given high importance in EBM, while opinions and conventional wisdom of experts are not taken into consideration in the practice of EBM.

Solution to Question 14:

The correct order of ranking of the hierarchy of evidence is:

Meta-analysis and systemic review > Randomised control trial > Cohort study > Case-control study > Case reports.

Solution to Question 15:

Meta-analysis is not always performed in systematic reviews.

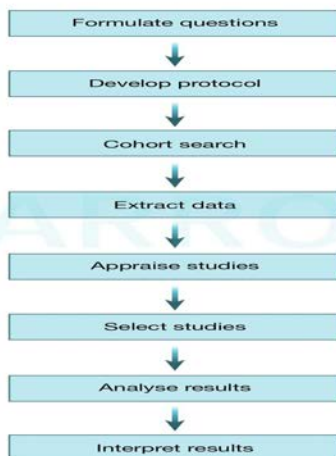
A systematic review is the process of locating and synthesizing the research on a particular question using organized, transparent, and replicable procedures. It ranks high in the hierarchy of evidence in medicine.

Steps of a systematic review:

- The research question is formulated. It is always focused.
- The literature on studies is always found and appraised after criteria of eligibility are established.
- Meta-analysis is usually done in a systematic review for quantification but it is not compulsory.

These steps are depicted in the diagram below.

Steps of systematic review



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Solution to Question 16:

The validity of the meta-analysis depends on the quality of the systematic review.

Solution to Question 17:

Ensuring community participation is not the use of epidemiology. Community participation is a principle of primary health care.

Uses of epidemiology:

- To study the rise and fall of disease in populations
- Community diagnosis
- Planning and evaluation
- Evaluation of individual's risks and chances
- Syndrome identification
- Completing the natural history of disease
- Searching for causes and risk factors

Solution to Question 18:

Knowledge, attitude and practice (KAP) studies were first used in India to study family planning. These studies help in evaluating the knowledge, attitude, and practices of a community. It also gives an educational diagnosis.

Solution to Question 19:

The quantitative method which is the best to study previous research studies of a selected topic/research question is meta-analysis.

A systematic review is a process of collecting, reviewing, and presenting information from earlier research to answer a question. It is a qualitative technique. It may or may not involve a meta-analysis.

Basic Definitions in Infectious Disease Epidemiology

Question 1:

What is the host in which a parasite passes through its sexual stages of maturity called?

- a) Obligate host
- b) Definitive host
- c) Intermediate host
- d) Secondary host

Question 2:

For which of the following diseases is man the dead-end host?

- a) Measles, yellow fever
- b) Tetanus, yellow fever
- c) Measles, rabies
- d) Tetanus, rabies

Question 3:

While studying a disease, you find that it affects both humans and lower vertebrates and can be transmitted in either direction. How would you classify this disease?

- a) Anthroponoses
- b) Zooanthroponoses
- c) Amphixenosis
- d) Epizootic

Question 4:

What is a disease imported to an area from another called?

- a) Epornithic disease

- b) Enzootic disease
- c) Epizootic disease
- d) Exotic disease

Question 5:

Which of the following best describes a disease that has a haphazard occurrence?

- a) Sporadic
- b) Endemic
- c) Epidemic
- d) Pandemic

Question 6:

In a community, there are usually 40-50 reported cases of dengue every week. This week, there were 48 reported cases. This would classify as:

- a) Epidemic
- b) Endemic
- c) Sporadic
- d) Outbreak

Question 7:

The following are the disease frequencies in a population. Based on the number of cases which of the following is the correct match?

- a) Pandemic, endemic, sporadic
- b) Sporadic, endemic, epidemic
- c) Endemic, epidemic, sporadic
- d) Endemic, sporadic, epidemic

Question 8:

Surveillance data of an endemic disease reveals that most of the children in the area had been affected and had to be hospitalised. However, most of the adults remained asymptomatic

carriers. Which category does this disease belong to?

- a) Hyperendemic
- b) Syndemic
- c) Quasi-endemic
- d) Holoendemic

Question 9:

In which of the following diseases is subclinical infection not seen?

- a) Hepatitis A
- b) Diphtheria
- c) Measles
- d) Japanese encephalitis

Question 10:

In which of the following states will the host not shed the infectious organism?

- a) Latent infection
- b) Convalescent carrier state
- c) Subclinical infection
- d) Healthy carrier state

Question 11:

While in a health camp, you notice a child with fever and rash. On subsequent history, you find out that similar symptoms occurred in her friend 6 days back. How will you epidemiologically categorise the child you examined?

- a) Primary case
- b) Index case
- c) Secondary case
- d) Tertiary case

Question 12:

You are part of a research programme for COVID-19 and are asked to define the median incubation period. How will you define it?

- a) Interval between onset of symptoms in primary and secondary cases
- b) Time required for 50% of cases to occur following exposure
- c) Interval between exposure and the occurrence of the first symptom
- d) Time required for the start of pathogenesis following exposure

Question 13:

In a school of 100 children, a child developed measles. Four days later, 3 of her classmates developed measles. What do these intervening four days correspond to?

- a) Lead interval
- b) Incubatory interval
- c) Serial interval
- d) Latent interval

Question 14:

The time interval between which of the following is called generation time?

- a) Development of primary case and secondary case
- b) Receipt of infection by host and maximum infectivity of the same host
- c) Infection and first sign or symptom
- d) Infection and initiation of pathogenesis inside the host

Question 15:

For which of the following diseases is soil not a reservoir?

- a) Coccidiomycosis
- b) Anthrax
- c) Mycetoma
- d) Brucellosis

Answer Key

Question No.	Correct Option
1	b
2	d
3	c
4	d
5	a
6	b
7	c
8	d
9	c
10	a
11	b
12	b
13	c
14	b
15	d

Detailed Explanations

Solution to Question 1:

A host in which a parasite passes through its sexual stage of maturity is called the definitive host or primary host.

A host in which a parasite exists in the larval or asexual form is called an intermediate host or secondary host.

A host which acts as the only source of development for a parasite is called the obligate host.

Note: A host in which the organism remains alive but no development occurs is called a transport host.

Solution to Question 2:

Among the given options, man is the dead-end host for tetanus and rabies.

A dead-end host is a host from which the infectious agent lacks a portal of exit to continue its propagation.

Humans are dead-end hosts for :

- Rabies
- Tetanus
- Bubonic Plague
- Trichinosis

Solution to Question 3:

A disease that affects both humans and lower vertebrates and which can be transmitted in either direction is classified as amphixenosis. E.g, Trypanosoma cruzi and Schistosoma japonicum.

It is a type of zoonoses. Zoonoses are diseases that are transmitted between vertebrate animals and humans.

Other types of zoonoses include:

- Anthroozoonoses - infections that are transmitted to man from vertebrates. E.g., Rabies, plague etc.
- Zooanthroponoses - infections that are transmitted to vertebrates from humans. E.g., Human tuberculosis in cattle.

Epizootic disease refers to an outbreak or epidemic of a disease in an animal population, often with the implication that it may also affect human populations. Enzootic disease is an endemic occurring in animals.

Note: Epornithic diseases are those diseases which are epidemic in bird populations. E.g., Psittacosis.

Solution to Question 4:

Exotic diseases are those imported into a country where they do not occur usually. E.g, rabies in the United Kingdom.

Enzootic diseases are those which are endemic in animals. E.g, anthrax, brucellosis.

Epornithic disease refers to an epidemic in birds. E.g, psittacosis.

Epizootic disease refers to an outbreak of a disease or an epidemic in animals with the implication that it may affect humans. E.g, Q fever, influenza.

Solution to Question 5:

Sporadic diseases occur in a haphazard and irregular manner from time to time.

Endemic refers to the constant presence and /or usual prevalence of a disease or infectious agent in a population within a geographic area.

Epidemic refers to an increase, often sudden, in the number of cases of a disease above what is normally expected in that population in that area.

Pandemic is an epidemic occurring over a wide area, crossing international boundaries, and usually affecting a large number of people.

Note: Opportunistic infections are those that occur in the presence of immune suppression or deficiency like cytomegalovirus.

Solution to Question 6:

The given scenario is an example of an endemic disease.

Endemic diseases are those that have a constant presence and/or usual prevalence of a disease or infectious agent in a population within a geographic area.

Epidemic refers to an increase, often sudden, in the number of cases of a disease above what is normally expected in that population in that area.

Outbreak carries the same definition of an epidemic, but is often used for a more limited geographic area.

Sporadic diseases occur in a haphazard and irregular manner from time to time.

Solution to Question 7:

Disease 1- the cases have been almost the same so it has to be endemic.

Disease 2- there is an alarming increase in the number of cases. It has to be an epidemic.

Disease 3- the number of cases has decreased without any explanation so it has to be sporadic.

Epidemic refers to the occurrence of a disease in excess of expected frequency.

A pandemic is an epidemic affecting a large population over a large geographical area crossing international boundaries.

An endemic disease is a regularly occurring disease in a population in expected frequency without any importation from outside.

Sporadic diseases present as scattered cases of infection without relation temporally/spatially.

Solution to Question 8:

The disease in question belongs to the category of holoendemic diseases. In such diseases, a higher prevalence and pathogenic response are seen among the children in an area. The adults tend to be either asymptomatic carriers or show reduced damage by the disease process. E.g, malaria.

An endemic disease is a disease that is present constantly in a population or in a given area without importation from outside. E.g, kala-azar in Bihar.

A hyperendemic disease is a disease that is constantly present at high incidence and prevalence among all groups in a population. E.g, dengue in India.

A syndemic is the aggregation of two or more concurrent or sequential epidemics or disease clusters in a population with biological interactions, which exacerbate the prognosis and burden of disease. It is also known as a synergistic epidemic.

Note: Quasiendemic is not a term used in infectious disease epidemiology.

Solution to Question 9:

Subclinical infection is not seen in measles. It always causes the development of overt signs and symptoms.

Subclinical infection refers to an inapparent, covert, missed or abortive case. The host harbours and sheds the infectious agent but does not develop signs and symptoms of the disease.

Subclinical infections are seen in diseases like:

- Hepatitis A
- Diphtheria
- Japanese encephalitis
- Rubella

Subclinical infections can only be detected by laboratory tests. They frequently occur during a person's lifetime and are responsible for the immunity shown by humans to a variety of disease-producing microbes.

Note: Subclinical cases should be differentiated from latent infections. In a latent infection, the host does not shed the organism but it lies dormant without signs and symptoms.

Solution to Question 10:

The host does not shed the infectious organism in case of a latent infection. This is defined as a state in which the host does not show signs and symptoms of the disease and does not shed the organism.

It can be seen in conditions like herpes simplex, ancylostomiasis, infections due to slow viruses, Brill-Zinsser disease, etc.

Subclinical infection is a state where the host harbours the infectious agent without developing signs and symptoms but sheds the infectious agent. This state is seen in conditions like hepatitis A, rubella, mumps, etc.

Healthy carriers are those who had subclinical infections and later developed a carrier state without suffering from the overt disease. This state is seen in: (mnemonic: CM - DPT)

- Cholera
- Meningitis
- Diphtheria
- Polio
- Typhoid

Convalescent carriers are those which shed the infectious agent during the convalescent phase/recovery phase. Such carriers are seen in diphtheria, typhoid, cholera, pertussis, etc.

Solution to Question 11:

In the given scenario, the child is categorised as the index case. It is the first case that comes to the notice of an investigator or physician.

The primary case is the first case of a communicable disease introduced into a population unit.

A secondary case is one that develops after contact with the primary case.

A suspect case refers to a patient who has all the features of the disease but has not been diagnosed with it yet.

Note: There is no such term as tertiary case.

Solution to Question 12:

The median incubation period is the time required for 50% of the cases to occur following exposure. The median incubation period of COVID-19 is approximately 5 days.

Incubation period is the time interval between invasion of the infectious agent and the appearance of the first symptom or sign in the body. It is termed as the latent period in non-infectious diseases.

Incubation period depends upon:

- Generation time of pathogen
- Portal of entry
- Infective dose
- Individual susceptibility

Incubation period is useful for :

- Tracing the source of infection and contacts.
- Determining the period of surveillance or quarantine.
- Planning immunization.
- Identification of point source or propagated epidemics.
- Determining the prognosis.

Solution to Question 13:

In the given scenario, the first child is the primary case and her classmates are the secondary cases. Hence, the interval between them developing the disease i.e., the intervening four days, is called the serial interval.

The time interval between the onset of the primary case and the secondary case is called serial interval.

By collecting information about a whole series of such disease onsets, the distribution of the secondary cases can be made. From this information, an approximation of the incubation period can be done.

Lead time/lead interval refers to the period between the first possible time of detection and the usual time of diagnosis.

Incubatory interval/ incubation period refers to the time period between invasion of a pathogen to appearance of the first sign or symptom.

Latent interval/ latent period refers to the period between disease initiation to detection of the disease. It is usually used for non-infectious diseases and is equivalent to the incubation period.

Note :

- Primary case is the first case of a communicable disease introduced into a population unit.
- Secondary case is one that develops after contact with the primary case.
- Index case is the first case that comes to the notice of an investigator or physician.

Solution to Question 14:

Generation time is the time interval between receipt of infectious agent by the host and the point of maximum infectivity of the same host.

Solution to Question 15:

The main reservoir of the coccobacilli belonging to the genus *Brucella* that cause brucellosis are animals like sheep, cattle, horses, goats, whales, and dogs.

A reservoir is the environment (e.g. soil, humans, animals, water, etc.) where the infectious agent grows and multiplies (natural habitat) and from where it can be transmitted to a susceptible host.

A source is an animal, object, human or substance which acts as an immediate mode of transmission and may or may not be a part of the reservoir.

Option A: Coccidiomycosis is also called Valley fever and the causative fungi belong to the Coccidiodes. The main reservoir of this organism is soil.

Option B: *Bacillus anthracis* is a soil commensal and is also found on the hides and dried skins of infected animals.

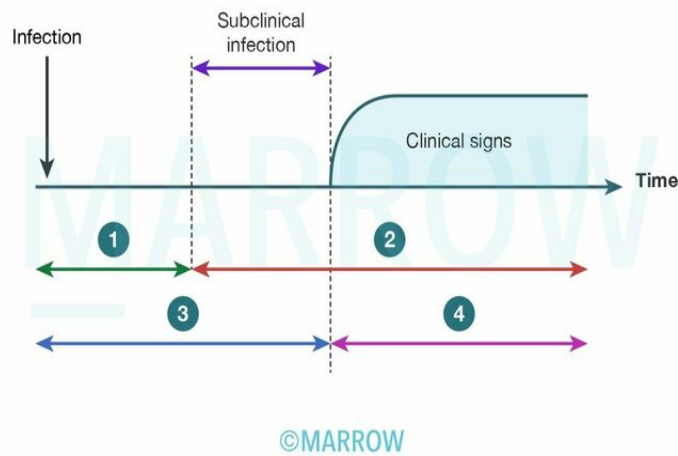
Option C: The natural reservoir for the fungi causing mycetoma is soil.

Sold by @Itachibot
If you purchased this from someone else,
you may have been scammed.

Dynamics of Disease Transmission

Question 1:

Which of the following is not communicable during the later part of the period marked 3 in the below image?



- a) Measles
- b) Whooping cough
- c) Diphtheria
- d) Rabies

Question 2:

Which of the following carriers shed the organism for indefinite periods?

- a) Convalescent carriers
- b) Healthy carriers
- c) Chronic carriers
- d) Paradoxical carriers

Question 3:

Which of the following diseases does not show vertical transmission?

- a) Tetanus
- b) Herpes
- c) Rubella
- d) Varicella

Question 4:

Which among the following is false about droplet transmission?

- a) All droplets can travel only as far as 3 feet
- b) Droplets are usually larger than 5 microns
- c) It is a type of direct transmission
- d) Basic ventilation decreases the potential for this type of transmission

Question 5:

What is the maximum size of airborne droplet nuclei?

- a) 5 microns
- b) 10 microns
- c) 3 microns
- d) 2 microns

Question 6:

Which of the following is not a method of direct transmission of communicable diseases?

- a) Vertical transmission
- b) Contact with soil
- c) Droplet nuclei
- d) Droplet transmission

Question 7:

Spread of disease via which of the following is an example of an indirect mode of transmission?

- a) Mother to child
- b) Droplet infection
- c) Dog bite
- d) Contaminated food

Question 8:

Which of the following statements about fomite-borne transmission is false?

- a) It is an indirect mode of transmission
- b) Transmission via contaminated food is an example of this
- c) Hepatitis A is transmitted in this way
- d) Routine disinfection can prevent such transmission

Question 9:

What is the lodgement and development of an arthropod on the body surface called?

- a) Infection
- b) Infestation
- c) Contamination
- d) Colonisation

Question 10:

An entomologist was studying a vector and the pathogen it transmits. He found that the pathogen underwent changes in number and morphology in the vector. What is this type of biological transmission called?

- a) Propagative
- b) Cyclo-developmental
- c) Cyclo-propagative
- d) Developmental

Question 11:

A patient with chronic lymphedema was found to have a positive DEC provocation test. Which type of biological transmission does the pathogen implicated here undergo?

- a) Propagative
- b) Cyclo-developmental
- c) Cyclo-propagative
- d) Developmental

Question 12:

You are a rural medical health officer investigating an epidemic. What is the correct sequence of steps you must follow?

- a) $a \rightarrow c \rightarrow b \rightarrow e \rightarrow d$
- b) $b \rightarrow e \rightarrow a \rightarrow d \rightarrow c$
- c) $a \rightarrow b \rightarrow e \rightarrow c \rightarrow d$
- d) $b \rightarrow a \rightarrow d \rightarrow e \rightarrow c$

Question 13:

You have been asked to write the final report following the investigation of a cholera epidemic. Which of the following will you not include in this report?

- a) History of previous epidemics
- b) Usual number of cases that existed before the epidemic
- c) Geographical distribution of the epidemic
- d) Incubation period

Question 14:

You have been sent to an area to study the ongoing typhoid epidemic. When will you declare the area to be free of the epidemic?

- a) No new cases for 14 days
- b) No new cases for 60 days
- c) No new cases for 45 days

- d) No new cases for 28 days

Question 15:

The probability that infection will occur among susceptible persons following contact with an infectious person within the incubation period is known as the_____.

- a) Secondary attack rate
- b) Case fatality rate
- c) Primary attack rate
- d) Tertiary attack rate

Question 16:

Following an outbreak of a disease, you are asked to calculate the secondary attack rate (SAR). What is the denominator you would use?

- a) All people having history of contact with a case
- b) All people susceptible in the entire area
- c) All the people in contact with the primary case
- d) All the close or exposed contacts those are susceptible

Question 17:

A village has 100 children under the age of 5. The coverage of measles vaccination is 60%. Following a case of measles, 26 children developed measles. What is the secondary attack rate?

- a) 41%
- b) 25%
- c) 69%
- d) 66%

Question 18:

A 28-year-old man is found to be positive for SARS-CoV-2. On contact tracing, 30 people have been identified as close contacts but none of them shows any symptoms or signs. What is the

time duration for which they must be quarantined?

- a) Maximum period of communicability
- b) Maximum incubation period
- c) Average period of communicability
- d) Average incubation period

Question 19:

Which of the following pairs of diseases and their periods of isolation is wrongly matched?

- a) Rubella – 3 days after onset of rash
- b) Chickenpox – 6 days after onset of rash
- c) Herpes zoster – 6 days after onset of rash
- d) Measles – Up to 5 days after onset of rash

Question 20:

A paradoxical carrier is defined as:

- a) A person who sheds pathogens during incubation period
- b) A carrier who acquired pathogen from another carrier
- c) A person who acquired pathogen from patient
- d) A patient who became a carrier

Answer Key

Question No.	Correct Option
1	d
2	c
3	a
4	a
5	a
6	c
7	d

8	b
9	b
10	c
11	b
12	d
13	d
14	d
15	a
16	d
17	d
18	b
19	a
20	b

Detailed Explanations

Solution to Question 1:

The period marked 3 in the figure is the incubation period while the later part of the period represents subclinical infection. Rabies is not communicable during its incubation period as it is transmitted from the saliva of a rabid animal through bite or scratch.

An incubatory carrier state is one in which carriers shed the organism during the incubation period of the disease, especially during the later parts of it.

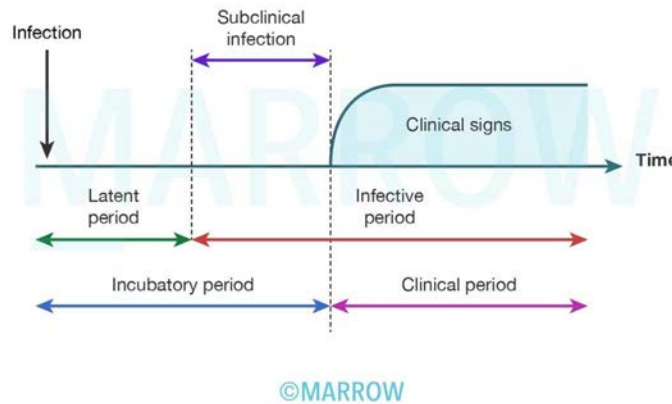
Examples of diseases showing incubatory carrier state are:

- Measles
- Polio
- Mumps
- Whooping cough
- Diphtheria
- Influenza
- Hepatitis A

Carriers of a disease are classified by the following:

- Type: Incubatory, convalescent or healthy.
- Duration: Temporary or chronic.
- Portal of exit: Urinary, respiratory, intestinal or others.

Periods of infectious diseases



Solution to Question 2:

Chronic carriers are those who shed or excrete the organism for indefinite periods.

Carriers are classified into chronic carriers and temporary carriers based on the duration.

Organisms may be shed continuously or intermittently in chronic carriers. The duration of the carrier state also depends on the disease. Chronic carrier state is seen in typhoid, hepatitis B, gonorrhea, malaria, etc.

Temporary carriers are classified into:

- Incubatory carriers - Those that shed the organism during the incubation period of the disease, especially during the later parts of it. It is seen in measles, mumps, polio, etc.
- Healthy carriers - Those who suffer from subclinical infections and develop a carrier state without developing the overt disease. It is seen in polio, meningococcal meningitis, cholera, diphtheria, etc.
- Convalescent carriers - Those that shed the organism during recovery or convalescence. It is seen in typhoid.

Paradoxical carrier refers to a carrier who acquires the pathogen from another carrier.

Note: Pseudo-carriers are carriers of avirulent organisms.

Solution to Question 3:

Tetanus does not show vertical or transplacental transmission.

Transmission of diseases or non-infectious agents transplacentally is called vertical transmission.

Diseases that show this type of transmission include:

- Herpes
- HIV
- Rubella
- Varicella

Apart from diseases, non-infectious agents like lithium, thalidomide, stilbestrol, etc. can cross the placenta and lead to the development of congenital anomalies in a foetus.

Note: Neonatal tetanus occurs as a result of the failure to use aseptic techniques in managing the umbilical stump, unclean hands, and instruments or contamination by dirt, straw, or other nonsterile materials in the delivery field. Immunisation of pregnant women can prevent this disease via the transplacental transport of immunoglobulins.

Solution to Question 4:

Droplet nuclei can travel as far as 6 feet, whereas a droplet (particle $> 5\mu\text{m}$) travels $\approx 30\text{-}60\text{cm}$, as recent studies have shown.

Droplets are a mode of direct transmission of infectious agents. They are formed from saliva and nasopharyngeal secretions released during coughing, sneezing, or speaking into the surrounding atmosphere. The expelled droplets may impinge directly upon the conjunctiva, oro-respiratory mucosa or skin of close contact. This mode of transmission is seen in common cold, diphtheria, etc.

Proper basic ventilation and prevention of overcrowding reduce the potential for this type of transmission.

Solution to Question 5:

Droplet nuclei are sized less than or equal to 5 microns.

They may be formed by evaporation of droplets coughed or sneezed into the air or be generated purposefully by a variety of atomizing devices (aerosols). They are capable of indirect airborne transmission of diseases like tuberculosis, influenza, chickenpox, measles, etc.

The droplet nuclei may remain airborne for long periods of time, some retaining and others losing infectivity or virulence. These small particles when inhaled are capable of reaching the alveoli and getting retained there.

Solution to Question 6:

Airborne transmission by droplet nuclei is not a method of direct transmission of diseases.

It is a method of indirect transmission of diseases.

Droplet nuclei are tiny particles of size 1-5 microns range that may be formed by either due to evaporation of droplets coughed or sneezed into the air or are formed from aerosols.

These droplet nuclei remain airborne for long periods of time and can transmit diseases like tuberculosis, influenza, chickenpox, measles, Q fever and other respiratory infections.

Droplet transmission, vertical transmission, and contact with soil are methods of direct transmission of diseases.

Differences between droplet and droplet nuclei are summarised below:

DROPLET NUCLEI	DROPLET
Indirect transmission	Direct transmission
It is the residue from evaporated droplets	It is produced while coughing, sneezing, talking
<5microns in diameter	≥5 microns in diameter
No need for close contact for transmission	Requires closeness and direct contact for transmission
Diseases transmitted: chickenpox, measles etc.	Diseases transmitted: diphtheria, meningococcal meningitis etc.

Solution to Question 7:

The spread of a disease through contaminated food is an example of vehicle-borne transmission. It is an indirect mode of transmission.

Vehicle-borne transmission is the transmission of infectious agents through food, water and biological products (blood, plasma etc.). Of these, food and water are more frequent vehicles of transmission because they are used by everyone.

Mother to child transmission, droplet infection and transmission via dog bite are examples of direct modes of transmission.

Solution to Question 8:

Fomites are objects other than water or food that can be contaminated by infectious agents and help in transmitting the disease. They include clothes, towels, linen, cups, plates, spoons, etc.

Fomites are an example of an indirect mode of transmission.

Some diseases transmitted by fomites include diphtheria, typhoid, hepatitis A, etc.

Good public and personal hygiene can prevent this type of transmission.

Solution to Question 9:

The lodgement and development of an arthropod on the body surface is called infestation.

Infection is the entry and development or multiplication of an infectious agent in a host's body.

Contamination is the presence of a disease agent on a body surface like clothes, toys, etc. and also in substances like food or water.

Colonization is the presence of a disease agent in or on a host, with growth and multiplication but without any overt clinical expression or detected immune response in the host at the time it is isolated.

Solution to Question 10:

This type of biological transmission where a pathogen undergoes a change in number and morphology in a vector is called cyclo-propagative transmission.

Malaria is a disease whose causative organism (*Plasmodium* sp.) undergoes cyclo-propagative transmission. In malaria, the parasite undergoes both developmental changes and multiplication in the *Anopheles* mosquito (vector).

Solution to Question 11:

In the given scenario, the patient most likely has lymphatic filariasis. The causative organism of this disease (*Wuchereria bancrofti*) undergoes cyclo-developmental transmission.

Cyclo-developmental transmission is a transmission where the agent undergoes only developmental changes and not multiplication within the vector.

DEC provocation test: In this test, a small dose of DEC is given 20-50 minutes before sample collection so that the microfilaria appears in the blood even during the daytime.

Solution to Question 12:

Steps for the investigation of an epidemic are:

- Verification of diagnosis
- Confirmation of the existence of an epidemic
- Defining the population at risk -
- Obtaining a map of the area
- Counting the population
- Rapid search for all the cases -
- Medical survey

- Epidemiological case sheets
- Searching for more cases
- Data analysis on the basis of time, place and person
- Formulation of hypothesis
- Testing of hypothesis
- Evaluation of ecological factors
- Further investigation of the population at risk
- Writing the report

Solution to Question 13:

The final report of an epidemic does not include the incubation period of the disease.

The final report following the investigation of an epidemic should include:

- Background information - Geography, climate, demography, socio-economic groups, health service organization, surveillance data and normal disease prevalence.
- Historical data - Previous occurrences of the epidemic, the occurrence of any related diseases and discovery of the primary case in the present outbreak.
- Methodology of investigations - Case definition, epidemiological questionnaires, collection and laboratory techniques.
- Analysis of data - Clinical and epidemiological data, modes of transmission, laboratory data and all their interpretations.
- Control measures - Strategy definitions, methodology of implementation, evaluation methods and preventive measures.

Solution to Question 14:

The area will be declared free of the typhoid epidemic when no new cases are reported for 28 days.

An area is declared free of an epidemic when there are no cases reported for twice the incubation period of the disease since the last case.

The incubation period of typhoid is 14 days. So, in the given scenario, the area will be declared free of the epidemic when no new cases are reported for twice the incubation period i.e., $14 \times 2 = 28$ days.

Solution to Question 15:

The probability that infection will occur among susceptible persons following contact with an infectious person within the incubation period is called secondary attack rate.

It is calculated by the following formula:

Secondary attack rate (SAR) = (Number of exposed or susceptible persons developing the disease within the incubation period x 100)/(Total number of close or exposed contacts that are susceptible)

Primary cases are usually excluded from the numerator and denominator.

Secondary attack rate measures the communicability of the disease.

Solution to Question 16:

The denominator used for calculating SAR is all the close or exposed contacts that are susceptible.

The secondary attack rate is defined as the probability that infection will occur among susceptible persons following contact with an infectious person within the incubation period.

It is calculated by the formula:

Secondary attack rate (SAR) = (Number of exposed or susceptible persons developing the disease within the incubation period x 100)/(Total number of close or exposed contacts that are susceptible)

Primary cases are usually excluded from the numerator and denominator.

Secondary attack rate measures the communicability of the disease.

Solution to Question 17:

The secondary attack rate for the given scenario is 66%.

The secondary attack rate (SAR) is given by:

(Number of exposed or susceptible persons developing the disease within the incubation period x 100)/(Total number of close or exposed contacts that are susceptible)

Here, number who developed the disease after exposure= 26

Susceptible population= 100 - 60 (vaccinated) - 1 (primary case) = 39

Therefore, SAR = 26 / 39 x 100 = 66%.

SAR is a measure of communicability.

Solution to Question 18:

Quarantine is the limitation in freedom of movement of healthy contacts of cases, who have been exposed for a period that lasts as long as the maximal incubation period of the disease.

It is a primary level of prevention and falls under specific protection measures. The various types of quarantine are:

- Absolute - Complete limitation of movement
- Modified - Selective partial limitation of movement
- Segregation - Observation of a group of persons in a population after separation

Solution to Question 19:

Patients with rubella should be isolated for 7 days after they develop the rash.

Isolation is defined as separation, for the period of communicability of infected persons from others to prevent or limit transmission of the infectious agent.

In settings where women who are in their 1st trimester or sexually active or non-immune without contraception usage may be exposed, outbreak control measures should begin as soon as rubella is suspected and should not be postponed until laboratory confirmation of cases.

Note: Multiple sources have confirmed the period of infectivity of measles which is from 4 days before the onset of rash to 5 days after the onset of rash.

Solution to Question 20:

A paradoxical carrier is a carrier that acquired a pathogen from another carrier.

Incubatory carriers are those who shed the infectious agents during the incubation period of the disease.

Convalescent carriers are those who continue to shed the disease agent during the period of convalescence.

Healthy carriers are those who had a subclinical infection and developed a carrier state without suffering from overt disease.

Principles of Immunization and Vaccination

Question 1:

Which of the following is false about herd immunity?

- a) It refers to group protection beyond what is afforded by the protection of immunized individuals
- b) Vaccination allows the possibility of 100% herd immunity against a disease
- c) It is affected by presence and distribution of alternative animal hosts
- d) It doesn't protect the individual in the case of tetanus

Question 2:

Herd immunity threshold does not vary with which of the following?

- a) Virulence of the disease
- b) Efficacy of the vaccine for the disease
- c) Number of vaccines for the disease
- d) Contact parameter for the population

Question 3:

Which of the following is false about normal human immunoglobulin (NHIG)?

- a) All live vaccines can be given simultaneously with NHIG.
- b) They are used against measles and hepatitis A.
- c) There should be a low IgA concentration in NHIG preparations.
- d) Each NHIG fraction is obtained from at least 1,000 donors.

Question 4:

Which of the following is false about specific human immunoglobulins?

- a) Used for Hepatitis B
- b) Used for chickenpox prophylaxis
- c) Obtained from random donors

- d) They are hyperimmune human immunoglobulins

Question 5:

Which of the following is not true about secondary booster immune response in comparison with primary response?

- a) Antibody elicited tends to have a greater avidity to bind to the antigen
- b) Antibody response is at a higher level
- c) Antibody response is maintained for a longer period of time
- d) Production of antibody is slower

Question 6:

Which of the following is false about live vaccines?

- a) Simultaneous administration of two live vaccines should be done in the same syringe
- b) Two live vaccines can be administered simultaneously
- c) Immuno-compromised state poses a contraindication to live vaccines
- d) Live vaccines contain both major and minor antigens

Question 7:

Which of the following is not a killed vaccine?

- a) Yellow fever vaccine
- b) Pertussis vaccine
- c) Rabies vaccine
- d) Salk vaccine

Question 8:

Which of the following is an absolute contraindication to administration of inactivated or killed vaccines?

- a) Immunocompromised state
- b) Egg allergy

- c) Reaction to previous dose
- d) Pregnancy

Question 9:

Which of the following are live vaccines?

- a) 3 and 5 only
- b) 2 and 5 only
- c) 1, 2 and 5 only
- d) 2 and 3 only

Question 10:

What should be the minimum interval between the administration of two live vaccines?

- a) 2 weeks
- b) 4 weeks
- c) 8 weeks
- d) 12 weeks

Question 11:

In which of the following diseases should active and passive immunisation not be done together?

- a) Measles
- b) Rabies
- c) Tetanus
- d) Diphtheria

Question 12:

What is the main drawback of polysaccharide-based vaccines?

- a) They cause severe pain at the injection site
- b) The immune responses elicited are serotype specific

- c) It cannot be conjugated with proteins to enhance immunogenicity
- d) The rate of adverse reactions is very high

Question 13:

Which of the following is a subunit vaccine?

- a) BCG vaccine
- b) Hepatitis B vaccine
- c) DPT vaccine
- d) IPV vaccine

Question 14:

You are a medical officer conducting an ANC camp in a rural village near Hubli, Karnataka. Which of the following vaccines would you not administer to a pregnant woman?

- a) Hepatitis B
- b) Rabies
- c) Diphtheria
- d) Yellow fever

Question 15:

A patient is diagnosed to have an allergy to eggs. Administration of which of the following vaccines is contraindicated in this patient?

- a) Pneumococcal
- b) MMR
- c) Varicella
- d) Influenza

Question 16:

A newborn baby is HIV + and symptomatic. Which of the following vaccines must not be given to her?

- a) Measles

- b) IPV
- c) BCG
- d) Mumps

Question 17:

Which of the following side effects is most likely to develop in a newborn, following BCG vaccination?

- a) Ulceration
- b) Fever
- c) Regional lymphadenitis
- d) Suppurative lymphadenopathy

Question 18:

A child who had recently received the BCG vaccine comes to you with complaints of fever, pain and swelling at the injection site. You suspect a localised abscess formation. Which of the following steps should be avoided in the treatment of this child?

- a) Aspiration
- b) Incision and application of INH powder
- c) Systemic INH therapy
- d) Incision and application of PAS powder

Question 19:

A complete blood count was obtained for a patient after they were administered vaccine X. The report is shown below. Which of the following options is most likely to be vaccine X?

- a) MMR vaccine
- b) Typhoid vaccine
- c) Hib vaccine
- d) DTP vaccine

Question 20:

Persistent inconsolable crying can be a result of administration of which of the following vaccines?

- a) Hepatitis B
- b) BCG
- c) DPT
- d) Yellow fever

Question 21:

Which of the following statements is false about the DPT vaccine?

- a) Adult preparation contains a lower dose of diphtheria than the pediatric version
- b) Whole killed bacteria of *Bordetella pertussis* has an adjuvant effect
- c) The whole-cell killed *B. pertussis* component is encephalogenic
- d) Presence of *H. influenzae* type b component increases its immunogenicity

Question 22:

A worried mother brings her baby to your PHC saying her child has gone limp. She mentions the child had recently been given some oral drops as part of a mass vaccination campaign. Which of the following is the most likely cause of the baby's symptoms?

- a) Rotavirus vaccine
- b) Oral polio vaccine
- c) Oral typhoid vaccine
- d) Oral cholera vaccine

Question 23:

A patient presents with complaints of severe aching pain and weakness in his left arm and shoulder. He mentions he had recently been administered a vaccine in his left arm, but he does not remember the name of the vaccine. Which of the following could be the most common cause of his symptoms?

- a) Measles vaccine
- b) Tetanus vaccine
- c) Pertussis vaccine

d) BCG vaccine

Answer Key

Question No.	Correct Option
1	b
2	c
3	a
4	c
5	d
6	a
7	a
8	c
9	b
10	b
11	a
12	b
13	b
14	d
15	d
16	c
17	a
18	c
19	a
20	c
21	d
22	b
23	b

Detailed Explanations

Solution to Question 1:

It is not possible to achieve 100% herd immunity against a certain disease but studies have shown this is not necessary for disease control.

Herd structure is never constant. It is subject to constant variation because of new births, deaths and population mobility. An ongoing immunization programme will keep up the herd immunity at a very high level.

Solution to Question 2:

Herd immunity threshold does not vary with the number of vaccines for the disease. It varies only due to the virulence of the disease, efficacy of the vaccine and contact parameter for the population.

Solution to Question 3:

Live vaccines should generally be avoided for 12 weeks following an injection of normal human immunoglobulin (Ig). If a live vaccine has already been administered, the injection of normal human Ig should be delayed for 2 weeks.

NHIG is an antibody-rich fraction that is usually obtained from a pool of at least 1,000 donors. AN NHIG preparation must contain at least 90% intact IgG and minimal IgG aggregates with low IgA concentration. NHIG is commonly used to prevent measles and hepatitis A infections.

Specific human immunoglobulins are made from the plasma of patients who have recently recovered from an infection or are obtained from individuals who have been immunized against a specific infection. These preparations have a high antibody content against an individual infection and provide immediate protection.

Solution to Question 4:

Specific human immunoglobulins or hyperimmune human Igs are obtained from selected donors that have high titres of the antibodies to the disease like those who have recovered from the disease.

It is commonly used in prophylaxis of chickenpox in highly susceptible individuals and in post-exposure prophylaxis of hepatitis B.

Solution to Question 5:

Production of antibodies is faster in secondary booster response.

Primary response is the immune response developed when an antigen is administered for the first time to a person who has never been exposed to it. There is a latent period of 3-10 days before antibodies appear in the blood.

Secondary booster response is the immune response elicited by a booster dose. In comparison to a primary immune response, it has a shorter latent period, faster production of antibodies, and

an abundance of antibodies. The antibody response is maintained at higher levels for a longer period of time, and the antibody elicited tends to have a greater avidity or capacity to bind to the antigen.

Solution to Question 6:

Simultaneous administration of vaccines is defined as administering more than one vaccine on the same clinic day, at different anatomic sites. They are not combined in the same syringe.

Simultaneous administration of the most widely used live and inactivated vaccines does not result in decreased antibody responses or increased rates of adverse reaction. However, in children with functional or anatomic asplenia pneumococcal conjugate vaccine (PCV13) and Menactra brand meningococcal conjugate vaccines should not be administered at the same visit; separate these vaccines by at least 4 weeks.

Advantages of simultaneous administration of vaccines:

- The organisms multiply in the host and thus, the antigenic dose is larger than what is injected.
- They have all the major and minor antigenic components.

Live vaccines should not be administered to those with an immunocompromised state, since they may cause severe systemic disease by way of viremia/bacteremia.

Note: If live vaccines are not given simultaneously, they must be spaced as follows-

- The administration of two live parenteral vaccines or live intranasal influenza vaccines must be separated by at least 4 weeks (28 days).
- If two live vaccines are given less than 4 weeks apart, the second vaccine should be repeated in four weeks. This is because the live vaccine given first can interfere with the second vaccine, and the person may not develop active immunity to the second vaccine.

Solution to Question 7:

Yellow fever vaccine is a live attenuated vaccine.

Injectable polio vaccine (Salk vaccine), pertussis vaccine, and rabies vaccine are killed vaccines.

Solution to Question 8:

The only absolute contraindication to administration of an inactivated or killed vaccine is severe local or general reaction to a previous dose of the vaccine.

Inactivated or killed vaccines are made from growing cultures of the organism and then inactivating them with heat or chemicals. They activate the channels of active immunity after injection and are less efficacious than live vaccines. Multiple doses or booster doses are usually required to produce an adequate antibody response.

Killed vaccines can be given to immune-compromised individuals and in pregnancy as the infective agent is inactivated and cannot grow.

Egg allergy is not an absolute contraindication to the administration of an inactivated or killed vaccine.

Solution to Question 9:

Yellow fever vaccine and Sabin vaccine (oral polio vaccine) are live vaccines.

HBV and HPV vaccines are recombinant vaccines. Tetanus vaccine is a toxoid vaccine.

Solution to Question 10:

The minimum interval between the administration of two live vaccines is 4 weeks.

This is because decreasing the interval between two doses may interfere with the antibody response and protection. Thus, live vaccines can either be given on the same day according to the schedule or 4 weeks apart.

Note: In Park 27th ed. the minimum interval between two live vaccines is given as 3 weeks. However, several other standard sources including CDC and UNDP suggested an interval of 4 weeks. When both are given in options, it is recommended to mark 4 weeks as the correct answer.

Solution to Question 11:

Active and passive immunization should not be done together for measles as the vaccine used is a live vaccine.

Live vaccines should generally be avoided for 12 weeks following an injection of normal human immunoglobulin (Ig). If a live vaccine has already been administered, the injection of normal human Ig should be delayed for 2 weeks.

Note: In the case of measles alone, the recommended interval between administration of Measles immunoglobulins and Live measles vaccine is about 6 months in a non-immunocompromised individual and 8 months in an immunocompromised individual.

Solution to Question 12:

Polysaccharide-based vaccines stimulate an antibody response against the surface polysaccharide of the pathogenic-capsulated bacteria. The polysaccharide structure varies not only between bacteria of different species but also between different strains within a single species. Thus, a limitation of these vaccines is that the immune responses are serotype-specific.

If these vaccines are conjugated with proteins, they are given the capability of eliciting a stronger immune response. Multivalent formulations of both polysaccharide-based vaccines and conjugated vaccines are required to achieve protection against multiple serotypes.

Solution to Question 13:

Hepatitis B vaccine is an example of a subunit vaccine made using HBsAg.

Subunit vaccines are produced from macromolecules derived from immunogenic components of organisms by recombinant DNA technology. E.g., HPV vaccine (uses L1 major capsid protein) and hepatitis B vaccine (uses HBsAg).

BCG is a live attenuated vaccine, IPV is an inactivated vaccine and DPT has diphtheria and tetanus toxoids and a killed pertussis component.

Solution to Question 14:

Yellow fever vaccines are contraindicated in pregnancy in non-endemic areas. It is a live vaccine and most live vaccines are contraindicated in pregnancy unless the risk of infection is much greater than the risk of harm to the fetus.

Note: Yellow fever vaccine is not an absolute contraindication as it is administered in certain endemic areas and in outbreaks

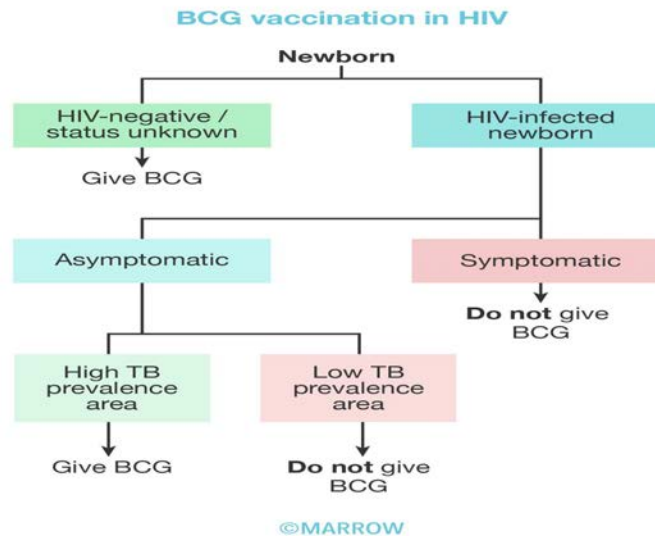
Solution to Question 15:

Influenza and yellow fever vaccines are contraindicated in patients having an allergy to eggs.

Solution to Question 16:

BCG vaccine must not be administered to HIV-infected infants who are symptomatic despite the presence of high TB prevalence. Since BCG is a live vaccine, it significantly increases the risk of disseminated TB in a symptomatic HIV+ infant.

However, asymptomatic infants who are HIV+ can be given BCG, provided the risk of contracting TB is high in that region. In areas of low TB prevalence or risk, the BCG vaccine is avoided.



WHO recommendations for administering live vaccines in symptomatic HIV+ patients:

- Measles vaccine (or MMR) is allowed even in the symptomatic stage of HIV infection (except in the terminal phase), so as to prevent serious and life-threatening wild measles virus disease. Avoid MMR only if CD4+ < 15% (severely immunocompromised).
- IPV is an inactivated vaccine and must be used as an alternative to OPV.

Solution to Question 17:

Ulceration is the most common side effect post BCG vaccination.

Following vaccination, the local tissue response begins in 2-3 weeks and results in scar formation and healing within 3 months. Side effects most commonly seen include ulceration at the vaccination site and regional lymphadenitis. Some vaccine strains may also rarely cause osteomyelitis.

In immunocompromised states, disseminated BCG infection can occur.

The image below shows the typical adverse effects seen following BCG vaccination:

- Regional lymphadenopathy
- Suppurative lymphadenopathy with fistula
- Ulcer at injection site
- Cold abscess at inject site



Solution to Question 18:

There is no need for systemic treatment with INH in case of a local abscess formation after BCG vaccination.

If there is a local abscess formation following BCG vaccination, it should be treated by aspiration. If it does not clear, then it should be incised and INH or PAS powder should be applied daily to the site. The patient should be assured of the harmless nature of the lesion.

Solution to Question 19:

The lab report shows a low platelet count or thrombocytopenia. This is a possible adverse effect of the MMR vaccine.

Solution to Question 20:

Persistent inconsolable crying is usually seen after DPT vaccine. It is mostly attributed to the pertussis fraction of the vaccine.

It usually lasts for 3 hours and settles after a day or so. Other adverse effects of the DTP vaccine include neurologic effects like encephalitis and encephalopathies.

Solution to Question 21:

H. influenzae type b (Hib) is not a component of the DPT vaccine and hence cannot affect its immunogenicity. It is a component of the pentavalent vaccine.

The DPT vaccine is made up of diphtheria, pertussis and tetanus vaccines. The pertussis component in DPT vaccine enhances the potency of the diphtheria toxoid. Both whole-cell killed *B. pertussis* and acellular pertussis component have this adjuvant effect. The only difference is that whole-cell killed *B. pertussis* is encephalitogenic and immunity is short-lasting, while the acellular pertussis component (DTaP) is devoid of neurological complications.

There are two types of DPT vaccines - plain and adsorbed. Adsorption is usually carried out on a mineral carrier like aluminium phosphate or hydroxide. Adsorbed vaccines are more immunogenic and recommended by WHO for national vaccination programmes.

Solution to Question 22:

The given scenario is suggestive of paralysis and the likely cause is vaccine-associated paralytic poliomyelitis following OPV administration.

This is a rare side effect of OPV. Acute-onset flaccid paralysis occurs within 4 to 30 days of receipt of the vaccine in these cases or within 4 to 75 days after contact with a vaccine recipient.

Solution to Question 23:

The given scenario is suggestive of brachial neuritis which is a known complication of the tetanus vaccine.

Brachial neuritis is a dysfunction of nerves supplying the arm and shoulder without the involvement of the rest of the nervous system. It presents with a deep, steady, severe aching pain in the shoulder and arm followed by weakness and wasting in arm and shoulder muscles. Sensory loss may also be present. It is treated symptomatically by analgesics.

Vaccine Production and Storage

Question 1:

Select the wrongly matched pair of excipients and vaccines.

- a) MMR- Neomycin
- b) OPV- Magnesium Chloride
- c) DPT- Thiomersal
- d) BCG- Formaldehyde

Question 2:

Which of the following is false about adjuvants in vaccines?

- a) Immune response is increased by degree or duration by their use
- b) They are used in HPV vaccines
- c) There is a high incidence of side effects with their use
- d) They help in the reduction of amount of immunogens in vaccines

Question 3:

Which adjuvant is used in the DPT vaccine?

- a) Hemagglutinin
- b) Alpha Tocopherol
- c) Aluminium
- d) Tween 80

Question 4:

A child has come to you for the administration of the IPV vaccine, as part of his routine immunisation schedule. Which of the following antibiotics may be found in this vaccine?

- a) Penicillin
- b) Streptomycin

- c) Amikacin
- d) Erythromycin

Question 5:

Which of the following are the most and least heat-sensitive vaccines respectively?

- a) Sabin vaccine and hep B vaccine
- b) Tetanus vaccine and meningococcal vaccine
- c) Japanese encephalitis vaccine and pneumococcal vaccine
- d) Salk vaccine and hep B vaccine

Question 6:

Which of the following vaccines should be protected from sub-zero temperatures?

- a) BCG
- b) OPV
- c) Japanese encephalitis
- d) Hepatitis B

Question 7:

What vaccines are supplied in the type of bottle given below?



- a) BCG and Measles
- b) HPV and Tetanus
- c) Rotavirus and JE
- d) IPV and Hepatitis B

Question 8:

Which of the following is a non-electrical cold-chain equipment?

- a) Solar refrigerator battery drive
- b) Cold box
- c) Walk in coolers
- d) Ice lined refrigerator

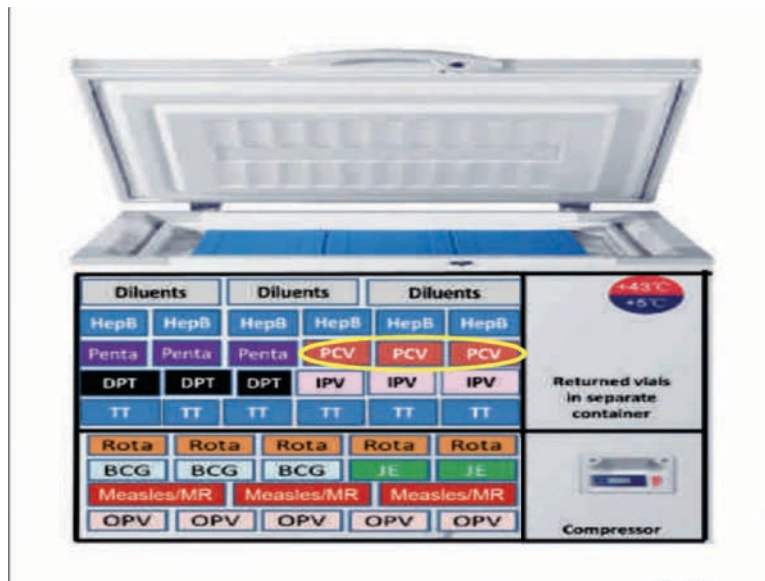
Question 9:

What is the temperature maintained in the freezer compartment of the solar refrigerator battery drive?

- a) -10°C
- b) 8°C
- c) -7°C
- d) 2°C

Question 10:

Which of the following statements is not true about the device shown in the image below?



- a) Smaller ones are given to PHCs and the larger ones to district vaccine stores
- b) They have temperature zones for different vaccines
- c) Kata thermometer is used to record temperature inside this device
- d) These devices are provided with baskets to store vaccines

Question 11:

While working as a medical officer in a CHC, you experience a sudden power cut in your area due to which the ice-lined refrigerator loses power. What is the time taken for the temperature inside this equipment to rise to the maximum recommended temperature limit?

- a) Heating time
- b) Hold over time
- c) Catch up time
- d) Watch down time

Question 12:

At what level is the risk of cold-chain failure the highest?

- a) District level
- b) Regional level
- c) PHC level
- d) Sub-centre level

Question 13:

Open vial policy is not applicable to which of the following vaccines?

- a) Hepatitis B
- b) BCG
- c) Pentavalent vaccine
- d) Rotoviral vaccine in New container

Question 14:

What are the devices given below used to measure?



- a) Temperature changes in atmosphere
- b) Cumulative heat exposure of vaccine
- c) Cumulative heat and freeze exposure
- d) Disintegration of vaccine constituents due to temperature

Question 15:

You are preparing to administer the BCG vaccine to a newborn at a PHC. Before administering the vaccine, you check the vaccine vial monitor. In which of the following observations on the VVM, will you discard the vaccine?

- a) Inner square is white
- b) Inner square has the same color as the outer circle
- c) Inner circle has the same color as the outer square
- d) Inner circle is lighter than the outer square

Question 16:

Which of the following vaccine vials can be used based on the vaccine vial monitor (VVM) color changes shown?



- a) Only vial 1 can be used
- b) Vials 1 and 2 can be used
- c) Vials 1, 2, and 3 can be used
- d) All vials can be used

Question 17:

You are volunteering in a vaccination camp. You perform a shake test before administering a particular vaccine. Which of the following is checked by this test?

- a) Freeze damage to vaccine
- b) Contamination of vaccine
- c) Expiry of vaccine
- d) Potency of vaccine

Answer Key

Question No.	Correct Option
1	d
2	c
3	c
4	b
5	a
6	d
7	a
8	b
9	c
10	c
11	b
12	d
13	b
14	b
15	b
16	b
17	a

Detailed Explanations

Solution to Question 1:

Formaldehyde is not used in the BCG vaccine. It is used in IPV and diphtheria and tetanus vaccines.

Excipients (other vaccine constituents) include: Adjuvants: serve to enhance the immune reaction to the vaccine , Stabilizers : stabilize and prolong the shelf-life of the vaccine and

Preservatives : preserve the components. Some examples are listed below:

- As stabilizing agents-
- MgCl₂ - OPV
- MgSO₄ - measles
- Lactose-sorbitol and sorbitol-gelatine

- To prevent microbial growth and contamination-
- Thiomersal- DPT, hepatitis B, Haemophilus influenzae type b (Hib), rabies, influenza, and meningococcal diseases. It contains very small amounts of organic mercury.
- Neomycin- MMR and IPV
- Other preservatives-
- 2-phenoxyethanol - IPV
- Phenol - Typhoid vaccine

Solution to Question 2:

Side effects are rare with the use of adjuvants in vaccines.

Adjuvants are substances that are added to vaccines to enhance their immunogenicity by degree and/or duration. They help in the reduction of the amount of immunogen in vaccines or in the number of doses required to achieve immunity.

Aluminium salts are the most commonly used adjuvants.

Adjuvants are used in many vaccines like:

- Anthrax vaccine
- Hep A vaccine
- Hep B vaccine
- Japanese encephalitis vaccine
- DT and DTaP
- HPV vaccine
- Pneumococcal vaccine

Solution to Question 3:

Aluminium is the adjuvant used in DPT vaccines.

The table below lists adjuvants used in human vaccines:

Name	Class	Components	Vaccine
Alum	Mineral salts	Aluminium phosphate, aluminium hydroxide	Diphtheria, tetanus, pneumococcus, HAV, HBV, anthrax, tick-borne encephalitis, Men C, HPV
MF59	Oil-in-water emulsion	Squalene, Tween 80, Span 85	Seasonal and pandemic influenza

Name	Class	Components	Vaccine
ASo3	Oil-in-water emulsion	Squalene, Tween 80, alpha tocopherol	Pandemic influenza
AFo3	Oil-in-water emulsion	Squalene, Montane 80, Eum ulgin B1PH	Pandemic influenza
Virosomes	Liposomes	Phospholipids, cholesterol, haemagglutinin	Seasonal influenza, HAV
ASo4	Alum-adsorbed TLR4 agonist	Aluminium hydroxide, monophosphoryl lipid A (MPL)	HBV, HPV
RC-529	Alum-adsorbed TLR4 agonist	Aluminium hydroxide, synthetic MPL	HBV

Solution to Question 4:

IPV may contain trace amounts of streptomycin, neomycin or polymyxin B.

MMR contain small amount of neomycin.

Antibiotics are used during vaccine manufacture to prevent bacterial contamination of tissue culture cells.

Solution to Question 5:

Oral polio (Sabin) vaccine is the most heat-sensitive vaccine. Hepatitis B vaccines are the least heat-sensitive alongside meningococcal A vaccine, freeze-dried Hib vaccine and pneumococcal vaccines.

Solution to Question 6:

Hepatitis B vaccine is sensitive to freezing and should be protected from sub-zero temperatures.

Vaccines to be protected from freezing are as follows:

- Hepatitis B
- Cholera
- DTaP-hepB-Hib-IPV (hexavalent)
- DTwP-hepB-Hib (pentavalent)
- Liquid Hib
- IPV
- Influenza, pneumococcal vaccines
- Tetanus, DT, Td

- Rotavirus (freeze-dried, least sensitive), the liquid form is resistant to freezing
- HPV

Solution to Question 7:

The image shows an amber-colored vial. Vaccines like BCG and measles vaccine are sensitive to light and thus are supplied in such vials.

Solution to Question 8:

Cold boxes, vaccine carriers, and ice packs are some of the non-electrical components of the cold-chain system.

Cold boxes are insulated containers used for transportation and temporary storage of vaccines and ice packs. Vaccine carriers are usually not used for the storage of vaccines but for their transportation. Ice packs generally have two holes in them for placing vaccines. They are filled with water.

The below image shows a cold box containing vaccines and ice packs.



Solution to Question 9:

The freezer compartment of the solar refrigerator battery drive is maintained at -7°C , and it is used to store mainly ice packs.

The vaccine storage compartment of this solar cold chain equipment is maintained at 2 to 8 $^{\circ}\text{C}$. The other solar cold-chain equipment commonly used is the solar refrigerators direct drive, which uses the sun's energy to freeze phase-change materials like water and then uses this formed

ice bank to keep the refrigerator cold.

Solution to Question 10:

The image shown in the question is of an ice-lined refrigerator (ILR). Dial or digital thermometers are used to record temperatures inside an ice-lined refrigerator.

ILRs are the main method of vaccine storage in primary health centers. They are an important part of the cold chain and they maintain a temperature of 2 to 8 °C for the storage of vaccines.

They have different temperature zones for different vaccines. The upper zone is for freeze-sensitive vaccines like cholera, hep B, rotavirus(liquid), etc while the lower zone is for heat-sensitive vaccines like OPV, measles, etc.

All vaccines are kept in the basket provided with ILR. Vaccines like OPV, BCG, measles, and JE can be kept at the bottom of the basket while, DPT, TT, Hep B, IPV, and pentavalent vaccines and diluents are kept in the upper part of the basket.

Solution to Question 11:

Hold-over time is the time taken by cold-chain equipment to raise the temperature inside it, during loss of power, to the maximum temperature limit in its recommended range. For an ice-lined refrigerator(ILR), this time depends on:

- Ambient temperature
- Frequency of opening the equipment
- Quantity of vaccines kept inside with spacing
- Condition of ice-pack lining

The below image shows an ILR.



Solution to Question 12:

The risk of cold chain failure is highest at the sub-centre and village levels. Thus, vaccines are not stored at the sub-centre and are supplied on the day of use.

Solution to Question 13:

Open vial policy is not applicable for BCG and measles vaccines.

It is applicable for:

- DPT
- TT
- Hepatitis B vaccine
- OPV
- Pentavalent vaccine
- Rotavirus vaccine (New version with the VVM on the side)

Solution to Question 14:

The image shown is of vaccine vial monitors (VVM), which measure the cumulative heat exposure of the vaccine.

VVMs are heat-sensitive indicators that are routinely used to monitor vaccine viability as the vaccine vial moves across the supply chain. They record cumulative heat exposure through a gradual change in color. If the color of the inner square is the same or is darker than the outer circle, the vaccine has been exposed to too much heat and should be discarded.

Solution to Question 15:

A vaccine must be discarded if the VVM shows the inner square is the same color or darker than the outer circle.

Solution to Question 16:

From the above image, only vials 1 and 2 can be used as the inner squares of the VVMs are lighter than the outer circles.

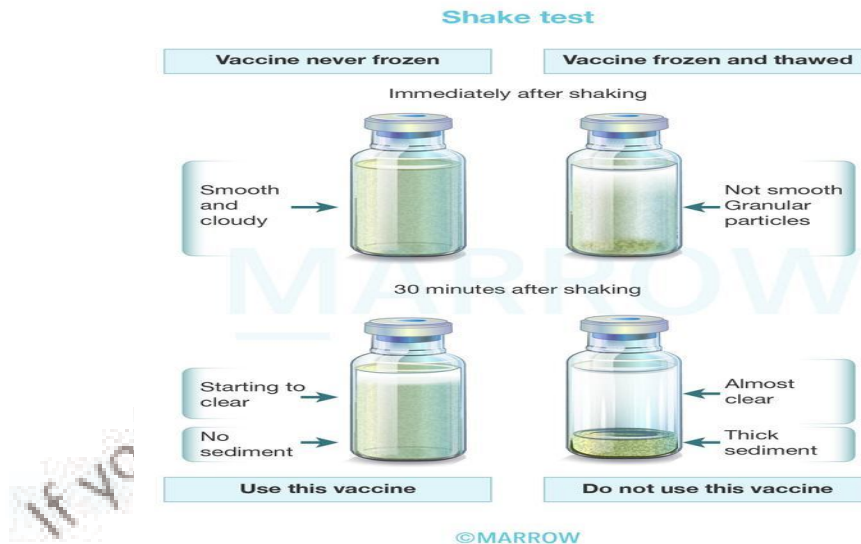
Vial 3 has crossed the discard point as the inner square and outer circle are the same color and thus cannot be used. Vial 4 cannot be used as the VVM shows a darker inner square compared to the outer circle.

Solution to Question 17:

The shake test is done to check for damage to vaccines that are sensitive to freezing temperatures.

Steps to perform the shake test:

- A frozen vaccine is used as a control vial and the vial suspected to be damaged by freezing is taken as the test vial.
- Both are shaken after thawing and the rate of sedimentation is compared.
- If the rate of sedimentation in the test vial is slower than that in the control vial, then it is usable and has not been damaged.



Some of the most widely used vaccines are freeze-sensitive. These include vaccines against:

- Diphtheria
- Tetanus
- Pertussis
- Haemophilus influenza type b (Liquid)
- Hepatitis B

Sterilization and Disinfection

Question 1:

Which is the process designed to prevent microorganisms from reaching a protected environment?

- a) Sterilization
- b) Decontamination
- c) Disinfection
- d) Asepsis

Question 2:

All of the following are methods of dry heat sterilisation except:

- a) Flaming
- b) Incineration
- c) Hot air oven
- d) Autoclaving

Question 3:

What is the temperature that has to be maintained for sterilization in the device shown below?



- a) 160°Celsius for 3 hours
- b) 160° Celsius for 2 hours
- c) 121° Celsius for 2 hours
- d) 121° Celsius for 3 hours

Question 4:

An OT technician is sterilizing surgical instruments in the following device. What temperature should be maintained?



- a) 121° C for 10–15 minutes

- b) 140° C for 90-120 minutes
- c) 134° C for 10-20 minutes
- d) 160° C for 15-20 minutes

Question 5:

Which of the following materials are sterilized by a hot air oven?

- a) 1, 3, 4 and 5
- b) 2, 4 and 5
- c) 3 and 4 only
- d) 1, 2 and 5

Question 6:

You have performed an upper GI endoscopy on a HbsAg positive patient. Which of the following should be used to sterilize the scope?

- a) Hypochlorite
- b) Glutaraldehyde
- c) Formaldehyde
- d) Ethyl alcohol

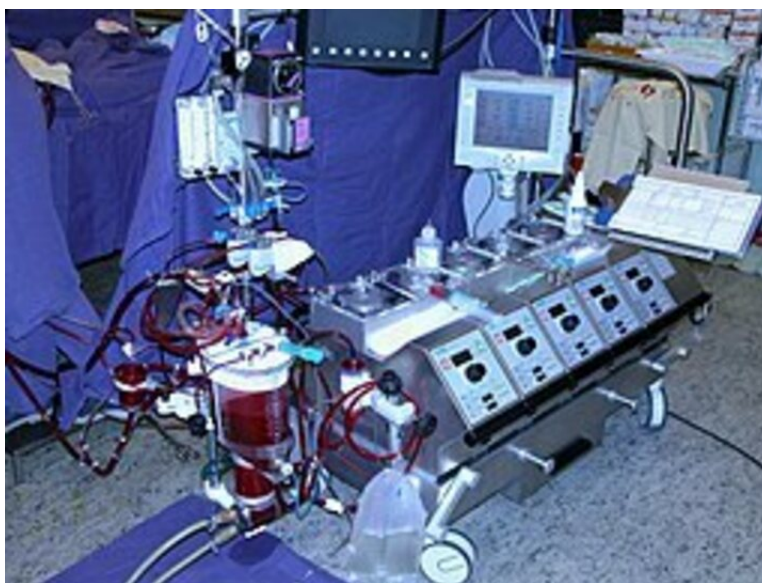
Question 7:

Which of the following is false regarding the use of ethylene oxide in sterilisation and disinfection?

- a) It is an inflammable and explosive gas
- b) It is used for heat-sensitive objects
- c) It is mixed with nitrogen for sterilization
- d) Temperature is maintained at 50 to 60 °C

Question 8:

The following machine was used during a cardiac bypass surgery. How is it best sterilized?



- a) Formaldehyde
- b) Isopropyl alcohol
- c) Hypochlorite
- d) Ethylene oxide

Question 9:

Povidone - iodine is an example of:

- a) Quaternary compounds
- b) Iodophors
- c) Iodine-derived substances
- d) Halophors

Question 10:

Which of the following is an example of precurrent disinfection?

- a) Chlorination of a well by village public health authorities
- b) Disinfection of kidney tray used during delivery of placenta
- c) Disinfection of bed linen after the death of a patient
- d) Disinfection of artery forceps following a surgery

Question 11:

Chlorine is the main disinfectant element in all of the following except:

- a) Superoxidized water
- b) Halazone tablets
- c) Cetrimide
- d) Hypochlorites

Question 12:

Hydrogen peroxide is:

- a) Only bactericidal
- b) Only bactericidal and sporicidal
- c) Bactericidal, fungicidal, sporicidal and virucidal
- d) Bactericidal, virucidal and sporicidal

Question 13:

Which among the following is a low-level disinfectant?

- a) Isopropyl alcohol
- b) Glutaraldehyde
- c) Hydrogen peroxide
- d) Benzalkonium chloride

Question 14:

Which among the following is not a sporicidal agent?

- a) Cresol
- b) Glutaraldehyde
- c) Chlorine dioxide
- d) Hydrogen peroxide

Question 15:

Which of the following disinfectants has anti-viral action?

- a) Chlorhexidine
- b) Hypochlorite
- c) Low concentration of phenol
- d) Cetrimide

Question 16:

A nurse in a PHC is using bleaching powder to disinfect bed linen contaminated with stool. What is the amount that she should use?

- a) 50 g/L
- b) 75 g/L
- c) 90 g/L
- d) 100 g/L

Question 17:

Which of the following methods of disinfection is not effective against prions?

- a) Autoclaving at 121 °C
- b) Sodium hydroxide
- c) Sodium hypochlorite
- d) Benzalkonium chloride

Question 18:

Which of the following methods is not effective for disinfection of sputum of patients?

- a) Autoclaving
- b) Boiling
- c) Cresol
- d) Chlorhexidine

Question 19:

Match the following.

- a) 1-a, 2-b, 3-d , 4-c
- b) 1-d, 2-c, 3-a, 4-b
- c) 1-c, 2-d, 3-b, 4-a
- d) 1-b, 2-a, 3-c, 4-d

Question 20:

Rideal-Walker co-efficient is employed for the assessment of which of the following?

- a) Sufficiency of pasteurization
- b) Effect of autoclaving
- c) Effect of dry heat sterilization
- d) Power of a disinfectant

Question 21:

Which of the following is false regarding the Spaulding classification?

- a) Semi-critical items require high level disinfection
- b) Non-critical items do not require disinfection
- c) Cardiac catheters are classified as critical items
- d) Critical items are associated with a high risk of infection

Question 22:

While collecting blood for samples, you have spilled some on the floor. What would be the immediate next step?

- a) Call the infection control team of the hospital
- b) Mop it
- c) Cover it with 1% sodium hypochlorite
- d) Clean with absorbable material

Question 23:

A blood spill in the operation theatre must be cleaned with _____.

- a) Chlorine compound
- b) Phenolic compound
- c) Quarternary ammonium compounds
- d) Alcoholic compounds

Answer Key

Question No.	Correct Option
1	d
2	d
3	b
4	a
5	d
6	b
7	c
8	d
9	b
10	a
11	c
12	c
13	d
14	a
15	b
16	a
17	d
18	d
19	b
20	d
21	b
22	d
23	a

Detailed Explanations

Solution to Question 1:

Asepsis is the process designed to prevent microorganisms from reaching a protected environment.

Sterilization is a process of killing or removal of all living microorganisms either in the vegetative or spore state from an area or surface or a medium.

Decontamination is the process of rendering an article or area free of contaminants, including microbial, chemical, radioactive and other hazardous materials.

Disinfection is the process by which the destruction of microorganisms is achieved with the use of physical or chemical agents. Microbial forms like spores and some viruses may be resistant to certain disinfectants.

Solution to Question 2:

Autoclaving is a method of moist heat sterilization.

Dry heat sterilization	Moist heat sterilization
FlamingIncinerationHot air oven	BoilingPasteurizationAutoclaving

Solution to Question 3:

The image shown is a sterilizing hot air oven. The temperature to be maintained is 160° C for 2 hours.

Exposure to dry heat in a sterilizing hot air oven ensures the carbonization of organic material and the destruction of microorganisms and spores. The following combinations of temperature and duration of sterilization can be applied in hot air ovens:

- 180°C for 30 mins
- 170°C for 60 mins
- 160°C for 2 hours

Solution to Question 4:

The given image shows an autoclave. Surgical instruments are sterilized at 121° C for 10-15 minutes.

Autoclaving is a form of moist heat sterilization which creates an increased temperature of steam under pressure. This method of sterilization is effective against spores also. Autoclaving is done for instruments that cannot withstand high temperatures of hot air oven or dry heat.

Solution to Question 5:

Glasswares, oils and powders, swabs, and dressings are sterilized by a hot air oven.

Surgical instruments and culture media are sterilized by autoclave.

Solution to Question 6:

2% glutaraldehyde is used to sterilize endoscopes and fibreoptic scopes.

Aldehydes in 2% solutions (like glutaraldehyde and formaldehyde) have sporicidal activities.

Glutaraldehyde is usually efficacious against tubercle bacteria, fungi & viruses.

Solution to Question 7:

A mixture of ethylene oxide with carbon dioxide is used for sterilization, not with nitrogen.

Ethylene oxide is an alkylating agent and has inflammable and explosive properties. It is often mixed with the following prior to use:

- Carbon dioxide - decreases its explosive nature
- Water vapor - increases the efficiency of the gas

Sterilization using this gas is usually done at 50-60 °C. and is suitable for heat-sensitive items like plastics, heart valves, dental equipment. It is effective against bacteria, spores, and some viruses.

Solution to Question 8:

The image shown is a heart-lung machine. It is best sterilized using ethylene oxide.

Ethylene oxide gas diffuses through many types of porous materials and readily penetrates some plastics and is used for sterilizing machines.

Solution to Question 9:

Povidone - iodine is an iodophor.

Iodophors are iodine-based disinfectants in which iodine is combined with solubilizing agents or carriers. This results in the formation of a complex that provides a sustained release reservoir of iodine and releases small amounts of free iodine in an aqueous solution.

Solution to Question 10:

Chlorination of water in a well is an example of precurrent disinfection.

Precurrent disinfection refers to prophylactic disinfection prior to discharge of infectious material. This includes pasteurization of milk, chlorination of water and hand washing.

Concurrent disinfection is the application of disinfection methods after discharge of infectious material from a person or soiling. This includes disinfection of urine, faeces, vomit, contaminated linen, clothes, hands.

Terminal disinfection is done after the death of a patient or transfer or after cessation of isolation.

Solution to Question 11:

Cetrimide (trade name - Cetavlon) is a quaternary ammonia compound (QAC) that does not contain chlorine. QACs are low-level disinfectants and are sporostatic but not sporicidal. It is a bactericidal agent against vegetative gram-positive organisms.

Other QACs include cetrimide (cetavlon) and benzalkonium chloride.

Savlon contains cetrimide and chlorhexidine and is mainly used for plastic items. Clinical thermometers may be best disinfected in Savlon mixed with spirit in a 1:6 ratio.

Chlorine compounds are potent disinfectants that are bactericidal, fungicidal, sporicidal, tuberculocidal, and virucidal.

Chlorine compounds	Mechanism of action and uses
1. Bleaching powder (chlorinated lime)	Disinfection of water, feces, and urine
2. Chloramine - T	Has a prolonged bactericidal effect, used in a health care setting
3. Hypochlorites	Disinfect infectious waste and surfaces
4. Halazone tablets	Disinfect small quantities of water
5. Superoxidized water	Saline is electrolyzed to release hypochlorous acid and chlorine, which act as disinfectants

Solution to Question 12:

Hydrogen peroxide is effective against bacteria, fungi, spores and viruses.

It is used in the hospital setting to disinfect surfaces. Sporicidal activities require higher concentrations and longer contact times. A 3% solution is also used as an antiseptic and for cleaning wounds and discharging ulcers.

Solution to Question 13:

Benzalkonium chloride is a quaternary ammonium compound that is used as a low-level disinfectant.

There are 3 types of disinfectants:

- Low-level disinfectant - kills only vegetative bacteria, fungi and lipid enveloped viruses (e.g. quaternary ammonium compounds)
- Intermediate level disinfectant: kills all microbes including mycobacteria and non-enveloped viruses, but not spores (e.g. alcohol, iodophors)
- High-level disinfectant - This is a chemical that kills all microbial pathogens and some spores if the contact time is increased (e.g. glutaraldehyde, hydrogen peroxide)

Solution to Question 14:

Cresol a phenolic compound is not sporicidal.

Disinfectants that are not sporicidal are:

- Biguanides
- Alcohols
- Heavy metal disinfectants
- Organic acids
- QACs
- Phenolic compounds

Sporicidal disinfectants include:

- Aldehydes
- Chlorine disinfectants
- Hydrogen peroxide
- Ethylene oxide

- Peracetic acids

Solution to Question 15:

Hypochlorite is an antiviral disinfectant.

Antiviral disinfectants include:

- Chlorine compounds
- Phenol (at high concentrations)
- Aldehydes
- Hydrogen peroxide

Solution to Question 16:

Bleaching powder of 50 g/L of water is required for disinfecting feces and urine.

The agents commonly used for disinfecting faeces and urine are:

Disinfectant	Amount per litre	Per cent
Bleaching powder	50 g	5
Crude phenol	100 ml	10
Cresol	50 ml	5
Formalin	100 ml	10

Solution to Question 17:

Benzalkonium chloride is not effective against prions.

Prions can be inactivated by following the given steps:

- Immersing the instruments in 1N NaOH or sodium hypochlorite for 1 hour
- Then they are transferred to water
- Then they are heated in a gravity displacement autoclave at 121°C for 1 hour
- After this, they are cleaned and subjected to routine sterilization

Solution to Question 18:

Chlorhexidine is not used to disinfect sputum.

Sputum can be disinfected by:

- Autoclaving or boiling - if a considerable amount of sputum is present (e.g. in TB hospitals)
- 5% cresol - patients are asked to spit in a sputum cup half-filled with cresol
- Burning - for samples received in gauze or paper handkerchiefs

Solution to Question 19:

The biological indicators used to check the efficiency of various sterilization methods are:

- Clostridium tetani - Hot air oven
- Geobacillus stearothermophilus - Autoclave
- Bacillus atrophaeus - Ethylene oxide
- Bacillus pumilus - Ionizing radiation

Sterilization technique	Bacteria
Hot air oven	Spores of Bacillus subtilis sub sp. Niger (reclassified as B. atrophaeus) Clostridium tetani
Autoclave	Geobacillus stearothermophilus
Filtration	Serratia marcescens Brevundimonas diminuta
Ionizing radiation	Bacillus pumilus
Ethylene oxide	Bacillus globigii (reclassified as B. atrophaeus)
Plasma sterilization	Geobacillus stearothermophilus Bacillus atrophaeus (Bacillus subtilis subsp. Niger)

Solution to Question 20:

The Rideal-Walker coefficient helps in assessing the power of a disinfectant relative to phenol.

It assesses this by determining the phenol coefficient (PC). A PC value of > 1 indicates that it is more effective than phenol. The higher the number, the greater its effectiveness in comparison to phenol and vice versa. Traditional organisms used to test the power of the disinfectant include S. aureus and Salmonella enteric subtype typhi.

Phenol coefficient = $\frac{\text{The dilution of test disinfectant which kills microorganisms in a given time}}{\text{The dilution of phenol which kills the microorganism in the same time under identical conditions}}$

Other tests that are used for disinfectants are as follows:

- Chick Martin test - modification of the Rideal-Walker test where disinfectants act in the presence of organic matter like feces or yeast to simulate natural conditions
- Kelsey-Skyes test (capacity test) - tests the capacity of a disinfectant to retain its activity when used repeatedly.
- In-use test - determines whether the chosen disinfectant is effective in actual hospital use

Joseph Lister first introduced phenol (carbolic acid) as an antiseptic to sterilize surgical instruments and clean wounds. Lister's work led to a reduction in postoperative infections and made surgery safer for patients, distinguishing him as the 'Father of Modern Surgery' and 'Father of Antiseptic Surgery'.

Solution to Question 21:

Under the Spaulding classification, non-critical items require intermediate or low-level disinfection.

Solution to Question 22:

The immediate next step in this scenario is to clean the spilled blood with absorbable material.

This absorbable material is then discarded in the yellow bin. Following this, sodium hypochlorite solution is used to decontaminate the surface. The concentrations of sodium hypochlorite used are:

- < 10 mL blood spill - 1:100 dilution to decontaminate non-porous surfaces
- > 10 mL blood spill or culture spill - 1:10 dilution for the first application, followed by 1:100 dilution

Solution to Question 23:

Sodium hypochlorite, a compound of chlorine, is the best disinfectant for surface blood spills. It is used with a concentration of 0.5-5%.

If the spill contains large amounts of blood or body fluids, clean the visible matter with disposable absorbent material and discard the contaminated materials in appropriate, labelled containment.

A 1:10–1:100 dilution of 5.25%–6.15% sodium hypochlorite (i.e., household bleach) or an EPA-registered tuberculocidal disinfectant has been recommended for decontaminating blood spills.

Screening

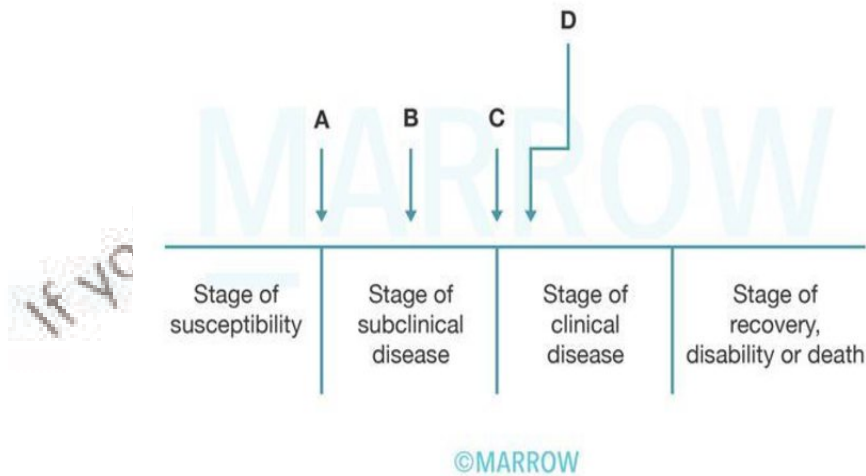
Question 1:

Which of the following are true for a screening test?

- a) 1 and 5
- b) 1, 3, 4
- c) 2, 4, 5
- d) 1, 4, 5

Question 2:

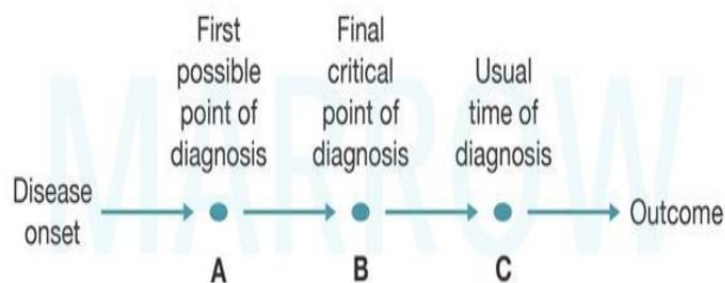
Below is a diagram depicting the natural history of the disease. Which point marks the onset of symptoms?



- a) Point A
- b) Point B
- c) Point C
- d) Point D

Question 3:

In the following natural history of disease flowchart, point A to C is called as?



- a) Screening time
- b) Lead time
- c) Lag time
- d) Generation time

Question 4:

In which of the following diseases, screening procedure increases the overall survival to the maximum?

- a) Lung cancer
- b) Prostate cancer
- c) Colon cancer
- d) Ovarian cancer

Question 5:

Which of the following is an example of prospective screening?

- a) Cervical Pap smear in a 40 year old patient
- b) Neonatal screening of a new-born baby for hypothyroidism
- c) Screening of immigrants to a country
- d) Urine for sugar screening in a 40 year old man

Question 6:

Which of the following is an example of prescriptive screening?

- a) Screening of immigrants for yellow fever
- b) Screening of blood donors
- c) Screening of infants for phenylketonuria
- d) HIV screening during antenatal visits

Question 7:

Which of the following is an example of multiphasic screening?

- a) Chest X-ray for TB in large populations
- b) Annual health checkups
- c) Pap smear in women above 40 years
- d) Random blood sugar during antenatal visits

Question 8:

A large number of tests were done in series. Which of the following will be true?

- a) Sensitivity and specificity will increase
- b) Sensitivity will decrease, specificity will increase
- c) Sensitivity and specificity will decrease
- d) Sensitivity will increase, specificity will decrease

Question 9:

Colon cancer screening is being done by a hemoccult stool test in a healthcare center. Patients who test positive, undergo a repeat stool hemoccult test. If the second test comes out positive, the individual is referred for more extensive evaluation. How is this method of screening affecting sensitivity and specificity?

- a) Net sensitivity and net specificity are both increasing
- b) Net sensitivity is decreasing and net specificity is increasing
- c) Net sensitivity is increasing and net specificity is decreasing

- d) Net sensitivity is decreasing and net specificity remains the same

Question 10:

Blood pressure is measured in an individual 5 times a day under the same conditions by a new screening test. All 5 readings obtained are the almost same. What is the attribute of a new screening test being described?

- a) Validity
- b) Reliability
- c) Sensitivity
- d) Predictive accuracy

Question 11:

Ten successive readings of hemoglobin level of a single sample measured by a new method are given below. The hemoglobin level measured by the previous standard test was 10.2. What is true regarding the new method?

- a) High validity, low reliability
- b) Low validity, low reliability
- c) Low validity, high reliability
- d) High validity, high reliability

Question 12:

Out of 100 people having lung cancer, 60 tested positive by a new screening test. Out of 900 people who don't have lung cancer, 300 tested positive. What will be the numerator while calculating sensitivity and its value?

- a) True positive, 60
- b) False positive, 300
- c) True negative, 600
- d) False negative, 40

Question 13:

A screening test was carried out among 114 people out of which 15 have a disease. The test showed 22 people tested positive for the disease among which 14 actually had the disease. What is the sensitivity of the test?

- a) 37%
- b) 63%
- c) 93%
- d) 97%

Question 14:

A researcher finds out a sensitive test to be positive for a person. For diagnosis, which of the following will hold true?

- a) It confirms that the person does not have the disease
- b) It suggests that if the disease is rare, the person surely has the disease
- c) It confirms that the person has the disease if the disease is common
- d) It is an expected finding in case the disease is prevalent

Question 15:

1000 patients from a healthcare setup were screened for disease 'X' with a new test (biopsy). The test showed positive results in 90 patients out of the 100 who actually had the disease. What is the sensitivity of the new test?

- a) 90/100
- b) 100/110
- c) (90-10)/1000
- d) 90/1000

Question 16:

Specificity of a screening test is the ability of a test to detect:

- a) True positives
- b) False positives
- c) True negatives
- d) False negatives

Question 17:

5000 persons underwent screening for a disease. Out of 500 diseased, 350 tested positive, and out of healthy people, 3000 tested negative. Which of the following is true?

- a) Sensitivity is 70% and Specificity is 80%
- b) Specificity is 70% and Sensitivity is 80%
- c) Specificity is 66.6% and Sensitivity is 70%
- d) Sensitivity is 87.5% and Specificity is 66.6%

Question 18:

If a patient tests positive for a disease, which of the following represents the probability of her actually having the disease?

- a) Sensitivity
- b) Specificity
- c) Positive predictive value
- d) Negative predictive value

Question 19:

To calculate the positive predictive value of a new screening test for breast carcinoma, what will you use for the denominator?

- a) True positives + false negatives
- b) True positives + true negatives
- c) True positives + false positives
- d) True negatives + false positives

Question 20:

Which of the following maximally affect the positive predictive value of a test?

- a) Sensitivity
- b) Prevalence
- c) Specificity

- d) Negative predictive value

Question 21:

Which of the following reflects the diagnostic power of a screening test?

- a) Predictive accuracy
- b) Sensitivity
- c) Reliability
- d) Specificity

Question 22:

Screening of 100 immigrants for COVID-19 with a new test showed 48 positive and 52 people negative. Those who tested negative underwent RTPCR out of which 25 came positive. Which of the following is lowest for the new test compared to RTPCR?

- a) Specificity
- b) Sensitivity
- c) Positive predictive value
- d) Negative predictive value

Question 23:

While screening a disease with a new screening test, the test is positive in many patients who actually don't have the disease. Which of the following is true?

- a) Prevalence is high
- b) Positive predictive value is high
- c) Specificity is low
- d) Sensitivity is low

Question 24:

ECG findings detecting the myocardial infarction obtained from the department of cardiology are given in the table below. Which of the following is true with regard to the below table?

	MI Present	MI Absent
Positive ECG	300	100
Negative ECG	25	75

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- a) Sensitivity is less than specificity
- b) Sensitivity is the same as specificity
- c) PPV is more than NPV
- d) PPV is the same as NPV

Question 25:

A new test for hepatitis C is performed on 200 patients with the biopsy-proven disease and 200 patients who are free of the disease. The test shows positive results on 180 of the patients with the disease and negative results on 150 of the patients without the disease. Which of the following is true?

- a) Positive predictive value is 90%
- b) Negative predictive value is 75%
- c) Specificity is 75%
- d) Sensitivity is 78%

Question 26:

Due to an effective prevention program, the prevalence of a disease in a community has been reduced by 90%. However, the physicians continue to use the same screening test for that disease. Which of the following changes is expected to be seen?

- a) Increase in negative predictive value
- b) Increase in positive predictive value

- c) Increase in specificity
- d) Increase in sensitivity

Question 27:

In a population of 10000, the prevalence of a disease is 20%. The sensitivity of a screening test is 95% and specificity is 80%. What is the positive predictive value of the test?

- a) 54.3%
- b) 45.7%
- c) 15.3%
- d) 98.5%

Question 28:

The prevalence of the disease 'X' in a population is 30%. The diagnostic test for this disease has a specificity of 0.80 and a sensitivity of 0.90. What is the negative predictive value of the test?

- a) 45%
- b) 62%
- c) 70%
- d) 95%

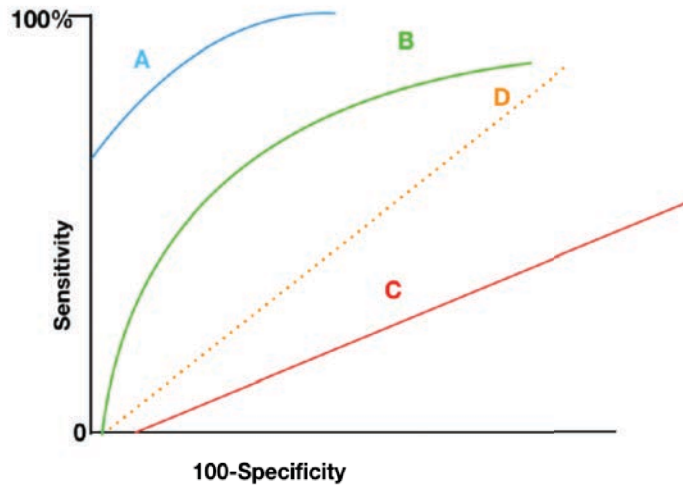
Question 29:

Which of the following is best for determining the threshold for diagnosis of a positive test?

- a) Pre test probability
- b) Receiver operator characteristic curve
- c) Pearson coefficient
- d) Analysis of Variance

Question 30:

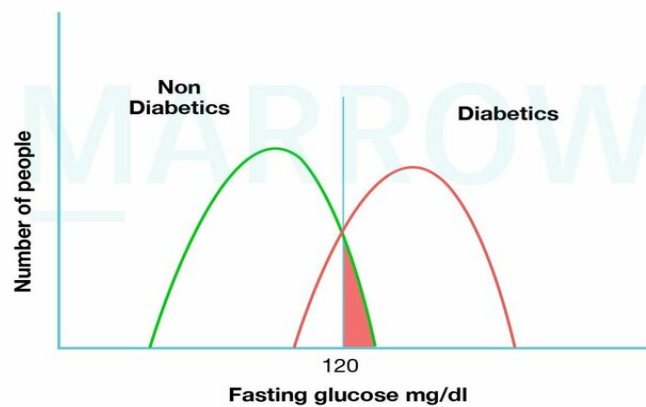
Which curve from the given image reflects a better test?



- a) Test A
- b) Test B
- c) Test C
- d) Test D

Question 31:

The fasting blood glucose levels of a population from two overlapping distributions are shown in the below graph. A test was done with the cutoff of 120mg/ dl for diagnosing diabetes. What does the shaded area represent?

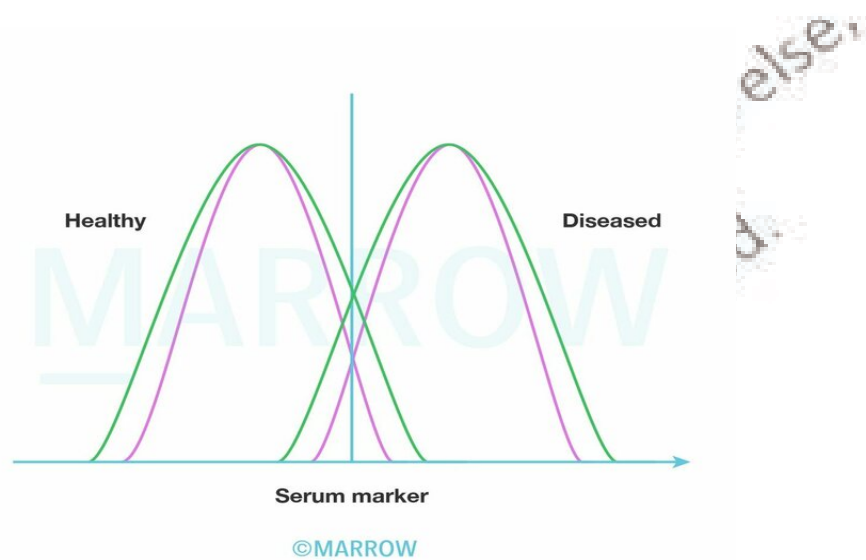


- a) True positives

- b) True negatives
- c) False positives
- d) False negatives

Question 32:

Observe the following curves, which of the following is a true statement about sensitivity and specificity when the curve changes from green to pink?



- a) Both sensitivity and specificity increase
- b) Both sensitivity and specificity decrease
- c) Sensitivity increases while specificity decreases
- d) Sensitivity decreases while specificity increases

Answer Key

Question No.	Correct Option
1	d
2	c
3	b
4	c
5	c

6	c
7	b
8	b
9	b
10	b
11	a
12	a
13	c
14	c
15	a
16	c
17	c
18	c
19	c
20	b
21	a
22	b
23	c
24	d
25	c
26	a
27	a
28	d
29	b
30	a
31	c
32	a

Detailed Explanations

Solution to Question 1:

Screening tests are done on apparently healthy individuals, the results of the test are relatively inaccurate, the tests are based on one criterion.

The difference between screening and diagnostic tests is given below:

Screening tests are recommended in the following conditions:

- Early diagnosis changes the disease course

- In a disease with an asymptomatic stage
- In a disease with a test for diagnosis and treatment
- The natural history of the disease is well known

Screening test	Diagnostic test
Done on apparently healthy	Done on patients with signs and symptoms
Applied to groups	Applied to single patients
Based on one criterion or cut-off point	Based on evaluation of a number of symptoms, signs, and laboratory findings
Less accurate	More accurate
Less expensive	More expensive
Not a basis for treatment	Used as a basis for treatment
The initiative comes from the investigator or agency providing care	The initiative comes from a patient with a complaint

Solution to Question 2:

Point C represents a point where signs and symptoms appear.

Various points marked in the image represent the following:

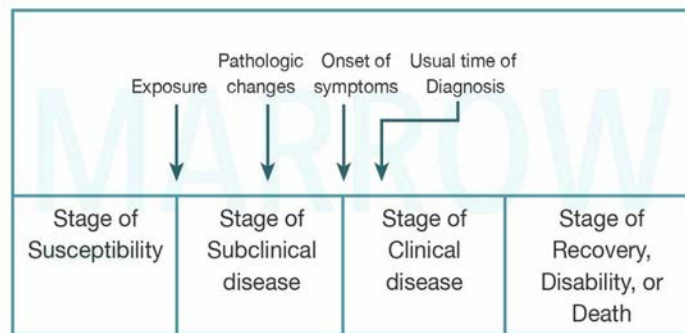
Point A: Entry of infectious agent into the host.

Point B: Pathological and physiological changes occur in the stage of the subclinical phase. These changes are induced by the causative agent but are not sufficient enough to cause clinical signs and symptoms.

Point C: Marks onset of clinical signs and symptoms.

Point D: Usual time of diagnosis at which clinical signs and symptoms are well evident.

Timeline of Natural history of disease



©MARROW

Solution to Question 3:

Lead time is the period between the first possible time of detection (point A) and the usual time of diagnosis (point C).

Screening time is the time difference between the first point of diagnosis and the critical point of diagnosis.

Diagnosis of a disease before the critical point has a good outcome and reduces the severity by reversing the disease process. Diagnosis beyond the critical point treatment would be unsuccessful and/or permanent damage would have occurred.

Screening must be done in conditions where there is a considerable time lag between the onset of disease and the usual time of diagnosis. That is conditions with a long lead time.

Generation time is defined as the interval of time between the onset of infection by a host and the maximal infectivity of that host.

Solution to Question 4:

Screening of colon cancer increases the overall survival to the maximum.

Colon cancer has a long lead time and can be detected early before pathological changes occur. It can be curable if treated early, otherwise fatal in half the cases.

Early diagnosis and treatment before the critical point have a good outcome and reduce the severity by reversing the disease process. Diagnosis beyond the critical point treatment would be unsuccessful and/or permanent damage would have occurred.

Other diseases that can be screened are:

- Cervical cancer

- Breast cancer
- Prostate cancer

Solution to Question 5:

Screening of immigrants to a country is an example of prospective screening.

Screening of disease is done to detect unrecognized disease or defect, in apparently healthy individuals, by means of rapidly applied tests, examinations, or other procedures.

The difference between prospective and prescriptive screening is given below:

Note: Most common neonatal disease to be screened for is neonatal hypothyroidism.

	Prospective screening	Prescriptive screening
Definition	People screened for others benefit	People screened solely for their own benefit
Essential purpose	Disease control	Case detection
Request for screening	Specific request from the authority	Request from physician
Example	Immigrants for infectious diseases like tuberculosis, syphilis, yellow fever, etc; HIV screening during antenatal visits, sex workers. Screening of blood donors.	Neonatal screening for hypothyroidism. Pap smear for detection of cervical cancer. Testing of urine sugar for diabetes mellitus.

Solution to Question 6:

Screening of infants for phenylketonuria is an example of prescriptive screening.

Screening of disease is done to detect unrecognized disease or defect, in apparently healthy individuals, by means of rapidly applied tests, examinations, or other procedures.

The difference between prospective and prescriptive screening is given below:

Note: Most common neonatal disease to be screened for is neonatal hypothyroidism.

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	Prospective screening	Prescriptive screening
Example	Immigrants for infectious diseases like tuberculosis syphilis, yellow fever, etc; HIV screening during antenatal visits, sex workers. Screening of blood donors.	Neonatal screening for hypothyroidism. Pap smear for detection of cervical cancer. Testing of urine sugar for diabetes mellitus.

Solution to Question 7:

Annual health checkups are an example of multiphasic screening.

Types of screening:

- Multiphasic screening is the application of two or more screening tests in combination with a large number of people at one time. Example- Annual health checkups.
- High-risk or selective screening is done on high-risk groups. Example- MRI screening of breast done annually in 25-year old women with BRCA 1, BRCA 2 mutations in a first-degree relative.
- Mass screening is a screening of a whole or large population like using a chest X-ray for TB screening in large populations. It is done for diseases with high prevalence.

Solution to Question 8:

When a large number of tests are done in series, net sensitivity is decreased and net specificity is increased.

Screening tests are done in series when the result of the first test determines whether to run the second test or not. e.g., HIV screening is done using consecutive ELISA.

Screening tests are done in parallel when the second test is independently performed regardless of what the first test result is. Both the tests are done at the same time. For e.g., In LFT, multiple tests are done together like SGOT, SGPT, total bilirubin, alkaline phosphatase, etc.

Following are the trends that are observed while evaluating screening tests:

	Tests in Series	Tests in parallel
Combined sensitivity	Decreases	Increases
Combined specificity	Increases	Decreases
Combined PPV	Increases	Decreases
Combined NPV	Decreases	Increases

Solution to Question 9:

In the given scenario, tests are being performed in series. Here, the net sensitivity is decreasing and net specificity is increasing.

Screening tests are done in series, where the result of the first test determines whether to run the second test or not. e.g. HIV screening is done using consecutive ELISA.

Screening tests are done in parallel, where the second test is independently performed regardless of what the first test result is. Both the tests are done at the same time. For e.g. In LFT, multiple tests are done together like SGOT, SGPT, total bilirubin, alkaline phosphatase, etc.

Following are the trends that are observed while evaluating screening tests:

	Tests in Series	Tests in parallel
Combined sensitivity	Decreases	Increases
Combined specificity	Increases	Decreases
Combined PPV	Increases	Decreases
Combined NPV	Decreases	Increases

Solution to Question 10:

The attribute of a new screening test to yield the same reading when repeated under the same conditions is called reliability.

Reliability is also called precision or reproducibility.

Solution to Question 11:

The new method for measuring hemoglobin has high validity and low reliability.

The new test doesn't give the same values on repeat testing, thus it has low reliability. The mean value of all the readings is 10. The mean is close to the value of 10.2, which is obtained from the standard test, making the validity high.

The validity of a test implies that the test must give correct and accurate results. It represents the closeness to a true/ actual value.

The reliability of a test implies that the test must give consistent results when repeated on the same individual or material, under the same conditions.

Solution to Question 12:

The numerator of sensitivity in a screening test is true positive and its value is 60.

True positives are those people who tested positive and actually have the disease. Out of 100 people with lung cancer, 60 are tested positive i.e. true positive.

The given data is represented below:

TP- true positive, FP- false positive, FN- false negative, TN- true negative

Sensitivity is the ability of a screening test to detect true positive(TP) out of the total number of people who have the disease (TP+FN).

$$\text{Sensitivity} = \text{TP} / \text{TP} + \text{FN} = 60 / 60 + 40 = 60\%$$

	Disease +	Disease -	Total
Test +	60(TP)	300(FP)	360
Test -	40(FN)	600(TN)	640
Total	100	900	1000

Solution to Question 13:

The sensitivity of the above screening test is 93%.

The given data is represented below

TP- true positive, FP- false positive, FN- false negative, TN- true negative

Sensitivity is given as $\text{TP} / (\text{TP} + \text{FN}) \times 100 = 14 / 15 = 93\%$.

	Disease +	Disease -	Total
Test +	14(TP)	8(FP)	22
Test -	1(FN)	91(TN)	92
Total	15(TP+FN)	99(FP+TN)	114

Solution to Question 14:

A sensitive test, if it is positive it confirms that the person has the disease if the disease is common.

The sensitivity of a screening test is the ability of the test to identify correctly those who have the disease. In other words, Sensitivity is the ability of a test to detect true positives (TP) out of total disease (TP+FN).

$$\text{Sensitivity} = \text{TP} / \text{TP} + \text{FN}$$

Solution to Question 15:

The sensitivity of new test is 90/100.

Biopsy gives the definitive number of cases for Disease X. Hence, the given data is represented below:

$$\text{Sensitivity} = \text{TP} / (\text{TP} + \text{FN}) = 90 / (90 + 10) = 90 / 100 = 90\%$$

True positive is defined as a positive test result in a person who has the disease.

	Disease +	Disease -
Test +	90(TP)	unknown
Test -	10(FN)	unknown
Total	100	

Solution to Question 16:

Specificity of a screening test is the ability of the test to identify correctly those who do not have the disease, i.e. true negative.

Sensitivity of a screening test is the ability of the test to identify correctly those who have the disease, i.e. true positive.

False-positive are those who don't have the disease but tested positive.

False-negative are those who have the disease but tested negative.

Solution to Question 17:

The sensitivity of the above screening test is 70% and Specificity is 66.6%.

The given data is represented below:

$$\text{Sensitivity} = \text{TP} / (\text{TP} + \text{FN}) \times 100 = 350 / 500 = 0.7 = 70\%.$$

$$\text{Specificity} = \text{TN} / (\text{FP} + \text{TN}) \times 100 = 3000 / 4500 = 0.6666 = 66.6\%.$$

	Disease +	Disease -	Total
Test +	350(TP)	1500(FP)	1850
Test -	150(FN)	3000(TN)	3150
Total	500(TP + FN)	4500(TN + FP)	5000

Solution to Question 18:

Positive predictive value implies the probability of a person with a positive test, actually having the disease.

It is the ability of a test to detect true positive(TP) out of total positive(TP+FP) and it is directly proportional to the prevalence of a disease.

$$PPV = TP / (TP + FP)$$

Solution to Question 19:

The denominator for the calculation of the positive predictive value of a screening test is true positives + false positives.

Positive predictive value (PPV) of a test is $TP / (TP + FP) \times 100$.

It indicates the probability that a person with a positive test actually has the disease. It is directly proportional to the prevalence of a disease.

Predictive value is affected by sensitivity, specificity, and prevalence. However, it is affected the most by prevalence.

Solution to Question 20:

Positive predictive value (PPV) is maximally affected by the prevalence of the disease.

If the prevalence of a disease in a population is high, the high will be the positive predictive value of a screening test, and the more accurate will be the test. If the prevalence declines, the PPV will be low.

PPV is directly proportional to the prevalence of a disease. It is affected the most by prevalence. However, both sensitivity and specificity can change the predictive value.

Solution to Question 21:

The predictive accuracy of a screening test reflects its performance and diagnostic power.

Predictive accuracy is given by:

- Positive predictive value (PPV): the probability of a person with a positive test, actually having the disease.
- Negative predictive value (NPV): the probability of a person with a negative test, definitely not having the disease.

Sensitivity is the ability of a screening test to detect true positive out of total disease.

Specificity is the ability of a screening test to detect true negative out of total nondiseased.

The reliability of a test implies that the test must give consistent results when repeated on the same individual or material, under the same conditions.

Solution to Question 22:

The sensitivity of the new screening test is low compared to RTPCR.

The new screening test showed 52 people negative, out of which 25 are detected positive by RTPCR. i.e. 25 are false negative.

The false-negative rate is the number of false-negative out of total people having the disease.

False-negative rate = $1 - \text{Sensitivity}$

The false-negative rate increases as sensitivity decreases.

In this scenario, because the new test has more false negatives, it would have low sensitivity as compared to RTPCR.

Solution to Question 23:

The given new screening test has low specificity.

The test is positive in many patients who actually don't have the disease, i.e. high false positives.

The false-positive rate is the number of false positives out of total non-diseased.

False-positive rate = $1 - \text{Specificity}$

The false-positive rate is inversely proportional to specificity. Hence, the given test with high false positives will have low specificity.

Positive predictive value (PPV) = $\frac{TP}{TP+FP}$. It shows the positive predictive value (PPV) of the test decreases with an increase in false positives.

An increase in the false-positive does not affect the prevalence of the disease. The prevalence of a disease depends on the incidence(I) and duration(D) of the disease.

Prevalance= $I \times D$

Solution to Question 24:

In the above scenario, PPV is the same as NPV.

Predictive value of a positive test (PPV)

$$= \text{TP}/(\text{TP} + \text{FP}) \times 100 = 300/(300+100) \times 100 = 75 \%$$

Predictive value of a negative test (NPV)

$$= \text{TN}/(\text{FN} + \text{TN}) \times 100 = 75/(25+75) \times 100 = 75 \%$$

Hence PPV = NPV

In this scenario, sensitivity is 92% and the specificity is 43% as per the following calculation,

$$\text{Sensitivity} = \text{TP}/(\text{TP} + \text{FN}) \times 100 = 300/(300+25) \times 100 = 92\%$$

$$\text{Specificity} = \text{TN}/(\text{FP} + \text{TN}) \times 100 = 75/(100+75) \times 100 = 43\%$$

	MI Pres ent	MI Abs ent
Positive E CG	300(TP)	100(FP)
Negative ECG	25(FN)	75(TN)

Solution to Question 25:

The specificity of the above screening test is 75%.

The given data is represented below:

TP- true positive, FP- false positive, FN- false negative, TN- true negative

$$\text{Specificity} = \text{TN}/(\text{FP} + \text{TN}) \times 100 = 150/200 = 75\%$$

$$\text{Sensitivity} = \text{TP}/(\text{TP} + \text{FN}) \times 100 = 180/200 = 90\%$$

$$\text{PPV} = \text{TP}/(\text{TP} + \text{FP}) \times 100 = 180/230 = 78\%$$

$$\text{NPV} = \text{TN}/(\text{FN} + \text{TN}) \times 100 = 150/170 = 88\%$$

	Hep C +	Hep C -	Total
Test +	180(TP)	50(FP)	230
Test -	20(FN)	150(TN)	170
Total	200(TP +FN)	200(FP +TN)	400

Solution to Question 26:

If the prevalence of the disease falls, the negative predictive value(NPV) increases.

The NPV of the test is inversely proportional to the prevalence of a disease.

The PPV of the test is directly proportional to the prevalence of a disease.

Sensitivity and specificity are independent of the prevalence of the disease in the area. They do not change with the change in prevalence.

Solution to Question 27:

The positive predictive value (PPV) of the test is 54.3%.

The prevalence of the disease is 20% in 10000, i.e., 2000 people have the disease(TP + FN) and 8000 are disease-free (FP + TN).

The sensitivity of the test is 95%, i.e. $TP/(TP + FN) = 95\%$.

Thus, the true positives (TP)= $95/100 \times (TP + FN) = 95/100 \times 2000 = 1900$.

The specificity of the test is 80%, i.e. $TN/(FP + TN) = 80\%$.

Thus, the true negatives (TN)= $80/100 \times (FP + TN) = 80/100 \times 8000 = 6400$.

The given data is represented below:

TP- true positive, FP- false positive, FN- false negative, TN- true negative

From the above table, PPV is calculated as $TP/(TP + FP) \times 100 = 1900/3500 = 54.3\%$.

An alternate formula(Bayes theorem) that can be used is $PPV = (Sn \times P) / (Sn \times P) + [(1-Sp)(1-P)]$

Sn- Sensitivity, P- Prevalence, Sp- Specificity

	Disease +	Disease -	Total
Test +	1900(TP)	1600(FP)	3500(TP +FP)
Test -	100(FN)	6400(TN)	6500(FN +TN)
Total	2000(TP +FN)	8000(FP +TN)	10000

Solution to Question 28:

The negative predictive value of the above test is 95%.

For convenience, let us assume that the total population in this scenario is 100.

The prevalence of the disease is 30%. So the total number of people with the disease is 30(TP+FN) and disease-free(FP+TN) is 70.

Since sensitivity is 0.90, i.e. $TP/(TP + FN) = 90\%$.

The total true positives (TP) = $90/100 \times (TP+FN) = 90/100 \times 30 = 27$

False-negative(FN) = $30-27 = 3$

Since specificity is 0.80, i.e. $TN/(FP+TN) = 80\%$.

Total true negatives (TN) = $(FP+TN) \times 80/100 = 70 \times 80/100 = 56$

False-positive(FP) = $70-56 = 14$

The given data is represented below:

Negative predictive value (NPV) = $TN/(FN + TN) \times 100 = 56/59 = 0.949 = 95\%$.

An alternate formula(Bayes theorem) that can be used is $NPV = (1-P) \times Sp / [P \times (1-Sn)] + [(1-P) \times Sp]$

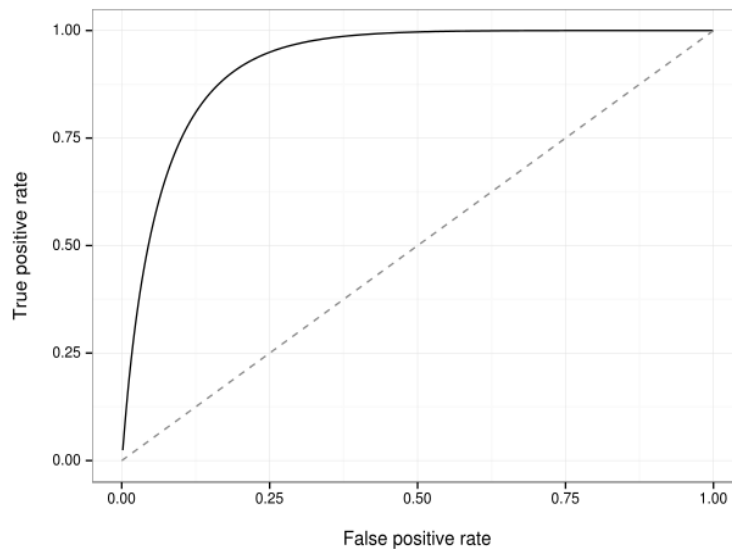
Sn- Sensitivity, P- Prevalence, Sp- Specificity

	Diseas e +	Diseas e -	Total
Test +	27 (TP)	14 (FP)	41
Test -	3 (FN)	56 (TN)	59
Total	30(TP +FN)	70(FP+ TN)	100

Solution to Question 29:

The receiver operator characteristic curve is the best for determining the threshold for the diagnosis of a positive test. It is also used for the comparison of two or more screening/ diagnostic tests of disease with each other.

The following is an example of a ROC curve for a screening test.



Solution to Question 30:

Test A is the best test among the four given, as it has the highest sensitivity and the lowest false positive rate (100-specificity).

A receiver operating characteristic curve (ROC) is obtained by plotting a test's true positive rate (sensitivity) against its false-positive rate (100-specificity) for different diagnostic cut-off points.

The shape of the curve reflects the quality of the test. The better the test, the more it moves to the upper left (more the area under the curve). For a good test, sensitivity comes close to 100% on the y-axis, and the false positive rate comes close to 0% on the x-axis.

Solution to Question 31:

The shaded area represents the false positives (Non-diabetics incorrectly classified as diabetics).

Solution to Question 32:

Both sensitivity and specificity are higher in the test represented in pink (Test B) compared to the test represented in green (Test A).

Sensitivity is the proportion of all people with the disease who test positive. It is the probability of the test being positive when the disease is present.

$\text{Sensitivity} = \text{TP} / (\text{TP} + \text{FN})$ or $\text{Sensitivity} = 1 - \text{false-negative rate}$.

Specificity is the proportion of all people without the disease who test negative. It is the probability of the test being negative when the disease is absent.

Specificity= $TN / (TN + FP)$ or specificity= $1 - \text{false-positive rate}$.

Test B has an increased number of true positives and true negatives and decreased number of false positives and negatives, compared to Test A.

Hence leading to increased sensitivity and specificity.

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If you purchased this from someone else,
you may have been scammed.

Viral Respiratory Infections

Question 1:

When did the WHO declare that smallpox has been globally eradicated?

- a) 24th March 1977
- b) 31st May 1975
- c) 7th April 1975
- d) 8th May 1980

Question 2:

Which of the following statements is not included in the case definition of smallpox?

- a) Acute onset fever and malaise 2-4 days before onset of rash
- b) Development of a maculopapular rash starting on the face and forearms
- c) Evolution of the rash to umbilicated pustules within 48 hours
- d) No alternative diagnosis explaining the illness

Question 3:

A 15-year-old boy presents to you with complaints of fever and malaise. He also complains of an itchy rash which first appeared on the trunk as shown below. When is this disease the most contagious?



- a) After formation of scab
- b) 2 days before and 5 days after the onset of rash
- c) 7 days before to 7 days after the onset of rash
- d) 4 days before to 4 days after the onset of rash

Question 4:

Which of the following is not characteristic of the rash of chicken pox?

- a) Pleomorphic
- b) Centripetal
- c) Deep seated
- d) Unilocular

Question 5:

Which of the following statements are true about chickenpox?

- a) 1, 4 & 5
- b) 2, 4 & 5
- c) 2, 3 & 5
- d) 1, 2 & 4

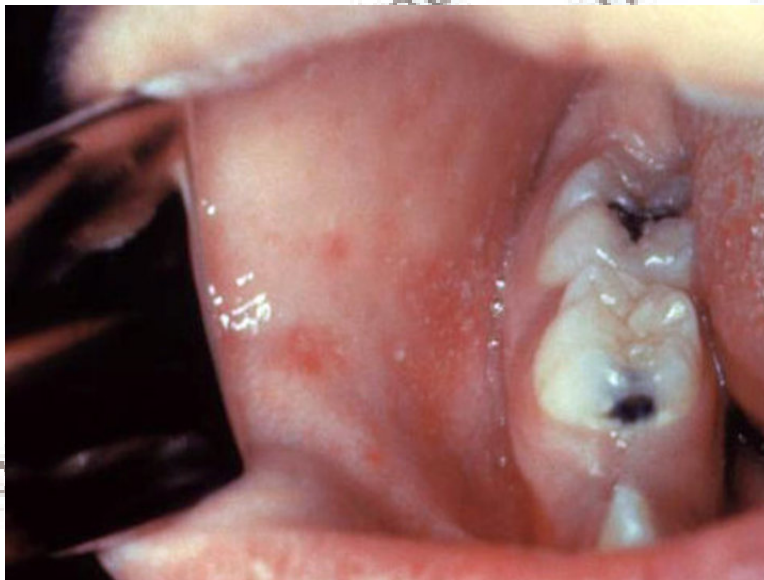
Question 6:

The varicella vaccine can be safely administered to which of the following individuals?

- a) A pregnant woman
- b) A patient allergic to neomycin
- c) A HIV infected patient with CD4+ T cell levels $\geq$ 15%
- d) A patient on long term corticosteroid therapy

Question 7:

A child with a fever and rash was brought to your OPD. Oral findings as shown in the image were observed. During which period can this disease spread to others?



- a) 4 days before and 2 days after the appearance of rash
- b) 2 days before and 2 days after the appearance of rash
- c) 4 days before and 4 days after the appearance of rash
- d) 2 days before and 4 days after the appearance of rash

Question 8:

Which of the following statements is true about measles?

- a) Incubation period is 5 days
- b) Otitis media is a common complication

- c) It is caused by Orthomyxovirus
- d) Large number of carriers are present

Question 9:

During a measles epidemic in the community, an eight-and-a-half-month-old child was vaccinated for measles. What is the correct immunisation schedule to be followed for her, regarding measles?

- a) The child should be given a second dose after 4 weeks.
- b) The child should be given a second dose as soon as she reaches 9 months of age.
- c) The child should directly be given the second dose of measles at 16 months.
- d) The child should be given the second dose of measles at 12 months of age.

Question 10:

A child was brought to you with complaints of fever, malaise and a rash. You diagnose the child with measles. What is the recommended period of isolation for this disease?

- a) From onset of rash to 3 days after rash
- b) 2 weeks before rash to 1 week after rash
- c) From onset of rash to 5 days after rash
- d) From catarrhal stage to 5 days after rash

Question 11:

Which of the following is correct about the criteria for elimination of measles?

- a) There should be an absence of transmission of epidemic measles
- b) The duration of absence of transmission should be ≥ 6 months
- c) A well-performing surveillance system should be present
- d) Absence of transmission should be seen in multiple geographical areas

Question 12:

Which of the following is not true about the Measles-Rubella campaign?

- a) It is a mass vaccination campaign organised to introduce MR vaccine

- b) Follow-up campaign is organised every 3-5 years
- c) It covers children from 9 months to 15 years of age
- d) Mop-up strategy is a part of the campaign

Question 13:

According to the WHO, what percentage of immunization coverage is needed for eradication of measles?

- a) 68%
- b) 72%
- c) 84%
- d) 96%

Question 14:

Which of the following is false about Rubella infection?

- a) It is a type of droplet infection
- b) Vertical transmission is possible
- c) Infection in early pregnancy causes a milder form of disease in the fetus
- d) Fetus affected in late pregnancy may have only deafness

Question 15:

According to the WHO, which of the following tests cannot be used to confirm a case of swine flu?

- a) Real-time PCR
- b) Viral culture
- c) 4-fold rise in virus specific neutralizing antibodies
- d) ELISA

Question 16:

What is the route of administration of avian influenza vaccine?

- a) Intranasal
- b) Intramuscular
- c) Subcutaneous
- d) Intradermal

Question 17:

Vaccine that requires strain change due to evolving organism is?

- a) Influenza
- b) Diphtheria
- c) Polio
- d) Typhoid

Question 18:

Pandemic that occurred in 2009 was due to which influenza virus?

- a) H1N1
- b) H5N1
- c) H5N7
- d) H3N2

Answer Key

Question No.	Correct Option
1	d
2	c
3	b
4	c
5	a
6	c
7	c
8	b
9	a

10	d
11	c
12	d
13	d
14	c
15	d
16	b
17	a
18	a

Detailed Explanations

Solution to Question 1:

The WHO declared the eradication of smallpox on 8th May 1980.

Solution to Question 2:

The rash of smallpox evolves within 48 hours, to first form deep-seated, firm, and well-circumscribed vesicles, which later become umbilicated or confluent pustules.

Case definition for notification of smallpox:

- An individual of any age with acute onset fever, malaise, severe prostration, headache, and backache 2-4 days before the onset of rash.
- Development of a maculopapular rash starting on the face and forearms and then spreading to the trunk and legs. Rash evolves within 48 hours to deep-seated, firm, and well-circumscribed vesicles, which later become umbilicated or confluent pustules.
- Lesions that appear at the same stage of development (i.e. all are vesicles or all are pustules) on any given part of the body.
- No alternative diagnosis explaining the illness.
- Laboratory confirmation: Identification of variola (smallpox) DNA with PCR or isolation of the virus from a clinical specimen

Solution to Question 3:

The rash (different stages at one time) in the image and the clinical history are suggestive of chickenpox. The period of communicability for chickenpox is 2 days before and 5 days after the onset of rash.

Solution to Question 4:

The rash of chickenpox is superficial and the rash of smallpox is deep-seated.

Difference between the rash of chickenpox and smallpox:

Chickenpox	Smallpox
Centripetal	Centrifugal
Spares palms and soles, rash mostly on flexor surfaces	Palms and soles involved, rash predominant on extensor surfaces
Superficial	Deep-seated
Vesicles are unilocular and give dewdrop on rose petal appearance	Vesicles are multilocular and umbilicated
Rash is pleomorphic and symmetrical	Rash is monomorphic
Rapid evolution	Slow evolution

Solution to Question 5:

Chickenpox is caused by the human herpes virus-3 (varicella-zoster virus). It remains latent in the sensory ganglion. Reactivation of the virus can cause shingles.

The secondary attack rate of chickenpox is very high, 85% (range 61%–100%).

The characteristic infective vesicle, surrounded by an erythematous halo, is described as a dewdrop on a rose petal. The scab of chickenpox is not infective.

The vaccine given for chickenpox prophylaxis is live attenuated vaccine.

Solution to Question 6:

Varicella vaccine can be considered for clinically stable HIV infected children or adults with CD4+ T cell level $\geq 15\%$, including those receiving active antiretroviral therapy.

Varicella vaccine is contraindicated in:

- Pregnant women
- Immunocompromised patients
- Patients allergic to neomycin

Following vaccination, salicylates should be avoided for 6 weeks to prevent Reye's syndrome.

Solution to Question 7:

The given image shows koplik spots, which are a pathognomonic sign of measles. The period of communicability for measles is 4 days before and 4 days after the appearance of rash. The patient is isolated for 7 days from the onset of rash.

Solution to Question 8:

The most common complication of measles is otitis media, occurring in about 10–15 % of patients.

Measles is caused by an RNA virus of the Paramyxovirus family. The incubation period of measles is 10 to 14 days. The only source of infection is a case of measles. Carriers are not known to occur.

The complications of measles include:

- Otitis media (most common)
- Measles-associated diarrhea
- Pneumonia
- Conjunctivitis
- Febrile seizures
- Subacute sclerosing pan encephalitis (rare but is fatal) - occurs several years after primary infection

Solution to Question 9:

In this child, the next dose of measles vaccine should be given after 4 weeks.

While the scheduled age of receiving a measles vaccine is 9 months, it can be lowered up to 6 months during a measles epidemic in the community.

For infants immunized between 6 to 9 months of age, a second dose should be administered as soon as the child reaches the age of 9 months. It is to ensure adequate immunity has developed, provided that 4 weeks have elapsed since the last dose.

In this child, the first dose was given at the age of 8 and a half months, and there must be a gap of 4 weeks between the two doses. If the child were to be vaccinated as soon as she turns 9 months old, only two weeks would have been elapsed.

Solution to Question 10:

The isolation period for measles is from the catarrhal (prodromal) stage to 5 days after the onset of the rash.

The period of communicability is approximately 4 days before and 5 days after the appearance of the rash. Isolation of the patient for a week from the onset of rash adequately covers the period of communicability.

Solution to Question 11:

Measles elimination is operationally defined as the absence of endemic measles transmission in a defined geographical area (region or country) for ≥ 12 months in the presence of a well-performing surveillance system.

Solution to Question 12:

Mop-up strategy is not a part of the Measles-Rubella campaign.

Measles-Rubella vaccine introduction campaign:

- A mass vaccination campaign to introduce an MR vaccine
- Targets a wide age group of children: 9 months to ≤ 15 years
- All children in the target age group receive an additional dose of the vaccine
- Regardless of previous vaccination status or history of illness.

Measles-Rubella follow-up campaign:

- A mass vaccination campaign as a periodic event every 3–5 years guided by country-specific surveillance data
- Target age group for immunization: all children aged ≥ 9 months, who were born after the previous MR vaccination campaign

Solution to Question 13:

According to the WHO, eradication of measles requires achieving an immunization coverage of at least 96% of children under 1 year of age.

Solution to Question 14:

Earlier the rubella infection occurs in pregnancy, greater is the damage to the foetus.

The major determinant of the extent of fetal infection in congenital rubella syndrome is the gestational age at which fetal transmission occurs.

Rubella infection in 1st trimester can cause:

- Abortions
- Still birth
- Skin lesions - blueberry muffin rash
- Congenital rubella syndrome

Solution to Question 15:

ELISA is not included in the criteria for confirming a case of H1N1.

The ELISA is an easy to perform, highly specific, and fairly sensitive diagnostic tool for the 2009 pandemic influenza A (H1N1) virus infections. However, it is not included in the WHO 's list of confirmation tests.

According to WHO, 2009 pandemic (H1N1) virus infection is confirmed by one or more of the following tests:

- Polymerase chain reaction (PCR)
- Viral culture
- 4■fold rise in H1N1 virus■specific neutralizing antibodies

Solution to Question 16:

The avian influenza H5N1 vaccine is administered intramuscularly.

H5N1 avian influenza vaccine was approved in 2017 for the age group of 18 to 64 years of age who are at increased risk of exposure to the H5N1 influenza virus subtype contained in the vaccine.

It is administered intramuscularly in the upper arm and given in 2 doses, 28 days apart.

Note: The live attenuated H1N1 influenza vaccine is given intranasally.

Solution to Question 17:

Influenza vaccine requires periodic strain change due to antigenic change in the surface glycoprotein of the virus.

The change may be gradual (antigenic drift) or drastic (antigenic shift).

Solution to Question 18:

Recent influenza pandemic occurred in 2009 due to the emergence of a novel H1N1 influenza A virus.

Past influenza pandemics:

- 1968 pandemic due to H3N2 virus
- 1957-58 pandemic due to H2N2 virus
- 1918 pandemic due to H1N1 virus

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Bacterial Respiratory Infections

Question 1:

Which of the following is the most important factor in reduction of diphtheria cases?

- a) Chemoprophylaxis
- b) Improved standard of living
- c) Vaccination
- d) Increased surveillance rates

Question 2:

Which of the following statements is true about pertussis?

- a) Most of the infections are subclinical
- b) Most infectious during incubation period
- c) Secondary attack rate is 90 % in unimmunized individuals
- d) Cerebellar ataxia is a common chronic complication

Question 3:

Droplet precaution and isolation is to be done for all of the following conditions except:

- a) Pertussis
- b) Diphtheria
- c) Localized varicella zoster
- d) Mycoplasma pneumonia

Question 4:

According to IMNCI, which of the following is a key clinical sign for pneumonia classification?

- a) Nasal flaring
- b) Wheezing

- c) Cough
- d) Fast breathing

Question 5:

According to IMNCI, a 6-month-old child is said to have pneumonia if the child has fast breathing with a respiratory rate more than:

- a) 40
- b) 60
- c) 30
- d) 50

Question 6:

A one-and-half-year-old child was brought to the hospital with a history of breathlessness, fever, cough, and cold. On examination, respiratory rate was 46/min, and there were no chest retractions. Which is the most appropriate management?

- a) Give IV antibiotics and observe
- b) Give IV antibiotics and refer to higher centre
- c) Oral antibiotics, warn of danger signs, and send home
- d) No need of treatment only home remedies

Question 7:

A polysaccharide vaccine against *Streptococcus pneumoniae* has 23 serotypes. In which of the following patients is this vaccine most beneficial?

- a) A child < 2 years
- b) A patient with sickle cell disease
- c) A patient with cystic fibrosis
- d) A patient with recurrent otitis media

Question 8:

Which of the following tests is used to determine the prevalence of TB infection?

- a) Sputum examination
- b) Mantoux test
- c) CB-NAAT test
- d) Chest X-ray

Question 9:

A 24-year-old patient has been given an intradermal injection of 0.1ml PPD (5 tuberculin units) over the forearm. After 72 hrs, the site is examined as shown in the image below. What is the interpretation of the result obtained?



- a) Tuberculin test negative, suggesting immunity from TB
- b) Tuberculin test positive, suggesting an active infection
- c) Tuberculin test negative, suggesting a recent or past infection
- d) Tuberculin test positive, suggesting a recent or past infection

Question 10:

Which of the following is the most appropriate method to assess the incidence of tuberculosis in the community?

- a) Identify all positives to tuberculin test
- b) Sputum examination of symptomatic patients
- c) Identify new converter to tuberculin test

- d) Screen all under 5 years children with tuberculin test

Question 11:

Tuberculin testing is done at which anatomical site?

- a) Dorsum of left arm
- b) Dorsum of right arm
- c) Ventral aspect of left forearm
- d) Ventral aspect of right forearm

Question 12:

The approach shown in the image is a part of which of the following?



- a) Joint TB Monitoring Mission 2017
- b) National Strategic Plan for TB, 2017 -2025
- c) Standards of TB Care
- d) National AIDS Control Organization

Question 13:

Which of the following is the most rapid diagnostic test for pulmonary TB?

- a) Quantiferon TB gold plus

- b) Sputum microscopy
- c) Radiometric BACTEC method
- d) CB-NAAT

Question 14:

BCG vaccine provides protection against which of the following?

- a) Pulmonary TB
- b) CNS and disseminated TB
- c) Pulmonary and CNS TB
- d) Pulmonary and skeletal TB

Question 15:

What is the next step in management of a TB smear positive patient with no significant findings on a chest X ray?

- a) Repeat sputum smear examination
- b) Start ATT
- c) Do CB-NAAT
- d) Do sputum culture

Question 16:

You are posted in a TB clinic. You need to start an intensive phase (HRZE 75/150/400/275) regimen for a patient weighing 70 kgs. How many tablets of the FDCs are to be taken by him on a daily basis?

- a) 2
- b) 3
- c) 4
- d) 5

Question 17:

According to NTEP, how will you classify a TB patient who had taken antitubercular drugs for only 3 weeks in the past?

- a) New case
- b) Previously treated case
- c) Treatment after loss to follow-up
- d) Recurrent TB case

Question 18:

Which of the following is the appropriate schedule for the treatment of multi-drug resistant tuberculosis?

- a) 6 drugs for 4-6 months followed by 4 drugs for 12 months
- b) 7 drugs for 4-6 months followed by 4 drugs for 5 months
- c) 4 drugs for 4 months followed by 6 drugs for 10-12 months
- d) 5 drugs for 6 months followed by 4 drugs for 12 months

Question 19:

Which of the following drugs is not used in the management of a patient with MDR-TB under NTEP?

- a) Levofloxacin
- b) PAS
- c) Ethionamide
- d) Cycloserine

Question 20:

Which of the following is true regarding the duration for treatment of XDR TB?

- a) Intensive phase is for 6 to 9 months
- b) Continuation phase is for 12 months
- c) Total duration is for 20 months
- d) A combination of oral and injection drugs are used

Question 21:

Which of the following is not a criteria for the bedaquiline conditional access programme under RNTCP?

- a) Patient below 5 years can be included
- b) Post menopausal females can be included
- c) Patients with stable controlled arrhythmia can be included
- d) Pregnant patients are excluded

Question 22:

Which of the following statements regarding TB is false?

- a) The Index TB guidelines are used for extra pulmonary TB.
- b) X-pert MTB/RIF assay is an adjunct for diagnosis of lymph node TB.
- c) Steroids may be used in patients with pleural TB without HIV for 4 weeks.
- d) Duration of ATT for TB meningitis is 9 months.

Question 23:

Where is the National Tuberculosis Institute located?

- a) Mumbai
- b) Bangalore
- c) Delhi
- d) Hyderabad

Answer Key

Question No.	Correct Option
1	c
2	c
3	c
4	d
5	d

6	c
7	b
8	b
9	d
10	c
11	c
12	b
13	b
14	b
15	c
16	d
17	a
18	b
19	b
20	c
21	a
22	c
23	b

Detailed Explanations

Solution to Question 1:

Vaccination is the most important factor responsible for the reduction of diphtheria cases over the past decade.

Solution to Question 2:

The secondary attack rate in pertussis is about 90% in unimmunized individuals.

The source of infection in pertussis is a case of pertussis. There is no evidence of subclinical infections. The patients are most infectious during the catarrhal stage.

The chief complications of pertussis are:

- Bronchopneumonia
- Bronchitis
- Bronchiectasis

Solution to Question 3:

Droplet precaution and isolation is not done for localized varicella zoster, as it is only transmitted through direct contact.

Varicella-zoster localized in the patient with an intact immune system with a lesion that can be contained/covered do not require droplet precaution. Disseminated varicella in any patient requires airborne and contact precautions.

Diphtheria, pertussis, and mycoplasma are transmitted by droplets and hence require droplet precaution and isolation.

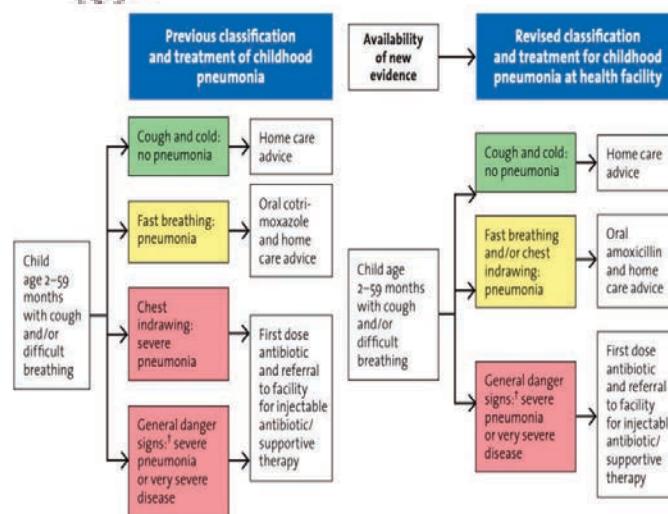
Although patients are not contagious starting 48 hours after antibiotic therapy, droplet precautions are typically maintained until the completion of the antibiotic course and are culture-negative on two cultures taken 24 hours apart. Droplet precautions include the use of masks, gowns, gloves and protective eye gear when providing direct patient care.

Solution to Question 4:

According to the IMNCI, three key clinical signs are used to assess a sick child with suspected pneumonia:

- Respiratory rate (RR): distinguishes children who have pneumonia from those who do not. Children are classified under pneumonia if:
 - > 50/min RR for infants 2 months to 12 months
 - > 40/min RR for children 12 months to 5 years of age
- Lower chest wall indrawing indicates pneumonia
- Stridor indicates those with severe pneumonia requiring hospital admission.

The following image gives the IMNCI classification for cough/difficulty in breathing in children.



[†] Not able to drink, persistent vomiting, convulsions, lethargic or unconscious, stridor in a calm child or severe malnutrition.

Signs of severe pneumonia are :

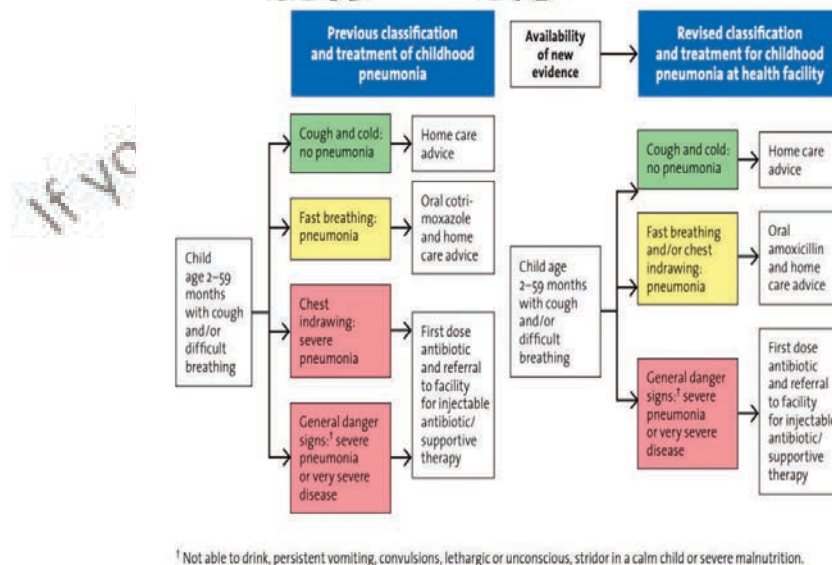
- Not able to drink
- Persistent vomiting
- Convulsions
- Lethargic or unconscious
- Stridor in a calm child
- Severe malnutrition

Solution to Question 5:

Three key clinical signs are used to assess a sick child with cough or difficult breathing:

- Respiratory rate (RR): distinguishes children who have pneumonia from those who do not. Children are classified as pneumonia if:
 - ≥ 50 /min RR for infants 2 months to 12 months
 - ≥ 40 /min RR for children 12 months to 5 years of age
- Lower chest wall indrawing indicates severe pneumonia
- Stridor indicates those with severe pneumonia requiring hospital admission.

The following image gives the IMNCI classification for cough/difficulty in breathing in children.



Signs of severe pneumonia are :

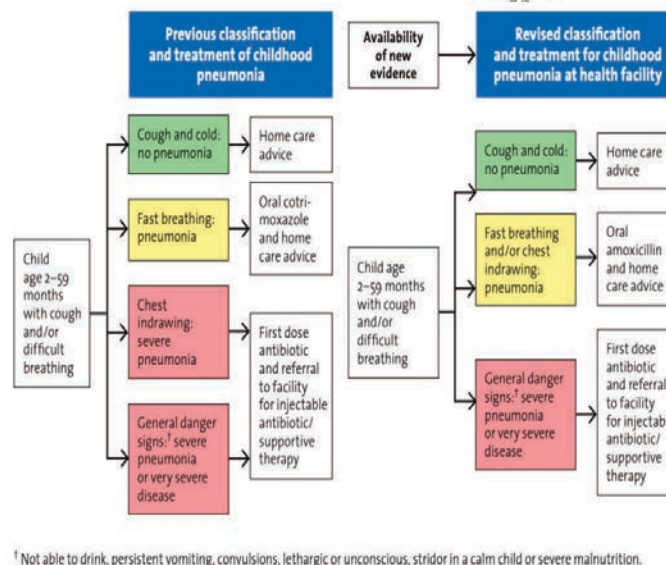
- Not able to drink
- Persistent vomiting
- Convulsions

- Lethargic or unconscious
- Stridor in a calm child
- Severe malnutrition

Solution to Question 6:

From the given clinical scenario, it can be easily inferred that the child has pneumonia. The treatment for pneumonia for a child between 2 months and 5 years of age is oral amoxicillin and home care advice.

The following image gives the IMNCI classification for cough/difficulty in breathing in children.



Solution to Question 7:

The polysaccharide vaccine against *Streptococcus pneumoniae* (PPSV-23) is most beneficial for patients with sickle cell disease.

S. pneumoniae is an encapsulated organism that is detected by the spleen and is removed. However, in sickle cell disease, the sickled cells block the vessels in the spleen which results in splenic infarction. This hyposplenism results in an increased risk of pneumococcal infection. Hence, pneumococcal vaccine (PPSV-23) is given to these patients as a prophylactic measure to reduce this risk.

Solution to Question 8:

Mantoux or tuberculin testing is used to detect the prevalence of TB infection.

Infection of tuberculosis does not always lead to disease. In latent infections, the Mantoux test can be positive without any clinical symptoms. Therefore, the prevalence of TB infection is detected by the Mantoux test.

However, the prevalence of TB disease is confirmed by a sputum smear.

Solution to Question 9:

The above image shows an induration of ≥ 10 mm after 72 hrs of intradermal PPD injection suggesting a positive tuberculin test (TST), indicating recent or past infection.

In TST, 0.1 ml of PPD (purified protein derivative) containing 5 TU (tuberculosis unit) is injected intradermally and induration is read after 48-72 hours. An induration of 10 mm or more is considered positive in patients from countries with a high prevalence of tuberculosis. Surrounding erythema is not considered.

A positive tuberculin test indicates hypersensitivity to tuberculo-protein, denoting either latent infection with tubercle bacilli, or BCG immunization, or recent or past infection with or without clinical disease.

False positives may be seen in infections with related mycobacteria and false negatives in miliary tuberculosis, severely immunosuppressed states, etc.

Solution to Question 10:

Identification of new converters to the tuberculin test is a marker of the incidence of tuberculosis. Identification of all positives to the tuberculin test is a marker of the prevalence of tuberculosis.

Tuberculin conversion refers to the situation where an individual's tuberculin skin test (TST) results in changes from negative (0-4 mm diameter induration) to positive (equal to or ≥ 10 mm diameter induration) within a 24-month period. TST conversion is considered presumptive evidence of acquired latent tuberculosis infection (LTBI) and potential risk for progression to active tuberculosis disease.

Solution to Question 11:

Mantoux/ Tuberculin testing is carried out by injecting 1 TU of PPD in 0.1 ml intradermally on the flexor (ventral) surface of the left forearm, midway between elbow and wrist.

The result of the test is usually read after 48-96 hours, but reading it after 72 hours is ideal.

The horizontal transverse diameter of induration is measured, based on which the following diagnosis can be inferred:

- < 6 mm – Negative
- 6-9 mm – Doubtful

- ≥ 10 mm – Positive

Solution to Question 12:

Detect – Treat – Prevent – Build approach for eliminating TB in India is part of the National Strategic Plan for TB (2017-2025). The components of this approach are:

- Detect - Early identification of presumptive TB cases, at the first point of care, be it private or public sectors, and prompt diagnosis using high sensitivity diagnostic tests.
- Treat - Provide sustained, equitable access to high quality TB treatment, care and support services.
- Prevent - Prevent the emergence of TB in susceptible populations.
- Build - Undertake critical management reforms, restructuring of HR and financial norms, pathways for private sector participation, in order to improve efficiency, effectiveness.

Solution to Question 13:

The most rapid diagnosis of pulmonary TB can be done by sputum microscopy, with results in 30 minutes.

It has the least sensitivity and specificity. Smears are stained by Ziel Neelson acid-fast stain.

CBNAAT gives results within 90 minutes. Cartridge-based nucleic acid amplification test is a Mycobacterium tuberculosis-specific automated, cartridge-based nucleic acid amplification assay, having fully integrated and automated amplification and detection using real-time PCR.

Quantiferon TB gold plus takes 24 hours to give the result. The radiometric BACTEC method takes approximately 2-3 weeks. Sputum culture takes approximately 2 to 8 weeks.

Solution to Question 14:

BCG vaccine provides protection against CNS and disseminated TB.

It does not prevent primary infection. More importantly, it does not prevent reactivation of latent pulmonary infection, the principal source of bacillary spread in the community. The impact of BCG vaccination on transmission of mycobacterium tuberculosis is therefore limited.

Solution to Question 15:

The next best step in a smear-positive patient with no findings on chest-xray is to perform a CB-NAAT. This patient is a microscopically confirmed case of tuberculosis (TB).

According to the TB annual report, CBNAAT has been tried to be established in all centres. Ideally, drug sensitivity status via NAAT is to be done following a sputum smear before initiation of treatment. This helps in the early detection and treatment of drug-resistant TB. This has been integrated into the regular TB services to detect rifampicin resistance at the time of diagnosis. Depending on the sensitivity of the tubercle bacilli, appropriate ATT is started.

If NAAT is not done and sputum is positive, patient is considered as Microbiologically positive with sensitivity status unknown.

Solution to Question 16:

An adult weighing 65-75kg needs to take 5 tablets of FDC (fixed-dose combinations) containing HRZE 75/150/400/275 in the intensive phase.

The new WHO guidelines suggest the number of FDC tablets to be taken daily according to five weight band categories.

Note: A TB patient kit contains the full course of treatment for a single patient and thus assures the TB patient that his or her medications will be available throughout treatment. The kit provides health workers with a container that has all the required drugs in the necessary strengths and quantities. This helps limit confusion and wastage and makes it easier to monitor the regularity of treatment; avoiding stock-outs also helps to maintain patient confidence in the health system.

Weight category	Number of tablets (FDCs)	
Intensive phaseHRZE75/150/400/275	Continuation phaseHRE75/150/275	
25-34 kg	2	2
35-49 kg	3	3
50-64 kg	4	4
65-75 kg	5	5
>75 kg	6	6

Solution to Question 17:

Under NTEP, a new case is defined as a TB patient who had never taken ATT previously or taken ATT for <1 month.

Solution to Question 18:

The appropriate schedule for the treatment of multi-drug resistant tuberculosis (MDR-TB) is 7 drugs for 4-6 months followed by 4 drugs for 5 months.

Note: Pearl 1720 has been updated according to NTEP 2021 guidelines.

Solution to Question 19:

PAS (para-aminosalicylic acid) is not included under the MDR-TB regimen.

Solution to Question 20:

All oral longer MDR-TB regimen is used for the treatment of XDR-TB patients for a duration of 20 months.

Regimen class	Drugs
All oral longer MDR TB regimen	Bdq(6 months) Lfx Lzd Cfz Cs for 18-20 months

Solution to Question 21:

Patients under 5 years are excluded from the bedaquiline conditional access programme.

The criteria to be met under the bedaquiline conditional access programme are as follows:

- Children aged 5 to <18 years of age and weighing at least 15 kg (in consultation with a paediatrician)
- Females on non-hormone-based contraceptive or post-menopausal females can be included.
- Patients with controlled stable cardiac arrhythmia can be considered after cardiac consultation.
- Pregnant women to be excluded

Solution to Question 22:

Steroids may be used in HIV-negative patients with TB meningitis and pericarditis for 4 weeks, but not for pleural TB.

The Index TB guidelines for extrapulmonary TB are as follows:

- Duration of ATT in TB meningitis is for 9 months and lymph node and abdominal TB is 6 months.
- X-pert MTB/RIF assay may be used as an adjunct in the diagnosis of lymph node TB and TB meningitis.
- Steroids may be used in HIV-negative patients with TB meningitis and pericarditis for 4 weeks.

Solution to Question 23:

National Tuberculosis Institute is located in Bangalore.

Important national centers for tuberculosis in India:

- Tuberculosis Research Center: Chennai
- National Tuberculosis Institute: Bangalore
- National Institute of Tuberculosis and Respiratory Diseases: Delhi
- JALMA Institute for Leprosy and Other Mycobacterial Diseases: Agra.

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Intestinal Infections

Question 1:

Which of the following is an example of disability limitation for polio?

- a) Polio immunization drives for infants to reduce the occurrence.
- b) Arranging for schooling for a child suffering from residual polio paralysis.
- c) Resting the limbs affected by polio in neutral position.
- d) Providing calipers for walking to patients affected by polio.

Question 2:

A person is involved in unsafe sexual practices. Which of the following viruses is least likely to get transmitted via this route?

- a) Hepatitis A
- b) Hepatitis E
- c) Hepatitis C
- d) Hepatitis D

Question 3:

For how long is Hepatitis A excreted in the feces?

- a) One week before the symptoms appear
- b) Two weeks after the symptoms appear
- c) Two weeks before the symptoms and two weeks thereafter
- d) One week before the symptoms and one week thereafter

Question 4:

An intern sustains a needlestick injury while drawing blood from a Hepatitis B positive patient. Assuming he has been infected, which among the following will be the first to rise?

- a) Anti- HBs
- b) Anti- HBc

- c) HBeAg
- d) HBsAg

Question 5:

A 30-year-old pregnant woman is rushed to the hospital with labor pains at 34 weeks of gestation. She did not attend any prior antenatal visits. On further evaluation, she is found to be positive for HbsAg. Which of the following should be administered to the newborn after delivery?

- a) Hepatitis B vaccine + Immunoglobulin
- b) Immunoglobulin only
- c) Hepatitis B vaccine only
- d) Immunoglobulin, followed by vaccine 1 month later

Question 6:

HCV screening is done for which of the following people?

- a) 1, 2, and 5
- b) 1, 2, 3, 4 and 5
- c) 3 and 5
- d) 1, 2 and 3

Question 7:

Which of the following statements is false about Rotavirus?

- a) Severe rotavirus infections are limited to children aged 5-7 years.
- b) Rotavirus is shed in high concentrations for several days in the stools and vomit of infected children.
- c) Rotavirus diarrhoea increases in frequency during drier, cooler months.
- d) Clean water supplies and good hygiene are unlikely to have a substantial effect on viral transmission.

Question 8:

A mother brought a 2 month old child with complaints of diarrhea and vomiting. As a part of management, you advise her regarding vaccination for rotavirus. Which is the false statement regarding the same?

- a) Rotarix is administered in 2 oral doses.
- b) It cannot be given in an immunocompromised child.
- c) The first dose of Rotarix is contraindicated for infants <12 weeks.
- d) It is not a part of catchup vaccination programmes due to its risk of intussusception.

Question 9:

Which of the following statements is false for carriers of cholera?

- a) A convalescent carrier may continue to excrete vibrios for 2-3 weeks.
- b) Incubatory carriers are potential patients.
- c) The gall bladder is infected in chronic carriers.
- d) The antibody titres of a contact carrier is normal while the stool culture is positive.

Question 10:

A convalescent carrier of cholera will remain infectious for how many days after recovery?

- a) <7 days
- b) 7-14 days
- c) 14-21 days
- d) 21-28 days

Question 11:

A child is brought to the PHC with acute diarrhea. The doctor based on the history suspects that the child may be suffering from an infection due to shigella over cholera. Which of the following points on the history, apart from bloody stools, favors the doctor's suspicion?

- a) Absence of rectal pain
- b) Presence of abdominal cramps
- c) Presence of vomiting
- d) Presence of fever

Question 12:

Which of the following guidelines will you follow to treat a child diagnosed with cholera?

- a) Start ciprofloxacin immediately after admission.
- b) A cardiotonic should be added if shock persists even after treating with iv fluids.
- c) If the diarrhoea persists after 48 hours of treatment, suspect resistance.
- d) Antidiarrheals and antiemetics should be started.

Question 13:

A 36-year-old man is diagnosed with cholera. Which of the following drug would you prescribe to his family members to prevent the spread of the disease?

- a) Tetracycline
- b) Azithromycin
- c) Ciprofloxacin
- d) Cotrimoxazole

Question 14:

Which of the following is the most effective prophylactic measure against cholera?

- a) Health education
- b) Vaccination
- c) Chemoprophylaxis
- d) Early case detection

Question 15:

You are a medical officer posted in a village 'X'. The village has been affected by the epidemic of cholera. What is the first step that should be taken to decrease the incidence of the disease?

- a) Safe water supply and sanitation
- b) Cholera vaccination to all uninfected individuals
- c) Mass chemoprophylaxis
- d) Treat everyone in the village with tetracycline

Question 16:

In the ward, a patient with cholera soils the bed with stools. Which chemical should be used to disinfect the bed?

- a) Bleaching powder
- b) Cresol
- c) Hydrogen Peroxide
- d) Chlorine

Question 17:

Cholera is a notifiable disease locally and nationally. Which of the following statements is true regarding the international health regulations for the notification of Cholera?

- a) Cholera is notifiable to the WHO by the National Government within 72 hours of its occurrence.
- b) The number of cases and deaths are to be reported daily and weekly till the area is declared free of cholera.
- c) An area is declared free of cholera when 1 month has elapsed since the last case.
- d) Mass chemoprophylaxis must be initiated immediately.

Question 18:

A 5-year-old boy has had 15 episodes of loose stools in the last 24 hours and vomited twice in the last 4 hours. The child is restless and has sunken eyes. On examination, he is irritable but is drinking fluids. What is the optimal therapy for the child?

- a) Intravenous fluids
- b) Oral rehydration therapy
- c) Intravenous fluid initially for 4 hours followed by oral fluids
- d) Intravenous fluids + antibiotic therapy

Question 19:

A mother brings her child with complaints of acute diarrhea. After an initial assessment, you diagnose that the child has some dehydration. You counsel the mother to give ORS. After how many hours shall she be advised to discard the old solution and prepare a new one?

- a) 4 hours
- b) 8 hours
- c) 12 hours
- d) 24 hours

Question 20:

A baby is brought with complaints of diarrhea and vomiting. The weight of the baby is 12 kgs. What is the estimated amount of fluid to be replaced in 4 hours?

- a) 0-400 ml
- b) 400-800 ml
- c) 800-1200 ml
- d) 1200-1600 ml

Question 21:

A 9-month-old baby with diarrhea is feeding well and is thirsty. Skin pinch when done returns in 2 seconds. What is the treatment?

- a) 200-400 ml of ORS to be given in the first four hours.
- b) 400-700 ml of ORS to be given in the first four hours.
- c) 700-900 ml of ORS to be given in the first 4 hours.
- d) In addition to the usual fluid intake, 100-200 ml should be given after each loose stool.

Question 22:

Which among the following home remedies are not advised for a child with diarrhea?

- a) A mixture of 1 teaspoon of salt and 6 teaspoons of sugar in 1 L of drinking water
- b) Unsalted water in which a cereal has been cooked
- c) Salted coconut water
- d) Soup with 3g/litre salt added to it

Question 23:

A child with severe dehydration is being treated with intravenous fluids for the past 2 hours. Which of the following will be used as an early indicator for adequate rehydration?

- a) Skin elasticity
- b) Presence of tears
- c) Urine flow rate
- d) Moistness of tongue

Question 24:

A 15-year-old boy is referred from a sub-center due to a feeble pulse and low BP. The patient has had diarrhea for 2 days. He has sunken eyes and is lethargic. On examination, BP is 90/60 mmHg. Which of the following fluids cannot be used for resuscitation in this patient?

- a) Hartmann's solution
- b) Diarrhoea treatment solution
- c) Normal saline
- d) Dextrose solution

Question 25:

What is the recommended dosage for zinc supplements in children with diarrhea?

- a) 20 mg of zinc daily for infants < 12 months of age for 10-14 days
- b) 20 mg of zinc for children > 6 months of age for 7 days
- c) 10 mg of zinc for children > 12 months of age for 20 days
- d) 10 mg of zinc for children < 6 months of age for 10-14 days

Question 26:

For a child with SAM, what should be the sodium content of his rehydration fluid?

- a) 90 mmol/L
- b) 60 mmol/L
- c) 45 mmol/L
- d) 75 mmol/L

Question 27:

What is the average incubation period of typhoid fever?

- a) 10-14 days
- b) 3-5 days
- c) 21-25 days
- d) 7-10 days

Question 28:

In a boys hostel, there has been an outbreak of typhoid. The hostel cook is identified as the carrier of the disease. Which is true among the following statements regarding her condition?

- a) Gallbladder is usually not involved in carrier state
- b) Chronic carriers are those who excrete the bacilli for more than a year
- c) Tetracycline is the drug of choice for carriers
- d) To treat the carrier state, treatment should be given for 4 weeks

Question 29:

In which of the following patients can the Ty21a vaccine be administered?

- a) A 50-year-old HIV patient who is asymptomatic with CD4+ count 240/mm³
- b) A 42-year-old patient with fever and cough for 3 days
- c) A 25-year-old woman suffering from amoebiasis
- d) A 64-years-old alcoholic on colchicine therapy for acute gout

Question 30:

Which of the following statements is false regarding dracunculiasis?

- a) India has eliminated the disease.
- b) Niridazole prevents transmission of the disease.
- c) The disease is limited to tropical and subtropical regions.
- d) No animal reservoir has been proved.

Question 31:

Which of the following statements is false about hookworms?

- a)** Blood loss due to *Necator americanus* infection is more compared to *Ancylostoma* infection.
- b)** Malnutrition is a predisposing factor for this chronic disabling disease.
- c)** Chandler's index is used in epidemiological studies of hookworm.
- d)** Chronic blood loss in pregnant women may lead to abortions, stillbirth and low birth weight babies.

Answer Key

Question No.	Correct Option
1	c
2	b
3	c
4	d
5	a
6	d
7	a
8	c
9	d
10	c
11	d
12	c
13	a
14	a
15	a
16	b
17	b
18	b
19	d
20	c
21	b
22	b
23	a

24	d
25	d
26	c
27	a
28	b
29	a
30	b
31	a

Detailed Explanations

Solution to Question 1:

Resting the limbs affected by polio in a neutral position is an example of disability limitation. It is an intervention done to either halt or prevent the transition of the disease process from impairment to handicap.

Rehabilitation has been defined as the use of medical, social, educational, and vocational measures for training the individual to the highest possible level of functional ability. Like providing calipers for walking to patients and arranging schooling after being affected by polio.

Immunization drives are examples of specific protection.

Solution to Question 2:

Hepatitis E has feco-oral or waterborne transmission. It does not transmit sexually. Transmission of Hepatitis A, C, and D can occur sexually.

Transmission of the hepatitis A virus can occur from any sexual activity with an infected person and is not limited to fecal-oral contact. People who are sexually active are considered at risk for hepatitis A if they are MSM, live with or are having sex with an infected person, or inject drugs.

Solution to Question 3:

HAV is excreted in the feces for about 2 weeks before the onset of jaundice and for up to 2 weeks thereafter.

The risk of transmitting is greatest at 2 weeks before to 1 week after the onset of jaundice.

Solution to Question 4:

HbsAg is the first serological marker to be detected in Hepatitis B infection.

Solution to Question 5:

The simultaneous administration of hepatitis B immunoglobulin (HBIG) and hepatitis B vaccine is more efficacious than HBIG alone for the prevention of the carrier state in newborn babies of HBV-positive mothers.

HBIG does not interfere with the antibody response to the hepatitis B vaccine. This combined procedure is ideal both for postexposure prophylaxis of persons accidentally exposed to blood known to contain hepatitis B virus and for the prevention of the carrier state in the newborn babies of carrier mothers.

HBIG (0.05–0.07 mL/kg) IM

- As early as possible, preferably within 24 hours of exposure

Hepatitis B vaccine (20 mcg/1.0 mL), IM

- 1st dose: within 7 days of exposure
- 2nd and 3rd dose: at one and six months, respectively

Solution to Question 6:

Patients with a history of IV drug abuse, HIV, and a history of blood transfusion 32 years ago are screened for HCV.

Since the screening of blood for Hepatitis C was started after 1992, hence people who have received a blood transfusion before 1992 (in this MCQ, the year is 1992 in point 3) have to be screened.

HCV screening is done for people who may be at risk for infection. These include:

- People who received blood, blood products, or organs before screening for hepatitis C virus was implemented
- Current or former injecting drug users
- People on long-term hemodialysis
- Health-care workers
- People living with HIV
- People with abnormal liver tests or liver disease
- Infants born to infected mothers.

Solution to Question 7:

Severe rotavirus gastroenteritis is largely limited to children aged 6-24 months, not 5-7 years.

Rotavirus is the leading cause of severe, dehydrating diarrhea in children aged <5 years globally. In developing countries, 75% of the children acquire their first rotavirus infection in the first 12 months.

The universal occurrence of rotavirus infections shows that clean water supply and good hygiene are unlikely to have a substantial effect on virus transmission. In tropical areas, rotavirus diarrhea occurs throughout the year, increasing in frequency in drier, cooler climates.

Transmission is through the feco-oral route, person-to-person contact, or indirectly through fomites. Infected individuals shed very high concentrations of the virus for many days in their stool and vomit.

Solution to Question 8:

The first dose of the Rotarix vaccine is contraindicated for infants > 12 weeks of age due to the potential risk of intussusception. Hence, it is not a part of catch-up immunization programs as the age is difficult to ascertain in such programs.

Rotavirus vaccines are live, attenuated, and are administered orally. So, it is not given to an immunocompromised child.

Given below are the different Rotavirus vaccines available along with their schedule:

Vaccines	Number of doses	Interval between doses	Maximum age limit
Rotarix	2	6 weeks10 weeks	12 weeks24 weeks
RotaTeq	3	8, 16, 24 weeks	32 weeks
Rotavac	3	6, 10, 14 weeks	32 weeks
Rotasil	3	6, 10, 14 weeks	1 year

Solution to Question 9:

The antibody titers of all carriers are risen and positive till the person harbors the organism.

A cholera carrier is defined as an apparently healthy person excreting V.cholerae O1 in stools. Selective media and enrichment is needed for their diagnosis. In cholera, 4 types of carriers are described.

- Incubatory/Pre-Clinical carriers: The duration of carrier state is 1-5 days (incubation period). They present with symptoms later.

- Convalescent carriers: Recovered from an attack of cholera, continues to excrete vibrios for 2-3 weeks, due to ineffective antibiotic treatment.
- Contact/Health carrier: Due to subclinical infection contracted through a case or an infected environment, may excrete for 10 days.
- Chronic carriers: Gallbladder is infected, duration may be as long as 10 years.

Solution to Question 10:

A convalescent carrier of cholera will remain infectious for 14-21 days after recovery.

This type of carrier continues to excrete vibrios after being recovered from an attack of cholera. It is seen due to ineffective antibiotic treatment. These carriers can often become chronic or long-term carriers.

Solution to Question 11:

Among the given options, the presence of fever favors the suspicion towards Shigella as the infectious agent.

The absence of rectal pain and the presence of vomiting favors cholera while abdominal cramps are present in both of these conditions.

Symptomatic differential diagnosis of shigella and cholera:

Symptoms	Cholera	Shigella
Diarrhea	Acute watery	Acute bloody
Fever	No	Yes
Abdominal cramps	Yes	Yes
Vomiting	Yes	No
Rectal pain	No	Yes
Stool	>3 loose stools per day, watery like rice water	> 3 stools/day, with blood or pus.

Solution to Question 12:

In patients with cholera, if diarrhea persists after 48 hours of antibiotic therapy, resistance to the antibiotic should be suspected.

Treatment for cholera:

- Rehydration therapy - is started immediately, either oral or intravenous.

- Antibiotic therapy - As soon as vomiting has stopped (usually after 3-4 hours of rehydration). Fluoroquinolones, tetracycline, azithromycin, ampicillin, and trimethoprim-sulfamethoxazole are commonly used.
 - Shock in diarrhea must be corrected by intravenous fluids only.
- Antidiarrhoeals, cardiotonics, antiemetics, antispasmodics, and corticosteroids are not used.

Solution to Question 13:

Tetracycline is the drug of choice for chemoprophylaxis of cholera.

It should be given for:

- Adults - a 3-day period in a twice-daily dose of 500mg
- Children (4-13 years) - 125mg
- Children (0-3 years) - 50mg

Alternatively, the long-acting tetracycline (doxycycline) may be used for chemoprophylaxis, if the prevailing strains are not resistant. A single oral dose of doxycycline (300 mg for adults and 6 mg/kg for children under 15 years) has proved to be effective.

Chemoprophylaxis is advised only for household contacts or a closed community in which cholera has occurred.

Solution to Question 14:

The most important effective measure against cholera is health education.

The prophylactic measures against cholera include:

- Health education
- Food hygiene practices
- Hand washing practices after defecation
- Drinking safe water
- Knowledge about ORS

Solution to Question 15:

Safe water supply and sanitation is the first step to be taken to decrease the incidence of cholera.

Steps taken to decrease the incidence of cholera include:

- Safe water supply
- Excreta disposal

- Food sanitation
- Disinfection
- Early case detection

Cholera Vaccines: available in oral forms

- Dukoral
- Sanchol and mORCVAX
- Euvichol
- Vaxchora
- CVD103-HgR - no longer produced

Sanchol and Euvichol are available for mass vaccination campaigns. The campaigns have been implemented in areas experiencing an outbreak, in areas at heightened vulnerability during humanitarian crises, and among populations living in highly endemic areas, known as hotspots.

Vaxchora is given for adults (18 – 64 years) who are traveling to an area of active cholera transmission, at least 10 days before the trip.

Mass chemoprophylaxis is not recommended for the control of cholera. This is because the drug's effect is only short-lived (a few days). Moreover, for preventing 1 case of cholera, around 10,000 individuals must be given the drug. Chemoprophylaxis is recommended for only household contacts of confirmed cases with tetracycline.

Solution to Question 16:

The most effective disinfectant for cholera stool is a coal tar disinfectant with Riedel-Walker (RW) coefficient of ≥ 10 such as cresol. Bleaching powder, if used, should be of good quality.

For disinfection, attention should be paid to the following: patient's stools and vomit; clothes and other contaminated personal items; the latrine, the patient's house, and neighborhood.

Solution to Question 17:

The number of cases and deaths due to cholera are to be reported daily and weekly till the area is declared free of cholera.

International health regulations for the notification of Cholera:

- Cholera is notifiable to the WHO by the National Government within 24 hours of its occurrence.
- The number of cases and deaths due to cholera are to be reported daily and weekly till the area is declared free of cholera.
- When twice the incubation period (5 days) i.e. 10 days has elapsed since the death, recovery, or isolation of the last case, an area is declared free of cholera.

- Mass prophylaxis is not advised in cholera, as it fails to stop the spread of the disease. Chemoprophylaxis is only advised for household contacts or a closed community.

Solution to Question 18:

Oral rehydration therapy is the optimal therapy to treat acute diarrheas in some dehydration patients.

Recommended formulation: Reduced osmolarity ORS solution, which has a reduced concentration of sodium (75 mmol/liter) and glucose (75 mmol/liter). It helps in reducing the adverse effect of hypertonicity on net fluid absorption.

According to IMNCI guidelines, irritable child who is able to drink fluids come under some dehydration category.

SIGNS	CLASSIFY AS	IDENTIFY TREATMENT (Urgent pre-referral treatments are in bold print)
Two of the following signs: • Lethargic or unconscious • Sunken eyes • Not able to drink or drinking poorly • Skin pinch goes back very slowly	SEVERE DEHYDRATION	▶ If child has no other severe classification - Give fluid for severe dehydration (Plan C). OR - If child also has another severe classification: Refer URGENTLY to hospital with mother giving frequent sips of ORS on the way. Advise the mother to continue breastfeeding. ▶ If child is 2 years or older, and there is cholera in your area, give antibiotic for cholera.
Two of the following signs: • Restless, irritable • Sunken eyes • Drinks eagerly, thirsty • Skin pinch goes back slowly.	SOME DEHYDRATION	▶ Give fluid, Zinc supplements and food for some dehydration (Plan B) ▶ If Child also has a severe classification: Refer URGENTLY to hospital with mother giving frequent sips of ORS on the way. Advise the mother to continue breastfeeding. ▶ Advise mother when to return immediately. ▶ Follow-up in 5 days if not improving. ▶ If confirmed/symptomatic HIV, follow-up in 2 days if not improving.
• Not enough signs to classify as some or severe dehydration	NO DEHYDRATION	▶ Give fluid, Zinc supplements and food to treat diarrhoea at home (Plan A) ▶ Advise mother when to return immediately. ▶ Follow-up in 5 days if not improving. ▶ If confirmed/symptomatic HIV, follow-up in 2 days if not improving.

Solution to Question 19:

The freshly prepared ORS should not be used after 24 hours.

Guidelines for administering ORS:

- After each loose stool, give:
- Children <2 years of age - 50-100 ml (a quarter to half a large cup) of fluid
- Children 2 to 10 years - 100-200 ml (a half to one large cup)
- Older children and adults - as much fluid as they want
- For children <2 years, give a teaspoon every 1 to 2 minutes and offer frequent sips out of a cup for older children. Try to give the estimated required amount within a 4-hour period.
- If the child vomits, wait for 10 minutes, then give a spoonful every 2 to 3 minutes.

- If the child wants to drink more ORS solution than the estimated amount and does not vomit, there is no harm in feeding him/her more.
- If the child is breastfeeding, continue breastfeeding with ORS solution.

Solution to Question 20:

According to the guidelines of ORS by WHO, a child with a weight of 11-15.9 kg has to be treated with 800-1200 ml of ORS solution within the first 4 hours.

Guidelines for oral rehydration therapy (for all ages) during the first four hours by WHO:

The patient's age should be used if weight is not known.

The approximate amount of ORS required in ml may also be calculated by $75 \times \text{body weight (kg)}$.

Age	Under 4 months	4-11 months	1-2 yrs	2-4yrs	5-14 yrs	15 yrs or over
Weight (kg)	under 5	5-7.9	8-10.9	11-15.9	16-29.9	30 or over
ORS solution (ml)	200-400	400-600	600-800	800-1200	1200-2200	2200-4000

Solution to Question 21:

According to the IMNCI classification, the baby has some dehydration and comes under Plan B. The amount of ORS given is 75ml/kg body weight.

Here, the weight of the child is not specified. Hence, 400-700 ml of ORS at home is to be given in the first four hours according to the age of the child. The below table shows the amount of ORS to be given according to the age of the child.

In addition to this, 50-100 ml extra fluid is given after each loose stool. After four hours, the baby is reassessed and the appropriate plan is selected again for the continuation of the treatment.

The four rules of home treatment are:

- Give extra fluid
- Give zinc supplements (1 tablet per day for 14 days, dissolved in a small amount of ORS or breast milk or clean water)
- Continue breastfeeding
- Counsel the mother to return if there is blood in stools and if the child is feeding poorly

Age	under 4 months	4-11 months	1-2 years	2-4 years	5-14 years	15 years and over
Weight(kg)	under 5	5-7.9	8-10.9	11-15.9	16-29.9	30 or over

Age	under 4 months	4-11 months	1-2 years	2-4 years	5-14 years	15 years and over
ORS solution (ml)	200-400	400-600	600-800	800-1200	1200-2200	2200-4000

SIGNS	CLASSIFY AS	IDENTIFY TREATMENT (Urgent pre-referral treatments are in bold print.)
Two of the following signs: • Lethargic or unconscious • Sunken eyes • Not able to drink or drinking poorly • Skin pinch goes back very slowly	SEVERE DEHYDRATION	➤ If child has no other severe classification: — Give fluid for severe dehydration (Plan C). OR - If child also has another severe classification: — Refer URGENTLY to hospital with mother giving frequent sips of ORS on the way. Advise the mother to continue breastfeeding ➤ If child is 2 years or older and there is cholera in your area, give antibiotic for cholera.
Two of the following signs: • Restless, irritable • Sunken eyes • Drinks eagerly, thirsty • Skin pinch goes back slowly	SOME DEHYDRATION	➤ Give fluid and food for some dehydration (Plan B). ➤ If child also has a severe classification: — Refer URGENTLY to hospital with mother giving frequent sips of ORS on the way. Advise the mother to continue breastfeeding ➤ Advise mother when to return immediately. ➤ Follow-up in 5 days if not improving.
Not enough signs to classify as some or severe dehydration.	NO DEHYDRATION	➤ Give fluid and food to treat diarrhoea at home (Plan A). ➤ Advise mother when to return immediately. ➤ Follow-up in 5 days if not improving.

Solution to Question 22:

Salted water in which cereal has been cooked can be given to a child with diarrhoea.

If the WHO ORS mixture is not available, a simple mixture of 1 teaspoon of salt and 6 teaspoons of sugar must be dissolved in 1 L of drinking water and given to the child.

Fluids that do not contain salt in water, for eg: rice water, unsalted soup, yoghurt drinks, green coconut water, etc, must not be given. 3g/litre of salt must be added to such drinks before giving it to the child with diarrhoea.

The infant's usual diet of cereals, vegetables and other foods should be continued and never be diluted. The child should be given as much nutrient-rich food as acceptable.

Avoid the following:

- Sweetened drinks with sugar, like commercial fruit juices and sweetened tea, as they lead to osmotic diarrhoea and hypernatremia.
- Fluids with a stimulant and purgative effect like coffee and medicinal tea infusions.

Solution to Question 23:

Skin elasticity and pulse strength are the most helpful indicators for adequate intravenous rehydration.

The urine output may be unchanged even after 3-4 hours of adequate rehydration. Hence, it cannot be a criterion for continuous examination.

The initial rehydration must be fast until an easily palpable pulse is present. Reassess the patient every 1-2 hours.

After 1-2 liters of fluid have been infused, the drip should be slowed down till the blood pressure and pulse return to normal. Rehydration must be continued otherwise till all signs of dehydration have disappeared. ORS should be given when the patient can drink oral fluids.

Puffy eyelids are a sign of excess rehydration and intravenous fluids must be stopped when it occurs.

Solution to Question 24:

This patient has severe dehydration which is treated by giving IV fluids. Plain glucose and dextrose solutions should not be used as they worsen the hyperosmolar state.

Intravenous rehydration: Used for initial rehydration for patients in shock or those who are unable to drink.

- Ringer's lactate solution/Hartmann's solution - Used for injection. Best commercially available solution.
- Diarrhea treatment solution- Recommended as an ideal polyelectrolyte solution for intravenous infusion.
- Normal saline- Used if nothing else is available. It is the poorest fluid, as it does not correct the acidosis and will not replace the potassium losses.

Solution to Question 25:

According to WHO and UNICEF, 10 mg of zinc is given to infants < 6 months of age for 10-14 days.

20 mg of Zinc is given to infants ≥ 6 months of age for 10-14 days during an episode of acute diarrhoea. This is done to reduce the duration and severity of the episode and prevent recurrence in the following months.

Intensified Diarrhea Control Fortnight : The goal is to attain zero child deaths due to childhood diarrhea. The overall objective of IDCF is to ensure high coverage of ORS and Zinc use rates in children with diarrhoea. It also complements awareness activities (including Swachh) for prevention and management of diarrhoea in under-five children.

Solution to Question 26:

The sodium content of the rehydration solution to treat SAM is 45mmol/L.

ReSoMal (rehydration solution for malnourished children) contains a lower amount of sodium (45mmol/L) and a higher amount of potassium (40 mmol/L) than the standard WHO-ORS.

ReSoMal is used in in-patient centers for the treatment of children with severe acute malnutrition (SAM). It is useful in correcting hypokalemia, and in lowering the risks of excess sodium and/or overhydration.

Composition of ReSoMal and ORS:

Component (mmol/l)	Standard ORS	ReSoMal
Glucose	75 (anhydrous)	125
Sodium	75	45
Potassium	20	40
Chloride	65	70
Citrate	10	7
Magnesium	-	3
Zinc	-	0.3
Copper	-	0.045
Osmolality	245	300

Solution to Question 27:

The average incubation period of typhoid is 10-14 days.

However, it can vary from 3 days to 3 weeks depending on the dose of bacilli ingested.

Solution to Question 28:

Chronic carriers of typhoid are those who excrete the bacilli for more than a year.

Convalescent carriers excrete bacilli for 6-8 weeks and chronic carriers excrete for more than a year. The carrier rate in females is higher. Gallbladder and biliary tract are affected.

To treat the carrier state, an intensive course of ampicillin or amoxicillin (4-6 g a day) together with Probenecid (2 g/day) for 6 weeks is given. Cholecystectomy with concomitant ampicillin therapy is the most successful treatment for the carrier state.

Solution to Question 29:

Ty21a vaccine can be administered in an HIV-positive asymptomatic patient as long as the CD4+ count is ≥ 200 /cubic mm.

Ty21a is an oral vaccine for individuals ≥ 5 years of age. It is given in 3 doses on the 1st, 3rd, and 5th day. Repeat every 3 years for endemic areas, and every year for traveling individuals.

It is not recommended in:

- Patients on antimitotic drugs and immunosuppressants
- Acute febrile illness
- Acute intestinal infestation

Proguanil and antibacterial drugs should be stopped from 3 days before to 3 days after the vaccine.

Solution to Question 30:

Niridazole does not prevent transmission of the disease.

Dracunculiasis is treated by removing the worm by sequentially rolling them out over a small stick. Wet compresses may relieve discomfort. Occlusive dressings limit shedding of infectious larvae.

Metronidazole and mebendazole are used to limit inflammation and facilitate worm removal.

Dracunculiasis or Guinea-worm disease is amenable to eradication. The eradication strategy comprises the following elements:

- Provision of safe drinking water.
- Control of cyclops.
- Health education regarding drinking boiled water.
- Surveillance: Active search for new cases.

Solution to Question 31:

The blood loss due to *Ancylostoma* (0.2 ml/day) is more compared to *Necator americanus* (0.03ml/day)

Malnutrition is a predisposing factor. The chronic disabling disease does not occur in individuals who are well-nourished and whose iron intake is adequate.

Chandler's index is used to identify importance of hookworm infestation as a public health problem. It is calculated as average number of hookworm eggs/gram of stool.

- < 200 : no significance
- 200-250: the potential danger
- 250-300: minor public health problem

- >300: major public health problem

Hookworm infection causes chronic blood loss and depletion of body's iron stores leading to iron deficiency anemia. This in pregnant women can lead to abortions, stillbirth and low birth weight babies.

Sold by @Itachibot
If you purchased this from someone else,
you may have been scammed.

Arthropod-Borne Infections

Question 1:

Which malarial parasite has the shortest incubation period?

- a) *P. vivax*
- b) *P. malariae*
- c) *P. ovale*
- d) *P. falciparum*

Question 2:

A patient from a hilly area was diagnosed with malaria. Which vector is most likely responsible for this?

- a) *Anopheles stephensi*
- b) *Anopheles fluviatilis*
- c) *Anopheles culicifacies*
- d) *Anopheles sundanicus*

Question 3:

Individuals with Duffy negative RBCs are resistant to:

- a) *P. vivax*
- b) *P. malariae*
- c) *P. ovale*
- d) *P. falciparum*

Question 4:

Which is the most sensitive index of recent transmission of malaria?

- a) Spleen rate
- b) Infant parasite rate

- c) Parasite density index
- d) Slide positivity rate

Question 5:

In a town with a population of 1,00,000, a total of 50,000 slides were examined for malaria. Among these, 100 cases of malaria were confirmed. What is the annual parasite incidence of this town?

- a) 0.1
- b) 0.5
- c) 1
- d) 2

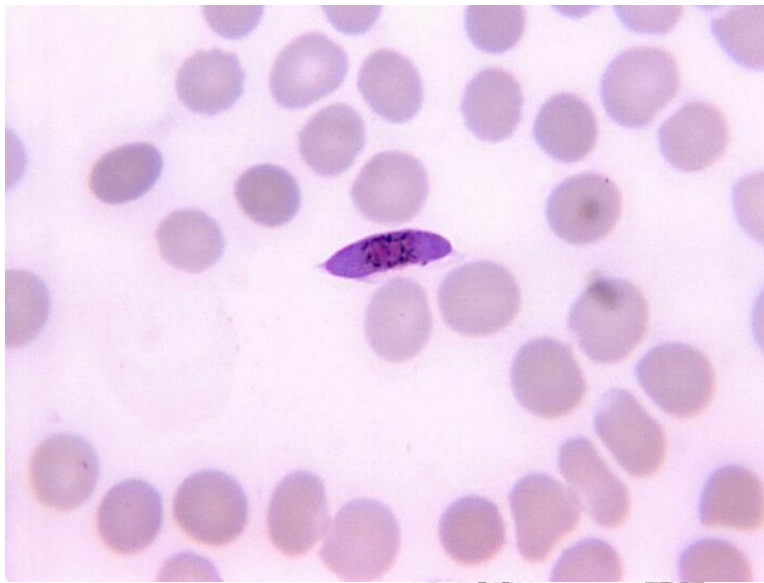
Question 6:

Which parameter is used to measure the operational efficiency of the National Vector Borne Disease Control Programme?

- a) Annual parasite index
- b) Slide positivity rate
- c) Annual falciparum incidence
- d) Annual blood examination rate

Question 7:

A 44-year-old woman from Assam presented with fever and chills recurring every 48 hours. On examination, she was conscious, oriented, and febrile. CBC, LFT, and RFT were normal. Peripheral smear revealed the following. What is the drug of choice?



- a) Quinine
- b) Artesunate and sulphadoxine-pyrimethamine
- c) Artemether and lumefantrine
- d) Chloroquine and pyrimethamine

Question 8:

A 23-year-old pregnant woman at 11 weeks of gestation was diagnosed with falciparum malaria. What is the drug of choice?

- a) Quinine
- b) Artemisinin-based combination therapy
- c) Chloroquine
- d) Primaquine

Question 9:

Which of the following patients with malaria can be treated with primaquine?

- a) A 24-year-old pregnant woman at 30 weeks of gestation
- b) A 10-year-old boy diagnosed with G6PD deficiency
- c) A 40-year-old woman recently treated for a flare-up of SLE
- d) A 60-year-old man who had a myocardial infarction 4 months ago

Question 10:

What is the dose of mefloquine given for chemoprophylaxis of malaria?

- a) 100 mg/week
- b) 200 mg /week
- c) 250 mg/week
- d) 300 mg/week

Question 11:

What is the most important strategy for the control of malaria in endemic areas?

- a) Stratification of the problem
- b) Chemoprophylaxis
- c) Vector control
- d) Avoiding relapses

Question 12:

Which of the following was a component of the Roll back malaria programme?

- a) IEC campaigns for community awareness
- b) Insecticide treated bed nets
- c) Development of larvivorous fish for eradication of larvae
- d) Presumptive treatment of malaria

Question 13:

Which sequence of infection poses the highest risk of dengue hemorrhagic fever?

- a) DENV-1 --> DENV-2
- b) DENV-3 --> DENV-2
- c) DENV-4 --> DENV-2
- d) DENV-1 --> DENV-3

Question 14:

What is the extrinsic incubation period of the dengue virus?

- a) 1-2 days
- b) 4-6 days
- c) 8-10 days
- d) 12-14 days

Question 15:

What is the cornerstone of treatment for a patient of dengue with warning signs?

- a) IV Fluids
- b) Platelets
- c) Cryoglobulins
- d) FFP

Question 16:

In patients with dengue fever, platelets should be transfused if the platelet count falls below which of the following?

- a) $< 50,000/\text{cu. mm}$
- b) $< 30,000/\text{cu. mm}$
- c) $< 20,000/\text{cu. mm}$
- d) $< 10,000/\text{cu. mm}$

Question 17:

When can a patient with dengue hemorrhagic shock be discharged?

- a) No episode of fever for more than 24 hours with paracetamol
- b) After the return of appetite
- c) Urine output more than 200ml
- d) After 24hrs from recovery of shock

Question 18:

Which of the following is a WHO-approved dengue vaccine for individuals seropositive for dengue virus?

- a) Dengvaxia
- b) DENVax
- c) Tetravax -DV
- d) TDENV PIV

Question 19:

The prepatent period in lymphatic filariasis is the time interval between:

- a) Invasion of infective larvae and blockage of lymphatics
- b) Inoculation of infective larvae and detection of microfilaria
- c) Ingestion of microfilaria and development of third stage larva
- d) Inoculation of infective larvae and maturation into adult worm

Question 20:

A 44-year-old man presented with high-grade fever, chills, and pain in his groin area. Examination revealed reddish streaks over his lower limbs and bilateral non-pitting edema to the level of the knee. Swelling and tenderness in his scrotum was also noted. To diagnose the condition, when will you draw blood from the patient?

- a) 9 pm to 11 pm
- b) 2 am to 5 am
- c) 10 pm to 2 am
- d) 8 pm to 10 pm

Question 21:

Which is the most sensitive method to detect low-density microfilaremia in blood?

- a) Thick film
- b) Thin film
- c) Membrane filter concentration
- d) DEC provocation test

Question 22:

What is the dose of diethylcarbamazine for the treatment of Bancroftian filariasis?

- a) 3 mg/kg/d
- b) 6 mg/kg/d
- c) 8 mg/kg/d
- d) 10 mg/kg/d

Answer Key

Question No.	Correct Option
1	d
2	b
3	a
4	b
5	c
6	d
7	c
8	a
9	d
10	c
11	c
12	b
13	a
14	c
15	a
16	d
17	b
18	a
19	b
20	c
21	c
22	b

Detailed Explanations

Solution to Question 1:

Plasmodium falciparum has the shortest incubation period of 12 days.

Solution to Question 2:

Anopheles fluviatilis is the vector for transmitting malaria in hilly areas, forests and forest fringes.

Vector	Area
<i>Anopheles stephensi</i>	Urban
<i>Anopheles culicifacies</i>	Rural
<i>Anopheles fluviatilis</i>	Hilly areas and forests

Solution to Question 3:

People with Duffy negative RBCs are resistant to *P. vivax*.

Resistance to *P. falciparum* is seen in.

- Individuals with sickle cell trait
- Neonates, as fetal hemoglobin suppresses the development of *P. falciparum*

Solution to Question 4:

The infant parasite rate is regarded as the most sensitive index for recent transmission of malaria in a locality.

It is defined as the % of infants showing malarial parasites in their blood films. The transmission of malaria is considered absent if the infant parasite rate is zero for 3 consecutive years in a locality (even though the Anopheline vectors responsible for the previous transmission may remain).

Solution to Question 5:

The annual parasite incidence of this town (API) is 1.

API = Confirmed cases in a year

----- x 1000

Population under surveillance

In this scenario, $\text{API} = 100 \times 1000 / 100000 = 1$

API is based on the number of slide-confirmed cases that are found through active and passive surveillance. It is a measure of malaria incidence in a community and helps in determining the burden of the disease. Areas with $\text{API} \geq 2 / 1000$ population are classified as high-risk areas and are eligible for vector control.

Note: The number of slides examined is used to determine the annual blood examination rate (ABER).

Solution to Question 6:

The annual blood examination rate (ABER) is used to measure the operational efficiency of the National Vector Borne Disease Control Programme (NVBDCP).

Solution to Question 7:

This clinical scenario and peripheral smear showing crescent-shaped gametocytes are suggestive of uncomplicated falciparum malaria. As this woman is from a North-Eastern state, the drug regimen of choice is ACT-AL (artemisinin combination therapy):

- Artemether (20mg)
- Lumefantrine (120mg)

This regimen is followed in the North-Eastern states of India due to documented resistance to sulphadoxine-pyrimethamine in these areas. ACT-AL is contraindicated in:

- 1st trimester of pregnancy
- Children < 5 kg

Solution to Question 8:

Quinine is the drug of choice for falciparum malaria during 1st trimester of pregnancy.

The treatment of falciparum malaria during pregnancy is as follows:

- 1st trimester - quinine
- 2nd and 3rd trimester - ACT-AL in North-Eastern states and ACT-SP in other states

Solution to Question 9:

Cardiac disease is not a contraindication for the use of primaquine. It is contraindicated in:

- Infants
- Pregnant and lactating women
- Patients with G6PD deficiency
- Patients at risk of developing granulocytopenia:
- Systemic diseases including active RA and SLE
- Drugs that cause hemolysis or bone marrow depression

Solution to Question 10:

Mefloquine 250 mg weekly is given for chemoprophylaxis against malaria in adults. It should be administered 2-3 weeks before travelling.

Chemoprophylaxis is recommended for travellers from non-endemic areas and as a short term measure for soldiers, police and labour forces serving in highly endemic areas. It should be complemented by personal protection and vector control.

The following drugs are used for malaria chemoprophylaxis:

- Chloroquine - 300 mg/week
- Proguanil - 200mg/day
- Mefloquine - 250mg/week
- Doxycycline - 100mg/day

Mefloquine is contraindicated in cases with a history of convulsions, neuropsychiatric problems and cardiac conditions.

Solution to Question 11:

Vector control is the most important strategy to control malaria in endemic areas. Proper measures should be taken to kill both adult mosquitoes and larvae.

Solution to Question 12:

Ensuring proper and expanded use of insecticide-treated mosquito nets was a component of the Roll Back Malaria (RBM) programme.

The other components of the RBM initiative included:

- Strengthening the health system to ensure better delivery of health care
- Ensuring adequate access to basic health care and training of health workers
- Encouraging the development of more effective and new anti-malaria drugs

Solution to Question 13:

The DENV-1 & DENV-2 sequence of infection is associated with the highest risk of dengue hemorrhagic fever (DHF).

Multiple infections with different serotypes of the dengue virus lead to severe forms of dengue such as DHF/DSS. The first infection probably sensitizes the patient, while the second infection with a different serotype appears to produce immunological catastrophe.

- DENV-1 & DENV-2 - 500-fold risk (compared to primary infection alone)
- DENV-3 & DENV-2 - 150-fold risk
- DENV-4 & DENV-2 - 50-fold risk

Solution to Question 14:

The extrinsic incubation period of the dengue virus is 8-10 days. This refers to the period necessary for the development of the disease agent in the arthropod host.

Aedes mosquitoes become infective when they feed on infected patients from 1 day before onset of illness to the 5th day (viremia stage) of illness. After 8-10 days of extrinsic incubation of the virus in a mosquito, Aedes becomes infective and can transmit the infection. Higher temperature results in shortening of the extrinsic incubation period, allowing mosquitoes to remain infectious for a longer duration.

Solution to Question 15:

The primary modality of treatment in patients with dengue with warning signs is IV fluids.

The main pathology in dengue is plasma leakage resulting in intravascular volume depletion and thereby shock. Warning signs in dengue include:

- Abdominal pain or tenderness
- Persistent vomiting
- Clinical fluid accumulation (ascites, pleural effusion)
- Mucosal bleeding
- Lethargy or restlessness
- Hepatomegaly >2 cm
- Increase in hematocrit concurrent with a rapid decrease in platelet count

Inpatient management is advised to all patients of dengue with warning signs. The management of dengue includes:

- IV crystalloids - mainstay of treatment

- Platelet transfusion - severe thrombocytopenia ($<10,000/\text{mm}^3$) and active bleeding
- FFP and cryoglobulins are not the preferred for treatment in dengue.

Solution to Question 16:

Prophylactic platelet transfusion in dengue is to be initiated when the platelet count is $<10,000/\text{cu.mm}$.

Indications for platelet transfusion in dengue are as follows:

- Platelet count $<10,000/\text{cu.mm}$
- Prolonged shock with coagulopathy
- Systemic massive bleeding

Solution to Question 17:

A patient with dengue hemorrhagic shock can be discharged after the return of appetite.

Criteria for the discharge of patients recovering from dengue:

- Absence of fever for at least 24 hours without the use of antipyretic drugs
- Platelet count $>50,000/\text{cu.mm}$
- Return of appetite
- Adequate urine output
- Visible clinical improvement
- Minimum of 2-3 days after recovery from shock
- No respiratory distress from pleural effusion or ascites

Solution to Question 18:

Dengvaxia is a WHO-approved vaccine for individuals seropositive for dengue virus. It is currently licensed in 20 countries.

A pre-vaccination screening strategy must be adopted so that only dengue-seropositive individuals are vaccinated. It is indicated for individuals between 9 to 16 years. It is given in 3 doses, 6 months apart. Booster doses are not required.

However, it carries an increased risk of severe dengue in those who experience their first natural dengue infection after vaccination (seronegative individuals).

Note: TV003, a live, attenuated, tetravalent dengue vaccine is currently in Phase-3 trials. It has shown efficacy in promoting immunity in both seropositive and seronegative individuals between 2-59 years of age.

Solution to Question 19:

The prepatent period in lymphatic filariasis is the time interval between inoculation of the infective larvae and the first appearance of detectable microfilaria.

The clinical incubation period is the time interval from the invasion of infective larvae and the development of clinical manifestations. It is usually 8-16 months.

Solution to Question 20:

This patient has features of lymphangitis, lymphedema, and epididymo-orchitis, which are suggestive of acute filariasis. Blood must be drawn between 10 pm to 2 am. This is because the maximum density of microfilaria are present in the peripheral blood at this time.

Solution to Question 21:

The most sensitive method to detect low-density microfilaremia in the blood is by membrane filter concentration.

The thick film is the most commonly used method for epidemiological assessment. The DEC provocative test is used to induce microfilaria to appear in the blood in the daytime.

Solution to Question 22:

The dose of diethylcarbamazine (DEC) for treatment of Bancroftian filariasis is 6 mg/kg/day orally for 12 days.

Medicated common salt with 1-4g DEC/kg is a special form of mass treatment, particularly in endemic areas. By this method, very low doses of DEC are given over a long period of time.

Zoonotic Infections - Viral

Question 1:

A man is referred for fever and agitation for the past 4 days. Examination reveals a disoriented man who is unable to swallow water due to involuntary inspiratory muscle spasms of the pharynx. He had been bitten by a dog 3 months ago but was not treated. What genus does the causative organism of this condition belong to?

- a) Flavivirus
- b) Lyssavirus
- c) Rubivirus
- d) Rhabdovirus

Question 2:

Which of the following is false about the fixed virus of rabies?

- a) Has a long incubation period
- b) Not form Negri bodies
- c) Does not multiply in extra neural tissues
- d) Used in vaccine preparation

Question 3:

Which of the following does not lead to the transmission of rabies to humans?

- a) Nibbling by an infected animal on abraded skin
- b) Bite of an infected bat
- c) Direct contact of infected saliva with the mucosa
- d) Ingestion of raw meat of infected animal

Question 4:

You have witnessed your neighbor being bitten by a rabid dog. You should advise him to wash the wound with soap and water for a minimum of:

- a) 25 minutes
- b) 5 minutes
- c) 10 minutes
- d) 15 minutes

Question 5:

A worker at a veterinary hospital has just received the 1st dose of rabies vaccine as a pre-exposure prophylactic measure. When should he return for the remaining doses?

- a) On day 3 only
- b) On days 3, 7, 28
- c) On days 3, 7, 14, 28
- d) On days 7, 21

Question 6:

A 7-year-old boy was bitten by a suspected rabid dog. On examination, there are minor scratches and abrasions without bleeding. Which of the following is true about the category and management of the wound?

- a) Category II - Local treatment of wound with povidone and follow up
- b) Category III - Local treatment of wound and give ARV and antibiotics
- c) Category II - Local treatment of wound and give ARV
- d) Category III - Give ARV and anti-rabies immunoglobulin

Question 7:

A public health worker was brought with multiple transdermal dog bites. He tells you that he had received pre-exposure prophylaxis 6 months ago. What is the next best step in management?

- a) Only wound treatment is required
- b) Only RIG should be given
- c) Wound washing and ARV with half dose of RIG
- d) Wound treatment and ARV; RIG is not required

Question 8:

An 8-year-old child was brought with an unprovoked dog bite. Which of the following measures should be taken?

- a) Withhold the vaccine and observe the dog for 10 days
- b) Give half the dose of the vaccine
- c) Give antibiotics only
- d) Give cell culture derived vaccine

Question 9:

A 5-year-old boy had received a full course rabies vaccine taken in December 2018. His pet dog which has been fully vaccinated now (November 2019) bit him on the toe. Bleeding was present at the site of the bite. What would be the line of treatment?

- a) No treatment
- b) 2 doses of vaccine on Day 0 & 3
- c) 5 doses of vaccine with Immunoglobulin
- d) 5 doses of vaccine

Question 10:

What is the incubation period of yellow fever?

- a) 3-6 days
- b) 3-4 weeks
- c) 1-2 weeks
- d) 8-10 weeks

Question 11:

According to the International Health Regulations, an Aedes aegypti index of which of the following must be maintained to achieve zero risk of spread of yellow fever?

- a) $\leq 1\%$
- b) $\leq 5\%$
- c) $\leq 0.5\%$

d) < 0.1%

Question 12:

A woman is planning a trip to Morocco, which is endemic for yellow fever. As per travel regulations, she receives the yellow fever vaccine. After how many days following vaccination is she eligible to travel?

- a) 5
- b) 10
- c) 15
- d) 7

Question 13:

Which of the following is the yellow fever reference center in India?

- a) Halfkin's institute
- b) Central Research institute, Kasauli
- c) National Institute of Nutrition, Hyderabad
- d) AIIMS, Delhi

Question 14:

An outbreak of Kyasanur forest disease occurred among a group of forest officers in Karnataka. Which of the following vectors should be targeted to prevent the further spread of the disease?

- a) Sand fly
- b) Tick
- c) Mosquito
- d) Mite

Question 15:

Which of the following organisms follow a bird-arthropod-human transmission route?

- a) Malaria

- b) Japanese encephalitis
- c) Plague
- d) Paragonimiasis

Question 16:

Which following is false about Japanese encephalitis?

- a) Man is the incidental dead-end host
- b) Culicines act as vectors for this condition
- c) The incubation period is 1-2 days
- d) It is caused by a group B arbovirus

Question 17:

Which of the following is true about Japanese encephalitis?

- a) SA-14-14-2 strain vaccine is commonly used
- b) Diagnosis is confirmed by CSF culture
- c) Patients usually recover without any sequelae
- d) Subclinical infection does not occur

Question 18:

The life cycle of which of the following is depicted in the image below?



- a) Japanese encephalitis
- b) Zika virus
- c) Rabies
- d) Nipah virus

Answer Key

Question No.	Correct Option
1	b
2	a
3	d
4	d
5	d
6	c
7	d
8	d
9	b
10	a
11	a
12	b
13	b
14	b

15	b
16	c
17	a
18	d

Detailed Explanations

Solution to Question 1:

This clinical scenario with hydrophobia and a history of an untreated dog bite is highly suggestive of rabies. The causative virus belongs to the genus *Lyssavirus*.

Rabies virus is an RNA virus that belongs to the family *Rhabdoviridae*. This family has 3 genera:

- *Lyssavirus*
- *Vesiculovirus*
- *Ephemerovirus*

Solution to Question 2:

The fixed virus has a short incubation period.

There are 2 types of rabies viruses:

Fixed virus	Street virus
Isolated after several intracerebral passages in rabbits	Isolated from naturally occurring cases of rabies
Short, fixed, and reproducible incubation period of 6 to 7 days	The incubation period is long and highly variable between 1 to 12 weeks
Does not form Negri bodies or multiply in extraneural tissues	Forms Negri bodies and multiplies in extraneural tissues
Used in the preparation of live-attenuated anti-rabies vaccine	It is the primary cause of rabies disease

Solution to Question 3:

Ingestion of the raw meat of the infected animal does not lead to the transmission of rabies.

Mammals like dogs and bats are important animal sources. The important modes of transmission of rabies are as follows:

- Scratches or deep bites
- Exposure of infected saliva to intact mucosa or fresh skin wounds

Human-to-human transmission through bites has never been confirmed.

Solution to Question 4:

Following dog bites, the bite wound should be washed with soap under running tap water for 15 minutes.

Local treatment of the bite wound is essential for the prevention of rabies and can reduce the infection rate by up to 80%. The following steps must be followed:

- Cleansing: Washing under running water with soap or irrigation with a catheter in case of deep wounds
- Chemical agent: Alcohol or tincture /0.01% aqueous solution of iodine or povidone-iodine is effective for inactivation of residual virus
- Suturing: Must be done only following 24-48 hours if necessary, under the cover of rabies immunoglobulin locally and minimal stitches
- Antibiotics and anti-tetanus measures when indicated

Solution to Question 5:

Pre-exposure prophylaxis for rabies consists of 3 doses to be given on days 0, 7, and 21 or 28. They are given intramuscularly or intradermally in the deltoid area of the arm.

High-risk groups like laboratory staff, wildlife wardens, veterinarians, animal handlers, and travelers to endemic regions are candidates for pre-exposure prophylaxis.

Solution to Question 6:

Minor scratches and abrasions without bleeding are classified under category II wounds. Category II wounds are managed by immediate local treatment of wound and vaccination (ARV).

Antibiotics are not required in case of minor scratches and abrasions.

Solution to Question 7:

This given clinical scenario of a patient with multiple transdermal dog bites would be categorised under category III. As the patient received pre-exposure prophylaxis 6 months ago, his line of

management includes the following:

- Wound treatment with soap and running water for 15 minutes and application of antiseptic
- Anti-rabies vaccine 2 doses on day 0 and day 3, with either intradermal vaccine(0.1ml, 1 site) or intramuscular vaccine(1 vial, 1site).

There is no requirement for Rabies Immunoglobulin (RIG), if previous pre-exposure prophylaxis or post exposure prophylaxis documentation is available. If documentation of PEP or PrEP is unavailable, patient should be treated as fresh case and given full PEP.

Solution to Question 8:

This child should be given immediate post-exposure prophylaxis (PEP) by administering a cell culture-derived vaccine to reduce the chances of getting rabies.

Solution to Question 9:

This boy should receive 2 doses of vaccine on days 0 & 3 as he has received a prior course of post-exposure prophylaxis (PEP) > 3 months ago.

Management of a new dog bite in a previously immunized individual with either pre-exposure prophylaxis (PrEP) or post-exposure prophylaxis (PEP) is as follows:

- Category I: Wound treatment only
- Category II/ III:
 - Wound treatment with soap and running water for 15 min and apply antiseptic
 - Receive Anti-rabies vaccine (ARV) 2 doses ID (0.1 ml, 1 site)/ 2 doses IM (1 VIAL, 1 site) on days 0 and 3.

*Note: In developing countries like India, the vaccination status of animals alone, doesn't play a significant role in determining the initiation of prophylaxis in humans.

Solution to Question 10:

The incubation period of yellow fever is 3–6 days.

Solution to Question 11:

The Aedes aegypti index should be < 1% in endemic areas to achieve zero risk of spread of yellow fever.

The Aedes aegypti index is a type of house index. It refers to the percentage of houses and premises in a limited well-defined area with actual breeding of Aedes aegypti larvae.

Solution to Question 12:

This woman is eligible to travel to the endemic area 10 days after being vaccinated for yellow fever. This period is required for the majority of vaccine recipients to develop immunity.

The 17D vaccine for yellow fever is given subcutaneously. Unvaccinated people are required to remain quarantined for 6 days (maximum period of incubation).

Solution to Question 13:

The reference centers for yellow fever in India are as follows:-

- Central Research Institute, Kasauli
- National Institute of Virology, Pune

Solution to Question 14:

Kyasanur forest disease (KFD) is transmitted by the bite of infective hard ticks of *Haemaphysalis* species. Hence, control of ticks must be done to prevent spread of the disease.

KFD is a hemorrhagic febrile disease caused by a flavivirus. Man is an incidental dead-end host in the life cycle of the vector (hard tick). Small mammals like rodents and squirrels are the reservoirs while monkeys serve as the amplifier hosts. KFD is endemic in 4 districts of Karnataka (Shimoga, North Kannada, South Kannada, Chikmagalur).

The incubation period is between 3 to 8 days. The disease presents with sudden onset of fever, headache, myalgia. In severe cases, gastrointestinal disturbances and hemorrhages from the nose, gums, stomach, and intestine may occur. The case fatality rate is 5 to 10%.

Solution to Question 15:

Japanese encephalitis follows bird-arthropod-human transmission.

JE virus infects several extra-human hosts, e.g., pigs and birds (like pond herons). The basic cycles of transmission are:

(a) Pig → Mosquito → Pig

(b) Bird → Mosquito → Bird

Humans are incidental "dead-end" hosts and are infected by the bite of infected *Culex* mosquitoes.

Solution to Question 16:

The incubation period of Japanese encephalitis (JE) in man following a mosquito bite is 5-15 days. JE is mosquito-borne encephalitis caused by a group B arbovirus (Flavivirus). It is transmitted by Culex (culicine) mosquitoes mainly C. tritaeniorhynchus and C. vishnui. Ardeid birds (herons) are the reservoirs while pigs are the amplifier hosts. Man is an incidental dead-end host. Man-to-man transmission has not been recorded.

Solution to Question 17:

SA-14-14-2 strain vaccine is used for the prevention of Japanese encephalitis (JE).

Japanese encephalitis (JE) is mosquito-borne encephalitis caused by a group B arbovirus (Flavivirus) and transmitted by Culex mosquitoes. Most individuals develop subclinical infection.

The course of disease in man divided into 3 stages:

- Prodromal stage - fever, headache, gastrointestinal disturbances, lethargy, and malaise
- Acute encephalitic stage - high-grade fever, nuchal rigidity, convulsions, signs of raised intracranial pressure, paralysis, altered sensorium
- Late-stage and sequelae - neurological or psychiatric sequelae seen in up to 50% of survivors

Diagnosis is serological, using blood or CSF samples. The virus rarely grows in culture. The case fatality rate varies between 20-40%. There are 3 types of JE vaccines available:

- Nakayama or Beijing strain - purified inactivated vaccine
- Beijing P-3 strain - inactivated vaccine
- SA-14-14-2 - live attenuated vaccine

Note: Killed JE vaccines (Inactivated SA 14-14-2 cell culture-derived and inactivated Vero cell culture-derived) are gradually replacing live attenuated vaccines. According to the national immunisation schedule, both live attenuated, killed-inactivated vaccines are now used for vaccination against the JE virus.

Solution to Question 18:

The given image depicts the life cycle of the Nipah virus.

Nipah virus is an RNA virus that belongs to the family Paramyxoviridae, genus Henipavirus. It is a zoonotic virus that shows both animal-to-human (bats, pigs) and human-to-human transmission. The natural hosts are fruit bats of the Pteropodidae family.

The clinical manifestations include asymptomatic subclinical infection, acute respiratory infection, and fatal encephalitis. There is no treatment or vaccine available for either people or animals. The primary treatment for humans is supportive care.

Nipah virus is on the WHO list of Blueprint priority diseases.

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Zoonotic Infections - Bacterial & Parasitic

Question 1:

Which of the following are transmitted biologically by invertebrate vectors?

- a) Direct zoonoses
- b) Cyclo-zoonoses
- c) Meta-zoonoses
- d) Sapro-zoonoses

Question 2:

Multiple members of the same community presented with sudden-onset fever, chills, and painful inguinal swellings as shown below. Further questioning leads to the identification of a rat infestation in the locality. Which of the following is the likely source of transmission of this disease?



- a) Hard tick
- b) Sand fly
- c) Rat flea
- d) Mite

Question 3:

Which of the following indices is the best determinant of a potential plague outbreak?

- a) Total flea index
- b) Cheopsis index
- c) Burrow index
- d) Special percentage of fleas

Question 4:

All of the following are causes of different types of typhus except:

- a) *R. rickettsii*
- b) *R. prowazekii*
- c) *R. tsutsugamushi*
- d) *R. typhi*

Question 5:

Which of the following rickettsial diseases is correctly matched with their respective vectors?

- a) Scrub typhus - hard tick
- b) Rocky mountain spotted fever - louse
- c) Endemic typhus - flea
- d) Epidemic typhus - mite

Question 6:

A 38-year-old woman is being evaluated for fever and rash. All of the following can be included in the differential except:

- a) Epidemic typhus
- b) Endemic typhus
- c) Scrub typhus
- d) Q-fever

Question 7:

Which of the following is false about scrub typhus?

- a) Mites are the vectors
- b) Adult mite feeds on vertebral host
- c) Caused by *R. tsutsugamushi*
- d) Treated with tetracyclines

Question 8:

Which of the following are the causative agent and vector of epidemic typhus respectively?

- a) *Rickettesia prowazekii* and louse
- b) *Rickettesia rickettsii* and louse
- c) *Rickettsia prowazekii* and tick
- d) *Rickettesia rickettsii* and tick

Question 9:

A patient presents with fever for 1 week, with abdominal distension, loss of appetite, and lymphadenopathy. There is a poor response to antibiotics and antimalarials. The Widal test is negative but rk39 is positive. Which drug can be used in the treatment of this patient?

- a) Liposomal Amphotericin B
- b) Fluconazole
- c) Linezolid
- d) Bedaquiline

Question 10:

Which of the following is not true about Leishmania?

- a) Leishmania is an extracellular parasite
- b) Extrinsic incubation period of kala-azar is 4 to 25 days
- c) It is transmitted by sandfly
- d) Post kala-azar dermal leishmaniasis is caused by *L. donovani*

Question 11:

Kala-Azar is endemic in all these areas except:

- a) West Bengal
- b) UP
- c) Bihar
- d) Assam

Question 12:

In patients with vagabond disease, which of the following is implicated?

- a) Louse
- b) Mite
- c) Tick
- d) Black fly

Answer Key

Question No.	Correct Option
1	c
2	c
3	b
4	a
5	c
6	d
7	b
8	a
9	a
10	a
11	d
12	a

Detailed Explanations

Solution to Question 1:

Metazoonoses are a type of zoonoses where the infective agents are transmitted by invertebrate vectors. They multiply and develop within the invertebrates (extrinsic incubation period) before transmission to a vertebrate host.

A zoonosis is any disease or infection that is naturally transmissible from vertebrate animals to humans. They can be classified based on their reservoir hosts into:

- Anthroozoonoses - transmitted from lower vertebrates to man
- Zooanthroponoses - transmitted from man to lower vertebrates
- Amphixenoses - may be transmitted in either direction

They can also be classified based on the life cycle of the infecting agent:

Category	Mode	Example
Direct zoonoses	Direct contact, fomites, mechanical vector No developmental or propagative changes during transmission	Rabies, brucellosis, trichinosis
Cyclo-zoonoses	Require > 1 vertebrate host to complete the developmental cycle	Human taeniasis, Echinococcosis, Pentastomid infections
Meta-zoonoses	Transmitted by invertebrate vectors Infective agent develops and multiplies within the invertebrate (extrinsic incubation period) after which it is transmitted to another vertebrate host	Arboviral infections, plague, schistosomiasis
Sapro-zoonoses	Has both a vertebrate host and a non-animal (food, soil, plants) developmental site or reservoir	Various forms of larva migrans, dermatomycoses

Solution to Question 2:

This clinical scenario and image showing buboes are suggestive of bubonic plague. It is transmitted by the bite of the infected rat flea, *Xenopsylla cheopis*. Rodents, including rats, are the reservoirs of infection.

On the other hand, the pneumonic plague spreads from man to man by droplet infection.

Solution to Question 3:

Cheopis index is the best determinant of potential plague outbreaks. If this index is ≥ 1 , it indicates there is a high likelihood of an outbreak.

Solution to Question 4:

R. rickettsii causes Rocky Mountain spotted fever. It is a type of spotted fever, not typhus.

*Note:- *Orientia tsutsugamushi* was previously known as *Rickettsia tsutsugamushi*.

Solution to Question 5:

Endemic typhus is caused by the bite of an infective flea.

This condition, also called murine typhus or flea-borne typhus, is caused by *Rickettsia typhi*. Rats serve as reservoirs. The incubation period is 1–2 weeks. The Weil Felix reaction is positive in 2nd week.

The drug of choice is tetracycline. Rodent control is most important and is best done by BHC and malathion.

Solution to Question 6:

Rashes and local lesions are not a feature of Q-fever.

It is caused by *Coxiella burnetii*. The mode of transmission is the inhalation of infected dust, entry through abrasions, or ingestion of contaminated milk. Rarely, ticks can be involved in transmission. The incubation period is 2 to 3 weeks.

The drug of choice is prolonged treatment with doxycycline. Pasteurization is advised to inactivate the causative agent.

Solution to Question 7:

Both adult and nymphal stages of the mite are free-living in nature. They do not feed on the vertebral host.

Scrub typhus is the most widespread rickettsial disease. It is caused by *Orientia tsutsugamushi* (previously known as *Rickettsia tsutsugamushi*) and transmitted by the bite of trombiculid mite. The incubation period is 10-12 days.

It presents with a typical eschar at the bite site. It is diagnosed by the Weil Felix reaction, which is strongly positive with proteus strain OXK. It is treated by tetracyclines.

Solution to Question 8:

The causative agent of epidemic typhus is *Rickettsia prowazekii* and the vector is a louse.

The incubation period is 10 to 14 days. Brill-Zinsser disease refers to the recurrent form of epidemic typhus.

Solution to Question 9:

This clinical scenario is suggestive of visceral leishmaniasis. This patient should be managed with liposomal amphotericin B.

This condition causes fever with rigor and chills associated with splenomegaly and hepatomegaly. Though lymphadenopathy is common in most endemic regions, it is rare in India.

It is diagnosed by ELISA, indirect immunofluorescent antibody test (IFAT), and rk39 which detects antibodies to *Leishmania donovani*.

The treatment is as follows:

- In most endemic regions:
- Drug of choice - pentavalent antimonial (sodium stibogluconate or meglumine antimoniate)
- If it fails or is unavailable - amphotericin B
- In India and Mediterranean countries, the drug of choice is liposomal AmB.
- Other drugs - paromomycin and miltefosine

Option B: Fluconazole is an antifungal and is not useful in visceral leishmaniasis

Option C: Linezolid is used in the treatment of MRSA and related infections.

Option D: Bedaquiline is a newer anti-TB drug used in MDR-TB.

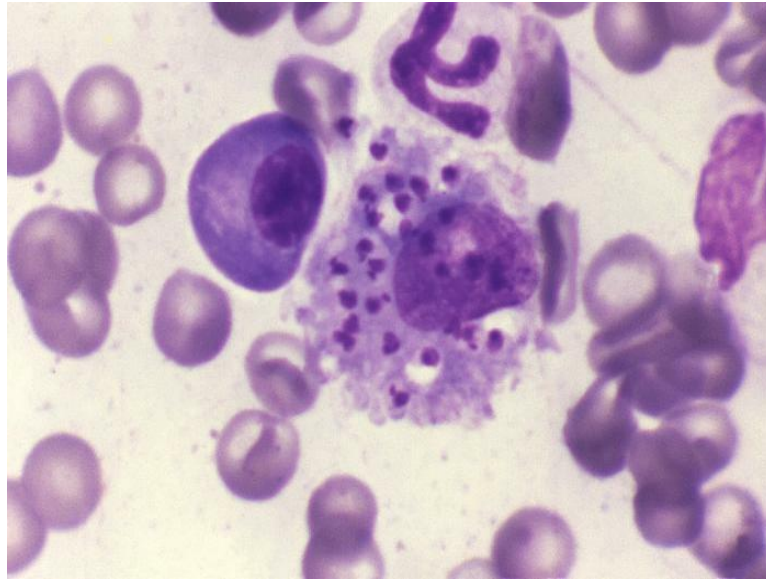
Solution to Question 10:

Leishmania is an intracellular parasite in the macrophages.

The vector is *Phlebotomus* (sandfly). Man is the reservoir in the Indian kala-azar. The incubation period is 10 days to 2 years, whereas the extrinsic incubation period is 4 to 25 days. The following conditions are caused by *Leishmania*:

- Visceral leishmaniasis or kala-azar – *L. donovani*
- Cutaneous leishmaniasis or oriental sore – *L. tropica*
- Mucocutaneous leishmaniasis - *L. braziliensis*
- Post-kala-azar dermal leishmaniasis - *L. donovani*

The image below is a microscopic image of the LD body.



According to the NVBDCP, the management is as follows:

- First-line - single-dose liposomal amphotericin B (10 mg/kg IV infusion)
- Alternate - paromomycin + miltefosine for 10 days

Monotherapy with antimonials (sodium stibogluconate) is no longer recommended due to resistance and toxicity. Synthetic pyrethroid is used for indoor residual spraying in all the endemic areas.

Solution to Question 11:

Kala-Azar is not endemic in Assam. It is endemic in the following states in 54 districts:

- Uttar Pradesh
- West Bengal
- Bihar
- Jharkhand

Solution to Question 12:

Vagabond's disease or pediculosis corporis is caused by the body louse.

The following image shows the body louse.



Sold by @Itachibot
If you purchased this from someone else,
you may have been scammed.

STDs and Surface Infections

Question 1:

When is World AIDS day observed?

- a) March 24th
- b) February 4th
- c) December 1st
- d) April 25th

Question 2:

What are the most common modes of transmission of HIV in India in descending order?

- a) Transplacental, homosexual, heterosexual
- b) Homosexual, heterosexual, transplacental
- c) Heterosexual, transplacental, homosexual
- d) Heterosexual, homosexual, transplacental

Question 3:

Health care workers will require post-exposure prophylaxis after exposure to which of the following body fluids of an HIV patient?

- a) Tears
- b) Urine
- c) Saliva
- d) Menstrual blood

Question 4:

As per WHO staging of HIV, which of the following HIV-positive patients cannot be classified as stage IV?

- a) A 29-year-old man diagnosed with cerebral toxoplasmosis
- b) A 56-year-old woman with significant weight loss and muscle wasting

- c) A 43-year-old lady being treated for *Pneumocystis pneumonia*
- d) A 32-year-old man with oral candidiasis

Question 5:

A 45-year man with a high-risk sexual history now presents with prolonged fever for the past 1.5 months. On examination, the following finding is noted. Which one of the following additional features is required to diagnose this patient with AIDS?



- a) Generalised lymphadenopathy > 1 month
- b) Generalized pruritic dermatitis
- c) Chronic diarrhea > 1 month
- d) Persistent cough > 1 month

Question 6:

Which of the following is incorrect regarding the epidemiology of HIV?

- a) HIV infected infants and children do not progress rapidly to AIDS
- b) Seminal secretions are more infectious than vaginal secretion
- c) People are infectious during the window period
- d) HIV prevalence has not exceeded 5% in any subpopulation

Question 7:

In individuals exposed to HIV, within how many hours should prophylaxis be initiated?

- a) 6 hours
- b) 24 hours
- c) 48 hours
- d) 72 hours

Question 8:

A nurse sustained a needle stick injury while sampling the blood of a patient who is HIV-positive. Which of the following is incorrect?

- a) Washing hands with soap and water
- b) Administration of tenofovir monotherapy
- c) Measure baseline viral markers of health care worker immediately
- d) Measure viral markers after 6 weeks

Question 9:

A resident doctor sustained a needlestick injury while sampling blood of a patient who is HIV positive. A decision is taken to offer him post-exposure prophylaxis. Which one of the following would be the best recommendation?

- a) Zidovudine + Lamivudine for 4 weeks
- b) Lamivudine + Tenofovir + Efavirenz for 4 weeks
- c) Lamivudine + Tenofovir + Dolutegravir for 4 weeks
- d) Zidovudine + Lamivudine + Nevirapine for 4 weeks

Question 10:

Which of the following HIV-infected patients would you start on cotrimoxazole prophylaxis?

- a) A 24-year-old pregnant woman with WHO stage 2 disease.
- b) A 47-year-old man in a malaria-endemic area with WHO stage 1 disease
- c) A 2-week-old HIV-exposed infant who is being exclusively breastfed
- d) A 7-year-old girl with a CD4 count of 400 cells/cubic mm

Question 11:

A 45-year-old man with HIV is being screened for symptoms of TB. He reports no cough, fever, night sweats, or weight loss. What is the next step in management?

- a) Start 5 mg/kg/day Isoniazid daily for 9 months
- b) Prophylaxis does not need to be started
- c) Rifabutin prophylaxis should be started
- d) Start weekly dose of rifapentine + isoniazid for 3 months

Question 12:

Which of the following is true about the 2025 targets of the Global AIDS Strategy for 2021 to 2026?

- a) B,E only
- b) A,B,C only
- c) A,B,D,E only
- d) C,D only

Question 13:

A 29-year-old man comes to you with complaints of painful swelling in his right groin, as shown below. He gives a history of 3-4 painless ulcers over his penis 4 weeks ago that had resolved on its own. What is the likely causative organism?



- a) *Haemophilus ducreyi*
- b) *Calymmatobacterium granulomatis*
- c) *Treponema pallidum*
- d) *Chlamydia trachomatis*

Question 14:

Match the following conditions with their causative agents:

- a) I-A, II-B, III-D, IV-C
- b) I-B, II-A, III-C, IV-D
- c) I-D, II-C, III-A, IV-B
- d) I-C, II-D, III-B, IV-A

Question 15:

Match the following STDs with their respective incubation periods:

- a) I-D, II-B, III-A, IV-C
- b) I-A, II-C, III-D, IV-B
- c) I-B, II-A, III-C, IV-D
- d) I-C, II-D, III-B, IV-A

Question 16:

Which of the following is not a part of case detection in STDs?

- a) Screening
- b) Contact tracing
- c) Cluster testing
- d) Notification

Question 17:

A 32-year-old man is diagnosed with syphilis. On probing, he gives you the information of other people in the same sociosexual environment as him. You then refer all these people for testing. What is this method of case detection?

- a) Contact testing
- b) High-risk screening
- c) Selective screening
- d) Cluster testing

Question 18:

A 29-year-old man with urethral discharge is treated with the grey kit. This treatment is effective for the control of which of the following organisms?

- a) *Neisseria gonorrhoeae* and *Herpes genitalis*
- b) *Chlamydia trachomatis* and *Herpes genitalis*
- c) *Neisseria gonorrhoeae* and *Chlamydia trachomatis*
- d) *Neisseria gonorrhoea* and *Treponema pallidum*

Question 19:

Which of the following is false about the epidemiology of leprosy?

- a) Increased prevalence in childhood indicates that the disease is active and spreading.
- b) Multibacillary cases are the most infectious in the community
- c) Bacterial load is highest in patients with tuberculoid leprosy
- d) Seldom may be transmitted by insect vectors

Question 20:

Elimination of leprosy in India is defined as the prevalence of :

- a) < 1 per 1000
- b) < 1 per 10,000
- c) < 5 per 1000
- d) < 5 per 10,000

Question 21:

What is the prevalence of leprosy in India per 10,000?

- a) More than 1
- b) 0.66
- c) 0.71
- d) 0.41

Question 22:

The Ridley Jopling classification for leprosy uses which of the following parameters?

- a) Clinical, bacteriological, immunological, epidemiological
- b) Clinical, bacteriological, immunological, therapeutic
- c) Clinical, bacteriological, immunological, histological
- d) Clinical, immunological, epidemiological, operational

Question 23:

A pathologist is studying the split skin smears taken from 4 sites of a leprosy patient. The following findings are noted in the samples. What is the bacteriological index?

- a) 3.5
- b) 4.5
- c) 5.5
- d) 2.5

Question 24:

The lepromin test was performed on a patient with suspected leprosy. When will you ask him to come back for interpretation of the test results?

- a) After 24 hrs and on 3rd day
- b) After 48hrs and on 7th day
- c) After 48 hrs and on 21st day
- d) After 24 hrs and on 45th day

Question 25:

Which of the following is false about the lepromin test?

- a) It is negative in most children in the first 4 months of life
- b) It can be used as a diagnostic test
- c) It is used to classify leprosy cases
- d) False positive may be seen in those who underwent BCG vaccination

Question 26:

A group of public health workers is trying to identify trachoma cases at an ophthalmology camp. Which of the following is the best index to assess for trachoma inflammation?

- a) Follicles on the upper tarsal conjunctiva
- b) Limbal follicles and Herbert's pits
- c) Presence of trichiasis and entropion
- d) Vascular pannus at the superior limbus

Question 27:

Which of the following is true about the organisms causing trachoma?

- a) Morax- Axenfeld bacillus is the most widespread
- b) Gonococcus is the most dangerous organism contributing to the disease process
- c) Koch-Weeks bacillus is the most innocuous
- d) C. trachomatis type D commonly causes permanent visual damage

Question 28:

The elimination of neonatal tetanus is defined as:

- a) < 1 case / 100 live births
- b) < 1 case / 1000 live births
- c) < 1 case / 10,000 live births
- d) < 1 case / 1,00,000 live births

Question 29:

Which of the following diseases is called the 8th day disease?

- a) Trench fever
- b) Pertussis
- c) Tetanus
- d) Scarlet fever

Question 30:

A 13-year-old girl is brought to the casualty a few hours after being injured by a rusty knife. According to her parents, her course of tetanus vaccination was completed 7 years ago. What is the next best step in management?

- a) Full course of tetanus toxoid
- b) Single dose of tetanus toxoid
- c) Human tetanus immunoglobulin
- d) Human tetanus immunoglobulin and single dose of toxoid

Answer Key

Question No.	Correct Option
1	c
2	c
3	d
4	d
5	c
6	a
7	d
8	b
9	c
10	b
11	d
12	c
13	d
14	c
15	d
16	d

17	d
18	c
19	c
20	b
21	d
22	c
23	a
24	c
25	b
26	a
27	b
28	b
29	c
30	b

Detailed Explanations

Solution to Question 1:

The World AIDS Day is observed on December 1st. The theme for 2021 is 'Ending the HIV epidemic: Equitable Access, Everyone's Voice'.

Option A: The World Tuberculosis Day is on March 24.

Option B: The World Cancer Day is on February 4.

Option D: The World Malaria Day is on April 25.

Solution to Question 2:

The most common mode of transmission of HIV in India is heterosexual transmission, followed by transplacental and homosexual contact, respectively.

The common modes of transmission of HIV are:

- Heterosexual (83.7%)
- Injecting drug users (5.6%)
- Transplacental transmission (3.8 %)
- Homosexual (2.5%)
- Blood and blood products (0.8%)
- Unknown (3.6%)

The highest risk of transmission is seen in homosexuals due to the increased risk of abrasions during anal sex.

Solution to Question 3:

Health care workers exposed to the menstrual blood of an HIV-positive patient will require post-exposure prophylaxis. The risk of transmission of HIV during vaginal intercourse is higher when the woman is menstruating.

Tears, non-blood-stained saliva, urine, and sweat do not pose a significant risk. Exposure to these fluids does not require post-exposure prophylaxis.

Solution to Question 4:

Oral candidiasis is seen in stage 3, not stage 4 of HIV.

The following conditions are included in the WHO stage IV of HIV:

- HIV wasting syndrome (Slim disease)
- HIV encephalopathy
- HIV-associated nephropathy or cardiomyopathy
- Infections:
 - Pneumocystis jiroveci pneumonia (previously known as Pneumocystis carinii)
 - Chronic herpes simplex infection, cytomegalovirus disease
 - Oesophageal/ tracheal candidiasis
 - Extrapulmonary TB or disseminated M. avium infection
 - CNS toxoplasmosis
 - Extrapulmonary cryptococcosis
 - Chronic cryptosporidiosis or isosporiasis
 - Disseminated fungal infections (histoplasmosis, coccidioidomycosis)
 - Recurrent severe bacterial pneumonia or septicemia
 - Atypical disseminated leishmaniasis
 - Progressive multifocal leukoencephalopathy
- Neoplasms:
 - Kaposi sarcoma
 - Lymphoma (cerebral or B cell non-Hodgkin)
 - Invasive cervical carcinoma

Solution to Question 5:

This clinical scenario is suspicious for HIV infection. As per the WHO definition, the presence of chronic diarrhea $\geq$ 1 month is needed to diagnose this patient with AIDS.

The WHO clinical definition is used for AIDS surveillance in the clinical setting. Patients $\geq$ 12 years of age are said to have AIDS if they have at least 2 major and 1 minor signs, provided other clinically similar conditions are ruled out.

Major signs:

- Weight loss $\geq$ 10% of body weight
- Chronic diarrhea $\geq$ 1 month
- Prolonged fever for $\geq$ 1 month (intermittent or constant)

Minor signs:

- A persistent cough $\geq$ 1 month
- Generalized pruritic dermatitis
- History of herpes zoster
- Oropharyngeal candidiasis
- Chronic progressive or disseminated herpes simplex infection
- Generalized lymphadenopathy

For surveillance purposes, the presence of either of the following is also considered sufficient to diagnose AIDS:

- Generalized Kaposi's sarcoma
- Cryptococcal meningitis

Solution to Question 6:

Infants and children infected with HIV progress rapidly to AIDS.

Semen is more infectious than vaginal or cervical fluids as it has a higher concentration of HIV.

HIV-infected patients are highly infectious during:

- Window period (very early stages before antibody production)
- Advanced stages of infection (due to higher levels of viremia)

HIV prevalence has consistently remained $\leq$ 5% in all defined subpopulations.

Solution to Question 7:

Post-exposure prophylaxis for HIV should ideally be initiated within 72 hours. It is recommended that it be started as even as early as $\leq$ 2 hours.

Solution to Question 8:

Triple therapy (tenofovir 300mg + lamivudine 300mg + dolutegravir 50mg) is recommended for HIV post-exposure prophylaxis (PEP), not monotherapy.

Steps to be undertaken after a needlestick injury are:

- Wash the wound and skin with water and soap. Do not scrub and avoid antiseptics.
- Assess exposed individual:
- Check baseline viral markers
- Monitor for acute seroconversion illness within 3-6 weeks after exposure
- If testing is positive, refer to ART center
- Assess exposure source and determine the risk of transmission
- Counseling and support
- Initiate PEP at the earliest - preferably < 2 hours and ideally < 72 hours
- 28-day prescription with drug information
- HIV test at 3 months after exposure

Solution to Question 9:

Triple therapy with tenofovir 300mg, lamivudine 300mg, and dolutegravir 50mg for 4 weeks is recommended for HIV post-exposure prophylaxis (PEP).

Steps to be undertaken after a needlestick injury are:

- Wash the wound and skin with water and soap. Do not scrub and avoid antiseptics.
- Assess exposed individual:
- Check baseline viral markers
- Monitor for acute seroconversion illness within 3-6 weeks after exposure
- If testing is positive, refer to ART center
- Assess exposure source and determine the risk of transmission
- Counseling and support
- Initiate PEP at the earliest - preferably < 2 hours and ideally < 72 hours
- 28-day prescription with drug information
- HIV test at 3 months after exposure

Solution to Question 10:

All HIV infected adults and children in malaria-endemic areas should be given cotrimoxazole prophylaxis regardless of CD4 count or WHO clinical stage.

Cotrimoxazole is also indicated in areas with a low prevalence of malaria as shown below:

Age group	Indication	When to discontinue
HIV-exposed infants > 4-6 weeks of age	All infants	After HIV infection has been ruled out
Children < 5 years	All children	At 5 years in areas with low prevalence of malaria(In high prevalence areas, continue till adulthood)
Children > 5 years	CD4 count <350 and/or WHO stage 3/4	When CD4 > 350 and they have been on ART > 6 months
Adults and pregnant women	CD4 count <350 and/or WHO stage 3/4	When CD4 count is rising and viral load is decreasing
HIV and TB coinfection	All patients	—

Solution to Question 11:

The next step in management in this patient is to provide tuberculosis preventive therapy (TPT) with rifampentine+ isoniazid weekly for 3 months.

According to the NTEP guidelines, TPT is indicated for the following groups:

IGRA - interferon-gamma release assay; TST - tuberculin skin test; ATT - antituberculous therapy

TPT is contraindicated under the following conditions:

- Active TB
- Active hepatitis
- Peripheral neuropathy
- Regular and heavy alcohol consumption

Risk group	Criteria
People living with HIV	Negative 4 symptom test - start TPT after ruling out TB Patients with active TB - start TPT after completion of the ATT regimen
Household contacts (HHC) of pulmonary TB patients < 5 years	Start TPT after ruling out active TB

Risk group	Criteria
Household contacts of pulmonary TB patients ≥ 5 years Other risk groups: Children/adults on immunosuppressive therapy Patients on anti-TNF treatment Patients on dialysis or post-transplantation Patients with silicosis	First rule out active TB Start TPT if IGRA or TST is positive with a normal chest x-ray

Target Population	Treatment option
People living with HIV (adults and children > 2 years) HIV co-infection of pulmonary TB patients	3-month weekly Isoniazid (300 mg) and Rifapentine (300 mg) in persons older than 2 years
Infants <12 months of age and Children < 2 years of age with HIV	6-months daily isoniazid (6H) : 10 mg/kg/day

Solution to Question 12:

95%, not 75% of HIV-exposed children are tested by 2 months of age and after stopping breastfeeding according to the 2025 targets.

Global AIDS Strategy 2021-2026 has the following targets for the year 2025:

Target %	Target population	Target intervention
95% of	People at risk of HIV infection	Access to effective prevention options
	Women of reproductive age	Have their HIV, sexual, reproductive health service needs met
	Pregnant and breastfeeding women with HIV	Suppressed viral loads
	HIV-exposed children	Tested by 2 months of age and again after stopping breastfeeding
90% of	People living with HIV	Preventive treatment for TB
	People living with HIV and people at risk	Linked to services for overall health and wellbeing including: Communicable and non-communicable diseases Sexual health Gender-based violence Mental health and substance abuse

Target %	Target population	Target intervention
75% of	All children living with HIV	Suppressed viral loads by 2023 (interim target)
95–95–95 testing and treatment targets to be achieved with in all subpopulations, age groups, and geographic settings, including children living with HIV.		

Solution to Question 13:

This clinical scenario and image showing the groove sign of Greenblatt are suggestive of lymphogranuloma venereum (LGV). The causative organism is *Chlamydia trachomatis*.

LGV commonly presents with multiple small, painless ulcers on the genitalia that are often unrecognized and resolve spontaneously. This is followed by painful lymphadenopathy and bubo formation after 4–6 weeks. If left untreated, it can lead to extensive lymphatic damage, elephantiasis, and deformity of the genitalia.

The treatment of choice is doxycycline.

Solution to Question 14:

The causative organisms of the given conditions are as follows:

India is the first country in the world to be officially acknowledged as being yaws-free. The WHO declared India free of yaws on 5th May 2016.

Condition	Organism
Pinta	<i>Treponema carateum</i>
Lymphogranuloma venereum	<i>Chlamydia trachomatis</i>
Yaws	<i>Treponema pertenue</i>
Granuloma inguinale	<i>Klebsiella granulomatis</i>

Solution to Question 15:

The incubation periods of various STDs are as follows:

STD	Incubation period
Syphilis	10-90 days
Chlamydia sis	7-21 days
Donovano sis	3-40 days
Chancroid	2-3 days

Solution to Question 16:

Notification is not part of the case detection strategy for STDs. Case detection in STDs includes the following:

- Screening
- Contact tracing
- Cluster testing

Solution to Question 17:

This method of case detection is known as cluster testing.

The methods employed in case detection of STDs are:

- Screening - testing of apparently healthy volunteers from the general population for early detection of disease
- Contact tracing - technique by which the sexual partners of diagnosed patients are identified, located, investigated, and treated
- Cluster testing - patients are asked to name other persons who move in the same socio-sexual environment, who are then tested

Solution to Question 18:

The grey kit is effective for the treatment of infections caused by *Neisseria gonorrhoeae* and *Chlamydia trachomatis*.

Solution to Question 19:

The bacterial load is highest in patients with lepromatous leprosy. 1 gram of leproma contains as many as 2-7 billion bacilli.

Multibacillary cases (lepromatous and borderline lepromatous) are the most important source of infection in the community. Infection can occur at any age. In general, cases rise to a peak between 10-20 years of age and then fall. In endemic areas, the disease is acquired commonly during childhood. A high prevalence in this age group indicates that the disease is active and spreading.

The organism may also be transmitted by insect vectors or by tattooing needles. However, this is not clinically significant.

Solution to Question 20:

Elimination of leprosy in India is defined as a prevalence of < 1 per 10,000.

Solution to Question 21:

The prevalence of leprosy in India is 0.41 per 10,000 as of March 2021.

Solution to Question 22:

The Ridley Jopling classification uses clinical, bacteriological, immunological, and histological parameters.

Solution to Question 23:

The bacteriological index (BI) of the given samples is 3.5

According to Ridley's logarithmic scale,

- 10 bacilli out of 100 oil immersion fields indicate an index of 1+
- > 1000 bacilli out of average fields imply an index of 6+

The bacteriological index is calculated by adding the index from each site examined and dividing the total by the number of sites examined.

Hence, $BI = (1+1+6+6) / 4$

$BI = 14/4$

$BI = 3.5$

Solution to Question 24:

The patient should be asked to come back after 48 hours and on the 21st day for interpretation of the test results.

The lepromin test is a skin test for delayed hypersensitivity and is the only method to study immunity in leprosy patients. It is performed by injecting 0.1ml of lepromin reagent intradermally into the inner aspect of the forearm. The following 2 reactions are observed:

Early reaction of Fernandez	Late reaction of Mitsuda
Develops in 24-48 hours	Starts at 7-10 days, peaks at 3-4 weeks (read on day 21)
Remains for 3-5 days	Gradually subsides over the next few weeks
Poorly defined erythema and induration. Carries less significance	Considered positive if there is a nodule > 5 mm in diameter

Solution to Question 25:

The lepromin test is not a diagnostic test. It is only used to assess the immune status, classify the cases, and estimate the prognosis.

The two main drawbacks of this test are:

- False-positive results in those who received BCG vaccination
- False-negative results in lepromatous and borderline lepromatous cases

Solution to Question 26:

In epidemiological studies, the upper tarsal conjunctiva is the best index of trachomatous inflammation in the eye.

For the purpose of diagnosis, any 2 of the following diagnostic criteria must be present:

- Follicles on the upper tarsal conjunctiva
- Limbal follicles and Herbert's pits
- Typical conjunctival scarring (trichiasis and entropion)
- Vascular pannus, most marked at the superior limbus

The following image shows follicles on the upper tarsal conjunctiva seen in trachomatous inflammation.



Solution to Question 27:

Gonococcus is the most dangerous organism contributing to the disease process in trachoma.

Classic endemic trachoma is caused by *Chlamydia trachomatis* types A, B, C. Other pathogenic organisms that contribute to the disease process are:

- Morax-Axenfeld diplobacillus - most innocuous
- Kochs-Weeks bacillus - most widespread
- Gonococcus - most dangerous

The types D-K of *C. trachomatis* are sexually transmitted and cause inclusion conjunctivitis. It rarely causes permanent vision loss.

Solution to Question 28:

The elimination of neonatal tetanus is defined as <1 case per 1000 live births in every district. Maternal and neonatal tetanus (MNT) was eliminated in India in 2015.

Since July 2018, TT vaccines have been replaced by the Td vaccine for dual protection.

Solution to Question 29:

Neonatal tetanus is known as the 8th-day disease. This is because the clinical features are often seen after the 7th day after birth due to infection of the umbilical cord.

Option A: Trench fever is also called the 5-day fever.

Option B: Pertussis is called the 100-day cough.

Option D: Scarlet fever is also known as the 2nd disease.

Solution to Question 30:

This clinical scenario is suggestive of a contaminated wound in a patient who received complete immunization against tetanus 7 years ago. She comes under category B of tetanus immunization and must be given a single dose of tetanus toxoid alone.

The WHO recommends a 3-dose primary vaccination series with tetanus-diphtheria-containing vaccine followed by 3 booster doses. Ideally, there should be at least 4 years between booster doses.

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Non-Communicable Diseases - Cardiovascular Diseases and Diabetes

Question 1:

The WHO has formulated a Global Action Plan for the prevention and control of non-communicable diseases. Which of the following is not a voluntary global target under this initiative?

- a) 10% relative reduction in harmful use of alcohol
- b) 30% relative reduction in mean population intake of sodium
- c) 10% relative reduction in prevalence of insufficient physical activity
- d) 10% relative reduction in use of trans-fat for cooking

Question 2:

Project MONICA is related to:

- a) Cardiovascular diseases
- b) Cervical Cancer
- c) Breast Cancer
- d) Road Traffic Accidents

Question 3:

Which of the following statements is false regarding the relationship between smoking and coronary heart disease (CHD)?

- a) Smoking can cause sudden death from CHD.
- b) Filters in cigarettes are protective against the development of CHD.
- c) Risk of developing CHD 10-20 years after the cessation of smoking is equal in smokers and non smokers.
- d) Degree of risk of developing CHD is directly related to the number of cigarettes smoked per day.

Question 4:

A chronic smoker was found to have a deranged lipid profile during his annual health checkup. Which of the following variables tested is most directly associated with the risk of developing coronary heart disease?

- a) HDL level
- b) VLDL level
- c) LDL level
- d) IDL level

Question 5:

A 45-year-old diabetic male is concerned about his risk of developing coronary heart disease. What cholesterol/HDL ratio would you aim to achieve in this patient?

- a) Cholesterol/HDL ratio > 2.5
- b) Cholesterol/HDL ratio < 2.5
- c) Cholesterol/HDL ratio > 3.5
- d) Cholesterol/HDL ratio < 3.5

Question 6:

While conducting a screening camp for coronary heart disease, you encounter the following people. Who among them is at the highest risk for developing this disease?

- a) A person who is competitive and ambitious
- b) A person who is calm and relaxed
- c) A person who tends to suppress their emotions
- d) An introverted person with negative affectivity

Question 7:

A 45-year-old obese woman enquires about ways she can reduce her risk of coronary artery disease. Which of the following dietary modifications would you suggest?

- a) Reduction of salt intake to less than 10g per day
- b) Consumption of saturated fats limited to less than 10% of the total energy intake
- c) Reduction of total fat intake to less than 5% of total energy intake
- d) Dietary cholesterol increased to above 100 mg/1000 kcal energy intake

Question 8:

Which of the following studies is not related to coronary heart disease?

- a) Stanford 3 community study
- b) Framingham study
- c) CATIE trial
- d) MRFIT

Question 9:

A 50-year-old man, under treatment for hypertension for the past two months, is found to have a blood pressure of 144/96 mmHg. How will you grade it according to the European society of cardiology guidelines-2018?

- a) Normal
- b) High Normal
- c) Grade 1 hypertension
- d) Grade 2 hypertension

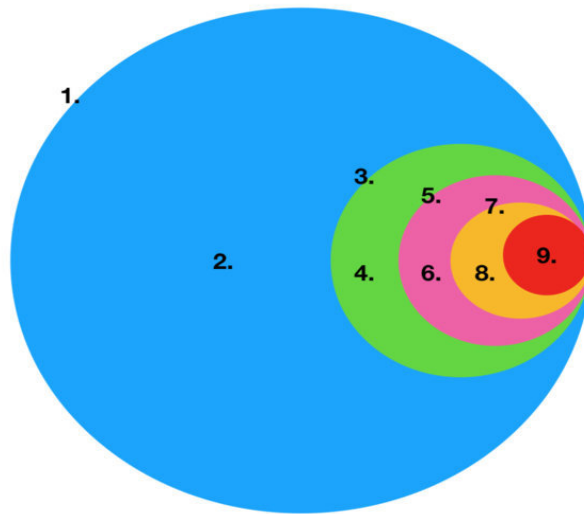
Question 10:

A doctor records the BP of a patient as 142/90mmHg. One hour later, another doctor records the BP of the same patient as 160/98 mmHg. Both the readings are measured with the same apparatus under the same emotional and physical state of the patient. What type of error is seen here?

- a) Subject error
- b) Observer error
- c) Instrument error
- d) Recording error

Question 11:

What does number 2 in the given diagram related to hypertension denote?



- a) Normotensive individuals
- b) Undiagnosed hypertensive subjects
- c) Diagnosed hypertensive subjects not on treatment
- d) Whole community

Question 12:

Which of the following statements is true regarding tracking of blood pressure?

- a) BP increases with age in all individuals
- b) BP falls steadily in the elderly
- c) Hypertensive children become hypotensive as they grow
- d) BP of hypotensive children tends to remain low in adult life

Question 13:

While formulating a diet plan for a hypertensive patient, you advise salt restriction. How much reduction in systolic blood pressure is expected by this intervention?

- a) 8-14 mm of Hg
- b) 2-8 mm of Hg
- c) 4-9 mm of Hg
- d) 5-20 mm of Hg

Question 14:

Which of the following interventions will cause the maximum decrease in systolic blood pressure in a hypertensive patient?

- a) Restricting intake of animal products
- b) Taking a brisk walk for at least 30 minutes a day
- c) Adopting the DASH eating plan
- d) Restricting sodium intake to less than 6g per day

Question 15:

Which of the following is a minor criterion for the diagnosis of rheumatic fever in a low risk population?

- a) Carditis
- b) Erythema marginatum
- c) Subcutaneous nodules
- d) Polyarthralgia

Question 16:

A child having strep throat is being prescribed benzathine penicillin G. Which level of prevention of rheumatic fever does this intervention fall under?

- a) Primary prevention
- b) Primordial prevention
- c) Secondary prevention
- d) Tertiary prevention

Question 17:

A 16-year-old boy with rheumatic fever developed carditis which subsequently healed in a couple of weeks. How long will you administer benzathine penicillin G for secondary prophylaxis to him?

- a) Till 21 years of age
- b) Till 24 years of age

- c) Till 26 years of age
- d) Lifelong

Question 18:

Which of the following is not a protective factor for diabetes mellitus?

- a) Intake of PUFAs
- b) Intake of non-starch polysaccharides
- c) Exclusive breastfeeding
- d) Intake of saturated fats

Question 19:

A patient's fasting plasma glucose is 122 mg/dl and postprandial plasma glucose is 138 mg/dl. According to the WHO diagnostic criteria for diabetes, which category does this patient fall under?

- a) Isolated post prandial hyperglycemia
- b) Impaired glucose tolerance
- c) Impaired fasting glucose
- d) Overt diabetes

Question 20:

Which of the following statements are incorrect regarding the benefits of various interventions in diabetes?

- a) 1 & 2
- b) 2, 3 & 4
- c) 3 & 5
- d) 1, 2 & 4

Question 21:

The community health officer of an area is instructed to assess the population for the risk factors of Non-alcoholic Fatty Liver Disease under NP-NCD. Which of the following will not

be a risk factor?

- a) Pedal edema
- b) Abdominal obesity
- c) History of cancer
- d) Low blood pressure

Answer Key

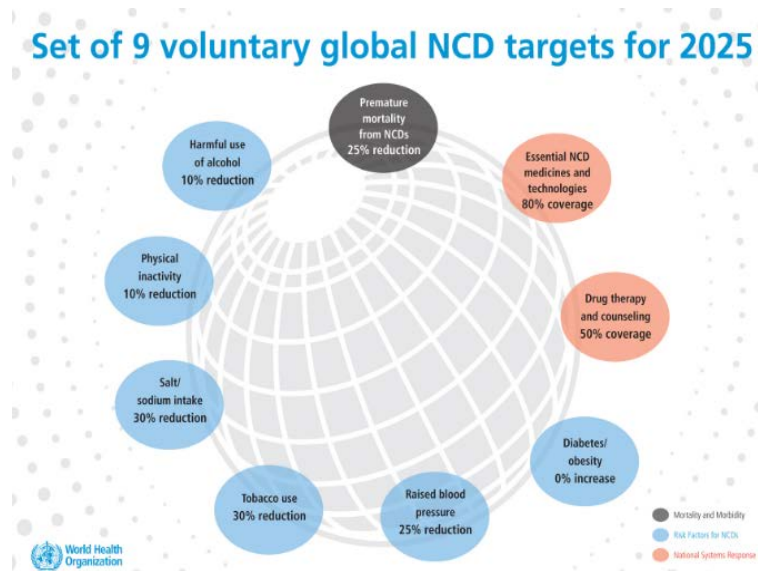
Question No.	Correct Option
1	d
2	a
3	b
4	c
5	d
6	a
7	b
8	c
9	c
10	b
11	a
12	d
13	b
14	c
15	d
16	a
17	c
18	d
19	c
20	c
21	d

Detailed Explanations

Solution to Question 1:

A 10 percent relative reduction in the use of trans-fat for cooking is not a voluntary global target under the WHO Global Action Plan for prevention and control of non-communicable diseases.

There are 9 targets under this plan which are to be achieved by 2025. These are shown in the image below.



Solution to Question 2:

Project MONICA is related to cardiovascular diseases.

It stands for multinational monitoring of trends and determinants in cardiovascular diseases. It was done to study the reasons for the changes in the trends of cardiovascular diseases in different countries.

Solution to Question 3:

Filters are probably not protective against the development of coronary heart disease in smokers.

Smoking is a modifiable risk factor for coronary heart disease (CHD). The degree of risk is directly related to the number of cigarettes smoked per day. Several mechanisms have been suggested for this:

- Carbon monoxide induced atherogenesis
- Nicotine stimulation of adrenergic drive
- Lipid metabolism with fall in levels of HDL

Cigarettes also increase the risk of sudden death in patients with CHD who are below 50 years of age. The risk of death from CHD decreases with the cessation of smoking. The risk declines quite

substantially within one year of stopping smoking and more gradually thereafter until, after 10-20 years, it is the same as that of non-smokers.

Risk factors for coronary heart disease are:

- Modifiable:
 - Cigarette smoking
 - High blood pressure
 - Elevated serum cholesterol
 - Diabetes
 - Obesity
 - Sedentary habits
 - Stress
- Non-modifiable:
 - Age
 - Sex
 - Family history
 - Genetic factors
 - Personality

Solution to Question 4:

The plasma LDL level is most directly associated with the risk of developing coronary heart disease (CHD).

HDL is protective against the development of CHD. Its levels should ideally be more than 40 mg/dl.

Recent evidence shows that levels of Apolipoprotein-A-1 and Apolipoprotein-B are better predictors of CHD than HDL or LDL.

VLDL is more strongly associated with the risk of developing peripheral vascular disease.

Solution to Question 5:

A cholesterol/HDL ratio < 3.5 is recommended for the prevention of coronary heart disease (CHD).

Solution to Question 6:

A person who is competitive and ambitious is said to have a type A personality. Such people are more prone to develop coronary heart disease (CHD). Personality type is a non-modifiable risk factor for CHD.

Risk factors for coronary heart disease are:

- Modifiable:
 - Cigarette smoking
 - High blood pressure
 - Elevated serum cholesterol
 - Diabetes
 - Obesity
 - Sedentary habits
 - Stress
- Non-modifiable:
 - Age
 - Sex
 - Family history
 - Genetic factors
 - Personality

Option B: A calm and relaxed person is said to have a type B personality.

Option C: A person who tends to suppress their emotions, especially negative emotions, is said to have a type C personality.

Option D: A person with negative affectivity and social inhibition is said to have a type D personality.

Solution to Question 7:

Limiting the consumption of saturated fats to less than 10% of total energy intake is recommended for the prevention of coronary artery disease.

The WHO'S Expert Committee suggested certain dietary modifications for the prevention of coronary artery disease. These include:

- Reduction of the total fat intake to 20–30% of total energy intake
- Consumption of saturated fats must be limited to less than 10% of total energy intake
- A reduction of dietary cholesterol to below 100mg per 1000 kcal per day
- Increase in the complex carbohydrate consumption (i.e vegetables, fruits, whole grains and legumes)
- Avoidance of alcohol consumption and reduction of salt intake to 5g daily or less

Solution to Question 8:

The CATIE trial is related to the study of antipsychotics and their effectiveness.

The trials related to coronary heart disease and their subject of study include:

Name of Study	Factors affecting CHD Studied
Oslo diet/Smoking intervention study	Effect of lowering serum lipids and smoking cessation
MRFIT trial	Effect of smoking cessation, blood pressure control and diet alteration in reduction of serum cholesterol
The North Karelia project	1. Effect of reduction of risk factors like smoking, blood pressure and serum cholesterol. 2. Promotion of early diagnosis, treatment and rehabilitation of patients
The Stanford-Three-Community trial	Effects of community health education
Lipid Research Clinics study	Effect of reducing serum cholesterol in type II hyperlipidemics

Solution to Question 9:

According to the European society of cardiology guidelines-2018, a patient having a systolic BP $\geq$ 140 and a diastolic BP $\geq$ 90 mmHg is classified as having grade 1 hypertension.

Solution to Question 10:

Such errors can arise due to differences in hearing acuity and interpretation of Korotkoff sounds.

The types of errors that can occur during the measurement of blood pressure include:

- Observer errors - due to difference in hearing acuity and interpretation of Korotkoff sounds
- Instrumental errors - due to a leaking valve or cuffs that do not encircle the arm completely
- Subject errors - due to anxiety, fear, etc. of the patient

Solution to Question 11:

The above diagram depicts the "rule of halves", which shows the distribution of hypertension in the community. The number 2 in this diagram denotes normotensive subjects.

The rule of halves is a way of depicting the iceberg phenomenon seen in hypertension. It depicts that only about half of the hypertensive subjects in the general population of most developed countries are aware of the condition and only half of those aware are being treated and only half of those being treated are considered to be adequately treated.

The circles marked by numbers in the given diagram correspond to the following parts of the population:

- Whole community
- Normotensive subjects
- Hypertensive subjects
- Undiagnosed hypertension
- Diagnosed hypertension
- Diagnosed but untreated
- Diagnosed and treated
- Inadequately treated
- Adequately treated

Solution to Question 12:

If the blood pressure of individuals is followed up over years from early childhood to adult life, low blood pressures tend to remain low and high blood pressures become higher as the individuals grow older. This phenomenon of persistence of the rank order of blood pressure is described as tracking.

This is helpful in identifying children and adolescents who are at risk of developing hypertension in the future.

Although blood pressure is known to increase with age in both sexes, some communities have been identified whose mean blood pressure does not rise with age. This is because their salt intake and calorie consumption are at a subsistence level.

Solution to Question 13:

Salt restriction or decreasing dietary sodium intake decreases the systolic blood pressure by 2-8 mmHg approximately.

Lifestyle modifications to manage hypertension:

Note: DASH stands for dietary approaches to stop hypertension.

Recommendation	Reduction in systolic blood pressure(approximate range)
Weight reduction (Target BMI:18.5-24.9)	5–20 mm Hg /10 kg weight loss
Consume diet rich in fruits, vegetables and low fat dairy products with decreased saturated fats and total fat (DASH diet)	8–14 mm Hg
Reducing intake of dietary sodium to no more than 100mEq/day(< 6 g of sodium chloride)	2–8 mm Hg
Physical activity like brisk walk (at least 30 min per day), most days of week.	4–9 mm Hg
Limit consumption of alcohol to no more than two drinks per day	2–4 mm Hg

Solution to Question 14:

Among the above options, adopting the dietary approaches to stop hypertension (DASH) eating plan causes maximum decrease in systolic BP. It decreases the systolic BP by 8-14 mm Hg.

Lifestyle modifications to manage hypertension:

Recommendation	Reduction in systolic blood pressure(approximate range)
Weight reduction (Target BMI:18.5-24.9)	5–20 mm Hg /10 kg weight loss
Consume diet rich in fruits, vegetables and low fat dairy products with decreased saturated fats and total fat (DASH diet)	8–14 mm Hg
Reducing intake of dietary sodium to no more than 100mEq/day(< 6 g of sodium chloride)	2–8 mm Hg
Physical activity like brisk walk (at least 30 min per day), most days of week.	4–9 mm Hg
Limit consumption of alcohol to no more than two drinks per day	2–4 mm Hg

Solution to Question 15:

Polyarthralgia is a minor criterion for the diagnosis of rheumatic fever in a low risk population.

Solution to Question 16:

The given scenario is an example of primary mode of prevention of rheumatic fever. This is because the patient only has strep throat and rheumatic fever has not developed yet.

In case benzathine penicillin G is given to a patient with rheumatic fever to prevent further complications, then it becomes secondary level of prevention.

Solution to Question 17:

In the given scenario, benzathine penicillin G for secondary prophylaxis should be given for 10 years from the last attack i.e. till 26 years of age. The given case is of rheumatic fever with carditis but with no residual valvular disease, for which secondary prophylaxis is given for 10 years from the last attack or till 21 years of age, whichever is longer.

Recommendations for the duration of secondary prophylaxis with penicillin:

Category of the patient	Duration of prophylaxis
Rheumatic fever without carditis	For 5 years after the last attack or until 21 years of age (whichever is longer)
Rheumatic fever with carditis but with no residual valvular disease	For 10 years after the last attack or until 21 yrs of age (whichever is longer)
Rheumatic fever with persistent valvular disease evident clinically or on echo	For 10 years after the last attack or until 40 years of age, or sometimes lifelong prophylaxis

Solution to Question 18:

A higher intake of saturated fats is associated with an increased risk of diabetes mellitus.

Dietary factors that are protective for diabetes mellitus are:

- Higher intake of unsaturated fatty acids
- A minimum daily intake of 20g of dietary fibre

- Exclusive breastfeeding

Solution to Question 19:

Fasting blood glucose of 122 mg/dl with a postprandial value of 138 mg/dl indicates the patient falls under the criteria for impaired fasting glucose.

The WHO diagnostic criteria for diabetes and intermediate hyperglycemia is as follows:

Fasting plasma glucose	Post prandial plasma glucose	Diagnosis
≥ 126 mg/dl	> 200 mg/dl	Diabetes
< 126 mg/dl	140-200 mg/dl	Impaired glucose tolerance
110-125 mg/dl	< 140 mg/dl	Impaired fasting glucose

Solution to Question 20:

The use of ACE inhibitors in diabetics reduces the risk of nephropathy by 42% and cardiovascular risk by 22%. Annual eye examination in diabetics reduces the incidence of serious visual loss by 60-70%.

Interventions in diabetes:

- Glycaemic control in diabetics with HbA1c > 9% reduces microvascular disease risk by 30% per 1% drop in HbA1c.
- Blood pressure control in patients with BP > 130/80 mmHg reduces microvascular and macrovascular disease risk by 35% per 10 mmHg drop in blood pressure.
- Lifestyle modification reduces the incidence of diabetes by 35-58%.
- Metformin administered as a preventive therapy reduces the incidence of diabetes by 25-31%.
- Proper foot care reduces the risk of serious foot disease by 50-60%.

Solution to Question 21:

Low blood pressure is not a risk factor for NAFLD according to the NP-NCD.

The National Programme for Prevention & Control of Non-Communicable Diseases (NP-NCD), which was earlier NPCDCS, has integrated guidelines for the management of Non-alcoholic Fatty Liver disease (NAFLD) into its framework. Through a community-based assessment checklist, the community health officer along with ASHA workers will assess for the presence of the following risk factors:

- Abdominal obesity (waist circumference of ≥ 90 cm in men and ≥ 80 cm in women)
- Personal or family history of diabetes, hypertension, cancers, heart diseases, liver diseases or gallstones
- Obesity (BMI ≥ 23 kg/m²)
- Presence of pedal edema

Level of healthcare	Provisions
Community level	ASHA workers assess for abdominal obesity, relevant personal or family history. Refer to the sub-health centre.
Sub-Health Centre	Validate ASHA workers' referrals. Further assess for pedal edema, BMI, and screen for other NCDs.
Primary Health Centre	Assessment of SHC referrals for signs of chronic liver disease. Basic treatment for DM/HTN.
Community Health Centre	Rule out other causes of liver disease, and perform CBC and LFT. Calculate FIB4 or NFS score. If low risk of fibrosis - Refer back to the PHC for conservative management, weight reduction etc. Reassess in 3 years. If indeterminate, or high risk of fibrosis - Refer to a higher centre.
District level	Perform Fibroscan for indeterminate risk of fibrosis. In case of low risk of fibrosis - Refer back to the PHC for conservative management, weight reduction, etc. Reassess in 3 years. In case of high risk of fibrosis - Management of fibrosis, screening for HCC and portal hypertension. Refer complicated cases to tertiary level.
Tertiary level	Management of complicated cases.

Non-Communicable Diseases - Cancer, Obesity and Blindness

Question 1:

Which of the following organisms is least likely to be associated with malignancies?

- a) Cytomegalovirus
- b) Epstein-Barr Virus
- c) HPV 6 and 11
- d) Schistosoma

Question 2:

Which among the following is not a primary method of prevention of carcinoma in the population?

- a) Screening
- b) Restricting the use of tobacco
- c) Immunization
- d) Avoiding alcohol

Question 3:

According to Globocan 2020 India, which of the following are the most common malignancies among men and women respectively?

- a) Ca lung and Ca cervix
- b) Ca lung and Ca breast
- c) Ca oral cavity and Ca breast
- d) Ca colon and Ca cervix

Question 4:

Which of the following women is not at an increased risk of developing breast cancer in future?

- a) A woman who attained menarche at 9 years of age
- b) A woman who has never breastfed in her lifetime
- c) A woman who attained menopause at 56 years of age
- d) A woman who had her first full term pregnancy at 26 years of age

Question 5:

A 50-year-old woman is at increased risk of breast cancer as she is on hormone replacement therapy. Which of the following is the most sensitive method of screening for this condition?

- a) Self breast examination
- b) Clinical breast examination
- c) Ultrasound of the breast
- d) Mammography

Question 6:

Which of the following is the best level of prevention of breast cancer?

- a) Specific protection
- b) Early diagnosis and treatment
- c) Disability limitation
- d) Rehabilitation

Question 7:

A patient with Prader-Willi syndrome needs to be evaluated for obesity but the scale for measuring height in your clinic is broken. Which of the following indices can you still use in this situation?

- a) Quetelet's index
- b) Corpulence index
- c) Lorentz formula
- d) Broca's index

Question 8:

Which of the following indices of obesity is obtained by dividing weight in kg by square of height in metres?

- a) Ponderal index
- b) Broca's index
- c) Lorentz's index
- d) Quetelet's index

Question 9:

The BMI of two Indian college students are 17 and 24.99 respectively. How will you classify them in terms of their BMI?

- a) Underweight and normal
- b) Underweight and overweight
- c) Normal and overweight
- d) Underweight and obese

Question 10:

The weight of a diabetic woman is 75 kg and her height is 150 cm. Calculate her BMI.

- a) 30
- b) 33
- c) 35
- d) 28

Question 11:

For which of the following conditions, an obese individual is at the greatest risk of developing in the future?

- a) Polycystic ovary syndrome
- b) Coronary artery disease
- c) Low back pain
- d) Type 2 diabetes

Question 12:

Which of the following statements is incorrect regarding waist circumference?

- a) The cut-off for normal waist circumference in Indian women is 90 cm
- b) It is measured at the midpoint between the lower border of the rib cage and the iliac crest
- c) It is an approximate index of intra-abdominal and total body fat
- d) It is an indicator of the risk of chronic diseases

Question 13:

According to the WHO, what is the cut-off for visual acuity for blindness?

- a) 6/60
- b) 3/60
- c) 1/60
- d) Perception of light

Question 14:

Which of the following is the most common cause of blindness in India?

- a) Refractive error
- b) Cataract
- c) Diabetic retinopathy
- d) Trachomatous corneal opacity

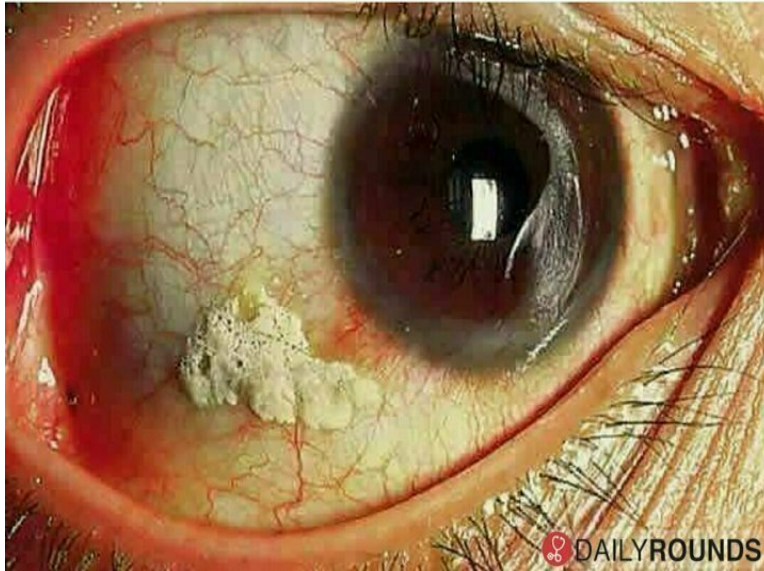
Question 15:

You administer vitamin A prophylactically to a 15-month-old child during a routine checkup. What is the level of prevention being achieved here?

- a) Primary
- b) Secondary
- c) Tertiary
- d) Primordial

Question 16:

A boy is brought to the ophthalmologist with complaints of dryness in both the eyes and difficulty in reading. On examination, his eye appears as shown in the below image. What is the stage of xerophthalmia according to WHO?



- a) X1 A
- b) X1 B
- c) X3 A
- d) X3 B

Question 17:

A 2-year-old child on examination shows signs of vitamin A deficiency. What is the dose of vitamin A to be administered to this child?

- a) 50000 IU
- b) 1 lakh IU
- c) 2 lakhs IU
- d) 2.5 lakh IU

Question 18:

What is the most common cause of injury-related death?

- a) Interpersonal violence
- b) Falls
- c) Road traffic accidents
- d) Suicide

Answer Key

Question No.	Correct Option
1	c
2	a
3	c
4	d
5	d
6	b
7	b
8	d
9	b
10	b
11	d
12	a
13	b
14	b
15	a
16	b
17	c
18	c

Detailed Explanations

Solution to Question 1:

HPV 6 and 11 are considered low-risk subtypes and are associated mainly with genital warts (condyloma acuminata) and respiratory papillomatosis in both genders.

HPV 16 and 18 are considered high-risk subtypes and are associated with cancer cervix and pre-cancerous cervical lesions. There is also evidence linking HPV 16 and 18 with cancers of the

anus, vulva, vagina, penis, and oropharynx.

Important agents associated with malignant conditions:

- HHV-8 (mainly), cytomegalovirus - Kaposi's sarcoma
- Epstein-Barr virus - Burkitt's lymphoma and nasopharyngeal carcinoma
- HPV 16 and 18 - Cancer cervix
- HBV and HCV - Hepatocellular carcinoma
- Schistosoma - Bladder carcinoma

Solution to Question 2:

Screening for cancer is a secondary level of prevention.

This is because it detects the disease at a pre-invasive or pre-malignant stage when the process of carcinogenesis has already started. It thereby aids in further treatment and improved survival.

Immunization, restricting the use of tobacco and avoiding alcohol are primary levels of prevention because they prevent the start of carcinogenesis.

Solution to Question 3:

According to Globocan 2020 India data, the most common malignancy among men is cancer lip and oral cavity, and among women is cancer breast.

The table below lists the top 5 common cancers according to the Globocan 2020 India.

Men	Women	Overall
1. Lip, oral cavity	1. Breast	1. Breast
2. Lung	2. Cervix uteri	2. Lip, oral cavity
3. Stomach	3. Ovary	3. Cervix uteri
4. Colorectum	4. Lip, Oral cavity	4. Lung
5. Oesophagus	5. Colorectum	5. Colorectum

Solution to Question 4:

Women who have their first full-term pregnancy after 30 years of age are at increased risk for breast cancer.

The risk of breast cancer increases with an increase in duration of exposure to estrogen. As progesterone levels are increased during pregnancy, it reduces the risks of conditions that occur due to hyperestrogenemia.

Risk factors for breast cancer:

- Modifiable risk factors:
 - Age at first live birth (First full-term pregnancy after 30 years)
 - Parity
 - Lack of breastfeeding
 - Obesity
 - Alcohol consumption
 - Tobacco smoking
 - Use of hormone replacement therapy
 - Decreased physical activity
- Non-modifiable risk factors:
 - Increasing age
 - Early age at menarche (onset of menses before 12 years of age)
 - Menstrual factors
 - Female sex
 - Older age at menopause (onset beyond 55 years of age)
 - Nulliparity
 - Race, ethnicity
 - Genetic predisposition
 - Family history
 - History of radiation exposure

Solution to Question 5:

Mammography is the most sensitive and specific investigation for detecting breast cancer.

It can detect even small tumors that are missed during a self-breast examination, clinical examination by the physician and ultrasonographic examination.

Solution to Question 6:

Early diagnosis and treatment is the best level of prevention of breast cancer. It is a secondary level of prevention.

The earlier a disease is diagnosed and treated, the better is the prognosis. The occurrence of secondary cases or any long term disability can be prevented. This is of particular importance in chronic diseases.

Early diagnosis and treatment is not as effective and economical as primary prevention. However, it may be critically important in reducing the high morbidity and mortality in diseases such as essential hypertension, cancer cervix and breast cancer.

Solution to Question 7:

The corpulence index can be used in this scenario as it is independent of height. The formula for corpulence index is actual weight/ desired weight.

Solution to Question 8:

Quetelet's index (Body Mass Index) is obtained by dividing weight in kg by square of height in metres.

Solution to Question 9:

An adult Indian with a BMI of 17 is classified as underweight and a BMI of 24.99 is classified as overweight.

Solution to Question 10:

The BMI of an individual with a weight of 75 kg and height of 150 cm is 33. It is calculated as follows,

$$\text{BMI} = \text{Weight (in kg)} / \text{Height}^2 \text{ (in metre}^2\text{)} = 75 / (1.5)^2 = 33.$$

Solution to Question 11:

Among the given options, an obese individual is at the greatest risk of developing type 2 diabetes. The relative risk of health problems associated with obesity:

Greatly increased risk	Moderately increased risk	Slightly increased risk
Type 2 diabetes	Coronary artery disease	Cancer (breast cancer in post menopausal women, endometrial cancer, colon cancer)
Gall bladder disease	Hypertension	Reproductive hormone abnormalities

Greatly increased risk	Moderately increased risk	Slightly increased risk
Dyslipidaemia	Osteoarthritis(knees)	Polycystic ovary syndrome
Insulin resistance	Hyperuricaemia and gout	Impaired fertility
Breathlessness		Low back pain due to obesity
Sleep apnea		Increased risk of anaesthesia complications
		Fetal defects associated with maternal obesity

Solution to Question 12:

The cut-off value for normal waist circumference in Indian women is 80cm, while in Indian men it is 90cm.

Waist circumference is an approximate indicator of intra-abdominal and total body fat. It is measured at the midpoint between the lower border of the rib cage and the iliac crest. It is an indicator of the risk of chronic diseases.

The cut-off for normal waist circumference is:

Category	Cut-off for normal waist circumference
Indian	
Men	90cm
Women	80cm
World	
Men	102cm
Women	88cm

Solution to Question 13:

According to the WHO, blindness is defined as visual acuity of less than 3/60, or a corresponding visual field loss to less than 10°, in the better eye with the best possible correction.

Solution to Question 14:

According to the National Survey on Blindness (2015-2019), cataract is the most common cause of blindness in India accounting for 66.2% of the total cases.

Solution to Question 15:

Prophylactic administration of vitamin A to a normal child is an example of primary level of prevention. It is done to prevent xerophthalmia.

Solution to Question 16:

The finding in the given image is a Bitot's spot. It belongs to WHO class XI B of vitamin A deficiency.

Solution to Question 17:

2 lakh IU of vitamin A should be administered to a 2-year-old child with vitamin A deficiency.

Dosage of vitamin A in children:

- 50,000 IU for children less than 6 months of age
- 1 lakh IU for children between 6 months to 12 months of age
- 2 lakh IU for children above 12 months of age.

These doses of vitamin A are to be administered on days 1, 2 and 14 for the treatment of vitamin A deficiency.

Solution to Question 18:

The most common cause of injury-related deaths are road traffic accidents, followed by suicide.

National Health Programmes I - NVBDCP

Question 1:

You are working as the PHC Medical Officer in your village. Which of the following diseases would you not target under the National Vector Borne Disease Control Programme?

- a) Chikungunya
- b) Japanese encephalitis
- c) Visceral leishmaniasis
- d) Cutaneous leishmaniasis

Question 2:

The insect shown below is a vector for a disease that is targetted under the National Vector Borne Disease Control Programme. Identify the disease.



- a) Chikungunya
- b) Visceral leishmaniasis
- c) Malayan filariasis
- d) Japanese encephalitis

Question 3:

Which of the following is carried out by the Drug Distribution Centres under the National Vector Borne Disease Control Programme for Malaria?

- a) 2 and 4
- b) 2 only
- c) 1, 2, and 3
- d) 1, 2, and 4

Question 4:

Which of the following statements is true regarding the surveillance system set up for malaria?

- a) Active case detection is carried out when patients report with fever to the peripheral health workers
- b) Blood smears are collected fortnightly during house visits by Accredited Social Health Activists
- c) Antibody detecting rapid diagnostic kits are used for passive case detection as part of surveillance
- d) Passive case detection at the Primary Health Centre is done by examination of blood smears

Question 5:

Which of the following measures for controlling malaria is the Urban Malaria Scheme mainly based on?

- a) Measurement of Annual Parasite Index
- b) Anti-adult measures
- c) Anti-larval measures
- d) Active surveillance

Question 6:

What is the most efficient anti-larval method to prevent urban malaria?

- a) Clean drainage systems and sewers
- b) Cover overhead tanks properly
- c) Filling cesspools and ditches
- d) Use of larvivorous fishes

Question 7:

The blood smear slides (A) and (B) prepared from a patient suspected with malaria are shown below. Which of the following statements is incorrect regarding these tests?

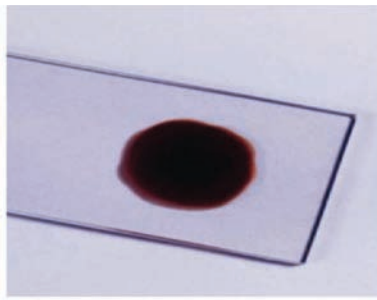


- a) Slide (A) is used for estimating the parasite concentration
- b) Slide (A) has been air dried for 10 minutes
- c) Slide (B) cannot be used for detecting the species of the parasite
- d) Jaswant Singh Bhattacharya stain is used for both

Question 8:

The following two diagnostic techniques (A) and (B) were performed in a medical camp conducted for diagnosing malaria. Which is a true statement regarding these techniques?

A.



B.



- a)** Image (A) - It is used for identifying the species of the malarial parasite
- b)** Image (A) - It is not the established method of choice for diagnosing malaria in urban areas
- c)** Image (B) - Antibody detection by this test is the method of choice for diagnosing malaria in inaccessible and tribal areas
- d)** Image (B) - Bivalent type of this test was used for detecting *Plasmodium falciparum* and *Plasmodium vivax*

Question 9:

Which of the following parameters would you use to check the efficiency of the surveillance system for malaria under the National Vector Borne Disease Control Programme?

- a)** Annual Parasite Index
- b)** Annual Blood Examination Rate
- c)** Slide positivity rate
- d)** Slide falciparum rate

Question 10:

According to the National Framework for Malaria Elimination in India, states are classified based on which of the following parameters?

- a)** Annual parasite incidence
- b)** Slide positivity rate
- c)** Annual falciparum incidence

- d) Annual blood examination rate

Question 11:

Which of the following is correct about the classification of malaria endemic states under the National Strategic Plan for Malaria?

- a) The states classified under Category 0 are in the elimination phase
- b) The states with API ≤ 1 and all their districts reporting API ≤ 1 are in the elimination phase
- c) The states with districts reporting API in between 1-2 per 1000 population are in the elimination phase
- d) The states classified under category 2 are in the phase of intensified control

Question 12:

Which of the following months is observed as the anti-malaria month in India?

- a) April
- b) May
- c) June
- d) July

Question 13:

A rural medical officer implements Integrated Vector Management in his village. Which of the following measures of vector control would you prefer in a rural area?

- a) Source reduction
- b) Indoor residual spray (IRS)
- c) Biological control using *Gambusia affinis*
- d) Chemical control using mineral oils

Question 14:

Select the correct option regarding Integrated Vector Management:

- a) Malathion is the insecticide of choice used for vector management
- b) Larvivorous fish and larvicides are the main methods of vector control in rural areas

- c) Long Lasting Insecticide Nets are provided in areas with API ≥ 5
- d) Insecticide treated bed nets are supplied in areas with API ≥ 3

Question 15:

You are working as a medical officer in Jharkhand, which is endemic to the disease transmitted by the vector shown below. Which of the following agents would you prefer to control this vector in view of resistance to commonly used agents?



- a) Dihydro Diphenyl Trichloro ethane
- b) Synthetic pyrethroid
- c) Malathion
- d) Paris green

Question 16:

According to the revised strategy for the eradication of kala-azar, which of the following drugs would you use across all age groups in order to reduce the human reservoir of infection?

- a) Miltefosine
- b) Sodium stibogluconate
- c) Liposomal amphotericin B
- d) Ivermectin

Question 17:

Which of the following statements regarding the elimination and eradication of lymphatic filariasis is true?

- a) It will be considered as eliminated when the number of microfilaria carriers are less than 5 percent
- b) It will be considered as eliminated when children born after initiation of the programme are free from antigenemia
- c) India has set the target of eliminating lymphatic filariasis by 2025
- d) WHO recommends dual therapy with diethylcarbamazine and albendazole for elimination of transmission

Question 18:

As the medical officer in a Japanese encephalitis endemic area, which of the following measures would you not advocate for controlling its transmission?

- a) The use of Indoor Residual Sprays
- b) The use of larvivorous fishes
- c) Keeping pigs away from human dwellings
- d) The use of malathion for outdoor fogging

Answer Key

Question No.	Correct Option
1	d
2	d
3	b
4	d
5	c
6	b
7	a
8	d
9	b
10	a

11	b
12	c
13	b
14	c
15	b
16	c
17	b
18	a

Detailed Explanations

Solution to Question 1:

The National Vector Borne Disease Control Programme (NVBDCP) does not target the prevention and control of cutaneous leishmaniasis (also known as Chiclero's ulcer).

NVBDCP is the nodal agency for the prevention and control of the following vector-borne diseases:

- Kala-azar (also called visceral leishmaniasis)
- Malaria
- Dengue
- Lymphatic filariasis
- Japanese encephalitis
- Chikungunya

The logo of the NVBDCP:



Solution to Question 2:

The image shown is of the Culex species of mosquitoes, which is a vector for Japanese encephalitis and Bancroftian filariasis.

Mansonia species is the vector for Malayan filariasis.

Solution to Question 3:

The Drug Distribution Centres are only for dispensing anti-malarial drugs, under the National Vector Borne Disease Control Programme (NVBDCP) for Malaria.

The organizational structure for the Anti-Malaria Programme is as follows:

- Central Government: supplies DDT and larvicides.
- State government: procures the Indoor Residual Sprays (IRS) for the districts.
- District Malaria Office: is the key unit for the implementation of the programme.
- District Malaria Officer: has the direct responsibility of IRS spraying operations and is assisted by one Assistant Malaria Officer and Malaria Inspectors.
- Medical officer of the PHC: has the overall responsibility for the surveillance and laboratory services.
- Multi-Purpose Workers and ASHAs: carry out case-based surveillance and case detection in the community.
- Drug Distribution Centres: only for dispensing the antimalarial drugs.
- Fever Treatment Depots: collect blood slides in addition to the distribution of antimalarial drugs.

Solution to Question 4:

Passive case detection at Primary Health Centres (PHCs) is done by examination of blood smears.

Malaria surveillance system:

Active case detection: blood smears are collected by the Multipurpose Health Workers (MPHW)/ Auxiliary Nurse Midwife (ANM) on house visits every fortnight.

Passive case detection: detection of fever case by collecting of blood smears of those that report to ASHAs or peripheral health volunteers.

Solution to Question 5:

The Urban Malaria Scheme is mainly based on anti-larval measures for controlling malaria. In Urban Malaria Scheme, urban areas with a population of $>50,000$ and a slide positivity rate of $\geq 5\%$ are included.

The vector in urban areas breeds in man-made containers. Therefore, anti-larval measures such as larvivorous fish and larvicides are used to control breeding as the primary method of vector control.

In rural areas, anti-adult measures, such as Indoor Residual Spray (IRS) is the primary method of vector control.

Solution to Question 6:

The most efficient anti-larval method to prevent urban malaria is to cover overhead tanks and other man-made containers properly. This is because the vector of malaria in urban areas breeds largely in such man-made containers.

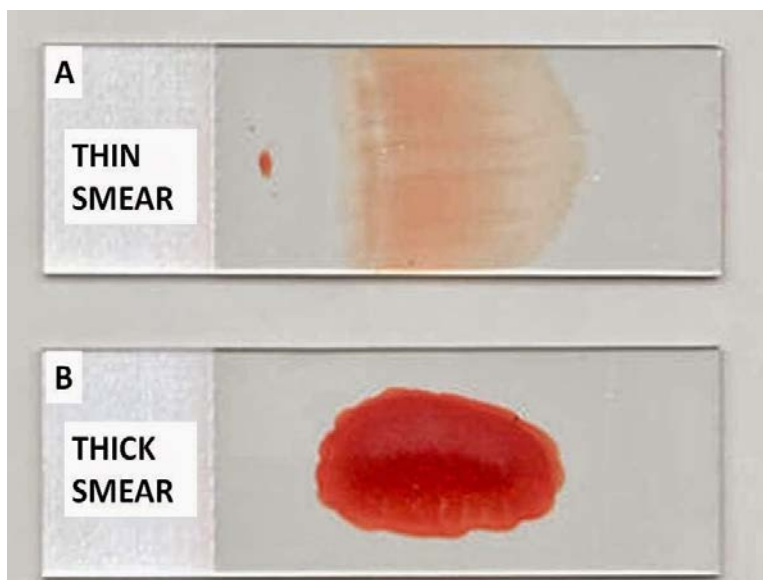
Solution to Question 7:

Slide (A) is a thin blood smear and slide (B) is a thick blood smear. Thin blood smears (slide A) are not used for estimating parasite concentration. They are used to detect the species of parasite.

Microscopy is the best method for the diagnosis of malaria due to its high sensitivity and specificity. Jaswant Singh Battacharya (JSB) stain is used for staining both thick and thin smears.

Thick smear is used to detect infection and estimate parasite concentration. The correct thickness is the thickness through which a newspaper is barely visible. It is prepared by slow-drying for 30 minutes, which leads to hemolysis and distortion of the plasmodia morphology. Hence, species cannot be detected.

Thin smear is used to identify the particular infecting species of Plasmodium. It is prepared by air drying for 10 minutes and fixed in methanol. Thus, the plasmodia morphology is not distorted.



Solution to Question 8:

Image (A) is a thick blood smear and Image (B) is the Rapid Diagnostic Test (RDT). Bivalent RDTs detect *P. falciparum* and *P. vivax*.

Thin blood smear is used to identify the particular infecting species. Thick blood smear is used to detect infection and estimate parasite concentration.

Antibody detecting Rapid Diagnostic Tests (RDTs) are not the method of choice anymore in diagnosing malaria in inaccessible and tribal areas. The Antigen detecting RDTs and blood smear examination are the recommended methods for diagnosis of malaria.

Note: According to the announcement by Union Health Ministry in April 2018, antibody detecting Rapid Diagnostic Tests (RDTs) for malaria is banned due to the high false-positive rates in endemic areas.

Malaria Antigen detecting Rapid Diagnostic Test kit:



Solution to Question 9:

The Annual Blood Examination Rate (ABER) is used to check the efficiency of the surveillance system for malaria. It takes into account the number of slides examined for the population under surveillance.

Solution to Question 10:

The annual parasite incidence (API) is used in classifying states/UTs according to the National Framework for Malaria Elimination in India.

The Annual Parasite Incidence is calculated as: $(\text{Cases in one year} \times 1000) / (\text{Population under surveillance})$.

Solution to Question 11:

The National Strategic Plan for Malaria classifies the states under Category 1 of the NVBDCP to be in the elimination phase.

Category 1 states are all states/UTs with API ≤ 1 and all their districts reporting API ≤ 1 .

Solution to Question 12:

The anti-malaria month is observed in the month of June every year in our country, just before the onset of monsoon (transmission season of malaria) to enhance the level of awareness.

World Malaria Day is observed on 25th April.

Health days	Dates
World Tuberculosis Day	24th March
World Health Day	7th April
World Malaria Day	25th April
National Anti Malaria Month	June
World Immunization Week	Last week of April
World No Tobacco Day	31st May
World Blood Donor Day	14th June
World Hepatitis Day	28th July
Breastfeeding Week	1st week of August
World AIDS Day	1st December
Sexual and Reproductive Health Awareness Day	12th February

Solution to Question 13:

The measure of vector control preferred for rural areas is indoor residual spray (IRS).

Under integrated vector management (IVM), the preferred measures of vector control for

- Rural areas include: Anti-adult measures, using indoor residual spray (IRS)
- Urban areas include: Anti-larval measures, like environmental control by source reduction, chemical control (mineral oils, Paris green, synthetic insecticides, etc.), and biological control (fish species *Gambusia affinis* and *Lebistes reticulatus* which feed on mosquito larvae).

Solution to Question 14:

Under the Integrated Vector Management initiative, Long Lasting Insecticide Nets (LLIN) are provided in areas with Annual Parasite Index (API) ≥ 5 .

Integrated Vector Management (IVM):

- For adult mosquitoes:
 - Indoor Residual Sprays (IRS) are used as the main method of vector control in rural areas. The insecticide of choice will be the insecticide to which the local vector is susceptible as per the best available information.
 - Two rounds of spraying are done with DDT, and in DDT resistant areas, three rounds of spraying are done with malathion.
- For larvae:
 - Larvivorous fish and larvicides are used. This is the main method of vector control in urban areas.
- For personal protection:
 - Insecticide-treated bed nets are supplied in areas with Annual Parasite Index (API) ≥ 2 .
 - Long-Lasting Insecticide Nets (LLIN) are provided in areas with API ≥ 5 .

Solution to Question 15:

The image given is that of a sandfly, which is the vector for kala-azar. In endemic states with increasing resistance to commonly used agents like DDT, synthetic pyrethroids are preferred now.

The four states endemic for kala-azar are Bihar, Jharkhand, West Bengal and Uttar Pradesh.

Solution to Question 16:

A single dose of intravenous treatment with liposomal amphotericin B is used across all age groups in order to reduce the human reservoir of infection. WHO will supply this drug free of cost.

Drugs for Kala Azar:

- Single-dose intravenous liposomal amphotericin B for all age groups and for pregnant and HIV co-infected patients.
 - Miltefosine- an oral drug, once daily for 28 days. Contraindicated during pregnancy, lactation, and in women who refuse contraception.
 - Combination of miltefosine capsules with paromomycin i.m, for 10 days. Contraindications are the same as miltefosine.
 - The drug of choice for Post Kala-Azar Dermal Leishmaniasis (PKDL) is miltefosine 100mg orally per day for 12 weeks.
- Ivermectin is used for lymphatic filariasis.

Solution to Question 17:

Lymphatic filariasis will be considered eliminated when the children born after the initiation of the programme are free from antigenemia. Antigenemia refers to the presence of adult filaria worms in the human body.

According to the Elimination of Lymphatic Filariasis (ELF) Programme, there are two criteria to consider elimination of lymphatic filariasis:

- Children born after the initiation of the elimination programme are free from circulating antigenemia.
- The number of microfilaria carriers is $\leq 1\%$.

Following the failure to achieve ELF by 2015, the target set to eliminate lymphatic filariasis by WHO was 2020. To achieve the same WHO recommended the use of triple therapy for lymphatic filariasis. It incorporated ivermectin along with diethylcarbamazine (DEC) and albendazole instead of the traditional DEC and albendazole therapy, for the elimination of transmission of lymphatic filariasis.

In 2021, WHO released the 'Road Map for Neglected Tropical Diseases (2021–2030)'. Under this, lymphatic filariasis is targeted for elimination as a public health problem by 2030.

In February 2023, the Ministry of Health & Family Welfare in India launched a nationwide Sarva Dawa Sevan or Mass Drug Administration (MDA) campaign to eliminate LF by 2027, aiming to achieve this goal three years ahead of the global target.

Solution to Question 18:

Indoor Residual Sprays (IRS) are not used in the vector control measures for Japanese Encephalitis (JE) as its vectors (Culex) are outdoor resters.

Prevention and control of JE include:

- Vaccination
- Keeping pigs away from human dwellings from dusk to dawn (biting time of the vectors). Pigs act as important amplifiers of the virus.

- Use of bed nets for personal protection.
- Use of larvivorous fishes.
- Using malathion for outdoor fogging to control an outbreak.

Sold by @Itachibot
If you purchased this from someone else,
you may have been scammed.

National Health Programmes II - NLEP, NTEP & NACO

Question 1:

Identify the organization that has the given emblem:



- a) National Vector Borne Disease Control Programme
- b) National Leprosy Eradication Programme
- c) Integrated Child Development Scheme
- d) National Programme for Prevention and Control of Cancer, Diabetes, Cardiovascular Disease, and Stroke

Question 2:

The following drug packet is used in:



- a) TB
- b) Leprosy
- c) Vaginal discharge
- d) Gonorrhea

Question 3:

The Sparsh campaign was launched to spread awareness about which of the following diseases?

- a) HIV
- b) Immunization
- c) Leprosy
- d) TB

Question 4:

A patient with leprosy comes to the PHC following a pinprick to his foot. He noticed it only after seeing bloodstains on the floor and complains of loss of sensation in his foot. After addressing the issue, the patient is referred to another centre. Where has he been referred to and why?

- a) For secondary prevention to a district hospital.
- b) For tertiary prevention to a district hospital.
- c) For secondary prevention to a district nucleus.

d) For tertiary prevention to a district nucleus.

Question 5:

A patient with leprosy, who has completed a course of multi-drug therapy, presents with complaints of developing painful, tender, and raised lesions on his forehead recently as shown in the image below. What is the first-line treatment for him?



- a) Restart multi drug therapy and start thalidomide
- b) Restart multi drug therapy and start prednisolone
- c) Start on clofazimine only
- d) Start on prednisolone only

Question 6:

ABSULS, which is an ASHA-based surveillance project, has been initiated for which of the following diseases?

- a) Leishmaniasis
- b) Diarrhoeal diseases
- c) Leprosy
- d) Sexually Transmitted Diseases (STDs)

Question 7:

The treatment of paucibacillary leprosy consists of MDT (which comprises rifampicin, dapsone, and clofazimine). The latest recommended regimen is:

- a) 2 drug therapy for 6 months
- b) 3 drug therapy for 12 months
- c) 2 drug therapy for 12 months
- d) 3 drug therapy for 6 months

Question 8:

A District Leprosy Officer is preparing the annual report as per NLEP 2020. Which of the following parameters would he not consider as an essential indicator?

- a) Treatment Completion Rate (TCR)
- b) Prevalence rate (PR)
- c) Rate of new cases with Grade II disabilities per 10 lakh population per year
- d) Proportion of new cases correctly diagnosed

Question 9:

As per NLEP 2020, which of the following is not a quality of service indicator?

- a) Patient month blister calendar pack stock
- b) Proportion of grade II disabilities among new cases.
- c) Number of relapses reported
- d) Proportion of cases who developed new or additional disability after starting MDT

Question 10:

You are part of a rural health camp organized under Project Aashya. Which of the following diseases does it aim to control?

- a) Diabetes
- b) Kala azar
- c) Tuberculosis
- d) Zika Virus disease

Question 11:

You are working as a medical officer in a PHC where you diagnose a patient with TB. Within how many days do you have to report it to the District Health Officer?

- a) 7 days
- b) 15 days
- c) 30 days
- d) 21 days

Question 12:

Which of the following is true about the Revised National Tuberculosis Control Programme?

- a) Follow-up sputum smear examination at the end of intensive phase only
- b) Directly Observed Treatment Short course strategy is applied
- c) Treatment is only given to smear positive cases
- d) MDR TB is resistance to either rifampicin or isoniazid

Question 13:

A patient presents with fever, productive cough, and weight loss. He gives a history of not having completed treatment for tuberculosis thrice in the past. Extensively drug-resistant TB is suspected. How is this entity defined?

- a) Resistance to INH + Amikacin + Ofloxacin + Ethambutol
- b) Resistance to INH + Rifampicin + Amikacin
- c) Resistance to Rifampicin + Amikacin + Ofloxacin
- d) Resistance to INH + Rifampicin + Ofloxacin + Linezolid

Question 14:

A patient on HAART presents with complaints of fever and expectorant cough for two weeks. What is the investigation to be ordered first?

- a) Sputum microscopy
- b) Interferon Gamma Release Assay
- c) CB- NAAT

- d) Sputum microscopy and Chest X-ray

Question 15:

Which of the following statements is false under the National Tuberculosis Elimination Programme?

- a) Continuation phase for new sputum smear positive: HRE for 16 weeks
- b) Intensive phase for sputum smear positive after loss to follow up: HRZE for 8 weeks
- c) Continuation phase for sputum smear positive recurrent case: HRE for 16 weeks
- d) Intensive phase for sputum smear positive after failure to treatment: HRZES for 12 weeks

Question 16:

Which of the following is the most peripheral centre under the Revised National Tuberculosis Control Programme organization structure?

- a) District TB centre
- b) Intermediate Reference Laboratory
- c) Tuberculosis Unit
- d) Designated Microscopy Centre

Question 17:

Which is the new online software launched under the national tuberculosis elimination program (NTEP) by the government to monitor TB patients?

- a) NIKSHAY
- b) NISCHAY
- c) E-DOTS
- d) NIKUSTH

Question 18:

Choose the false statement regarding the initiatives and strategies for tuberculosis.

- a) The Stop TB strategy was announced by the World Health Organisation

- b) The STOP TB strategy has a target of ending TB by achieving an incidence rate of <10 people per 10,000 population per year
- c) The "Practical approach to lung health" was launched by the WHO for syndromic management of patients with respiratory symptoms
- d) The End TB strategy is a 20 year programme

Question 19:

Which of the following group of individuals would you not target under the programme represented by the logo shown below?



- a) Sex workers
- b) Migrants
- c) Transgenders
- d) Industrial workers

Question 20:

Which of the following statements is false regarding the target interventions under National AIDS Control Organisation?

- a) Lubricants are given to homosexual men
- b) Clean needles and syringes are given to injecting drug users
- c) Condoms are not given to bridge population
- d) Opioid substitution therapy is given to Injecting Drug Users

Question 21:

A physician follows the two test protocol for the diagnosis of HIV in a patient. It consists of:

- a) Western blot as the first test
- b) Line immunoassay as the second test
- c) HIV ELISA as the second test
- d) Western blot as the second test

Question 22:

As a medical officer involved in the sentinel surveillance for HIV under the National AIDS Control Programme, your duties include all of the following except:

- a) Estimation of total infection in the community
- b) Estimation of total cases in hospitals
- c) Estimation of trend of disease
- d) Classification of districts

Question 23:

Which of the following is targetted by the establishment bearing the logo shown below:



- a) Family planning

- b) Immunisation
- c) Sexually Transmitted Diseases
- d) Maternal and Child Health Services

Question 24:

Choose the wrong combination of the colour-coded STI kits among the given options.

- a) Kit 6 – yellow
- b) Kit 3 – white
- c) Kit 4 – black
- d) Kit 5 – red

Answer Key

Question No.	Correct Option
1	b
2	b
3	c
4	d
5	d
6	c
7	d
8	d
9	b
10	c
11	c
12	b
13	d
14	c
15	d
16	d
17	a
18	b
19	d

20	c
21	c
22	b
23	c
24	c

Detailed Explanations

Solution to Question 1:

The given image is the emblem of National Leprosy Eradication Programme (NLEP).

Characteristics of NLEP emblem:

- The lotus in the NLEP emblem symbolizes beauty and purity.
- A partially affected thumb, a normal forefinger, and the shape of a house symbolize that leprosy can be cured and a patient with leprosy can be a useful member of the society.
- The rising sun symbolizes hope and optimism.
- The emblem captures the spirit of hope and positivity in the eradication of leprosy.

NLEP is a centrally sponsored health scheme of the Ministry of Health and Family Welfare, Govt. of India.

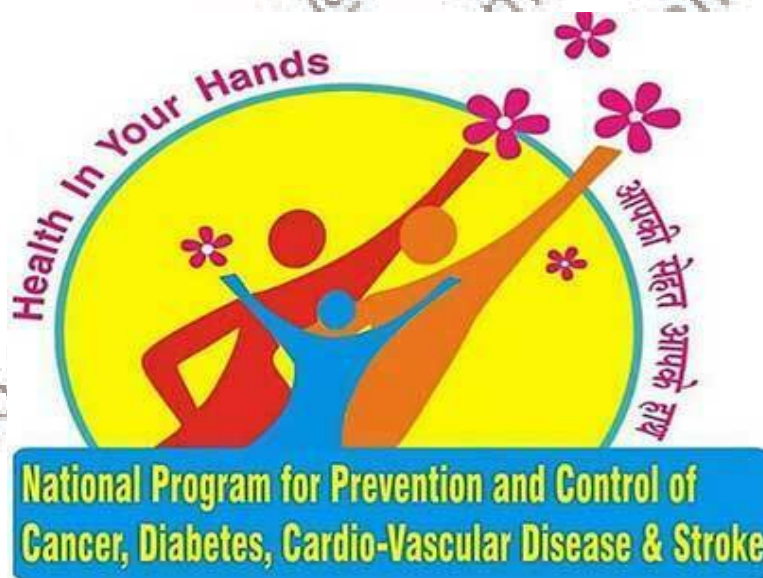
Option A: National Vector Borne Disease Control Programme (NVBDCP)



Option C: Integrated Child Development Services (ICDS)



Option D: National Programme for Prevention and Control of Cancer, Diabetes, Cardiovascular Disease, and Stroke (NPCDCS).



Solution to Question 2:

The given image shows the adult blister packet used in the treatment of leprosy.

Solution to Question 3:

Sparsh Leprosy Awareness Campaign (SLAC) was introduced in 2017 on the occasion of the anti-leprosy day (observed on the last Sunday of January) to promote awareness about various aspects of leprosy.

Sapna refers to a local school-going girl who will help to spread awareness in the community.

The other developments under NLEP are:

- Mycobacterium Indicus Prani (MiP) vaccine for multibacillary leprosy patients and their contacts (it is in a project mode).
- Nikusth: A web-based system for the data management of leprosy cases.

Note: Meena is a nine-year-old girl, who is the symbol of UNICEF's Girl Empowerment programme to impart important messages on gender, child rights, education, health and development in the South Asian region.

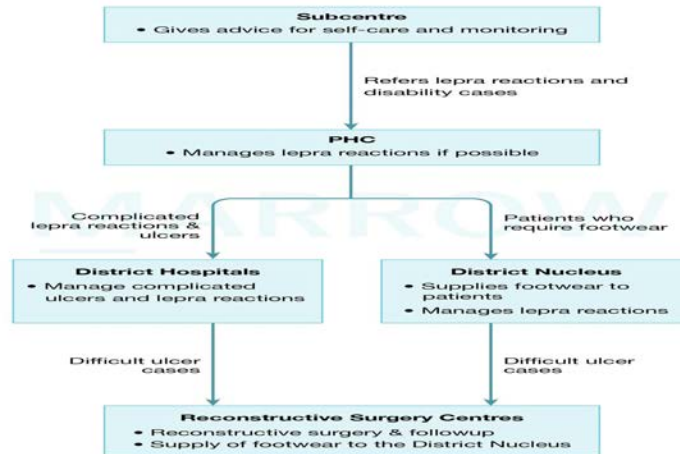
The posters below depict the role play by Sapna:



Solution to Question 4:

The given clinical scenario of a patient with leprosy having neuropathy needs Micro Cellular Rubber (MCR) footwear, which is a form of tertiary prevention. He is to be referred to a district nucleus which supplies MCR footwear to patients.

System of levels of Implementation & Referral in NLEP



©Marrow

Solution to Question 5:

The presence of new erythematous and tender subcutaneous nodules in a patient with leprosy indicates that he has developed type II lepra reaction. It is also known as Erythema Nodosum Leprosum (ENL) and the first line of treatment is prednisolone.

Multi-drug therapy (MDT) should not be restarted in patients who develop lepra reactions after the completion of treatment.

Prednisolone is started at a dose of 40 mg per day and gradually tapered by 5 mg every 2 weeks. 'Prednisolone card' is given when treatment is started. If the patient becomes steroid-dependent, then clofazimine is added.

The key targets of the Global Leprosy Strategy 2021-2023 announced by WHO (Accelerating towards a leprosy-free world) are:

- 120 countries reporting zero new autochthonous cases.
- 70% reduction in the annual number of new cases detected.
- 90% reduction in rate per million population of new cases with Grade 2 disability.
- 90% reduction in rate per million children of new child cases with leprosy.

The current goal is the elimination of leprosy.

Solution to Question 6:

ABSULS which stands for Asha Based Surveillance for Leprosy Suspects has been initiated to prioritize leprosy case detection by ASHAs.

Solution to Question 7:

The latest recommendation is 3 drug therapy for 6 months to treat paucibacillary (PB) leprosy by WHO. The regimen consists of rifampicin, dapsone, and clofazimine.

Meanwhile, multibacillary (MB) leprosy is treated using the same 3 drugs for a duration of 12 months. While the NLEP guidelines earlier mentioned a 2-drug MDT for PB leprosy, the National Strategic Plan and Roadmap for Leprosy (2023-2027) now recommends a uniform MDT of 3 drugs for both PB and MB leprosy.

Note: The central government has approved a new treatment regimen for paucibacillary (PB) leprosy. The Ministry of Health and Family Welfare has introduced a 3-drug regimen for PB leprosy instead of a 2-drug regimen. This revised drug regimen will be applicable in India from April 1, 2025.

Solution to Question 8:

The proportion of new cases correctly diagnosed is not included under essential indicators. It is a part of the quality of service indicators as per NLEP 2020.

The essential indicators include:

- Annual new case detection rate (ANCDR) per 1 lakh population.
- Rate of new cases with grade II disabilities (G2D) per 10 lakh population per year.
- Treatment completion rate (TCR) as a proxy to cure rate.
- Prevalence rate (PR).

Additional epidemiological indicators include:

- Proportion of Grade II disabilities among new cases.
- Proportion of females among new cases.
- Proportion of multibacillary (MB) among new cases.
- Proportion of child (0- 14 years) among new cases.
- Child rate per 100,000 population.
- Scheduled Caste New Case Detection Rate.

Solution to Question 9:

Proportion of grade II disabilities among new cases is not a quality of service indicator. It is an additional epidemiological indicator as per NLEP 2020.

The quality of service indicators are:

- Patient month blister calendar pack stock.
- Absolute number of patients made released from treatment (RFT).

- Number of relapse reported.
- Proportion of cases who developed new or additional disability after starting MDT.
- Proportion of treatment defaulters.
- Proportion of new cases correctly diagnosed.

Solution to Question 10:

Project Axshya was launched in 2010 to control TB. Axshya means free from TB which is the ultimate goal of the project and the National TB Elimination Programme.

Project Axshya is being implemented by the International Union Against Tuberculosis and Lung Disease (known simply as The Union). The project works to improve access to quality TB care and control through a partnership between the government and civil society.

The guiding principles of project Axshya are:

- Universal access to quality TB services
- Community participation in TB care and control
- Sustainable interventions and equitable distribution of project benefits with social and gender sensitivity

Solution to Question 11:

The guidelines for TB notification under the RNTCP is within 30 days (a month) of receiving a new case. The notification must be done to the District Health Officer or the Chief Medical Officer of a District Hospital by the practitioner who diagnosed it.

Note: RNTCP has now been renamed to National Tuberculosis Elimination Program (NTEP).

Solution to Question 12:

DOTS stands for Directly Observed Treatment Short-course and is given by peripheral and voluntary health workers.

Recent updates in RNTCP:

- Active case finding for TB has been started.
- Follow up:
 - On treatment: monthly follow up.
 - After treatment: Follow up at 6, 12, 18 and 24 months.
- A sputum smear examination must be done at the end of intensive phase (IP) and at the end of treatment.

Solution to Question 13:

XDR-TB (extensively drug-resistant TB) is defined as multi-drug resistant TB (resistant to rifampicin and isoniazid) which is also resistant to one fluoroquinolone (eg. ofloxacin) AND resistant to at least one additional group A drug (as of 2021, these are bedaquiline and linezolid)

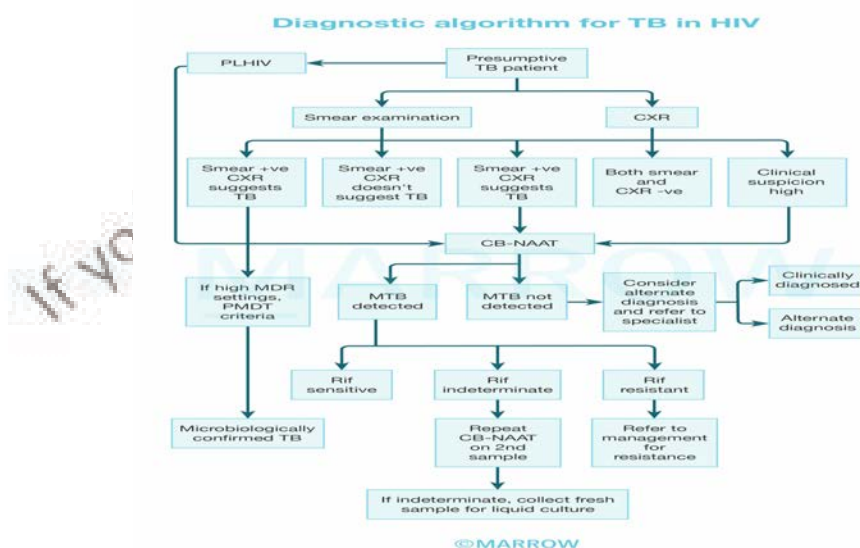
It is usually seen in patients who have previously received treatment for TB and have either failed to complete treatment or relapsed.

Solution to Question 14:

An HIV patient suspected to have TB is directly investigated by CB-NAAT (Cartridge Based Nucleic Acid Amplification Test).

TB is a very serious infection in the immunocompromised and can lead to its most severe form as miliary TB and cause death. Since TB in HIV does not have its typical presentation, early suspicion and advanced diagnosis is the key to preventing mortality.

Diagnostic algorithm for TB:



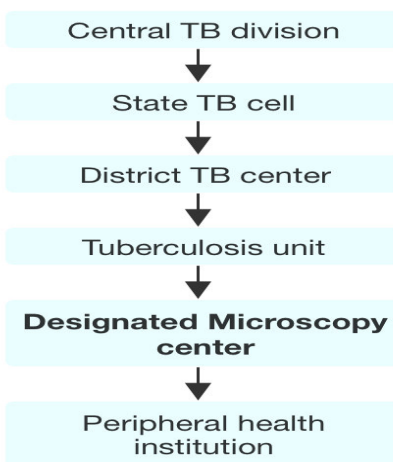
Solution to Question 15:

Under National Tuberculosis Elimination Programme (NTEP) guidelines Initiation phase (IP) for both new and previously treated cases which are H and R sensitive is HRZE for 8 weeks.

Solution to Question 16:

Among the options given, Designated Microscopy Centre is the most peripheral centre.

Structure of the RNTCP
(Revised National Tuberculosis Control Programme)



Solution to Question 17:

Nikshay is the new online software launched under the national tuberculosis elimination program for monitoring TB patients.

Nikusth is the new online software for monitoring leprosy patients.

Nischay is a pregnancy test kit.

Solution to Question 18:

The Stop TB strategy was announced by World Health Organisation to reduce the global burden of TB by 2015 (and not ending TB).

The End TB strategy is a 20-year programme with tagline "A world free of TB - Zero TB deaths, Zero TB disease, and Zero TB suffering". It aims to:

- end TB by achieving an incidence rate of <10 people/1 lakh population per year by 2035.
- reduce TB deaths by 95% and cut new cases by 90% by 2035.

The "Practical approach to lung health initiative" was launched by the WHO as a syndromic approach to the management of patients who attend primary health care services for respiratory symptoms.

Ending the TB epidemic is also one of the targets of the Sustainable Development Goals (SDG) specifically under Goal 3 (to "ensure healthy lives and promote well-being for all at all ages").

WHO defines that "Tuberculosis control" would be achieved when the prevalence of the natural order of infection in the age group of 0-14 years is of the order of 1% (40% in India currently).

Vision of End TB: A world free of TB- zero deaths, disease and suffering due to TB		
Goal: End the global TB epidemic		
Indicators	Targets	
	SDG 2030	End TB 2035
Reduction in no. of TB deaths compared with 2015	90%	95%
Reduction in TB incidence rate compared with 2015	80%(<20/1,00,000)	90%(<10/1,00,000)
TB-affected families facing catastrophic costs due to TB	0%	0%

Solution to Question 19:

The red ribbon shown is a part of the logo of the National AIDS Control Organization (NACO) programme. Industrial workers are not included in the target population under NACO.

Target interventions for the control of AIDS transmission are done for:

- High-risk groups like sex workers
- Men who have sex with men
- Transgender
- Injecting drug users
- Bridge population like migrants and long-distance truckers

The services offered through targeted interventions include detection and treatment for sexually transmitted diseases and condom distribution.

The logo of NACO:



Solution to Question 20:

Bridge populations are supplied with condoms under the NACO target interventions.

Target interventions:

General:

- Detection and treatment for sexually transmitted diseases
- Condom distribution

For injecting drug users:

- Distribution of clean needles and syringes.
- Abscess prevention and management.
- Opioid substitution therapy.
- Linkage with detoxification and rehabilitation services.

For homosexual men and transgenders:

- Provision of lubricants.
- Provision of project-based clinics for sexually transmitted infections.

Solution to Question 21:

The two test protocol for the diagnosis of HIV consists of ELISA as the second test. If the initial ELISA is positive, a second ELISA is carried out. If the second ELISA is also positive, only then the case is reported to be positive.

Investigations involved in HIV testing strategies:

- ELISA/ rapid tests.
- Confirmatory tests with high specificity like western blot and line immunoassays are used in problem cases eg: indeterminate/ discordant cases of rapid tests.

In India, NACO provides western blot (WB) confirmatory testing only to resolve indeterminate results on previous serological testing. It is not used for routine confirmation after a positive result on screening tests. Drawbacks of WB include low rates of detection during acute HIV infection and misclassification of HIV-2 infection.

Note: The western blot is used as a confirmatory test after 2 indeterminate primary tests (ELISA). It is essentially the third test used and does not fit into the two test protocol.

Solution to Question 22:

As a medical officer involved in the sentinel surveillance, the estimation of total cases in hospitals is not included in the sentinel surveillance for HIV under National AIDS Control Programme (NACO).

The aims of the HIV sentinel surveillance include:

- Determine the level of HIV infection in the community.
- Understand the trends of HIV epidemic.
- Understand the geographical spread of HIV infection and classify districts according to epidemiological and vulnerability criteria.

These objectives are achieved by conducting annual cross-sectional surveys of the risk group in the same geographical place.

Solution to Question 23:

The logo shown represents Suraksha Clinics which are established for the management of sexually transmitted diseases (STDs) where patients are given color-coded kits.

Solution to Question 24:

Kit 4 is blue in colour and is used for genital ulcers. These kits are given at Suraksha clinics.

National Health Programmes III - NIS, JSY, RBSK and Others

Question 1:

A 9-month-old infant who has not been previously vaccinated is brought to the OPD. The mother is however adamant and is allowing you to inject the baby with only one vaccine. Which of the following vaccines would you choose to administer in this situation?

- a) DPT
- b) Measles
- c) IPV
- d) Pentavalent

Question 2:

An intern posted in the labour room is administering birth dose vaccines to a newborn baby. According to the National Immunization Schedule, which of the following vaccines would he not administer?

- a) 1, 3, and 4
- b) 1 and 5 only
- c) 2, 4, and 5
- d) 2 and 5 only

Question 3:

A farmer residing in West Bengal brings his 16-month-old child to the OPD for routine immunization. The child has been adequately immunized till now and has already received a measles vaccine during a supplementary immunization activity drive a few months back. What will you administer to him now?

- a) OPV booster, DPT booster and Vitamin A
- b) OPV Booster, DPT booster, MR, JE and Vitamin A
- c) OPV booster, DPT booster, JE vaccine and Vitamin A
- d) OPV Booster, DPT booster, MR and Vitamin A

Question 4:

A three and a half-year-old child, who is completely unimmunized, is brought to a PHC. Which of the following vaccines can be given to the child now?

- a) DPT-1, OPV-1 and Vit A
- b) DT-1, OPV-1, Measles, Vit A
- c) DPT-1, OPV-1, Measles, Vit A
- d) BCG, DPT-1, OPV-1, Measles, Vit A

Question 5:

A 5-year-old girl is brought to your clinic for routine immunization. According to the National Immunisation Schedule, which of the following would you administer?

- a) DT booster + vitamin A
- b) DPT booster only
- c) DPT booster + OPV
- d) DPT booster + vitamin A

Question 6:

A 9-year-old child is brought to the PHC for receiving the JE vaccine. The nurse cleans the injection site with a spirit swab and injects the live attenuated JE vaccine subcutaneously on the left upper arm. Which of the following statements is true regarding this scenario?

- a) The procedure of administering the vaccine is wrong
- b) The site of administering the JE vaccine is inappropriate
- c) The JE vaccine cannot be administered under routine immunization in a PHC after 2 years of age
- d) Only live attenuated JE vaccine is currently under use

Question 7:

Which of the following was launched to ensure complete immunization of children?

- a) Mission Dhanush
- b) Mission Indradhanush

- c) Mission Indira Yojana
- d) Mission Suraksha Yojana

Question 8:

You are visiting a rural area for a vaccination drive conducted under the Intensified Mission Indradhanush scheme. Which of the following vaccines would you not administer in this vaccination drive?

- a) H. influenzae B vaccine
- b) Rubella vaccine
- c) Pneumococcal conjugate vaccine
- d) Typhoid vaccine

Question 9:

Which of the following statements is correct regarding the National Deworming Programme?

- a) It is conducted in the months of March and September
- b) Free tablets are given to children up to the age of 15 years
- c) It is carried out in schools and anganwadis
- d) Age appropriate dose of tablet metronidazole is recommended

Question 10:

Which of the following is true about the National Programme for Prevention and Control of Cancer, Diabetes, Cardiovascular Diseases and Stroke?

- a) No separate centres for stroke, DM and cancer
- b) Implementation in over 10 districts in 5 states
- c) District hospital has specialized facilities
- d) Subcentre has facility for diagnosis and treatment

Question 11:

Which of the following is a false statement about the Indian Hypertension Control Initiative (IHCI)?

- a) The IHCI monitoring personnel includes cardiovascular health officers and senior treatment supervisors
- b) Provision of real-time blood pressure monitoring for all patients via information systems
- c) The IHCI provides 30-day medication for individuals registered for hypertension management
- d) The target is a 90% reduction in the prevalence of hypertension by the year 2025

Question 12:

Under the National Programme for Control of Blindness & Visual Impairment, NGO eye hospitals are considered as:

- a) Primary level centres
- b) Secondary level centres
- c) Tertiary level centres
- d) Intermediate level centres

Question 13:

Which of the following is used to measure the output of the National Programme for Control of Blindness & Visual Impairment?

- a) Number of cases of refractory errors corrected
- b) Number of cataract surgeries leading to sight restoration
- c) Number of cases of childhood blindness detected and cured
- d) Prevalence of blindness reduced by 0.5% per year

Question 14:

Which of the following diseases is not a part of the Vision 2020 programme in India?

- a) Refractive errors
- b) Diabetic retinopathy
- c) Corneal blindness
- d) Onchocerciasis

Question 15:

A pregnant woman from a rural area registers under Janani Suraksha Yojana. Which of the following services would not be provided to her under this scheme?

- a) Safe motherhood services under Vandematharam scheme
- b) Cash disbursement for institutional delivery
- c) STD counseling
- d) Safe abortion services

Question 16:

Where and how often is the Village Health Nutrition Day conducted?

- a) Once a month in PHC
- b) Once a month in Anganwadi
- c) Once every two weeks in PHC
- d) Once every two weeks in Anganwadi

Question 17:

A 20-day old baby is gasping for air and not feeding properly. The mother has no money to go to the nearby hospital. Which national programme has provisions to benefit this woman to take her baby to the health facility?

- a) Janani Suraksha Yojana
- b) Rashtriya Bal Swasthya Karyakram
- c) Janani Shishu Suraksha Karyakram
- d) Navjaat Shishu Suraksha Karyakram

Question 18:

The Navjat Shishu Suraksha Karyakram was designed for which of the following demographic groups?

- a) Lactating mothers
- b) Children
- c) Health personnel
- d) Pregnant women

Question 19:

A 10-year-old girl has just started menstruating. Under which of the following national programmes, she will be provided with a pack of sanitary napkins by the ASHA worker in her village?

- a) Janani Suraksha Yojana
- b) Rashtriya Kishor Swasthya Karyakram
- c) Sukanya Samriddhi Yojana
- d) Kishori Shakti Yojana

Question 20:

Which among the following is not true about POSHAN Abhiyan?

- a) It is a programme of the Ministry of Women and Child Development
- b) The goals are to achieve improvement in neonatal mortality
- c) Launched on the date of International Women's Day 2018
- d) Targets to reduce low birth weight by 2% per annum

Question 21:

Which of the following is the objective of the Ujjawala Scheme?

- a) Provide LPG connections to families below poverty line
- b) Eliminate violence against children
- c) Prevent trafficking
- d) Provide LED bulbs in every household

Question 22:

An ASHA worker is offering services under the Swachh Bharat Abhiyan in her village. All of the following services will be provided by her except:

- a) Water pollution control
- b) Solid waste management
- c) Household toilets and public toilets
- d) Information, education, communication and public awareness

Question 23:

Which of the following statements is false about the Ayushman Bharat-National Health Protection Mission?

- a) Government provides health insurance cover of up to Rs. 5,00,000 per family per year
- b) The limit of the family size that is covered is up to 7 members
- c) All pre-existing diseases of the individual are covered
- d) It covers secondary and tertiary care hospitalization

Question 24:

Which of the following statements is true regarding Rajiv Gandhi Shramik Kalyan Yojana?

- a) Unemployment allowance is not given in case of closure of factory
- b) Unemployment allowance is given in case of employment injury
- c) Allowance is given at 50% of the daily wage for a maximum of 2 years
- d) To qualify for the scheme, worker must have worked for more than 2 years

Question 25:

Which of the following does the Laqshya initiative launched recently by the Government of India refer to?

- a) Initiative for providing free sanitary pads
- b) Initiative to provide vaccination to under-5 children
- c) Initiative for labour room quality improvement
- d) Initiative to provide affordable healthcare to all

Question 26:

The image below shows the logo of which of the following national programmes?



- a) Beti bachao, Beti padhao programme
- b) India New Born Action Plan
- c) Janani Shishu Suraksha Karyakram
- d) Family Planning Campaign

Question 27:

What is the main objective of the Pradhan Mantri Bhartiya Jan-aushadhi Pariyojana?

- a) Improve maternal and child health services
- b) Make quality, generic medicines affordable to all
- c) Enhance quality of life and increase employment for the differently abled
- d) Promote cleanliness and enhance quality of healthcare facilities in India

Question 28:

The Mid Day Meal Programme aims to supplement:

- a) $\frac{1}{2}$ protein and $\frac{1}{3}$ calorie requirements
- b) $\frac{1}{3}$ protein and $\frac{1}{3}$ calorie requirements
- c) $\frac{1}{2}$ protein and $\frac{1}{2}$ calorie requirements
- d) $\frac{1}{3}$ protein and $\frac{1}{2}$ calorie requirements

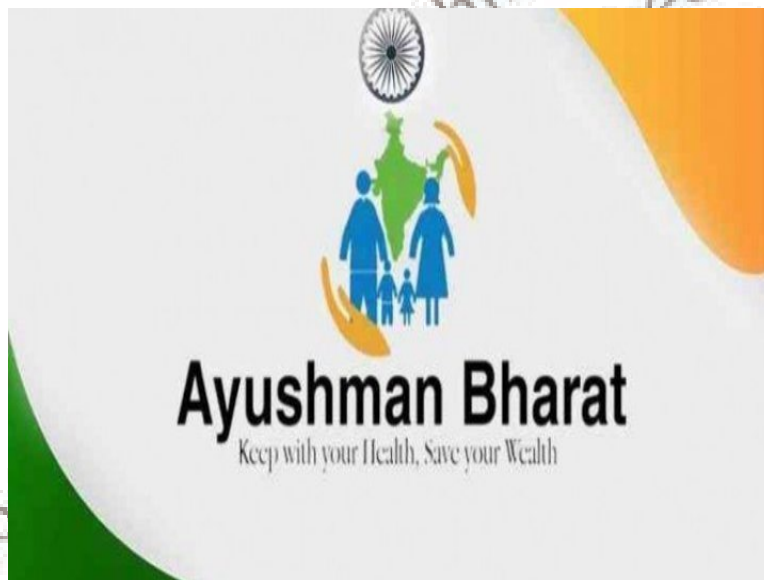
Question 29:

Which of the following statements is not true regarding the Vandematharam scheme?

- a) Free treatment in private clinics
- b) Its aim is to provide safe motherhood services
- c) An MBBS graduate or an obstetrician can volunteer
- d) IFA tablets and TT injections are provided by District Medical Officers

Question 30:

Which of the following is a component of the scheme depicted by the logo given below?



- a) National Health Protection Mission
- b) Pradhan Mantri Surakshit Matritva Abhiyaan
- c) National Programme for Family Planning
- d) National Rural Health Mission

Question 31:

Which of the following vaccines has not been introduced to the immunization programme in India?

- a) Adult Japanese encephalitis
- b) Influenza

- c) Measles-Rubella
- d) Rotavirus

Question 32:

The POCSO Act is related to which of the following?

- a) Women empowerment
- b) Human trafficking
- c) Juvenile delinquency
- d) Child sexual abuse

Question 33:

Mass drug administration is not done for which of the following conditions?

- a) Worm infestation
- b) Vitamin A deficiency
- c) Scabies
- d) Filariasis

Question 34:

Monetary benefit is measured in which of the following analyses?

- a) Cost effective analysis
- b) Cost benefit analysis
- c) Network analysis
- d) Input Output analysis

Question 35:

As per NCEP- ATP III, which of the following is not a criterion for metabolic syndrome?

- a) Hypertriglyceridemia
- b) High LDL
- c) Central obesity

d) Hypertension

Question 36:

In Sustainable Development Goal 3, the target 3.1 aims to reduce the Maternal Mortality Ratio by 2030 to ____ (per 1 lakh live births):

- a) <70
- b) <100
- c) <50
- d) <130

Question 37:

As per the new National Policy for Rare Diseases, financial support of up to Rs. 20 lakhs can be given to which group of rare diseases?

- a) Group 1
- b) Group 2
- c) Group 3
- d) Group 4

Question 38:

You are a medical officer at a PHC. Which of the following would you advise for mass prophylaxis/ treatment for anemia?

- a) FeSO_4 + Vit C + Folate + Vit B12
- b) Iron (III) hydroxide polymaltose
- c) Ferrous fumarate
- d) Inj. iron sucrose

Answer Key

Question No.	Correct Option
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1	b
2	d
3	b
4	c
5	d
6	a
7	b
8	d
9	c
10	c
11	d
12	b
13	b
14	d
15	c
16	b
17	c
18	c
19	b
20	b
21	c
22	a
23	b
24	c
25	c
26	d
27	b
28	a
29	a
30	a
31	b
32	d
33	b
34	b
35	b
36	a
37	a

38	a
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Detailed Explanations

Solution to Question 1:

In this situation, measles vaccine should be administered subcutaneously because, at 9 months of age, it is a priority to give the measles vaccine. OPV and vitamin A should be given orally.

Solution to Question 2:

Inactivated Polio Vaccine (IPV) and pentavalent vaccine are not given at birth. The vaccines given at birth are BCG, OPV (0 dose) and hepatitis B vaccines.

Solution to Question 3:

Vaccines to be given at 16 months of age are OPV booster, DPT booster, MR, JE, and Vitamin A.

The child should receive routine doses of the measles vaccine according to the schedule irrespective of the measles supplementary immunisation activity (SIA) dose. So, the second dose of the measles vaccine is also given.

Japanese Encephalitis (JE) is endemic in rural agricultural areas, often associated with rice cultivation and flood irrigation. Vaccination for JE is given in endemic states only. Two doses of a live attenuated JE vaccine are given under the Universal Immunisation Programme (UIP) schedule. Under the Indian Academy of Pediatrics (IAP) schedule, two doses of a new vero cell-derived inactivated JE vaccine are given.

Solution to Question 4:

DPT-1, OPV-1, measles and vitamin A can be given to this child of 3.5 years of age in his first immunization visit.

BCG can only be given up to 1 year of age, according to the National Immunisation Schedule (NIS).

Note: BCG can be given up to 5 years of age, as part of the catch-up immunization schedule which is by the Indian Academy of Paediatrics (IAP) and is not followed in government setups.

OPV can be given till 5 years of age.

Measles can be given till 5 years of age.

DPT can be given till 7 years of age (with booster doses).

Solution to Question 5:

At 5 years of age, the 2nd booster dose of DPT must be given along with the 9th (final) dose of vitamin A.

1st booster dose of DPT is given at 16 – 24 months of age.

1st dose of vitamin A is given at 9th month and the 2nd dose is given at 16 months followed by one dose every 6 months thereafter until the age of 5.

Note: DPT booster + Vitamin A is a better answer than DPT booster + OPV because the 9th (final) dose of vitamin A must be given at 5 years but OPV at 5 years is an optional vaccine.

Solution to Question 6:

The procedure of administration is wrong as it is not advisable to clean the injection site with a spirit swab because some of the live components of the vaccine are killed if they come in contact with spirit.

The child is eligible to receive the Japanese Encephalitis (JE) vaccine through routine immunization till 15 years of age. Live attenuated JE vaccine is administered in the left upper arm subcutaneously.

JENVAC is a new inactivated vero cell culture derived JE vaccine produced by Bharath Biotech International limited in India. It is a Kolar strain-derived vaccine. The recommended dose is 0.5ml which is given intramuscularly at left anterolateral aspect of mid thigh, 4 weeks apart starting from 1 year of age. Although a booster dose is recommended at a later period, the appropriate time for booster dose administration is not clear.

JEEV is another inactivated vero cell culture derived vaccine from the SA-14-14-2 strain. The recommended dose is:

- 1-3 years: 2 doses of 0.25 ml on 0 and 28 days
- >3 years, adolescents, and adults: 2 doses of 0.5ml on 0 and 28 days
- A booster dose of this vaccine is suggested to be given after 12 months of primary vaccination in non-endemic areas.

The inactivated vaccines are currently under the recommendation of IAP. Under the Universal Immunization Programme (UIP), a live attenuated vaccine is given.

Solution to Question 7:

Mission Indradhanush was launched by the Ministry of Health and Family Welfare in 2014 to fully immunize children (>90%) who are either partially immunized or completely unimmunized by 2020. It was later modified in 2017 to the Intensified Mission Indradhanush, to achieve the same goal by December 2018.

Intensified Mission Indradhanush includes 12 vaccine-preventable diseases (updated from the 7 diseases that Mission Indradhanush focussed on). 4 rounds of immunization are proposed to be held for 7 days each month (from the 7th- 13th of each month).

The diseases included in Mission Indradhanush are:

- TB
- Polio
- Hep B
- Diphtheria
- Pertussis
- Tetanus
- Measles

The diseases included recently in Intensified Mission Indradhanush are:

- Rotavirus diarrhoea
- Rubella (as MR vaccine)
- H. influenzae B diseases (as part of the pentavalent vaccine)
- Pneumococcal meningitis
- Japanese Encephalitis

The Intensified Mission Indradhanush 2.0:

The government of India introduced IMI 2.0 to accelerate the vaccine drive and extend the coverage to children and pregnant women in identified districts and blocks from December 2019-March 2020.

Note: Currently, the government has completed Intensified Mission Indradhanush 5.0:

Intensified Mission Indradhanush (IMI 5.0), the flagship routine immunization campaign of the Union Ministry of Health and Family Welfare concluded all 3 rounds in October 2023. This is the first campaign to be conducted across all the districts in the country and includes children up to 5 years of age (Previous campaigns included children up to 2 years of age). It also had a special focus on improvement of Measles and Rubella vaccination coverage.



Solution to Question 8:

Typhoid vaccine is not a part of the Intensified Mission Indradhanush scheme.

The Intensified Mission Indradhanush scheme has been introduced to ensure full immunization (>90%) by December 2018. It now includes 12 vaccine-preventable diseases (VPDs) (updated from the 7 that Mission Indradhanush focussed on). 4 rounds of immunization are proposed to be held for 7 days each month (from the 7th-13th of each month).

The diseases included in Mission Indradhanush are:

- TB
- Polio
- Hep B
- Diphtheria
- Pertussis
- Tetanus
- Measles

The diseases included recently in Intensified Mission Indradhanush are:

- Rotavirus diarrhoea
- Rubella (as MR vaccine)
- H. influenzae B diseases (as part of the pentavalent vaccine)
- Pneumococcal meningitis
- Japanese Encephalitis

The purpose of Intensified Mission Indradhanush (IMI) is to reach each and every child under 2 years of age and all those pregnant women who have been left uncovered under the routine

immunization coverage.



Solution to Question 9:

The National Deworming programme is carried out in schools and anganwadis.

It is conducted twice a year, in the months of February and August, covering all the children from the age of 1 to 19 years. All the children are given age-appropriate dose of tablet albendazole.

In the states of Rajasthan and Madhya Pradesh, it is carried out only once a year (as the prevalence is $\leq 20\%$ in these states).

Solution to Question 10:

District hospital has specialized facilities for the National Programme for Prevention and Control of Cancer, Diabetes, Cardiovascular Diseases and Stroke (NPCDCS).

The NPCDCS has separate components for cancer and other non-communicable diseases like diabetes, cardiovascular diseases, and stroke. It has been implemented in over 100 districts in 21 states.

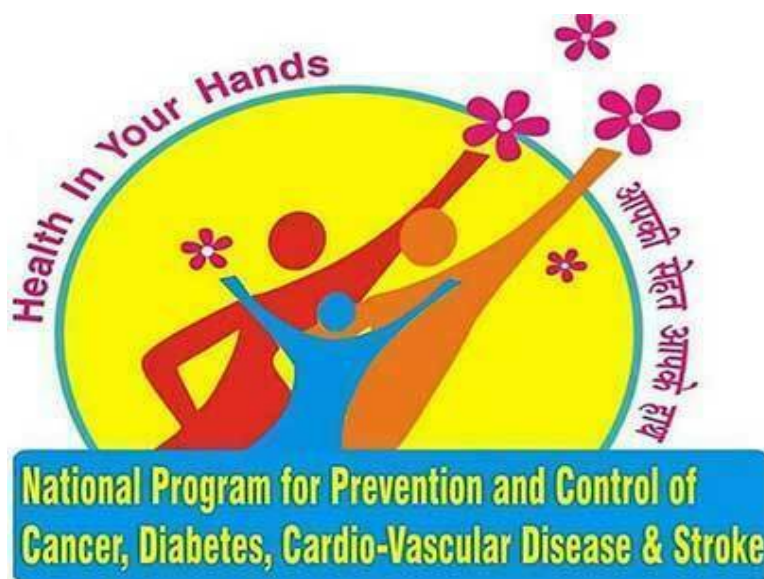
At the subcentre level under NPCDCS, screening and referral are done.

Note: The National Programme for Prevention and Control of Cancer, Diabetes, Cardiovascular Diseases and Stroke (NPCDCS) has been changed to the National Programme for Prevention and Control of Non-Communicable Diseases (NP-NCD) to include new diseases and new health initiatives.

The new diseases included in the NP-NCD are:

- Non-alcoholic fatty liver disease
- Chronic Kidney disease

- Asthma and Chronic obstructive pulmonary disease
- ST elevation myocardial infarction



Solution to Question 11:

The false statement about the Indian Hypertension Control Initiative is a 90% reduction in the prevalence of hypertension by the year 2025.

The IHCI is an initiative by the MoHFW, WHO, ICMR, and state governments to control hypertension in India. It has adopted the goal of “25 by 25” to reduce the prevalence of hypertension in India by 25% by 2025. The activities of IHCI are monitored by the cardiovascular health officers (CVHO's) at the state and district level and by the senior treatment supervisors (STS's) at the sub-district level. The IHCI has strengthened the prevention of hypertension by ensuring the provision of antihypertensive medications for 30 days to all patients registered under the IHCI.

The 5 components of IHCI are:

- Use of standardized hypertension treatment protocols approved by national and global policies.
- Availability of recommended protocol drugs to provide a regular and uninterrupted supply of medications.
- Task-sharing and team-based care involving nurses, health workers, and ASHAs to improve treatment compliance.
- Patient-centered services provide easy access to BP monitoring at Health & wellness centers and PHCs.
- Information systems that provide real-time BP monitoring to improve patient follow-ups: Use of the “Simple App” in all IHCI states.

*Note: Amlodipine, telmisartan, and chlorthalidone are the drugs of choice in the standardized treatment protocols of the IHCI.

Solution to Question 12:

NGO eye hospitals are considered secondary level centres under the National Programme for Control of Blindness and Visual Impairment (NPCBVI).

Levels of eye care centres according to NPCBVI:

- Primary level: Sub-district level hospitals, community health centres, mobile ophthalmic units, upgraded PHCs, and link workers.
- Secondary level: District hospitals and NGO eye hospitals.
- Tertiary level: Regional institutes of ophthalmology and medical colleges.

The goal of NPCBVI is to reduce the prevalence of blindness to 0.25% by 2025.



Solution to Question 13:

The measure of the output of the National Programme for Control of Blindness & Visual Impairment (NPCBVI) is indicated by the number of cataract surgeries leading to sight restoration.

The most common cause of low vision in India is cataract (low vision is visual acuity $\leq 6/18$ - $6/60$).

Solution to Question 14:

Onchocerciasis is not a part of the Vision 2020 programme in India. It is a part of the Global Vision 2020 programme.

Global Vision 2020	India Vision 2020
CataractRefractive errors and low visionChildhood blindnessTrachoma	CataractRefractive errors and low visionChildhood blindness
Onchocerciasis	Glaucoma
	Diabetic Retinopathy
	Corneal blindness

Solution to Question 15:

The Janani Suraksha Yojana (JSY) does not provide STD counseling.

Janani Suraksha Yojana is a safe motherhood intervention launched under National Rural Health Mission (NRHM) to reduce maternal and neonatal mortality by promoting institutional delivery among poor pregnant women. The services provided by the JSY are:

- Cash is disbursed to the mother at the institution for having an institutional delivery.
- Safe motherhood services (IFA tablets, TT vaccination) under the Vandemataram scheme.
- Village health and nutrition day
- Maternal death review
- Pregnancy tracking: ASHA worker/Anganwadi Worker (AWW) or any other link worker help in monitoring antenatal check-ups, and post-delivery care.
- Safe abortion services: mifepristone followed by misoprostol and manual vacuum aspiration.

Solution to Question 16:

Village Health Nutrition Day is conducted once a month in the Anganwadi centres in the village itself. The villagers interact with health workers and are provided with basic services and information about health care.

Solution to Question 17:

The Janani Shishu Suraksha Karyakram (JSSK) has provisions for free transport of pregnant women and sick infants from:

- home to the health facilities
- between facilities in case of a referral
- back home from the health facilities

The provisions of JSSK include:

- Free transport
- Free treatment
- Free drugs and consumables
- Free diagnostics and diet
- Free and cashless delivery
- Free C-Section
- Free provision of blood

Note: Previously under JSSK, the free entitlements were provided only for sick newborns till 30 days after birth. Now it is expanded to cover infants also.

Solution to Question 18:

The Navjat Shishu Suraksha Karyakram (NSSK) was designed to train health personnel in basic newborn care and resuscitation.

The training is aimed to improve the delivery of excellent newborn care at every point and is usually done for 2 days.

Solution to Question 19:

The girl will be provided with a pack of 6 sanitary napkins for 6 Rupees (called Freedays) by the ASHA worker through the Menstrual Hygiene Scheme under the Rashtriya Kishor Swasthya Karyakram (RKSK).

RKSK focuses on:

- Development of adolescent population (10-19 years).
- Weekly Iron and Folic Acid supplementation (100mg Fe + 500 mcg Folic Acid)
- Menstrual hygiene: A pack of 6 sanitary napkins for 6 Rupees (called Freedays) by the ASHA.
- It also includes enhancement of mental health and prevention of substance abuse and violence.

Sukanya Samriddhi Yojana is a government-sponsored savings program for parents of girl children <10 years age. It helps parents to set aside money for their female child's potential college and marriage expenses by opening an account at any post office branch or any commercial bank approved by the central government.

Kishori Shakti Yojana (KSY) seeks to empower adolescent girls, so as to enable them to take charge of their lives. The objectives are to:

- improve the nutritional, health and development status of adolescent girls
- promote awareness of health, hygiene, nutrition and family care
- opportunities for learning life skills

- going back to school
- help them gain a better understanding of their social environment
- take initiatives to become productive members of society.

Recent update: Department of Pharmaceuticals under Ministry of Chemicals and Fertilizers has recently launched 'Suvidha' (100% Oxo-biodegradable Sanitary Napkin), under the Pradhan Mantri Bhartiya Janaushadhi Pariyojana (PMBJP) on World menstrual hygiene day. 'Swachhta, Swasthya and Suvidha' will be available for Rs. 1 per pad over 3200 Janaushadhi Kendras for the underprivileged women of India.



Solution to Question 20:

The goal of POSHAN Abhiyaan /National Nutrition Mission (NNM) is to achieve an improvement in the nutritional status during the next three years beginning 2017-18. It is a flagship programme of the Ministry of Women and Child Development (MWCD). Its target population include:

- children from 0-6 years
- adolescent girls
- pregnant women
- lactating mothers

The logo is:

POSHAN Abhiyaan

PM's Overarching
Scheme for Holistic
Nourishment



सही पोषण - देश रोशन

The following image depicts the targets of this mission:

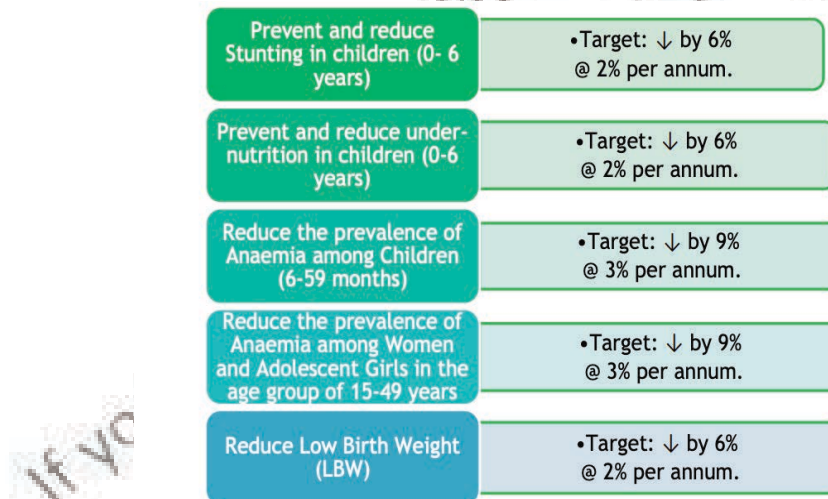


Figure 1: Targets of the POSHAN Abhiyaan

Note: Nikshay Poshan Yojana is a government scheme in India to provide Rs.500 per month for TB patients to buy food.

Solution to Question 21:

The Ujjawala Scheme is for the prevention of trafficking and for the rescue, rehabilitation, and reintegration of victims of trafficking for commercial sexual exploitation.

The Ujjwala Yojana (note spelling without "a") is for providing 5 crores of free cooking LPG connections to all families below the poverty line till 2019.

The Ujala Scheme (Unnat Jyoti by Affordable LEDs and Appliances for All) is the largest domestic lighting project in the world, launched under the Ministry of Power. It aims at promoting the use

of energy-efficient LED bulbs in every household in the country.

INSPIRE strategy is a set of 7 evidence-based strategies introduced to select, implement, monitor, and evaluate effective policies, programs, and services to prevent and respond to violence against children.

Solution to Question 22:

Water pollution is not managed under the Swachh Bharat Abhiyan.

The Swachh Bharat Abhiyan was launched in 2014 with an aim to eradicate open-air defecation by the year 2019 through the construction of 12 million toilets. The two sub-missions of Swachh Bharat Abhiyan are Swachh Bharat Mission Gramin (SBM-G) and Swachh Bharat Mission Urban (SBM-U).

- Components of this mission include:
- Household toilets including improving unsanitary latrines
- Community toilets
- Public toilets
- Solid waste management
- Information, education, communication, and public awareness
- Capacity building and administrative & office expenses

Solution to Question 23:

There is no cap on family size or the age of the members. All the families listed under the Socio-Economic Caste Census (SECC) database are covered as per the defined criteria.

The following are the benefits provided under this program:

- Health insurance cover of up to Rs. 5,00,000 per family per year.
- All families listed in the SECC database, as per the defined criteria, will be covered. There is no cap on family size and age of members.
- Free treatment is available at all public and empanelled private hospitals
- Both secondary and tertiary care hospitalization is covered.
- All pre-existing diseases are covered. Hospitals cannot deny treatment.
- To control costs, the payments for treatment will be done on a package rate (to be defined by the Government in advance) basis.
- Cashless and paperless access to quality health care services.

Logo:



Solution to Question 24:

Unemployment allowance under Rajiv Gandhi Shramik Kalyan Yojana (RGSKY) is given at the rate of 50% of daily wage for a maximum of 2 years.

RGSKY is a scheme of ESI that provides unemployment allowance to an insured person going out of employment involuntarily, on account of:

- closure of a factory or establishment
- retrenchment (laying off employees due to economic difficulty)
- permanent disability due to nonemployment injury

Allowance and eligibility criteria of Rajiv Gandhi Shramik Kalyan Yojana (updated in 2019):

- The allowance is given at the rate of 50% of the daily wage for a maximum of 2 years.

- To qualify for the scheme the worker must have worked in that setup for >3 years.

Note: Disability due to employment injury is covered under disablement benefits of ESI Act, 1948.

- Temporary disability: 90% of wages until disablement lasts.
- Permanent disability: 90% of wages as a life pension on the basis of loss of earning capacity.
- Dependants benefit: 90% of wage as monthly payment to the dependants of the deceased person, in cases where death occurs due to employment injury or occupational hazards.

Solution to Question 25:

The Laqshya initiative launched recently by the Government of India refers to the initiative for labour room quality improvement.

There is enough evidence to support the fact that improving the quality of care in labour rooms, especially on the day of birth, is crucial to maternal and neonatal survival.

Laqshya is expected to improve the quality of care being provided to the pregnant mother in the labour room and maternity operation theatres, thereby preventing the undesirable adverse outcomes associated with childbirth. This initiative is implemented in government medical colleges besides district hospitals, and high delivery load sub-district hospitals, and community health centres.

It plans to conduct quality certification of labour rooms and also incentivize facilities achieving the targets outlined. The goal is to reduce preventable maternal and new-born mortality, morbidity, and stillbirths associated with the care around delivery in labour room and maternity OT, and ensure respectful maternity care.

Solution to Question 26:

The image shown is the new logo of the 360-degree holistic Family Planning Campaign.

Logo of other options:



"Beti bachao Beti padhao."



India Newborn Action Plan (INAP)



Janani Shishu Suraksha Karyakaram

The logo of Janani Shishu Suraksha Karyakaram (JSSK)

Solution to Question 27:

The Pradhan Mantri Bhartiya Jan-aushadhi Pariyojana (PMBJP) carries the objective of making quality, generic medicines affordable for everyone, especially the lower socio-economic groups.

BPPI (Bureau of Pharma PSUs of India), under the Ministry of Chemicals & Fertilizers, is the implementation agency for PMBJP. The mission of this programme includes creating awareness and demand for generic medicines through education and supply of such medications.

Kayakalp programme is a national initiative launched by the Ministry of Health and Family Welfare (MoHFW) under Swachh Bharat Abhiyan to promote cleanliness and enhance the quality of healthcare facilities in India.



Solution to Question 28:

Under the Mid-Day Meal (MDM) scheme, supplementary nutrition provided must cover at least $\frac{1}{2}$ of total daily protein requirement and $\frac{1}{3}$ rd of total daily energy requirement of children in schools.

The MDM scheme is also called School Lunch Programme and has been operational since 1961 in India. Currently, it functions under the broad coverage of the Minimum Needs Programme.

Some other principles of the MDM are:

- The meal provided under the MDM should only be a supplement and not a substitute for home diets.
- The cost of meals should be reasonably low.
- The meal should be easy to cook in schools.
- Locally available foods must be used, for as far as possible.
- The menu should be regularly changed to avoid monotony.

Note: In September 2021, the Mid-Day Meal Scheme was renamed PM POSHAN or Pradhan Mantri Poshan Shakti Nirman.

Solution to Question 29:

The Vandematharam scheme does not include free treatment in private clinics.

Vandematharam scheme is a voluntary scheme for providing safe motherhood services. It can be volunteered by an obstetrician, maternity home, nursing home, lady doctor or an MBBS graduate.

IFA tablets, OCPs, and TT injections are provided by the respective District Medical Officers (DMOs) to the Vandematharam doctors or clinics for free distribution to beneficiaries. Cases needing special care and treatment can be referred to government hospitals.

Solution to Question 30:

National Health Protection Mission is a component of the Ayushman Bharat scheme.

Ayushman Bharat-National Health Protection Mission (AB-NHPM) aims to build a comprehensive need-based health care service and targets to build a 'New India 2022'. Its components are:

- Creation of 1,50,000 Health and Wellness Centres- these centres will provide Comprehensive Primary Health Care (CPHC), covering both maternal and child health services and non-communicable diseases, including free essential drugs and diagnostic services.
- National Health Protection Mission (NHPM) provides health protection cover to poor and vulnerable families, screening for non-communicable diseases, and follow-up of hospitalization cases among others.

Beneficiary Level:

- The government provides health insurance cover of up to Rs. 5,00,000 per family per year.
- All families listed in the Socio-Economic Caste Census (SECC) database as per defined criteria will be covered. No cap on family size and age of members.
- Priority to a girl child, women, and senior citizens.
- Free treatment is available at all public and empanelled private hospitals in times of need.
- Covers secondary and tertiary care hospitalization.
- 1,350 medical packages covering surgery, medical and daycare treatments, cost of medicines, and diagnostics.
- All pre-existing diseases are covered. Hospitals cannot deny treatment.
- Cashless and paperless access to quality health care services.
- Hospitals will not be allowed to charge any additional money from beneficiaries for the treatment.
- Can reach out for information, assistance, complaints, and grievances to a 24X7 helpline number.





Solution to Question 31:

Influenza vaccine (Flu shot) has not been introduced to the National Immunization Programme in India.

The Influenza vaccine provides protection against viral flu caused by the influenza virus.

Note: Bacterial Hemophilus influenzae type-b (Hib) vaccine is different from the viral influenza (Flu) vaccine. Hib is a part of the pentavalent vaccine.

Solution to Question 32:

The POCSO (Protection of Children from Sexual Offences) Act is related to child sexual abuse.

POCSO Act defines a child as any person below eighteen years of age. It defines different forms of sexual abuse, including penetrative and non-penetrative assault, as well as sexual harassment and pornography.

The Act deems a sexual assault to be "aggravated" under certain circumstances, such as when the abused child is mentally ill or when the abuse is committed by a person in a position of trust or authority vis-à-vis the child, like a family member, police officer, teacher, or doctor.

People who traffic children for sexual purposes are also punishable under the provisions relating to abetment in the Act. The Act prescribes stringent punishment graded as per the gravity of the offence, with a maximum term of rigorous imprisonment for life and a fine.

Solution to Question 33:

Mass drug administration is not done in case of vitamin A deficiency.

Mass Drug Administration (MDA) is the administration of a drug to an entire concerned population irrespective of age and gender. For example, antimalarials are given as MDA in malaria-endemic island communities and in low-endemic non-island settings approaching elimination, where there is minimal risk of re-introduction of infection, good access to treatment, and implementation of vector control and surveillance.

Vitamin A prophylaxis is given under National Immunization Schedule (NIS) only to children up to 5 years. It is given as mass prophylaxis and not as a drug for treatment. Vitamin A as a treatment is given only to a child with features of vitamin A deficiency and such doses are not instituted to everyone in the community. Therefore, it is not considered under MDA.

On the other hand, MDA is done in scabies (ivermectin), worm infestation (albendazole), and filariasis (DEC + albendazole).

Solution to Question 34:

The monetary benefit is measured in a cost-benefit analysis.

In cost-benefit analysis, the economic benefits of any program are compared with the cost of the program. The benefits are expressed in monetary terms to determine whether a given programme is economically sound and to select the best out of several alternate programmes.

The main drawback with this technique is that the benefits in the health field, as a result of a particular programme, cannot always be expressed in monetary terms. We generally express the benefit in terms of births or deaths prevented, or illness avoided or overcome.

In cost-effective analysis, the benefit is expressed in terms of results achieved. (Eg: number of lives saved or number of days free from disease).

Input-output analysis: how much input is needed to produce a unit amount of output. In the health field, input refers to all health service activities which consume resources (manpower, money, materials, and time); and output refers to useful outcomes such as cases treated, lives saved, or inoculations performed.

Network analysis: a network is a graphic plan of all events and activities to be completed in order to reach an end objective.

Management Methods in Health Programmes:

Methods based on behavioral sciences	Quantitative methods
Organisational design Personnel management Communication Information systems Management by objectives (MBO)	Cost-Benefit analysis Cost-Effective analysis Cost-Accounting Input-Output analysis Model Systems analysis Network analysis- Includes Programme Evaluation and Review Technique (PERT) and Critical Path Method (CPM) Planning-Programming-Budgeting System (PPBS) Work sampling Decision making

Solution to Question 35:

As per NCEP- ATP III, high LDL is not included in the definition criteria of metabolic syndrome.

Metabolic syndrome is present if three or more of the following five criteria are met, which is given by NCEP-ATP III (National Cholesterol Education Program and Adult Treatment Panel III). It includes:

- Central obesity: waist circumference ≥ 102 cm (M), ≥ 88 cm (F)
- Hypertriglyceridemia: triglyceride level ≥ 150 mg/dL
- Low HDL cholesterol: ≤ 40 mg/dL and ≤ 50 mg/dL for men and women, respectively.
- Hypertension: blood pressure ≥ 130 mmHg systolic or ≥ 85 mmHg diastolic.
- Fasting plasma glucose level: ≥ 100 mg/dL or previously diagnosed type 2 diabetes.

Solution to Question 36:

In Sustainable Development Goal 3, the target 3.1 aims to reduce the Maternal Mortality Ratio by 2030 to ≤ 70 per 1 lakh live births.

There are 17 global goals and 169 targets set for 2030 as part of the official Sustainable Development Goals of the Agenda 2030.



Solution to Question 37:

As per the new National Policy for Rare Diseases, a provision for financial support of up to Rs. 20 lakhs under the umbrella scheme of Rashtriya Arogya Nidhi is proposed for the treatment of rare diseases that require a one-time treatment (listed under Group 1 in the rare disease policy).

National Policy for Rare Diseases is a comprehensive policy for the prevention and management of rare diseases. The policy lists rare diseases under 3 groups:

- Group 1: Disorders amenable to one-time curative treatment:
 - Disorders amenable to hematopoietic stem cell transplantation eg: lysosomal storage disorders
 - Disorders amenable to organ transplantation eg: glycogen storage disorders, urea cycle disorders, ADPKD, etc.
- Group 2: Diseases requiring long term/ lifelong treatment having a relatively lower cost of treatment:
 - This group includes disorders managed with special dietary formulae/ hormone/ specific drugs. eg: phenylketonuria, osteogenesis imperfecta, etc.
- Group 3: Diseases for which definitive treatment is available but treatment is lifelong or expensive:
 - This group includes diseases such as Pompe disease, Duchenne muscular dystrophy, etc.

The objectives of the policy are:

- Lowering the high cost of treatment for rare diseases with an increased focus on indigenous research.
- Creating a national hospital-based registry of rare diseases.
- Early screening and prevention through primary and secondary health care infrastructure, and through counselling for the high-risk parents.
- Crowdfunding mechanism in which corporates and individuals will be encouraged to extend financial support.

Solution to Question 38:

Among the given options, FeSO_4 + vitamin C + Folate + vitamin B12 may be considered for mass prophylaxis against anemia.

Vit C enhances the absorption of iron from the stomach by acting as a reducing substance.

Subclinical folate deficiency is seen in up to 30% of North Indian pregnant women. Moreover, it may be masked by widespread iron deficiency. It is, for this reason, iron and folate supplementation is carried out in national programmes.

Clinical vitamin B12 deficiency is not commonly encountered, but subclinical cases might be seen in up to 30% of adults and children due to widespread restriction to consumption of vegetarian products only. Hence supplementing vitamin B12 is a good idea provided the cost factor is not an issue.

Option B: Iron (III) hydroxide polymaltose was marketed as being superior to iron sulfate in terms of efficacy, but it lacked any data to back up the evidence.

Option C: Ferrous fumarate may be considered for the treatment of iron deficiency anemia with similar efficacy as that of ferrous sulfate.

Option D: Injection iron sucrose or any parenteral form is not used as a first-line option. It is only indicated if the oral form is not tolerated.

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If you purchased this from someone else,
you may have been scammed.

Demography I: Demographic Cycle, Annual Growth Rate and Age Pyramid

Question 1:

Which of the following is not a demographic process?

- a) Fertility
- b) Morbidity
- c) Marriage
- d) Social mobility

Question 2:

In which stage of the demographic cycle does the death rate start declining?

- a) Stage 2
- b) Stage 3
- c) Stage 4
- d) Stage 1

Question 3:

In which stage of the demographic cycle does the birth rate start declining?

- a) Stage 2
- b) Stage 3
- c) Stage 4
- d) Stage 5

Question 4:

A group of demographers studied the changes in the birth and death rates of two countries, A and B from the year 2001 to 2011. The collected data is given below. In which stages of the demographic cycle are countries A and B in respectively?

- a) Stage 2, stage 3

- b) Stage 2, stage 4
- c) Stage 2, stage 5
- d) Stage 3, stage 4

Question 5:

Which of the following statements is true regarding the fifth stage of the demographic cycle?

- a) Birth rate remains unchanged
- b) Death rate increases
- c) Demographic gap is positive
- d) Birth rate falls below death rate

Question 6:

While studying the birth and death rates of a country, a demographer found it to be in the early expanding stage of the demographic cycle. Which of the following findings would lead him to such a conclusion?

- a) Increased birth rate and decreased death rates
- b) Increased birth and death rates
- c) Decreased birth rate and increased death rates
- d) Unchanged birth rate and decreased death rates

Question 7:

A researcher is studying the demographic cycle of various countries. In which stages of the cycle will she notice an increase and a decrease in the demographic gap respectively?

- a) Stage 1, stage 4
- b) Stage 2, stage 3
- c) Stage 3, stage 4
- d) Stage 3, stage 5

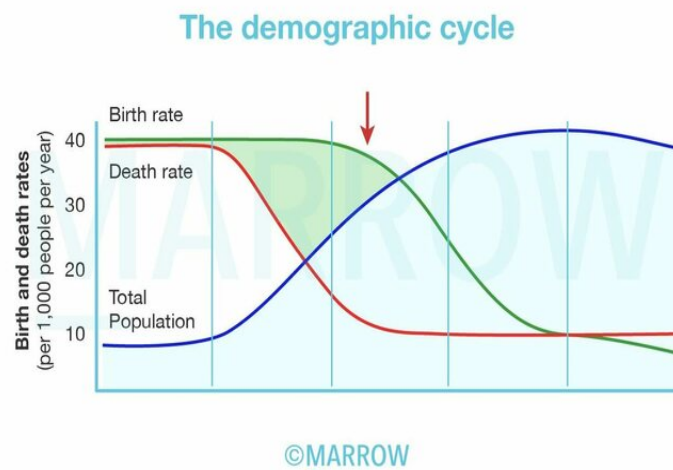
Question 8:

At which stage of the demographic cycle will the demographic gap be maximum?

- a) High stationary
- b) Early expanding
- c) Late expanding
- d) Declining

Question 9:

Which of the following statements is correct regarding the stage of demographic cycle marked by the red arrow?



- a) Known as high stationary stage
- b) Demographic gift is seen
- c) Demographic gap becomes maximum
- d) Birth rate remains unchanged

Question 10:

In which stage of the demographic cycle will the phenomenon of a demographic trap be maximally seen?

- a) Early expanding
- b) Late expanding
- c) Low stationary
- d) Declining

Question 11:

While conducting a field study in a village, you found the birth and death rates of the population to be 42 and 31 respectively. What is the current annual growth rate of this population?

- a) 11%
- b) 1.1%
- c) 0.11%
- d) 1.01%

Question 12:

The population of a country is 1 lakh in the year 2000 and its annual growth rate is 2%. By which year will the population of this country become 2 lakhs?

- a) 2025
- b) 2035
- c) 2045
- d) 2050

Question 13:

What was the population of India, as of 2021?

- a) 1.54 billion
- b) 1.15 billion
- c) 1.39 billion
- d) 1.2 billion

Question 14:

According to the 2021 UN data, what is India's population growth rate?

- a) 2.5
- b) 1
- c) 3.5

d) 2

Question 15:

You have been tasked with studying the age and sex structure of a population with high rates of female infanticide. Which of the following methods will you prefer to use?

- a) Life table analysis
- b) Correlation coefficient
- c) Component bar chart
- d) Population pyramid

Question 16:

What does the age pyramid of India look like?

- a) Broad at base and narrow at apex
- b) Broad at base and apex
- c) Broad at apex and narrow at base
- d) Broad in the middle and narrow at base

Answer Key

Question No.	Correct Option
1	b
2	a
3	b
4	b
5	d
6	d
7	b
8	b
9	b
10	a
11	b

12	b
13	c
14	b
15	d
16	a

Detailed Explanations

Solution to Question 1:

Morbidity is not a demographic process.

Demography refers to the study of the population. Demographic processes include:

- Fertility
- Mortality
- Marriage
- Migration
- Social mobility

These processes continuously bring about changes in the population size, composition, and distribution.

Solution to Question 2:

The death rate starts declining in stage 2 or the early expanding phase of the demographic cycle.

Solution to Question 3:

The birth rate starts declining in stage 3 or the late expanding phase of the demographic cycle.

India is currently in this stage of the demographic cycle.

The year 1921 is called the year of the great divide in India. This is because, since 1921, the absolute number of people getting added to the population during each decade has been on the rise. This is the year in which India moved from stage 1 to stage 2 of the demographic cycle.

Solution to Question 4:

In the given scenario, country A is in stage 2 and country B is in stage 4 of the demographic cycle.

According to the given data, in country A, the high birth rate has remained unchanged with a decline in the death rate. This will result in an exponential increase in the population. This means country A is in the early expanding stage of the demographic cycle.

In country B, the low birth rate and death rates have remained unchanged. Thus, the population will attain a plateau phase and remain stationary. This means country B is in the low stationary stage of the demographic cycle.

Solution to Question 5:

In stage 5 of the demographic cycle, the birth rate falls below the death rate.

In this stage, the death rate attains a low stable rate while the birth rate keeps declining. Thus, the curve of the birth rate will cross the curve of the death rate in a graph of the demographic cycle and falls further below.

The demographic gap is negative and the population is declining. So, this stage is known as the declining stage.

Solution to Question 6:

Early expanding stage (stage 2) of the demographic cycle is characterized by an unchanged birth rate and decreasing death rate.

Solution to Question 7:

The demographic gap starts increasing in stage 2 and starts decreasing in stage 3.

The demographic gap refers to the difference between birth rate and death rate which develops when a country undergoes a demographic transition.

In stage 2 of the demographic cycle, the death rate declines but the birth rate remains unchanged. Hence, the demographic gap increases and reaches a maximum by the end of stage 2. In stage 3, both the death rate and birth rate are declining so the demographic gap starts decreasing.

Solution to Question 8:

The demographic gap starts increasing and becomes maximum in the early expanding phase of the demographic cycle.

Solution to Question 9:

The marked stage is the late expanding phase (stage 3) of the demographic cycle. A demographic gift or demographic dividend is seen in this stage.

This refers to the presence of a higher number of economically productive individuals (young and middle-aged adults) in the population due to a higher birth rate in the previous demographic phase (phase 2). India is currently experiencing the demographic gift.

The birth rate starts declining alongside a death rate that continues to fall in stage 3 causing a decline in the demographic gap. The death rate declines further and the birth rate falls in this stage.

Solution to Question 10:

The phenomenon of a demographic trap is seen maximally in the early expanding stage (stage 2).

This phenomenon refers to a period in which there is high fertility and birth rate and a declining death rate. The population grows exponentially in this stage and this growing population exceeds the national or regional carrying capacity. Countries in this stage are said to be caught in a demographic trap.

Solution to Question 11:

In the given scenario, the current annual growth rate is 1.1%.

Here, the birth rate is 42 i.e 42 live births per 1000 population, and the death rate is 31 i.e 31 deaths per 1000 population.

Current annual growth rate = crude birth rate – crude death rate. So, current annual growth rate = $42 - 31 = 11$ per 1000 population or 1.1%.

Solution to Question 12:

For the given country, its population would become 2 lakhs i.e, double the population from the year 2000, by the year 2035.

As per the Malthusian growth model, when the annual growth rate is 1%, the population doubles in 70 years. Therefore, when the annual growth rate is 2%, the population will double in 35 years.

The Malthusian growth model/ rule of 70:

The growth rate of India is estimated at 1% as per 2020 UN data. Therefore, it falls in the rapidly growing phase of population growth.

Rating	Annual growth rate	Population doubles in
Slow growth	<0.5%	>139 years

Rating	Annual growth rate	Population doubles in
Moderate growth	from 0.5%	~139 years
Rapid growth	from 1%	70 years
Very rapid growth	from 1.5%	approximately 50 years
Explosive growth	from 2%	35 years
Explosive growth	from 3%	20-25 years

Solution to Question 13:

The population of India, as of 2021, was around 1.39 billion. This is as per UN 2021 data.

Solution to Question 14:

According to UN data, India's population growth rate in 2021 was 1%.

As per the last published SRS data of the Indian government, it was 1.2% in 2018.

India's population is currently increasing at a rate of 16 million each year. It currently falls in the rapidly growing phase of population growth.

Solution to Question 15:

The age and sex structure of a population may be best represented using a population pyramid.

It is a histogram that shows the distribution of ages across a population. It is divided down the center between male and female members of the population.

Solution to Question 16:

The age pyramid of India is broad at the base and narrow at the top. This is typical of most developing nations.

Demography II: Demographic indicators

Question 1:

Which of the following is not a vital statistic?

- a) Natural growth rate
- b) Fertility rate
- c) Sex ratio
- d) Life expectancy at birth

Question 2:

While conducting a demographic study, you need to calculate the total dependency ratio of the study population. Which of the following will you include in the numerator while calculating this?

- a) Population under 10 years and above 60 years
- b) Population under 15 years and above 60 years
- c) Population under 10 years and above 65 years
- d) Population under 15 years and above 65 years

Question 3:

While studying a community, the researcher found 30% of the population to be below 15 years of age, and 10% was over 65 years of age. What is the total dependency ratio of this population?

- a) 20.4%
- b) 40%
- c) 66.6%
- d) 50%

Question 4:

What is potential support ratio?

- a) Number of persons aged 15 to 64 per child below 15 years or less
- b) Number of persons aged 15 to 64 per older person aged 65 or more
- c) Number of persons aged 15 to 64 per each older person aged ≥ 65 and younger person aged ≤ 15
- d) Number of persons aged 15 to 64 per each older person aged ≥ 60 and younger person aged ≤ 15

Question 5:

Which of the following is the major contributor to demographic burden?

- a) Increased potential support ratio
- b) Increased young age dependency ratio
- c) Decreased fertility rates
- d) Increased old age dependency ratio

Question 6:

Which of the following determines the carrying capacity of a population?

- a) Population growth rate
- b) Birth rate
- c) Fertility rate
- d) Limiting resource

Question 7:

Which of the following is false about crude birth rate?

- a) Simplest measure of fertility
- b) Unsatisfactory usage
- c) Age independent
- d) Numerator includes still births

Question 8:

As part of the national maternity benefit scheme, you have been tasked with conducting a study in a community of 5000 people. What is the expected number of pregnant women in this community if the crude birth rate is 30 per 1000 people?

- a) 175
- b) 150
- c) 165
- d) 145

Question 9:

According to NFHS-5, what is the sex ratio in India?

- a) 1037 females per 1000 males
- b) 985 females per 1000 males
- c) 940 females per 1000 males
- d) 1020 females per 1000 males

Question 10:

For a population of 10,000, what does a sex ratio of more than 1000 mean?

- a) Males are less than 500
- b) Females are less than 500
- c) Males are less than 5000
- d) Females are less than 5000

Question 11:

What is the definition of the net reproduction rate?

- a) No. of children that would be born to married women assuming fixed age specific fertility and mortality rate
- b) No. of girls that would be born to married women assuming fixed age specific fertility and mortality rate
- c) No. of female children a new born girl will bear during her lifetime assuming fixed age specific fertility and mortality rate
- d) No. of children a new born girl will bear during her lifetime assuming fixed age specific fertility and mortality rate

Question 12:

An epidemiologist would consider the growth of his study population to be inadequate if he found the NRR to be:

- a) <1
- b) $=1$
- c) >1
- d) >2

Question 13:

What should the net reproductive rate of a population be to attain a 2 child norm?

- a) <1
- b) 1
- c) <2
- d) 2

Question 14:

In order to reach the demographic goal of $NRR=1$, the couple protection rate should exceed?

- a) 50%
- b) 40%
- c) 70%
- d) 60%

Question 15:

What is the denominator used in the formula for calculating the general fertility rate?

- a) Total number of women in the population
- b) Mid-year population
- c) Mid-year population of women in the reproductive age group
- d) Total number of married women in the age group 15 – 45

Question 16:

To assess the effects of the national population policy, data is being collected at the PHC level. You have been asked to submit data regarding the approximate magnitude of completed family size in the population under your PHC. Which of the following indicators will you calculate?

- a) General fertility rate
- b) Total fertility rate
- c) Gross reproduction rate
- d) Net reproduction rate

Question 17:

What is the infant mortality rate in India as per the SRS Statistical Report (2020)?

- a) 35.2 per 1000 live births
- b) 28 per 1000 live births
- c) 30 per 1000 live births
- d) 32.5 per 1000 live births

Question 18:

Maternal mortality ratio (MMR) for a country is best given as _____

- a) Maternal deaths per 100 live births
- b) Maternal deaths per 10,000 live births
- c) Maternal deaths per 100,000 live births
- d) Maternal deaths per 1,000 live births

Question 19:

What is the maternal mortality ratio in India according to the Special Bulletin on Maternal Mortality (2018-2020) based on SRS data?

- a) 122
- b) 97

- c) 113
- d) 110

Question 20:

While collecting data for the census, which of the following people would you consider to be literate?

- a) A person who is able to read, write and understand in any language
- b) A person who is not able to read or write but can sign legal documents
- c) A person who is able to speak fluently in his native language but is not able to read or write
- d) A person who is able to read, but may not be able to write

Question 21:

What is the denominator of the formula used to calculate effective literacy rate?

- a) Total mid-year population
- b) Population aged 7 years or more
- c) Population less than 15 years of age
- d) Population aged 18 years or more

Question 22:

You are studying the literacy rate of a community with a total population of 10000. Out of these, there are 2000 children from 0-6 years of age, and 4000 literate individuals above 7 years of age. What is the effective literacy rate of this population?

- a) 20%
- b) 40%
- c) 50%
- d) 30%

Question 23:

Which of the following is false regarding the criteria for an urban Indian city?

- a) Having 5000 or more inhabitants

- b) Having a density of 390 persons per square km
- c) At least half of the adult male population employed in occupations other than agriculture
- d) Having pronounced urban characteristics

Answer Key

Question No.	Correct Option
1	c
2	d
3	c
4	b
5	d
6	d
7	d
8	c
9	d
10	c
11	c
12	a
13	b
14	d
15	c
16	b
17	b
18	c
19	b
20	a
21	b
22	c
23	c

Detailed Explanations

Solution to Question 1:

Sex ratio is not a vital statistic, it is a population statistic.

Demographic indicators have been divided into 2 parts:

- Population statistics include -
- Population size
- Sex ratio
- Density
- Dependency ratio
- Vital statistics include -
- Birth rate
- Death rate
- Natural growth rate
- Life expectancy at birth
- Mortality rate
- Fertility rate

Solution to Question 2:

Population under 15 years and above 65 years are part of the numerator used to calculate the total dependency ratio. This is because this population is considered the dependant population.

Total dependency ratio

$$\text{Total dependency ratio} = \frac{\text{Total population below 15 years and above 65 years}}{\text{Total population in the age group of 15-64 years}} \times 100$$

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The dependant population includes children below 15 years of age and adults above 65 years of age. The total dependency ratio is the ratio of the combined dependent population to those in the age group of 15 - 65 years. It is also called the societal dependency ratio. It can also be divided into the young-age and old-age dependency ratios.

Note: As of mid-year 2020, the dependency ratio of India is 48.7%.

Solution to Question 3:

The total dependency ratio in this population is 66.6%.

This can be calculated as follows.

Total dependency ratio

$$\text{Total dependency ratio} = \frac{\text{Total population below 15 years and above 65 years}}{\text{Total population in the age group of 15-64 years}} \times 100$$

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Here, the total population below 15 years of age is = 30%

Total population above 65 years of age = 10%

So, total population between 15-65 years of age = $100 - (30 + 10) = 60\%$

Hence, dependency ratio in this population = $100 \times (30\% + 10\%) / 60\% = 66.6\%$

Solution to Question 4:

The potential support ratio is the number of persons aged 15 to 64 per one older person aged 65 or more. It is the inverse of the old-age dependency ratio.

Solution to Question 5:

Demographic burden refers to an increase in the total dependency ratio. It is mostly caused by an increased old-age dependency ratio.

The term demographic bonus denotes the decline in dependency ratio due to declining fertility. A decline in dependency ratio especially the young-age dependency ratio has been identified to be a key factor in the underlying rapid economic development.

The potential support ratio is the number of persons aged 15 to 64 per one older person aged 65 or more. This is the inverse of the old-age dependency ratio.

Solution to Question 6:

The carrying capacity is the number of individuals of a population that can be supported for a period of time given the resources of that area. Thus, the carrying capacity is determined by the limiting resources of an area.

Solution to Question 7:

The numerator of crude birth rate only includes live births and does not include stillbirths.

Crude birth rate = No. of live births during the year x 1000/ Estimated mid-year population

Solution to Question 8:

The expected number of pregnant females in the given community would be 165.

Step 1: In the given question, the crude birth rate is 30 per 1000 people i.e 30 children are born from pregnant mothers for every 1000 people. Therefore there are 30 pregnant females per 1000 population.

Step 2: Since the total population is 5000, the total no of pregnant females = $5000 \times 30/1000 = 150$

Step 3: Since some pregnancies may not result in live births (abortions and stillbirths), a correction factor of 10% should be added to avoid the underestimation of the number of pregnancies. Pregnancy wastage (i.e 10% of total live births) = $10 \times 150/100 = 15$.

Step 4: Total pregnant females = total live births + pregnancy wastage = $150 + 15 = 165$.

Solution to Question 9:

According to the NFHS-5 (2019-21), the sex ratio in India is 1020 females per 1000 males. It has increased from the 2011 Census which had found a sex ratio of 940 females per 1000 males.

The sex ratio at birth is 929 females per 1000 males.

Solution to Question 10:

For a population of 10000, if the sex ratio is more than 1000, it means males are less than 5000 i.e., less than half the population.

The sex ratio refers to the number of females per 1000 males. An ideal sex ratio would be 1000, indicating the population comprises an equal number of males and females.

Sex ratio >1000 indicates that there are more females than males in a population. Sex ratio <1000 indicates that there are fewer females compared to males in the population.

Solution to Question 11:

Net reproduction rate (NRR) is defined as the number of daughters a newborn girl will bear during her lifetime assuming fixed age-specific fertility and mortality rate.

India is aiming for an NRR of 1 as it is equivalent to the two-child norm.

Solution to Question 12:

If the NRR is less than 1, then the population growth of a community is said to be inadequate.

Net reproduction rate (NRR) is the number of daughters a newborn girl will bear during her lifetime assuming age-specific fertility and mortality rates.

Solution to Question 13:

The net reproductive rate (NRR) should be 1 to attain the 2 child norm.

Solution to Question 14:

In order to reach the demographic goal of $NRR=1$, the couple protection rate (CPR) should exceed 60%. This coincides with every couple having 2 children.

Solution to Question 15:

The denominator used to calculate the general fertility rate is mid-year population of women in the reproductive age group.

Solution to Question 16:

The total fertility rate is an indicator of the approximate magnitude of completed family size.

Solution to Question 17:

The infant mortality rate (IMR) in India as per the SRS statistical report (2020) is 28 per 1000 live births.

IMR is defined as the number of infant deaths for every 1000 live births. As per the National Family Health Survey - 5 (2019-2021), the IMR is 35.2 per 1000 live births.

IMR is considered the best indicator of the availability, utilization, and effectiveness of health care.

Solution to Question 18:

Maternal mortality ratio is the number of maternal deaths in a given period per 100,000 live births during the same period.

Solution to Question 19:

The maternal mortality ratio (MMR) in India according to the Special Bulletin (2018-2020) based on SRS data is 97.

MMR is defined as the number of maternal deaths per 100,000 live births.

The most common causes of maternal mortality in India are postpartum hemorrhage and infections.

Solution to Question 20:

A person is considered literate if he or she can read, write and understand in any language.

Solution to Question 21:

The denominator used to calculate the effective literacy rate is the population aged 7 years or more. It is calculated as follows.

Effective literacy rate = $\frac{\text{Number of literate persons aged 7 and above} \times 100}{\text{Population aged 7 and above in a given year}}$

This rate is generally used in Indian censuses including those in 1991, 2001, and 2011 for estimating literacy.

Solution to Question 22:

The effective literacy rate of this population is 50%.

It is calculated as follows,

Effective literacy rate = (Number of literate persons aged 7 and above x 100) / population aged 7 and above in a given year.

In the given scenario, people aged 7 years and above = 10000 - 2000 = 8000. Out of these 8000 people, 4000 are literate.

Thus, effective literacy rate = $4000 \times 100 / 8000 = 50\%$.

Solution to Question 23:

In an urban area, at least three-fourths of the adult male population are employed in occupations other than agriculture.

In the Indian context, the urban areas are places:

- having 5,000 or more inhabitants
- having a density of not less than 1,000 persons per square mile or 390 per square kilometer
- having pronounced urban characteristics
- having at least three-fourths of the adult male population employed in pursuits other than agriculture

Family Planning

Question 1:

What is the number of eligible couples per 1000 population in India?

- a) 50 – 70
- b) 100 – 120
- c) 150 – 180
- d) 120-150

Question 2:

Scope of modern concept of family planning services include all of the following except:

- a) Screening for cervical cancer
- b) Providing services for unmarried mothers
- c) Screening for HIV infection
- d) Providing adoption services

Question 3:

A study was conducted in a village with 180 eligible couples, to gather data on the different contraceptive methods used. The data obtained are given below. What is the couple protection rate in the village?

- a) 60%
- b) 33%
- c) 45%
- d) 50%

Question 4:

Which of the following is not a barrier method of contraception?

- a) C - film
- b) Foam tablet

- c) Cervical diaphragm
- d) Lippes loop

Question 5:

A sexually active woman has been using Today sponge as a contraceptive. What does this contain?

- a) Prostaglandin F₂
- b) Norethisterone
- c) 9 – Nonoxynol
- d) Levonorgestrel

Question 6:

Which of the following is a second-generation intra-uterine device?

- a) Only 5
- b) 1 & 3
- c) 1,2 & 4
- d) 3 & 5

Question 7:

Which of the following women will not be considered ideal candidates for IUD insertion?

- a) A multiparous woman
- b) A woman in a monogamous relationship
- c) A woman with no access to follow up
- d) A woman who has at least one living child

Question 8:

A patient who underwent an IUD insertion is most likely to develop which of the following in the future?

- a) Vaginal bleeding

- b) Abdominal pain
- c) Uterine perforation
- d) Pelvic infection

Question 9:

Insertion of an IUD is not contraindicated in patients with which of the following conditions?

- a) 2 & 4
- b) Only 2
- c) 3, 4 & 5
- d) 1, 3 & 5

Question 10:

Which of the following has the longest period of use?

- a) Progestasert
- b) LNG IUD
- c) Cu-T-380-A
- d) Nova - T

Question 11:

You have prescribed an IUCD to a multiparous woman. Which of the following is not a mechanism of action for this type of contraceptive?

- a) Cellular and biochemical reactions
- b) Changes cervical mucus consistency
- c) Prevents implantation
- d) Prevents ovulation

Question 12:

All of the following are non-contraceptive benefits of oral contraceptive pills except?

- a) Protection against fibroadenoma

- b) Protection against pelvic inflammatory disease
- c) Protection against ectopic pregnancy
- d) Protection against thyroid dysfunction

Question 13:

In patients with which of the following conditions can OCPs be safely prescribed?

- a) Breast cancer
- b) Cardiac abnormalities
- c) PID
- d) Liver disease

Question 14:

A sexually active woman who was using barrier contraceptives is planning on switching to DMPA injectables. Which of the following is the correct dosage of this contraceptive?

- a) 200 mg every 3 months
- b) 250 mg every 2 months
- c) 100 mg every 2 month
- d) 150 mg every 3 months

Question 15:

Which of the following is also known as the Billings method of contraception?

- a) Rhythm method
- b) Cervical mucus method
- c) Symptothermic method
- d) Calendar method

Question 16:

Which of the following is the most effective natural method of contraception?

- a) Basal body temperature method

- b) Cervical mucus method
- c) Symptothermic method
- d) Calendar method

Question 17:

Which of the following is not a condition to be fulfilled by a male patient undergoing sterilization?

- a) Has to be married
- b) Has at least one living child
- c) Age should be more than 60 years
- d) Has given consent without coercion

Question 18:

Which of the following is the most common cause of failure of vasectomy?

- a) Mistaken identification of vas
- b) Recanalization
- c) Surgical failure
- d) Idiopathic

Question 19:

Which of the following is important in the calculation of the pearl index?

- a) Number of abortions
- b) Total accidental pregnancies
- c) Total number of women in the population
- d) Number of live births

Question 20:

As part of the National family planning program, you are conducting a study in a village to gather data on the efficacy of a new contraceptive. 20 out of 100 women became pregnant in the village despite using this contraceptive for a period of 2 years continuously. What is the

pearl index of this contraceptive?

- a) 0.1 per HWY
- b) 5 per HWY
- c) 10 per HWY
- d) 1000 per HWY

Question 21:

Out of 100 women who took oral contraceptive pills for contraception, 10 women got pregnant within 24 months. What is the Pearl index?

- a) 10 per HWY
- b) 5 per HWY
- c) 15 per HWY
- d) 20 per HWY

Answer Key

Question No.	Correct Option
1	c
2	c
3	b
4	d
5	c
6	d
7	c
8	a
9	b
10	c
11	d
12	d
13	c
14	d
15	b

16	c
17	c
18	a
19	b
20	c
21	b

Detailed Explanations

Solution to Question 1:

There are approximately 150 – 180 eligible couples per 1000 population in India.

An eligible couple refers to a currently married couple wherein the wife belongs to the reproductive age group (15-49 years of age).

Solution to Question 2:

Screening for HIV infection does not come under the purview of family planning.

The modern concepts of family planning services include:

- Proper spacing and limitation of birth
- Advice on sterility
- Education for parenthood
- Sex education
- Screening for pathological conditions related to the reproductive system (e.g. cervical cancer)
- Genetic counseling
- Premarital consultation and examination
- Pregnancy tests
- Marriage counseling
- Preparation of couples for the first child
- Providing services to unmarried mothers
- Teaching home economics and nutrition
- Providing adoption services

Solution to Question 3:

Couple protection rate (CPR) is defined as the percentage of eligible couples effectively protected against childbirth by one or the other approved methods of family planning.

In the given scenario, 60 out of 180 eligible couples are using different contraceptive methods. Therefore, $CPR = (60/180) \times 100 = 33\%$.

Solution to Question 4:

Lippes loop is an intrauterine device and not a barrier method of contraception.

Barrier methods of contraception include:

- Physical methods – male and female condoms, diaphragm, vaginal sponge (Today)
- Chemical methods – foam tablets, aerosols, creams, jellies, suppositories, and soluble films (C-films)

Solution to Question 5:

Today is a type of vaginal sponge that contains 9 – nonoxynol.

Today is a barrier contraceptive inserted vaginally. It has to be left in place for 6 hours after coitus. It can lead to toxic shock syndrome if left in place for more than 24 hours (rarely).

'Today' vaginal sponge



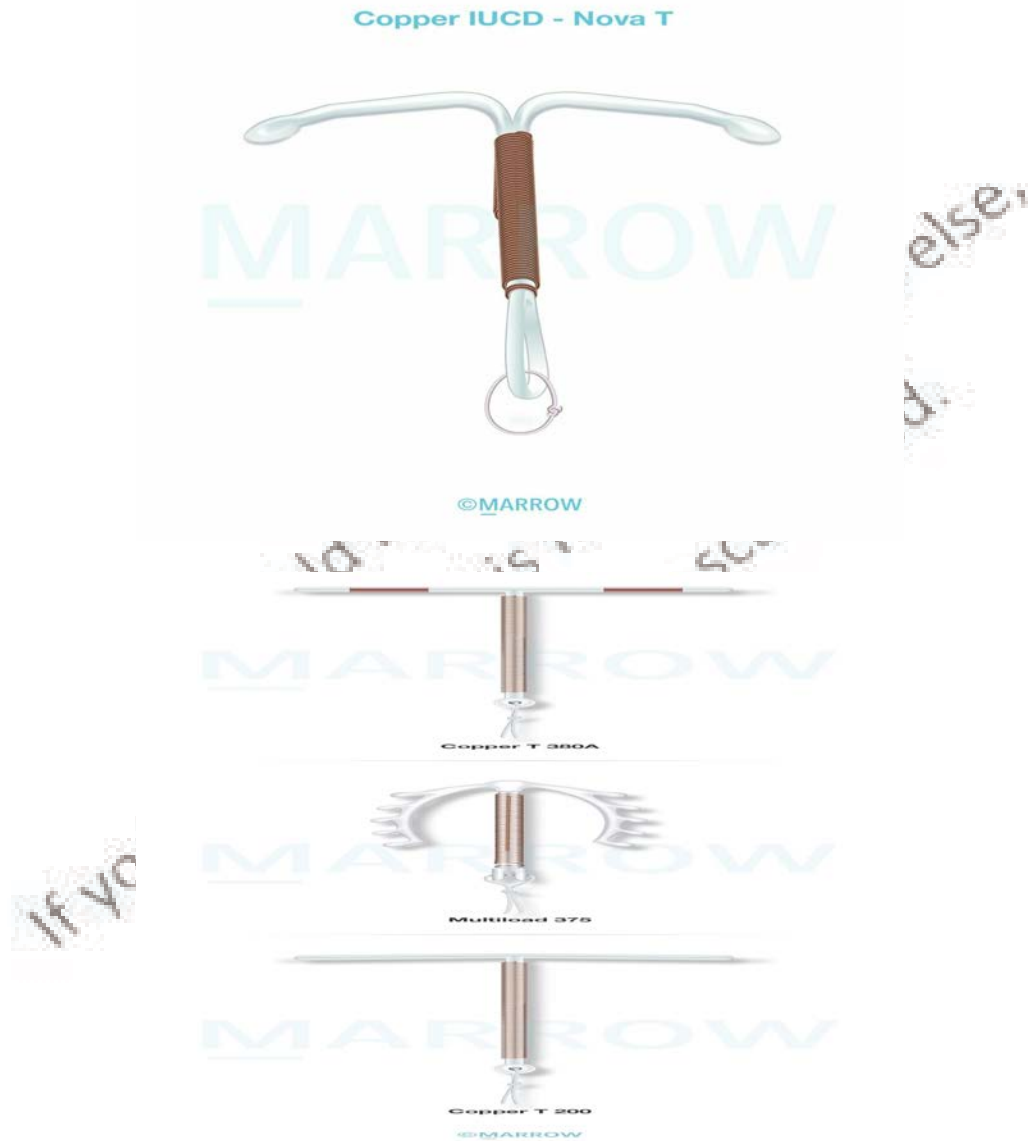
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Solution to Question 6:

Nova T and Cu-T-380 Ag are second-generation intrauterine devices (IUD).

IUD is a contraceptive method inserted into the uterine cavity. The different generations of IUDs include:

- First-generation IUDs – inert or non-medicated devices like Lippes loop
- Second-generation IUDs - medicated with metallic elements like copper eg: Cu-T-220 C, Nova T, Multi-load Cu-250, Multi-load Cu-375
- Third-generation IUDs – hormone-containing IUDs like progestasert, Mirena (LNG-20)



Solution to Question 7:

A woman with no access to follow-up is not considered an ideal candidate for IUD insertion.

An ideal candidate for an IUD must have the following attributes:

- Has at least one child
- No history of pelvic disease

- Normal menstrual periods
- Must be willing to check IUD tail
- Has access to follow up and for treatment of potential problems
- Is in a monogamous relationship

Solution to Question 8:

Patients who underwent IUD insertion are most likely to develop vaginal bleeding in the future.

Side effects and complications of IUDs include:

- Vaginal bleeding – most common
- Abdominal pain – second most common
- Pelvic infection
- Uterine perforation
- Pregnancy - due to failure of contraception
- Ectopic pregnancy
- Expulsion of the IUD
- Cancer and teratogenicity

Solution to Question 9:

Cardiac problems are not a contraindication for IUD usage.

Contraindications for IUD usage include:

Absolute contraindications	Relative contraindications
Suspected pregnancy PID Vaginal bleeding of unknown etiology Carcinoma cervix/uterus/adnexa /other pelvic tumors	Anemia Menorrhagia History of PID since the last pregnancy Purulent cervical discharge Distortions of the uterine cavity due to fibroids Uterine malformations An unmotivated person

Solution to Question 10:

Cu-T-380 A has the longest period of use (10 years).

IUD	Years of use
Progestasert	1
Cu-T-200	4
Nova T	5
LNG IUD	7-10
Cu-T-380A	10

Solution to Question 11:

Prevention of ovulation is not a mechanism of action of IUCDs.

Mechanisms of action of IUCDs include:

- Causing cellular and biochemical changes in the uterus and endometrium.
- Prevention of sperm penetration due to changes in cervical mucus consistency.
- Prevention of implantation by making the uterus hostile to it.

Solution to Question 12:

Oral contraceptive pills (OCPs) do not confer protection against thyroid dysfunction.

OCP usage can confer protection against:

- Benign breast diseases like fibroadenoma and fibrocystic disease
- Ovarian cysts
- Iron-deficiency anemia
- Pelvic inflammatory disease
- Ectopic pregnancy
- Ovarian cancer

Solution to Question 13:

Pelvic Inflammatory Disease (PID) is not a contraindication for prescribing OCPs.

Absolute contraindications of OCPs: (CULT)

- Cancers - breast and genital
- Cardiac abnormalities

- Congenital hyperlipidemia
- Undiagnosed AUB
- Liver disease
- Thromboembolism - past or present history.

Medical surveillance is required while prescribing OCPs if the patient is/or has:

- Age more than 40 years
- Chronic smoker and age over 35 years
- Amenorrhea
- Chronic renal disease
- Diabetes mellitus
- Epilepsy
- In the first 6 months of nursing
- Gall bladder disease
- History of infrequent bleeding
- Mild hypertension
- Migraine

Solution to Question 14:

The dosage for injectable depot-medroxyprogesterone acetate (DMPA) is 150 mg every 3 months.

It acts by:

- Bringing changes in the cervical mucus
- Making the endometrium hostile
- Preventing ovulation

Disadvantages include spotting and intermediate per vaginal bleeding.

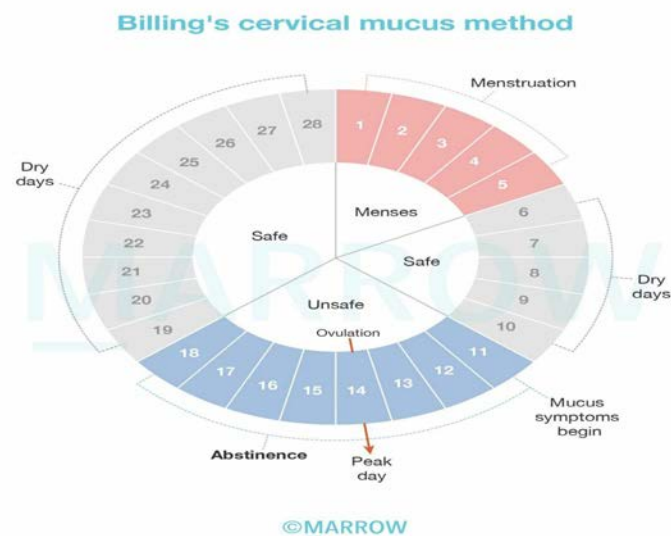


Solution to Question 15:

Billings method of contraception is also known as the cervical mucus method (or ovulation method).

It is a natural method of contraception. It is based on the observation of changes in the cervical mucus. At ovulation, the mucus is watery clear, smooth, slippery and profuse. After ovulation, it becomes thicker and scarce.

The pattern of cervical mucus secretion during a menstrual cycle is given below:



Solution to Question 16:

The most effective natural method of contraception is the sympto-thermic method.

It combines:

- Calendar method
- Cervical mucus method
- Basal body method.

In this method, if a woman is not able to identify one sign, she can double-check with her interpretation of the other signs.

Solution to Question 17:

A man should be below 60 years of age to be eligible for sterilization.

Eligibility for male sterilization:

- Clients should be at least 22 years old
- Clients should be below the age of 60 years
- Clients should be married
- The couple should have at least one child, whose age is above one year unless the sterilization is medically indicated
- Clients or their spouses must not have undergone sterilization in the past (not applicable in the cases of failure of previous sterilization)
- Clients must be in a sound state of mind, so as to understand the full implications of sterilization

- Mentally ill clients must be certified by a psychiatrist, and a statement should be given by the legal guardian/spouse regarding the soundness of the client's state of mind

Solution to Question 18:

The most common cause of failure of vasectomy is the mistaken identification of the vas.

This can be avoided by examining a stained smear prepared by squeezing the vas on a glass slide. If facilities are available a histological examination is preferred.

Causes of failure of vasectomy include:

- Mistaken identification of the vas - most common
- Spontaneous recanalization of the vas
- Presence of more than one vas on one side
- Sexual intercourse before the disappearance of sperms from the reproductive tracts

Solution to Question 19:

The numerator used for calculation of the pearl index is total accidental pregnancies.

Pearl index = Total accidental pregnancies x 1200/Total months of exposure

Solution to Question 20:

In the given scenario, the pearl index is 10 per HWY.

Here, the total number of accidental pregnancies is 20. The total months of exposure are 24 months for each woman.

Hence, the total months of exposure for 100 women are $24 \times 100 = 2400$.

Pearl index = Total accidental pregnancies x 1200/ Total months of exposure

Hence, pearl index= $20 \times 1200/2400 = 10$ per HWY

Solution to Question 21:

In the given scenario, the pearl index is 5 per HWY.

Here, the total number of accidental pregnancies is 10. There are 100 couples who are exposed to the method.

So, the total months of exposure are $24 \text{ months} \times 100 = 2400$.

Pearl index = Total accidental pregnancies x 1200/Total months of exposure

Hence, pearl index = $10 \times 1200/2400 = 5$ per HWY.

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Preventive Obstetrics, Pediatrics and Geriatrics

Question 1:

A pregnant woman has come for her first antenatal visit at 10 weeks of gestation. How many more antenatal visits does she require to fulfill the minimum requirements as per the Maternal and Child Health program?

- a) 7
- b) 2
- c) 3
- d) 8

Question 2:

You are a medical officer in a PHC. A pregnant woman comes to you for regular follow-up during her second trimester. Which of the following will you prescribe for her?

- a) 100 mg elemental iron + 500 mcg folic acid
- b) 100 mg elemental iron + 100 mcg folic acid
- c) 50 mg elemental iron + 500 mcg folic acid
- d) 60 mg elemental iron + 500 mcg of folic acid

Question 3:

Which of the following vaccine is contraindicated in pregnancy?

- a) Hepatitis A
- b) Hepatitis B
- c) Rabies
- d) Varicella

Question 4:

What is the average weight gain during pregnancy in Indian women with inadequate calorie intake?

- a) 7 - 8 kg
- b) 9 - 11 kg
- c) 8 - 9 kg
- d) 5 - 6 kg

Question 5:

Which of the following is incorrect about congenital syphilis?

- a) It can be prevented by procaine penicillin administration
- b) Infection of fetus most commonly occurs in the 1st trimester
- c) Neurological damage with mental retardation is a serious complication
- d) Chances of transmission are lower if mother has late syphilis

Question 6:

The District Maternal Death Review Committee is reviewing the number of maternal deaths in the district during 2020. In order to calculate the maternal mortality ratio, what other parameter is required?

- a) Total live births
- b) Birth Rate
- c) Total pregnant women
- d) Total women in child bearing age

Question 7:

Which of the following maternal deaths is not included when calculating the maternal mortality ratio?

- a) Death of a 30-year-old woman following spontaneous abortion at 11 weeks of gestation
- b) Death of a 24-year-old woman due to ruptured ectopic pregnancy
- c) Death of a 26-year-old woman following a road traffic accident at 36 weeks of gestation
- d) Death of a 21-year-old woman following secondary PPH on postpartum day 14

Question 8:

What is the most common cause of maternal deaths in India?

- a) Hemorrhage
- b) Eclampsia
- c) Anemia
- d) Unsafe abortion

Question 9:

In a town, there were 2500 live births in a period of 6 months. During the same period, 15 women died due to the following causes. What is the maternal mortality ratio?

- a) 400 per 100000 live births
- b) 6 per 1000 live births
- c) 40 per 1000 live births
- d) 60 per 1000 live births

Question 10:

A baby was delivered at home in a village. Until which day of life should an ASHA worker visit this baby?

- a) 14
- b) 28
- c) 21
- d) 42

Question 11:

What is the best indicator of obstetric and neonatal care given before and around the time of birth?

- a) Infant mortality rate
- b) Perinatal mortality rate
- c) Under 5 mortality rate
- d) Early neonatal mortality rate

Question 12:

What is the 2030 target of the India Newborn Action Plan with regards to the stillbirth rate?

- a) < 10 per 1000 live births
- b) < 20 per 1000 live births
- c) < 30 per 10000 live births
- d) < 40 per 10000 live births

Question 13:

A group of health workers are conducting a perinatal mortality audit in a hospital. Which of the following must they take into account to calculate the perinatal mortality rate?

- a) 1,3,4
- b) 1,2,4
- c) 1,4,5
- d) 2,4,5

Question 14:

In a given population, 4050 births were recorded over a period of 1 year. Among these, 50 were stillbirths, 50 died in the 1st week of life and 150 died within 8-28 days of life. What is the neonatal mortality rate of this population?

- a) 50
- b) 12.5
- c) 100
- d) 62.5

Question 15:

Which of the following is the leading cause of neonatal mortality?

- a) Prematurity
- b) Birth Asphyxia
- c) Sepsis
- d) Pneumonia

Question 16:

In children aged 1 month to 1 year, all of the following are common causes of mortality except:

- a) Tetanus
- b) Malnutrition
- c) Diarrheal diseases
- d) Acute respiratory infections

Question 17:

In a given population, there were 5100 births during the year 2020. Of these, 100 babies were stillborn. Among the living children, the following deaths were recorded. Calculate the infant mortality rate for this population.

- a) 100 per 1000 live births
- b) 10 per 1000 live births
- c) 160 per 1000 live births
- d) 16 per 1000 live births

Question 18:

Which of the following is true for a newborn stabilization unit?

- a) It is located in the delivery room.
- b) It is for sick babies and LBW babies ≤ 1800 g
- c) It is present in all community health centres
- d) All cases except major surgeries are managed here

Question 19:

You are a doctor at a child health clinic. You will consider all of the following children as at-risk infants except:

- a) A 2-month-old baby with 3 older siblings
- b) A 15-day-old baby with birth weight of 2.4 kgs

- c) An 1-year-old girl with protein-energy malnutrition
- d) An 8-month-old baby being raised by a working mother

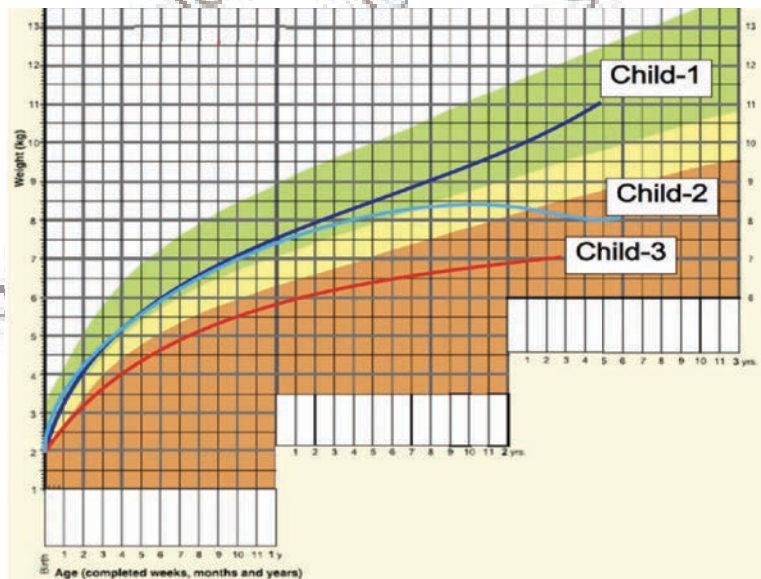
Question 20:

A researcher is calculating the child survival indices for different populations. Which of the following should he use for calculating the same?

- a) Maternal mortality rate
- b) Infant mortality rate
- c) Perinatal mortality rate
- d) Under-5 mortality rate

Question 21:

Identify the false statement about the following tool:



- a) Malnutrition can be detected even before signs and symptoms appear
- b) It provides an objective basis for policy making on child healthcare
- c) It can be used to teach the importance of adequate feeding
- d) It can be used to indicate proof of government aid

Question 22:

A juvenile delinquent was sentenced by the court to be placed in a center under the care of a doctor and psychiatrist. Which of the following fulfills this requirement?

- a) Borstal
- b) Foster home
- c) Remand home
- d) Shelter home

Question 23:

The police authorities rescued a girl from being trafficked for sexual exploitation. She is identified to be a refugee. As a part of the Ujjawala scheme, which of the following benefits will she not receive?

- a) Rehabilitation
- b) Re-integration
- c) Remuneration
- d) Repatriation

Question 24:

A 4-year-old girl was brought by her parents who tell you that she fell down while playing. On examination, bruises and ecchymoses of varying ages are noted all over the body. An x-ray showed a spiral fracture of the right femur. Which of the following programs can provide support services to this child and prevent further such incidents?

- a) Ujjawala scheme
- b) INSPIRE strategy
- c) SABLA scheme
- d) POCSO act

Question 25:

As a PHC medical officer, which of the following groups would you target as a part of Kishori Shakti Yojana?

- a) Girls aged 9-11 years
- b) Adolescent boys and girls

- c) Girls aged 3-6 years
- d) Adolescent girls

Question 26:

The school health programs are managed by which of the following?

- a) Primary health center
- b) District hospital
- c) Sub-centre
- d) Sub-divisional hospital

Question 27:

A 68-year-old man visits your outpatient clinic for a routine follow-up. Which of the following protocols would you use to assess his health literacy?

- a) GATHER
- b) SPEAK
- c) SPIKES
- d) S-BAR

Question 28:

Which of the following comes under the Rashtriya Varishth Jan Swasthya Yojana?

- a) Domiciliary visits to homes of elderly in the community by ANMs or male HCWs
- b) Rehabilitation units with physiotherapy at CHCs
- c) Super-specialty geriatric departments for training, guidelines and referral
- d) Weekly geriatric clinics arranged at PHCs by medical officer

Answer Key

Question No.	Correct Option
1	c

2	d
3	d
4	d
5	b
6	a
7	c
8	a
9	a
10	d
11	b
12	a
13	b
14	a
15	a
16	a
17	a
18	c
19	a
20	d
21	d
22	c
23	c
24	b
25	d
26	a
27	b
28	c

Detailed Explanations

Solution to Question 1:

This patient requires 3 more antenatal visits in order to fulfill the minimum requirements according to the Maternal and Child Health Program (MCH).

The minimum number of antenatal visits according to MCH is 4.

- 1st visit – within 12 weeks, preferably as soon as the pregnancy is suspected.
- 2nd visit – between 14 and 26 weeks

- 3rd visit – between 28 and 34 weeks
- 4th visit – between 36 weeks and term

The minimum number of antenatal visits according to the WHO is 8. Ideally, pregnant women should attend antenatal visits according to the following schedule:

- Once a month during the first 7 months
- Twice a month during the 8th month
- Once a week, thereafter if everything is normal.

Solution to Question 2:

As per the Anemia Mukht Bharat program, pregnant women should be given 60 mg of elemental iron and 500 mcg of folic acid in the II trimester. It should be started from the 4th month of pregnancy and continued for 180 days post-partum.

Solution to Question 3:

Varicella vaccine is contraindicated in pregnant women as it is a live vaccine.

The live attenuated varicella virus vaccine is currently recommended for children between 12 to 18 months of age who have not had chickenpox.

Solution to Question 4:

Weight gain during pregnancy in women with inadequate calorie intake is around 5 - 6 kgs.

Normally, a woman is expected to gain around 9-11 kgs of weight during pregnancy. An inadequate dietary intake can be suspected if the woman gains < 2 kg/ month.

Solution to Question 5:

The chances of transmission of syphilis during pregnancy is highest after 6th month of gestation. It does not occur before the 4th month.

It is routine to test the mother for syphilis in antenatal clinics during her first visit. The chance of infection of the fetus is higher when the mother is suffering from early (primary or secondary) syphilis and are lower in case of late syphilis. Neurological damage with mental retardation is a serious complication of congenital syphilis. It can be prevented by procaine penicillin administration.

Solution to Question 6:

The maternal mortality ratio is calculated using the number of maternal deaths and total live births.

Solution to Question 7:

The death of a pregnant woman following a road traffic accident is not included when calculating the maternal mortality ratio. It includes all deaths anytime during pregnancy or within 42 days of termination from any cause related to or aggravated by the pregnancy.

Solution to Question 8:

The most common cause of maternal deaths in India is hemorrhage.

Solution to Question 9:

The maternal mortality ratio (MMR) of this town is 400 per 100,000 live births.

Maternal mortality refers to deaths during pregnancy or 42 days within termination, from causes related to pregnancy. Hence, in this town, there were 10 maternal deaths (peripartum infection, PPH, obstructed labor).

$$\text{MMR} = \frac{\text{Number of maternal deaths in a period} \times 100,000}{\text{Number of live births in the same period}}$$

Hence, $\text{MMR} = \frac{10 \times 100000}{2500} = 400$ per 100,000 live births

Solution to Question 10:

Under the Home Based Newborn Care (HBNC) guidelines, ASHA workers should visit all newborns up to 42 days of life, according to the following schedule.

During her visits, ASHA workers must record weight, ensure vaccination, and registration of births. On completion of all these visits, they receive an incentive of Rs. 250.

Mode of delivery	No. of visits	Schedule of visits
Home delivery	7	Days 1, 3, 7, 14, 21, 28, 42
Hospital-based normal vaginal delivery	6	Days 3, 7, 14, 21, 28, 42(except day 1)
Hospital-based LSCS delivery	5	Days 7, 14, 21, 28, 42(except days 1, 3)

Solution to Question 11:

The perinatal mortality rate is the best indicator to evaluate obstetric and neonatal care given before and around the time of birth.

Solution to Question 12:

The India Newborn Action Plan targets to achieve a stillbirth rate of $< 10/1000$ live births and neonatal mortality rate of $< 10/1000$ live births (single-digit SBR and NMR) by 2030.

This program was launched to reduce preventable newborn deaths and stillbirths. It provides the following:

- Pre-conception and antenatal care
- Care during labor and childbirth
- Immediate newborn care
- Care of healthy newborn
- Care of small and sick newborn
- Care beyond newborn survival

Solution to Question 13:

The perinatal mortality rate (PMR) is calculated using the following parameters during a particular time period:

- Late fetal deaths or stillbirths ≥ 1000 g - correlates to 28 weeks of gestation
- Early neonatal deaths - from birth to 7 days of life
- Live births ≥ 1000 g

The neonatal mortality rate (NMR) refers to the deaths within the first 28 days of life. It is calculated using the following parameters:

- Early neonatal deaths
- Late neonatal deaths - after 7th day but before 28th completed day of life

Solution to Question 14:

The neonatal mortality rate (NMR) of the given population is 50 per 1000 live births.

Neonatal mortality rate = No. of deaths of neonates < 28 days within a year x 1000 / Total live births in the same year

In this population, there were 4000 live births (no. of total births - no. of stillbirths).

$$\text{NMR} = (50+150) \times 1000/4000 = 100/2 = 50$$

Solution to Question 15:

Prematurity is the leading cause of neonatal mortality.

Solution to Question 16:

Tetanus is not a common cause of mortality in children between the age of 1 month to 1 year (post-neonatal period). The main causes of post-neonatal deaths are diarrhea, respiratory infections, and malnutrition.

Solution to Question 17:

The infant mortality rate (IMR) for this population is 100 / 1000 live births.

IMR = No. of deaths within 1st year of life over a period of time x 1000 / No. of live births in the same period

In this population, there were 5000 live births (5100 – 100) and 500 deaths within the 1st year of life.

$$\text{Hence, IMR} = 500 \times 1000/5000 = 100 \text{ per } 1000 \text{ live births}$$

Solution to Question 18:

A newborn stabilization unit is present in all community health centers (CHC).

India accounts for 30% of neonatal deaths globally and 73.3% of these deaths occur in 1st week of life. To address this issue, the following facility-based newborn care services were introduced:

	Newborn care corner	Newborn stabilization unit	Special newborn care units
Present in	Delivery room of all centers and hospitals	Within or close to the maternity ward, in CHCs and first referral units	District hospitals and facilities with >3000 deliveries/ year
Admits	Immediate care to all newborns	Sick and LBW >1800 gms	Sick and LBW < 1800 gms

	Newborn care corner	Newborn stabilization unit	Special newborn care units
Service	Immediate resuscitation, early initiation of breastfeeding, identification, and referral of at-risk newborns	Hyperbilirubinemia, asphyxia, pneumonia, diarrhea, sepsis	All care except assisted ventilation and major surgery
Training of health personnel appointed	2 days, under Navjat Shishu Suraksha Karyakram	11 days in IMNCI	4 days of training and two weeks of observership
Staff	1 doctor, 1 nurse, 1 auxiliary midwife	1 doctor, 4 nurses	3-4 doctors and 10 nurses

Solution to Question 19:

The 2-month-old baby with 3 older siblings has a birth order of 4. Infants with birth order of 5 or more are considered at-risk infants.

The following are considered at-risk infants:

- Birth weight $\leq$ 2.5 kgs
- Twins
- Birth order of 5 or more
- Artificial feeding
- Weight below 70% of the expected weight
- Failure to gain weight during 3 successive months
- Children with protein-energy malnutrition (PEM), diarrhea
- Working mother or single parent

Solution to Question 20:

The under-5 mortality rate is used in the calculation of the child survival index. It is calculated as $(1000 - \text{Under-5 mortality rate})/10$.

Solution to Question 21:

The given image shows a growth chart. It is not used as proof of government aid.

The uses of growth charts are as follows:

- Growth monitoring and diagnosis - can detect malnutrition before clinical features appear
- Educational tool - to teach parents about the importance of adequate feeding.

- Objective basis for planning and policy-making on child healthcares

Solution to Question 22:

A remand home is an establishment where children are placed under the care of a doctor, psychiatrist and other trained personnel.

An orphanage or shelter home is an institution where children who have no home or parents or for some reason can't be cared for are placed.

Foster homes are arrangements where a family or a couple takes care of a child on a temporary basis.

Borstal is a place that falls between a certified institution and a prison. It is where boys > 16 years who have misbehaved or are too difficult to handle are sent.

Solution to Question 23:

The girl will not receive remuneration (monetary compensation) under the Ujjawala scheme.

The Ujjawala scheme was launched to combat women and child trafficking in India. The components of the scheme include:

- Prevention of trafficking by awareness, vigilance, education
- Rescue of victims from the place of exploitation
- Rehabilitation of victims
- Re-integration of victims into their family and society
- Repatriation of cross-border victims to their place of origin

Some other legislations associated with children in India are:

- The Right of Children to Free and Compulsory Education Act, 2009
- Juvenile Justice (Care and Protection of Children) Act, 2000
- Child Labour (Prohibition and Regulation) Act, 1986 (amended in 2006 and 2016)
- Protection of Children from Sexual Offences (POCSO), 2012 under the National Commission for Protection of Child Rights (NCPCR)
- Prohibition of Child Marriage Act, 2006

Solution to Question 24:

The given clinical scenario points towards child abuse. The INSPIRE strategy was introduced to eliminate such violence against children and also provide support services.

It refers to a set of 7 evidence-based strategies to eliminate violence against children:

Strategies		Approach
I	Implementation and enforcement of laws	Banning violent punishment of children and criminalizing sexual abuse Preventing alcohol misuse and limiting youth access to weapons
N	Norms and values	Changing perceptions about harmful gender and social norms
S	Safe environments	Reducing the spread of violence
P	Parent and caregiver support	Delivered through home visits and in groups in community settings
I	Income and economic strengthening	Cash transfers Providing small loans
R	Response and support services	Counseling and therapy Treatment programs for juvenile offenders Foster care interventions
E	Education and life skills	Increasing enrolment in schools Establishing safe school environments Improving sexual education for children and teaching self-defense strategies

Solution to Question 25:

The Kishori Shakti Yojana (KSY) program is aimed at adolescent girls (11-18 years).

This program is a component of the Integrated Child Development Services (ICDS) scheme and is a redesign of the already existing Adolescent Girls (AG) scheme.

The objectives of KSY are:

- To improve the nutrition, health, and development status
- To promote awareness on health, hygiene, nutrition, and family care
- To teach life skills and encourage school attendance
- To help understand their social environment
- Aspects aimed at empowerment and enhanced self-perception
- To help them take initiatives to become productive members of society

It also fosters convergence with other sectoral programs, addressing the interrelated needs of adolescent girls and women.

Solution to Question 26:

The school health programs are managed by the primary health center (PHC).

The School Health Committee (1961) in India recommended that the school health services should be a part of the general health services administered through PHCs. Formation of school health committees at the village, block, district, state, and national levels was recommended. The National School Health Council was created as an advisory and coordinating body.

PHCs provide the following school health services:

- Screening and general assessments for -
- Anemia/nutritional status
- Visual acuity and hearing problems
- Dental check-up
- Physical disabilities
- Learning disorders and behavior problems
- Basic medicines for common ailments
- Immunization
- Micronutrient management (vitamin A, iron, and folic acid)
- Deworming
- Mid-day meal

Solution to Question 27:

The SPEAK protocol is used to assess health literacy in the geriatric population in a clinical setting.

Health literacy is defined as the degree to which individuals have the capacity to obtain, process, and understand basic health information needed to make appropriate health decisions. The SPEAK protocol provides a simple framework that healthcare providers can use to assess literacy. SPEAK stands for:

- S - Speech
- P - Perception
- E - Education
- A - Access
- K - Knowledge

Option A: The GATHER approach is mainly used for counseling (usually, family planning and reproductive counseling) to help people make decisions about health. It stands for:

- G – Greet
- A – Ask
- T – Tell
- H – Help
- E – Explain
- R – Return

Option C: The SPIKES protocol is used to convey distressing information and break bad news in an organized manner to patients and families. It stands for:

- S - Setting
- P - Perception
- I - Invitation or information
- K - Knowledge
- E - Empathy
- S - Summarizing or strategizing.

Option D: The S-BAR technique is used for prompt and appropriate communication among health care workers in hospital settings. It stands for:

- S - Situation
- B - Background
- A - Assessment
- R - Recommendation

Solution to Question 28:

The establishment of super-specialty geriatric departments for training, guidelines, and referral comes under the Rashtriya Varishth Jan Swasthya Yojna (RVJSY). It refers to the tertiary component of the National Programme for Healthcare of the Elderly (NPHCE).

The NPHCE was established for geriatric care and includes provisions at primary, secondary, and tertiary levels.

The tertiary component of the program has been renamed as RVJSY and has the following provisions:

- Geriatrics departments at 20 super specialty institutions-
- 30-bedded inpatient facility and referral treatment
- Research and manpower development,
- Development of training, IEC material, and guidelines
- Geriatric units at district hospitals (tertiary) -
- 10 bedded geriatric ward

- Dedicated geriatric OPD

Centers	Services according to NPHCE
PHCs(primary)	Weeklygeriatricclinicsat PHC Treatment of minor ailments Referralservices to higher cen ters
Sub-centres(primar y)	ANMs/ male health workers makedomiciliary visitsto the elderlyArrange supportive de vices for the disabledTreatme nt of minor ailments
CHCs(secondary)	Dedicatedgeriatricclinicstwic e weeklyRehabilitation unit w ith trainedphysiotherapists

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Energy Metabolism and Macronutrients

Question 1:

Which of the following parameters is not indicative of reference Indian adult woman?

- a) 19-39 years
- b) 1.62 m height
- c) 50 kg weight
- d) BMI 20.95

Question 2:

The energy required by an adult for basal metabolism is:

- a) 1 kcal/ hour/ kg body weight
- b) 2 kcal/ hour/ kg body weight
- c) 3 kcal/ hour/ kg body weight
- d) 4 kcal/ hour/ kg body weight

Question 3:

What is the likely energy requirement of a pregnant woman who leads a moderately active lifestyle?

- a) 3000 kcal/ day
- b) 2500 kcal/ day
- c) 2000 kcal/ day
- d) 1500 kcal/ day

Question 4:

A nutritionist is preparing a meal plan for a healthy 32-year-old man. Which of the following principles must she follow?

- a) Proteins must constitute 15-30% of the daily energy intake
- b) Fats must constitute 10-15% of the daily energy intake

- c) Carbohydrate requirement must be met first followed by protein and fat
- d) The proportion of nutrients in the diet must be carbohydrates & fats & proteins

Question 5:

What is the daily requirement of high-quality protein for a 40-year-old man weighing 60 kgs?

- a) 40 g/day
- b) 50 g/day
- c) 72 g/day
- d) 85 g/day

Question 6:

A dietary protein-energy percentage of which of the following is the threshold for inadequate protein intake?

- a) < 2%
- b) < 4%
- c) < 20%
- d) < 40%

Question 7:

Which of the following is the best indicator of protein quality?

- a) Protein digestibility corrected amino acid score
- b) Digestible indispensable amino acid score
- c) Biological value
- d) Protein efficiency ratio

Question 8:

Which of the following has the highest net protein utilization value?

- a) Meat
- b) Milk

- c) Rice
- d) Soya

Question 9:

A farmer has been consuming a maize-rich diet for many years. Which of the following would be lacking in his diet?

- a) Only tryptophan
- b) Lysine and threonine
- c) Lysine and tryptophan
- d) Methionine and cysteine

Question 10:

Which of the following has the highest content of essential fatty acids?

- a) Soybean oil
- b) Safflower oil
- c) Groundnut oil
- d) Coconut oil

Question 11:

You are counseling a diabetic patient to consume foods with low glycemic index. Which of the following would you advise him to eat?

- a) White bread
- b) Brown rice
- c) Corn flakes
- d) Whole grains

Question 12:

The renal feed of a patient suffering from end-stage renal disease (stage 5) should ideally be:

- a) Low volume, high calories

- b) High volume, low calories
- c) High volume, high calories
- d) Low volume, low calories

Question 13:

Which of the following is not used in the assessment of body fat?

- a) Quetelet's index
- b) Total body potassium
- c) Total body water
- d) Breslow index

Question 14:

In the Indian population, individuals with BMI falling between which range are defined as overweight?

- a) 23 - 24.99
- b) 23 - 26.99
- c) 25 - 29.99
- d) ≥ 25

Question 15:

What value of BMI is considered as 'lethal' in men?

- a) < 12
- b) < 13
- c) < 14
- d) < 15

Question 16:

Which of the following indicators are used in Waterlow's classification?

- a) 1, 2, 3 only

- b) 1, 4 only
- c) 2, 3 only
- d) 4, 5 only

Answer Key

Question No.	Correct Option
1	c
2	a
3	b
4	d
5	b
6	b
7	b
8	a
9	c
10	b
11	d
12	a
13	d
14	a
15	b
16	c

Detailed Explanations

Solution to Question 1:

According to the 2020 guidelines of the National Institute of Nutrition, the weight of the reference Indian adult woman is 55 kg.

Solution to Question 2:

The energy required by an adult for basal metabolism is 1 kcal/ hour/ kg of body weight.

Solution to Question 3:

The energy requirement of a pregnant woman who leads a moderately active lifestyle is around 2500 kcal/day.

An adult woman leading a moderately active lifestyle requires 2130 kcal daily and an additional 350 kcal is needed in pregnancy. Hence, the total requirement for a moderately working pregnant woman is 2480 kcal/ day (approximately 2500 kcal/ day).

Solution to Question 4:

The proportion of nutrients in the diet must be carbohydrates > fats > proteins.

While constructing a balanced diet, nutrient requirements must be calculated in the following sequence:

- Protein requirements - 10-15% of daily energy intake
- Fat requirement - 15-30% of daily energy intake
- Carbohydrates - remaining food energy

Solution to Question 5:

This man requires 50 g/day of high-quality protein.

High-quality protein refers to proteins containing all the essential amino acids. According to the FAO/WHO 2007 guidelines, the recommended high-quality protein intake for adult men and women is 0.83 g/kg/day. This man weighs 60 kgs. Hence he requires around 50 g ($0.83 \times 60 = 49.8$) of high-quality protein per day.

Recommended high-quality protein intake by FAO/WHO:

However, the mixed Indian diet does not meet the requirement for high-quality protein. Hence, the protein requirement in Indian adults (men and women) is 1g/kg/day of mixed Indian diet protein according to the ICMR 2010 guidelines.

Age	Protein in g/kg/day
1-5 years	0.94
6-10 years	0.91
Adolescent male	0.88
Adolescent female	0.86

Age	Protein in g/kg/day
Adult man/woman	0.83

Solution to Question 6:

Dietary protein-energy percentage (PE %) < 4% indicates inadequate protein intake. The PE % indicates the proportion of total energy in the diet that is derived from proteins.

$$\text{PE \%} = (\text{Energy from protein} / \text{Total energy in the diet}) * 100$$

Some of the food items and their protein-energy percentage are given below:

Food	PE%
Fish	80
Cow milk	20
Dal	24
Rice	8
Potato	6
Banana	4
Tapio ca	2

Solution to Question 7:

Digestible indispensable amino acid score (DIAAS) is the best indicator for protein quality.

DIAAS reflects the digestibility of individual amino acids in the protein. Total protein digestibility does not always reflect the digestibility of individual dietary indispensable amino acids. Hence, DIAAS has replaced the PDCAAS (protein digestibility corrected amino acid score) as the preferred index for the evaluation of protein quality.

Solution to Question 8:

Among the given options, meat has the highest net protein utilization (NPU).

The NPU values of various food items in descending order are:

- Egg - 100
- Meat - 80
- Milk - 75
- Rice - 65
- Soya - 61

Low NPU indicates a high protein requirement. The NPU of Indian diets is usually between 50 and 80.

Solution to Question 9:

His diet would be lacking lysine and tryptophan as maize is deficient in these amino acids.

Principal Food	Amino acid Deficient
Cereals (in general)	Lysine
Maize	Lysine and Tryptophan
Wheat	Lysine and Threonine
Pulses (Poor man's meat)	Methionine and cysteine

Solution to Question 10:

Safflower oil contains the highest content of essential fatty acids.

Linoleic acid is the most important essential fatty acid.

Milk is also rich in linoleic acid and oleic acid.

Dietary source	% content of linoleic acid
Safflower oil	75
Corn oil	50
Sunflower oil	62
Soybean oil	53

Solution to Question 11:

The patient should be advised to eat whole grains since it has a low glycemic index (GI).

GI is a number representing the relative ability of a carbohydrate food to increase the level of glucose in the blood. It is calculated by comparing the test carbohydrate to the standard (glucose or white bread). This measure has practical use in the management of diabetes and obesity.

- Low GI foods – most fruits and vegetables, whole grains, pasta foods, beans, and lentils.
- Moderate GI foods – sucrose, basmati rice, and brown rice.
- High GI foods – corn flakes, baked potatoes, white bread, potatoes, watermelon, candies, and syrupy foods.

Solution to Question 12:

The renal feed of a patient suffering from end-stage renal disease should ideally be low volume, high calories.

Dietary factors may have an effect on the progression of kidney disease and its complications. A low-volume feed is preferred in order to avoid:

- Sodium and volume overload
- Hyperkalemia
- Hyperphosphatemia
- Accumulation of toxic metabolites of protein degradation

A high-calorie feed is preferred in order to combat the patient's risk of malnutrition and wasting seen in ESRD.

Solution to Question 13:

Breslow's staging is used for the staging of malignant melanoma, not for the assessment of body fat.

Solution to Question 14:

In the Indian (Asian) population, a BMI of 23-24.99 is classified as overweight.

However, in the global population, a BMI of 25-29.99 is considered overweight.

Solution to Question 15:

A body mass index (BMI) of 13 or less is considered lethal in men.

However, females can survive even at a lower BMI up to 11.

Solution to Question 16:

Waterlow's classification makes use of height for age (HFA) and weight for height (WFH) to assess the nutritional status of children.

HFA indicates stunting and denotes the chronicity of malnutrition. On the other hand, WFH indicates wasting and denotes the severity of malnutrition.

The Gomez classification uses weight for age as an indicator of nutritional status. The disadvantage is that it is difficult to determine whether the low weight is due to an acute attack of malnutrition or chronic undernutrition.

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Micronutrients and Water

Question 1:

A patient presents with the following skin finding. You would advise supplementation with which of the following?



- a) Safflower oil only
- b) Safflower oil and vitamin K
- c) Vitamin B complex and vitamin K
- d) Safflower oil and vitamin B complex

Question 2:

Which of the following has the highest content of thiamine?

- a) Whole wheat
- b) Milled rice
- c) Ground nut
- d) Gingelly seeds

Question 3:

A patient presented with loose stools and skin rashes over the sun-exposed areas. Diet history revealed that the patient had been consuming a jowar-rich diet since childhood. This condition results due to excess of which amino acid in the diet?

- a) Lysine
- b) Leucine
- c) Tryptophan
- d) Methionine

Question 4:

What is the recommended daily allowance of vitamin A for a 9-month old baby?

- a) 350 mcg
- b) 600 mcg
- c) 800 mcg
- d) 1000 mcg

Question 5:

What is the dosage of vitamin A given for a 2-year-old baby with keratomalacia?

- a) 2,00,000 IU immediately, followed by same dose after 24 hours
- b) 1,00,000 IU immediately, followed by the same dose after 1 week
- c) 1,00,000 IU immediately, followed by the same dose after 24 hours and after a week
- d) 2,00,000 IU immediately, followed by the same dose 24 hours later and after 2 weeks

Question 6:

Vitamin A deficiency is considered a public health problem if the prevalence of corneal xerosis among children aged 6 months - 6 years is more than:

- a) 5%
- b) 1%
- c) 0.5%
- d) 0.01%

Question 7:

The dose of iodine for lactating women in India according to the recommended dietary allowance is:

- a) 200 µg/day
- b) 220 µg/day
- c) 250 µg/day
- d) 280 µg/day

Question 8:

Which of the following is incorrect regarding the epidemiological assessment of iodine deficiency?

- a) Median urinary iodine excretion < 200 µg/ L indicates insufficient iodine intake
- b) Neonatal hypothyroidism is a sensitive indicator of environmental iodine deficiency
- c) Serum T4 level is a more sensitive indicator of thyroid insufficiency than T3
- d) Quality control of salt is performed by rapid testing with MBI kit

Question 9:

What is the recommended dietary allowance of folic acid for a 5-year-old child?

- a) 40µg
- b) 135µg
- c) 200µg
- d) 500µg

Question 10:

Under the Anemia Mukht Bharat programme, what dose of iron and folic acid tablets will you prescribe for a 6-year-old child?

- a) 20 mg iron and 100 mcg folic acid
- b) 45 mg iron and 400 mcg folic acid
- c) 50 mg iron and 350 mcg folic acid
- d) 100 mg iron and 500 mcg folic acid

Question 11:

How much calcium does a premenopausal woman require on a daily basis?

- a) 1000 mg/d
- b) 800 mg/d
- c) 600 mg/d
- d) 1200 mg/d

Question 12:

Individuals with fluoride intake above which of the following are at risk of developing crippling fluorosis?

- a) ≥ 6 mg/L
- b) ≥ 1.5 mg/L
- c) ≥ 3 mg/L
- d) ≥ 10 mg/L

Question 13:

What is the daily recommended intake of water for an elderly woman who is moderately active ?

- a) 27ml/kg
- b) 38ml/kg
- c) 43ml/kg
- d) 25ml/kg

Question 14:

What is the adequate daily requirement of water for a 12-year-old girl?

- a) 2.4 litres
- b) 2.7 litres
- c) 2.1 litres

d) 1.5 litres

Question 15:

A 45-year-old man visits your OPD for a general health check-up. His wife informs that he drinks only 4-5 glasses of water a day. What is the daily adequate intake of water that you would advise him to take according to the WHO guidelines?

- a) 3.7 litres
- b) 2.7 litres
- c) 2.2 litres
- d) 2 litres

Question 16:

What is the normal daily requirement of magnesium in adults?

- a) 100 mg/day
- b) 200 mg/day
- c) 400 mg/day
- d) 500 mg/day

Answer Key

Question No.	Correct Option
1	d
2	d
3	b
4	a
5	d
6	d
7	d
8	a
9	b
10	b

11	a
12	d
13	b
14	c
15	a
16	c

Detailed Explanations

Solution to Question 1:

The image shows rough, dry skin suggestive of phrynoderma. It is treated by supplementation of safflower oil (rich in essential fatty acids) and vitamin B complex.

Phrynoderma or toad skin is a type of follicular hyperkeratosis. It is caused by a deficiency of vitamin E, B, A, C, and essential fatty acids.

Solution to Question 2:

Gingelly (sesame) seeds have the highest content of thiamine.

The thiamine content in various foods in decreasing order are as follows:

- Gingelly seeds
- Groundnut
- Bengal gram
- Whole wheat
- Raw rice

Thiamine is lost from rice during the process of milling. As it is a water-soluble vitamin, washing, and cooking rice cause further losses. Hence, it is advisable to avoid highly polished rice and eat parboiled or under-milled rice.

Solution to Question 3:

This clinical scenario is suggestive of pellagra. It is caused by the excess of leucine in jowar.

Excess of leucine in jowar (sorghum) and maize interferes with the conversion of tryptophan to niacin causing pellagra. It presents with the following features:

- Diarrhea
- Dermatitis (sun-exposed areas)

- Dementia

People consuming jowar also have a higher risk of developing genu valgum and osteoporosis due to greater retention of ingested fluoride.

Solution to Question 4:

The RDA of vitamin A for infants is 350 mcg.

Solution to Question 5:

The recommended vitamin A dosage for a 2-year-old child with keratomalacia is 2,00,000 IU given immediately, followed by the same age-specific dose 24 hours later and after 2 weeks.

Solution to Question 6:

Vitamin A deficiency is considered a public health problem if the prevalence rate of corneal xerosis among children between 6 months to 6 years (high-risk category) is more than 0.01%.

The presence of any 1 of the following criteria should be considered as evidence of a xerophthalmia problem in the community:

Criteria	Prevalence in population at risk (6 months - 6 years)
Night blindness	> 1%
Bitot's spots	> 0.5%
Corneal xerosis / Corneal ulcer / Keratomalacia	> 0.01%
Corneal scar	> 0.05%
Serum retinol < 10 mcg/dl	>5 %

Solution to Question 7:

According to the National Institute of Nutrition (NIN) 2020 guidelines, the recommended dietary allowance (RDA) for dietary iodine in lactating women in India is 280 µg/day.

NIN guidelines on recommended dietary allowance for iodine per day:

Note: According to the WHO guidelines, the RDA during both pregnancy and lactation is 250 mcg.

Age	Male	Female
Birth to 6 months	100 mcg	100 mcg
7–12 months	130 mcg	130 mcg
1–9 years	90-120 mcg	90-120 mcg
10–12 years	150 mcg	150 mcg
14–18 years	150 mcg	150 mcg
> 19 years	150 mcg	150 mcg
Pregnancy	—	250 mcg
Lactation	—	280 mcg

Solution to Question 8:

Median urinary iodine excretion level $< 100 \mu\text{g/L}$ (in non-pregnant population) indicates insufficient iodine intake.

Urinary excretion of iodine is used by the WHO for surveillance and epidemiological studies to classify countries according to the level of iodine deficiency. It is an effective indicator of national programmes against iodine deficiency.

Neonatal hypothyroidism is a sensitive indicator of environmental iodine deficiency. It is also an effective indicator to monitor the impact of a programme. Serum T₄ level is a more sensitive indicator of thyroid insufficiency in the population than T₃.

Determination of iodine in salt is a part of the quality control of the product. The UNICEF uses MBI kits to rapidly assess the levels of iodine in salt.

Solution to Question 9:

The recommended dietary allowance of folic acid in a 5-year-old child is $135 \mu\text{g}$.

Solution to Question 10:

Under the Anemia Mukht Bharat programme, a 6-year-old child should be given 45 mg elemental iron + 400 mcg folic acid (1 iron and folic acid tablet weekly). The tablet is sugar-coated and pink in color.

Solution to Question 11:

A premenopausal woman requires 1000 mg calcium on a daily basis.

Solution to Question 12:

Individuals with fluoride intake > 10 mg/L are at risk of developing crippling fluorosis.

Complications of fluorosis seen with corresponding levels of fluoride intake are:

- Dental fluorosis - 1.5 mg/L
- Skeletal fluorosis - 3-6 mg/L
- Crippling fluorosis - > 10 mg/L

Genu valgum may occur in people who consume jowar (sorghum), as it promotes greater retention of ingested fluoride.

The Nalgonda technique is used for the defluoridation of water bodies. It involves the addition of lime and alum followed by flocculation, sedimentation, and filtration.

Solution to Question 13:

A moderately active elderly woman requires 38ml of water/kg/day.

Water requirement	
Adult	Male 32-58 ml/kg Female 27-52 ml/kg
Children	> 60 ml/kg
Adolescent	40-60 ml/kg
Pregnant woman	2.1 to 3.2 litres/day
Elderly	Sedentary 33ml/kg moderately active 38ml/kg.

Solution to Question 14:

According to WHO guidelines, for a 12-year-old girl, the adequate daily intake of water is 2.1 litres.

Age Group	Daily Adequate Water Intake (Litres/Day)
0 – 6 months	0.7, assumed to be from human milk.
7 – 12 months	0.8, assumed to be from human milk and complementary foods and beverages
1 – 3 years	1.3
3 – 8 years	1.7
9 – 13 years Boys Girls	2.4 2.1
14 – 18 years Boys Girls	3.3 2.3
19 – 70+ years Men Women	3.7 2.7

Solution to Question 15:

According to WHO guidelines, for an adult male, the daily adequate intake of water is 3.7 litres.

Age Group	Daily Adequate Water Intake (Litres/Day)
0 – 6 months	0.7, assumed to be from human milk.
7 – 12 months	0.8, assumed to be from human milk and complementary foods and beverages
1 – 3 years	1.3
3 – 8 years	1.7
9 – 13 years Boys Girls	2.4 2.1
14 – 18 years Boys Girls	3.3 2.3
19 – 70+ years Men Women	3.7 2.7

Solution to Question 16:

The normal daily requirement of magnesium in adults is 400 mg/day

The following are the approximate daily requirements of some electrolytes in adults:

- Sodium: 1.2 to 1.5 grams (g)/day (up to 2g/day)

- Potassium: 3500 mg/day
- Calcium: 1000-1200 mg/day
- Magnesium: 400 mg/day

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Food Quality and Processing

Question 1:

Which of the following formulates food standards for the international food market?

- a) ISI standards
- b) AGMARK standards
- c) Codex Alimentarius
- d) PFA standards

Question 2:

What is the dose of vitamin A added during fortification of vanaspati?

- a) 1500IU
- b) 2500IU
- c) 1400IU
- d) 2000IU

Question 3:

Which of the following contributes to a significant loss of beta carotene from food?

- a) Boiling
- b) Canning
- c) Freezing
- d) Dehydration

Question 4:

In order to effectively prevent endemic goiter, what is the recommended iodine content in salt at the consumer level?

- a) 10 ppm
- b) 15 ppm

- c) 20 ppm
- d) 30 ppm

Question 5:

In a milk factory, the milk is heated at 63°C for 30 minutes and then cooled quickly to less than 13°C . Which bacteria can survive this method?

- a) Salmonella typhii
- b) Mycobacterium bovis
- c) Coxiella burnetti
- d) Brucella melitensis

Question 6:

Which of the following tests is not used to check the efficiency of pasteurization?

- a) Phosphatase test
- b) Standard plate count
- c) Methylene blue reduction test
- d) Coliform count

Question 7:

In a rice processing factory, paddy is soaked in hot water. After this, the water is drained and the paddy is steamed and dried. Which of the following statements is false about the above process?

- a) This method is used for partial cooking of rice
- b) It makes the rice grains resistant to insects
- c) Vitamins and minerals are driven into endosperm during this process
- d) It softens the rice grains

Question 8:

A 35-year-old man presented with gradually progressive muscle weakness and leg spasms for 2 months. He says that he had been consuming an inexpensive dal resembling red gram for

the past 2 years. Pure motor paresis was seen on examination. Which of the following could have prevented this?

- a) Vit A
- b) Vit B12
- c) Vit C
- d) Vit D

Question 9:

A 45-year-old woman is being evaluated for sudden onset of hypersecretory glaucoma, diarrhea, and cardiac failure. Further probing reveals that she had been purchasing groceries from a dealer who was recently arrested for food adulteration. What toxin is likely responsible for her presentation?

- a) BOAA
- b) Sanguinarine
- c) Aflatoxins
- d) Pyrrolizidine alkaloids

Answer Key

Question No.	Correct Option
1	c
2	b
3	d
4	b
5	c
6	c
7	d
8	c
9	b

Detailed Explanations

Solution to Question 1:

The Codex Alimentarius Commission formulates food standards for the international market. It is the principal organ of the FAO/WHO food standards program.

The ISI standards (Indian Standards Institute) and AGMARK are the Indian food standards. They are based on the standards of Codex Alimentarius.

The PFA (Prevention of Food Adulteration) Act was introduced to provide protection from adulteration or contamination of food.

Solution to Question 2:

Vanaspati is fortified with 2500 IU of vitamin A and 175 IU of vitamin D.

Note: 1g of double fortified salt contains 40 mcg iodine and 1 mg iron.

Solution to Question 3:

Dehydration contributes to a significant loss in the beta carotene levels in food.

Processes that do not affect beta carotene levels in the food are:

- Boiling
- Canning
- Freezing

Solution to Question 4:

The recommended level of iodine content in salt is 15 ppm at the consumer level and 30 ppm at the production level.

The iodization of salt is used as a prophylactic measure against endemic goiter. In India, the level of iodization is fixed under the Prevention of Food Adulteration (PFA) Act.

Solution to Question 5:

The method described above is the holder method of pasteurization. *Coxiella burnetti* can survive this method since it is highly heat-resistant. It causes Q fever.

The various methods of pasteurization are:

- Holder/Vat method - 63-66 °C for 30 minutes
- HTST method (high-temperature short time) - 72°C for 15 seconds

- UHT method (ultra-high temperature) - 125 °C for a few seconds

Solution to Question 6:

Methylene blue reduction test is not used to check the efficiency of pasteurization. This test is used for grading raw milk quality prior to pasteurization.

The following tests are used to check the efficiency of pasteurization:

- Phosphatase test
- Standard plate count
- Coliform count

Solution to Question 7:

The procedure mentioned above is parboiling. It hardens the rice grains.

Parboiling is the process of partial cooking of rice in steam. It is done by the hot soaking process that was initially developed by the Central Food Technological Research Institute (CFTRI), Mysore.

The advantages of parboiling are:

- Preserves nutritive quality by driving the vitamins and minerals into the endosperm.
- Hardens the grains and makes them insect resistant.

Solution to Question 8:

The given clinical features are suggestive of neurolathyrism. It can be prevented and treated by the administration of vitamin C at doses of 500-1000 mg.

Lathyrism is a paralyzing disease of humans and animals. It is due to the consumption of khesari dal or Lathyrus sativus. When the husk is removed, it looks similar to red gram or Bengal gram.

BOAA (beta oxalyl amino alanine) is the toxic compound causing this disease. It is water-soluble and can be removed by the following methods:

- Soaking pulses in hot water and discarding the water
- Overnight soaking in lime water followed by boiling
- Parboiling - for large-scale detoxification

Solution to Question 9:

This clinical scenario is suggestive of epidemic dropsy. It is caused by the toxic compound sanguinarine, which is present in argemone oil. It is commonly used as an adulterant in mustard oil.

Epidemic dropsy results in sudden onset of swelling of lower limbs, diarrhea, and breathlessness. Some patients may develop hypersecretory glaucoma. It can lead to death due to cardiac failure.

Adulteration with argemone oil can be detected by the following tests:

- Paper chromatography - most sensitive test
- Nitric acid test

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Concepts of Sociology and Psychology

Question 1:

What is meant by the term sociology?

- a) Study of developments in the society
- b) Study of formation of social organisations and government
- c) Study of human relationships and behavior
- d) Study of social and cultural history of man

Question 2:

Which of the following social sciences does not deal directly with human behavior?

- a) Political science
- b) Anthropology
- c) Social psychology
- d) Sociology

Question 3:

Match the following:

- a) 1-i, 2-iv, 3-ii, 4-iii
- b) 1-i, 2-ii, 3-iii, 4-iv
- c) 1-ii, 2-i, 3-iv, 4-iv
- d) 1-iv, 2-iii, 3-i, 4-ii

Question 4:

'Putting profit ahead of health' as a cause of disease belongs to which theory of sociology of health?

- a) Eliasan
- b) Marxist

- c) Foucauldian
- d) Parsonian

Question 5:

Which of the following refers to a system of social relationships between individuals?

- a) Family
- b) Community
- c) Society
- d) Social institution

Question 6:

Which of the following options is correctly matched?

- a) Social institution- the pattern of inter-relationships between persons in a society
- b) Socialism- organised complex patterns of behaviour in people with a common interest
- c) Social structure- system of governance based on social ownership
- d) Socialization- the process by which a person imbibes the social beliefs and customs

Question 7:

Which of the following terms is used to describe the preventive, therapeutic, and rehabilitative services for the protection of society from antisocial, criminal, or deviant conduct of man?

- a) Social defence
- b) Social medicine
- c) Social democracy
- d) Social dynamics

Question 8:

Match the following.

- a) 1-ii, 2-i, 3-iii, 4-iv

- b) 1-ii, 2-i, 3-iv, 4-iii
- c) 1-iv, 2-i, 3-iii, 4-ii
- d) 1-iv, 2-ii, 3-iii, 4-i

Question 9:

Which of the following does not bring about acculturation?

- a) Trade and commerce
- b) Conquest
- c) Religious propagation
- d) Folklore

Question 10:

Which of the following is the correct ascending order of the Maslow's hierarchy of needs?

- a) Safety, self-actualization, belonging, esteem, physiological
- b) Safety, physiological, belonging, esteem, self-actualization
- c) Belonging, physiological, safety, self-actualization, esteem
- d) Physiological, safety, belonging, esteem, self-actualization

Question 11:

Which of the following could be classified as an ego-integrative need of man?

- a) Fellowship
- b) Love and affection
- c) Money
- d) Prestige

Question 12:

What is the benefit that comes from solving one's own problems termed?

- a) Intrinsic incentive
- b) Material incentive

- c) Interior incentive
- d) Internalization

Question 13:

Which of the following is false about attitudes in human behaviour?

- a) It is acquired from social interactions
- b) It includes an affective or feeling element
- c) It pre-disposes one to respond in a preferential manner
- d) It is a temporary and fleeting aspect of human behaviour

Question 14:

Which of the following terms represent the temporary and provisional views held by people at a point of dispute?

- a) Interests
- b) Beliefs
- c) Opinions
- d) Thoughts

Question 15:

Match the following types of learning:

- a) 1 – B, 2 – A, 3 – C
- b) 1 – B, 2 – C, 3 – A
- c) 1 – A, 2 – C, 3 – B
- d) 1 – A, 2 – B, 3 – C

Question 16:

A 6-yr-old boy is brought to the clinical psychologist for evaluation of intellectual disability. Which formula will she use to calculate the IQ of the child?

- a) $\text{Chronological age} - \text{mental age} / 100$

- b) $\text{Mental age} \times 100 / \text{Chronological age}$
- c) $\text{Chronological age} \times 100 / \text{Mental age}$
- d) $(\text{Chronological age} - \text{mental age}) \times 100$

Question 17:

Evaluation for mental retardation was done for a Down syndrome patient. His IQ was found to be 56. According to Levine and Mark's classification, you would classify him as?

- a) Idiot
- b) Imbecile
- c) Moron
- d) Dull

Question 18:

A clinical psychologist was asked to ascertain the IQ of a patient. She categorized the patient as having mild mental retardation. Which of the following represent's the patient's IQ range?

- a) 70 – 85
- b) 50 – 70
- c) 35 – 50
- d) 20 - 35

Question 19:

According to Wechsler's scale, which of the following persons would be categorized to have average intelligence?

- a) A child whose IQ is 125
- b) A teacher whose IQ is 115
- c) A banker with an IQ of 95
- d) An alcoholic man with an IQ of 85

Question 20:

Which of the following is an important epidemiological tool to assess disability in children?

- a) Activities of Daily Living scale
- b) Wing's Comprehensive Handicaps, Behaviour and Skills Schedule
- c) Binet and Simon's IQ test
- d) Physical Quality of Life Index

Answer Key

Question No.	Correct Option
1	c
2	a
3	a
4	b
5	c
6	d
7	a
8	c
9	d
10	d
11	d
12	a
13	d
14	c
15	b
16	b
17	c
18	b
19	c
20	b

Detailed Explanations

Solution to Question 1:

Sociology is considered to be both the study of human relationships and of human behavior.

Solution to Question 2:

Political science refers to the study of the judicial systems and institutions that constitute the government. It does not deal directly with human behavior.

Anthropology is the study of the physical, social, and cultural history of man. Social anthropology is extensively used in preventive and social medicine.

Social psychology is aimed at understanding the basis of perception, thoughts, ideas, etc., among people in a society or group.

Sociology is the study of human relationships and human behavior.

Solution to Question 3:

Social pathology refers to the study of social factors, which influence human disease and incite social disorganization. This includes crime, poverty, delinquency, abuse, etc.

Social medicine is an organized investigation of the social, environmental, and genetic factors affecting disease patterns and disability. It also deals with the promotion of methods for the prevention of disease and health measures.

Socialized medicine is the provision of medical service and education by the State. It is operated and regulated by professional groups, and not the government.

Social psychology is the effect of the social environment on people, their attitudes, and motivation. It deals with the study of human behavior, thoughts, and feelings and how they are influenced by the perception of others in society.

Solution to Question 4:

Putting profit ahead of health is termed as the cause of disease in the Marxist theory of sociology of health.

Elias' theory focused on the relationship between power, behavior, emotion, and knowledge over time.

Theory	Marxist	Parsonian	Foucauldian
Description	Emphasized causal role of economics in disease and the role of medical knowledge in sustaining class structure	Emphasized role of medicine in maintaining social harmony	Emphasized the role of helping professions in inducing individuals to comply with social roles
Model of Society	Conflictual and exploitative	Harmonious and stable set of interlinked social roles and structures	Net of power relations with no one dominant source

Theory	Marxist	Parsonian	Foucauldian
Cause of Disease	Putting profit ahead of health	Social strain caused by meeting the demands of social roles	Diseases are labels used to sort and segregate the population to make it easier to control
Role of Medical Professional	To discipline and control the working class To provide individualized explanations of the disease	Rehabilitate individuals to carry out their social roles	To enforce compliance with the normal social roles and to ensure internalization of these norms.

Solution to Question 5:

Society is the system of social relationships between people in an area.

A community is a social group determined by geographical boundaries and/or common values and interests.

A family is a group of biologically related individuals living together under the same roof and eating from a common kitchen.

Social Institutions are organized complex patterns of behavior in which a number of people engage to further a common interest.

Solution to Question 6:

Socialization is the process by which a person in a society imbibes its beliefs, customs, etc. through daily interaction with its members.

Social institutions are organized complex patterns of behaviour in which a number of people engage to further a common interest. Examples are families, schools, hospitals, etc.

Socialism is a system of governance in which production and distribution is based on social ownership (and not private ownership). It is aimed at improving the living standard.

Social structure is the pattern of inter-relations between persons in a society.

Solution to Question 7:

Preventive, therapeutic and rehabilitative services for the protection of society from antisocial, criminal or deviant conduct of man is known as social defence.

This is in the form of acts, laws and judicial instructions for elements like juvenile delinquency, begging, prison reforms, gambling, etc.

Solution to Question 8:

Culture is the socially acquired learned behaviour. It is passed onto generations through formal and informal learning. It is the representation of customs, beliefs, laws, arts, and moral values of a society. Cultural factors have a large part to play in health.

Customs are the traditionally accepted ways of behaving that is specific to a society. They can be folkways, which are referred to as the right ways of doing things in less vital areas of human conduct. Mores are stringent customs followed by a society.

Standard of living refers to the scale of the expenditure of a society, the goods they consume and the services they enjoy.

Role is the function or duties assumed by a person of a particular society.

Solution to Question 9:

Acculturation is the process by which there is the diffusion of cultures between two culturally different groups of people. Acculturation does not take place by folklore. Folklore is the passage of customs and beliefs of a community through generations, merely by word of mouth.

Acculturation can happen in the following ways:

- Trade and commerce
- Industrialization
- Religious propagation
- Education
- Conquest

This phenomenon is also known as culture contact.

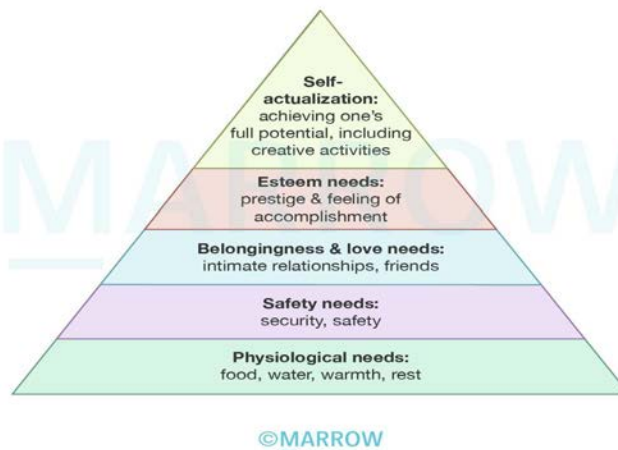
Solution to Question 10:

Maslow's hierarchy of needs was suggested by psychologist Abraham Maslow. It is a motivational theory in psychology where human needs are arranged according to hierarchy in a five-level pyramid.

He suggested that the higher needs can be fulfilled only once the lower needs are fulfilled.

The ascending order of the needs is shown in the image below.

Maslow's hierarchy of needs



Solution to Question 11:

Ego-integrative needs of man include power, prestige and self-respect.

The needs of man are as follows:

- Biological needs- water, food, warmth, medicines, etc.
- Social needs- love and affection, fellowship, education, etc.
- Economic needs- money and security
- Ego-integrative needs- power, prestige and self-respect.

Solution to Question 12:

Intrinsic incentives are the benefits that come from solving one's own problems.

Incentives are entities that motivate or encourage a specific behaviour. They are generally positive motivators. The various types of incentives are:

- Intrinsic incentives are the benefits that come from solving one's own problems.
- Extrinsic incentives are rewards that do not relate directly to the goal of the desired behaviour like financial compensation for sterilization.
- Psychological incentives include satisfaction, increased self-esteem, etc.
- Material incentives include goods or services or financial compensation.

Solution to Question 13:

Attitude is usually a permanent aspect of a human's behaviour and is difficult to change.

It is the enduring organization of beliefs around something that pre-disposes one to respond in a preferential manner. It is usually acquired through social interactions.

Attitude includes three components:

- A cognitive or knowledge element
- An affective or feeling element
- A tendency to act

Solution to Question 14:

Opinions are temporary and provisional views held by people at a point of dispute.

Beliefs are permanent and stable inner subjective thoughts of a person towards things that are without proof. These are usually acquired from family and other close contacts.

Solution to Question 15:

Learning is any relative or permanent change in behaviour as a result of practice or experience. It includes acquiring knowledge and skills, formation of habits and development of perception.

Types of learning:

- Cognitive learning – knowledge
- Affective learning – attitudes
- Psychomotor learning – skills

The conditions that affect learning are as follows:

- Intelligence
- Age
- Learning situation
- Motivation
- Physical and mental health

Theories of learning:

- Learning by conditioned reflex
- Trial and error
- Learning by observation and imitation
- Learning by doing
- Learning by remembering
- Learning by insight

- Demonstration
- Field experience
- Problem solving

Solution to Question 16:

The intelligence quotient is a score that is obtained from several standardized tests to assess human intelligence.

Intelligence quotient (IQ) = Mental age x 100 / Chronological age

Solution to Question 17:

According to the Levine and Mark's classification, a moron would have an IQ of 50–74.

Solution to Question 18:

Mild mental retardation based on IQ score would fall in the range of 50–70.

Solution to Question 19:

According to the Wechsler's Scale, average intelligence would fall in the range of 91–110.

Solution to Question 20:

Wing's Comprehensive Handicaps, Behaviour and Skills Schedule (HBS) has been used in epidemiological studies to assess the total child population in terms of detailed scales of specific abilities and disabilities.

HBS is not useful for children with mental retardation.

Social Organization and Economics

Question 1:

Which of the following is the most powerful example of social cohesion in social organisation?

- a) The village
- b) The family
- c) The state
- d) The community

Question 2:

Which of the following is an unstable and emotional temporary social group with a leader?

- a) Crowd
- b) Band
- c) Herd
- d) Mob

Question 3:

What is the correct order of a family cycle?

- a) Formation, extension, complete extension, dissolution, contraction, complete contraction
- b) Formation, contraction, extension, complete extension, dissolution, complete dissolution
- c) Formation, contraction, complete contraction, extension, complete extension, dissolution
- d) Formation, extension, complete extension, contraction, complete contraction, dissolution

Question 4:

What are the constituents of a nuclear family?

- a) Father, mother, son and his wife
- b) Husband, wife and 4 year old daughter

- c) Husband and wife only, without their children
- d) Grandfather, father, mother and daughter

Question 5:

Match the following:

- a) A-1, B-2, C-3, D-4, E-5
- b) A-4, B-1, C-3, D-5, E-2
- c) A-1, B-2, C-3, D-5, E-4
- d) A-4, B-1, C-5, D-3, E-2

Question 6:

In urbanized and industrialized countries, occupation is a major determinant of all of the following except:

- a) Extent of authority
- b) Values and lifestyle
- c) Extent of obligation
- d) Community participation

Question 7:

Which of the following is not a limitation of classifying social class according to occupation?

- a) Heterogeneous grouping of different occupations
- b) Inequality between men and women in same jobs
- c) Rarely taking skills into consideration
- d) Occupational mobility

Question 8:

You are tasked with conducting a health and wellness survey of the employees of your hospital. Which of the following scales will you not use to classify their socio-economic statuses?

- a) Jalota scale

- b) Kulshetra scale
- c) Likert scale
- d) Hollingshead scale

Question 9:

As a part of a community health survey team, you interview 50 families from a nearby village in your district. Which of the following scales would you use to classify the socio-economic statuses of these families?

- a) Uday Pareek scale
- b) Adson's scale
- c) Kuppuswamy scale
- d) Hollingshead scale

Question 10:

Which of the following components are included under the modified Kuppuswamy scale for socio-economic status ?

- a) 1,2 and 3
- b) 1,3 and 4
- c) 1,2 and 4
- d) 3,4 and 5

Question 11:

Match the following scores in accordance with Modified Kuppuswamy's scale.

- a) 1-a, 2-d, 3-c, 4-b
- b) 1-a, 2-d, 3-b, 4-c
- c) 1-d, 2-b, 3-a, 4-c
- d) 1-d, 2-a, 3-b, 4-c

Question 12:

In the doctor-patient relationship, which of the following is not a level of communication?

- a) Communication on an emotional plane
- b) Communication on an intellectual plane
- c) Communication on a psychological plane
- d) Communication on a cultural plane

Question 13:

Which of the following is the best suited role for a medical social worker?

- a) Physiotherapy and drug administration
- b) Designing coping strategies, interviewing and case work
- c) Providing economic support to patients
- d) Treating patients

Question 14:

Which of the following practices is covered under the COPRA act?

- a) Children and old age population
- b) Observation of criminals and ex-convicts
- c) Corporate and office practices
- d) Consumer rights

Question 15:

Which of the following is used to describe an interview used to note the gradual influence of some social or psychological process?

- a) Structured interview
- b) Non directive interview
- c) Focused interview
- d) Repetitive interview

Question 16:

Arrange the following steps of an interview in the correct order in a sequence.

- a) Starting an interview, making a report, establishing contact, securing rapport
- b) Establishing contact, making a report, starting an interview, securing rapport
- c) Establishing contact, starting an interview, securing rapport, making a report
- d) Starting an interview, securing rapport, making a report, establishing contact

Question 17:

What is the application of scientific methods to investigate the study of complex human organizations, services and management called?

- a) Operational research
- b) Applied research
- c) Pure research
- d) Anthropological research

Question 18:

What was the purpose of setting up the Nirbhaya Fund?

- a) Promoting safety of women
- b) Promoting literacy among women
- c) Promoting self-help groups among women
- d) Providing free medical care for women

Question 19:

Choose the correct match.

- a) Gross Domestic Product- gross income generated only from within the country
- b) Gross National Product- gross domestic product minus the depreciation on a country's capital goods
- c) Net National Product- income generated from within the country and income received from abroad
- d) Net Domestic Product- market value of all final goods and services in a country during a year after allowing for depreciation

Question 20:

What percentage of GDP was spent on public health in 2020?

- a) 1.25%
- b) 4.70%
- c) 3.21%
- d) 2.50%

Question 21:

A mother brings her 4-year-old boy for a well-baby check-up. Upon examination the paediatrician diagnoses her son with hidden hunger. What is meant by the term hidden hunger?

- a) Child's diet is deficient in carbohydrates
- b) Child suffers from micronutrient deficiency
- c) Child is taking a diet that is deficient in essential amino acids
- d) Child suffers from deficiency of omega 3 fatty acids

Question 22:

Which of the following statements is correct with regards to the estimation of poverty line?

- a) Per capita expenditure of Rs. 32 per day in urban areas
- b) Per capita expenditure of Rs. 47 per day in rural areas
- c) Daily calorie intake of 2100 calories in urban areas
- d) Daily calorie intake of 2700 calories in rural areas

Question 23:

Which of the following legislations does not offer social security for industrial workers?

- a) Workmen's Compensation Act
- b) Family Pension Scheme
- c) Central Maternity Benefit Act
- d) Labour Charter Act

Question 24:

Which of the following is not a component of the global hunger index (GHI)?

- a) Infant mortality rate
- b) Under 5 mortality
- c) Percentage of undernourished population
- d) Percentage of undernourished children under 5 years

Answer Key

Question No.	Correct Option
1	b
2	d
3	d
4	b
5	d
6	d
7	c
8	c
9	a
10	b
11	c
12	c
13	b
14	d
15	d
16	c
17	a
18	a
19	a
20	a
21	b
22	c
23	d
24	a

Detailed Explanations

Solution to Question 1:

The most powerful example of social cohesion in social organization is the family.

It is the most basic and primary unit in all societies.

Solution to Question 2:

A mob is an unstable and emotional temporary social group with a leader.

Temporary social groups can be classified into:

- Crowd- It is a temporary group motivated by a common interest or curiosity. It lacks an internal organization and has no leader.
- Mob- It is a crowd with a leader, and is more emotional than a crowd. It is unstable and has no internal organization.
- Herd- This social group has a leader and the members have to follow his/her orders without question.

A band is a permanent spatial group that constitutes a few families living together. Other permanent spatial groups are villages, cities, towns and the state.

Solution to Question 3:

The order of the family cycle is formation, extension, complete extension, contraction, complete contraction and dissolution.

Solution to Question 4:

A nuclear family consists of a married couple and their children when they are dependant.

They are also called elementary families.

Solution to Question 5:

The correct match is A-4, B-1, C-5, D-3, E-2.

Term	Definition
Problemfamily	A family whichlagsbehind the rest of the community

Term	Definition
Communal family	A family in which all the members play a part in its management
Broken family	A family in which the parental system has been disrupted by separation, divorce or death
New family	A nuclear family which is less than 10 years consisting of parents and children
Joint family	A number of married couples and their children who live together in the same household with usually the men being related by blood

Solution to Question 6:

Occupation is not a determinant of community participation.

Occupation in most urbanized and industrialized communities is a major determinant of:

- Economic rewards
- Extent of authority
- Extent of obligations
- Degree of status
- Values and lifestyles

Solution to Question 7:

Most classifications of social class based on occupational take skills into consideration.

Occupation is a useful measure of social differentiation particularly in communities highly industrialised and urbanized.

Limitations of occupational classification of social class:

- Heterogenous grouping- occupations grouped together are not usually related to each other
- Occupational mobility- occupations practiced by the members of a social class as well as the perception of a certain occupation in the hierarchy may change over a period of time
- Gender biases- the same occupation may be considered as belonging to a superior class for men as compared to women

Solution to Question 8:

Likert scale is not a scale used for socio-economic status (SES). It is a psychometric scale used to understand the attitude of people from their responses. It is a scale widely used in questionnaire-based research.

Scales that have been used for SES are:

- Udai Pareek scale
- Kuppuswamy scale
- Radhukar scale
- Jalota scale
- Shirvastava scale
- Bharadwaj scale
- Kulshetra scale
- Hollingshead scale.

Solution to Question 9:

The Uday Pareek scale is used for the classification of socio-economic status in rural areas.

Solution to Question 10:

The modified Kuppuswamy scale uses income of the family, education and occupation of head of family.

Solution to Question 11:

The correct match is 1-d, 2-b, 3-a, 4-c.

Socioeconomic class according to Modified Kuppuswamy Scale:

Socioeconomic class	Scores
Upper	25-29
Upper Middle	16-25
Lower Middle	11-15

Socioeconomic classes	Scores
Upper	05-10
Lower	01-04

Solution to Question 12:

Communication on a psychological plane is not a level of communication in a doctor-patient relationship.

For a successful doctor-patient relationship, a doctor must have in his/her regard communication on the following levels:

- Communication on an emotional plane: To build rapport, ensure openness and incite trust from the patient.
- Communication on a cultural plane: To understand the community to which the patient belongs, his/her cultural factors that may influence health and disease and also for economic understanding of the patient.
- Communication on an intellectual plane: To give adequate explanation for their conditions and help the patients develop insight into their ailments.

Solution to Question 13:

A medical social worker is a person who is trained mainly in social work and interviewing.

He/she carries out duties that augment public and community healthcare by way of identifying, understanding and relaying important information about a patient's determinants of health.

They are not involved in active treatment, prescribing or providing economic support.

Solution to Question 14:

Consumer Protection Act or COPRA was instituted in 1986 and deals with consumer rights.

It recognizes six rights of the consumers, which are:

- Right to safety
- Right to be informed
- Right to be heard
- Right to choose
- Right to seek redressal

- Right to consumer education

Solution to Question 15:

A repetitive interview is used to note the gradual influence of some social or psychological process. A record is maintained to understand the changes.

Interviews are methods used widely in investigations and in research. A proper interview technique must be employed to achieve the objective.

Aims of an interview:

- To secure information usually through face to face association and to understand the individual/group being interviewed.
- To form a hypothesis.
- To collect personal data for quantitative purposes.
- To collect data from secondary sources of information.

Types of interviews:

- A direct or structured interview consists of predetermined questions and sequence.
- A non-directive or unstructured interview doesn't have any sequence of questions set beforehand. The interviewer listens and allows the speaker to narrate his/her own experiences or opinions.
- A focused interview is usually used to study effects of mass communication like radio broadcasts or film. In this, the interviewer/researcher focuses on a particular aspect on the problem and draws responses from people.
- A repetitive interview is used to note the gradual influence of some social or psychological process. A record is maintained to understand the changes.

Solution to Question 16:

The correct sequence of steps of interviewing are establishing contact, starting an interview, securing rapport, making a report.

Technique of interviewing involves:

- Establishing contact
- Starting an interview
- Securing rapport
- Recall
- Probe questions
- Encouragement

- Guiding the interview
- Recording
- Closing the interview
- Report

Solution to Question 17:

Operational research is the application of scientific methods of investigation to study complex human organizations, services and management.

Pure research is also called fundamental research or basic research. It is aimed at improving scientific theories or understanding of them.

Applied research is a methodology used for a specific, practical problem like diseases or business. It involves the practical application of science.

Solution to Question 18:

The Nirbhaya Fund was set up by the government of India for promoting the safety of women.

Solution to Question 19:

Gross Domestic Product (GDP) is the income generated only from within the country and does not include income received from abroad.

Solution to Question 20:

The total public health expenditure is around 1.25% percentage of India's GDP (National Health Profile 2020 report).

The National Health Policy, 2017 states the aim to increase this to around 2.5% by 2025.

Solution to Question 21:

Hidden hunger refers to micronutrient deficiency. It occurs when there is deficiency in intake or absorption of micronutrients like vitamins, iron, zinc, etc.

The health effects of these deficiencies are less obvious or invisible and thus referred to as hidden hunger. The consequences of hidden hunger are seen in all ages.

Solution to Question 22:

An estimate of the poverty line is the expenditure required for a daily calorie intake of 2100 in urban areas and 2400 in rural areas, per person.

Income-based poverty line in India is per capita expenditure of Rs. 47 in urban areas and Rs. 32 in rural areas per day.

Solution to Question 23:

The Labour Charter Act does not offer social security for industrial workers.

Social security is the security that society furnishes through appropriate organization, against certain risks to which its members are exposed. These risks include sickness, maternity, old age and death.

Social security for industrial workers is given by:

- Workmen's Compensation Act
- The Family Pension Scheme
- Central Maternity Benefit Act
- Employees' State Insurance Act

Social security for civil servants is given by governments in the form of pension, gratuity, provident fund, etc. The Central Government Health Scheme provides security to employees of the Central Government.

Insurance schemes from the likes of Life Insurance Corporation of India, etc. have security schemes for the general public.

Solution to Question 24:

The infant mortality rate is not a component of the global hunger index.

The Global Hunger Index (GHI) has three dimensions. These dimensions are measured by indicators.

$$\text{GHI score} = (1/3 \times \text{standardized PUN}) + (1/6 \times \text{standardised CWA}) + (1/6 \times \text{standardised CST}) + (1/3 \times \text{Standardised CM})$$

Where,

- PUN is the proportion of the population that is undernourished
- CWA is the prevalence of wasting in children under 5 years of age
- CST is the prevalence of stunting in children under 5 years of age

- CM is the proportion of children dying under the age of 5 years of age.

The result lies on a scale from 0 to 100, with 0 meaning no hunger and 100 being the worst score. The GHI is given by the International Food Policy Research Institute. In 2021, India's GHI was 27.5. It ranks 101 among 116 countries. In the 2022, India's GHI stands at 29.1, and India ranks 107th out of the 121 countries.

Dimensions	Indicators
Inadequate food supply	Proportion of the population undernourished
Child mortality	Under 5 mortality rate
Child undernutrition	Wasting in children under 5 years Stunting in children under 5 years

Water - I: Sources and purification of water

Question 1:

Which of the following is not a component of the definition of safe and wholesome water?

- a) Water should be odourless
- b) Usable for domestic purposes
- c) Not necessarily colorless
- d) Pleasant to taste

Question 2:

In an urban PHC, an increasing number of skin diseases are being reported due to inadequate water supply in the region. What would be the adequate supply of water for them to meet their needs for all domestic purposes?

- a) 200-250 litres per capita per day
- b) 150-200 litres per capita per day
- c) 250-300 litres per capita per day
- d) 100-150 litres per capita per day

Question 3:

What are the norms for the required litres of water per capita per day in rural areas?

- a) 75
- b) 45
- c) 55
- d) 65

Question 4:

What is the purest form of water in nature?

- a) Rain water
- b) Rivers and streams

- c) Ponds and lakes
- d) Water from deep wells

Question 5:

Which of the following is not a source of surface water?

- a) Streams
- b) Tanks
- c) Shallow wells
- d) Impounding reservoirs

Question 6:

What is the area from which water drains into impounding reservoirs known as?

- a) Embankment ponds
- b) Watershed area
- c) Catchment area
- d) Dam

Question 7:

What is the most common type of well in India?

- a) Tube well
- b) Dug well
- c) Artesian well
- d) Bore well

Question 8:

Which of the following statements is false about artesian wells?

- a) It is not common in India
- b) Water is put under pressure between two impervious layers
- c) No pump is required to bring water to surface

- d) It is a type of shallow well

Question 9:

Step wells belong to which of the following types of wells?

- a) Bore wells
- b) Katcha wells
- c) Pucca wells
- d) Driven wells

Question 10:

An outbreak of guinea worm disease was reported among villagers who used step wells for drinking water. In order to arrest the spread of disease, a decision was made to replace step wells with sanitary and safe wells. All of the following measures would be considered during its construction except:

- a) Must be located at least 15 metres away from sources of contamination
- b) A partition space for washing clothes and vessels
- c) Lining must be made to a depth of 6 metres or more
- d) Hand-pump must be made to handle rough use

Question 11:

What is the minimum length that the platform of a well should extend around it in all directions?

- a) 1 m
- b) 1.5 m
- c) 2 m
- d) 2.5 m

Question 12:

Which of the following is not a water-washed disease?

- a) Scabies

- b) Trachoma
- c) Cholera
- d) Conjunctivitis

Question 13:

A multipurpose worker is spraying herbicides on aquatic plants as part of vector control measures in your area. Which of the following water-related diseases can be controlled by this approach?

- a) Weil's disease
- b) Filariasis
- c) Enterobiasis
- d) Giardiasis

Question 14:

Which of the following methods of purification of water cannot be used on a large scale?

- a) Berkefeld filter
- b) Rapid sand filter
- c) Ozonation
- d) Chlorination

Question 15:

During your posting as a rural medical officer, you conduct an awareness drive prior to the monsoon season in order to highlight the necessity of water purification. Which of the following methods would you not recommend for a household setup?

- a) Bleaching powder
- b) Katadyn filter
- c) Slow sand filter
- d) Multi-stage reverse osmosis

Question 16:

When river water is stored properly, by how much percent does the bacterial count drop in the first 5-7 days?

- a) 25
- b) 50
- c) 75
- d) 90

Question 17:

What is the other name for slow sand filters?

- a) Biological filters
- b) Mechanical filters
- c) Biochemical filters
- d) Bacteriological filters

Question 18:

Which of the following is a part of a slow sand filter?

- a) 1,2,3 and 4
- b) 2,3 and 4
- c) 1,2,3,4 and 5
- d) 2,3,4 and 5

Question 19:

Which of the following statements is true about slow sand filters?

- a) It occupies less space
- b) The thickness of sand bed is 2 metres
- c) It takes a longer duration for water purification
- d) The sand grain size is 0.4 to 0.7 mm

Question 20:

Which of the following best describes Schmutzdecke?

- a) Layer of alum flocculate on supernatant water
- b) Layer of algae and diatoms on supernatant water
- c) Layer of alum flocculate on surface of sand bed filter
- d) Layer of algae and diatoms on surface of sand bed filter

Question 21:

Which of the following is not the vital layer of a slow sand filter?

- a) Zoogeleal layer
- b) Schmutzdecke
- c) Biological layer
- d) Supernatant layer

Question 22:

Where is the vital layer of the slow sand filter seen?

- a) On top of the raw water
- b) On the sand bed
- c) Near the filter valves
- d) Next to the Venturi meter

Question 23:

What does the venturi meter measure in a slow sand filter?

- a) Loss of head
- b) Loss of Schmutzdecke
- c) Loss of resistance
- d) Loss of organic material

Question 24:

Which of the following is an incorrect statement about rapid sand filters?

- a) No preliminary storage of raw water is required
- b) Operation requires highly skilled personnel
- c) Frequent washing is not required
- d) Can be gravity or pressure type

Question 25:

What is the effective size of sand grains in a rapid sand filter?

- a) 0.2 – 0.3 mm
- b) 0.4 – 0.7 mm
- c) 0.5 – 0.6 mm
- d) 0.7 - 1.0 mm

Question 26:

Rapid sand filter is different from the slow sand filter by which of the following features?

- a) Bacteria removed more effectively
- b) Occupies more space
- c) Sand particles are of smaller size
- d) Skilled labor is needed

Question 27:

Which of the following is not a ceramic filter that can be used for small scale purification of water?

- a) Chamberland filter
- b) Zhu filter
- c) Berkefeld filter
- d) Katadyn filter

Question 28:

Which of the following statements is incorrect about ceramic filters?

- a) Surface of Berkefeld filter is lined with a silver catalyst
- b) Chamberland filter has a porcelain filter candle
- c) Oligodynamic action is seen with Katadyn filter
- d) They are not widely used in India

Answer Key

Question No.	Correct Option
1	c
2	b
3	c
4	a
5	c
6	c
7	b
8	d
9	c
10	b
11	a
12	c
13	b
14	a
15	c
16	d
17	a
18	c
19	c
20	d
21	d
22	b
23	a
24	c
25	b
26	d
27	b

Detailed Explanations

Solution to Question 1:

Water intended for human consumption must be safe and wholesome. It must be colorless.

Safe and wholesome water is defined as:

- Free from pathogenic agents
- Free from harmful chemicals
- Pleasant to taste, free from colour and odour
- Usable for domestic purposes

Water is said to be polluted when it does not fulfill the above criteria.

Solution to Question 2:

The given scenario of skin diseases is suggestive of water-washed diseases caused by improper hygiene related to an inadequate water supply. In urban areas, 150–200 litres per capita per day is the recommended allowance to meet the needs for all kinds of domestic purposes.

In rural areas, the set target of water supply to be achieved in India is around 50-55 litres per capita per day.

Solution to Question 3:

According to the National Rural Drinking Water Programme, the requirement of water in rural areas (areas with public water posts) is 55 litres per capita per day. This is in accordance with the 12th Five Year Plan.

For reasons of comparability, the source of water provision in rural areas should be within a distance of 100 metres from the household or the time taken to fetch water should not be > 30 minutes.

Note: Park textbook mentions 40-50 litres per capita per day as the norm. However, in the 12th five year plan period, the Government of India enhanced the rural drinking water supply norms from 40 litres to 55 litres per capita per day.

Solution to Question 4:

Rain water is the purest form of water in nature. It contains a very low amount of total dissolved solids (TDS). Rain water is the origin of both surface and ground water.

Sources of water:

- Rain
- Surface water:
- Impounding reservoirs
- Rivers and streams
- Tanks, ponds and lakes
- Ground water:
- Shallow wells
- Deep wells
- Springs

Solution to Question 5:

Shallow wells are sources of ground water.

Sources of water:

- Rain
- Surface water:
- Impounding reservoirs
- Rivers and streams
- Tanks, ponds and lakes
- Ground water:
- Shallow wells
- Deep wells
- Springs

Solution to Question 6:

The area draining into the impounding reservoirs are known as catchment areas.

Impounding reservoirs are also called embankment ponds. These reservoirs are artificial lake-like water bodies. They are made by building an embankment or earth fill in an area into which natural water flows and can be stored for usage. Dams are also built to create these reservoirs.

Water from these reservoirs is second in purity to rainwater but is not completely free from impurities. Catchment areas are the main sources of impurities in the water in impounding

reservoirs.

The image given below shows catchment area:



Solution to Question 7:

Dug wells are the most common type of wells in India.

Solution to Question 8:

Artesian wells are types of deep wells.

Solution to Question 9:

Step wells belong to the type of pucca dug wells.

Solution to Question 10:

Washing clothes, vessels, and bathing must be prohibited around wells to avoid contamination of the water.

Step wells are a type of pucca wells which enable people to descend into the well to fetch water. It is obsolete now, as there is considerable personal contact between the user and the water which can pose a health hazard to the community.

Solution to Question 11:

The platform of a sanitary well should stretch for at least 1 metre around the well in all directions.

Solution to Question 12:

Cholera is a water-borne disease and not a water-washed disease.

The diseases which are transmitted by improper hygiene and are widely related to circumstances where there is inadequate water supply are called water-washed diseases.

Some examples of water-washed diseases are:

- Conjunctivitis
- Trachoma
- Scabies
- Typhus

Solution to Question 13:

Filariasis (caused by *B. malayi*) is transmitted by *Mansonia* species of mosquito that breeds in water bodies containing aquatic vegetation. It can be controlled by the use of herbicides to destroy aquatic vegetation.

Water-related diseases are classified as:

Category	Examples
Water-borne diseases	Caused by ingestion of water contaminated by human or animal faeces, or urine containing pathogenic bacteria or viruses.ex: Cholera, typhoid, amoebic and bacillary dysentery, and other diarrhoeal diseases.
Water-washed diseases	Caused by poor personal hygiene.ex: Scabies, trachoma, flea-, lice- and tick-borne diseases, in addition to the majority of waterborne diseases (which are also water-washed).
Water-based diseases	Caused by parasites found in intermediate organisms living in water.ex: Dracunculiasis, schistosomiasis and some other helminths.

Category	Examples
Water-related diseases transmitted by insect vectors	Caused by vectors that breed in water: Dengue, filariasis, malaria, onchocerciasis, trypanosomiasis and yellow fever.

Solution to Question 14:

Berkefeld filter is used for the purification of water on a small scale (ex: in households).

Solution to Question 15:

Slow sand filter is a method of filtration used for the purification of water on a large scale.

Solution to Question 16:

When river water is stored properly, the bacterial count drops by as much as 90% in the first 5 – 7 days.

Purification during the storage of water is a natural process that takes place by physical, chemical and biological phenomena. Components of water purification include storage, filtration and disinfection.

Solution to Question 17:

Slow sand filters are also called biological filters.

Solution to Question 18:

Components of a slow sand filter include:

- Under-drainage system
- Sand bed and a layer of organic matter
- Filter control system
- Supernatant water
- Filter box

Solution to Question 19:

Slow sand filters take a longer duration than most other methods for purification of water.

Solution to Question 20:

Schmutzdecke refers to the vital layer which is formed by algae, plankton, diatoms and bacteria on the surface of the sand bed filter.

Solution to Question 21:

The vital layer of a slow sand filter is also called the zoogeal layer, Schmutzdecke, or the biological layer.

Solution to Question 22:

The vital layer of the slow sand filter is seen on the sand bed.

Solution to Question 23:

Venturi meters measure the bed resistance or loss of head in a slow sand filter.

The following image shows a venturi meter:



Solution to Question 24:

Frequent washing of the filter bed is required in rapid sand filters. This is done by backwashing.

Solution to Question 25:

Effective size of sand grains in a rapid sand filter is 0.4 – 0.7 mm.

Effective size of sand grains in a slow sand filter is 0.2 – 0.3 mm.

Solution to Question 26:

Skilled labor is required for rapid sand filters. Unskilled labor is enough for slow sand filters.

Comparison of rapid and slow sand filters:

	Rapid sand filter	Slow sand filter
Space	Occupies very little space	Occupies large area
Rate of filtration (million gallons per acre per day)	200 m.g.a.d	2-3 m.g.a.d.
Effective size of sand	0.4-0.7 mm	0.2-0.3 mm
Preliminary treatment	Chemical coagulation and sedimentation	Plain sedimentation
Washing	By back-washing	By scraping the sand bed
Operation	Highly skilled	Less skilled
Loss of head allowed	6-8 feet (2- 2.5 m)	4 feet (1.5 m)
Removal of bacteria	98-99%	99.9- 99.99%

Solution to Question 27:

Filtration of water on a small scale is done by ceramic filters like Pasteur Chamberland, Berkefeld, and Katadyn filters.

Ceramic filters are not widely used in India. The most important parts of these filters are the filter candles. Filter candles of the fine type help in removing bacteria but not viruses. These filter candles need to be cleaned periodically as they get clogged with impurities and bacteria.

Types of ceramic filters include:

- Pasteur Chamberland filter – It has a candle made of porcelain.

Image of a Pasteur Chamberland filter is shown below:



- Berkefeld filter - it has a candle made of infusorial earth or kieselguhr.

Image of a Berkefeld filter is shown below:



- Katadyn filter - the surface of this filter is coated with a silver catalyst. The incoming bacteria are killed on contact with the surface by the oligodynamic action of silver ions which are liberated into the water.

Image of a Katadyn filter is shown below:



Solution to Question 28:

Berkefeld filter has a candle made of infusorial earth or kieselguhr.

Ceramic filters are used for small scale purification of water. They are not widely used in India. The most important parts of these filters are the filter candles.

Types of ceramic filters include:

- Pasteur Chamberland filter – It has a candle made of porcelain.
- Berkefeld filter - it has a candle made of infusorial earth or kieselguhr.
- Katadyn filter - the surface of this filter is coated with a silver catalyst. The incoming bacteria are killed on contact with the surface by the oligodynamic action of silver ions which are liberated into the water.

Water- II: Disinfection of water

Question 1:

Which of the following is the method of choice for the purification of highly polluted water on a large scale?

- a) Multi-stage reverse osmosis
- b) Superchlorination with dechlorination
- c) Treatment with ultraviolet light
- d) Filtration using Pasteur Chamberland filter

Question 2:

Which of the following is true regarding chlorination of water?

- a) Orthotolidine test measures only combined chlorine
- b) Chlorine acts best when the pH of water is around seven
- c) It kills bacteria, viruses, spores and slime organisms
- d) Hypochlorite ions are mainly responsible for its disinfectant activity

Question 3:

You assign a male health worker at a sub-centre with the task of chlorinating public water resources. What is the minimum concentration of free residual chlorine that should be maintained for effective chlorination?

- a) 0.5 mg/litre for a contact period of 1 hour
- b) 0.5 mg/litre for a contact period of $\frac{1}{2}$ hour
- c) 0.8 mg/litre for a contact period of 1 hour
- d) 0.8 mg/litre for a contact period of $\frac{1}{2}$ hour

Question 4:

As an epidemiologist, you are investigating an outbreak of cholera. What is the minimum free residual chlorine level that should be maintained in pipelines to prevent further spread of the

outbreak?

- a) 2.0 mg/litre
- b) 1.0 mg/litre
- c) 1.5 mg/litre
- d) 0.5 mg/litre

Question 5:

The municipal corporation in your area is performing routine chlorination of drinking water supplies. Which of the following statements about chlorination is incorrect?

- a) Free chlorine is released in water after breakpoint chlorination
- b) Chlorine demand is the amount needed to kill spores, protozoal cysts and helminthic ova
- c) 0.5 ppm of free chlorine must be present in water after the breakpoint has been reached
- d) Contact period of 1 hour is necessary

Question 6:

Which of the following tests identifies free and combined chlorine in water separately?

- a) Chick Martin test
- b) Orthotolidine test
- c) Orthotolidine-arsenite test
- d) Garrod's test

Question 7:

Which of the following compounds is predominantly responsible for the disinfecting property of bleaching powder?

- a) Hypochlorite ion
- b) Hydrochloric acid
- c) Hypochlorous acid
- d) Chloride ion

Question 8:

A village health and sanitation team is planning to disinfect 100 wells. Which of the following is the most cost-effective method in this scenario?

- a) Disinfection using lime
- b) Disinfection using potassium permanganate
- c) Disinfection using bleaching powder
- d) Disinfection using alum

Question 9:

An outbreak of viral hepatitis A has been reported in a village. On investigating, you find a contaminated well in the locality to be the source of the outbreak. Which of the following methods would you advise for chlorinating the well in this scenario?

- a) Disinfection using Horrock's apparatus
- b) Double pot method
- c) Hypochlorite testing method
- d) Disinfection using chloronome

Question 10:

In your community medicine practicals, you are given an apparatus as shown in the image given below. Which of the following statements regarding the apparatus is true?



- a) The indicator used is bromocresol purple

- b) The indicator used is starch-iodide
- c) It is used to estimate free chlorine levels
- d) It is used to estimate combined chlorine levels

Question 11:

What is the amount of bleaching powder required to disinfect 455 litres of water if the 4th, 5th, and 6th cups of a Horrock's apparatus show distinct blue color?

- a) 2 grams
- b) 4 grams
- c) 8 grams
- d) 6 grams

Question 12:

The water sample from a well is tested using Horrock's apparatus. It is observed that the 3rd cup shows a blue color after the addition of the starch-iodine indicator. How much bleaching powder will be required to disinfect 2275 litres of the well water?

- a) 6 grams
- b) 15 grams
- c) 25 grams
- d) 30 grams

Question 13:

Which of the following agents have residual germicidal effect when used for treatment of water?

- a) 3 only
- b) 1, 4 and 5
- c) 2 and 3
- d) 1 only

Question 14:

Which of the following properties of ozone makes it less suitable for the disinfection of water on a large scale?

- a) Ozone requires a long time period for action
- b) It has no virucidal effect
- c) It is teratogenic
- d) It has no residual disinfectant action

Question 15:

Which of the following is a lower-pressure membrane process for treatment of water?

- a) Reverse osmosis
- b) Nanofiltration
- c) Microfiltration
- d) Ozonation

Answer Key

Question No.	Correct Option
1	b
2	b
3	a
4	d
5	b
6	c
7	c
8	c
9	b
10	b
11	c
12	d
13	a
14	d
15	c

Detailed Explanations

Solution to Question 1:

Superchlorination followed by dechlorination is done for purification of highly polluted water on a large scale.

Solution to Question 2:

Chlorine acts best when the pH of water is around seven because of the predominance of the hypochlorous acid. Most water sources have a pH around 6.5–7, making chlorination an effective form of disinfection.

Option A: Orthotolidine test measures both free and combined chlorine in water.

Option C: Chlorination kills pathogenic bacteria, but has no effect on spores and certain viruses.

Option D: The disinfecting action of chlorine is mainly due to hypochlorous acid.

Solution to Question 3:

The minimum recommended concentration of free residual chlorine for routine chlorination is 0.5 mg/litre for a contact period of 1 hour.

Solution to Question 4:

The minimum level of free residual chlorine to be maintained in pipelines following an outbreak of cholera is 0.5 mg/litre.

After a disaster (such as a flood), the water bodies are contaminated with sewage and drainage. The mixing of faeces in drinking bodies increases the risk of numerous microbial infections. The advisable level of free residual chlorine in water bodies after a disaster is, therefore, higher i.e. 0.7–0.9 mg/litre.

Solution to Question 5:

Chlorine demand is the amount of chlorine that is needed to kill bacteria, and to oxidize all the organic matter and ammoniacal substances in the water. It is the difference between the amount of chlorine added to the water and the amount of residual chlorine remaining at the end of a specific period of contact (usually 1 hour) at a given temperature and pH of the water.

0.5 ppm (0.5 mg) of free chlorine must be present in water after the breakpoint has been reached.

Solution to Question 6:

Orthotolidine-arsenite test determines both free and combined chlorine in water separately.

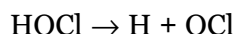
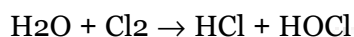
Orthotolidine test determines free and combined chlorine together.

Chick Martin test and Garrod's test are used to test the efficacy of disinfectants.

Solution to Question 7:

The disinfectant property of bleaching powder is predominantly due to the action of hypochlorous acid. It is 70-80 times more effective than hypochlorite ions.

Bleaching powder, also known as chlorinated lime (CaOCl_2), has 33 per cent of chlorine in it. When chlorine is added to water, there is the formation of hydrochloric and hypochlorous acids. The hydrochloric acid is neutralized by the alkalinity of the water. The hypochlorous acid ionizes to form hydrogen ions and hypochlorite ions, as follows:



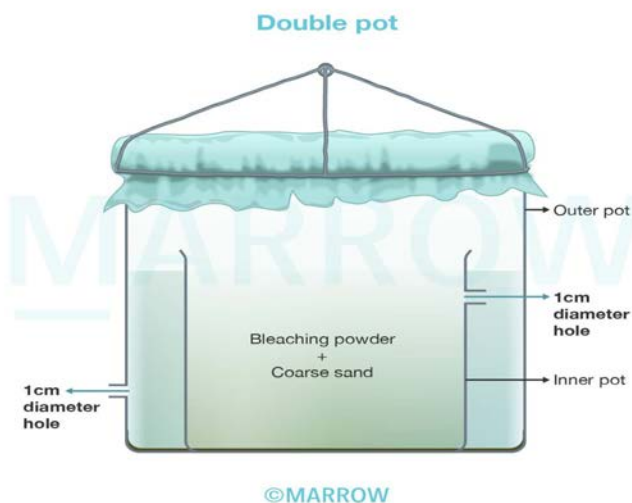
Solution to Question 8:

Disinfection of wells using bleaching powder is the most widely used and cheapest method.

Solution to Question 9:

The double pot method is used for disinfecting wells by chlorination in emergencies like hepatitis A outbreak in a village.

Double pot method has an inner and outer pot, each with a hole in it and covered with a polyethylene foil. The inner pot is filled with bleaching powder and coarse sand. The double pot system is then lowered into a well. This device can work satisfactorily for 2 – 3 weeks in small wells containing about 4500 litres of water.



Solution to Question 10:

The image given is that of a Horrock's apparatus. The indicator used in Horrock's apparatus is starch iodide.

Horrock's apparatus contains 6 white cups (200 ml each), 1 black cup with a circular mark, 2 metal spoons with 2 grams of bleaching powder in each, 7 glass stirring rods, 1 special pipette and 2 droppers. It is not useful in measuring free and combined chlorine. It estimates the chlorine demand of water and is used to calculate the amount of bleaching powder used to disinfect the well.

Solution to Question 11:

The dose of bleaching powder required is 8 grams.

It is given by $(n \times 2)$ gms for 455 litres of water, where n is the number of the first cup that shows distinct blue color. In this scenario, the dose of bleaching powder required to disinfect 455 litres of water is $(4 \times 2) = 8$ grams.

Solution to Question 12:

The dose of bleaching powder required is 30 grams.

It is given by $(n \times 2)$ gms for 455 liters of water, where n is the number of the first cup that shows a distinct blue color.

In this scenario, the dose of bleaching powder required to disinfect 455 liters of water is $(3 \times 2) = 6$ grams.

Since $2275/455 = 5$. The amount of bleaching powder required to disinfect 2275 litres of water = $6 \times 5 = 30$ grams.

Solution to Question 13:

Chlorine has residual germicidal effect when used for treatment of water.

UV radiation, ozone gas, iodine and potassium permanganate have good germicidal action but do not have residual germicidal effect.

Solution to Question 14:

Ozone has no residual disinfectant action.

Ozone gas can be used as a primary disinfectant of water in a large scale. Ozone disinfection is usually followed by biological filtration or granular activated carbon to prevent further contamination during distribution. Chlorine residual treatment is also done as ozone does not provide for a residual disinfectant action.

Solution to Question 15:

Microfiltration is a lower-pressure membrane process used for the treatment of water.

Membrane processes are employed in the large scale treatment of water. They are classified as high-pressure processes and lower-pressure processes.

- High-pressure processes:
 - Reverse osmosis - Rejects monovalent ions and organic molecules of molecular weight > 50 daltons. It is most commonly used for desalination of brackish water and sea water. The membrane pore size is $< 0.002 \mu\text{m}$.
 - Nanofiltration - Allows monovalent ions such as sodium or potassium to pass but rejects a high proportion of divalent ions like calcium and magnesium and some higher molecular weight organics. The membrane pore size ranges from $0.001 - 0.01 \mu\text{m}$.
- Lower-pressure processes:
 - Ultrafiltration - Rejects organic molecules of molecular weight > 800 daltons. The membrane pore size ranges from $0.002 - 0.003 \mu\text{m}$.
 - Microfiltration - Rejects colloidal and suspended material. It does not separate molecules and is usually used before reverse osmosis. The membrane pore size ranges from $0.01 - 12 \mu\text{m}$.

Water - III: Water Quality and Standards

Question 1:

Above what level does the turbidity of water become visible to the naked eye?

- a) 5 NTU
- b) 6 NTU
- c) 3 NTU
- d) 4 NTU

Question 2:

What is the maximum permissible level of chloride in drinking water?

- a) 200 mg/L
- b) 300 mg/L
- c) 500 mg/L
- d) 600 mg/L

Question 3:

A detectable taste begins to appear in drinking water if chloride levels in it exceed:

- a) 150 mg/L
- b) 200 mg/L
- c) 250 mg/L
- d) 300 mg/L

Question 4:

A large number of cases of methemoglobinemia were reported among infants from a municipal area. On detailed investigation, the nitrate levels in drinking water were found to be elevated in that area. This complication can occur when nitrate level in drinking water exceeds:

- a) 15 mg/L

- b) 20 mg/L
- c) 30 mg/L
- d) 45 mg/L

Question 5:

Which is the most undesirable metal in drinking water?

- a) Lead
- b) Iron
- c) Arsenic
- d) Mercury

Question 6:

What is the acceptable level of total dissolved solids in drinking water in India?

- a) 300 mg/L
- b) 800 mg/L
- c) 500 mg/L
- d) 1000 mg/L

Question 7:

What is the proposed guideline for the permissible radioactivity values for drinking water?

- a) Gross alpha activity up to 0.1 Bq/L and gross beta activity up to 1 Bq/L
- b) Gross alpha activity up to 1 Bq/L and gross beta activity up to 0.5 Bq/L
- c) Gross alpha activity up to 0.5 Bq/L and gross beta activity up to 1 Bq/L
- d) Gross alpha activity up to 0.1 Bq/L and gross beta activity up to 0.5 Bq/L

Question 8:

In view of a hepatitis A outbreak in a hostel, samples of drinking water are collected to detect fecal contamination. Presence of which of the following organisms serves as the most reliable evidence of fecal contamination?

- a) Escherichia coli
- b) Clostridium perfringens
- c) Streptococcus faecalis
- d) Klebsiella aerogens

Question 9:

A microbiologist is testing a sample of drinking water to detect fecal contamination. The presence of all of the following organisms indicates a positive test result except

- a) 1 and 4
- b) 1, 4 and 5
- c) 2 and 5
- d) 2, 3, and 5

Question 10:

Which of the following statements is incorrect about water purification and pollution?

- a) Presence of clostridial spores indicates recent contamination
- b) Sodium thiosulphate can be used to dechlorinate water
- c) Coliforms must not be detectable in any 100 ml sample of drinking water
- d) Coliforms can be detected by the multiple tube method

Question 11:

Multiple tube method is done for which of the following?

- a) Detecting the presence of coliform organisms in a sample of water
- b) Detecting the presence of enteric viruses in a sample of water
- c) Estimating the levels of mercury in a sample of water
- d) Estimating turbidity of a sample of water

Question 12:

What is the acceptable bacteriological quality of drinking water in small communities?

- a) No coliform bacteria in 100 ml of water
- b) Upto 10 coliform bacteria in 100 ml of water
- c) Upto 10 coliform bacteria in 1000 ml of water
- d) Upto 100 coliform bacteria in 1000 ml of water

Question 13:

Which of the following criteria is true for safe drinking water according to the WHO guidelines?

- a) pH – 6.5 to 8.5
- b) Chloride – 0.8 g/L
- c) Nitrate – 0.2 g/L
- d) Lead - 0.1 mg/L

Question 14:

Which of the following is a criterion included in the guidelines for drinking water quality recommended by WHO?

- a) Color < 15 TCU
- b) Nitrites < 20 mg/l
- c) Turbidity < 20 NTU
- d) Arsenic < 0.2 mg/l

Question 15:

Temporary hardness of water is primarily due to the presence of which of the following?

- a) Calcium and magnesium sulphates
- b) Calcium and magnesium chlorides
- c) Calcium and magnesium bicarbonates
- d) Calcium and magnesium carbonates

Question 16:

Which of the following does not cause hardness of water?

- a) Calcium carbonate
- b) Calcium sulphate
- c) Calcium bicarbonate
- d) Magnesium bicarbonate

Question 17:

What is the level of hardness in soft water?

- a) < 1 mEq/L
- b) 1 – 2 mEq/L
- c) 2 – 3 mEq/L
- d) > 3 mEq/L

Question 18:

How would you classify a water sample that has a level of hardness of 200 mg/litre?

- a) Very hard water
- b) Hard water
- c) Moderately hard water
- d) Soft water

Question 19:

A health worker is conducting periodic quality test for water sources in your area. What should be the hardness level of water used for drinking purposes?

- a) > 5 mEq/L
- b) < 1 mEq/L
- c) 1 – 3 mEq/L
- d) 3 – 5 mEq/L

Question 20:

Above what level of hardness is softening of water recommended?

- a) 50 mg/l
- b) 125 mg/l
- c) 100 mg/l
- d) 150 mg/l

Question 21:

Which of the following is not a method of removal of hardness of water?

- a) Base exchange process
- b) Addition of sodium carbonate
- c) Boiling
- d) Filtration

Question 22:

What is the composition of permutit which is used for the base exchange process to remove hardness of water?

- a) Sodium aluminium silicate
- b) Hexachlorocyclohexane
- c) Calcium magnesium silicate
- d) Calcium aluminium silicate

Answer Key

Question No.	Correct Option
1	d
2	d
3	c
4	d
5	a
6	c
7	c
8	a

9	a
10	a
11	a
12	a
13	a
14	a
15	c
16	a
17	a
18	b
19	c
20	d
21	d
22	a

Detailed Explanations

Solution to Question 1:

Turbidity of water becomes visible to the naked eye above 4 nephelometric turbidity units (NTU).

The ideal level of turbidity is ≤ 1.0 NTU for household drinking purposes. However, in situations where this is not achievable, the maximum permissible limit is ≤ 5 NTU.

Solution to Question 2:

The maximum permissible level of chloride in drinking water is 600 mg/litre.

Solution to Question 3:

A detectable taste begins to appear in drinking water if chloride levels in it exceed 250 mg/litre.

Solution to Question 4:

Nitrates in the excess of 45 mg/L may cause methaemoglobinemia in infants.

Solution to Question 5:

The most undesirable metal in drinking water is lead.

Solution to Question 6:

According to the Bureau of Indian Standards (BIS), the acceptable level of total dissolved solids (TDS) in drinking water in India is 500 mg/L.

Note: There is a variation in the value of the acceptable level of total dissolved solids in drinking water provided by WHO and the Bureau of Indian Standards (BIS). While the WHO sets the value at 600 mg/L, the BIS has set it at 500 mg/L for India.

Solution to Question 7:

The proposed guideline for the permissible radioactivity values for drinking water is:

Gross alpha activity up to 0.5 Bq/litre.

Gross beta activity up to 1.0 Bq/litre.

Solution to Question 8:

The presence of *Escherichia coli* is the most suitable and reliable evidence of fecal contamination of water.

Klebsiella aerogens is a typical example of the non-fecal group of coliform organisms.

Solution to Question 9:

Presence of *Staphylococcus* and *Klebsiella aerogens* does not indicate fecal contamination of water.

Biological indicators of fecal pollution of water are:

- *Streptococcus faecalis*
- *Escherichia coli*
- *Clostridium perfringens*

Solution to Question 10:

The presence of clostridial spores indicates remote or intermittent contamination of water.

Solution to Question 11:

The multiple tube method is done to detect the presence of coliform organisms in a sample of water.

In order to estimate the presumptive coliform count, water to be tested is inoculated into tubes containing a suitable liquid medium. These tubes are then incubated and examined after an appropriate period of time. The test is called presumptive because the reaction observed in the tube may occasionally be due to other organisms.

The proportion of false-positive reactions obtained depends both on the bacterial flora of the water under examination and the medium used. Lactose broth, or MacConkey's broth, with bromocresol purple as an indicator and a standardized concentration of bile salts incubated between 35°C–37°C for 48 hours, seems to be the most suitable media for presumptive tests.

Solution to Question 12:

There must be no coliform bacteria in any 100 ml sample of drinking water as its presence indicates fecal contamination.

Solution to Question 13:

The desirable pH of safe drinking water ranges from 6.5 to 8.5.

The maximum permissible level of chloride in drinking water is 0.6 g/L (600 mg/L), and that of nitrates is 0.05 g/L (50mg/l).

The concentration of lead must be kept at ≤ 0.01 mg/L. It is the most undesirable metal in drinking water.

Solution to Question 14:

According to WHO, the permissible level of color in drinking water is ≤ 15 true color units (TCU).

The permissible level of turbidity in drinking water is ≤ 1 NTU. The permissible level of arsenic is up to 0.01 mg/l and nitrites is up to 3 mg/l.

Solution to Question 15:

The temporary hardness of water is primarily due to the presence of calcium and magnesium bicarbonates.

Solution to Question 16:

Calcium carbonate does not cause hardness of water. Hardness of water is caused by bicarbonates and sulphates of calcium and magnesium.

There are two types of hardness of water:

- Temporary (carbonate) hardness is due to bicarbonates of calcium and magnesium.
- Permanent (non-carbonate) hardness is due to sulphates, chlorides and nitrates.

Solution to Question 17:

The level of hardness in soft water is < 1 mEq/litre.

Solution to Question 18:

When the level of hardness of water is in the range of 150 – 300 mg/litre, it is classified as hard water.

Solution to Question 19:

Water used for drinking purposes must have a hardness level of 1 – 3 mEq/L (moderately hard).

Solution to Question 20:

Softening of water is recommended when the hardness exceeds 3 mEq/l (150 mg/litre).

Solution to Question 21:

Filtration is not a method of removal of hardness of water.

Methods for removal of hardness of water are:

- Boiling
- Addition of lime

- Addition of sodium carbonate (soda ash)
- Base exchange process

Solution to Question 22:

Permutit is a complex chemical compound made of sodium, aluminium and silica. It is used to remove hardness of water by base exchange process.

Permutit has the property of ion exchange and removes calcium and magnesium from water. It removes both permanent and temporary hardness.

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you may have been scammed.

Environmental Meteorology

Question 1:

The water vapor and pollutants tend to get trapped in the lowermost layers of the atmosphere in winters. What is the reason behind this occurrence?

- a) Green house effect
- b) Ocean effect
- c) Precipitation
- d) Temperature inversion

Question 2:

Which of the following is not an indicator of air pollution?

- a) Soiling index
- b) Heavy metal pollution index
- c) Suspended particle matter
- d) Sulphur dioxide

Question 3:

Which agency monitors air quality in India?

- a) Ministry of Earth Sciences
- b) Indian Meteorological Department
- c) Central Power Research Institute
- d) Central Pollution Control Board

Question 4:

Greenhouse gases include all of the following except:

- a) Carbon monoxide
- b) Carbon dioxide

- c) Water vapor
- d) Methane

Question 5:

Which of the following protocols set the standard to reduce the emission of greenhouse gases internationally?

- a) Tokyo Protocol
- b) Beijing Protocol
- c) Kyoto Protocol
- d) Montreal Protocol

Question 6:

Which of the following is true about global warming?

- a) Carbon dioxide is the major greenhouse gas
- b) Stratospheric ozone is harmful
- c) CFC increases ozone levels in the stratosphere
- d) Kyoto Protocol called for 10% reduction in greenhouse emissions

Question 7:

Which of the following is incorrect regarding indoor air pollutants?

- a) Carbon monoxide is produced by fuel combustion
- b) Pesticides are indoor air pollutants
- c) Radon is an indoor air pollutant
- d) Mercury vapor is a source of indoor air pollution

Question 8:

Which of the following is unlikely to be seen in individuals affected by indoor air pollution?

- a) Chronic lung disease
- b) Pregnancy problems

- c) Neuro-developmental disorders
- d) Childhood pneumonia

Question 9:

The best indicators for routine monitoring of air pollution are:

- a) Sulfur dioxide, smoke, and suspended particles
- b) Sulfur dioxide, lead, and particulate matter
- c) Sulfur dioxide and carbon monoxide
- d) Carbon monoxide and hydrogen sulfide

Question 10:

Which of the following is not an index of thermal comfort?

- a) Air temperature and humidity
- b) Corrected effective temperature
- c) Cooling power
- d) Heat transfer coefficient

Question 11:

What are the instruments used to measure atmospheric pressure called?

- a) Barometers
- b) Anemometers
- c) Hygrometers
- d) Psychrometers

Question 12:

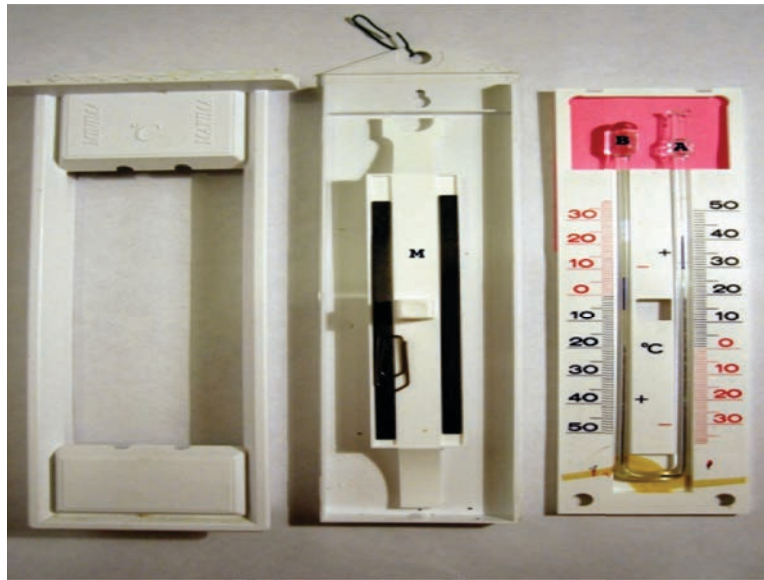
What sweat excretion rate in acclimatized, healthy young men is indicative of intolerable air temperature?

- a) > 4 litres/4 hours
- b) > 4 litres/6 hours

- c) > 4.5 litres/4 hours
- d) > 4.5 litres/8 hours

Question 13:

Identify the following instrument:



- a) Globe thermometer
- b) Jacen's psychrometer
- c) Wet bulb thermometer
- d) Six's thermometer

Question 14:

The mean radiant temperature is measured by the:

- a) Dry bulb thermometer
- b) Wet bulb thermometer
- c) Maximum-minimum thermometer
- d) Globe thermometer

Question 15:

The given instrument can be used to measure all of the following except:



- a) High air velocities
- b) Air temperature
- c) Humidity
- d) Air movement

Question 16:

Which of the following is not an index of heat stress?

- a) Predicted four-hour sweat rate
- b) Air velocity index
- c) Equatorial comfort index
- d) Heat stress index

Question 17:

Which of the following is unlikely to be seen in individuals suffering from heat stress?

- a) Hyperpyrexia
- b) Syncope
- c) Cramps
- d) Numbness

Question 18:

What level of heat stress index correlates with severe heat strain in individuals?

- a) 20 – 40
- b) 40 – 60
- c) 60 – 80
- d) 80 – 100

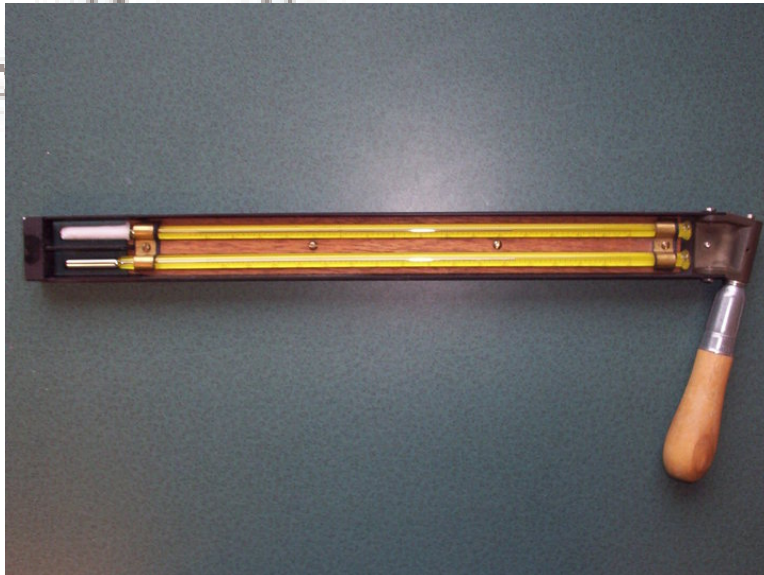
Question 19:

Which of the following is incorrect about cold stress?

- a) Wet-cold conditions cause trench foot
- b) Dry-cold conditions occur at temperatures above freezing
- c) Cold stress can cause desire to sleep
- d) Frostbite is common at high altitudes

Question 20:

Identify the following instrument shown below:



- a) Sling psychrometer
- b) Dry and wet bulb hygrometer
- c) Anemometer

d) Assmann psychrometer

Question 21:

Which is the most widely used instrument to measure humidity?

- a) Sling psychrometer
- b) Aspirating psychrometer
- c) Dry and wet bulb hygrometer
- d) Assmann psychrometer

Question 22:

What is the following instrument used to measure?



- a) Humidity
- b) Air velocity
- c) Radiant heat
- d) Air pollution index

Question 23:

Which is the most widely used rain gauge in India?

- a) Recording rain gauge

- b) Symon's rain gauge
- c) Tipping bucket rain gauge
- d) Float type rain gauge

Question 24:

Which of the following instruments measures the air velocity?

- a) Wind vanes
- b) Anemometers
- c) Psychrometers
- d) Barometers

Answer Key

Question No.	Correct Option
1	d
2	b
3	d
4	a
5	c
6	a
7	d
8	c
9	a
10	d
11	a
12	c
13	d
14	d
15	a
16	b
17	d
18	b
19	b

20	a
21	c
22	a
23	b
24	b

Detailed Explanations

Solution to Question 1:

In winters, water vapor and pollutants get trapped in the lowermost layers of the atmosphere due to temperature inversion.

This results in the formation of smog, especially in the winter months, and this is a possible threat to human health.

Solution to Question 2:

The heavy metal pollution index (HPI) is not an indicator of air pollution. It is a rating method and an effective tool to assess the water quality with respect to heavy metals.

Indicators of air pollution:

- Sulfur dioxide - best indicator
- Smoke or soiling index
- Grit and dust measurement
- The co-efficient of haze
- Air pollution index
- Suspended particle matter (SPM)

Solution to Question 3:

Central Pollution Control Board is the main body that monitors air quality in India and sets standards.

Solution to Question 4:

Carbon monoxide is not a greenhouse gas.

Greenhouse gases include:

- Water vapor
- Carbon dioxide
- Methane
- Tropospheric ozone
- Nitrous oxide
- Chlorofluorocarbons (non-natural)

Solution to Question 5:

The Kyoto Protocol was an international agreement to set internationally binding emission reduction targets.

It is linked to the United Nations Framework Convention on Climate Change. The parties committed to reduce GHGs by 18% in the period between 2013-2020.

Solution to Question 6:

Carbon dioxide is the major greenhouse gas.

Tropospheric ozone and chlorofluorocarbons are also important greenhouse gases. Water vapor is the most abundant greenhouse gas, but the water vapor produced directly by human activity contributes very little to the amount of water vapor in the atmosphere. Hence, its emissions have not been estimated.

Greenhouse gases lead to global warming, which is the major cause of climate change. Climate change accelerates species extinction on earth.

Solution to Question 7:

Mercury is not a source of indoor air pollution.

Indoor air pollutants commonly arise from coal or fuel combustion. They include:

- Carbon monoxide
- Polyaromatic hydrocarbons
- Nitrogen
- Sulfur oxides
- Pesticides
- Asbestos

- Lead
- Radon

Solution to Question 8:

Indoor air pollution has not been linked with neurodevelopmental disorders. It can result in a higher incidence of the following:

- Pregnancy complications
- Chronic lung disease
- Childhood pneumonia

Solution to Question 9:

The best indicators for routine monitoring of air pollution are sulfur dioxide, smoke, and suspended particles.

Solution to Question 10:

The heat transfer coefficient is a measure of the heat transfer by convection or phase transition. It is not an index of thermal comfort.

Thermal comfort is the subjective feeling of temperature in an environment. Optimum levels of thermal comfort help to increase productivity. Indices of thermal comfort include the following:

Index	Application
Air temperature	Considered an inadequate indicator
Air temperature and humidity	Considered an inadequate indicator
Cooling power- it measures: Air temperature Humidity Air movement	Measured using akata thermometer (dry and wet) Regarded as unreliable
Effective temperature- it measures: Air temperature Humidity Air movement	It ignores the effects of radiation from surrounding structures
Corrected effective temperature- it measures: Air temperature Humidity Air movement Mean radiant heat	It is measured using a globe thermometer
McArdle's maximum allowable sweat rate	It involves calculating the 4-hour sweat rate

Solution to Question 11:

Atmospheric pressure or barometric pressure is measured using barometers.

The most commonly known barometers are Fortin's barometer, Kew pattern station barometer, and the barograph. The Indian Meteorological Department uses the Kew pattern station barometer.

The pressure at Earth's surface, close to sea level averages around 760 mmHg, and man is adapted to live around this.

Solution to Question 12:

A sweat excretion rate of $> 4.5 \text{ L/4 hours}$ in acclimatized, healthy young men is indicative of intolerable air temperature. This is known as McArdle's maximum allowable sweat rate.

This is the maximum allowable sweat rate that is compatible with the normal physiological reaction of acclimatized, healthy young men who are exposed to heat. The predicted four-hour sweat rate (P4SR) is obtained from readings of air and mean radiant temperature and air velocity under different work intensities. Based on this value, the zone of comfort can be determined.

Zone of comfort	Predicted P4SR
Comfort zone	1 – 3 litres
Just tolerable	3 – 4.5 litres
Intolerable	4.5 + litres

Solution to Question 13:

The given image is of a Six's thermometer.

Solution to Question 14:

The mean radiant temperature is measured by the globe thermometer.

Solution to Question 15:

The image shown above is of the kata thermometer. It was originally used to measure the cooling power of air (air temperature, humidity and air movement) but currently, it is only used to measure low air velocities.

Solution to Question 16:

Air velocity index is not an index of heat stress.

Heat stress is the load of heat that must be dissipated from the body if it is to remain in thermal equilibrium. Factors that affect heat stress are metabolic rate, air temperature, humidity, air movement, and radiant temperature.

Indices for heat stress are:

- Predicted four-hour sweat rate
- Equatorial comfort index
- Heat stress index.

Solution to Question 17:

Numbness is an effect of cold stress, not heat stress.

Effects of heat stress include:

- Heatstroke
- Heat hyperpyrexia
- Heat exhaustion
- Heat cramps
- Heat syncope

Solution to Question 18:

A heat stress index of 40–60 indicates severe heat strain. It can pose a threat to health and greatly impair work if not fit or acclimatized.

Heat stress indices include:

- Equatorial comfort index
- Predicted four-hour sweat rate (P4SR)
- Heat stress index (HSI)

HSI	Thermal stress	Work and effects
0	No thermal strain	No work impairment
10 – 30	Moderate to mild heat strain	Minimal work impairment
40 – 60	Severe heat strain	Threat to health and greatly impaired work
70 – 90	Very severe heat strain	Only a few who are fit and acclimatized can sustain work
100	Upper limit of heat tolerance	Only fit and acclimatized can sustain work, although with impairment Varying degrees of heat stress occurs beyond this

Solution to Question 19:

Dry-cold conditions occur at temperatures below freezing and are common at high altitudes. It results in complications like frostbite.

Wet-cold conditions occur at temperatures above freezing. It results in complications like trench foot. Cold stress also causes hypothermia which can induce numbness, loss of sensations, muscular weakness, desire to sleep, coma, and death.

Solution to Question 20:

The image shown is of a sling psychrometer. Psychrometers are used to measure humidity.

Solution to Question 21:

The most widely used instrument to measure humidity is the dry and wet bulb hygrometer.

Solution to Question 22:

The image given is that of an Assmann psychrometer which is used to measure humidity.

They consist of two thermometers each inside a double tube for minimizing radiative heating.

Solution to Question 23:

The Government of India prescribes the use of Symon's rain gauge which is a non-recording type of rain gauge. It is used to measure the amount of rainfall.

The image below shows the Symon's rain gauge.



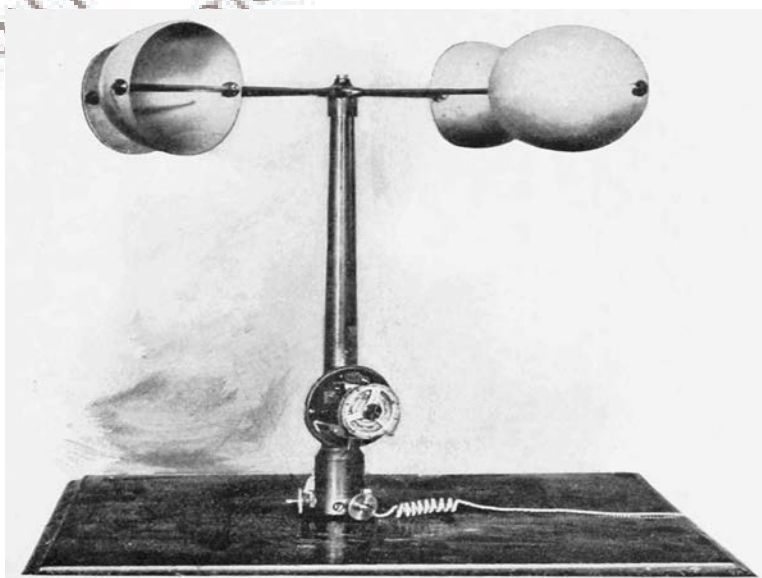
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Solution to Question 24:

Anemometers are used to measure air velocity.

Kata thermometers are used as anemometers for low air velocities.

The image below shows an anemometer.



Wind vanes are used to identify the wind direction (shown below).



The image below shows an anemometer wind vane.



Housing and Ventilation

Question 1:

Which of the following is not true about ventilation?

- a) Involves replacement of vitiated air
- b) Involves quality control of outgoing air
- c) Provides comfortable thermal environment
- d) Reduces risk of infection

Question 2:

What is the ideal number of air changes in a drawing-room?

- a) 2 – 3 per hour
- b) 3 – 4 per hour
- c) 4 – 5 per hour
- d) 1 – 2 per hour

Question 3:

What is ventilation provided by wind blowing through a space called?

- a) Perflation
- b) Aspiration
- c) Cross-ventilation
- d) Plenum ventilation

Question 4:

Which of the following is not a type of mechanical ventilation?

- a) Perflation ventilation
- b) Balanced ventilation
- c) Plenum ventilation

- d) Air conditioning

Question 5:

What is the mechanical system of ventilation which employs only centrifugal fans that create positive pressure inside a space by blowing fresh air into it called?

- a) Balanced ventilation
- b) Air conditioning
- c) Exhaust ventilation
- d) Plenum ventilation

Question 6:

Which of the following is not a social goal of housing standards?

- a) Economic stability
- b) Family participation in community life
- c) Family life
- d) Prevention of non-infectious diseases

Question 7:

What is the optimum floor area recommended per adult in a living room?

- a) 50 – 100 sq. ft
- b) 400 – 450 sq. ft
- c) 150 – 200 sq. ft
- d) 100 – 150 sq. ft

Question 8:

During a field visit, you explain about ideal living standards to the general public. Which of the following is false regarding windows in housing standards?

- a) At least 2 windows in each room
- b) At a height more than 5 feet in a living room

- c) One window should open into open space
- d) Window area should be 1/5th of floor area

Question 9:

Which of the following is incorrect about the housing standards in India?

- a) Minimum wall thickness is 9 inches
- b) Roof not less than 10 feet from the ground
- c) Daylight factor should be $\geq 1\%$ over half the floor area
- d) Air space per capita must be 250 – 500 cubic feet

Question 10:

Which of the following is not a rural housing standard in India?

- a) Ample verandah space
- b) One room with sufficient space
- c) Built-up area should not exceed 1/3rd of total area
- d) Window area should be 10% of floor area

Question 11:

While recording the detailed history of a patient with symptoms of tuberculosis, you note that his home is overcrowded. According to the accepted standard against overcrowding, how many people can 3 rooms accommodate?

- a) 4
- b) 6
- c) 5
- d) 7

Question 12:

To avoid overcrowding, what is the minimum area required per person in a house?

- a) 50 – 70 sq. ft.

- b) 70 - 90 sq. ft.
- c) 90 – 100 sq. ft.
- d) 110 – 120 sq. ft.

Question 13:

Which of the following is not an organization related to housing in India?

- a) National Building Organization
- b) Housing and Urban Development Corporation
- c) National Buildings Construction Corporation Ltd
- d) National Rural Housing Services

Question 14:

Which of the following was launched to address rural housing in India?

- a) Rajiv Gandhi Shramik Kalyan Yojana
- b) Pradhan Mantri Kaushal Vikas Yojana
- c) Rashtriya Krishi Vikas Yojana
- d) Indira Awaas Yojana

Answer Key

Question No.	Correct Option
1	b
2	a
3	a
4	a
5	d
6	d
7	a
8	b
9	d
10	b

11	c
12	b
13	d
14	d

Detailed Explanations

Solution to Question 1:

Ventilation involves the quality control of incoming air but not quality control of outgoing air.

It also involves the replacement of vitiated air (air with decreased oxygen content) by a supply of fresh outdoor air. It aims to provide a comfortable thermal environment and reduce the risk of infections.

Solution to Question 2:

The number of air changes in drawing rooms or living rooms must be 2 – 3 per hour.

The number of air changes in work rooms and assemblies must be 4 – 6 per hour.

Solution to Question 3:

Ventilation provided by wind blowing through a space is called perflation.

Ventilation can be broadly classified as natural and mechanical:

Natural ventilation includes:

- Perflation is the ventilation of a space achieved by allowing wind to blow through it and removing the obstacles for this to happen (e.g., opening windows and doors).
- Aspiration occurs when there is an obstruction as wind bypasses it and creates a suction action
- Diffusion is the passage of air through small spaces or openings
- Cross-ventilation is when air is allowed a natural path through the space by providing openings at appropriate places, opposite each other.

The drawback of natural ventilation is that regulation of air quality is not possible.

Mechanical ventilation includes:

- Exhaust ventilation utilizes exhaust fans placed high up near the roof on the external walls. A vacuum is created by which fresh air enters the ventilated space.
- Plenum ventilation uses centrifugal fans to blow fresh air into the room and create a positive pressure, by which vitiated air is pushed out.

- Balanced ventilation is the combination of both exhaust and plenum methods of ventilation.
- Air conditioning can be done through mechanical ventilation. It is the simultaneous control of all or at least three factors that influence air quality in a confined room or space. These factors include temperature, humidity, air movement, distribution, dust, bacteria, odours and toxic gases.

Solution to Question 4:

Perflation is not a type of mechanical ventilation. It is a type of natural ventilation.

Ventilation can be broadly classified as natural and mechanical:

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Solution to Question 5:

The mechanical system of ventilation that employs only centrifugal fans is plenum ventilation.

Ventilation can be broadly classified as natural and mechanical:

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Solution to Question 6:

Prevention of non-infectious diseases is not a social goal of housing standards.

The social goals of housing include:

- Shelter
- Family life
- Access to community facilities
- Family participation in community life
- Economic stability

Solution to Question 7:

The optimum floor area recommended per adult person in a living room is 50 – 100 sq. ft.

Non-Rural Housing Standards in India:

Housing parameters	Description
Site	Elevated to protect from rains and floods. Independent access to the street. Away from vector breeding sites and other pollutants like smoke, noise, dust. The soil on which it is founded must be dry and safe.

Housing parameters	Description
Floor	Apucca floor that is impermeable, easy to clean, free from cracks, damp-proof. Height of plinth must be 2 – 3 feet.
Walls	9 inch thick, strong, low heat capacity, weather-resistant, unsuitable for harbourage of rats/vermin.
Roof	Height not less than 10 feet (3 m).
Rooms	The number of rooms must be according to the size of the family. However, the basic number is two.
Floor area	A living room should have at least 120 sq. ft. of area for occupancy of more than 1 person and 100 sq. ft. for occupancy of only one person. 50 - 100 sq. ft. floor area per person is optimum in a living room.
Cubic space	500 – 1000 cubic feet of air space per capita.
Windows	At least 2 windows per room with one opening into an open space. Windows must be at a height of not more than 3 feet above the ground. The window area must be 1/5th of the floor area.
Lighting	The daylight factor should be >1% over half the floor area.
Sanitary privy	Every house must have a sanitary privy.
Others	Proper facilities for garbage disposal, bathing and washing and a good water supply.

Solution to Question 8:

The windows in the living room must be at a height of not more than 3 feet above the floor.

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Sanitary privy	Every house must have a sanitary privy.
Others	Proper facilities for garbage disposal, bathing and washing and a good water supply.

Solution to Question 9:

Air space per capita must be 500 - 1000 cubic feet.

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Housing parameters	Description
Site	Elevated to protect from rains and floods. Independent access to the street. Away from vector breeding sites and other pollutants like smoke, noise, dust. The soil on which it is founded must be dry and safe.
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Walls	9 inch thick, strong, low heat capacity, weather-resistant, unsuitable for harbourage of rats/vermin.
Roof	Height not less than 10 feet (3 m).
Rooms	The number of rooms must be according to the size of the family but the basic number is two.
Floor area	A living room should have at least 120 sq. ft. of area for occupancy of more than 1 person and 100 sq. ft. for occupancy of only one person. 50 - 100 sq. ft. floor area per person is optimum in a living room.
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Windows	At least 2 windows per room with one opening into an open space. Windows must be at a height of not more than 3 feet above the ground. The window area must be 1/5th of the floor area.
Lighting	The daylight factor should be >1% over half the floor area.
Sanitary privy	Every house must have a sanitary privy.
Others	Proper facilities for garbage disposal, bathing and washing and a good water supply.

Solution to Question 10:

One room with sufficient space is not a rural housing standard in India.

According to rural housing standards, there must be at least two living rooms.

Rural housing standards are:

- At least two living rooms
- Ample verandah space
- Separate kitchen with space for washing utensils
- Built-up area should not exceed 1/3rd of total area
- Sanitary latrine
- Window area should be 10% of floor area
- Sanitary well or tube well within quarter mile(400 m) of house
- Cattle sheds must be 25 feet away from the house
- Adequate arrangements for disposal of waste water, refuse and garbage

Solution to Question 11:

The accepted standard against overcrowding for 3 rooms is 5 persons.

Overcrowding is considered to exist if 2 persons over 9 years of age, not husband and wife, of opposite sexes are obliged to sleep in the same room.

The accepted standards against overcrowding are as follows:

The floor space standards against overcrowding are as follows:

No of rooms	No of persons
1 room	2 persons
2 rooms	3 persons
3 rooms	5 persons
4 rooms	7 persons
5 rooms	10 persons (1 room for every 2 additional persons)

Area	No of persons
110 sq. ft. and more	2 persons

Area	No of persons
90 – 100 sq. ft.	1 ½ persons (Children from 1 – 10 are counted as a half unit)
70 – 90 sq. ft.	1 person
50 – 70 sq. ft.	½ person

Solution to Question 12:

To avoid overcrowding, the minimum area required per person in a house is 70 – 90 sq. ft.

The accepted standards against overcrowding are as follows.

The floor space standards against overcrowding are as follows.

No of rooms	No of persons
1 room	2 persons
2 rooms	3 persons
3 rooms	5 persons
4 rooms	7 persons
5 rooms	10 persons (1 room for every 2 additional persons)

Area	No of persons
110 sq. ft. and more	2 persons
90 – 100 sq. ft.	1 ½ persons (Children from 1 – 10 are counted as a half unit)
70 – 90 sq. ft.	1 person
50 – 70 sq. ft.	½ person

Solution to Question 13:

National Rural Housing Services is not an organization related to housing in India.

Rural Housing Services is an agency of the United States Department of Agriculture located in Washington, D.C.

The organizations related to housing in India are:

- National Buildings Organization (NBO).

- National Buildings Construction Corporation Ltd.
- Housing and Urban Development Corporation (HUDCO).
- Hindustan Housing Factory.

They work under the Union Ministry of works and Housing to deal with various aspects of housing.

Solution to Question 14:

The Indira Awas Yojana was launched in 1985 to address rural housing in India.

It is now called the Pradhan Mantri Gramin Awaas Yojana (2015), which is a social welfare flagship programme created to provide housing to the rural poor in India.

A similar scheme called Housing for All by 2022 or Pradhan Mantri Awaas Yojana (Urban) was begun to address housing for the urban poor.

Option A: Rajiv Gandhi Shramik Kalyan Yojana is an unemployment protection scheme.

Option B: Pradhan Mantri Kaushal Vikas Yojana was begun for the recognition and standardization of skills among the youth.

Option C: Rashtriya Krishi Vikas Yojana is an agricultural scheme.

The logo of Pradhan Mantri Awas Yojana is given below.



Light, Sound and Radiation

Question 1:

Which of the following is incorrect about the requirements of good lighting?

- a) Colour of light should approximate daylight
- b) Sharp shadows should be absent
- c) Ceilings and roofs must have a 50% reflection factor
- d) Glare should be absent

Question 2:

What is the SI unit for luminous intensity?

- a) Candela
- b) Lumen
- c) Lux
- d) Lambert

Question 3:

Correctly match the following variables with their SI unit.

- a) 1-A, 2-B, 3-C, 4-D
- b) 1-B, 2-A, 3-D, 4-C
- c) 1-D, 2-C, 3-A, 4-B
- d) 1-B, 2-C, 3-D, 4-A

Question 4:

What is the basic minimum illumination for satisfactory vision?

- a) 10 – 12 foot candles
- b) 15 – 20 foot candles
- c) 20 – 30 foot candles

d) 30 – 40 foot candles

Question 5:

Which of the following is correct about daylight factor?

- a) D.F. = Instantaneous illumination outdoors x 100/ Simultaneously occurring illumination indoors
- b) D.F. = Instantaneous illumination indoors x 100/ Simultaneously occurring illumination outdoors
- c) D.F. = Delayed illumination indoors x 100/ Simultaneously occurring illumination outdoors
- d) D.F. = Delayed illumination outdoors x 100/ Simultaneously occurring illumination indoors

Question 6:

You have been requested to inspect the working conditions of an office for clerks. What is the standard illumination recommended for such a space?

- a) 100 lux
- b) 1100 lux
- c) 900 lux
- d) 400 lux

Question 7:

A watchmaker presents with symptoms of eye strain. What is the optimum illumination recommended at his workplace?

- a) 2000 – 3000 lux
- b) 1300 – 2000 lux
- c) 1000 – 1200 lux
- d) 3000 – 3800 lux

Question 8:

What is the intensity of sound produced during whispering?

- a) 10 - 20 dB

- b) 30 - 40 dB
- c) 20 - 30 dB
- d) 0 - 10 dB

Question 9:

What is the upper limit of daily exposure to noise which people can tolerate without hearing damage?

- a) 95 dB
- b) 65 dB
- c) 75 dB
- d) 85 dB

Question 10:

What is the acceptable noise level in a hospital ward?

- a) 40 – 50 dBA
- b) 35 – 40 dBA
- c) 50 – 60 dBA
- d) 20 – 35 dBA

Question 11:

A steel plant worker presented with sudden onset of bilateral hearing loss after hearing a blast in the plant. On examination, both his tympanic membranes were ruptured. What would have been the minimum intensity of the blast to cause this injury?

- a) 90 dB
- b) 110 dB
- c) 130 dB
- d) 160 dB

Question 12:

A 32-year-old patient presents with complaints of whistling and buzzing in both his ears since yesterday. He gives a history of being exposed to continuous noise due to construction near his house. At what intensity of noise do such conditions begin to occur?

- a) 75 dB
- b) 80 dB
- c) 90 dB
- d) 100 dB

Question 13:

An airline ground staff officer with difficulty hearing is found to have permanent hearing loss on follow up. Continuous exposure to sound above what intensity can cause this condition?

- a) 160 dB
- b) 120 dB
- c) 110 dB
- d) 100 dB

Question 14:

The WHO defines disabling hearing loss in adults as a hearing loss of more than:

- a) 60 dB
- b) 45 dB
- c) 35 dB
- d) 25 dB

Question 15:

Which of the following is a unit of absorbed dose of radiation?

- a) Roentgen
- b) Rad
- c) Rem
- d) Sievert

Question 16:

What is the SI unit of radiation dose equivalent?

- a) Rad
- b) Gray
- c) Sievert
- d) Roentgen

Question 17:

What is the permissible dose of manmade radiation per year?

- a) 3 rads
- b) 5 rads
- c) 7 rads
- d) 4 rads

Question 18:

A young girl who is 1 month pregnant is concerned about a chest X-ray that she had taken 3 weeks back. What is the dose of radiation below which deterministic effects on the fetus are not seen?

- a) 0.5 mSv
- b) 2 mSv
- c) 3 RAD
- d) 5 RAD

Question 19:

What was the 10 days rule in radiation related to?

- a) Occupational health
- b) Pregnancy
- c) Industrial radiation
- d) Radiotherapy

Answer Key

Question No.	Correct Option
1	c
2	a
3	b
4	b
5	b
6	d
7	a
8	c
9	d
10	d
11	d
12	c
13	d
14	c
15	b
16	c
17	b
18	d
19	b

Detailed Explanations

Solution to Question 1:

For good lighting in a room, ceilings and roofs must have an 80% reflection factor.

The requirements of good lighting are:

- Sufficient light to prevent eye strain. An illumination of 15 to 20-foot candles is accepted for satisfactory vision.
- Distribution of light must be uniform with the same intensity over the whole field of work
- Absence of glare
- Absence of sharp shadows

- The source of light should be constant and steady
- The colour of light should approximate the colour of daylight
- Ceilings and roofs must have an 80% reflection factor. Walls must have 50 to 60%, furniture 30 to 40% and floors 15 to 20%.

Solution to Question 2:

The SI unit for luminous intensity is candela.

The various units used for measuring light are given below.

Quantity Measured	Description	SI Unit	Other Units Used
Luminous intensity	Brightness of point source	Candela	Candle power
Luminous flux	Flow of light	Lumen	_____
Illumination(Illuminance)	Amount of light reaching a surface	Lux	Foot candle, Phot or Lumen/cm ²
Luminance	Amount of light reflected from a surface	Lambert	Foot lambert, Candles/cm ²

Solution to Question 3:

The correctly matched pairs are as follows:

The various units used for measuring light are given below.

Variable	SI Unit
Luminous intensity	Candela
Luminous flux	Lumen
Illumination	Lux
Luminance	Lambert

Quantity Measured	Description	SI Unit	Other Units Used
Luminous intensity	Power of light source	Candela	Candle power
Luminous flux	Flow of light	Lumen	_____
Illumination(Illuminance)	Amount of light reaching a surface	Lux	Foot candle, Phot or Lumen/cm ²
Luminance(Brightness)	Amount of light reflected from a surface	Lambert	Foot lambert, Candles/cm ²

Solution to Question 4:

The basic minimum illumination for satisfactory vision is accepted to be 15 - 20 foot candles. The illumination requirements vary from as little as 5 foot candles in stairways and corridors to 100 foot candles in some industries.

Solution to Question 5:

Daylight factor (DF)

$$\text{Daylight factor} = \frac{\text{Instantaneous illumination indoors}}{\text{Simultaneously occurring illumination outdoors}} \times 100$$

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Daylight factor (D.F.) is the ratio of the illumination at a given point to illumination at a point exposed simultaneously to the whole hemisphere of the sky (taken as 500-foot candles) excluding direct sunlight.

The D.F. is used for the measurement of daylight. It is determined by a modified photo-electric meter called daylight factor meter.

In living rooms, the daylight factor should be at least 8% and in kitchens, it should be 10%.

Solution to Question 6:

The standard illumination recommended for general office work is 400 lux.

The following standards have been recommended by the Illuminating Engineer Society.

Visual Task	Illumination
Casual reading	100 lux
General office work	400 lux
Fine assembly	900 lux
Very severe tasks	1300 – 2000 lux
Watchmaking	2000 – 3000 lux

Solution to Question 7:

The optimum illumination recommended for watchmaking is 2000 – 3000 lux.

The following standards have been recommended by the Illuminating Engineer Society.

Visual Task	Illumination
Casual reading	100 lux
General office work	400 lux
Fine assembly	900 lux
Very severe tasks	1300 – 2000 lux
Watchmaking	2000 – 3000 lux

Solution to Question 8:

Whispering produces a sound of 20 – 30 dB.

Loudness or intensity depends upon the amplitude of vibrations. It is expressed in decibels (dB).

The intensity of sounds produced during various activities is listed below.

The frequency of sound is expressed in hertz (Hz). The human ear can hear frequencies from about 20 to 20,000 Hz.

Source	Loudness
Whispering	20 – 30 dB
Normal conversation	60 – 65 dB
Heavy street traffic	60 – 80 dB
Boiler factories	120 dB

Solution to Question 9:

The upper limit of daily exposure to noise which people can tolerate without hearing damage is 85 dB. This is the maximum acceptable noise level.

It has been observed that the human ear responds in a non-uniform way to different sound-pressure levels. A weighting curve, called curve A has been constructed which takes into account the subjective effects of sound. Sound-pressure levels are therefore expressed in dBA because this reflects the perception of that sound emission by the normal human ear.

Acceptable noise levels in various surroundings are listed below.

Locality	Place	Acceptable noise levels (dBA)
Residential	Bedroom	25
	Living room	40
Commercial	Office	35 – 45
	Conference	40 – 45
	Restaurants	40 – 60
Industrial	Workshop	40 – 60
	Laboratory	40 – 50
Educational	Classroom	30 – 40
	Library	35 – 40
Hospitals	Wards	20 – 35

Solution to Question 10:

The acceptable noise level in a hospital ward is 20 – 35 dBA.

Acceptable noise levels in various surroundings are listed below.

Locality	Place	Acceptable noise levels (dBA)
Residential	Bedroom	25
	Living room	40
Commercial	Office	35 – 45
	Conference	40 – 45
	Restaurants	40 – 60
Industrial	Workshop	40 – 60
	Laboratory	40 – 50
Educational	Classroom	30 – 40
	Library	35 – 40
Hospitals	Wards	20 – 35

Solution to Question 11:

A minimum intensity of 160 dB can lead to rupture of the tympanic membrane, resulting in hearing loss.

Solution to Question 12:

The given clinical scenario is suggestive of auditory fatigue that begins to appear with exposure to noise at 90 dB.

Solution to Question 13:

Continuous exposure to sound above 100 dB can cause permanent hearing loss.

Solution to Question 14:

According to WHO, disabling hearing loss refers to hearing loss greater than 35 dB in the better hearing ear in adults.

WHO grades of hearing loss:

Grade of hearing loss range	Range of hearing loss
0: No impairment	< 20 dB
1: Mild	20 - 34 dB
2: Moderate	35 - 49 dB
3: Moderately Severe	50 - 64 dB
4: Severe	65 - 79 dB
5: Profound	>80 dB

Solution to Question 15:

Rad is a unit of absorbed dose of radiation by tissue or material. It has been replaced by the SI unit which is gray (Gy).

Solution to Question 16:

The SI unit of radiation dose equivalent is sievert (Sv).

Dose equivalent (H) = Absorbed dose (D) x Quality factor (Q)

The quality factor depends upon the density of ionization produced in the tissue by radiation. The factor Q for X-rays, γ -rays and electrons is 1 and that for an alpha particle is 20.

Solution to Question 17:

The permissible dose of manmade radiation is 5 rads per year.

Solution to Question 18:

Deterministic effects on the fetus are not seen below 5 rads or 50 mSv of radiation.

Dose limits of radiation exposure are:

- Occupational exposure - Effective dose should not exceed 20 mSv/year averaged over 5 consecutive years / 30 mSv in a year.
- General public exposure - Effective dose should not exceed 1 mSv/year
- Occupational exposure for pregnant staff - Effective dose to the fetus should not exceed 1mSv/year (same as the general public).

For pregnant patients who require diagnostics with exposure to ionising radiation, the radiation protection principles are followed:

- The exposure should be justified
- The dose should be as low as reasonably achievable (ALARA principle)
- The dose limit is not particularly specified as it involves a single time exposure to diagnostic radiation. Usually, the effective dose to the fetus is $\leq 1\text{mSv}$ and deterministic effects to the fetus are not seen below 50 mSv (5 Rad).

Solution to Question 19:

The 10 days rule in radiation was related to pregnancy. It is aimed at preventing unintentional exposure of the embryo or fetus to radiation.

This rule states that if a radiograph is being done on the abdomen or pelvis of a woman in her childbearing age, then it should be done during the 10 days following the onset of menstruation.

In most situations, there is growing evidence that strict adherence to the "ten-day rule" may be unnecessarily restrictive. Since organogenesis starts 3 to 5 weeks post-conception, it was felt that radiation exposure in early pregnancy couldn't result in malformation. Based on this, it was suggested to do away with the 10-day rule and replace it with a 28-day rule. This means that radiological examination if justified, can be carried throughout the cycle until a period is missed.

Waste and Sewage Disposal

Question 1:

Solid wastes generally should not contain:

- a) Food wastes
- b) Nightsoil
- c) Demolition products
- d) Sewage residue

Question 2:

Which of the following is not a method of solid waste or refuse disposal?

- a) Oxidation ponds
- b) Controlled tipping
- c) Composting
- d) Incineration

Question 3:

Which of the following is the most insanitary method of waste disposal?

- a) Dumping
- b) Controlled tipping
- c) Manure pits
- d) Composting

Question 4:

Which method is most appropriate when suitable land for waste disposal is available?

- a) Dumping
- b) Controlled tipping
- c) Incineration

- d) Manure pits

Question 5:

In controlled tipping, the amount of land required for the trench method to be practiced for a population of 10,000 is:

- a) 4 acres
- b) 5 acres
- c) 2 acres
- d) 1 acre

Question 6:

Which of the following is false about composting?

- a) Bangalore method is an aerobic process
- b) Mechanical composting uses a controlled environment
- c) Nightsoil can also be disposed with composting
- d) Compost has few or no disease-causing pathogens

Question 7:

All of the following are methods of disposing only excreta except:

- a) River outfall
- b) Aqua privy
- c) Oxidation ponds
- d) Bangalore method

Question 8:

A military camp is being set up in an area for one month. Which of the following latrines will not be suitable here?

- a) Shallow trench latrine
- b) Pit latrine

- c) Borehole latrine
- d) Septic tank

Question 9:

Which of the following is the oldest non-service type of latrines in India?

- a) Borehole latrines
- b) Dug well latrines
- c) Water-seal latrines
- d) Aqua privy

Question 10:

There is an outbreak of hookworm infection in your locality. On investigation, it is found that the outbreak is a result of open-air defecation. The district council decides to construct an RCA type of latrine in every household. What is the optimum water seal required?

- a) 5 cm
- b) 3 cm
- c) 2.5 cm
- d) 4 cm

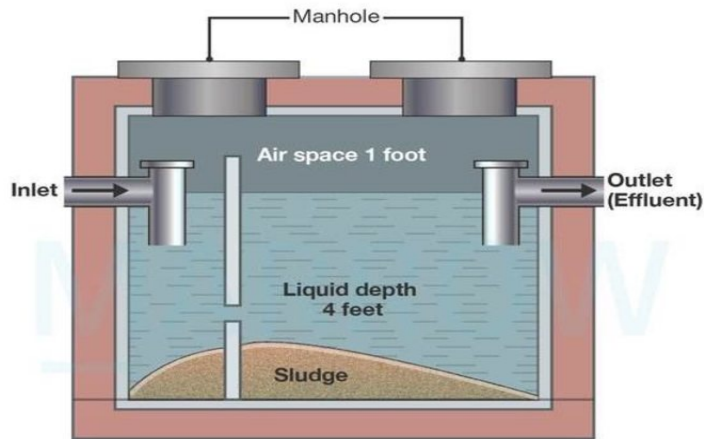
Question 11:

A health committee formed under Swachh Bharat Mission is involved in constructing septic tanks in every household. Which of the following is true about a septic tank?

- a) Always double chambered
- b) Minimum 200 gallon capacity
- c) Depth is 3-4 m
- d) Retention period is 24 hours

Question 12:

Which of the following is false about the sewage treatment shown in the image below?



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- a) Desludging must be done at least once a year
- b) Useful for very large communities
- c) Anaerobic digestion occurs inside the tank
- d) Soap and disinfectants like phenol should not be used inside the tank

Question 13:

What is wastewater without human excreta called?

- a) Sewage
- b) Refuse water
- c) Sullage
- d) Effluent

Question 14:

What is the average amount of sewage which flows through the sewerage system in 24 hours known as?

- a) Sewage rate
- b) Dry weather flow
- c) RCA index
- d) Sludge rate

Question 15:

Which of the following is not used to express the strength of sewage?

- a) E. coli count
- b) Suspended solids
- c) Biochemical oxygen demand
- d) Chemical oxygen demand

Question 16:

What does the biochemical oxygen demand of sewage indicate?

- a) Organic content
- b) Bacterial content
- c) Anaerobic bacteria
- d) Toxic substances

Question 17:

Which of the following values of biochemical oxygen demand would indicate that the sewage is weak?

- a) 100 mg/l
- b) 500 mg/l
- c) 300 mg/l
- d) 250mg/l

Question 18:

Which of the following is not a part of the primary treatment of sewage?

- a) Screening
- b) Grit chamber
- c) Aeration tank
- d) Primary sedimentation

Question 19:

What is the function of the grit chamber in modern sewage treatment plants?

- a) Formation of sludge
- b) Removal of large floating objects
- c) Settlement of heavier particles
- d) Allow anaerobic auto-digestion of organic matter

Question 20:

Trickling filter method is used in which of the following?

- a) Primary sewage treatment
- b) Secondary sewage treatment
- c) Sewage effluent disposal
- d) Screening of sewage

Question 21:

Which of the following is the heart of the activated sludge process?

- a) Aeration tank
- b) Primary sedimentation tanks
- c) Sludge digestion tank
- d) Secondary sedimentation tanks

Question 22:

Which of the following is not a requirement for an oxidation pond?

- a) Sunlight
- b) Algae
- c) Aeration
- d) Bacteria

Answer Key

Question No.	Correct Option
1	b
2	a
3	a
4	b
5	d
6	a
7	d
8	d
9	a
10	c
11	d
12	b
13	c
14	b
15	a
16	a
17	a
18	c
19	c
20	b
21	a
22	c

Detailed Explanations

Solution to Question 1:

Solid wastes should not contain nightsoil, which is untreated human waste (feces).

Solid wastes generally contain:

- Garbage (food wastes)
- Rubbish waste (papers, plastics, etc.)
- Demolition products (bricks, masonry, etc.)
- Sewage treatment residue

- Dead animals and other waste materials

Solution to Question 2:

Oxidation ponds are not used for solid waste or refuse disposal. It is used for sewage treatment.

Methods of solid waste disposal are:

- Dumping
- Controlled tipping
- Incineration
- Composting
- Manure pits
- Burial

Solution to Question 3:

The most insanitary method of waste disposal is dumping.

Methods of waste disposal include:

- Controlled tipping or sanitary landfill - Most satisfactory method of refuse disposal where suitable land is available
- Incineration - Method of choice where suitable land is not available
- Composting
- Manure pits
- Burial
- Dumping

Solution to Question 4:

Controlled tipping is the most satisfactory method of refuse disposal when suitable land is available.

In this method, waste materials are placed in a trench, compacted, and covered with earth at the end of the day. This method is also called a sanitary landfill. Chemical, bacteriological and physical changes occur in the buried refuse with temperatures rising to over 60 degrees Celsius within 7 days.

Methods used for controlled tipping are:

- Trench method

- Ramp method
- Area method

Sewage farming is also a method of sewage disposal in places where sufficient and suitable land is available. It is also called land treatment or broad irrigation.

Solution to Question 5:

In controlled tipping, one acre of land is required by the trench method to be practiced for a population of 10,000.

It is also known as a sanitary landfill. It is the most satisfactory method of refuse disposal where suitable land is available. In this method, waste material is placed in a trench or other prepared area. Then, it is adequately compacted and covered with earth at the end of the day.

Methods used in controlled tipping are:

- Trench method: Done in places where level ground is available
- Ramp method: Well suited where the terrain is moderately sloping
- Area method: Used for filling land depressions, disused quarries, and clay pits

Solution to Question 6:

The Bangalore method of composting is an anaerobic process.

Composting is a method of combined disposal of refuse and nightsoil. In this process, organic matter is broken down into compost. It has manure value and few or no disease-causing pathogens.

Bangalore method is an anaerobic method and is also called hot fermentation process. Trenches with a depth of not more than 90cm are used for this. This is done by spreading a layer of refuse (15 cm thick) at the bottom of the trench and nightsoil (5cm thick) is added next. Layers of refuse and nightsoil are added alternatively till the heap rises to 30cm above ground level. The heap is then covered with excavated earth. At the end of 4 – 6 months, decomposition will be completed.

Mechanical composting is an aerobic process. In this method, compost is manufactured on a large scale by processing raw materials and turning out a finished product. The refuse is first cleared of salvageable materials. It is then pulverised to reduce the size of particles to less than 2 inches. The pulverised refuse is then mixed with sewage, sludge or nightsoil in a rotating machine and incubated.

Solution to Question 7:

Bangalore method, also known as the hot fermentation process, is a form of composting that is used for disposing of both solid waste and nightsoil (excreta).

Methods of excreta disposal:

- Unsewered areas:
- Service type latrines (conservancy system) - In this, nightsoil is collected manually from pail or bucket type latrines and then disposed of by burying or composting
- Non-service type latrines (sanitary latrines)
- Borehole latrine
- Dug well or pit latrine
- Water-seal latrines – Examples are P.R.A.I. type, R.C.A. type, and Sulabh shauchalaya
- Septic tanks
- Aqua privy
- Latrines for camps and temporary use
- Shallow and deep trench latrine
- Pit latrine
- Borehole latrine
- Sewered areas:
- Water carriage system and sewage treatment
- Primary treatment – Screening, removal of grit and plain sedimentation
- Secondary treatment – Trickling filters, activated sludge process
- Other methods – sea and river outfalls, sewage farming, oxidation ponds

Solution to Question 8:

Septic tanks cannot be used for camps and temporary purposes.

Methods of excreta disposal:

- Unsewered areas:
- Service type latrines (conservancy system) - In this, nightsoil is collected manually from pail or bucket type latrines and then disposed of by burying or composting
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- Septic tanks
- Aqua privy

- Latrines for camps and temporary use
- Shallow and deep trench latrine
- Pit latrine
- Borehole latrine
- Sewered areas:
 - Water carriage system and sewage treatment
 - Primary treatment – Screening, removal of grit and plain sedimentation
 - Secondary treatment – Trickling filters, activated sludge process
 - Other methods – sea and river outfalls, sewage farming, oxidation ponds

Solution to Question 9:

The oldest non-service type of latrines in India is the borehole latrine. It was introduced in the 1930s to combat hookworm infections.

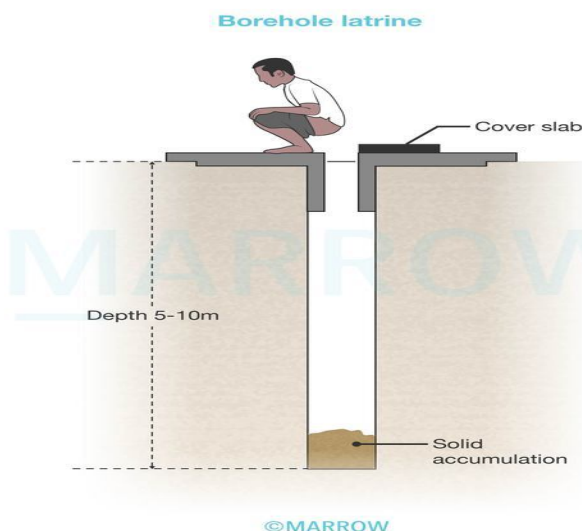
Special equipment called auger is used to dig a borehole latrine.

Advantages:

- No special staff is needed for upkeep as after using the latrine is closed and another hole is dug elsewhere
- Breeding of flies is less
- No water pollution if located far from a water source

Demerits:

- Fills up rapidly
- Augers are not readily available
- Water pollution is still a risk



Solution to Question 10:

The optimum water seal required for an RCA type of latrine is 2 – 2.5 cm.

It was designed by the Research-cum-Action Projects (RCA) in Environmental Sanitation of the Ministry of Health. Water seals aids in preventing flies and other organisms from reaching the waste. It also prevents the escape of odors and foul gases, eliminating the nuisance from the smell.

Features of RCA type of latrine	
Location	More than 15m away from the water source
Squatting plate	Should be impervious and should allow for drainage of water
Pan	Receives the nightsoil, urine, and washes water.
Trap	Holds the water and provides the water seal. A good trap has an effective seal of around 2 – 2.5 cms.
Connecting pipe	Present in indirect type and absent in direct type
Dug well	Has to be changed after it is filled
Superstructure	Built for privacy and shelter

Solution to Question 11:

The recommended retention period for a septic tank is 24 hours.

If the retention period is too short, there is insufficient treatment of sewage and if it is too long, then undue septicity results.

There are single and double-chambered septic tanks. The minimum capacity recommended for a septic tank is 500 gallons. The depth of a septic tank is usually 1.5 to 2m.

Solution to Question 12:

The given image is of a septic tank. It is not useful for very large communities and is used for household sewage treatment.

Solution to Question 13:

Wastewater that doesn't contain human excreta is called sullage. It is also called greywater. E.g., waste water from the kitchen.

Sewage is wastewater from a community that contains solid and liquid excreta along with street and yard washings. It is also called blackwater.

Solution to Question 14:

The average amount of sewage which flows through the sewerage system in 24 hours is known as dry weather flow.

Solution to Question 15:

E. coli count is not used to express the strength of sewage.

The strength of sewage is expressed only in terms of:

- Biochemical oxygen demand
- Chemical oxygen demand
- Suspended solids

Solution to Question 16:

Biochemical oxygen demand (BOD) indicates the organic content of sewage.

It is defined as the amount of oxygen absorbed by a sample of sewage for aerobic destruction by living microorganisms.

If wastewater contains toxic substances, the organic load is measured by the chemical oxygen demand.

Solution to Question 17:

Sewage with biochemical oxygen demand (BOD) of 100mg/L would indicate that it is weak.

If the BOD is ≥ 300 mg/L, sewage is said to be strong.

Solution to Question 18:

Aeration tank is not a part of primary treatment of sewage. It is a part of secondary treatment of sewage.

Solution to Question 19:

The function of the grit chamber in modern sewage treatment plants is to allow heavier particles in the sewage to settle.

Solution to Question 20:

The trickling filter method is used in secondary sewage treatment.

Solution to Question 21:

The heart of the activated sludge process is the aeration tanks.

Solution to Question 22:

Aeration is not a requirement for an oxidation pond.

Oxidation ponds are also called waste stabilization ponds or redox ponds or sewage lagoons. They have become a popular method of purifying sewage for small communities. It is usually an open, shallow pool of around 1 – 1.5m depth with an inlet and outlet.

Requisites for an oxidation pond include:

- Algae
- Saprophytic bacteria
- Sunlight

All of the above factors help in the treatment of sewage waste through aerobic and anaerobic processes. The effluent formed can be used for land irrigation or can be forwarded for further treatment.

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Medical Entomology - Mosquitoes and Flies

Question 1:

Which of the following is false regarding mosquitoes?

- a) It is the definitive host in malaria
- b) *Mansonia* is a vector for Malayan filariasis
- c) Life span of mosquitoes is around 8 to 34 days
- d) All mosquitoes can travel only for 100 metres

Question 2:

Which of the following species of mosquitoes breed in dirty water and water bodies containing aquatic vegetation respectively?

- a) *Anopheles* and *Aedes*
- b) *Culex* and *Mansonia*
- c) *Aedes* and *Anopheles*
- d) *Mansonia* and *Culex*

Question 3:

Which of the following statements is false about larva of mosquitoes?

- a) *Anopheles* larva has siphon tube
- b) *Culex* larva lies in water with head faced downwards
- c) *Anopheles* larva lies parallel to the undersurface of water
- d) *Culex* larva has siphon tube

Question 4:

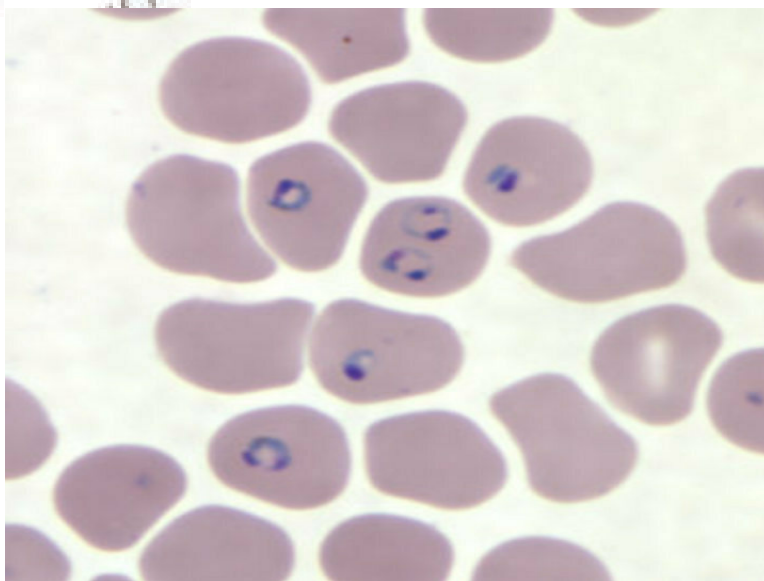
Identify the species of mosquito shown in the image below.



- a) Anopheles
- b) Culex
- c) Aedes
- d) Mansonia

Question 5:

A 25-year-old lady from a tribal area presented with fever, chills, vomiting, and abdominal pain for 4 days. The peripheral smear of the patient is given below. Which of the following is true about the mosquito which transmits this disease?



- a) Wings are spotted

- b) Breeds in polluted water
- c) Larva has siphon tube
- d) Keeps its body horizontal while feeding

Question 6:

A patient presented with fever, headache, and retro-orbital pain for 3 days. He tested positive for the NS1 antigen. Which of the following mosquitoes transmits the disease he's suffering from?

- a) Tiger mosquito
- b) Nuisance mosquito
- c) Anopheles stephensi
- d) Anopheles culicifacies

Question 7:

Which of the following diseases is not transmitted by Aedes mosquito?

- a) 2, 4, and 5
- b) 3 and 5 only
- c) 1 and 3 only
- d) 1, 2, and 4

Question 8:

A banker who is scheduled to travel to Congo is advised to take the yellow fever vaccine. What is the flight range of the mosquito which transmits this disease?

- a) 1 kilometre
- b) 100 metres
- c) 400 metres
- d) 11 kilometres

Question 9:

A patient with fever and mucosal bleeding is diagnosed with dengue fever. Which of the following is false about the mosquito which transmits this disease?

- a) It is a day biter
- b) Development of larva from egg does not require water
- c) It also transmits yellow fever
- d) It has a striped body pattern

Question 10:

A 10-year-old boy is brought to the OPD with complaints of fever and headache for 5 days. Serological examination shows the presence of IgM antibodies to the JE virus. Which of the following mosquitoes transmits this disease?

- a) Anopheles sundaicus
- b) Culex tritaeniorhynchus
- c) Aedes albopictus
- d) Anopheles culicifacies

Question 11:

Match the following diseases and the mosquitoes which transmit them.

- a) I - c, II - a, III - d, IV - b
- b) I - a, II - b, III - c, IV - d
- c) I - b, II - c, III - d, IV - a
- d) I - d, II - a, III - b, IV - c

Question 12:

Which of the following is true about pathogenic mosquitoes?

- a) Anopheles has spotted abdomen
- b) Mansonia lays eggs singly
- c) Culex transmits yellow fever
- d) Aedes lays eggs singly

Question 13:

You are posted in a PHC in a coastal region as a part of your rural postings. Which of the following *Anopheles* species is most commonly found here?

- a) *Anopheles philippinensis*
- b) *Anopheles stephensi*
- c) *Anopheles fluviatilis*
- d) *Anopheles culicifacies*

Question 14:

Which of the following species of *Anopheles* mosquitoes transmits urban malaria?

- a) *Anopheles philippinensis*
- b) *Anopheles stephensi*
- c) *Anopheles fluviatilis*
- d) *Anopheles culicifacies*

Question 15:

What is the distance around an airport or seaport that has to be kept free of all mosquitoes?

- a) 1 kilometre
- b) 500 metres
- c) 400 metres
- d) 100 metres

Question 16:

Identify the following vector.



- a) *Phlebotomus argentipes*
- b) *Phlebotomus papatasi*
- c) *Musca domestica*
- d) *Triatoma infestans*

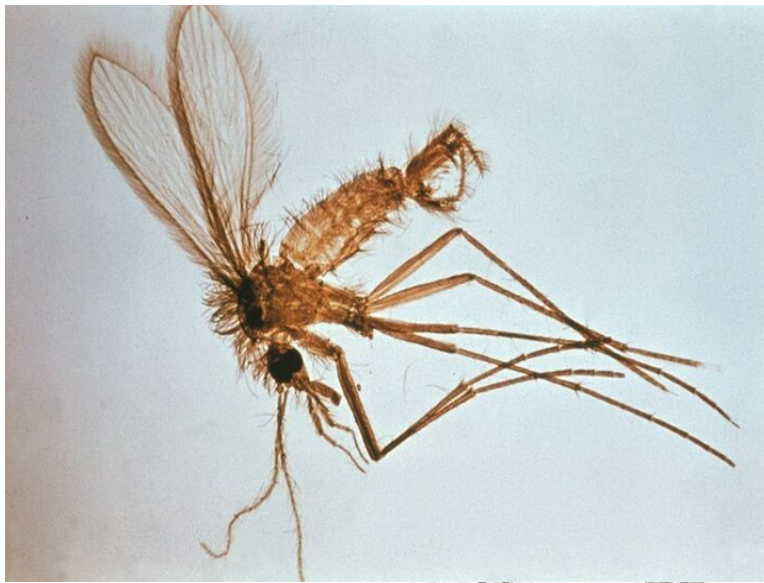
Question 17:

Which of the following is not a method of disease transmission by houseflies?

- a) Biting
- b) Defecation
- c) Vomit drop
- d) Mechanical transmission

Question 18:

Which disease is transmitted by the following vector that is found in cracks and crevices of walls in the day time?



- a) Japanese encephalitis
- b) Visceral leishmaniasis
- c) Dengue
- d) Trypanosomiasis

Question 19:

Sandfly does not transmit:

- a) Kala azar
- b) Oriental sore
- c) Oraya fever
- d) Trench fever

Question 20:

Which of the following diseases is transmitted by the vector shown in the image given below?



- a) Onchocerciasis
- b) Kyasanur forest disease
- c) Oroya fever
- d) Sleeping sickness

Question 21:

A patient presents with blurring of vision and a painless swelling over his left elbow which has gradually been increasing in size over the past 1 year. On examination, he is noted to have patchy depigmentation on his legs. Which of the following vectors transmits the disease he's suffering from?

- a) Sandfly
- b) Blackfly
- c) Rat flea
- d) Tsetse fly

Question 22:

Spot the wrongly matched arthropod and disease.

- a) Housefly – Poliomyelitis
- b) Louse – Epidemic typhus
- c) Sandfly – Oriental sore

d) Blackfly – Chagas disease

Answer Key

Question No.	Correct Option
1	d
2	b
3	a
4	c
5	a
6	a
7	b
8	b
9	b
10	b
11	c
12	d
13	b
14	b
15	c
16	c
17	a
18	b
19	d
20	d
21	b
22	d

Detailed Explanations

Solution to Question 1:

Most mosquitoes can travel or fly for distances up to 11 kms.

Solution to Question 2:

Culex species of mosquitoes breed mostly in dirty and polluted water. Mansonia species of mosquitoes breed in water bodies containing aquatic vegetation.

Solution to Question 3:

Anopheles larva has no siphon tube as shown in the image below.



The image given below shows Culex larva having a siphon tube.



Solution to Question 4:

The given image is of an Aedes mosquito.

They can be identified with the stripes on the legs and body. They are also called tiger mosquitoes because of their stripes.

Solution to Question 5:

The given clinical scenario and peripheral smear showing ring form/ early trophozoites in erythrocytes are suggestive of Plasmodium falciparum malaria. It is transmitted by Anopheles mosquitoes. They have spotted wings.

They breed in clean water. Their larva has no siphon tube and lies parallel to the undersurface of water. They tend to sit with their heads low and their rear ends raised high, especially when feeding.

Solution to Question 6:

The given clinical scenario along with the positive NS1 antigen test is suggestive of dengue fever. It is transmitted by the Aedes mosquito, which is also called as tiger mosquito.

Solution to Question 7:

Japanese encephalitis and West Nile fever are transmitted by Culex mosquitoes.

Chikungunya, yellow fever, and rift valley fever are transmitted by Aedes mosquitoes.

Solution to Question 8:

Yellow fever is transmitted by the Aedes mosquito. Its flight range is 100 metres.

Solution to Question 9:

Dengue fever is transmitted by the Aedes mosquito. The conversion from the egg stage to the larval stage of this mosquito requires water.

Eggs of any mosquito species do not survive without water. But the eggs of Aedes mosquitoes lie dormant in dry conditions for up to about nine months. They can hatch if exposed to favourable conditions such as water and food.

Solution to Question 10:

The given clinical scenario is suggestive of Japanese encephalitis. It is transmitted by Culex mosquitoes like Culex tritaeniorhynchus

Solution to Question 11:

The correctly matched diseases with the mosquitoes which transmit them are:

Disease	Mosquito
Yellow fever	Aedes
Bancroftian filariasis	Culex
Malaria	Anopheles
Malayan filariasis	Mansonia

Solution to Question 12:

Aedes and Anopheles lay eggs singly.

Culex lays eggs in rafts and Mansonia lays eggs in star-shaped clusters.

Anopheles has spotted wings while the other pathogenic species have unspotted wings.

Aedes transmits yellow fever, dengue and chikungunya. Culex is the main vector for JE and filariasis.

Solution to Question 13:

Anopheles stephensi and Anopheles sundaicus are found in coastal regions.

Anopheles stephensi is also the main vector of malaria in urban regions. Anopheles culicifacies and Anopheles philippinensis are found in the plains. Anopheles fluviatilis and Anopheles minimus are found in foot-hill regions.

Solution to Question 14:

Urban malaria is transmitted by Anopheles stephensi.

Solution to Question 15:

The distance around an airport or seaport that has to be kept free of all mosquitoes is 400 metres. This helps in limiting the spread of vector-borne diseases to new geographic regions.

Solution to Question 16:

The given image is of a housefly, whose scientific name is *Musca domestica*.

Solution to Question 17:

Houseflies do not transmit diseases by biting. They transmit diseases by mechanical transmission by their legs or feet, vomit drop, and defecation.

They transmit a variety of diseases like:

- Typhoid
- Polio
- Dysentery
- Amoebiasis
- Anthrax
- Yaws

Solution to Question 18:

The given image is of a sandfly. It transmits visceral leishmaniasis.

Note: Sandfly acts as a vector for the following diseases:

- Kala-azar
- Oriental sore
- Sandfly fever
- Oroya fever

Japanese encephalitis is transmitted by the *Culex* mosquito. Dengue is transmitted by the *Aedes* mosquito.

Trypanosomiasis is transmitted by the tsetse fly.

Solution to Question 19:

Trench fever is not transmitted by sandflies. It is transmitted by lice and caused by Bartonella quintana.

Solution to Question 20:

The vector shown in the image is a tsetse fly. It transmits African sleeping sickness (African human trypanosomiasis).

Tsetse flies belong to the genus Glossina. They have not demonstrated insecticide resistance. Methods of tsetse fly control include spraying of DDT, clearing of vegetation, destruction of wild game and genetic control.

Solution to Question 21:

The given clinical scenario is suggestive of onchocerciasis, also known as river blindness. It is transmitted by blackfly.

The usual presentation of onchocerciasis is the formation of a subcutaneous nodule or onchocercoma. It is a firm non-tender nodule arising from bony sites like the scalp, elbow, ribs, and iliac crest. It is associated with pruritus and rashes. Patients also present with patchy depigmentation on the legs called leopard skin.

The presence of microfilaria in the eye causes photophobia and gradual loss of vision. The most severe manifestations are sclerosing keratitis and chorioretinitis which can lead to blindness.

Blackfly is also called Simulium. It has a flight range of almost 100 miles, making control of adult flies difficult. Abate is widely used as a larvicidal insecticide. The image given below shows a blackfly.



Solution to Question 22:

Blackflies (Simulium) are the vector for onchocerciasis and not Chaga's disease.

An arthropod or an invertebrate that transmits infection is called a vector. It transmits infection by:

- Inoculation into or through the skin or mucous membrane
- Biting
- Deposit of infective materials on skin or food or other objects

Note: Chagas disease is transmitted by the reduviid bugs or kissing bugs or triatomines.

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Medical Entomology - Ticks, Fleas and Mites

Question 1:

A veterinarian with fever, chills, and myalgia is diagnosed with Q fever. Which of the following is the most common reservoir of this disease?

- a) Ticks
- b) Mites
- c) Louse
- d) Housefly

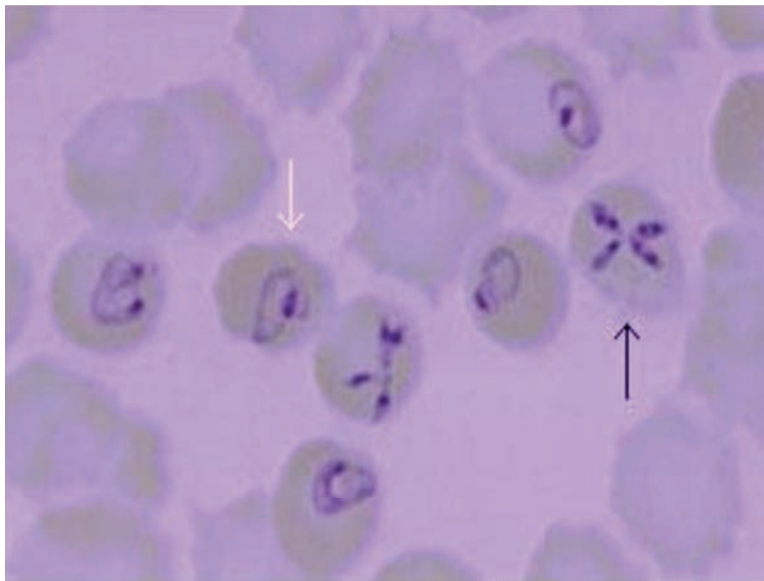
Question 2:

Which of the following is not transmitted by hard ticks?

- a) Relapsing fever
- b) Kyasanur forest disease
- c) Rocky Mountain spotted fever
- d) Tularemia

Question 3:

Peripheral smear of a patient with high grade fever, arthralgia, and myalgia is given below. Which of the following transmits this disease?



- a) Ticks
- b) Mites
- c) Fleas
- d) Sandfly

Question 4:

A patient with fever and joint pain presented with the following lesion on his left leg. Which of the following transmits this disease?



- a) Trombiculid mite

- b) Louse
- c) Soft tick
- d) Ixodes tick

Question 5:

Which of the following is not true about the vector in the given image?



- a) Cannot withstand starvation
- b) Head lies ventrally
- c) Reservoir for Q fever
- d) Scutum is absent

Question 6:

Which of the following is false about the vector in the given image?



- a) Scutum is present
- b) It cannot withstand starvation
- c) Its nymphal stage has seven stages
- d) It lays hundreds of eggs at once

Question 7:

Which of the following are also known as chiggers?

- a) Trombiculid mites
- b) Sand fleas
- c) Rat fleas
- d) Soft ticks

Question 8:

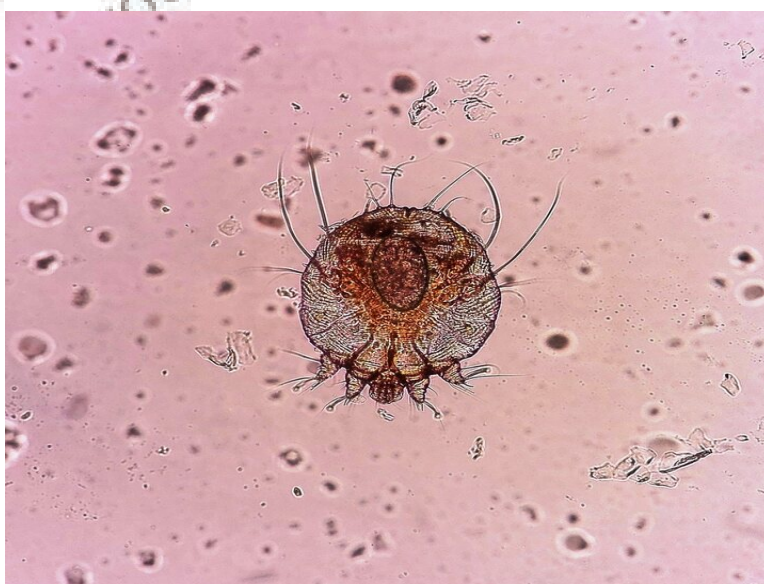
Which of the following diseases is transmitted by the vector shown in the image given below?



- a) American trypanosomiasis
- b) African trypanosomiasis
- c) Onchocerciasis
- d) Visceral leishmaniasis

Question 9:

A 10-year-old boy presents with complaints of nocturnal pruritus. Examination shows burrows over the skin of his hands. KOH mount prepared from these lesions is given below. Which of the following vectors transmits this disease?



- a) Trombiculid mite

- b) Itch mite
- c) Cyclops
- d) Ixodes tick

Question 10:

Which of the following diseases is not transmitted by mites?

- a) Scabies
- b) Scrub typhus
- c) Rickettsial pox
- d) Kyasanur forest disease

Question 11:

You are given the following slide during your community medicine practicals. Which of the following diseases is not transmitted by this vector?



- a) Endemic typhus
- b) Relapsing fever
- c) Pediculosis
- d) Epidemic typhus

Question 12:

Which of the following diseases is transmitted by lice?

- a) Murine typhus
- b) Epidemic typhus
- c) Tick typhus
- d) Scrub typhus

Question 13:

Which of the following is not transmitted by lice?

- a) Relapsing fever
- b) Q fever
- c) Trench fever
- d) Epidemic typhus

Question 14:

Identify the following vector which transmits murine typhus.



- a) Sand flea
- b) Rat flea
- c) Trombiculid mite

- d) House mouse mite

Question 15:

Which of the following diseases is not transmitted by rat fleas?

- a) Murine typhus
- b) Pneumonic plague
- c) Endemic typhus
- d) Hymenolepiasis

Question 16:

A flea survey is being conducted in an area prone to outbreaks of plague. Which of the following indices is not used for this purpose?

- a) Percentage incidence of flea species
- b) Rodent infestation rate
- c) General flea index
- d) Incidence of plague

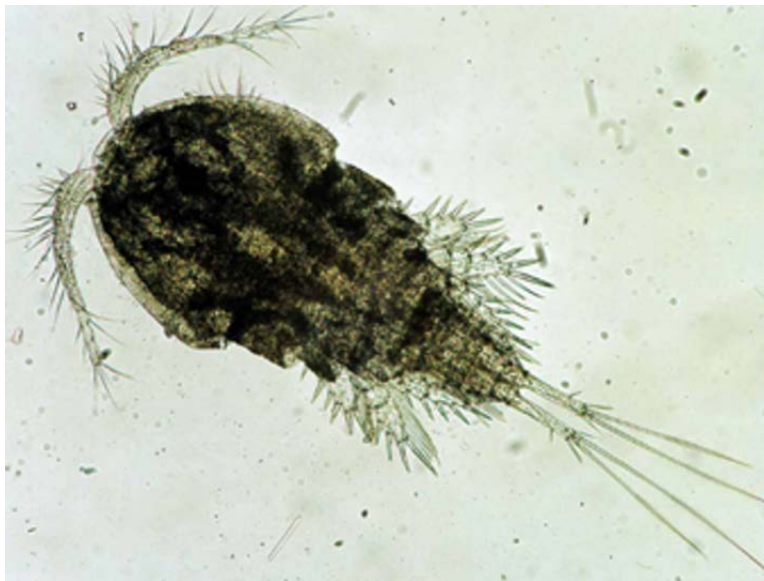
Question 17:

Which of the following is true about sand fleas?

- a) Vectors for kala-azar
- b) Cause foot ulcers
- c) Not found in India
- d) Cause buboes in the groin

Question 18:

Which of the following is transmitted by the vector shown in the image given below?



- a) Threadworm infection
- b) Guinea worm disease
- c) Rocky Mountain spotted fever
- d) Tularemia

Question 19:

Which of the following diseases is not associated with rodents?

- a) Trichinosis
- b) Lassa fever
- c) Tularemia
- d) Ascariasis

Answer Key

Question No.	Correct Option
1	a
2	a
3	a
4	d
5	a

6	c
7	a
8	a
9	b
10	d
11	a
12	b
13	b
14	b
15	b
16	d
17	b
18	b
19	d

Detailed Explanations

Solution to Question 1:

Ticks are the most common reservoir of Q fever.

Coxiella burnetii is the causative pathogen. The most commonly identified sources of human infection are farm animals, such as cattle, goats, and sheep.

Infected mammals shed *C. burnetii* in their urine, feces, milk, and placenta. Human exposure to *C. burnetii* may occur as a result of inhalation of contaminated aerosols arising from the placenta or parturient fluids of infected livestock or from contaminated secretions on the surface of newborn animals.

People working with farm animals and veterinarians are at a higher risk of infection.

Solution to Question 2:

Relapsing fever is not transmitted by hard ticks, but by soft ticks.

Solution to Question 3:

The given clinical scenario with peripheral smear showing maltese cross appearance is suggestive of babesiosis. It is transmitted by ticks.

Solution to Question 4:

The given clinical scenario along with the image showing bulls eye rash is suggestive of Lyme disease. It is transmitted by Ixodes tick.

Solution to Question 5:

The given image is of a soft tick. It can withstand starvation for over a year.

The head of the soft tick lies ventrally and it does not have a scutum. It serves as a reservoir for Q fever and transmits relapsing fever.

Solution to Question 6:

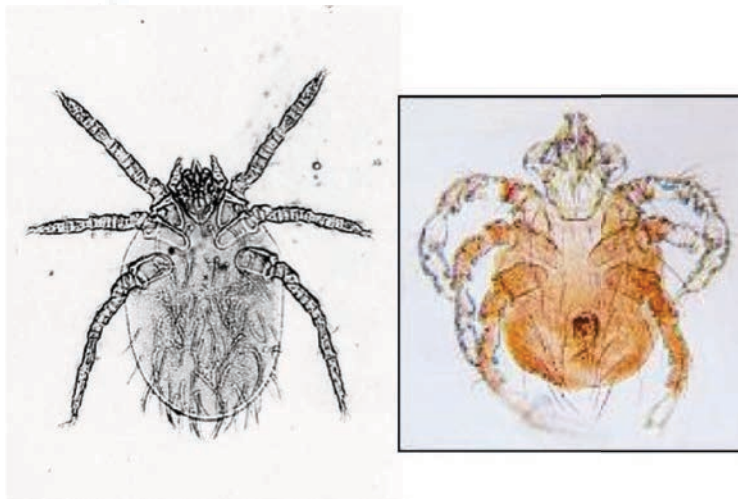
The given image is of a hard tick. The hard tick has only one nymphal stage.

It has a hard chitinous shield called the scutum which covers the entire back in males a small portion in the front in females. It feeds night and day and cannot withstand starvation. It lays several hundreds or thousands of eggs in one sitting.

Solution to Question 7:

Larval forms of trombiculid mites are also known as chiggers.

They are spider-like arthropods which are vectors for scrub typhus. The image given below shows trombiculid mite.



Solution to Question 8:

The given image is of a reduviid bug. It transmits American trypanosomiasis, also known as Chagas disease.

Reduviid bug is also known as the kissing bug or triatomine bug. It lives in cracks, fissures and other spaces in the walls and ceilings of human dwellings. Residual spraying with HCH or dieldrin is used for the control of these bugs.

Option B: African trypanosomiasis (sleeping sickness) is transmitted by the tsetse fly.

Option C: Onchocerciasis is transmitted by the blackfly (*Simulium*).

Option D: Visceral leishmaniasis (kala-azar) is transmitted by the sandfly.

Solution to Question 9:

The given clinical scenario and KOH mount showing the itch mite are suggestive of scabies. It is transmitted by itch mite or *Sarcoptes scabiei*.

The female parasite burrows into the epidermis where it breeds. It is a small parasite, barely visible to the naked eye, with 4 pairs of legs.

Its life cycle consists of four stages—egg, larva, nymph, and adult stages. Its spread can be through close contact or contaminated clothes.

Benzyl benzoate is very effective as a sarcopticide. HCH, tetmosol, and sulphur ointments are also used.

The image given below shows itch mite.



Solution to Question 10:

Kyasanur forest disease is not transmitted by mites. It is a viral hemorrhagic fever transmitted by hard ticks.

Scabies, scrub typhus, and rickettsial pox are transmitted by mites.

Solution to Question 11:

The above image is of a head louse. It does not transmit endemic typhus which is transmitted by fleas.

Lice transmit epidemic typhus, relapsing fever, and trench fever.

Solution to Question 12:

Epidemic typhus is transmitted by lice and caused by *Rickettsia prowazekii*.

Murine typhus is transmitted by the rat flea. Tick typhus is transmitted by a hard tick. Scrub typhus is transmitted by trombiculid mite.

Solution to Question 13:

Q fever is not transmitted by lice. It is spread by the inhalation of contaminated aerosols containing *Coxiella burnetii*.

Soft ticks are the most common reservoirs of Q fever. The aerosols arise from the placenta or parturient fluids of infected livestock or from contaminated secretions on the surface of newborn animals.

Solution to Question 14:

The given image is of a rat flea-*Xenopsylla cheopis*. It transmits murine typhus.

Solution to Question 15:

Pneumonic plague is not transmitted by rat fleas. It is transmitted by inhalation of aerosolized droplets.

Rat fleas are vectors for murine or endemic typhus, bubonic plague, and hymenolepiasis caused by *Hymenolepis nana* and *Hymenolepis diminuta*.

Solution to Question 16:

Incidence of plague is not an index used in flea surveys. Flea indices that are used in flea surveys are:

- General flea index
- Specific flea index
- Percentage incidence of flea species
- Rodent infestation rate

Solution to Question 17:

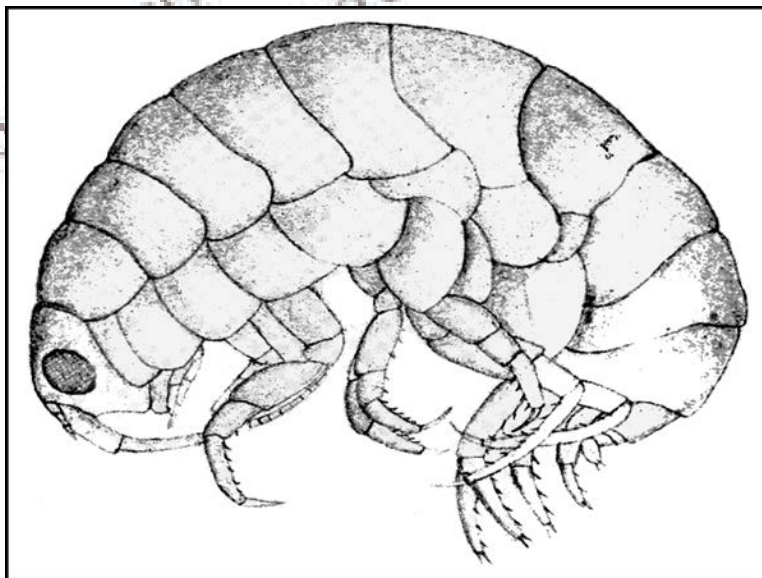
Sand fleas cause ulcers in the feet.

They are also called jigger or chigoe fleas. The species called *Tunga penetrans* is of particular importance as it causes tungiasis.

The fertilized female burrows into the skin of the feet, often beneath the nail and causes ulcers. Tetanus and gas gangrene frequently occur because of secondary infection.

They flourish best in sandy soil and are found in the western parts of India.

The below image shows a sand flea.



Solution to Question 18:

The given image is of a cyclops or water flea. It is the vector for guinea worm disease or dracunculiasis and fish tapeworm infection.

Humans are infected with these diseases by drinking water that contains cyclops. Its average life span is 3 months.

Physical methods like straining and boiling of water can ensure the killing of this organism. Chemicals that can be used to control cyclops are chlorine, lime, and abate. Biological methods include Gambusia fish and barbel fish.

Solution to Question 19:

Ascariasis is not associated with rodents.

Diseases related to rodents are:

- Bacterial -
- Plague
- Tularemia
- Salmonellosis
- Leptospirosis
- Rat-bite fever
- Viral -
- Lassa fever
- Encephalitis
- Hemorrhagic fever
- Rickettsial -
- Scrub typhus
- Endemic typhus
- Rickettsial pox
- Parasitic -
- Hymenolepis diminuta
- Leishmaniasis
- Amoebiasis
- Trichinosis
- Chagas disease
- Fungal -
- Histoplasmosis
- Ringworm

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Methods of Pest Control

Question 1:

Which of the following would you advise as the best approach to control arthropods?

- a) Environmental control
- b) Chemical control
- c) Biological control
- d) Genetic control

Question 2:

Which of the following is the best method to control houseflies?

- a) Eliminating breeding places
- b) Using space sprays
- c) Raising larvivorous fishes
- d) Using residual sprays

Question 3:

Which of the following is not an organo-phosphate insecticide?

- a) Dieldrin
- b) Fenthion
- c) Malathion
- d) Diazinon

Question 4:

Which of the following is not an aryl organo-phosphate insecticide?

- a) Malathion
- b) Parathion
- c) Chlorthion

d) Diazinon

Question 5:

Which of the following insecticides is least toxic to man?

- a) Malathion
- b) Diazinon
- c) Parathion
- d) Chlorpyrifos

Question 6:

Which of the following is false about fenthion?

- a) It is used as a residual spray
- b) It is used as a repellent
- c) It is a stomach poison
- d) It has larvicidal action

Question 7:

Which of the following is false about pyrethrum?

- a) Residual action as good as DDT
- b) Vegetable origin
- c) Contact poison
- d) Can be used synergistically with DDT

Question 8:

Which of the following cannot be used to describe pyrethrum?

- a) Nerve poison
- b) Contact poison
- c) Space spray
- d) Residual spray

Question 9:

Which of the following is a synthetic pyrethroid insecticide?

- a) Propoxur
- b) Permethrin
- c) Parathion
- d) Pyrolan

Question 10:

Which of the following vectors has not shown resistance to synthetic pyrethroids, in a widespread manner in India yet?

- a) *Musca domestica*
- b) *Anopheles stephensi*
- c) *Aedes aegypti*
- d) *Phlebotomus argentipes*

Question 11:

Which of the following best describes DDT?

- a) Stomach poison
- b) Repellant
- c) Contact poison
- d) Fumigant

Question 12:

During a dengue outbreak, the municipal corporation plans a DDT spraying initiative. Which of the following is false about this insecticide?

- a) Para-para isomer is the most active fraction
- b) It acts as a neurotoxin
- c) It has residual action up to 18 months

- d) It causes immediate death of insects

Question 13:

Which of the following is true about the usage of Paris green for mosquito control?

- a) Anti-adult measure
- b) Effective against Anopheles
- c) Harmful in normal doses to man and animals
- d) Nerve poison

Question 14:

Which of the following mosquitoes can be controlled by destroying aquatic plants with herbicides?

- a) Mansonia
- b) Culex
- c) Aedes
- d) Anopheles

Question 15:

Which of the following is not a mosquito anti-larval measure?

- a) Mineral oils
- b) Gambusia fish
- c) Paris green
- d) Pyrethrum

Question 16:

Which among the following insecticides is most widely used as a space spray?

- a) HCH
- b) Malathion
- c) DDT

d) Pyrethrum

Question 17:

Match the following methods of mosquito control to the appropriate examples.

- a) I – a, II – b, III – c, IV – d
- b) I – b, II – d, III – c, IV – a
- c) I – c, II – a, III – b, IV – d
- d) I – d, II – b, III – a, IV – c

Question 18:

To curb an outbreak of plague in your locality, the municipal corporation decides to use rodenticides. Which of the following can be given as a single dose for this purpose?

- a) Warfarin
- b) Pindone
- c) Coumafuryl
- d) Zinc phosphide

Question 19:

Which of the following color labels is used to indicate the most hazardous pesticides?

- a) Red
- b) Green
- c) Yellow
- d) Black

Answer Key

Question No.	Correct Option
1	a
2	a

3	a
4	a
5	a
6	b
7	a
8	d
9	b
10	d
11	c
12	d
13	b
14	a
15	d
16	d
17	d
18	d
19	a

Detailed Explanations

Solution to Question 1:

Environmental control is the best approach to control arthropods.

The principles of arthropod control include:

- Environmental control - Best approach
- Chemical control - Use of insecticides, pesticides, etc.
- Biological control - Use of larvivorous fish like Gambusia
- Genetic control
- Newer methods -
- Chemosterilants
- Insect growth regulators
- Sex attractants
- Integrated approach involves usage of more than one modality of arthropod control

Solution to Question 2:

The best method to control houseflies is environmental control by the elimination of breeding places.

They have shown resistance to a number of insecticides, including DDT.

Solution to Question 3:

Dieldrin is an organo-chlorine insecticide. It is a broad-spectrum insecticide that acts as a contact, stomach, and inhalation poison.

Solution to Question 4:

Malathion is an alkyl organophosphate insecticide. It is used as an ultra-low volume spray in mosquito control.

Solution to Question 5:

Among the given options, malathion is the least toxic to man. It is the least toxic organo-phosphate insecticide.

Solution to Question 6:

Fenthion is not used as a repellent. A commonly used repellent is N,N-diethyl-m-toluamide (DEET) that is used for protection from mosquitoes and ticks.

Fenthion is a broad-spectrum insecticide that acts as both contact and stomach poison. It is used as a larvicide in mosquito control and also as a residual spray.

Solution to Question 7:

Pyrethrum does not have a residual action like DDT or other synthetic insecticides. So, most space sprays contain pyrethrum and DDT for synergistic action.

Pyrethrum is a natural insecticide of vegetable origin. It is extracted from the flowers of *Chrysanthemum cinerariaefolium*. It kills adult mosquitoes and other insects instantly on mere contact.

Solution to Question 8:

Pyrethrum is not used as a residual spray. It is a contact and nerve poison that is also used as a space spray.

Solution to Question 9:

Permethrin is a synthetic pyrethroid. Pyrethroids are synthetic insecticides, which are derived from naturally occurring pyrethrins. Pyrethroids modify the sodium channels in neuronal membranes and thereby disrupt the electrical signalling in the nervous system.

Solution to Question 10:

Sandflies (*Phlebotomus*) have not shown resistance to synthetic pyrethroids, in a widespread manner in India yet.

Solution to Question 11:

DDT is a contact poison that belongs to the organochlorine family. It has been widely used but a number of organisms have become resistant to its effects now.

Solution to Question 12:

DDT does not cause the immediate death of insects. It takes several hours for its action to be complete.

It contains 70 – 80% of para-para isomer that is the most active fraction of DDT. It acts on the nervous system of insects. The residual action of DDT may last as long as 18 months depending upon the treated surface.

Solution to Question 13:

Paris green is very effective against surface-feeding Anopheles.

It is an anti-larval measure and is a stomach poison. It is not harmful in normal doses to man and animals.

Solution to Question 14:

Mansonia mosquitoes breed in water with aquatic vegetation, and destroying this vegetation with herbicides would control this mosquito.

Solution to Question 15:

Pyrethrum is an anti-adult measure and not an anti-larval measure.

Mineral oils, Gambusia, and Paris green are anti-larval measures used in mosquito control.

Solution to Question 16:

The insecticide most widely used as a space spray is pyrethrum.

Solution to Question 17:

The correctly matched methods of mosquito control are:

Method of control	Example
Biological control	Barbados millions
Genetic control	Sex distortion
Chemical control	Larvicidal oil
Environmental control	Source reduction

Solution to Question 18:

Zinc phosphide is a single-dose rodenticide that causes death in a one-time exposure.

Acute or single-dose rodenticides are classified according to the precautions that need to be taken before their use:

- Those requiring ordinary care - Red squill, norbormide, zinc phosphide
- Those requiring maximal precaution - Sodium fluoroacetate, fluoroacetamide and strychnine
- Too dangerous for use - Arsenic trioxide, phosphorus, thallium sulphate

Multiple-dose or cumulative rodenticides cause death after multiple times of exposure. Examples include:

- Warfarin
- Coumafuryl
- Diphacinone

- Pindone

Solution to Question 19:

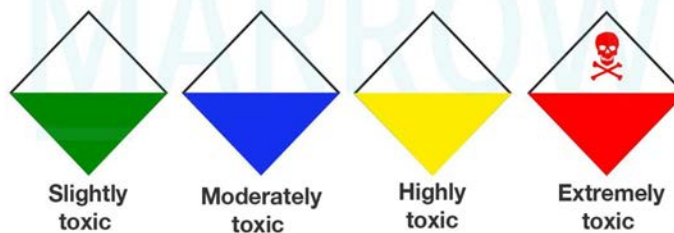
The most hazardous pesticides must have labels in red color.

Pesticide toxicity labels:

These labels are depicted in the image below.

Category of Pesticide	Color of Label
Extremely toxic	Bright red
Highly toxic	Bright yellow
Moderately toxic	Bright blue
Slightly toxic	Bright green

Degree of Pesticide Toxicity



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Biomedical Waste Management

Question 1:

What proportion of healthcare waste is hazardous waste?

- a) 15%
- b) 25%
- c) 50%
- d) 85%

Question 2:

Cytotoxic and expired drugs are disposed by:

- a) Dumping
- b) Autoclave
- c) Incineration
- d) Chemical disinfection

Question 3:

Which of the following category of wastes are treated by autoclaving and microwaving?

- a) Placenta, organs, blood bags
- b) Catheters, urine bags, gloves
- c) Anti neoplastic and expired drugs
- d) Culture plates, slides and tubes

Question 4:

While working as a 1st-year resident in the labor ward, where would you dispose of the placenta following a delivery?

- a) Red bag
- b) Blue bag

- c) Yellow bag
- d) Black bag

Question 5:

Which of the following statements are true about yellow waste disposal bags?

- a) 1 & 2
- b) 2 & 4
- c) 3, 4 & 5
- d) 1, 2 & 4

Question 6:

Which of the following should not be incinerated?

- a) 1, 2 & 3
- b) 3, 4 & 5
- c) 1, 2 & 4
- d) 2 & 4

Question 7:

An orthopedic resident has removed a K-wire from the fractured olecranon of a 13-year-old boy. Where should he discard this?

- a) Cadmium-free red colored bag
- b) Cardboard box with blue marking
- c) Puncture-proof and tamper-proof white container
- d) Leak-proof and tamper-proof yellow bag

Question 8:

Blood bag is to be disposed in:

- a) Red bin
- b) Yellow bin

- c) Blue bin
- d) Black bin

Question 9:

An HIV-positive man is brought to you following an accident. How would you discard his blood-soaked dressing material?

- a) Pour 1% hypochlorite on the dressing material and send for incineration
- b) Pour 5% hypochlorite on the dressing material and send for incineration
- c) Put the dressing material directly in an appropriate bag and send for incineration
- d) Pour 10% non chlorinated chemicals on the dressing material and send for incineration

Question 10:

Which of the following disposal methods is not as per biomedical waste management guidelines?

- a) Disposal of a dead fetus of 21 weeks in a yellow bag by deep burial in rural areas
- b) Pre-treatment of laboratory cultures with non-chlorinated disinfectant
- c) Disposal of a glass ampoule with its cytotoxic drug through incineration
- d) Disposal of radiological waste by deep burial

Question 11:

You are asked to catheterize an HBsAg-positive man before hernia surgery. According to the BMW management rules, where would you dispose of the plastic cover of the catheter?

- a) Yellow bag
- b) Blue bag
- c) Red bag
- d) Black bag

Question 12:

Which of the following diseases should healthcare workers involved in handling biomedical waste be vaccinated against?

- a) Hepatitis B alone
- b) Tetanus alone
- c) Hepatitis B and tetanus
- d) Hepatitis B, tetanus, and anthrax

Question 13:

An on-site waste treatment and disposal facility should be built in a hospital if the nearest common biomedical treatment facility is:

- a) > 25 km away
- b) > 50 km away
- c) > 75 km away
- d) > 45 km away

Question 14:

Which of the following is not a high-heat system for the treatment of biomedical waste?

- a) Hydroclaving
- b) Incineration
- c) Microwaving
- d) Plasma pyrolysis

Question 15:

Which of the following disposal methods uses high temperature and dry oxidation?

- a) Microwaving
- b) Autoclaving
- c) Incineration
- d) Hydroclaving

Question 16:

In a treatment facility, which of the following cannot be used to verify the treatment of biomedical waste done by incineration?

- a) *Bacillus atropheus*
- b) *Bacillus subtilis*
- c) *Geobacillus stearothermophilus*
- d) *Bacillus thuringiensis israelensis*

Question 17:

At a waste treatment facility, pharmaceutical waste was mixed with cement, lime, and water before disposal. What is this method known as?

- a) Hydroclaving
- b) Grounding
- c) Inertization
- d) Compacting

Question 18:

What is the correct combination of temperature and duration used in a hot air oven?

- a) 170 °C for 30 mins
- b) 160 °C for 60 mins
- c) 180 °C for 30 mins
- d) 190 °C for 15 mins

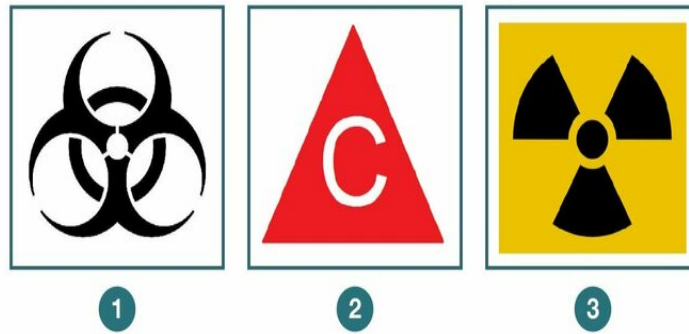
Question 19:

All the following are the characteristics of waste that can be incinerated except:

- a) Combustible matter $\geq$ 60%
- b) Low heating volume
- c) Non-combustible fines $\leq$ 20%
- d) Non-combustible solids $\geq$ 5%

Question 20:

Correctly match the following logos.



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- a) 1-c, 2-b, 3-a
- b) 1-c, 2-a, 3-b
- c) 1-a, 2-b, 3-c
- d) 1-b, 2-a, 3-c

Question 21:

You are working in a covid ward. After doffing, where will you dispose of the plastic coverall?

- a) Red bin
- b) Black bag
- c) Yellow bin
- d) Blue bin

Question 22:

An intern asks for your advice regarding the disposal of needles from the needle tip cutter after a vaccine drive. Which of the following bins will you advise her to discard them in?

- a) Yellow
- b) Red
- c) White
- d) Blue

Question 23:

Chemical waste is disposed in _____ bags.

- a) Yellow
- b) White
- c) Red
- d) Green

Answer Key

Question No.	Correct Option
1	a
2	c
3	b
4	c
5	a
6	c
7	b
8	b
9	c
10	d
11	d
12	c
13	c
14	c
15	c
16	d
17	c
18	c
19	d
20	b
21	a
22	c
23	a

Detailed Explanations

Solution to Question 1:

15% of health care waste is hazardous waste. It may be infectious, toxic, or radioactive in nature. The remaining 85% is general, non-hazardous waste.

Bio-medical waste means any waste that is generated during:

- Diagnosis, treatment, or immunization of human beings or animals
- Research activities pertaining thereto
- Production or testing of biologicals

Bio-medical waste is handled as per the Bio-medical Waste Management Rules, 2016. However, these rules do not apply to the following:

- Radioactive wastes
- Hazardous chemicals
- Solid wastes
- Lead-acid batteries
- Hazardous wastes
- e-Waste
- Hazardous microorganisms

Solution to Question 2:

Cytotoxic and expired drugs are collected in a yellow bag and then disposed of by incineration.

Expired cytotoxic drugs are to be returned to the manufacturer and subjected to incineration, encapsulation, or plasma pyrolysis. Other discarded medicines can be disposed by incineration.

Solution to Question 3:

The final treatment of catheters, urine bags, and gloves is done by autoclaving and microwaving.

They belong to recyclable contaminated waste and are disposed of in a red bag.

Option A: Human anatomical waste is disposed of by incineration.

Option C: Cytotoxic and expired/discarded medicine is disposed of by incineration/plasma pyrolysis/disposal in a secured landfill.

Option D: Microbiological waste such as culture plates and slides are pre-treated with nonchlorinated chemicals and later disposed of by incineration.

Solution to Question 4:

Human anatomical wastes, like the placenta, are discarded in a yellow bag and disposed of by incineration/ plasma pyrolysis/ deep burial.

Solution to Question 5:

Yellow waste disposal bags are made from non-chlorinated plastic material and chemical liquid wastes are collected in them.

Chlorinated items, glass, metal, and rubber are not sent for incineration as incinerating them would produce carcinogenic gases.

The right disposal methods of the other options are:

- Animal anatomical waste - Collected in yellow bags and later sent for incineration.
- Metallic body implants - Collected in cardboard boxes with blue coloured marking and sent for recycling following autoclaving.
- PVC gloves - Collected in red bags and sent for autoclaving followed by shredding.

Solution to Question 6:

Specimens of microorganisms and live or attenuated vaccines can be incinerated.

Wastes that should not be incinerated are:

- Sharps
- Pressurized gas containers
- Reactive chemical waste
- Waste with heavy metals like mercury and cadmium
- PVC
- Photographic or radiographic waste

Solution to Question 7:

Metallic body implants are disposed of in cardboard boxes with blue-colored markings.

The contents of the blue box are sent to Common Biomedical Waste Treatment and Disposal Facility (CBWTF). In the facility, these items are sterilized by autoclaving or microwaving and then sent for recycling.

Solution to Question 8:

Blood bag should be disposed off in the yellow bin.

Solution to Question 9:

The blood-soaked dressing material should be disposed of in a non-chlorinated yellow bag and sent for incineration. Pretreatment is not a requisite for incinerating soiled waste.

Items that require pretreatment with non-chlorinated chemicals before incineration:

- Used blood bags
- Other infectious plastic and liquid wastes
- Laboratory cultures, stocks, or specimens of microorganisms
- Live or attenuated vaccines
- Human and animal cell cultures used in research
- Dishes, and devices used for cultures

Solution to Question 10:

Radiological waste is not considered biomedical waste. Hence, deep burial is not done.

Disposal of radiological is covered under the provisions of the Atomic Energy Act. The radiological effluents are incinerated in a dedicated incinerator having off-gas cleaning facility. Another method is to convert the organic base into an aqueous form by treating it with sodium hydroxide.

Option A: Dead fetuses under the age of viability are considered human anatomical waste. Anatomical waste can be disposed of by deep burial in rural areas, where a common biomedical treatment facility and incineration is not available.

Option B: Laboratory cultures are classified under the category of microbiology, biotechnology, and other clinical laboratory waste. They are pretreated with a non-chlorinated disinfectant before they are sent for incineration.

Option C: The cytotoxic drugs are disposed of along with their packaging, through incineration.

Solution to Question 11:

The outer plastic cover of the Foleys catheter is discarded in a black bag.

Foleys catheters are packaged in plastic covers to keep them sterile. The plastic cover is not biomedical waste but is a part of general (municipal) waste. Hence, it is discarded in the black bag. The foleys catheter is disposed of in a red bag.

As per Bio-Medical Waste Management Rules, the general waste must be disposed of in green bins (for recyclable waste) and blue bins (for non-recyclable waste) containing black plastic bags.

Solution to Question 12:

Healthcare workers involved in handling biomedical waste should be vaccinated against hepatitis B and tetanus.

These are most likely to be transmitted by the handling of biomedical waste. Yearly health check-ups should be done, and a booster dose should be given if needed.

Solution to Question 13:

According to biomedical waste management rules of 2016, healthcare establishments shall establish an on-site treatment and disposal facility only when a common biomedical treatment facility (CBMWTF) is more than 75 km away.

According to these rules, biomedical waste (BMW) from healthcare facilities is to be transported to a CMBWTF. The establishment of CBMWTF reduces the need for transport and cost of treatment/ kg of BMW.

Solution to Question 14:

Microwaving is a low-heat system for the treatment of biomedical waste.

The heating systems used for the treatment and disposal of biomedical waste are:

Low-heat system	High-heat system
Autoclaving: Uses steam. It is of two types-Gravity flow autoclave Vacuum autoclave Microwaving: Uses moist heat and steam generated by microwave energy	Incineration: Waste is destroyed/denatured under high temperatures Plasma pyrolysis: Waste is treated at high temperatures under controlled conditions to form gases that are subjected to combustion in the secondary chamber Hydroclaving: The waste is subjected to indirect heating by applying steam to the outer jacket Dry heat sterilization: Waste sharps are treated using dry heat at a temperature >185°C for 150 minutes

Solution to Question 15:

Incineration is a high-temperature dry oxidation process.

Waste that is incinerated need not undergo any type of pre-treatment. It helps to reduce weight as well as volume. The temperature in the incinerator is 800°C in the primary chamber and 1050°C in the secondary chamber.

Types of incinerators:

- Rotary kilns
- Single chamber pyrolytic
- Double chamber pyrolytic

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Solution to Question 16:

Bacillus thuringiensis israelensis is not used to verify thermal methods of biomedical waste disposal. It is a biological larvicide, used in nonpotable water for killing mosquito larvae.

The following can be used to verify the adequacy of thermal methods of biomedical waste disposal:

- *Bacillus atropheus*
- *Bacillus subtilis*
- *Geobacillus stearothermophilus*

Solution to Question 17:

Inertization is the process of mixing biomedical waste with non-toxic substances to reduce the toxicity of the waste and to reduce the contamination of groundwater. It is most commonly done for incineration ash with high metal content.

The typical composition of the inertization mixture is:

- 65% pharmaceutical waste
- 15% cement
- 15% lime
- 5% water

Solution to Question 18:

The correct combinations of temperature and duration of sterilisation in a hot air oven is:

- 180°C for 30 mins
- 170°C for 60 mins
- 160°C for 2 hours

It can be used to sterilize powders, glasswares, and oils.

An autoclave is another apparatus that is used for sterilization. It uses steam at a temperature of 121°C for 30 mins under 15 lbs of pressure.

Solution to Question 19:

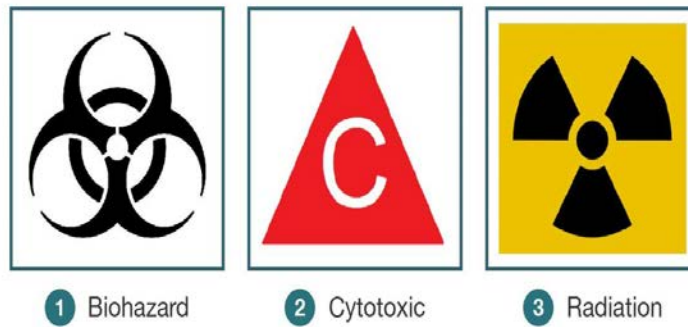
Only waste material containing $\leq 5\%$ of noncombustible solids can be incinerated. Wastes that are incinerated go in the yellow bag.

Waste products with the following characteristics can be incinerated:

- Combustible matter $\geq 60\%$
- Non-combustible solid $\leq 5\%$
- Non-combustible fine $\leq 20\%$
- Low heating volume
- Moisture content $\leq 30\%$.

Solution to Question 20:

The correctly matched logos are as shown below.



©Marrow

Solution to Question 21:

The plastic coverall used by a healthcare worker during the COVID-19 pandemic should be disposed of in the red bin.

Solution to Question 22:

Waste sharps including metals (both used, discarded, and contaminated metal sharps) are to be discarded in the white bin.

Solution to Question 23:

Chemical waste is disposed of in yellow-colored containers or non-chlorinated plastic bags.

Disaster Management

Question 1:

Which of the following is false regarding a disaster?

- a) Occurrence that causes damage to life and property
- b) Occurrence that has the potential to cause disruption or damage to people
- c) Occurrence that causes deterioration of health and health services
- d) Occurrence that can cause substantial economic harm

Question 2:

Which type of natural disaster causes the most deaths?

- a) Meteorological
- b) Fires
- c) Geological
- d) Hydrological

Question 3:

Which of the following steps of disaster management is in correct order?

- a) Disaster impact – Mitigation – Rehabilitation – Response
- b) Response – Mitigation – Disaster impact – Rehabilitation
- c) Disaster impact – Response – Rehabilitation – Mitigation
- d) Response – Disaster impact – Rehabilitation – Mitigation

Question 4:

Which of the following is false about disaster management?

- a) Measures to prevent hazards are designed in mitigation phase
- b) Rescue and evacuation occur in recovery phase
- c) Measures on vector control are done in rehabilitation phase

- d) Field care and tagging are done in disaster impact and response phase

Question 5:

Which of the following is the most common disease after a disaster?

- a) Acute respiratory infections
- b) Malaria
- c) Dengue
- d) Gastroenteritis

Question 6:

Which of the following diseases is not commonly seen in the post-disaster period?

- a) Leishmaniasis
- b) Leptospirosis
- c) Acute respiratory infections
- d) Rickettsia

Question 7:

A district in Andhra Pradesh reported severe drought. Which of the following vitamin deficiencies would you expect to see in this population?

- a) Vitamin A deficiency
- b) Vitamin B deficiency
- c) Vitamin C deficiency
- d) Vitamin D deficiency

Question 8:

Following a drought in Bihar, the government launched an immediate mass vaccination program. Which of the following vaccines is recommended in such cases?

- a) Cholera
- b) Hepatitis A

- c) Tetanus
- d) Measles

Question 9:

Immediately following a disaster, mass vaccination of which of the following vaccines does the WHO recommend?

- a) Typhoid
- b) Cholera
- c) Rabies
- d) None of the above

Question 10:

You are a field doctor at the site of an earthquake. You assist in rapidly classifying the injured on the basis of the likelihood of their survival and need for intervention. What is this process called?

- a) Field care
- b) Tagging
- c) Rehabilitation
- d) Triage

Question 11:

Which of the following is true about triage?

- a) Yellow – least priority
- b) Red – mortality
- c) Green – ambulatory
- d) Blue – ambulatory

Question 12:

Which of the following is correct about reverse triage?

- a) Classifying the victims based on the severity of their injuries and their survival with treatment

- b) Highest preference is given to severely injured patients
- c) More injured victims are given the lowest preference
- d) Color coding is done with red as the highest priority.

Question 13:

Triage is done during disaster management to ensure treatment of casualties is done properly. What is true about triage done at the disaster site?

- a) First come first treated basis
- b) Green labels are for those who need to be transported on priority
- c) Moribund patients receive the lowest priority
- d) Most commonly used system is three color code system

Question 14:

What is the level of residual chlorine to be maintained in the water bodies after a disaster?

- a) 0.2 - 0.5 mg/litre
- b) 0.1 - 0.4 mg/litre
- c) 0.4 - 0.6 mg/litre
- d) 0.7 - 0.9 mg/litre

Question 15:

Which of the following gas was responsible for the Bhopal Gas Tragedy?

- a) Methyl isocyanate
- b) Potassium isothiocyanate
- c) Sodium isothiocyanate
- d) Methyl isothiocyanate

Question 16:

Which of the following emission occurred in Chernobyl explosion accident of 1986?

- a) ^{235}U and ^{210}Po

- b) Methyl isocyanate
- c) Potassium isocyanate
- d) I131, Cs134, Sr90

Question 17:

What is the nodal ministry for disaster management in India?

- a) Ministry of Home Affairs
- b) Ministry of Environment, Forest, and Climate change
- c) Ministry of Agriculture
- d) Ministry of Earth Sciences

Question 18:

Which of the following is a nodal centre in case of disaster management?

- a) Primary health centre
- b) District hospital
- c) Community health centre
- d) District management authority

Question 19:

The NIDM is involved in the mitigation and management of disasters in India. Who is the president of this institution?

- a) Prime Minister of India
- b) President of India
- c) Union Home Minister
- d) Health Minister of India

Question 20:

Who is the chairman of the National Disaster Management Authority of India?

- a) Minister of Health and Family Welfare

- b) Prime Minister of India
- c) Union Home Minister
- d) Defence Minister

Answer Key

Question No.	Correct Option
1	b
2	d
3	c
4	b
5	d
6	a
7	a
8	d
9	d
10	d
11	c
12	c
13	c
14	d
15	a
16	d
17	a
18	d
19	c
20	b

Detailed Explanations

Solution to Question 1:

A disaster is defined as any occurrence that causes damage, ecological disruption, loss of human life, or deterioration of health and health services. It usually requires an extraordinary response from outside the affected community or area.

A hazard is any phenomenon that has the potential to cause disruption or damage to people and their environment.

Solution to Question 2:

Hydrological natural disasters cause the most deaths.

The Disaster Information Management System of DesInventar classifies natural disasters into five major categories. They are:

- Geophysical - earthquakes, tsunamis, and volcano eruptions
- Hydrological - floods, debris flow, and landslides
- Meteorological - cyclones, tornadoes, extreme temperatures, sandstorms, blizzards etc.
- Climatological - drought, forest wildfire, glacial lake outburst etc.
- Biological - epidemics and insect infestations

Solution to Question 3:

The correct order is Disaster impact – Response – Rehabilitation – Mitigation.

Disaster management includes three aspects:

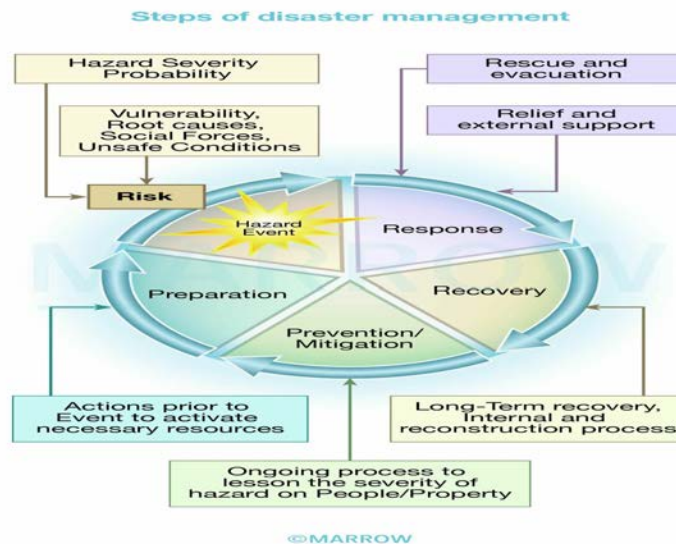
1. Disaster impact and response: It involves immediate management of mass casualties as follows:

- Search, rescue, and first-aid
- Field care
- Triage
- Tagging
- Identification of dead

2. Rehabilitation or recovery: It involves the restoration of the pre-disaster conditions by taking steps such as:

- Water supply
- Basic sanitation and personal hygiene
- Food safety
- Vector control

3. Mitigation: Measures designed either to prevent hazards from causing disaster or to reduce the effects of the disaster. This also includes preparedness for any impending disasters or in disaster-prone areas.



Solution to Question 4:

Rescue and evacuation occur in the response phase, not in the recovery phase.

Disaster management includes three aspects:

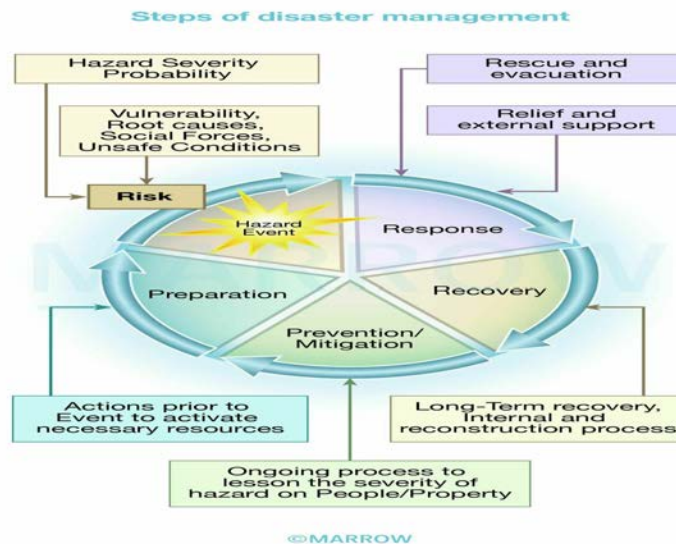
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- Food safety
- Vector control

3. Mitigation: Measures designed either to prevent hazards from causing disaster or to reduce the effects of the disaster. This also includes preparedness for any impending disasters or disaster-prone areas.



Solution to Question 5:

Gastroenteritis is the most commonly reported disease after a disaster.

It occurs due to contamination of water supply, damage to the sewerage system, overcrowding and poor sanitation. Therefore, supply of safe drinking water and proper disposal of excreta are the most practical and effective strategies of disease prevention in the post-disaster period.

Solution to Question 6:

Leishmaniasis is not commonly seen in the post-disaster period.

Diseases that are commonly seen in the post-disaster phase are:

- Gastroenteritis (most common)
- Acute respiratory infections
- Leptospirosis
- Rickettsiosis
- Rabies
- Equine encephalitis

Solution to Question 7:

Vitamin A deficiency is the most common deficiency in areas of famines and droughts. It leads to xerophthalmia and blindness.

Solution to Question 8:

Measles vaccination is recommended for infants and children, immediately following a disaster. This should be given along with vitamin A supplementation as measles can exacerbate vitamin A deficiency.

The vaccination of infants and children against measles is one of the most important public health response measures during a humanitarian crisis when less than 90% of the children or an unknown percentage have already been vaccinated against measles.

Measles and polio vaccines are recommended immediately after an emergency.

Option A: Cholera vaccine is not recommended.

Option B: Hepatitis A vaccine is recommended at the outset of an outbreak.

Option C: Tetanus vaccine is recommended in the early stages and post-emergency stage.

Solution to Question 9:

The WHO does not recommend typhoid, cholera, and rabies vaccines for mass vaccination after a disaster. However, high-risk populations are recommended to take the cholera vaccine. The rabies vaccine is given only for post-exposure prophylaxis.

Solution to Question 10:

The process of rapidly classifying the injured in a disaster on the basis of severity of injury and likelihood of survival with prompt medical intervention is called triage. It is carried out at the site of disaster.

Tagging is the identification of patients in a disaster with tags stating their name, age, place of origin, triage category, diagnosis, and initial treatment.

Solution to Question 11:

In triage, green signifies low priority or ambulatory patients and those with minor illnesses.

An internationally accepted four-color coded system, with red having the highest priority is used commonly here.

Solution to Question 12:

Reverse triage is a situation where the less injured are treated earlier than the ones that are injured more.

This is used in wars and military settings, where soldiers may need to return to the field of battle soon after treatment. It is also done when there is a scarcity of medical resources after a disaster.

The surge capacity of hospitals is increased to accommodate for the natural disaster, by discharging non-emergent existing patients. This enables hospitals to allow for an acute increase in incoming patients.

Solution to Question 13:

Moribund patients receive the lowest priority.

This is because despite paying quite a lot of attention, the benefit is still questionable.

Solution to Question 14:

The advisable level of residual chlorine in water bodies after a disaster is 0.7-0.9 mg/L.

Chlorine is a disinfectant that is added to water bodies. In normal conditions, water should have free residual chlorine (FRC) of 0.2 - 0.5 mg/L. However, during a disaster, there is an increased risk of fecal contamination of water sources from the sewage systems. Therefore, it is necessary to increase FRC to 0.7 - 0.9 mg/L.

Solution to Question 15:

The gas responsible for the Bhopal Gas Tragedy, which occurred on 3rd December 1984 at the Union Carbide Pesticide Plant, was methyl isocyanate.

Solution to Question 16:

Chernobyl nuclear explosion occurred on 26th April, 1986, in the Soviet Union. It caused the release of Iodine-I131, Cesium-Cs134, Cs137, and Strontium-Sr90, and other isotopes into the atmosphere.

Solution to Question 17:

The Ministry of Home Affairs is the nodal ministry for disaster management at the Centre except for dealing with droughts which are under the purview of the Ministry of Agriculture.

Other parts of the Indian government involved in disaster management are the Emergency Medical Relief Wing under the Ministry of Health and Family Welfare.

The Indian Meteorological Department plays a role in mitigation of disaster and its preparedness.

Solution to Question 18:

The district disaster management authority is the nodal center for disaster management.

It is chaired by either, the District Collector, District Magistrate or the Deputy Commissioner. The Chief Medical Officer of the district is a part of this authority.

Solution to Question 19:

The National Institute of Disaster Management (NIDM) is under the Ministry of Home Affairs of India. Its president is the Union Home Minister.

National Institute of Disaster Management is a premier institute for training and capacity development programs for managing natural disasters in India, on a national as well as regional basis.

Solution to Question 20:

National Disaster Management Authority (NDMA) is an apex body of the Government of India, with a mandate to lay down policies for disaster management. It is chaired by the Prime Minister of India.

The image given below shows the logo of the NDMA.



Sold by @Itachibot
If you purchased this from someone else,
you may have been scammed.

Occupational Health Diseases

Question 1:

You are working as a medical officer in a ceramic factory. What particle size of the dust inhaled would you be most concerned about?

- a) 5-8 microns
- b) 8-10 microns
- c) 3-5 microns
- d) 0.5 to 3 microns

Question 2:

Match the following causative agents with the disease caused by them:

- a) 1-b, 2-c, 3-a, 4-d
- b) 1-d, 2-c, 3-b, 4-a
- c) 1-b, 2-c, 3-d, 4-a
- d) 1-b, 2-d, 3-a, 4-c

Question 3:

While working as an MO, you noticed an increased number of bagassosis patients in the area. Which of the following fungal control measures would you advise the sugarcane farmers to implement?

- a) 10% propionic acid
- b) 10% acetic acid
- c) 2% propionic acid
- d) 2.5% acetic acid

Question 4:

A shipyard worker is diagnosed with asbestosis. Which of the following is false about his condition?

- a) The symptoms usually occur after 5-10 years of exposure
- b) Sputum sample contains asbestos bodies
- c) Peribronchial fibrosis is a characteristic finding
- d) The most dangerous form of asbestos is chrysolite type

Question 5:

A patient presents with abdominal colic and constipation. On examination, pallor and blue stippled lines on gums are observed. What is the most common mode of exposure to this particular toxicity?

- a) Ingestion of paint
- b) Inhalation of fumes
- c) Absorption through skin
- d) Drinking contaminated water

Question 6:

Which of the following statements is true about lead poisoning?

- a) Signs and symptoms occur when blood lead levels $>70\text{mcg/dL}$
- b) It inhibits ALA synthase enzyme
- c) Wrist drop is the earliest sign
- d) Lead competes with magnesium ions

Question 7:

You are screening the workers in a battery manufacturing factory for exposure to lead. Which of the following screening test will you use?

- a) Blood lead levels
- b) Coproporphyrin in urine
- c) Cotinine levels in urine
- d) Blood ALA levels

Question 8:

Barosinusitis is likely to be seen in which of the following patients?

- a) A person who is suffering from mountain sickness
- b) A patient Infected by gas-forming bacteria
- c) A patient diagnosed with Aspergillus sinusitis
- d) A diver who developed decompression sickness

Question 9:

Workers working at which of the following industries require periodical examination on a daily basis?

- a) Dichromate
- b) Lead
- c) Dyes
- d) Radon

Question 10:

Serial chest X-rays of an elderly man reveal progressive massive fibrosis. Which of the following pneumoconiosis is associated with this?

- a) Asbestosis
- b) Byssinosis
- c) Anthracosis
- d) Bagassosis

Question 11:

Bronchoalveolar lavage from a patient with pneumoconiosis showed Micropolyspora faeni. What is the diagnosis?

- a) Woodworker's lung
- b) Bird fancier's lung
- c) Farmer's lung
- d) Malt worker's lung

Question 12:

Which of the following does not belong to the 4 P's of occupational dermatitis?

- a) Pre-selection check-up
- b) Post-selection check-up
- c) Protection
- d) Personal hygiene

Question 13:

Which of the following statements is true about occupational cancer?

- a) Age of incidence is same as compared to general population for the same cancer
- b) The localization of the tumor is variable within a particular industry
- c) The most common occupational cancer is squamous cell carcinoma
- d) Onset of cancer does not occur after the person has left the industry

Question 14:

Cystoscopy performed on a patient with painless hematuria showed a fungating mass in the bladder. Exposure to all of the following agents can cause this disease except:

- a) Benzol
- b) Cyclophosphomide
- c) Auramine
- d) Magenta

Question 15:

A young woman is being evaluated for infertility. She gives a history of being employed in the printing industry. Exposure to which of the following is likely to cause this situation?

- a) Lead
- b) Selenium
- c) Mercury
- d) Beryllium

Question 16:

Which of the following is not a personal protective equipment?

- a) Goggles
- b) Gloves
- c) Face shield
- d) Lab coat

Answer Key

Question No.	Correct Option
1	d
2	c
3	c
4	d
5	b
6	a
7	b
8	d
9	a
10	c
11	c
12	b
13	c
14	a
15	a
16	d

Detailed Explanations**Solution to Question 1:**

Inhalation of dust particles in the size range of 0.5 to 3 microns is considered to be the most dangerous. It can lead to various occupational lung diseases called pneumoconioses.

Solution to Question 2:

The correct match is:

- Coal dust - Anthracosis
- Hay dust - Farmer's lung
- Cotton fibre - Byssinosis
- Cane fibre - Bagassosis

Solution to Question 3:

Fungal control on bagasse can be done by using 2% propionic acid and keeping the moisture content above 20%.

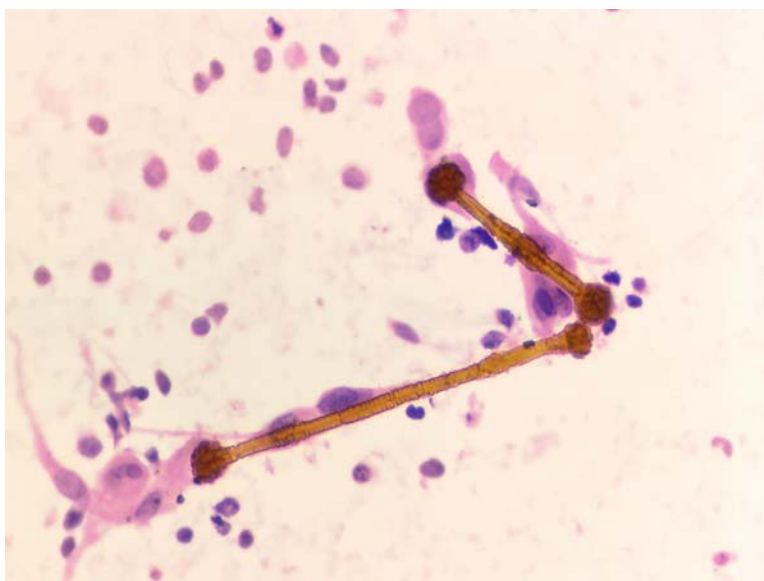
Bagassosis is a hypersensitivity pneumonitis caused by inhaling bagasse dust contaminated by the fungus *Thermoactinomyces sacchari*. It is a reaction to the various antigens present in sugarcane molasses.

Solution to Question 4:

Amphibole is the most dangerous form of asbestos. Chrysolite is a safer form of asbestos compared to amphibole.

Asbestosis is a pneumoconiosis that occurs after 5-10 years of exposure to asbestos. It progresses even after the removal of contact with asbestos. Asbestosis affects the basal part of the lung causing diffuse peribronchial fibrosis.

Sputum shows asbestos bodies, which are asbestos fibers coated with fibrin. The image given below shows asbestos bodies:



The complications that can occur are:

- Pulmonary fibrosis
- Carcinoma of the lung
- Mesothelioma of the pleura
- Malignancy of the gastrointestinal tract

Solution to Question 5:

A patient presenting with constipation, pallor, and Burton's lines (blue stippled lines on gums), most likely has lead poisoning. The most common mode of exposure to lead toxicity is the inhalation of lead fumes.

Lead poisoning occurs by lead exposure via the following modes:

- Inhalation- lead fumes in gasoline and petroleum products
- Ingestion- lead paint
- Absorption through the skin– seen only with organic leads

The most toxic forms of lead are lead oxide, lead carbonate, and lead arsenate. The least toxic form is lead sulfide.

Solution to Question 6:

The signs and symptoms of lead poisoning occur only when the blood lead levels exceed 70mcg/dL, which is known as the threshold phenomenon.

Lead poisoning is also known as plumbism/ Saturnism/ painter's colic.

Solution to Question 7:

The screening test used for estimating lead exposure is the measurement of coproporphyrin in urine.

Diagnosis of lead poisoning

Note: Cotinine is the primary metabolite of nicotine and is currently considered the best biomarker for tobacco smoke exposure (both active and passive).

Laboratory parameter	Value	Remarks
Coproporphyrin in urine	>150 mcg/L	Screening test- indicates exposure
Aminolevulinic acid (ALA) in urine	>5 mg/L	Indicates lead absorption
Lead in blood	>70 mcg/dL	Clinical symptoms appear
Lead in urine	0.8 mg/L	Indicates lead absorption
Basophilic stippling of RBCs		Sensitive indicator

Solution to Question 8:

Barosinusitis occurs in decompression sickness or Caisson's disease. It is due to the sudden expansion of air in the sinuses.

It can be seen in:

- Divers who ascend rapidly from deep-sea (high pressure) to the surface of the earth (low pressure)
- Pilots of unpressurized aircraft or mountaineers who rapidly ascended from the surface of the earth (high-pressure zone) to a high altitude (low-pressure zone)

Under high pressure, gases get compressed, and their solubility increases. When a person is in deep-sea (high-pressure zone) for a long time, an increased amount of nitrogen gets dissolved in his body. When he suddenly comes back to the surface (atmospheric pressure or a low-pressure zone), the solubility of gases decreases and they escape from the dissolved state forming nitrogen bubbles.

These bubbles can cause sudden expansion of air in various parts of the body such as the middle ear (barotitis), sinuses (barosinusitis), beneath teeth (barodontalgia), intestinal loops (abdominal distension), and emphysema.

Nitrogen bubbles can also cause chokes (dyspnoea), bends (joint pain), and prickles (nerve irritation).

The definitive treatment for Caisson's disease is recompression.

Solution to Question 9:

Workers handling irritant chemicals like dichromates require periodical examination on a daily basis.

Most occupational exposures require an annual periodical examination. Certain exposures like lead, radon, and dyes require monthly periodical examination.

Solution to Question 10:

Progressive massive fibrosis (PMF) is associated with anthracosis. PMF classically develops in anthracosis and silicosis.

Exposure to coal dust causes progressive lung injury called anthracosis. Initially, it manifests as simple coal worker pneumoconiosis, which occurs after exposure for 10–12 years and is reversible. Later, it can progress to progressive massive fibrosis (PMF) which is irreversible.

Solution to Question 11:

Farmer's lung is a hypersensitivity pneumonitis caused by thermophilic actinomycetes like *Saccharopolyspora rectivirgula* (formerly known as *Micropolyspora faeni*). It occurs due to inhalation of moldy hay and grains causing an immunological reaction.

Woodworker's lung is a hypersensitivity pneumonitis caused by *Alternaria* species, and *Bacillus subtilis* in persons exposed to oak, cedar, pine, and mahogany dust.

Bird fancier's lung is caused by proteins derived from birds like pigeons and parakeets in persons exposed to their feathers and droppings.

Malt worker's lung is caused by *Aspergillus* species in persons exposed to barley.

Solution to Question 12:

Post-selection check-up does not belong to the 4 P's of occupational dermatitis.

The 4 P's help in the prevention of occupational dermatitis, which include:

- Pre-selection check-up
- Protection
- Personal hygiene
- Periodic inspection

Solution to Question 13:

The most common occupational cancer is squamous cell carcinoma of the skin. It is associated with exposure to radiation, dyes, and coal tar.

In occupational cancers, the age of incidence of cancer is earlier than that in the general population, affected by the same cancer type. The onset of cancer can occur even after leaving the industry. The localization of the cancer is remarkably constant for a particular industry.

Solution to Question 14:

The given clinical scenario is suggestive of bladder carcinoma. Benzol does not cause bladder cancer, it causes leukemia.

Bladder carcinoma is associated with exposure to aniline dyes like benzidine, beta-naphthylamines, auramine, magenta, and drugs like cyclophosphamide.

Solution to Question 15:

A woman working in the printing industry is likely to suffer from infertility due to lead toxicity.

More industrial workers are exposed to lead than to any other toxic metal. Occupational exposure to lead is seen in several industries including the manufacturing of storage batteries, glass, shipbuilding, printing, potteries, and rubber industries.

The effects of lead poisoning include abdominal colic, blue-line on the gums (Burton's line), stippling of red cells, anemia, wrist & foot drop, headache, and convulsions. In young or pregnant women, prolonged lead exposure can cause miscarriage, stillbirth, premature birth, and low birth weight.

Solution to Question 16:

A lab coat is not a personal protective equipment.

Personal Protective Equipment (PPE) are:

- Gloves – protect hands.
- Gowns/ aprons – protect skin and clothing.
- Masks protect the mouth/nose and respirators protect the respiratory tract from airborne infectious agents.
- Goggles – protect eyes.
- Face shields – protect face, mouth, nose, and eyes.

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ESI and Factories Act

Question 1:

What is the definition of ergonomics?

- a) Fitting the job to the worker
- b) Study of human behavior
- c) Process of improving industrial safety standards
- d) Prevention of health problems of factory workers

Question 2:

Who is the Chairman of ESI corporation in India?

- a) The Prime Minister
- b) Union Minister of Health & Family Welfare
- c) Union Minister of Labour
- d) Union Minister of Human Resources Development

Question 3:

You are working as a medical officer in a factory. You are expected to notify all of the following occupational diseases, under the Factories Act, except:

- a) Occupational dermatitis
- b) Bagassosis
- c) Noise induced hearing loss
- d) Byssinosis

Question 4:

According to the Factories Act 1948, what is the maximum work hours per week including overtime work hours?

- a) 48 hours
- b) 72 hours

- c) 60 hours
- d) 66 hours

Question 5:

Which of the following is true about Factories Act?

- a) Children under 14 years of age must not be employed in factories
- b) A child of 16 years can be employed only between 6 AM and 8 PM
- c) The maximum working hours for an adult is 72 hours per week
- d) A child of 16 years can work till 5 hours per day

Question 6:

What is the maximum daily working hours for an adolescent employee, according to the Factories Act of 1948?

- a) 6 hours
- b) 8 hours
- c) 6 and a half hours
- d) 4 and a half hours

Question 7:

According to Factories Act of 1948, a welfare officer has to be employed if the number of employees is:

- a) ≥ 1000
- b) ≥ 500
- c) ≥ 250
- d) ≥ 750

Question 8:

Which of the following establishments is covered by the Employees' State Insurance Act?

- a) Defence

- b) Private medical and educational institutions
- c) Mines
- d) Railways

Question 9:

What percentage of the total wage bill is contributed by the employer under the ESI Act?

- a) 3.75%
- b) 3.25%
- c) 4.75%
- d) 0.75%

Question 10:

Under the ESI Act, which of the following is not given as a cash benefit?

- a) Sickness benefit
- b) Medical benefit
- c) Maternity benefit
- d) Dependant's benefit

Question 11:

What is enhanced sickness benefit?

- a) Benefit given at 70% of the wage for 91 days
- b) Benefit given at 80% of the wage for 120 days
- c) Benefit given for a duration of 2 years for 34 specified diseases
- d) Benefit given at full wages for undergoing sterilization procedure

Question 12:

Which of the following is not a benefit given under the ESI Act?

- a) Dependence benefit
- b) Welfare benefit

- c) Rehabilitation benefit
- d) Funeral benefit

Question 13:

An employee diagnosed with TB gets extended sickness benefit for:

- a) 1 year
- b) 2 years
- c) 3 years
- d) 4 years

Question 14:

Which of the following statements is true about ESI Act?

- a) Funeral benefit is upto Rs.20,000.
- b) State government covers 1/8th and ESI corporation covers 7/8th of the cost of medical care
- c) Maternity benefit is payable at full wages for upto 24 weeks
- d) It covers all employees getting income up to Rs.18,000/- per month

Question 15:

Who is the beneficiary under the Rajiv Gandhi Shramik Kalyan Yojana scheme of ESI?

- a) An insured employee who retired
- b) An insured person rendered unemployed, involuntarily
- c) An insured employee who is pregnant
- d) Employees of a new establishment

Answer Key

Question No.	Correct Option
1	a
2	c

3	b
4	c
5	a
6	d
7	b
8	b
9	b
10	b
11	d
12	b
13	b
14	b
15	b

Detailed Explanations

Solution to Question 1:

Ergonomics is defined as fitting the job to the worker. The aim of ergonomics is to achieve the best mutual adjustment of man and his work, for the improvement of human efficacy and well-being.

Emporiiatrics, also called travel medicine, deals with the prevention and management of health problems of international travelers.

Solution to Question 2:

The Union Minister of Labour is the Chairman of the Employees' State Insurance Corporation (ESIC) in India.

The Secretary to the Ministry of Labour is the Vice-Chairman of ESIC.

Solution to Question 3:

Bagassosis is not a notifiable disease, under the Factories Act.

Solution to Question 4:

Maximum work hours per week including overtime work hours is 60 hours.

Solution to Question 5:

Factories act prohibits children of <14 years from being employed.

Children between 15–18 years can be employed only in certain industries and work only between 6 AM–7 PM. They should not be made to work for more than four and a half hours.

The maximum working hours for an adult industrial worker should be not more than 48 hours per week, not exceeding 9 hours a day; if overtime is paid, then it should be not more than 60 hrs per week.

Solution to Question 6:

According to the Factories Act of 1948, adolescents should not be made to work for more than four and a half hours a day.

Solution to Question 7:

A welfare officer has to be employed if the number of employees is > 500 according to the Factories Act of 1948.

According to the Factories Act, safety and welfare measures include the provision of the following:

- > 1000 employees – Safety Officer
- > 500 employees – Welfare Officer
- > 250 employees – Canteen
- > 50 employees - Crèche

Solution to Question 8:

Currently, private medical and educational institutes employing >20 people are covered under the ESI Act.

The ESI Act is applicable to all industries except:

- Defence
- Railway
- Mines

Solution to Question 9:

The employer contributes 3.25% of the total wage bill under the ESI Act.

The government has revised the rates of ESI contribution by employers (from 4.75% to 3.25%) and employees (from 1.75% to 0.75%) in 2019.

The distribution of the financial contribution is given below:

- Employee: 0.75% of the total wage bill
- Employer: 3.25% of the total wage bill
- Medical benefit: 1/8th by the state government and 7/8th by the central government

Solution to Question 10:

Under the ESI Act, medical benefit is not given as a cash benefit. It is provided in the form of consultations, investigations and treatment by means of the ESI hospitals or ESI dispensaries.

Solution to Question 11:

Enhanced sickness benefit under the ESI Act is the benefit given at full wages for undergoing sterilization procedures.

Under the ESI Act, sickness benefit is given at 70% of the daily wage for a duration of 91 days in one year. There are two special types of sickness benefits given:

- Enhanced sickness benefit: Benefit given at full wages for sterilization procedures; for a duration of 7 days post-vasectomy and 14 days post-tubectomy.
- Extended sickness benefit: Benefit given for a duration of 2 years for 34 specified diseases.

Solution to Question 12:

Welfare benefit is not given under the ESI act.

The following benefits are given under the ESI act:

- Medical benefit
- Sickness benefit
- Maternity benefit
- Dependence benefit
- Disablement benefit
- Funeral benefit
- Rehabilitation benefit

Solution to Question 13:

Extended sickness benefit is given for a period of 2 years.

Solution to Question 14:

The state government's share of expenditure on medical care is 1/8th, and the ESI corporation's share of expenditure on medical care is 7/8th of the total cost.

Few important points about the ESI Act:

- It covers all employees getting income up to Rs.21,000/- per month.
- The employer contributes 3.25% of the total wage bill.
- An employee contributes 0.75% of wages.
- State and Central government share medical expenditure in the ratio 1:7.
- Medical benefit: Full medical care
- Sickness benefit: 70% of the average daily wages and is payable for 91 days.
- Extended sickness benefit: Payable for 2 years for a set of 34 diseases.
- Enhanced sickness benefit: Full average daily wage for a duration up to 7 days in case of vasectomy and up to 14 days in case of tubectomy.
- Maternity benefit: Full average daily wage for a duration up to 26 weeks (confinement) or 6 weeks (miscarriage/MTP) or 4 weeks (sickness arising out of pregnancy).
- Funeral expenses: Cash not exceeding Rs.15,000.

Solution to Question 15:

Rajiv Gandhi Shramik Kalyan Yojana (RGSKY) is a scheme of ESI that benefits an insured person rendered unemployed, involuntarily.

Unemployment allowance under RGSKY is given when an insured person is going out of employment involuntarily, on account of:

- Closure of a factory or establishment
- Retrenchment (laying off employees due to economic difficulty)
- Permanent disability due to nonemployment injury

Allowance and eligibility criteria of Rajiv Gandhi Shramik Kalyan Yojana (updated in 2019):

- The allowance is given at the rate of 50% of the daily wage for a maximum of 2 years.
- To qualify for the scheme the worker must have worked in that setup for more than 3 years.

Note: Disability due to employment injury is covered under disablement benefits of ESI Act, 1948.

- Temporary disability: 90% of wages until disablement lasts.
- Permanent disability: 90% of wages as a life pension on the basis of loss of earning capacity.
- Dependants benefit: 90% of wage as monthly payment to the dependants of the deceased person, in cases where death occurs due to employment injury or occupational hazards.

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Communication for Health Education

Question 1:

What is the correct order for the process of communication?

- a) Message->Sender->Channel->Feedback->Receiver
- b) Sender->Message->Channel->Receiver->Feedback
- c) Message->Sender->Receiver->Channel->Feedback
- d) Sender->Channel->Message->Feedback->Receiver

Question 2:

Which of the following is not a didactic type of communication?

- a) Newspapers
- b) Radio
- c) Lectures
- d) Workshop

Question 3:

Which of the following is a Socratic type of communication?

- a) Panel discussion
- b) Lectures
- c) Demonstrations
- d) Journals

Question 4:

Which of the following statements is not true about mass media?

- a) It has increased influence only in below-average educated population
- b) It gives greater support for concentrated programmes
- c) Mass media alone is inadequate in changing human behavior

- d) Mass media reaches the widest population

Question 5:

Which of the following is not a function of health communication?

- a) Community participation
- b) Motivation
- c) Organization
- d) Counselling

Question 6:

Which of the following is not an approach to health education?

- a) Regulatory approach
- b) Service approach
- c) Socio-environmental approach
- d) Primary care approach

Question 7:

In which model of health education does internalization occur?

- a) Medical model
- b) Socio-environmental model
- c) Motivation model
- d) Service model

Question 8:

Which of the following is not a principle of health education?

- a) Participation
- b) Learning by doing
- c) Setting an example
- d) Drawing conclusions

Question 9:

Which of the following statements about health education is false?

- a) It aids in developing a reflective behavior
- b) It appeals to emotion
- c) It is behavior centered
- d) It is actively acquired

Question 10:

Which is the best method for counseling a patient who has tested positive for HIV?

- a) Demonstrations
- b) Role play
- c) Group discussions
- d) Individual approach

Question 11:

Flannelgraphs, exhibits, and flipcharts are used to demonstrate the effect of injudicious use of antibiotics. What is this method of group communication known as?

- a) Chalk and Talk
- b) Symposium
- c) Mass media
- d) Demonstration

Question 12:

A group discussion was conducted among villagers to assess their knowledge on personal hygiene. Which of the following statements is false regarding this format of health education?

- a) Many different topics can be spoken about in one discussion
- b) A group leader guides the conversation
- c) It can bring about change in the attitude of participants

- d) It usually consists of 6 to 12 people

Question 13:

Which of the following is false about panel discussion?

- a) It usually consists of four to eight experts
- b) No moderator is required
- c) No specific agenda is set
- d) Members of the audience are invited to take part at the end

Question 14:

Match the following:

- a) 1-B, 2-C, 3-D, 4-A
- b) 1-C, 2-A, 3-D, 4-B
- c) 1-A, 2-B, 3-C, 4-D
- d) 1-D, 2-A, 3-B, 4-C

Question 15:

The Indian Association of Paediatrics is organizing a workshop on neonatal resuscitation. Which of the following best describes this program?

- a) Discussion of 4-8 experts in front of an audience
- b) Discussion between 6-12 members
- c) Series of four or more interactive meetings
- d) Series of speeches on a given subject

Question 16:

All of the following are types of combined audio-visual aids except:

- a) 1, 2, and 5
- b) 1 and 4 only
- c) 1 and 2 only

- d) 2, 3, and 4

Question 17:

A Norovirus outbreak occurs in an urban slum where you are posted as the PHC medical officer. Which is the best method to teach the people about ORS?

- a) Group discussions
- b) Role plays
- c) Home visits
- d) Demonstrations

Question 18:

Which of the following is used to obtain consensus forecasts among a large number of people?

- a) Television
- b) GATHER approach
- c) Conferences
- d) Delphi method

Question 19:

What is total communication used for?

- a) Advertising
- b) School teaching
- c) Community participation
- d) Hearing-impaired children

Question 20:

In which of the given scenarios would you use the GATHER approach?

- a) Alcohol de-addiction
- b) Contraceptive counseling

- c) Awareness programmes
- d) Screening for drug abuse

Question 21:

SPIKES protocol is for:

- a) Triage
- b) Consent for study participation
- c) Hospital infection control
- d) Breaking bad news

Answer Key

Question No.	Correct Option
1	b
2	d
3	a
4	a
5	a
6	c
7	c
8	d
9	b
10	d
11	a
12	a
13	b
14	b
15	c
16	a
17	d
18	d
19	d
20	b

Detailed Explanations

Solution to Question 1:

The correct order for the process of communication is:

Sender→Message→Channel→Receiver→Feedback

The sender (communicator) is the originator of the message. A message is the information which the communicator transmits to his audience. Channel is the media of communication between the sender and the receiver (e.g., interpersonal communication, mass media, traditional or folk media). The audience who receives, understands, and acts upon the message is called the receiver. Feedback is the reaction to the message which flows from the receiver to the sender.

Solution to Question 2:

Workshop is an example of Socratic type of communication.

Examples of didactic communication include:

- Lectures
- Demonstrations
- Television
- Radio
- Magazines
- Newspapers

Didactic communication	Socratic communication
One-way communication	Two-way communication
Knowledge is imposed and learning is authoritative	Learning is active and democratic
No audience participation	Audience share opinions, ideas, and questions
Does not influence human behavior	Influences human behavior
Feedback absent	Feedback present

Solution to Question 3:

A panel discussion is an example of Socratic type of communication.

Examples of Socratic communication include:

- Symposium
- Panel discussions
- Workshops

Didactic communication	Socratic communication
One-way communication	Two-way communication
Knowledge is imposed and learning is authoritative	Learning is active and democratic
No audience participation	Audience share opinions, ideas, and questions
Does not influence human behavior	Influences human behavior
Feedback absent	Feedback present

Solution to Question 4:

The advantage of mass media includes that it has increased influence across all educational levels, especially in above-average educational level.

Mass media includes TV, newspaper, radio, etc.

- Advantages:
- It reaches people even in the most remote areas.
- It gets public attention.
- It has increased influence across all educational levels, especially in above-average educational level.
- It gives greater support for concentrated programmes.
- Disadvantages:
- One-way communication
- Impersonal method and no face-to-face contact
- Mass media alone is inadequate in changing human behavior.

Solution to Question 5:

Community participation is not a function of health communication.

The functions of health communication are:

- Information
- Education
- Motivation
- Persuasion
- Counselling
- Raising morals
- Health development
- Organization

Methods of Health Communication		
Individual approach	Group approach	Mass approach
Personal contact Home visits Personal letters	Lectures Demonstrations Discussion methods Group discussions Panel discussion Symposium Workshop Conferences Seminars Role play	TV Radio Newspaper Printed material Direct mailing Posters Health museums Health exhibitions Folk method Internet

Solution to Question 6:

Socio-environmental approach is not an approach to health education.

There are four approaches to health education:

Regulatory approach- external control or laws placed on people by the government. Ex: Child Marriage Act, No smoking in public places, Helmet law

Service approach- government aims at providing health services to the people at their doorsteps. Its drawback is that people will not accept a service, even if it is offered free of cost unless it is based on their felt needs. Ex: Government provided latrines free of cost yet people did not use them because its need was not felt

Health education approach- creates awareness of health needs and problems. Awareness leads to motivation which includes the stages of interest, evaluation, and decision-making. It is followed by acceptance of the new idea which becomes part of his existing values, known as internalization. Ex: Adoption of ORS for diarrhea

Primary health care approach- delivery of health services with the active involvement of people. Principles of community participation and intersectoral coordination are stressed here. Its objective is to help individuals become self-reliant in health matters. Ex: People receive guidance from healthcare providers (Rogi Kalyana Samiti, Mahila Arogya Samiti).

Solution to Question 7:

Internalization belongs to the motivation model.

There are three models of health education:

- Medical model: transfer of proven medical information into education. It does not bridge the gap between knowledge and behavior.
- Motivation model: creates awareness of health needs and problems. Awareness leads to motivation which includes stages of interest, evaluation, and decision-making. It is followed by acceptance of the new idea which becomes part of his existing values, known as internalization.
- Social intervention model: focuses on changing the social environment which subsequently shapes the behavior of individual and the community.

Solution to Question 8:

Drawing conclusions is not a principle of health education.

Principles of Health Education:

- Participation
- Learning by doing
- Setting an example
- Good human relations
- Feedback
- Credibility
- Leaders
- Interest
- Comprehension
- Known to unknown
- Motivation
- Reinforcement

Known to unknown means to start where the people are and with what they understand and then proceed to new knowledge.

Solution to Question 9:

Health education appeals to people's reason. Health propaganda appeals to people's emotions.

Health Education vs. Propaganda:

	Health education	Health propaganda
Knowledge & Skills	Knowledge and skills are actively acquired	Knowledge is instilled in the mind
Primitive desires	Disciplined	Aroused
Action	Reflective behavior- trains people to use judgement before acting	Reflexive behavior- aims at impulsive action
Emotion	Does not appeal to emotion but appeals to reason	Appeals to emotion
Process	Behavior centered	Information centered

Solution to Question 10:

Individual approach is the preferred counseling method for HIV pre- and post-testing. This approach will equip the individual and family to deal with the situation more effectively.

Methods of Health Communication		
Individual approach	Group approach	Mass approach
Personal contact Home visits Personal letters	Lectures Demonstrations Discussion methods Group discussions Panel discussion Symposium Workshop Conferences Seminars Role play	TV Radio Newspaper Printed material Direct mailing Posters Health museums Health exhibitions Folk method Internet

Solution to Question 11:

Flannelgraphs, exhibits, and flipcharts belong to the chalk and talk method of group communication, in which the chalk is the visual component.

It is also known as lectures. Flannelgraphs, flipcharts, exhibits, films, and charts come under this method. Disadvantages of lectures are:

- Minimum interaction
- Do not stimulate thinking or problem solving
- Comprehension is dependent on the individual
- Do not influence behavior directly

Methods of Health Communication		
Individual approach	Group approach	Mass approach

Methods of Health Communication		
Personal contact Home visits Personal letters	Lectures Demonstrations Discussion methods Group discussions Panel discussion Symposium Workshop Conferences Seminars Role play	TV Radio Newspaper Printed material Direct mailing Posters Health museums Health exhibitions Folk method Internet

Solution to Question 12:

A group discussion discusses one topic on its agenda.

Group discussion is an aggregation of people interacting in a face-to-face manner. Exchange of ideas, knowledge, and opinions is done by participants which influence a change in their attitudes.

It usually involves 6–12 people with a group leader who guides the conversation, giving equal opportunity to everyone to voice themselves. In a group discussion, everyone must speak and be open to criticism without interrupting others.

A limitation to this form of health education is that not everyone is open to speaking in public. There is also a high possibility of diversion of the topic of discussion when many people are involved.

Methods of Health Communication		
Individual approach	Group approach	Mass approach
Personal contact Home visits Personal letters	Lectures Demonstrations Discussion methods Group discussions Panel discussion Symposium Workshop Conferences Seminars Role play	TV Radio Newspaper Printed material Direct mailing Posters Health museums Health exhibitions Folk method Internet

Solution to Question 13:

A moderator is required to regulate the panel discussion.

In a panel discussion, 4 to 8 experts have a discussion on a topic in front of an audience.

No speeches or order is set beforehand. The audience is invited to take part after the panelists speak and ask questions.

Solution to Question 14:

The correct match is 1-C, 2-A, 3-D, 4-B

A workshop is a series of meetings with an emphasis on individual work within the group. External consultants or experts may be involved.

A symposium is a series of lectures on a selected subject by experts after which the audience is allowed to ask questions (no discussion is involved).

Role play is the dramatization of social situations for health education by those taking part in it.

Seminars are similar to conferences and incorporate a variety of methods like speeches, discussions for health education, which may last for half a day to a week.

Methods of Health Communication		
Individual approach	Group approach	Mass approach
Personal contact Home visits Personal letters	Lectures Demonstrations Discussion methods Group discussions Panel discussion Symposium Workshop Conferences Seminars Role play	TV Radio Newspaper Printed material Direct mailing Posters Health museums Health exhibitions Folk method Internet

Solution to Question 15:

A workshop is a series of meetings with an emphasis on individual work within the group, using external consultants or experts. It helps increase individual efficiency as a professional worker.

Solution to Question 16:

Flannelgraphs, podcasts, and tape recorders are not combined audio-visual (A-V) aids.

Flannelgraph is a rough flannel fixed over a wooden board used for displaying pictures, cut-outs, etc. It is only a visual aid. Podcasts and tape recorders are auditory aids.

Audio-visual aids are classified into three groups:

- Auditory aids: Radio, tape recorders, podcasts
- Visual aids: Chalkboards, flannelgraphs, charts, posters, slides
- Combined A-V aids: Cinema, television, slide-tape combination

Solution to Question 17:

The best method to teach an urban slum about ORS is by demonstrations.

Demonstration is a carefully prepared presentation on how to perform a certain task or skill or procedure. It shows the step-by-step process of a task. It arouses interest and incites 'learning by doing,' and 'seeing is believing' philosophies.

Methods of Health Communication		
Individual approach	Group approach	Mass approach
Personal contact Home visits Personal letters	Lectures Demonstrations Discussion methods Group discussions Panel discussion Symposium Workshop Conferences Seminars Role play	TV Radio Newspaper Printed material Direct mailing Posters Health museums Health exhibitions Folk method Internet

Solution to Question 18:

Delphi method is a systematic interactive forecasting method, which is used to obtain a consensus forecast among a large number of people. It determines the degree of agreement on a topic.

This can help in setting priorities or selecting alternatives in planning. This method uses questionnaires, which are answered by experts in two or more rounds.

Solution to Question 19:

The philosophy of total communication is used in the education of hearing-impaired children.

It requires the incorporation of aural, oral, and manual modes of communication for effective communication with and among hearing-impaired people.

Solution to Question 20:

The GATHER approach is mainly used for counseling, like in contraceptive counseling, wherein a person is helped to make a decision about health.

GATHER stands for:

- G – Greet
- A – Ask
- T – Tell options or give answers
- H – Help in making choices
- E – Explain everything about the decision
- R – Return for follow-up

Solution to Question 21:

SPIKES protocol is for breaking bad news.

Each letter represents a phase in the six-step sequence.

- S - setting
- P - perception
- I - invitation or information
- K - knowledge
- E - empathy
- S - summarizing or strategizing

Key components of the SPIKES strategy include:

- Demonstrating empathy
- Acknowledging and validating the patient's feelings
- Exploring the patient's understanding and acceptance of the bad news
- Providing information about possible interventions

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Health Planning and Management

Question 1:

Which of the following is the ultimate long-term desired state towards which all the resources are directed?

- a) Target
- b) Mission
- c) Goal
- d) Objective

Question 2:

Which of the following is a set of rules that helps ensure maximum use of the resources and efforts?

- a) Procedure
- b) Programme
- c) Schedule
- d) Target

Question 3:

A team of doctors is planning to conduct a survey on the prevalence of infective diarrhea in a village. What is the correct sequence of steps involved in the planning of this activity?

- a) 1, 2, 3, 4
- b) 4, 1, 2, 3
- c) 3, 2, 4, 1
- d) 3, 1, 4, 2

Question 4:

Which of the following is not a component of the evaluation of health programs?

- a) Accessibility

- b) Relevance
- c) Efficiency
- d) Repeatability

Question 5:

Which of the following techniques is based on behavioral sciences?

- a) Decision making
- b) Systems analysis
- c) Network analysis
- d) Management by objective

Question 6:

Which of the following is a quantitative method of management?

- a) Cost-benefit analysis
- b) Organizational design
- c) Human resource management
- d) Management by objectives

Question 7:

In which of the following methods of management is the benefit measured in natural units?

- a) Cost minimization analysis
- b) Cost-benefit analysis
- c) Cost accounting
- d) Cost-effective analysis

Question 8:

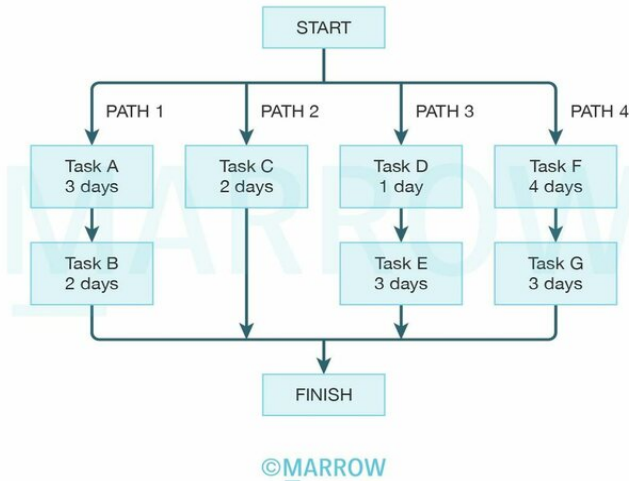
What is the programme evaluation and review technique used for?

- a) Network analysis
- b) Input–output analysis

- c) Systems analysis
- d) Cost-accounting analysis

Question 9:

Which of the following depicts the critical path in the project plan shown below?



- a) Path 2
- b) Path 1
- c) Path 4
- d) Path 3

Question 10:

Which of the following is true about the zero-based budgeting approach?

- a) It does not rely on data from previous budget
- b) It is a management method based on behavioral sciences
- c) It proceeds from resource to target
- d) It is not a priority-based budgeting

Question 11:

A hospital administrator is systematically recording the activities of his team at regular intervals. What is this known as?

- a) Organizational design
- b) Decision making
- c) Systems analysis
- d) Work sampling

Question 12:

Which committee had recommended a bachelor degree in medical education for rural health services?

- a) Krishan committee
- b) High-level expert group on universal health coverage
- c) Shrivastav committee
- d) Committee on Integration of Health Services

Question 13:

Who was appointed as the chairman of the Health Survey and Development Committee in 1943?

- a) Dr. A. L. Mudaliar
- b) Shri Mukerji
- c) Sir Joseph Bhore
- d) Dr. M. S. Chadah

Question 14:

Which health planning committee proposed the 3 million plan?

- a) Kartar Singh committee
- b) Mudaliar committee
- c) Chadah committee
- d) Bhore committee

Question 15:

What is the provision of preventive, curative, and promotional health services to every individual called?

- a) Appropriateness
- b) Acceptability
- c) Adequacy
- d) Comprehensiveness

Question 16:

In the inventory management of drug stores, which of the following is a component of the VED method?

- a) Desirable
- b) Discrete
- c) Definite
- d) Discard

Question 17:

A drug store owner uses the following technique of inventory management to procure drugs. What is this method known as?

- a) FSN
- b) VED
- c) HML
- d) ABC

Question 18:

Which of the following is not true regarding the National Health Policy of 2017?

- a) Increase public health expenditure to 5% of GDP by 2025
- b) Reduce infant mortality to 28 by 2019
- c) Access to safe water and sanitation to all by 2020
- d) Achieve elimination of lymphatic filariasis by 2017

Question 19:

Which article of the Indian Constitution confers Right to Life to the citizens of India?

- a) Article 11
- b) Article 21
- c) Article 23
- d) Article 25

Question 20:

Which of the following is part of the concurrent list of the Indian Constitution?

- a) Adulteration of food
- b) Fishing and fisheries beyond territorial waters
- c) Regulating labor in mines
- d) Public health and sanitation

Answer Key

Question No.	Correct Option
1	c
2	a
3	d
4	d
5	d
6	a
7	d
8	a
9	c
10	a
11	d
12	b
13	c

14	d
15	d
16	a
17	d
18	a
19	b
20	a

Detailed Explanations

Solution to Question 1:

The goal is the ultimate desired state towards which objectives and resources are directed. It is long-term and not constrained by time or resources.

The plan is the blueprint for taking action in order to achieve the goal.

Option A: Target refers to the quantifiable and achievable mid-term endpoints in the plan.

Option B: Mission describes the cause of the existence of the programme.

Option D: The objectives are short-term endpoints. They are precise points in the plan. They are either achieved or unachieved.

Solution to Question 2:

Procedure is a set of rules for carrying out work. It helps to ensure the maximum use of resources and efforts.

Option B: Programme is a sequence of activities designed to implement policies and accomplish objectives.

Option C: Schedule is a time sequence for the work to be done.

Option D: Target refers to the quantifiable and achievable mid-term endpoints in the plan.

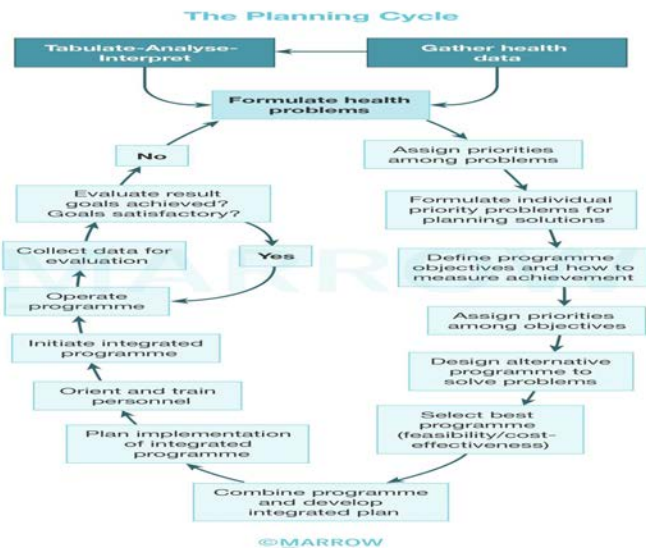
Solution to Question 3:

The planning cycle includes the following steps:

- Analysis of the health situation
- Establishment of objectives and goals
- Assessment of resources
- Fixing priorities

- Writing-up of a formulated plan
- Programming and implementation
- Monitoring
- Evaluation

The planning cycle has been summarized in the following image:



Solution to Question 4:

Repeatability is not a component of the evaluation of health programs.

Components of evaluation are:

(Mnemonic: Ryan Always Ate An Egg Every Evening)

- Relevance
- Adequacy
- Accessibility
- Acceptability
- Effectiveness
- Efficiency
- Effect or impact

The steps of evaluation of a health programme are:

- Determine what is to be evaluated: structure, process, or outcome
- Establish standards and criteria
- Plan the methodology to be applied

- Gather information
- Analyze the results
- Take action
- Re-evaluate

Solution to Question 5:

Management by objective is based on behavioral sciences.

Management methods used in health programmes are either quantitative or based on behavioral sciences.

Methods based on behavioral sciences include:

- Organizational design
- Personnel management
- Communication
- Information systems
- Management by objectives (MBO)

Quantitative methods include:

- Cost-benefit analysis
- Cost-effective analysis
- Cost-accounting
- Input-output analysis
- Model
- Systems analysis
- Network analysis - includes programme evaluation and review technique (PERT) and critical path method (CPM)
- Planning-programming-budgeting system (PPBS)
- Work sampling
- Decision making

Solution to Question 6:

Cost-benefit analysis is a quantitative method of management.

In this method, the economic benefits of any programme are compared with the costs incurred. The benefit of the programme is expressed in monetary units. The main drawback here is that the benefits of a health programme cannot always be expressed in monetary terms.

Management methods used in health programmes are either quantitative or based on behavioral sciences.

Methods based on behavioral sciences include:

- Organizational design
- Personnel management
- Communication
- Information systems
- Management by objectives (MBO)

Quantitative methods include:

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- Network analysis - includes programme evaluation and review technique (PERT) and critical path method (CPM)
- Planning-programming-budgeting system (PPBS)
- Work sampling
- Decision making

Solution to Question 7:

In cost-effective analysis (CEA), benefits are measured in natural units of the outcomes of the programs.

	Cost-effective analysis	Cost-benefit analysis
Cost measured in	Health units	Monetary units
Outcome measured in	Natural (health) units like life -years gained, cases prevented, etc. Most comprehensive indicator - quality-adjusted life years (QALYs)	Monetary gain
Uses	Evaluation of activities under a programme and comparing clinical strategies to deliver the same	At executive level of government (eg, budgeting, etc.)

Solution to Question 8:

The programme evaluation and review technique (PERT) is used for network analysis.

It is a type of quantitative management technique which helps in detailed planning and comprehensive supervision.

Solution to Question 9:

The critical path in the critical path method (CPM) is the longest path denoted by path 4 (7 days) in the plan shown above.

Any delay in the critical path will cause a delay in the entire project.

Solution to Question 10:

In the zero-based budgeting approach (ZBB), the budget for each new period is created starting from a zero base. It does not rely on the data from the previous budget.

It is a quantitative method of management.

Traditional Budgeting	Zero-based Budgeting
Previous year's budget used as a template	Start from zero base with newly set targets, followed by priority-based allocation of resources (target to resource)
Only new expenses need to be justified	Each expense needs to be justified, both old and recurring expenses
Used to study changes in expenditure	Used to study the cost-benefit analysis
Activities are directed towards monitoring of expenditure	Activities are directed towards achievements of objectives

Solution to Question 11:

Work sampling is a systematic observation and recording of activities of one or more individuals carried out at regular or random intervals.

It is a type of quantitative method of management. It helps in standardizing the methods of performing the jobs and determining the need for manpower.

Management methods used in health programmes are either quantitative or based on behavioral sciences.

Methods based on behavioral sciences include:

- Organizational design
- Personnel management
- Communication
- Information systems
- Management by objectives (MBO)

Quantitative methods include:

- Cost-benefit analysis
- Cost-effective analysis
- Cost-accounting
- Input-output analysis
- Model
- Systems analysis
- Network analysis - includes programme evaluation and review technique (PERT) and critical path method (CPM)
- Planning-programming-budgeting system (PPBS)
- Work sampling
- Decision making

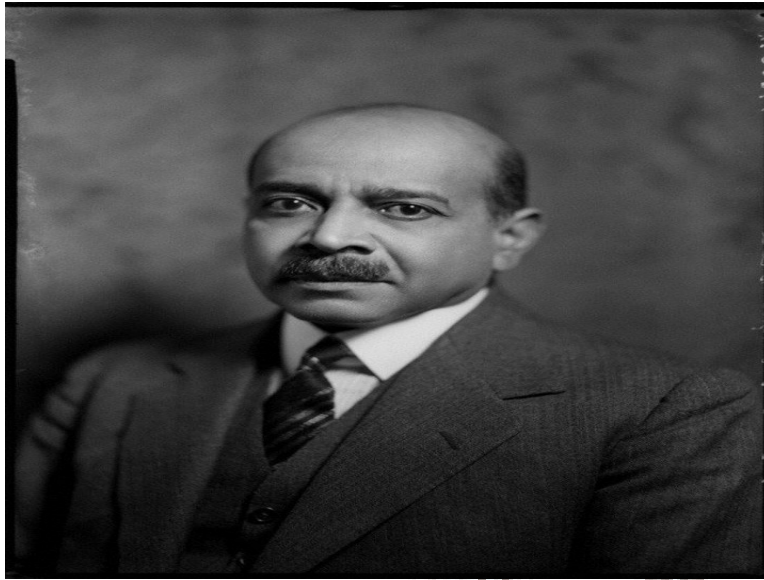
Solution to Question 12:

The high-level expert group (HLEG) on universal health coverage under the aegis of the Planning Commission of India suggested a 3.5 year bachelor degree course in medical education for rural health services.

This was planned for the 12th five-year plan (2012-2017). The degree given was B.Sc Community Health, and these rural health care practitioners were to be called community health officers.

Solution to Question 13:

Sir Joseph William Bhore was appointed as the chairman of the Health Survey and Development Committee in 1943. This committee was hence named the Bhore committee.



Solution to Question 14:

The 3 million plan was proposed by the Bhoré Committee.

It proposed to set up the following in the long-term:

- Primary health units with 75-bedded hospitals for each 10,000 to 20,000 population
- Secondary units with 650 beds
- District hospitals with 2,500 beds

As a short-term measure, it was proposed that each PHC must cater to 40,000 people and have:

- 2 medical officers, 4 public health nurses, 1 nurse
- 4 midwives, 4 dais, 2 sanitary inspectors, 2 health assistants
- 1 pharmacist
- 15 other class IV employees

Solution to Question 15:

Comprehensive health care as stated by the Bhoré committee means giving integrated preventive, curative, and promotional health care to every individual, irrespective of their ability to pay.

Solution to Question 16:

The 'D' in the VED method stands for desirable.

VED is a method of inventory management used mainly in drug stores and pharmacies.

The drugs are classified in three categories as follows:

- V - Vital drugs. Items whose absence cannot be tolerated (10% of total items).
- E - Essential drugs. Items whose absence can be tolerated for some time (40% of total items).
- D - Desirable drugs. Items whose absence can be tolerated for a long time (50% of total items).

Solution to Question 17:

The method of analysis depicted above is the ABC analysis (always better control). It refers to a type of inventory management in which items are divided into 3 categories according to their cost.

This type of analysis aims to calculate expenditures of large proportions on a small number of items and vice versa. It makes it easier to understand how to control the majority of expenditure by analyzing the small numbers of items it is being spent on.

The table below explains ABC analysis:

Other types of inventory management are:

- FSN - Fast-slow-non moving drugs.
- VED - Vital-essential-desirable drugs
- HML - High-cost medium-cost low-cost drugs. In HML analysis the items are categories into H, M, L category based on unit cost.

Category	Label	Cost (%)	Items (%)
Class A - items	Outstandingly important	70	15
Class B - items	Average importance	25	30
Class C - items	Relatively unimportant	5	55

Solution to Question 18:

National Health Policy (NHP) 2017 aims to increase the public health expenditure from existing 1.15% to 2.5% of the GDP by 2025.

The draft of the policy from 2015 was passed by the government in 2017 and is called the National Health Policy (NHP) 2017.

Components of NHP 2017	Quantitative goals and objectives
Health status and programme impact	↑life expectancy at birth from 67.5 to 70 by 2025 ↓infant mortality rate to 28 by 2019 ↓neonatal mortality to 16 and stillbirth rate to a single digit by 2025 ↓maternal mortality ratio from current levels to 100 by 2020 Elimination of lymphatic filariasis by 2017
Health systems performance	↑ utilization of public health facilities by 50% by 2025 ↑ Antenatal care coverage above 90% >90% of newborns to be fully immunized by 1 year of age by 2025 Access to safe water and sanitation to all by 2020 (Swachh Bharat Mission)
Health systems financing	↑ public health expenditure to 2.5% of GDP by 2025 ↑ state sector health spending to > 8% of budget by 2020
Priority actions to improve the environment for health	Swachh Bharat Abhiyan Balanced healthy diets and regular exercise Addressing tobacco, alcohol, and substance abuse Yatri Suraksha to reduce death due to accidents Nirbhaya Nari for action against gender violence Reduced stress and improved safety in workplace Reducing indoor and outdoor air pollution
Other provisions	NEET exam for UG ↑ provisions for treatment and diagnosis in HIV-centered communities Establishment of nursing schools in every district Food safety

Solution to Question 19:

The Right to Life is given in Article 21 of the Indian Constitution.

Option A: Article 11 defines the right of the parliament to regulate the right of citizenship by law.

Option C: Article 23 defines the prohibition of traffic in human beings and forced labor.

Option D: Article 25 defines the freedom of conscience and free profession, practice, and propagation of religion.

Solution to Question 20:

Adulteration of food is included under the concurrent list.

The legislative section of the seventh schedule is divided into three lists:

- Union list - a list of 100 items on which the parliament has exclusive power to legislate
- State list - a list of 61 items on which the state government has exclusive power to legislate
- Concurrent list - a list of 52 items on which both central and state governments have the power to legislate

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you may have been scammed.

Healthcare In India

Question 1:

Which of the following is not included in the secondary level of the healthcare system?

- a) Community health centres
- b) Medical colleges
- c) District hospitals
- d) Sub-district hospitals

Question 2:

Which of the following is not an element of primary healthcare?

- a) Prevention of locally endemic diseases
- b) Safe water supply and sanitation
- c) Providing physical rehabilitation
- d) Providing essential drugs

Question 3:

All of the following are pillars of primary health care except:

- a) Health education
- b) Intersectoral coordination
- c) Appropriate technology
- d) Equitable distribution

Question 4:

Which of the following statement is false regarding ASHA workers?

- a) Few of their responsibilities are integrated with Anganwadi workers
- b) They should have formal education at least up to class 10
- c) A population of 1000 should have at least 1 ASHA

- d) They have to procure ORS, OCPs, IFA tablets from CHC

Question 5:

Which of the following is incorrect regarding Anganwadi workers?

- a) One AWW serves a population of 5000
- b) They are appointed under the ICDS scheme
- c) They link the community with the health services
- d) Their responsibilities include counsel for immunization, health education, and nutrition

Question 6:

One male health assistant is posted for a population of:

- a) 1000
- b) 5000
- c) 10,000
- d) 30,000

Question 7:

How many people does a Mahila Arogya Samiti member cater to?

- a) 50-100
- b) 100-200
- c) 250-500
- d) 500-1000

Question 8:

Which of the following duties would be performed by a male health worker?

- a) 2,4
- b) 1,3,4
- c) 1,3,5
- d) 2,5

Question 9:

Which of the following is not a duty of a female health worker?

- a) DOTS activities
- b) Malaria surveillance
- c) Distribution of condoms
- d) Immunization

Question 10:

Which of the following are not involved in providing antenatal support to pregnant women?

- a) Female health workers
- b) Trained birth attendants
- c) Female health assistants
- d) Anganwadi workers

Question 11:

How many people does one sub-center in a hilly or remote area cater to?

- a) 7000
- b) 10,000
- c) 30,000
- d) 3000

Question 12:

What should be the maximum number of female health workers at a sub-center?

- a) 1
- b) 2
- c) 3
- d) None

Question 13:

As a PHC medical officer, you would be responsible for all of the following activities except:

- a) Mainstreaming AYUSH and PMR
- b) Performing selected surgical procedures
- c) Formal education of children < 6 years of age
- d) Collection and recording of vital events

Question 14:

Which of the following category of workers are not part of the workforce at PHCs?

- a) Accountant cum data entry operator
- b) Radiographers
- c) Health Educator
- d) Clerks

Question 15:

What is the minimum number of beds a HWC-PHC is required to have?

- a) 2
- b) 4
- c) 6
- d) 10

Question 16:

One community health center (CHC) covers a population of:

- a) 30,000 to 50,000
- b) 40,000 to 60,000
- c) 80,000 to 1,20,000
- d) 1,00,000 to 2,00,000

Question 17:

What is the minimum number of beds required in a community health center?

- a) 25
- b) 34
- c) 30
- d) 20

Question 18:

One eye surgeon can cater to how many CHCs according to Indian Public Health Standard norms?

- a) 2 CHCs
- b) 5 CHCs
- c) 6 CHCs
- d) 3 CHCs

Question 19:

Under the National Health Mission, an urban CHC for a metropolitan city caters to a population of how many persons?

- a) 50,000
- b) 1 lakh
- c) 2.5 lakh
- d) 5 lakh

Question 20:

Which of the following is incorrect regarding the staffing at urban primary health centers?

- a) 2 doctors
- b) 2 staff nurses
- c) 1 lab technician
- d) 1 pharmacist

Question 21:

Which of the following is the principal unit of administration in India?

- a) Village
- b) Centre
- c) District
- d) Taluk

Question 22:

Which of the following is incorrectly matched?

- a) Panchayat - Village level
- b) Panchayat Samiti - Block level
- c) Zilla Parishad - District level
- d) Janpada Panchayat - Village level

Question 23:

Community or village health guides are selected by:

- a) ASHA
- b) Medical officer-in-charge
- c) Female health assistant
- d) Zila parishad

Question 24:

As a PHC medical officer, you would keep all of the following in mind while selecting a village health guide except:

- a) They must be above 45 years of age
- b) They must be a resident of the community that they will serve
- c) They must be educated up to 6th grade
- d) They should be able to spare 2-3 hours in a day for community work

Answer Key

Question No.	Correct Option
1	b
2	c
3	a
4	d
5	a
6	d
7	c
8	c
9	b
10	b
11	d
12	b
13	c
14	b
15	a
16	c
17	c
18	b
19	d
20	b
21	c
22	d
23	b
24	a

Detailed Explanations

Solution to Question 1:

Medical colleges are included in the tertiary level of the health care system, which is also the second referral centre.

Solution to Question 2:

Providing physical rehabilitation is not an element of primary health care. Primary health centers (PHCs) are only responsible for the popularization and mainstreaming of PMR in the community.

Primary health care refers to essential health care made universally available to individuals and acceptable to them, through their full participation and at a cost the community and country can afford.

Solution to Question 3:

Health education is an element of primary health care, not a pillar.

Solution to Question 4:

Accredited social health activist (ASHA) workers do not need to procure supplies from CHCs. They only act as depot holders for certain essential supplies so that they are readily available to their community.

These essential supplies may include ORS, contraceptives, iron-folic acid tablets, chloroquine, disposable delivery kits, oral contraceptive pills, condoms, etc.

Solution to Question 5:

The norm is one Anganwadi worker (AWW) per 400-800 population.

Anganwadi workers (AWW) are allotted under the integrated child development services (ICDS). AWW are selected from the community they are to serve and act as links between the community and health services. They are mostly part-time workers.

Responsibilities of an AWW include:

- Health checkups and growth chart maintenance
- Counsel women on birth preparedness, importance of safe delivery, immunization and contraception and prevention of common infections including STD and RTIs.
- Nutrition
- Health education
- Non-formal school education
- Referral services

Solution to Question 6:

One male and female health assistant are required for a population of 30,000 in plain areas and of 20,000 in hilly or remote areas.

Solution to Question 7:

1 Mahila Arogya Samiti member is appointed for a population of 250-500 (covering about 50-100 households).

Solution to Question 8:

The duties of a male health worker include chlorination of water bodies, distribution of conventional contraceptives, and reproductive and child health services.

Maintenance of eligible couple registers (part of family planning activities) and immunization are duties of female health workers.

Health workers (male and female) are 1 per 5000 in plains and 1 per 3000 in remote or hilly areas.

Solution to Question 9:

Malaria surveillance activities, in accordance with the National Vector Borne Disease Control Programme, are the duty of a male health worker.

Health workers (male and female) are 1 per 5000 in plains and 1 per 3000 in remote or hilly areas.

Solution to Question 10:

Trained birth attendants or dais provide medical services during delivery and in the post-partum period. They cater to a population of 1000 in villages.

Antenatal support is provided by the following:

- Health workers (male and female) - 1 per 5000 in plains and 1 per 3000 in hilly or remote areas.
- Health assistants (male and female) - 1 per 30,000 in plains and 1 per 20,000 in remote or hilly areas.
- Anganwadi workers - 1 per 400-800 in plains and 1 per 300-800 in hilly or remote areas.

Solution to Question 11:

A sub-center caters to a population of 5000 in general areas and 3000 in hilly or remote areas.

Services provided at sub-centers:

- Maternal and child health, family planning
- Counseling for abortion and adolescent health care
- Water quality monitoring and sanitation promotion
- Assistance to school health services
- Field visits and treating minor ailments
- Training of ASHA and birth attendants
- Disease surveillance and National Health Programme functions

Solution to Question 12:

Two female health workers are to be at a sub-center as proposed by the Indian Public Health Standards (IPHS).

There are two types of sub-centers:

- Type A - It has all facilities except for conducting deliveries. It can be upgraded to Type B.
- Type B - Also called MCH sub-centers. It has delivery facilities.

Indian Public Health Standards (IPHS) norms for manpower at sub-centers:

*To be outsourced

**If the number of deliveries ≥ 2

Type of sub-center	Sub-centre A	Sub-centre B		
Staff	Essential	Desirable	Essential	Desirable
ANM/Health worker (female)	1	+1	2	
Health worker(male)	1		1	
Staff nurse (or ANM, if staff nurse is not available)				1**
Safai karamchari*	1(part time)		1(full time)	

Solution to Question 13:

Formal education is not a function of primary health centers in India.

Solution to Question 14:

A radiographer is not a part of the manpower at a PHC. They are present in CHCs.

Accountants cum data entry operators, health educators, and clerks are appointed at PHCs.

Solution to Question 15:

The minimum number of beds a HWC-PHC is required to have is 2. Per the IPHS 2022 guidelines, the desirable number of beds in a HWC-PHC is 4.

Note: The desirable beds will be over and above the essential beds.

Solution to Question 16:

Each community (CHC) covers a population of 80,000-1,20,000.

Services provided by CHC:

- Care of routine and emergency cases in medicine and surgery
- Maternal, newborn, and child health
- Family planning
- All the functions of national health programmes
- Physical medicine and rehabilitation
- Oral health
- School and adolescent health services
- Blood storage facility
- Diagnostic services
- Referral services
- Maternal death review (MDR)

Solution to Question 17:

The minimum number of beds in a CHC is 30.

The following manpower must be maintained at a CHC:

Personnel and Strength at CHCs	
Medical	General duty MO (6 - at least 2 female doctors)Public health manager (1)General surgeon (1)Physician (1)OBG (1)Pediatrician (1)Anesthetist (1)Dental surgeon (1)Ophthalmologist (1 for 5 CHCs)AYUSH MO (1)AYUSH specialist (1)
Total: 15/16	
Nursing	Staff nurse (19)Public health nurse* (1)ANM* (1)*Appointed through ASHA scheme
Paramedical	Pharmacist (3)AYUSH Pharmacist (1)
Support staff	Lab technician (3)Radiographer (2)Ophthalmic assistant (1)Dresser (2)Ward boys/Nursing Orderly (5)Accountant (1)OT technician (1)Statistical assistant (2)Registration clerk (2)OPD attendant (1)Group D staff (19)
Total: 64	

Solution to Question 18:

One ophthalmologist is recommended by the IPHS for 5 CHCs.

Personnel and Strength at CHCs	
Medical	General duty MO (6 - at least 2 female doctors)Public health manager (1)General surgeon (1)Physician (1)OBG (1)Pediatrician (1)Anesthetist (1)Dental surgeon (1)Ophthalmologist (1 for 5 CHCs)AYUSH MO (1)AYUSH specialist (1)
Total medical personnel: 15/16	
Nursing support staff	Staff nurse (19)Public health nurse* (1)ANM* (1)*Appointed through ASHA scheme

Personnel and Strength at CHCs	
Paramedical support staff	Pharmacist (3) AYUSH Pharmacist (1)
Other support staff	Lab technician (3) Radiographer (2) Ophthalmic assistant (1) Dresser (2) Ward boys/Nursing Orderly (5) Accountant (1) OT technician (1) Statistical assistant (2) Registration clerk (2) OPD attendant (1) Group D staff (19)
Total support personnel: 64	

Solution to Question 19:

An urban CHC caters to a population of 5 lakh in a metropolitan city and 2.5 lakh in other cities.
An urban PHC caters to a population of 50,000.

Solution to Question 20:

An urban primary health center will cover 50,000-60,000 population and will be staffed with:

- Two doctors (one regular and one on a part-time basis)
- 3 staff nurses
- 1 pharmacist
- 1 lab technician
- 1 lady health visitor
- 4-5 ANMs (depending upon the population covered)
- Other clerical and support staff and one program manager

Basic functions provided are:

- OPD
- Basic lab diagnosis
- Drug/contraceptive dispensing
- Health education along with referral services

*Note: The above data is according to IPHS 2017 guidelines. The revised guidelines as per IPHS 2022 have been listed in the pearl attached below.

Solution to Question 21:

The principal unit of administration in India is the district under a collector.

Under districts are the sub-divisions, tahsils or talukas, municipalities or corporations, villages, panchayats, etc. As of 2021, there are 748 districts in India.

Solution to Question 22:

The Panchayat Samiti or Janpada Panchayat operates at the block level.

Panchayati Raj is a three-tier structure of rural self-government in India. It operates at the following levels:

- Panchayat at the village level (includes Gram Sabha, Gram Panchayat, and Nyaya Panchayat)
- Panchayat Samiti or Janpada Panchayat at the block level
- Zilla Parishad at the district level

Solution to Question 23:

Village health guides (VHG) are selected by the medical officer-in-charge (MO).

2-3 candidates are put forward by the community for the position of a VHG. The MO selects one of them after consultation with the block development officer and field staff.

Solution to Question 24:

Village health guides (VHG) are chosen by the medical officer-in-charge (including PHC MO). Age is not a criterion for selection.

The VHG scheme was introduced in 1977. Guidelines for choosing a VHG include:

- They can be of either sex but must be residents of the village
- They must know to read and write and have education up to 6th grade
- They must be able to spare around 2-3 hours a day for community work
- They must be acceptable to all parts of the community

VHGs are trained for 2-3 months before beginning work. The norm is to have 1 VHG per 1000 population.

Healthcare in India- IPHS 2022 Update

Question 1:

Which of the following public health centers do First Referral Units (FRUs) and non-FRUs belong to?

- a) Health and Wellness Center - PHC
- b) Health and Wellness Center - Sub-Center
- c) Sub-District Hospital
- d) Community Health Center

Question 2:

A district has a population of 20 lakhs. According to the IPHS 2022 guidelines, approximately how many beds should the district public health system have for this population?

- a) 2000 essential, 4000 desirable
- b) 1000 essential, 2000 desirable
- c) 4000 essential, 8000 desirable
- d) 3000 essential, 6000 desirable

Question 3:

You are involved in the planning for a hospital in your district. If you decide that the minimum number of essential beds is 50, what should the population of your district be, according to the IPHS 2022 guidelines?

- a) 4 lakhs
- b) 5–10 lakhs
- c) Less than 2 lakhs
- d) 3 lakhs

Question 4:

Which of the following is not true about district and sub-district hospitals?

- a) All districts need at least two district hospitals.
- b) The minimum essential number of beds in a sub-district hospital is 100.
- c) All districts do not need a sub-district hospital.
- d) DNB training can be done in district hospitals.

Question 5:

Which of the following is the correct statement regarding Community Health Centers (CHCs)?

- a) Non-FRU CHCs should have 50 essential beds.
- b) Non-FRU CHCs can be located in urban areas.
- c) FRU-CHCs can be located in rural areas.
- d) Elective surgeries will not be performed at FRU-CHCs.

Question 6:

What is the essential number of beds in a non-First Referral Unit CHC (non-FRU-CHC)?

- a) 60
- b) 50
- c) 20
- d) 30

Question 7:

A rural hilly area has a population of 40,000. How many Primary Health Centers (PHCs) should it have?

- a) 1
- b) 2
- c) 3
- d) 4

Question 8:

Which of the following health centers serves a population of 2.5-3 lakhs?

- a) Urban HWC-PHC
- b) HWC-PHC in plain areas
- c) Polyclinic
- d) HWC-PHC in hilly areas

Question 9:

All of the following surgical procedures can be conducted in a Primary Health Center (PHC) except _____.

- a) Vasectomy
- b) Cholecystectomy
- c) Incision and drainage of abscesses
- d) Hydrocelectomy

Question 10:

You are a medical officer in an Urban Health and Wellness Center (UHC). The population of the place is 10,000. How many ASHA workers are needed for your UHC?

- a) 3
- b) 4
- c) 5
- d) 6

Question 11:

You are posted as a medical officer in a rural Health and Wellness Center - Primary Health Center (HWC-PHC). According to the IPHS 2022 guidelines, which of the following is incorrect regarding the essential manpower required for this PHC?

- a) 2 staff nurses
- b) 1 pharmacist
- c) 1 laboratory technician
- d) 1 ophthalmologist

Answer Key

Question No.	Correct Option
1	d
2	a
3	c
4	a
5	c
6	d
7	b
8	c
9	b
10	c
11	d

Detailed Explanations

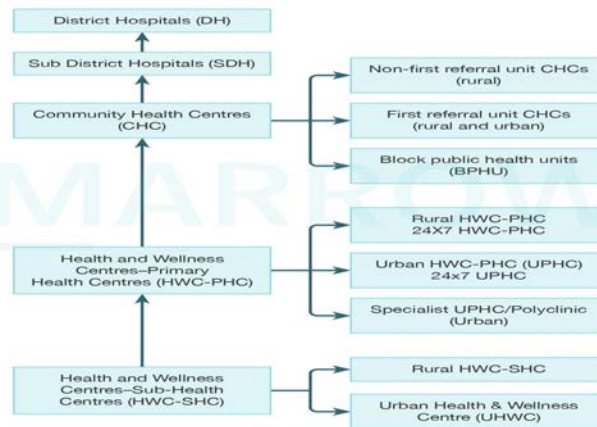
Solution to Question 1:

First Referral Units (FRUs) and non-FRUs belong to the Community Health Centers (CHCs).

The Indian Public Health Standards (IPHS) guidelines are a code of standards for uniformity in health facilities in the Indian public health system, under the National Health Mission (NHM), Ministry of Health and Family Welfare (MoHFW), India. It aims to provide guidance on the infrastructure, human resources, drugs, diagnostics, equipment, quality, and governance requirements for delivering health services at all levels of facilities in the system.

The 2022 edition of IPHS mainly concentrates on comprehensive primary health care, increasing the human workforce, decentralized access to healthcare, and better implementation of universal health coverage.

Organisation of Public Health Systems according to IPHS 2022



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Solution to Question 2:

In a district having a population of 20 lakhs, the number of beds the district public health system should have is given as follows: 2000 essential and 4000 desirable.

According to IPHS 2022 norms for the number of beds in district hospitals, 1 bed per 1000 population is essential and 2 beds per thousand population would be desirable.

Therefore, in a population of 20,00,000 the district public health system should have 2000 essential and 4000 desirable beds.

The essential number of beds should be achieved through the public health system of tertiary care (medical colleges), secondary care (DH, SDH, CHCs), and primary care in the district (PHCs and CHCs).

Solution to Question 3:

According to the IPHS 2022 guidelines, a minimum number of 50 essential beds would be required for a district with a population of less than 2 lakhs.

Categorization and grading for district health services

Population	Essential Beds + Additional Beds	Desirable
Less than 2 lakhs	50 + 15 emergency & daycare beds	100
2–5 lakhs	100 + 25 emergency & daycare beds	200

Population	Essential Beds + Additional Beds	Desirable
5–10 lakhs	200 + 38 emergency & day care beds	300
10–20 lakhs	300 + 49 emergency & day care beds	400
20–30 lakhs	400 + 60 emergency & day care beds	500
More than 30 lakhs	500 + 65 emergency & day care beds	700

Solution to Question 4:

Every district should have at least one comprehensively functioning district hospital (DH).

District hospitals and subdistrict hospitals provide secondary care health services. Every district should have at least one comprehensively functioning DH. Districts with a population of less than 5 lakhs and with a functional DH do not need a subdistrict hospital (SDH). Districts with a population of 5–10 lakhs can have one SDH. For a population of more than 10 lakhs, 1 SDH is needed for every 10 lakh population.

The three pillars of district hospital services are providing clinical care, providing a knowledge hub for capacity development, and public health programs.

Important functions of DHs:

- Providing curative services
- Palliative and rehabilitative services
- Implementation of health programs
- Drugs and diagnostic services
- Administrative and other support services
- Health promotion and prevention, education
- Providing educational services, like Diplomate of National Board (DNB) training; training courses for healthcare professionals (HCPs) in auxiliary nursing and midwifery (ANM), telemedicine, and technology; and partnerships with universities

Solution to Question 5:

First Referral Unit - Community Health centers (FRU-CHCs) can be located in rural areas.

Community health centers (CHC) can be classified into the following types:

- Non-FRU-CHCs - They are located in rural areas. They provide preventive, promotive, palliative, and rehabilitative services. They also have curative services for normal delivery, stabilization of

common emergencies, etc.

- FRU-CHCs are the first centers of referral from smaller centers such as sub-centers or primary health centers. They can be located in rural as well as urban areas. Apart from providing the abovementioned essential services, they also provide specialized services and elective surgeries. All the CHCs located in urban areas are FRU-CHCs.

In addition to the above centers, CHCs called Block Health Public Units (BHPU) may be located in the block headquarters. Apart from clinical and public health services, they have a health management information system (HMIS). They also help in decentralizing health services.

Solution to Question 6:

The number of essential beds in a non-First Referral Unit CHC (non-FRU CHC) is 30.

HWC: Health and Wellness Center

UPHC: Urban Primary Health Center

CHC: Community Health Centre

Essential bed counts in various health systems according to IPHS 2022	
District norms	1 bed per 1000 population
FRU-Urban CHCs	50 beds
Rural FRU CHCs	30 beds
Non-FRU CHCs	30 beds
24x7 HWC-PHC/UPHC	6 beds
Polyclinic	2 daycare beds
HWC-UPHC	2 daycare beds
HWC-PHC	2 beds

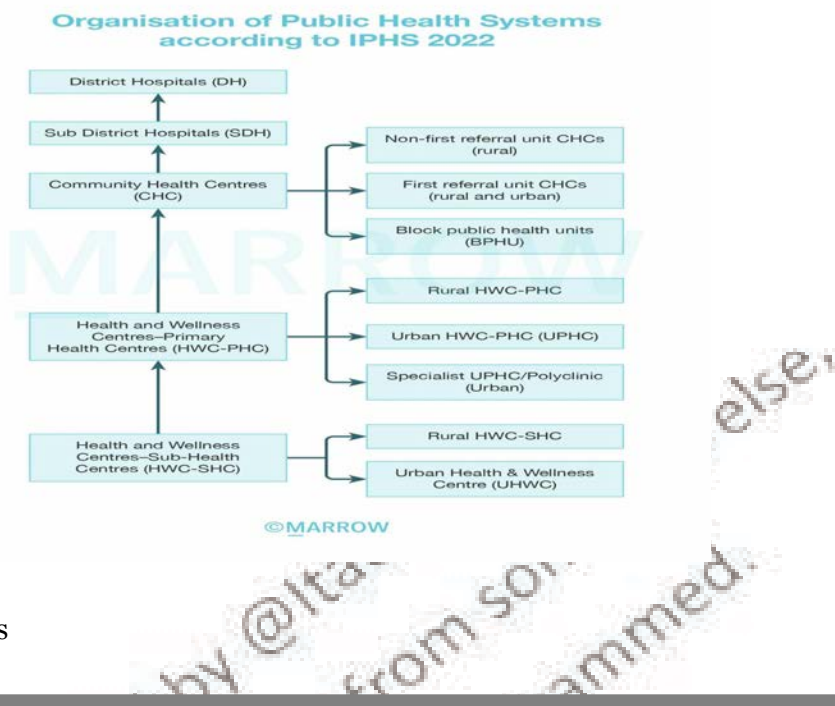
Solution to Question 7:

A rural hilly area with a population of 40,000 should have 2 Primary Health Centers (PHCs).

According to the latest IPHS 2022 guidelines, in hilly areas, there should be one PHC for every 20,000 population. Since this village has a population of 40,000, it should have 2 PHCs.

Solution to Question 8:

A polyclinic serves a population of 2.5 to 3 lakhs.



Types of PHCs/UPHCs

HWC-PHC	Urban HWC-UPHCs (UPHC)	Specialist UPHC/ Polyclinic
Provide primary care services in rural areas, ideally	Provide primary care services in urban areas	Provide primary care services in urban areas
Should aspire to become 24x7 centers	Provide assured round-the-clock emergency and secondary care services, along with other basic services	Provide specialist services on an ambulatory/daycare basis, closer to the urban community
PHCs with delivery services to function as 24x7 HWC-PHCs	UPHCs with IP beds and delivery services to function as 24x7 HWC-UPHCs	Also called Multispecialty UPHC/Polyclinics, mainly six specialties
Population norms: In hilly/tribal areas= 1 per 20,000 population In plain areas= 1 per 30,000 population	Population norms: 1 for every 50,000 population	Population norms: Serves a population of 2.5 to 3 lakhs, covering a catchment population of 5-6 UPHCs

Solution to Question 9:

Cholecystectomy is not a surgical procedure that is conducted in a PHC. The surgical procedures done in PHC include suturing of wounds, incision and drainage of abscesses, vasectomy, tubectomy, and hydrocelectomy.

Services provided under PHC:

- General services: OPD, teleconsultation services
- Emergency services
- 24x7 labor rooms

- Antenatal clinics, non-communicable diseases (NCD) clinics
- Rotational specialist clinics
- Surgical procedures:
 - Suturing of wounds
 - Incision and drainage of abscesses
 - Vasectomy, tubectomy, hydrocelectomy in PHCs with a functional minor OT, and standard infection-prevention protocols
- Antenatal care and services:
 - Detecting pregnancy
 - Provision of ANC and counseling services
 - Identification, management, and referral of high-risk pregnancies
 - Detection of alarming signs during pregnancy and labor with timely and appropriate referral
 - Monitoring high-risk pregnancies
 - Management of malaria for pregnant women, as per National Center for Vector Borne Diseases Control (NCVBDC) guidelines
 - Management of tuberculosis during pregnancy
 - Management of mild/moderate anemia
 - Management of gestational diabetes mellitus, as per Government of India guidelines

Solution to Question 10:

In a place with a population of 10,000, 5 ASHA workers are needed.

There are two forms of Health and Wellness Centers in Sub-Health Centers.

- Health and Wellness Centre - Sub-Health Center (HWC-SHC) in rural areas: It requires 1 ASHA per 1000 population or 1 ASHA per habitation in hilly/tribal areas. It has a total staff of 3 + ASHAs.
- Urban Health and Wellness Center (UHC): It requires 1 ASHA per 2000 population and 1 ANM per 10000 population. Health and wellness center- Sub-Center 1 ASHA per 1,000 population or 1 ASHA per habitation in hilly/tribal areas. It has a total staff of 4 + ASHAs.

Functions of HWCs under Sub-Health Centers:

- Provide primary care services
- Deliver the 12 packages under CPHC
- Serve as a platform for teleconsultation and widening the range of diagnostics
- Serve as a fulcrum for services in National Health Programmes

Solution to Question 11:

Specialists like ophthalmologists are not included as essential manpower for a PHC. Specialists for medicine, obstetrics & gynecology, pediatrics, ophthalmology, etc. are essential manpower included under polyclinics.

According to the Indian Public Health Standards (IPHS) 2022 guidelines, a total of 12 staff is considered essential manpower for a PHC. They include 1 Medical Officer (MBBS), 2 staff nurses, 1 pharmacist, 1 health worker female/ANM, 1 health worker/health assistant (male), 1 health assistant (female)/lady health visitor, 1 laboratory technician, 1 data entry operator, 1 dresser, 1 accountant, and 1 sanitation staff.

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International Health

Question 1:

Which of the following was the world's first international health agency?

- a) Pan American Sanitary Bureau
- b) World Health Organization
- c) Health Organization of the League of Nations
- d) United States Agency for International Development

Question 2:

When is breastfeeding week celebrated?

- a) 1st week of March
- b) 1st week of July
- c) 1st week of August
- d) 1st week of May

Question 3:

What was the theme for World Health Day 2023?

- a) Build a fairer, healthier world
- b) Health for all
- c) Our planet, our health
- d) Support nurses and midwives

Question 4:

When is World No Tobacco Day observed?

- a) 2nd August
- b) 31st May
- c) 14th June

d) 20th April

Question 5:

What was the theme of World TB Day 2023?

- a) Yes! We can end TB!
- b) It's time to end TB
- c) Invest to End TB. Save Lives
- d) The clock is ticking

Question 6:

Where is the headquarters of the WHO located?

- a) Copenhagen
- b) New York
- c) Geneva
- d) Glasgow

Question 7:

Which of the following is not one of the duties of the WHO?

- a) Family health
- b) Health literature and information
- c) Biomedical research
- d) State health

Question 8:

Where is the South East Asian Regional Office of the WHO located?

- a) Kuala Lumpur
- b) Colombo
- c) Singapore
- d) New Delhi

Question 9:

What is the surveillance reporting system used by the WHO?

- a) Technical Report Series
- b) Weekly Epidemiological Record
- c) Morbidity and Mortality Weekly Report
- d) WHO Bulletin

Question 10:

As a part of the 'Clean care is safer care' campaign, you have been asked to give a presentation to the nurses in the obstetrics and gynecology department. Which of the following areas is targeted in this campaign?

- a) Hand hygiene
- b) Obstetric care
- c) Cord care
- d) Breastfeeding care

Question 11:

Which agency of the United Nations has the given logo?



- a) UNICEF
- b) UNESCO
- c) UNFPA
- d) UNIFEM

Question 12:

Which of the following are components of UNICEF's GOBI FFF strategy?

- a) 1,2 and 4
- b) 2,3 and 5
- c) 1,3 and 5
- d) 4 and 5

Question 13:

Which agency of the UN gives funds for technical assistance and has the following as part of its logo?



- a) UNESCO
- b) UNDP
- c) UNEP
- d) UNHCR

Question 14:

Which organisation gives the highest funding for reproductive health?

- a) UNESCO
- b) UNFPA
- c) WHO
- d) UNIDO

Question 15:

Which of the following is not an objective of the FAO?

- a) To better the condition of rural people
- b) Improve nutrition of people of all countries
- c) Help nations raise living standards
- d) Improve access to nutritious foods

Question 16:

Which agency of the United Nations has the following logo?



- a) International Labor Organization
- b) World Food Programme
- c) United States Agency for International Development

d) World Bank

Question 17:

What does the branch of medicine known as nosology deal with?

- a) Cosmetic surgery
- b) Religious practices in medicine
- c) Prion diseases
- d) Classification of diseases

Question 18:

Study of which of the following is known as epidemiology?

- a) Occupational health
- b) Health of travelers
- c) Health of a community
- d) Environmental factors in health

Question 19:

Quarantine measures were implemented for the first time to prevent the spread of:

- a) Spanish flu
- b) Tuberculosis
- c) Plague
- d) Leprosy

Question 20:

Which of the following is not included in the Special Programme for Research and Training in Tropical Disease (TDR) strategy of the WHO?

- a) Malaria
- b) Lymphatic filariasis
- c) Leishmaniasis

- d) Tick-borne encephalitis

Question 21:

According to the International Health Regulations, on diagnosing a patient with which of the following diseases is notification not mandatory in all countries?

- a) Cholera
- b) Tuberculosis
- c) Yellow fever
- d) Smallpox

Question 22:

Which of the following disease is not under international surveillance by the WHO?

- a) Louse borne typhus
- b) Yellow fever
- c) Chagas disease
- d) Relapsing fever

Question 23:

Patients with which of the following diseases do not require isolation or quarantine?

- a) West Nile fever
- b) Diphtheria
- c) Cholera
- d) Tuberculosis

Question 24:

You are asked to investigate a possible act of terrorism using anthrax. Which of the following is false regarding this category of bioterrorism agent?

- a) High priority agents
- b) Have high mortality rates

- c) Moderately difficult to disseminate
- d) Require specialized public health responses

Question 25:

How many goals does the Sustainable Development Goals (SDG) comprise of?

- a) 17
- b) 28
- c) 8
- d) 14

Answer Key

Question No.	Correct Option
1	a
2	c
3	b
4	b
5	a
6	c
7	d
8	d
9	b
10	a
11	a
12	c
13	b
14	b
15	d
16	d
17	d
18	b
19	c
20	d

21	b
22	c
23	a
24	c
25	a

Detailed Explanations

Solution to Question 1:

The first international health agency to be established was the Pan American Sanitary Bureau (PASB) in 1902.

This was followed by the establishment of the Health Organization of the League of Nations in 1923.

The World Health Organisation (WHO) was established on 7th April 1948.

Solution to Question 2:

Breastfeeding week is celebrated in the 1st week of August.

Health days	Dates
World Tuberculosis Day	24th March
World Health Day	7th April
World Malaria Day	25th April
National Anti Malaria Month	June
World Immunization Week	Last week of April
World No Tobacco Day	31st May
World Blood Donor Day	14th June
World Hepatitis Day	28th July
Breastfeeding Week	1st week of August
World AIDS Day	1st December
Sexual and Reproductive Health Awareness Day	12th February

Solution to Question 3:

The theme for World Health Day 2023 was Health for all.

World Health Day is celebrated on 7th April and each year a different theme is adopted such as:

- 2024- My health, my right
- 2022- Our planet, our health
- 2021- Build a fairer, healthier world for everyone, everywhere
- 2020 - Support nurses and midwives
- 2018 & 2019 - Universal health coverage: everyone, everywhere
- 2017 - Depression: Let's talk
- 2016 - Beat diabetes
- 2015 - Food safety

Health days	Dates
World Tuberculosis Day	24thMarch
World Health Day	7thApril
World Malaria Day	25thApril
National Anti Malaria Month	June
World Immunization Week	Last week of April
World No Tobacco Day	31stMay
World Blood Donor Day	14thJune
World Hepatitis Day	28thJuly
Breastfeeding Week	1stweek of August
World AIDS Day	1stDecember
Sexual and Reproductive Health Awareness Day	12th February

Solution to Question 4:

World No Tobacco Day is observed on 31st May.

The theme of World No Tobacco Day 2021 is 'commit to quit'. It conveys that the COVID-19 pandemic has led to millions of tobacco users saying they want to quit.

The theme of World No Tobacco Day 2020 was #TobaccoExposed. It aimed to provide a counter-marketing campaign that shows smokers have a higher risk of getting a severe case of COVID-19 and to empower young people to engage in the fight against tobacco.

Solution to Question 5:

The theme of world TB day 2023 was 'Yes! We can end TB!'.
World Tuberculosis Day is observed on 24th March of every year.

The themes for the other world TB days are:

The theme of 2024 continues to be, 'Yes! We can end TB!'

2022: Invest to End TB. Save Lives

2021: The clock is ticking

2020: It's time to End TB

2019: It's time for action

2018: Wanted: Leaders for a TB-free world

Solution to Question 6:

The headquarters of the WHO is in Geneva, Switzerland.

The WHO is a specialized, non-political health agency of the United Nations. It came into force on 7th April 1948.

Its objective is the attainment of the highest level of health by all people. The structure of the WHO includes:

- The World Health Assembly
- The Executive Board
- The Secretariat

Important headquarters of international organisations include:

- The headquarters of UNICEF and UNDP is in New York, USA.
- Food and Agriculture Organization (FAO) headquarters is in Rome, Italy.
- Headquarters of the International Labor Organization (ILO) and the International Red Cross are also in Geneva.

The image below shows the WHO emblem.



Solution to Question 7:

State health is not one of the duties of the WHO. It is the responsibility of the government.

Work and duties of the WHO:

- Prevention and control of specific diseases
- Development of comprehensive health services
- Family health
- Environmental health
- Health statistics
- Biomedical research
- Health literature and information
- Cooperation with other organizations

Solution to Question 8:

The South-East Asian Regional Office (SEARO) of the WHO is located in New Delhi. Duties of the SEARO include planning for malaria eradication, tuberculosis control, and rural health services.

Members of SEARO are:

- Bangladesh
- India
- Bhutan
- Indonesia

- Democratic People's Republic of Korea
- The Republic of Maldives
- Myanmar
- Nepal
- Sri Lanka
- Thailand
- Timor-Leste

The regional offices of the WHO are listed below.

Regional office	Location
South-East Asian office	New Delhi, India
African regional office	Brazzaville, Republic of Congo
American regional office	Washington D. C., USA
European regional head office	Copenhagen, Denmark
Eastern Mediterranean office	Alexandria, Egypt
Western Pacific regional office	Manila, Philippines

Solution to Question 9:

The surveillance reporting system used by the WHO is the Weekly Epidemiological Record (WER).

The WHO collects and disseminates epidemiological information on diseases subject to International Health Regulations. It is done through an Automatic Telex Reply Service (ATRS) and the Weekly Epidemiological Record (WER).

Solution to Question 10:

'Clean care is safer care' campaign was started by the WHO to improve hand hygiene practices.

In 2005, WHO developed this campaign aimed at the reduction of healthcare-associated infections (HCAI) through the worldwide promotion of appropriate hand hygiene practices.

Solution to Question 11:

The given logo is that of the United Nations Children's Fund (UNICEF). It works closely with the WHO on health matters.

UNICEF's content of services include:

- Child health
- Child nutrition
- Family and child welfare
- Formal and informal education

UNICEF participates in urban basic services (UBS) with the aim to promote access to basic services like nutrition, water supply, and sanitation.

UNICEF started the Baby-Friendly Hospital Initiative (BFHI) to promote breastfeeding and improve maternal and child care. In 2016, the Ministry of Health and Family Welfare launched the Mothers Absolute Affection (MAA) programme to revitalize efforts towards protection, promotion and support of breastfeeding practices.

Solution to Question 12:

The components of the GOBI FFF strategy by the UNICEF are:

- Growth monitoring with growth charts
- Oral rehydration therapy for diarrhoea and dehydration
- Breastfeeding
- Immunization of children
- Family planning and birth spacing
- Food supplementation
- Female literacy promotion

The GOBI FFF strategy is aimed at revolutionising child health and survival. Initially, it was called GOBI strategy and later the FFF aspect was added to it.

Solution to Question 13:

United Nations Development Programme (UNDP) is an agency of the UN that gives funds for technical assistance. It has the following logo.

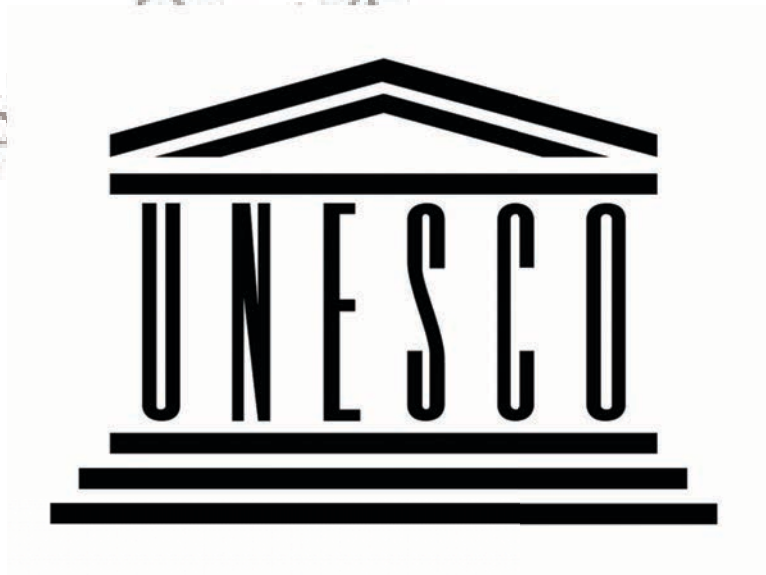


Its main objective is to help poorer nations develop their human and natural resources to their potential.

Some other agencies of the UN include:

- UNESCO - United Nations Educational, Scientific and Cultural Organization
- UNEP - United Nations Environment Programme
- UNHCR - United Nations High Commissioner for Refugees

Their logos are given below.





Solution to Question 14:

United Nations Fund for Population Activities (UNFPA) gives the highest funding for reproductive health. It works in the area of health and family welfare.

It mainly works by:

- Developing the national capability in manufacturing contraceptives,
- Promoting population education programmes
- Introducing innovative approaches to family planning and MCH care.

Sexual and Reproductive Health Awareness Day is organized on February 12th every year to raise awareness about sexual and reproductive health.

The logo of the UNFPA is given below.



Solution to Question 15:

Improving access to nutritious foods is not one of the objectives of the FAO.

Aims of the Food and Agriculture Organization (FAO):

- To help nations raise living standards
- To improve the nutrition of people of all countries
- To increase the efficiency of farming, forestry, and fisheries
- To better the condition of rural people

The FAO organized a Freedom from Hunger Campaign (FFHC) in 1960. Its main objective was to combat malnutrition and to disseminate information and education.

The FAO logo is given below.



Solution to Question 16:

The given logo is of the World Bank.

It helps less developed countries raise their living standards with economic support. The allocation of funds is decided by its Board of Governors.

Solution to Question 17:

Nosology is the branch of medicine that deals with the classification of diseases.

Solution to Question 18:

Emporiatrics is the study of the health of travelers.

It is also called travel medicine. It deals with the prevention and management of health problems of international travelers.

Solution to Question 19:

Quarantine was introduced for the first time to prevent the spread of the plague, also known as the black death.

Quarantine is the restriction on the movement of people, animals, and goods that may have been exposed to a communicable disease. It is intended to prevent the spread of communicable diseases.

Quarantine is distinct from medical isolation, in which those confirmed to be infected with a communicable disease are isolated from the healthy population.

Solution to Question 20:

Tick-borne encephalitis is not included in the Special Programme for Research and Training in Tropical Disease (TDR) strategy of the WHO. The diseases included under this programme are:

- Chagas disease
- Dengue
- Ebola
- Helminthiasis

- Leishmaniasis
- Leprosy
- Lymphatic filariasis
- Malaria
- Onchocerciasis
- Schistosomiasis
- Trypanosomiasis
- Tuberculosis

The programme is sponsored by the WHO, UNICEF, UNDP, and World Bank.

Solution to Question 21:

According to the International Health Regulations(IHR), notification is not mandatory if a patient is diagnosed with tuberculosis in all countries.

It needs to be notified only by the countries where the public health impact is serious and there is a risk of international spread.

The government of India made the reporting of new cases of TB mandatory for all medical practitioners from 2012. An online web-based application named Nikshay is available for this purpose. Treatment providers are given a nominal cash incentive of Rs.1000/- per TB patient reported.

Solution to Question 22:

Chagas disease is not under international surveillance by the WHO.

Surveillance is usually done for infectious diseases and diseases that have the potential to become epidemics or pandemics.

Solution to Question 23:

Patients with West Nile fever do not require isolation or quarantine.

Isolation and quarantine are used for highly infectious diseases that are at risk of becoming epidemics or pandemics.

Solution to Question 24:

Anthrax falls under category A of bioterrorism agents. These organisms are usually easy to disseminate.

Category A refers to high priority agents. They have high case-fatality rates and high transmissibility. Thus, a specialised public health response is required to combat such attacks.

Solution to Question 25:

The Sustainable Development Goals (SDG) are comprised of 17 goals.

These were set as part of the official Sustainable Development Goals of the Agenda 2030. These goals consist of a total of 169 targets. SDGs replaced the Millennium Development Goals on 1st January 2016.

The 2030 agenda consists of five P's:

- People
- Planet
- Prosperity
- Peace
- Partnership



Descriptive Statistics I - Probability and Data

Question 1:

You are conducting a study to determine the mean height of children visiting the local PHC. Which type of statistical study is being done here?

- a) Descriptive
- b) Analytical
- c) Inferential
- d) Observational

Question 2:

Which of the following statements about statistical terms is incorrect?

- a) Population is the universe about which an investigator wishes to draw conclusions
- b) A sample is a representative subset of the population that is being studied
- c) An element is a group of observations in the population
- d) A population need not be only people but can be measurements too

Question 3:

Which of the following statements about probability is true? (Probability of event occurring is p)

- a) 2,3,4 & 5
- b) 1,3 & 5
- c) 1,2 & 3
- d) 2 & 4

Question 4:

In which of the following processes is binomial distribution widely used?

- a) Marriage counseling
- b) Genetic counseling

- c) Treatment counseling
- d) Differential diagnosis

Question 5:

If the events A and B are mutually exclusive events, what is the probability of either A or B occurring?

- a) $p(A) + p(B)$
- b) $p(A) \times p(B)$
- c) $p(A) + p(B) - p(A \text{ and } B)$
- d) $1 - [p(A) + p(B)]$

Question 6:

You are conducting a study on birth weight. As per your data, the probability of a baby having birth weight ≤ 2500 grams is 0.50 and of having birth weight of 2500-2999 grams is 0.20. What is the probability that a baby will have a birth weight of $\leq 3\text{kg}$?

- a) 0.70
- b) 0.60
- c) 0.10
- d) 0.3

Question 7:

You select 3 people from a population where the prevalence of anemia is 20%. What is the probability that all the three randomly selected people from the population would have anemia?

- a) 0.002
- b) 0.004
- c) 0.6
- d) 0.008

Question 8:

There are 50 patients admitted in a ward of your medical college. Here 20 are girls and 30 are boys. Out of this, 10 girls and 20 boys require surgery. What is the probability that a randomly selected boy and girl both require surgery?

- a) $\frac{1}{6}$
- b) $\frac{3}{5}$
- c) $\frac{6}{25}$
- d) $\frac{1}{3}$

Question 9:

The chance of passing a genetic disease X by affected parents to their first child is 0.16. If they plan to have two children, what is the probability that both the children will have the disease?

- a) 0.2944
- b) 0.32
- c) 0.0032
- d) 0.0256

Question 10:

Which of the following is not a measurement scale for data?

- a) Nominal
- b) Ordinal
- c) Ratio
- d) Proportion

Question 11:

Which of the following scales of data measurement do blood groups belong?

- a) Interval scale data
- b) Ratio scale data
- c) Categorical data
- d) Ordinal Data

Question 12:

Which of the following are examples of nominal scale data?

- a) 1 & 3
- b) 4 & 5
- c) 1,2 & 5
- d) 1,2 & 3

Question 13:

A physician classifies a group of patients based on disease severity as normal, mild, moderate and severe. Which of the following scales of data does this classification system belong to?

- a) Nominal
- b) Ordinal
- c) Ratio
- d) Interval

Question 14:

Which of the following are representative of ordinal scale data?

- a) 1,2,3,4 & 5
- b) 2,3 & 5
- c) 1 & 4
- d) 2,3,4 & 5

Question 15:

Which of the following is not a continuous variable?

- a) Blood glucose value of a diabetic patient
- b) Mid-arm circumference of a child
- c) Number of visits to the hospital in a year
- d) Birth weight of a baby

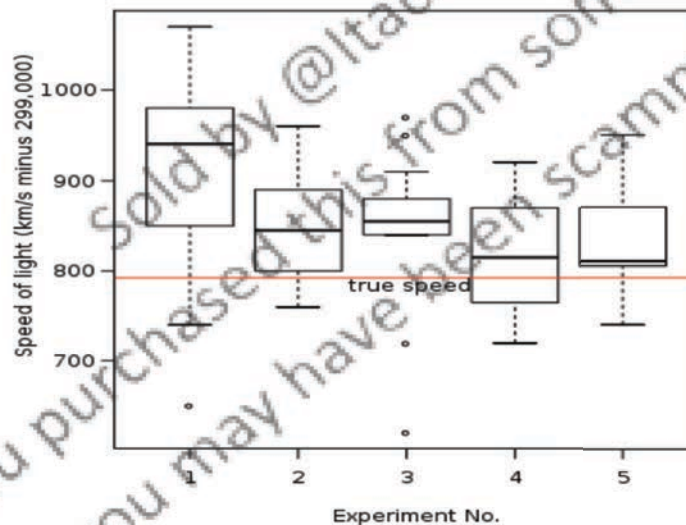
Question 16:

Match the following.

- a) 1-a, 2-b, 3-c, 4-b, 5-c
- b) 1-a, 2-c, 3-b, 4-c, 5-b
- c) 1-a, 2-c, 3-c, 4-b, 5-c
- d) 1-b, 2-c, 3-b, 4-c, 5-a

Question 17:

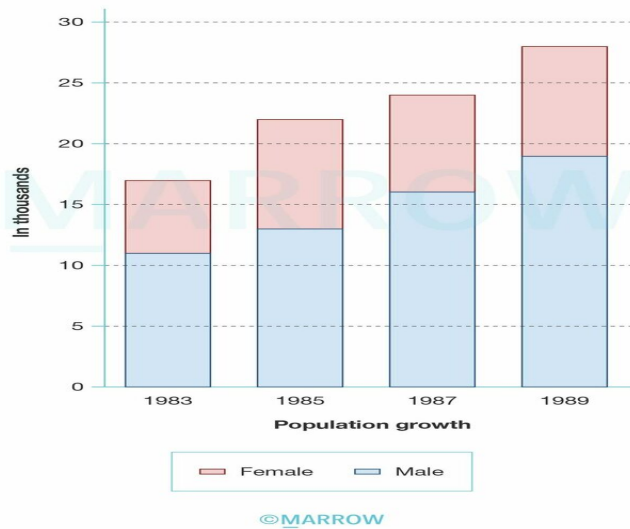
Identify the following representation of data.



- a) Pictogram
- b) Boxplot
- c) Histogram
- d) Stem and leaf plot

Question 18:

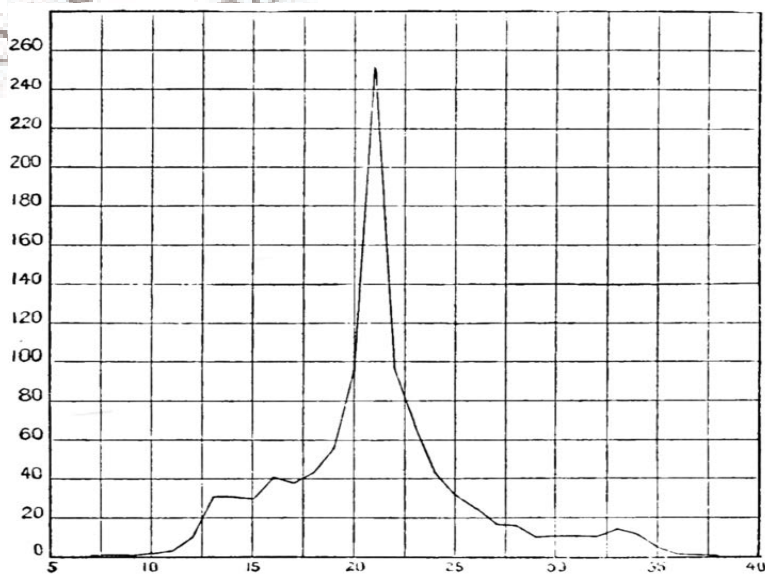
The following diagram charts the population change over years from 1983 to 1989 in a given place. What is the statistical diagram used here?



- a) Histogram
- b) Line diagram
- c) Component bar graph
- d) Multiple bar graphs

Question 19:

Identify the following graphical representation of data.



- a) Ogive
- b) Scattergram

- c) Frequency polygon
- d) Line diagram

Question 20:

During a cholera outbreak, you are asked to map the geographical distribution of the disease. What is the best method of doing this?

- a) Dot-density map
- b) Bar diagram
- c) Line diagram
- d) Pie chart

Question 21:

What is the best way to plot the changes in incidence of a disease in a given area over time?

- a) Line graph
- b) Histogram
- c) Ogive
- d) Tree diagram

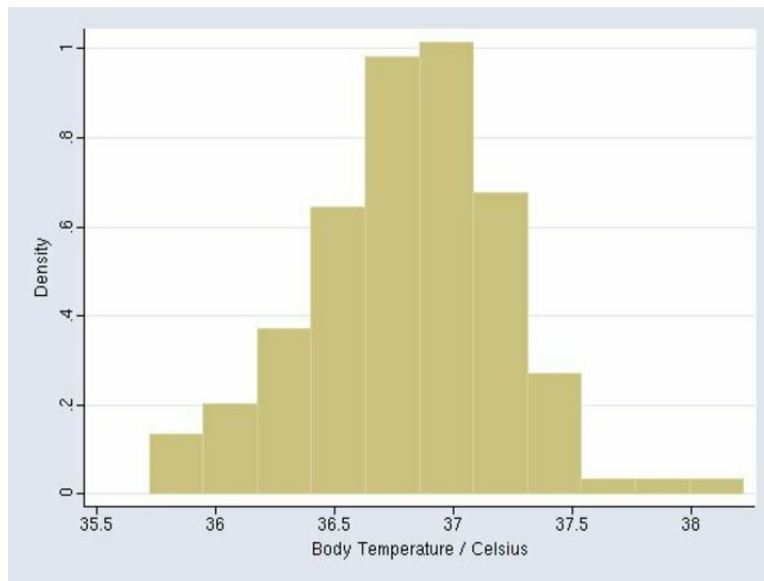
Question 22:

A doctor is taking history from a diabetic patient and decides to make a chart to assess joint involvement. Which chart should he make to explain this?

- a) Histogram
- b) Line diagram
- c) Pie chart
- d) Tree diagram

Question 23:

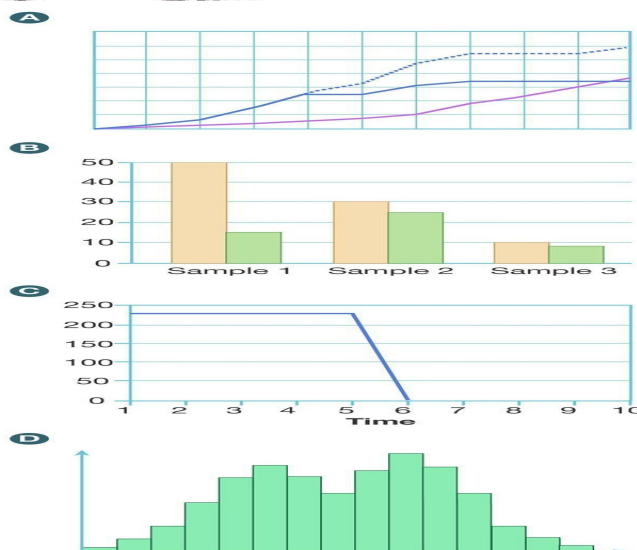
What type of data does the image given below represent?



- a) Nominal
- b) Discrete
- c) Continuous
- d) Ordinal

Question 24:

Which of the following can be used to represent quantitative data?



- a) Only C
- b) B and D

- c) A, C and D
d) A, B and C

Answer Key

Question No.	Correct Option
1	a
2	c
3	b
4	b
5	a
6	a
7	d
8	d
9	d
10	d
11	c
12	c
13	b
14	b
15	c
16	c
17	b
18	c
19	c
20	a
21	a
22	d
23	c
24	c

Detailed Explanations

Solution to Question 1:

A study to determine the mean height of children is an example of a descriptive statistical study. Descriptive statistics are the statistical methods that merely organize, describe or summarize the actual data available. For example, a study determining the mean blood pressure of patients visiting a PHC on a single day.

Analytical statistics are the second major category of epidemiological studies, after descriptive statistics. The objective of an analytical study is to test the hypothesis.

An observational study is an empiric investigation in which the objective is to establish a relationship between cause and effect. It falls under the category of analytical statistics.

It is performed when controlled experimentation cannot be feasibly used to determine a hypothesis. For example, in the case of cigarette smoking causing lung cancer, it would be unethical and not feasible to conduct a controlled experiment. So, an observational study is preferred.

Inferential statistics involve inductive reasoning i.e., generalizing observations to a population after having observed only a sample. An example of such a study would be a study determining the expected success rate of vasectomy

Theoretical statistics is a mathematical approach to describe something, predict events, or analyze the relationship between things. Some of the concepts of theoretical statistics are mean, median, mode, standard deviation and correlation.

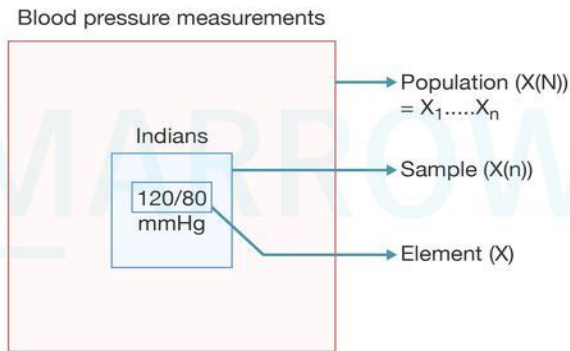
Solution to Question 2:

An element is a single observation in a study. For example, in a study of blood pressure values, the blood pressure value of one person is an element.

In a statistical study, the population refers to the group about which information is to be gathered. The composition of the population in terms of its age, sex, occupation, cultural characters, etc need to be defined before a study is conducted. The population need not consist of just people but can also be made of measurements. For example, in a study of blood glucose measurements in Indians, the population consists of the blood glucose measurements.

The part/subset of the population, which is being studied is known as a sample. For example, in a questionnaire-based study on cervical cancer in a population, the sample would include females attending the out-patient departments of a hospital. This is a subset of the entire population.

Statistical terms



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Solution to Question 3:

If the probability of an event occurring is 1, it means there is an absolute certainty that the event will occur.

The addition rule and multiplication rule can be used to calculate the probability of an event. The addition rule is used to calculate the combined probability of the occurrence of two or more mutually exclusive events. The multiplication rule is used to calculate the combined probability of two or more statistically independent events occurring together.

Probability values range from 0 (no probability) to 1 (absolute certainty). So negative value of p is not possible.

Since p is the probability of an event occurring and q is the probability of the event not occurring,
 $q = 1 - p$

Solution to Question 4:

Binomial distribution is widely used in genetic counseling.

Binomial distribution helps determine the probability of occurrence of a specific combination of mutually exclusive independent events. In a binomial distribution, there are only two possible events like Yes/No or Male/Female or Alive/Dead. The results of the distribution gives the probabilities of obtaining an exact number of successes in a series of independent trials.

For example, the inheritance of cystic fibrosis follows a binomial distribution. There are:

- Two possible events - inheriting the disease/ not inheriting it
- Events are mutually exclusive - one person cannot both have the disease and not have it

c. Independent possibilities - if one child has inherited it, this will not affect if the other child will inherit it.

The physician using this technique can explain to the patients the probabilities of their children suffering from the disease.

Solution to Question 5:

If the events A and B are mutually exclusive, then the probability of either A or B occurring is given by,

$$p(A \text{ or } B) = p(A) + p(B)$$

This is as per the addition rule of probability.

The probability of both the mutually exclusive events happening at the same time is given by,

$$p(A \text{ and } B) = p(A) \times p(B)$$

The probability of any of two mutually non-exclusive events occurring at a point of time is given by,

$$p(A \text{ or } B) = p(A) + p(B) - p(A \text{ and } B)$$

If $p(A)$ and $p(B)$ are the only two outcomes possible in a given event then,

$$p(A) + p(B) = 1$$

For example, probability of patient suffering from lung cancer dying within 5 years and probability of patient suffering from lung cancer surviving these 5 years adds up to 1.

Solution to Question 6:

The probability that a baby will have a birth weight of $\leq 3\text{kg}$ is 0.70.

In this scenario, the probability of having a baby with BW ≤ 2500 grams, $p(A) = 0.50$ and that of having a baby with BW between 2500 – 2999 grams, $p(B) = 0.20$.

Since both events are mutually exclusive, i.e., they cannot happen together, it satisfies the condition for the addition rule of probability.

Thus, the probability of having a baby that weighs less than 3 kilos is,

$$p(A \text{ or } B) = p(A) + p(B) = 0.50 + 0.20 = 0.70.$$

If $p(A)$ and $p(B)$ were non-mutually exclusive events, then the probability would have been calculated by

$$p(A \text{ or } B) = p(A) + p(B) - p(A \text{ and } B).$$

Solution to Question 7:

In the given scenario, the probability of all 3 randomly selected individuals having anemia is 0.008.

In this scenario, the prevalence of anemia is 20%.

So, the probability of a person having anemia in the given population (p) = $20/100 = 0.2$.

If selecting 3 people from the population randomly, the chance of the first person having anemia is independent of the second or third person also having it and vice versa. This satisfies the condition for the multiplication rule of probability which states that the probability of two or more statistically independent events all occurring together is equal to the product of their individual probabilities.

Thus, here the probability that all the 3 randomly selected people from the population would have anemia = $0.2 \times 0.2 \times 0.2 = 0.008$.

Solution to Question 8:

In the given scenario, the probability that a randomly selected boy and girl both require surgery is $1/3$.

First, we have to calculate the probability of picking the boys requiring surgery and the girls requiring surgery separately. These are mutually exclusive events.

In this scenario, the probability of correctly picking a girl requiring surgery is $p(A) = 10/20 = 1/2$

The probability of correctly picking a boy requiring surgery is $p(B) = 20/30 = 2/3$

Next, we apply the multiplication rule of probability.

The probability of picking boys and girls that require surgery,

$$p(A \text{ and } B) = p(A) \times p(B) = \frac{1}{2} \times \frac{2}{3} = \frac{2}{6} = \frac{1}{3}.$$

Solution to Question 9:

The probability of both children having the disease is 0.0256.

In this scenario, the chance of each child having the disease is 0.16. The second child having the disease is independent of the first child having it. So, the events are statistically independent.

Applying the multiplication rule of probability, the probability of both children having the disease is,

$$0.16 \times 0.16 = 0.0256.$$

These are non-mutually exclusive events as the first child having the trait does not rule out the chance of the second child having it.

On the other hand, the probability of one of the two children having the disease is given by,

$$p(A \text{ or } B) = p(A) + p(B) - p(A \text{ and } B) = 0.16 + 0.16 - (0.16 \times 0.16) = 0.32 - 0.0256 = 0.2944$$

Solution to Question 10:

A proportion is not a measurement scale for data. It is a ratio, where the numerator is always a part of the denominator. It indicates a value as a part of the whole. It is usually expressed as a percentage.

The four scales of measurement used for data can be remembered with the mnemonic NOIR. This stands for nominal, ordinal, interval and ratio data scales.

Solution to Question 11:

Blood groups are nominal scale data or categorical data. This is because they are just labels and have no numerical value or meaningful order.

Solution to Question 12:

Genders, types of anemia and types of hospitals come under nominal scale data.

There is no meaningful order or quantifiable variable in this kind of data. It is arranged on the basis of labels, titles or categories.

Kuppuswamy scale is an ordinal scale data. It is a scale representing socio-economic data.

Height is a ratio scale data as it can be arranged in a meaningful order with a well-defined size interval.

Solution to Question 13:

The classification of this disease as normal, mild, moderate and severe belongs to the ordinal scale. This is because this arrangement is not only qualitative but also has a meaningful order with no interval in between.

Solution to Question 14:

TNM staging, Likert scale and severity of pain are measured on the ordinal scale.

In such cases, the arrangement is qualitative, with a meaningful order but there is no way of quantifying the difference between the individual variables. For example, grading of pain as mild, moderate, and severe.

The Likert scale is used in collecting self-reported data in questionnaire-based research. It is a psychometric scale used to understand people's attitudes from responses. Usually, a 5-point scale of agreement is used which includes:

- 1 – Strongly disagree

- 2 – Disagree
- 3 – Neither agree nor disagree
- 4 – Agree
- 5 – Strongly agree

TNM staging is done for the staging of cancer. It is based on tumor size, lymph node status and metastasis. Lower stages indicate better prognosis than higher stages.

Hemoglobin level is an example of ratio scale data.

HIV positive or negative status falls under the category of nominal scale data.

Solution to Question 15:

The number of visits to the hospital in a year is a discrete variable. The only values it can take are whole numbers and none in between. For example, you can't visit the hospital 2.5 times.

Blood glucose value, mid-arm circumference and birth weight are continuous variables. They can have any value within a limit or a range.

Solution to Question 16:

The number of cancer patients in a hospital is an example of discrete data. They can only take certain values and none in between.

The temperature measured in the Celsius scale is an example of interval scale data. Such data are arranged in a meaningful order but they can not have an absolute zero value. For example, 80 degree celcius is not twice as hot as 40 degree celcius and 0 degree celcius does not indicate an absolute absence of heat.

Height and weight, blood pressure and temperature in the Kelvin scale are examples of ratio scale data. Such data are arranged in a meaningful order and the scales they are measured in have absolute zero value. For example, a temperature of 0 kelvin indicates an absolute absence of heat.

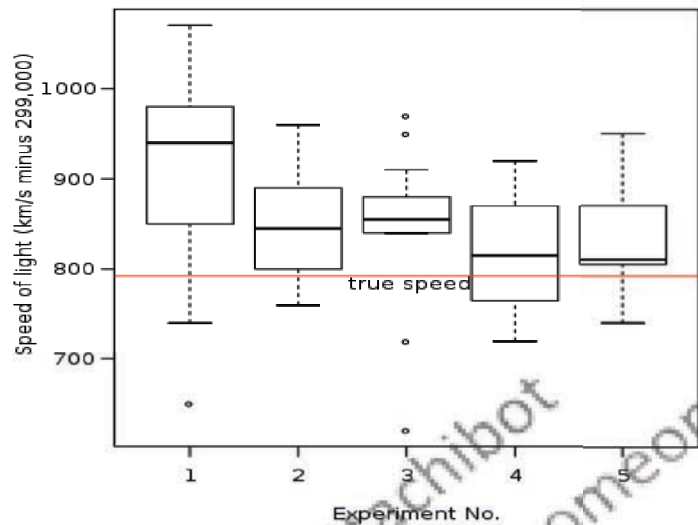
Height and weight, blood pressure, and temperature are examples of continuous data as they can take any values between limits or a range.

Solution to Question 17:

The image shown is of a boxplot. This is a method for graphically depicting groups of numerical data through their quartiles.

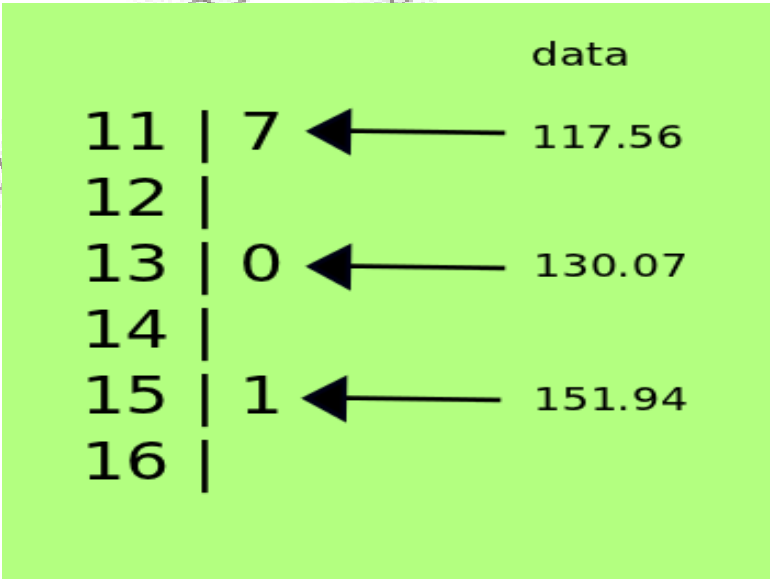
Box plots may also have lines extending vertically from the boxes (i.e., whiskers) indicating variability outside the upper and lower quartiles. Hence, they are also known as box-and-whisker plots and box-and-whisker diagrams.

The below image shows a boxplot depicting a series of experiments done to calculate the velocity of light.



A stem and leaf plot is used to depict frequency distribution and individual values for numerical data. The stem is are usually arranged in ascending or descending order vertically, and a vertical line is drawn to separate it from the leaf. The number of leaf items total up to the number of observations. It is difficult to represent a large number of observations using this method.

The below image shows a stem and leaf plot.

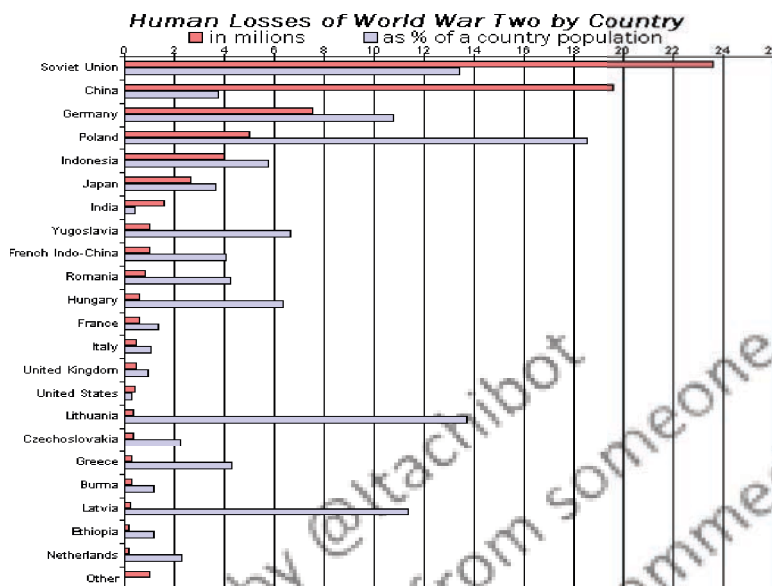


Solution to Question 18:

The given image shows a component bar graph. It is used to represent data in which the total magnitude is divided into different subsets or components.

A bar chart or bar graph is used mainly for nominal or ordinal scale data. They can be simple, multiple or component bar graphs.

The image given below shows a multiple bar graph showing the human losses of world war II by country. In a multiple bar chart, 2 or more sets of data are shown simultaneously.



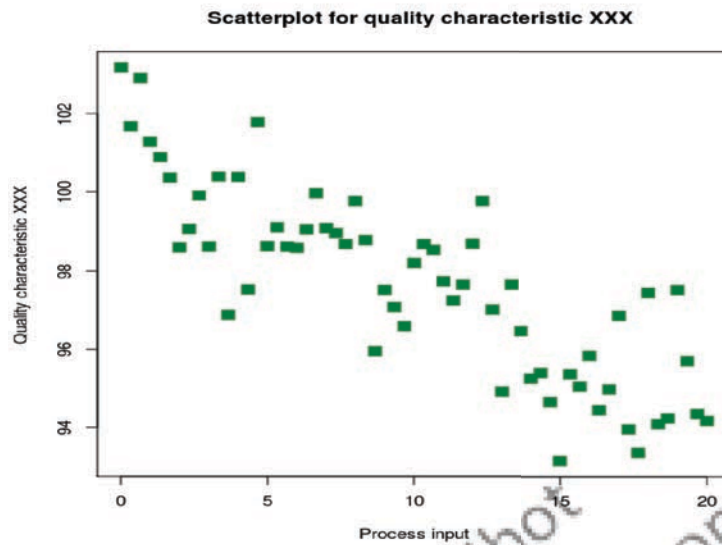
A bar chart can be distinguished from a histogram because the rectangles are separated by the spaces. This indicates that the data forms discrete categories.

In a histogram, the rectangles are not separated by spaces as the data represented forms a continuous set instead of discrete groups.

Solution to Question 19:

The given image is of a frequency polygon. It is obtained by joining the midpoints of histogram blocks. It is used to diagrammatically represent frequency distribution.

Scattergram is a graph in which the values of two variables are plotted along two axes. The pattern of the resulting points reveals any correlation that may be present. The below image shows a scattergram.

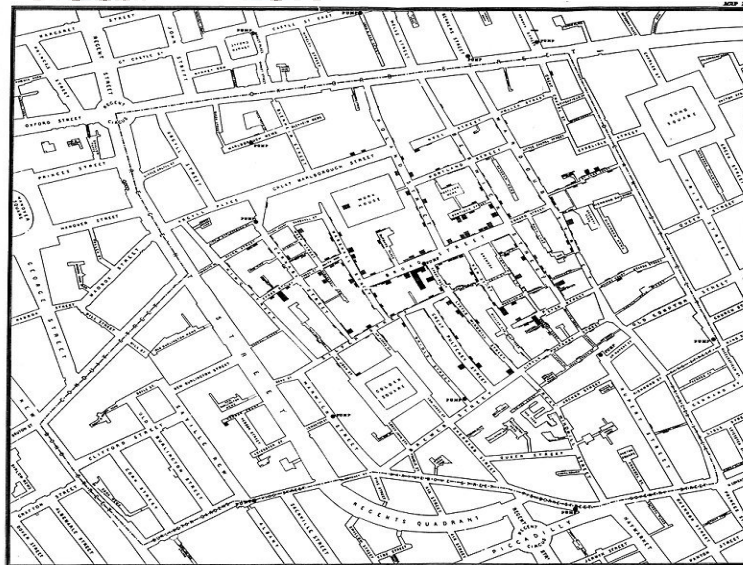


A line diagram is a type of chart that displays information as a series of data points called 'markers' connected by straight line segments. It is usually used to study time trends.

Solution to Question 20:

Dot-density maps are the best way to plot the geographical distribution of diseases.

The image given below shows a dot-density map of London that was plotted during the 1854 outbreak of cholera.



Solution to Question 21:

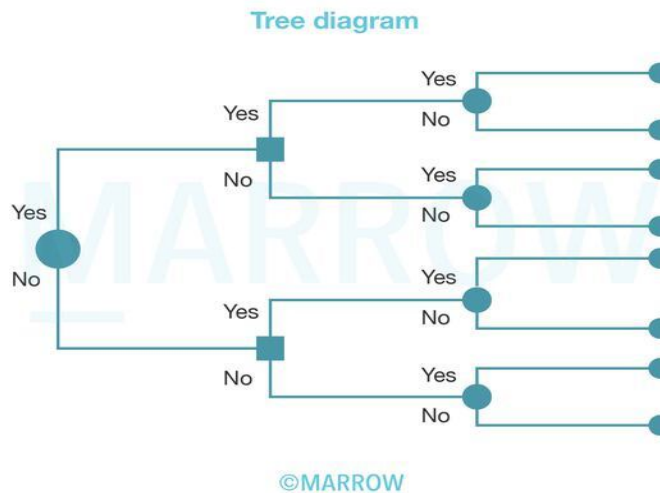
Line graphs or line diagrams are the best way to plot the changes in the incidence of a disease in a given area over time.

Solution to Question 22:

In this case, a tree diagram would help the doctor in explaining the possible outcomes to the patient's diabetes, with the perspective of joint involvement.

A tree diagram is used mainly to chart the possible outcomes of a sample event.

The following image shows the outline of a tree diagram.



Solution to Question 23:

The given image is of a histogram. It is used to represent frequency distributions of continuous quantitative data.

Nominal and ordinal data can be represented by a bar chart.

Solution to Question 24:

An ogive, a line diagram and a histogram can be used to represent quantitative data.

The images shown above are:

- A – Ogive
- B – Bar graph
- C – Line diagram

D – Histogram

Quantitative data cannot be represented by a bar graph. It is used mostly for nominal scale or qualitative data.

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If you purchased this from someone else,
you may have been scammed.

Descriptive Statistics II - Measures of Location

Question 1:

Which of the following is not a measure of location?

- a) Mean
- b) Median
- c) Mode
- d) Percentile

Question 2:

The mean can be calculated only if data is on:

- a) Interval scale
- b) Dichotomous scale
- c) Nominal scale
- d) Ordinal scale

Question 3:

During a school health survey on malnutrition, the children are arranged according to ascending order of their heights. Which of the following can be obtained by this arrangement?

- a) Arithmetic mean
- b) 50th centile
- c) Geometric mean
- d) Mean deviation

Question 4:

Which of the following can have more than one value?

- a) Median
- b) Harmonic mean

- c) Mode
- d) Geometric mean

Question 5:

While collecting data on which of the following variables would median not be the preferred measure of central tendency?

- a) Blood pressure
- b) Survival time
- c) Incubation period
- d) Health expenses

Question 6:

Which of the following is true for the given distribution of ages of patients who are visiting a family planning clinic in a day?

- a) Mean is 30
- b) Mode is 15
- c) Median is 15
- d) Median is 28

Question 7:

You are conducting a study on birth weights in your hospital. Out of 11 births, 5 babies weighed over 2.5 kg, one baby weighed 2.5 kg and 5 weighed less than 2.5 kg. What does the value 2.5 represent in this distribution?

- a) Arithmetic Mean
- b) Geometric Mean
- c) Median
- d) 3rd quartile

Question 8:

You are conducting an epidemiological study on malaria in a town X. The number of cases reported during the last 10 years is given below. What is the most appropriate measure of central tendency for this data?

- a) Arithmetic Mean
- b) Mode
- c) Median
- d) Geometric Mean

Question 9:

During a school health assessment, the mean value of weight for a group of 10 boys was found to be 18.2 kg. Later, it was found that the weight of one of the boys was wrongly recorded as 2.0 kg instead of 20 kg. What is the true mean weight of the group?

- a) 20 kg
- b) 20.2 kg
- c) 16.4 kg
- d) 18.2 kg

Question 10:

Which of the following statements is correct about the given distribution of ages of patients attending a geriatric medicine clinic?

- a) Mode is 70, 76
- b) Median is 80
- c) Mode is 73
- d) Median is 76, 80

Question 11:

The smoking history for the past 6 years of a patient with emphysema is given in the table below. If a pack has 20 cigarettes, what is the mean, median and mode of the given data?

- a) 16, 10, 15
- b) 16, 10, 10
- c) 10, 10, 15

d) 16, 15, 10

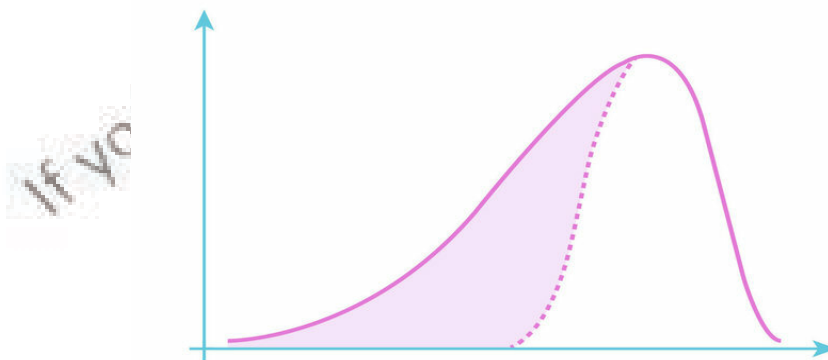
Question 12:

While reviewing a data distribution, you find that the highest value of 58 was wrongly recorded as 85 and so you correct it. How will this correction change the measures of central tendency?

- a) No change in mean, decreased median
- b) Decreased mean, no change in median
- c) Decreased mean, Decreased median
- d) Decreased mean, increased median

Question 13:

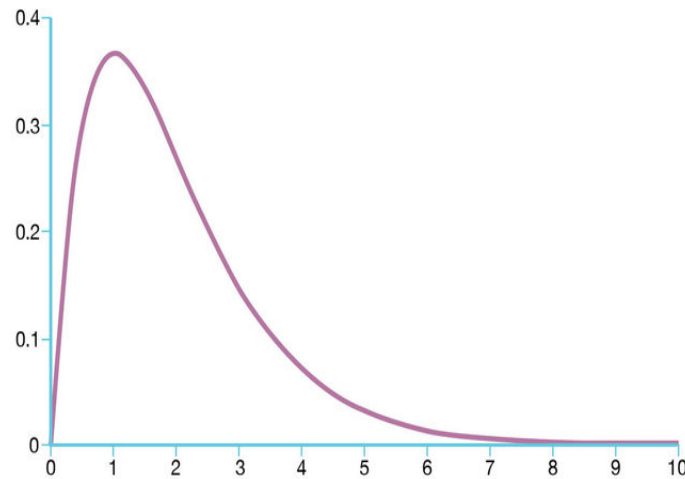
While conducting a study on the age at death in males of a population Y, the distribution curve given below is obtained. Which of the following is true?



- a) 2 & 5
- b) 1 & 3
- c) 1, 3 & 4
- d) 2, 4 & 5

Question 14:

Which of the following sets of measures of central tendency is likely to match the distribution depicted in the given image?



- a) Median = 3, mode = 1, mean = 2
- b) Mean = 3, median = 2, mode = 1
- c) Mode = 3, median = 2, mean = 1
- d) Mean = 3, mode = 2, median = 1

Question 15:

You are conducting a study on blood pressure in a population. The systolic blood pressure values have a mean of 130 mmHg, a median of 140 mmHg and mode of 160 mmHg. What type of distribution is this?

- a) Unimodal symmetrical distribution
- b) Positively skewed distribution
- c) Negatively skewed distribution
- d) Bimodal asymmetric distribution

Question 16:

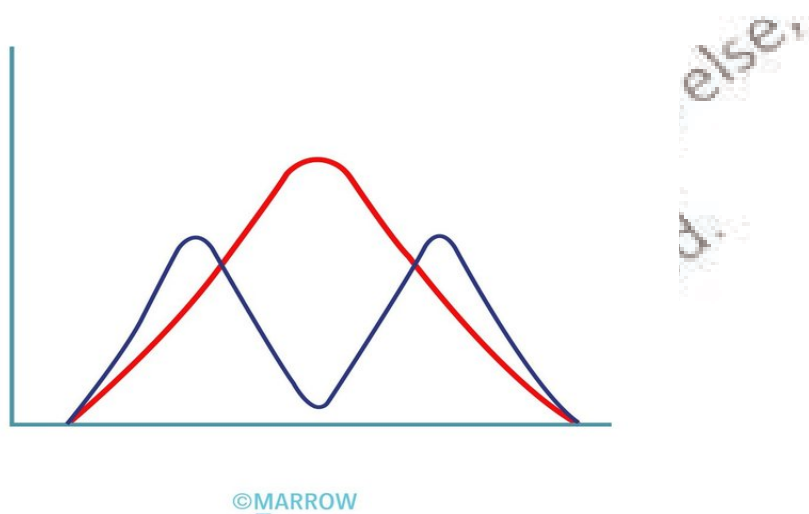
The mean of a dataset is 2 and the median is 3. What is the mode?

- a) 5
- b) 2.5

- c) 4
- d) 3

Question 17:

Study A(red) and Study B(blue) done in a given population yields two different distribution curves as shown below. Comparing both the curves, which of the following will be true?



- a) Mean = mode = median
- b) Mean = median, not equal to mode
- c) Mean, median, mode not equal
- d) Mode = median, not equal to mean

Answer Key

Question No.	Correct Option
1	d
2	a
3	b
4	c
5	a
6	a

7	c
8	c
9	a
10	a
11	b
12	b
13	a
14	b
15	c
16	a
17	b

Detailed Explanations

Solution to Question 1:

Percentile is a measure of position. It can be defined as a number that states the percentage of observations that fall below any particular score.

Measures of location help in summarizing the data by locating its centre or middle. The three main measures of location include mean, median and mode. They are also called measures of central tendency.

Solution to Question 2:

The mean can be worked out only for quantitative values. Thus, it can be calculated for interval or ratio scale data.

A dichotomous scale is a type of nominal scale data that can take only two values. Example: Male/Female. For nominal scale data, the preferred measure of central tendency is mode. For ordinal scale data, both median and mode can be calculated.

Median can be derived from ordinal, interval and ratio data scales. Mode can be derived from all scales of data- nominal, ordinal, interval and ratio.

Solution to Question 3:

The 50th centile can be obtained when values are arranged in order and the central value is picked out. This is the same as the median.

In a distribution that has an odd number of elements, the median is the middle value.

In a distribution that has an even number of elements, the median is the average of the middle values, i.e., the arithmetic mean of the two middle values.

An example of how to calculate the median of an odd and an even number of elements is given below.

1, 3, 3, **6**, 7, 8, 9

Median = **6**

1, 2, 3, **4**, **5**, 6, 8, 9

Median = $(4 + 5) \div 2$
= **4.5**

Option C: Geometric mean is obtained by taking the arithmetic mean of values on a logarithmic scale.

Option D: Mean deviation is a measure of dispersion. The average of deviations from the arithmetic mean gives the mean deviation.

Solution to Question 4:

The mode of a given sample can have more than one value. It is the most frequently occurring value in a distribution.

For example, in the sample [1, 3, 6, 6, 6, 6, 7, 7, 12, 12, 17], the most repeated value i.e., the mode is 6.

For the list of data [1, 1, 2, 4, 4], the mode is 1 and 4. Therefore, the mode is not unique. This dataset may be said to be bimodal, while a set with more than two modes may be described as multimodal.

Median can have only one value, obtained by arranging values in order and then picking out the central value. In a sample with an even number of elements, median is obtained by taking the average of the middle 2 elements.

Mean can only have a single value. The harmonic mean is the reciprocal of the arithmetic mean of the reciprocals of the data values. The geometric mean is obtained by taking the arithmetic mean of values on a logarithmic scale.

Solution to Question 5:

Blood pressure usually forms a normal symmetrical distribution for which mean is the preferred measure of central tendency.

Distributions of survival time, incubation period and health expenses have the probability of having extreme values, i.e., outliers. Thus, they form skewed distributions and in such cases, median is the preferred measure of central tendency.

Solution to Question 6:

The arithmetic mean from the following data is given by the sum of all the values, 270 divided by the number of values, $9 = 30$.

The data can be arranged in ascending order as 20, 25, 28, 31, 31, 31, 31, 35, 38.

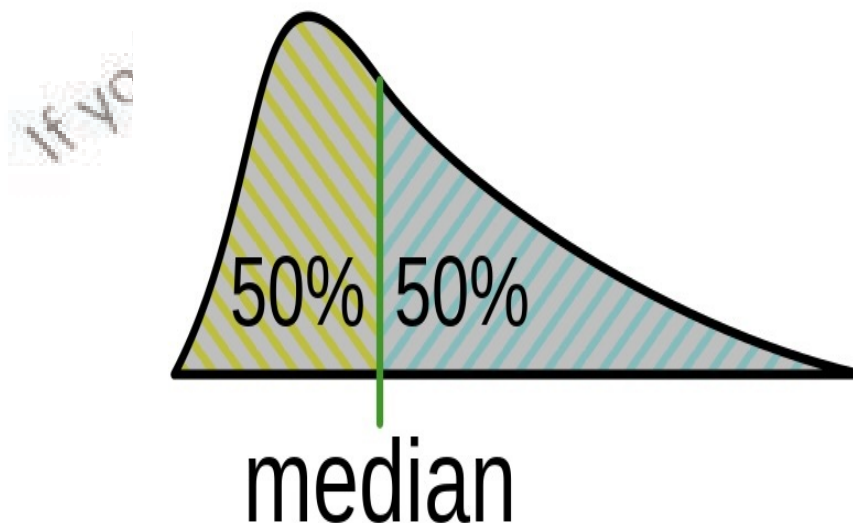
After arranging in order, the median is obtained by picking out the central value which is 31.

The mode is the most frequently occurring value, which is 31 here.

Solution to Question 7:

Out of the 11 birth weights, 2.5 kg would represent the middle value or the median. 5 babies weigh more than 2.5 kg and 5 less than that.

This can be understood from the image below.



Option A: Mean or arithmetic mean is the average value. In this case, the mean is the average of all the 11 values. It need not be 2.5.

Option B: Geometric mean is obtained by taking the arithmetic mean of values on a logarithmic scale.

Option D: Quartile is a measure of position, where the data is divided into four equal sets. $\frac{3}{4}$ of the values are below the 3rd quartile reading. Hence, it cannot be the central reading.

Solution to Question 8:

In this scenario, median is the preferred measure of central tendency as there are outliers.

The value of 5000 is an outlier in the given set, i.e., an observed value which is very distant from other observations in the distribution. This indicates the usage of median as the measure of central tendency as it is not sensitive to extreme values, unlike the mean.

By arranging the data in order: 100, 160, 190, 250, 260, 300, 320, 320, 350, 5000, we get the median as $(260 + 300) / 2 = 280$.

Options A and B: Mean is the most affected by the presence of outliers. Mode is the least affected by outliers but it is used mainly for measuring the central tendency of nominal data.

Option D: Geometric mean can be used for datasets with extreme values but it is not used when the values can be zero or negative. In the given scenario, the number of malaria cases reported in a town can also have a zero value.

Solution to Question 9:

The true mean weight of the group is 20 kg.

Mean is given as $\bar{x} = \sum x / n$, where n is the number of elements and $\sum x$ denotes the sum of all the elements in the series.

In this scenario, the incorrect mean is 18.2, i.e.,

$$\sum x / 10 = 18.2, \text{ making } \sum x = 182.$$

But this is incorrect as one of the 10 weights was wrongly recorded as 2 instead of 20.

The corrected $\sum x = (\text{Incorrect } \sum x - \text{incorrect value}) + \text{correct value}$.

The incorrect sum of all elements = 182, incorrect value = 2 and the correct value = 20.

Thus, the correct sum of all the 10 weights would be,

$$\sum x = (182 - 2) + 20 = 200.$$

From this, the correct mean value of the 10 weights can be given by:

$$\sum x / 10 = 200 / 10 = 20.$$

Solution to Question 10:

For the given dataset, the modes are 70 and 76. The mode is the most frequently occurring value. It can have multiple values. In this scenario, the values 70 and 76 have been repeated twice.

Median of this distribution is the central value, $(76 + 80) / 2 = 78$.

Solution to Question 11:

The distribution in this scenario in terms of the number of cigarettes smoked is 5, 10, 10, 10, 20, 40.

The mean here is $\Sigma x/n = 95/6 = 15.833$ rounded off to 16.

The median here is the central value = average of 3rd and 4th values since the distribution has an even number of values. This comes to $(10 + 10) / 2 = 10$.

The mode here is 10 which has a frequency of 3 in the given dataset.

Solution to Question 12:

The given change in a dataset would cause a decrease in the mean but the median would remain unchanged.

The mean is sensitive to every value in the distribution. Since the correct value of 58 is much lesser than the wrongly recorded value of 85, the final mean will be decreased compared to the earlier recorded value.

However, as only the highest value was wrongly recorded, the median i.e., the central value, will remain the same.

Solution to Question 13:

The given image shows a negatively skewed or a left-skewed distribution curve. Here, mean < median < mode.

Mode is the most frequently occurring value in the distribution and is given by the highest point of the graph. Since the given graph has only one peak, it is unimodal.

A positively skewed distribution refers to a right-skewed distribution. In such cases, mean > median > mode.

Solution to Question 14:

The given image is of a right-skewed or positively skewed distribution. So, among the given options, the most likely values of mean = 3, median = 2, and mode = 1.

We can tell the direction of skew of a distribution by observing the direction of the tail. Here, the tail of the distribution is towards the positive direction and thus it is positively skewed or right-skewed. In a positively skewed distribution, mean > median > mode.

If the tail of the distribution is towards the negative direction, the distribution would be negatively skewed or left-skewed. In such distributions, mode > median > mean.

Solution to Question 15:

This is a unimodal negatively skewed distribution.

In this scenario, the mean of 130 mmHg is less than the median of 140 mmHg which is less than mode of 160mmHg. Since mean < median < mode, this is an asymmetrical negatively skewed or left-skewed distribution.

It is a unimodal distribution as there is only a single value for mode.

In a symmetrical distribution, the mean = median = mode.

In a positively skewed distribution, mean > median > mode.

Solution to Question 16:

As the mean $\neq$ median, this is an asymmetric distribution and so we can use the formula:

Mode = 3 median - 2 mean.

Using the given values, mode = $(3 \times 3) - (2 \times 2) = 9 - 4 = 5$.

A distribution is termed symmetric when the mean, median and mode coincide i.e., mean = median = mode.

Solution to Question 17:

On comparing the 2 graphs, we can say mean = median for both study A and B but the mode is different.

Study A is a unimodal symmetrical distribution, so mean = median = mode.

Study B is a bimodal distribution. Here, mean = median, but the curve has two highest points, therefore it has 2 modes.

Descriptive Statistics III - Measures of Dispersion

Question 1:

Which of the following is not a measure of dispersion?

- a) Standard deviation
- b) Range
- c) Variance
- d) Z scores

Question 2:

What is standard deviation a measure of?

- a) Central tendency
- b) Spread
- c) Position
- d) Range

Question 3:

Which of the following is the correct relationship between variance and standard deviation?

- a) Variance = square root of standard deviation
- b) Standard deviation = square root of variance
- c) Variance = 2 x standard deviation
- d) Variance = standard deviation / 2

Question 4:

You are conducting a study on chronic kidney disease in a population. Your sample size is 28. What formula for standard deviation will you use here?

- a) $\sqrt{\sum(x - \bar{x})/n}$
- b) $\sqrt{\sum(x - \bar{x})/n-1}$

- c) $\sqrt{\sum(x - x_{\text{mean}})^2/n}$
- d) $\sqrt{\sum(x - x_{\text{mean}})^2/n-1}$

Question 5:

Which among the following is true about standard deviation?

- a) One standard deviation covers 95% of the values
- b) Indicates distribution of variables
- c) It can be used only in normal distributions
- d) It is a measure of central tendency

Question 6:

You are using the Kuppuswamy scale to calculate the socio-economic status of 10 families in the railway staff quarters. You find that every family has a total monthly income of Rs. 73,355. What is the standard deviation of this sample?

- a) 73.355
- b) 0
- c) 1
- d) 733.55

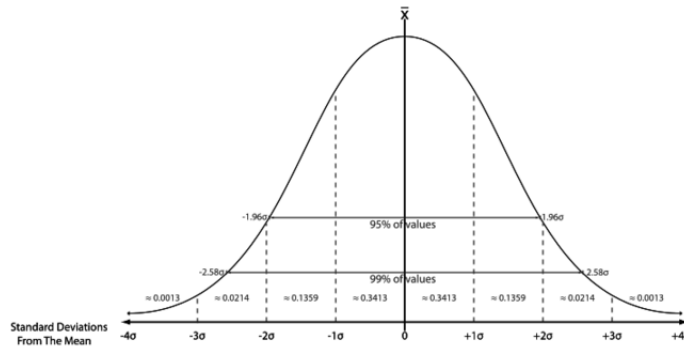
Question 7:

Each value of a study sample is multiplied by 10. What would be the new standard deviation of the observations?

- a) Original standard deviation x 10
- b) Original standard deviation x100
- c) Original standard deviation /10
- d) Original standard deviation itself

Question 8:

On plotting the blood pressure values of a population on a graph, you obtain the curve given below. Which of the following is true?



- a) 1, 2 & 5
- b) 1 & 3
- c) 2 & 4
- d) 1, 3 & 4

Question 9:

Which of the following is true about the standard normal distribution?

- a) 1, 2 and 3
- b) 1 and 3
- c) 2 and 5
- d) 2, 4 and 5

Question 10:

In a normal curve, what is the percentage of distribution of one standard deviation?

- a) 68%
- b) 34%
- c) 99%
- d) 95%

Question 11:

During a study on anaemia in a population, a normal distribution curve with a mean of 13.5 g% is obtained. What proportion of the population constitutes people with haemoglobin values above 13.5 g%?

- a) 0.25
- b) 0.50
- c) 1
- d) 0.34

Question 12:

The IQ scores of your study population form a bell-shaped curve. How much of the population falls between the median (Mdn) and $Mdn + 1 \text{ S.D}$?

- a) 0.34
- b) 0.68
- c) 0.17
- d) 0.95

Question 13:

The fasting blood glucose levels of adults in a population forms a normal distribution. The statistical measures obtained are given below. Within what limits would 95% of the population have their blood glucose values?

- a) 75 and 135 mg
- b) 85 and 125 mg
- c) 95 and 115 mg
- d) 80 and 120 mg

Question 14:

While conducting a study on malnutrition, you record the following data. If it forms a normal distribution curve, which of the following is true for this sample?

- a) Weight of 95 children is between 12 and 18 kg
- b) Weight of 95 children is between 10.5 and 19.5 kg

- c) Weight of 99 children is between 10 and 19 kg
- d) Weight of 34 children is between 13.5 and 16.5 kg

Question 15:

The distribution of intraocular pressure (IOP) in 100 glaucoma patients form a normal distribution curve. It has a mean of 30 mm of Hg with a standard deviation (SD) of 1 mm of Hg. 97.5% of patients would have an IOP more than what value?

- a) 28
- b) 26
- c) 32
- d) 29

Question 16:

A study is being conducted on anaemia in pregnancy. A normal distribution curve with a mean haemoglobin level of 10.6 gm/dL with a standard deviation of 2 gm/dL is obtained. Below what level would 5% of the women in the group have their haemoglobin values?

- a) 8.6 g/dL
- b) 7.31 g/dL
- c) 6.6 g/dL
- d) 5.0 g/dL

Question 17:

The peak expiratory flow rate (PEFR) values of a group of 11-year-old girls have a mean of 300 l/min and a standard deviation of 20 l/min. What are the deviation scores for the PEFR values of two girls recorded as 280 l/min and 330 l/min respectively?

- a) -20 and 30
- b) 20 and 30
- c) 20 and -30
- d) -20 and -30

Question 18:

The shape of a normal distribution curve depends on?

- a) Mean only
- b) Standard deviation and variance
- c) Mean and standard deviation
- d) Mean and median

Question 19:

Which of the following is a property of a normal distribution curve?

- a) Mean = 2 standard deviations
- b) Mean = median
- c) mean $>$ median
- d) mean $<$ median

Question 20:

Which of the following is incorrect about the standard normal distribution?

- a) Area under curve = 1
- b) Standard deviation = 1
- c) It is bell shaped
- d) Mean and median = 1

Question 21:

A population study showed a mean glucose of 86 mg/dL. In a sample of 100 showing normal distribution, what percentage of people will have blood glucose above 86 mg/dL?

- a) 34
- b) 50
- c) 95
- d) 68

Question 22:

The mean value of marks in a distribution of 100 students in a class is 105 and standard deviation is 10. How many students will have their marks between 85 and 125?

- a) 68%
- b) 50%
- c) 99.7%
- d) 95%

Question 23:

The systolic blood pressure of a group of normal people between the ages of 25–27 years was taken and a mean of 120 mmHg was found. What will be the expected number of individuals among a group of 100 people taken for the study whose BP will be below 120 mmHg?

- a) 25
- b) 50
- c) 99.7
- d) 95

Question 24:

In a normal distribution, what percentage of the distribution falls within 2 standard deviations of the mean?

- a) 68%
- b) 95%
- c) 99%
- d) 47.5%

Question 25:

In a population study, the value of the mean is 200. If the standard deviation is 20, 68% of the population will lie between?

- a) 180- 220
- b) 160- 240
- c) 170- 230
- d) 190- 210

Answer Key

Question No.	Correct Option
1	d
2	b
3	b
4	d
5	b
6	b
7	a
8	d
9	d
10	a
11	b
12	a
13	b
14	a
15	a
16	b
17	a
18	c
19	b
20	d
21	b
22	d
23	b
24	b
25	a

Detailed Explanations

Solution to Question 1:

Z scores are not a measure of dispersion, they are measures of position. It indicates the position of any element in a distribution in terms of how many standard deviations it lies from the mean of the distribution.

The most common measures of spread/dispersion are range, standard deviation and variance. They help in determining the extent to which data is stretched/scattered or squeezed/clustered.

They are also called measures of spread or scatter or variability.

Solution to Question 2:

Standard deviation is a measure of spread or dispersion or variability.

Range and variability are other measures of spread.

Solution to Question 3:

Standard deviation is the square root of variance.

Solution to Question 4:

In this scenario, the sample size is 28, which is ≤ 30 . For a small sample, usually less than 30, the formula for standard deviation used is:

$$\sqrt{\sum \frac{(x - \bar{x})^2}{n-1}}$$

It is better to use $n-1$ in the denominator in the case of a small sample. But in the case of a very large sample subtracting 1 from a huge number will not make much of a significant correction. In those cases, we may choose to ignore subtracting 1.

Solution to Question 5:

The standard deviation (S.D) indicates the distribution of variables. It is a measure of dispersion.

It can be used in normal and skewed distributions, but it is particularly useful in normal distributions.

1 S.D covers 68% of the values. 2 S.D covers 95% of the values.

Measures of central tendency are mean, median and mode.

Solution to Question 6:

In this scenario, the standard deviation will be zero.

The monthly income of every family is Rs. 73,355. So, all the elements in this distribution are the same. Hence, there is no variability. Both the standard deviation and variance which is the square of standard deviation will be zero.

Solution to Question 7:

When each element of a sample is multiplied by 10, the new standard deviation = 10 x original standard deviation.

The formula for standard deviation, $SD = \sqrt{\sum(x - \bar{x})^2/n-1}$.

If each element of the distribution is multiplied by 10, the new mean will become 10 times the original mean.

If all observations are multiplied by 10, then the new $SD = \sqrt{\sum(10x - 10\bar{x})^2/n-1}$

$$= 10 \sqrt{\sum(x - \bar{x})^2/n-1}$$

$$= 10 \times \text{Original SD}$$

Change in variance: variance = square of standard deviation, so when SD becomes 10 times, the variance will be $10^2 = 100$ times the original variance.

Solution to Question 8:

The given image shows a normal distribution, also called bell curve or Gaussian distribution. It has no skew. It is defined by the mean and standard deviation. In a normal distribution, mean = median = mode.

The total area under the curve of a normal distribution is 1.

Solution to Question 9:

A standard normal distribution is a type of normal distribution. It is bell-shaped, has a mean of 0 and a standard deviation of 1. Thus the variance, the square of standard deviation is also 1 = the standard deviation.

Solution to Question 10:

Approximately 68% of a normal distribution falls within ± 1 standard deviations of the mean. This is as per the empirical rule of a normal distribution.

Approximately 95% of the distribution falls within ± 2 standard deviations of the mean.

Approximately 99.7% of the distribution falls within ± 3 standard deviations of the mean.

Solution to Question 11:

In this scenario, 0.50, i.e., 50% of the population would have haemoglobin more than the mean, i.e., 13.5 gm%. In a normal distribution, 50% of the values lie above the mean.

Solution to Question 12:

34% or 0.34 of the population in a normal distribution falls between the median and median plus one standard deviation.

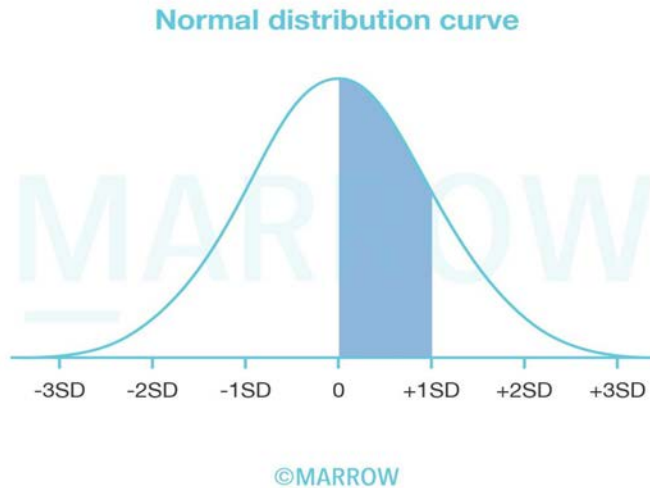
The IQ values form a bell-shaped curve, which is seen in normal symmetric distributions.

According to the empirical rule of normal distributions, 68% of the distribution falls within 1 standard deviation of the mean, i.e., $\text{mean} \pm 1 \text{ SD}$ covers 68% of the distribution.

Now, in a normal distribution, $\text{mean} = \text{median}$. Therefore, this 68% includes values from median -1SD to median +1SD.

The question asks about the percentage that falls under median +1SD, which would be half of 68%, that is 34% or 0.34.

Below is the graphical representation of the question. The question is asking to calculate the shaded region in the graph.



Solution to Question 13:

In this scenario, 95% of the population would have their blood sugar within the limits of 85 and 125 mg. Here, the mean blood sugar is 105 mg and the standard deviation is 10 mg.

According to the empirical rule of normal distribution, 95% of the distribution falls within two standard deviations of the mean.

In this question, $2 \text{ S.D} = 10 \times 2 = 20$. We do, + and - 2 S.D from the mean to arrive at the limits within which 95% of the population would have their blood sugar, i.e., 85 and 125 mg.

+ and - 1 S.D = 10 from the mean, we get the limits within which 68% of the population would have their blood sugar value, i.e., 95 and 115 mg.

+ and - 3 S.D = 30, from the mean, we get limits within which 99% would have their values, i.e., 75 and 135 mg.

Solution to Question 14:

In this scenario, the weight of 95 children will be between 12 and 18 kg as the mean is 15 kg and the standard deviation is 1.5 kg.

According to the empirical rule of normal distribution, 95% of the distribution falls within two standard deviations of the mean. Two standard deviations = $2 \times 1.5 = 3 \text{ kg}$

The weight of 99.7% of children will be between 10.5-19.5 Kg, i.e., within three standard deviations.

The weight of 68% of children will be between 13.5 to 16.5 Kg, i.e., within one standard deviation. Half will weigh between 13.5 and 15kg and the other half between 15 and 16.5kg

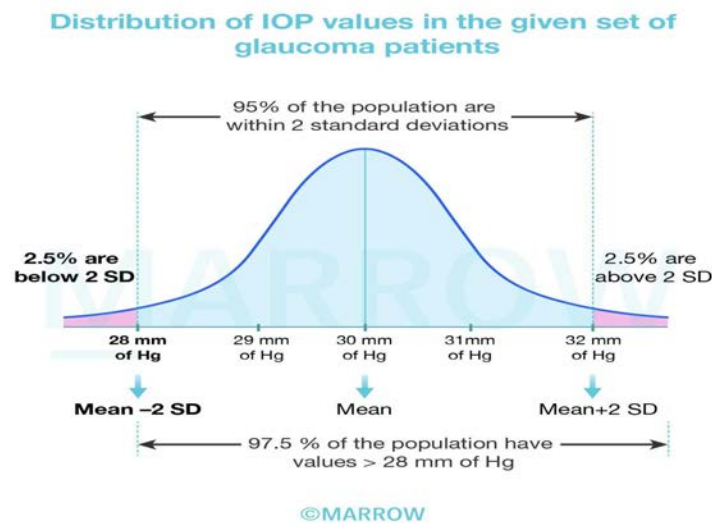
Solution to Question 15:

In this scenario, 97.5% of patients would have an IOP of more than 28 mm of Hg.

The mean IOP is 30mm of Hg and SD is 1.0 mm of Hg. According to the empirical rule of the normal distribution, 95% of the distribution falls within two standard deviations of the mean, i.e., $\text{mean} \pm 2 \text{ SD}$. So 95% of patients will have their IOP between $30 \pm 2 = 28$ to 32 mm of Hg.

Out of the remaining 5%, half will have IOP < 28 and the other half will have > 32 . So half of 5% = 2.5% has IOP < 28 mm Hg. Thus the remaining 97.5% has IOP > 28 mm of Hg.

This is shown in the image given below



Solution to Question 16:

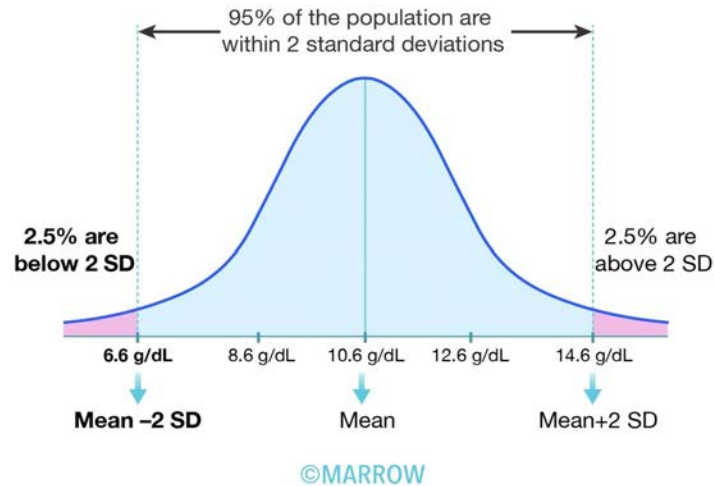
5% of the pregnant women in this group will have their haemoglobin levels below 7.31 g/dL.

Solution:

Given Data :

- Mean: 10.6 g/dL
- Standard deviation : 2 g/dL

Step 1:



According to the empirical rule of a normal distribution, 95% of the population fall within 2 standard deviations of the mean, i.e., $\text{mean} \pm 2\text{SD}$.

This implies that the remaining 5% lie outside 2 SD, i.e. 2.5% below 2 SD and 2.5% above 2 SD.

Therefore, $\text{Mean} - 2\text{SD} = 10.6 - 2(2)$

$$= 10.6 - 4$$

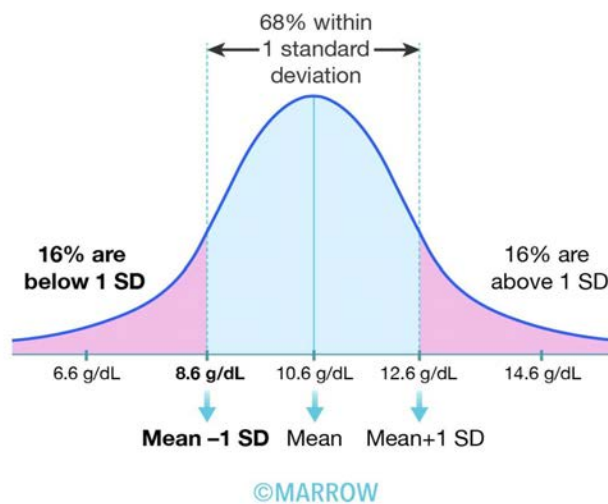
$$= 6.6 \text{ g/dL}$$

Thus, 2.5% of pregnant women have Hb levels below 6.6 g/dL.

However, the question asks about 5% and not 2.5%. So options C and D have been ruled out.

Now Let's move to the next step.

Step 2:



According to the empirical rule of a normal distribution, 68% of the population fall within 1 standard deviation of the mean, i.e., $\text{mean} \pm 1\text{SD}$.

This implies that the remaining 32% lie outside 1 SD, i.e., 16% below 1 SD and 16% above 1 SD.

The value of $\text{mean} - 1\text{SD}$ is 8.6 g/dL (i.e. $10.6 - 2$)

Thus, 16% of pregnant women have Hb levels below 8.6 g/dL.

so now option A is ruled out

Step 3:

The value for 5% must lie between the values where 2.5% and 16% of the population lie, i.e., between 6.6g/dL and 8g/dL. Only option B, 7.31 g/dL, satisfies this condition.

Hence, 5% of pregnant women have their Hb levels below 7.31 g/dL.

Solution to Question 17:

The deviation scores for PEFR values 280 l/min and 330 l/min are -20 and 30 respectively.

The Deviation score of an element in a distribution is the difference between the mean of the distribution and the element itself, i.e., it is obtained by subtracting the distribution's mean from the element.

Applying this here, deviation score of the value 280 l/min = the element - the mean = $280 - 300 = -20$.

For the value 330 l/min, deviation score = $330 - 300 = 30$.

Solution to Question 18:

The shape of a normal distribution curve depends upon its mean and standard deviation.

For example, for a particular mean value, a smaller standard deviation value gives a taller and narrower curve whereas, a larger value gives a flatter curve.

The curve is centered at the mean μ which coincides with the median and the mode and is located at the point beneath the peak of the curve. The mean, median and mode are equal.

Solution to Question 19:

In a normal distribution curve, mean = median.

mean $>$ median in positively/ rightward skewed distribution.

mean $<$ median in negatively/ leftward skewed distribution.

Solution to Question 20:

In a standard normal distribution, mean = 0.

Solution to Question 21:

In a normal distribution, 50% of the values lie above the mean.

In this scenario, 50% of the people will have a blood glucose level of more than 86 mg/dL.

Solution to Question 22:

According to the empirical rule of normal distribution, 95% of the distribution falls within two standard deviations of the mean, i.e., mean $\pm$ 2 SD.

In this scenario, the marks between 85 and 125 fall within 2 standard deviation of the mean which is 105. Thus 95% of the students will have the marks in this range.

Solution to Question 23:

In a normal distribution, 50% of the values lie below the mean.

In this scenario, 50%, i.e., 50 of the 100 people will have a systolic BP of less than 120 mmHg.

Solution to Question 24:

According to the empirical rule of normal distributions, 95% of the distribution falls within 2 standard deviations of the mean, i.e., mean $\pm$ 2 SD covers 95% of the distribution.

Solution to Question 25:

68% of the population will lie between 180- 220.

In a normal distribution curve, 68% of the values lie within the mean $\pm$ 1 standard deviation and the mean=median=mode. Since the mean = 200 and the standard deviation = 20:

68% of population will lie between Mean $\pm$ 1SD, that is, $200 \pm 20 = 180-220$

95% of population lies between Mean $\pm$ 2SD, $200 \pm 2*20 = 160- 240$

99% of population lies between Mean $\pm$ 3SD, $200 \pm 3*20 = 140- 260$

Descriptive Statistics IV - Measures of Position

Question 1:

Which of the following is not a measure of position?

- a) Z scores
- b) Range
- c) Centiles
- d) Quantiles

Question 2:

Which of the following statements about quantiles are true?

- a) 1, 4 and 5
- b) 1, 3 and 5
- c) 2, 3 and 4
- d) 1, 2, 4 and 5

Question 3:

Centiles divide data into _____ equal parts?

- a) 100
- b) 4
- c) 99
- d) 10

Question 4:

During a school health camp, a group of 250 students are arranged in the ascending order of their height. Which student would have the 40th percentile value for height?

- a) 80th
- b) 40th

- c) 100th
- d) 140th

Question 5:

By calculating which of the following can we obtain the central value of a sample with 180 elements?

- a) 90th percentile
- b) 2nd decile
- c) 9th decile
- d) 2nd quartile

Question 6:

Which measure of position divides the distribution in a ratio of 3:1?

- a) 33rd centile
- b) First quartile
- c) 7th decile
- d) Third quartile

Question 7:

You are recording the blood pressure values of 200 adults in a population. The first quartile diastolic BP is 68 mmHg and the third quartile is 105 mmHg. How many patients are there in the 3rd and 4th quarters?

- a) 50
- b) 100
- c) 75
- d) 150

Question 8:

The forced expiratory volume (FEV) of 300 patients visiting the respiratory medicine outpatient department is recorded. The data has a median value of 2.5 litres with the first and

third quartiles being 1.5 litres and 4.5 litres respectively. How many persons in the sample are expected to have an FEV between 1.5 and 4.5 litres?

- a) 75
- b) 150
- c) 225
- d) 200

Question 9:

Which of the following distributions is preferred for the calculation of z scores?

- a) Normal distribution
- b) Right skewed distribution
- c) Left skewed distribution
- d) Bimodal distribution

Question 10:

In a study on haemoglobin levels in healthy women, the data recorded has a mean of 13.5 gm/dL and a standard deviation of 1.5 gm/dL. What is the z score for an element with a haemoglobin value of 15.0 gm/dL?

- a) 1.5
- b) 10.0
- c) 2.0
- d) 1.0

Question 11:

The diastolic BP of 125 women aged between 18 and 60 years was recorded. The mean of the data is 70 mmHg with a standard deviation of 10 mmHg. What will be the 5th percentile value?

- a) 40
- b) 53.6
- c) 67.5
- d) 89.4

Answer Key

Question No.	Correct Option
1	b
2	b
3	a
4	c
5	d
6	d
7	b
8	b
9	a
10	d
11	b

Detailed Explanations

Solution to Question 1:

Range is not a measure of position, it is a measure of dispersion or spread.

The most common measures of position are quantiles, quartiles, centiles and z-scores (standard scores). Measures of position give the position of specific data in relation to other data in the set.

Solution to Question 2:

Quantiles are elements in a distribution that divide data into equal groups. They are also known as fractiles. Median divides a distribution into two equal parts, so it is considered as a quantile.

Quantiles are used for continuous, quantitative data.

Quintiles are a type of quantile that divides data into 5 equal groups.

Solution to Question 3:

Centiles divide data into 100 equal parts.

Solution to Question 4:

Here, the 100th student will have the 40th percentile value for height.

To arrive at this, we use the formula $R = P/100 \times (n + 1)$.

where R is the rank of the desired observation, P is the quantile/percentile desired and n is the number of variables in the data.

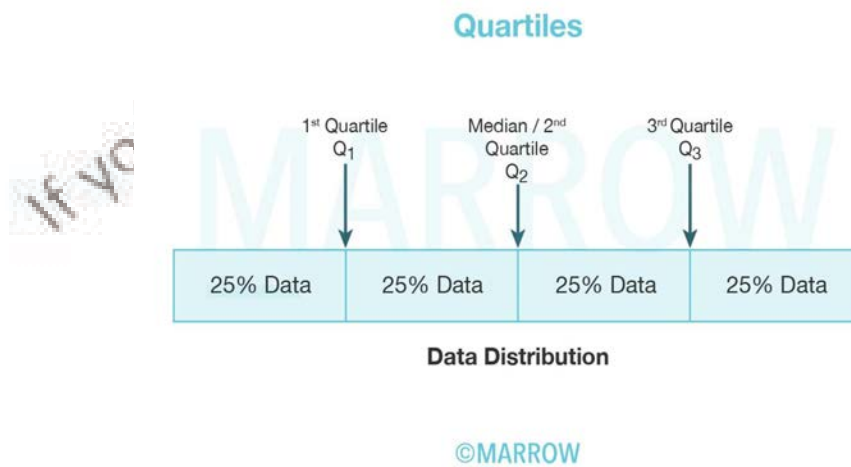
Thus, $R = 40/100 \times (250 + 1) = 0.4 \times 251 = 100.4$.

This is rounded up to the closest number making $R = 100$.

Solution to Question 5:

The 2nd quartile gives the central value of 180 elements. The central value of a distribution is given by the median. It divides the distribution into the ratio of 1:1.

In quartiles elements of the distribution are divided into 4 equal parts by 3 values. Each part contains a quarter, i.e., 25% of the values. So two quarters would give half of the values. So the element present at the position of the 2nd quartile would be the central value, i.e., the median, with half the values before and half the values after it. This is shown in the image given below.



The 5th decile and 50th percentile represent the median values of a distribution in decile and percentile respectively.

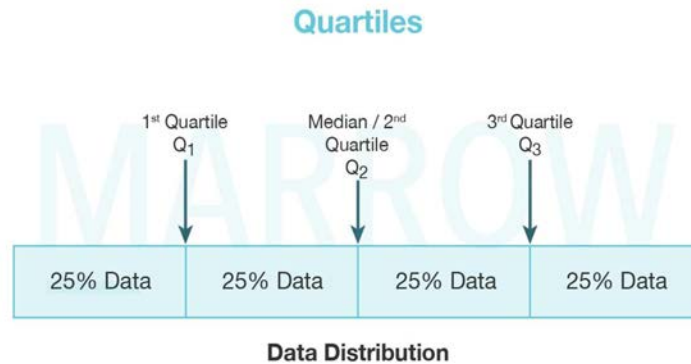
Solution to Question 6:

Third quartile Q₃ divides the distribution into the ratio of 3:1.

In quartiles:

- the first quartile Q_1 divides the distribution in the ratio of 1:3
- the second quartile Q_2 divides it in the ratio of 1:1.

The division of data by quartiles is shown in the below image.



Centiles divide data into 100 equal parts. 33rd centile divides the distribution in the ratio of 33:67.

Deciles divide data into 10 equal parts. 7th decile divides the distribution in the ratio of 7:3.

Solution to Question 7:

In a quartile, the data is divided into 4 equal parts, i.e., 4 quarters. Therefore, in this scenario, there are 50% of patients in the 3rd and 4th quarters, i.e 100 patients.

The division of data by quartiles is shown in the image given below.

Quartiles



Data Distribution

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Solution to Question 8:

150 persons are expected to have FEV between 1.5 and 4.5 litres.

In a quartile, the data is divided by 3 values into 4 equal parts. Thus, each part will have 25% of the values in the distribution.

In this scenario, each part will have 75 persons. Here, 1.5 is the first quartile Q₁, 2.5 is the second quartile Q₂ and 4.5 is the third quartile Q₃. Between Q₁ and Q₃ there are two quarters, thus there will be 150 people with FEV between 1.5 and 4.5 litres.

Data division by quartiles is shown in the below image.

Quartiles



Data Distribution

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Solution to Question 9:

Z scores are usually calculated in normal distributions.

Solution to Question 10:

The Z score for the element with a haemoglobin value of 15 gm/dL is 1.0.

The formula for z score, $z = (X - \mu) / \sigma$.

X is the value of the element, μ is the mean and σ is the standard deviation.

In this scenario, $z = (15 - 13.5) / 1.5 = 1.5 / 1.5 = 1$.

Solution to Question 11:

The z value for the 5th percentile is -1.645.

Using the formula for z scores, $z = (X - \mu) / \sigma$, we get $X = z\sigma + \mu$.

In this scenario, $X = (-1.645 \times 10) + 70 = -16.45 + 70 = 53.55 \approx 53.6$.

Inferential Statistics

Question 1:

The statistical methods that go further than merely summarizing data are a part of:

- a) Descriptive statistics
- b) Inferential statistics
- c) Observational statistics
- d) Analytical statistics

Question 2:

Which of the following measures relative variation between two different units?

- a) Variance
- b) Coefficient of variation
- c) Standard deviation
- d) Standard error

Question 3:

The weight of the 100 children that presented in the pediatric OPD in the last week was analyzed. The median weight was 12 kg and the standard deviation was 3. This data formed a normal distribution. What is the coefficient of variation?

- a) 25%
- b) 35%
- c) 40%
- d) 45%

Question 4:

The means of all possible samples from a population are plotted on a linear graph. What is this called?

- a) Population Distribution

- b) Parametric Distribution
- c) Sampling distribution
- d) Exponential distribution

Question 5:

Central limit theorem is related to:

- a) Random sampling distribution
- b) Skewness of a distribution
- c) Graphical representation of nominal data
- d) Statistical tests between two groups

Question 6:

A researcher wanted to calculate the mean age of onset of type II diabetes mellitus in a population. He got a sample mean of 45. To measure the extent of deviation from the population mean, which of the following should he use?

- a) Geometric mean
- b) Range
- c) Deviation scores
- d) Standard error

Question 7:

Standard deviation of mean measures variations caused by:

- a) Non-sampling errors
- b) Sampling errors
- c) Random errors
- d) Observational errors

Question 8:

Consider a sample of 100 women with an average Hb of 10 g% and standard deviation of 1. What is the standard error of this sample?

- a) 0.01
- b) 0.1
- c) 1
- d) 10

Question 9:

All of the following are true about standard error except:

- a) It increases as the sample size increases.
- b) It is based on a normal distribution
- c) It depends on the standard deviation of means
- d) It is used to estimate confidence limits

Question 10:

The incubation period for measles in a group of 25 children was calculated. Following findings were recorded. What is the standard error?

- a) 0.4
- b) 1.0
- c) 2.0
- d) 0.5

Question 11:

The uppermost and lowermost values of standard errors within which a parameter value is expected to lie are called the:

- a) Confidence interval
- b) Precision levels
- c) Confidence limits
- d) Range

Question 12:

Which of the following parameters are used to calculate the confidence limit?

- a) Mean and standard error
- b) Mean and standard deviation
- c) Median and standard deviation
- d) Median and standard error

Question 13:

During screening of a population, the mean cholesterol levels were found to be 230mg/dL. The standard error was 10. Calculate the 95% confidence limits.

- a) 210–250
- b) 250–290
- c) 290–330
- d) 190–210

Answer Key

Question No.	Correct Option
1	b
2	b
3	a
4	c
5	a
6	d
7	b
8	b
9	a
10	a
11	c
12	a
13	a

Detailed Explanations

Solution to Question 1:

Inferential statistics are those statistical methods that go further than just summarizing data.

They involve inductive reasoning and using a statistic to estimate a parameter. E.g., the expected success rate of vasectomy or mean blood pressure of all Indians.

One of the most common problems faced in research is using information gathered by researching a sample and then drawing conclusions about the population from which the sample is drawn.

Here, sample statistics (sample mean and SD) are known and population parameters (population mean and SD) are unknown.

Inferential statistical methods are used to draw conclusions about the population parameters from sample statistics.

Solution to Question 2:

Co-efficient of variation (CV) measures the relative variability between two different units, by using the mean and standard deviation.

Solution to Question 3:

In the given scenario, coefficient of variance (CV) = 25%.

Coefficient of variation, $CV = \text{standard deviation} \times 100 / \text{mean} = 3 \times 100 / 12 = 25\%$.

In a normal distribution, mean = median.

Solution to Question 4:

The random sampling distribution of means is the distribution formed when all possible sample means from a population are plotted in a frequency distribution.

Solution to Question 5:

The central limit theorem is related to the random sampling distribution of means.

According to this theorem, as the sample size increases:

- The sampling distribution tends towards a normal distribution curve (bell curve)
- Regardless of the population's actual distribution shape
- Assuming all samples are identical in size
- The sample mean and standard deviations become closer to the population mean and standard deviations

Solution to Question 6:

The standard error is a measure of the extent to which the sample means deviate from the true population means.

Solution to Question 7:

The standard deviation of mean or standard error of means measures variations caused by sampling errors.

Solution to Question 8:

In this scenario, the standard error $SE = 0.1$.

The formula for standard error:

$$SE = \text{Standard deviation} / \sqrt{\text{sample size}} = 1 / \sqrt{100} = 1/10 = 0.1$$

Solution to Question 9:

As the sample size increases, the standard error will decrease, not increase.

The standard error is inversely related to the square root of the sample size.

Solution to Question 10:

In this scenario, the standard error $= 0.4$.

The formula for standard error:

$$SE = \text{Standard deviation} / \sqrt{\text{sample size}} = 2 / \sqrt{25} = 2/5 = 0.4$$

Solution to Question 11:

The uppermost and lowermost limits of standard errors within which a parameter value is expected to lie are called the confidence limits.

The difference between upper and lower confidence limits is called the confidence interval.

Solution to Question 12:

Confidence limits are calculated with the sample mean and standard error.

Solution to Question 13:

In the given scenario, 95% confidence limits are 210 – 250.

In inferential statistics, 95% confidence limits = Sample mean ($\bar{x}$) $\pm$ 2 standard errors.

In this case, mean is 230 and the standard error is 10. Hence, confidence limits are 210-250.

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Concepts in Hypothesis Testing

Question 1:

What is the hypothesis which states that an apparent effect is only due to chance, called as?

- a) Experimental hypothesis
- b) Null hypothesis
- c) Inferential hypothesis
- d) Alternative hypothesis

Question 2:

What is the significance level in hypothesis testing called as?

- a) Probability level
- b) Alpha level
- c) Beta level
- d) Rejection level

Question 3:

What is the p value?

- a) Probability of declaring a significant difference when there is none in reality
- b) Probability of declaring a significant difference when it is actually present
- c) Probability of not declaring a significant difference when actually it is not present
- d) Probability of not declaring a significant difference when actually it is present

Question 4:

Which of the following statements is false about the p-value?

- a) It is equal to the probability of committing a type I error
- b) It is equal to $1 - \beta$
- c) It is the chance that the presence of difference is concluded when actually there may be none

- d) When the p value is less than α , the result is statistically significant

Question 5:

A randomized control trial comparing the efficacy of two chemotherapy regimens for Hodgkin's lymphoma showed that their difference is statistically significant with $p < 0.001$. However, in reality the two drugs do not differ in their efficacy. What type of error is this an example of?

- a) Type I error
- b) Type II error
- c) Alpha II error
- d) Beta I error

Question 6:

An investigator is conducting a study to assess the effect of a mood stabiliser on patients with bipolar I disorder. What does it mean if the difference is significant in the results?

- a) $p > 0.05$ and it is likely by chance
- b) $p > 0.05$ and it is unlikely by chance
- c) $p < 0.05$ and it is unlikely by chance
- d) $p < 0.05$ and it is likely by chance

Question 7:

While assessing the effect of a new intervention for type II DM, you obtain the p-value of the study as 0.023. Based on this observation, what is your inference?

- a) Null hypothesis is accepted and the study is rejected
- b) Null hypothesis is rejected and the study is accepted
- c) Null hypothesis is accepted and the study is accepted
- d) Null hypothesis is rejected and the study is also rejected

Question 8:

Which of the following is a false statement regarding hypothesis testing?

- a) Alpha is the maximum tolerable probability of a type 1 error being made
- b) Beta is the probability of a type II error
- c) When the null hypothesis is true but rejected, it is a type II error
- d) p value can be more or less than alpha

Question 9:

You are conducting a randomised clinical trial to see if a new drug 'A' has better efficacy than the standard drug 'B', in patients with grade III hypertension. While analysing the results of this study, you conclude that no difference exists between the two. But in reality, a difference between them truly exists. What type of error are you making?

- a) Type I error
- b) Type II error
- c) Sampling error
- d) Interpretation error

Question 10:

Power of a study is the ability of a test to _____

- a) Decrease type I error
- b) Decrease beta error
- c) Increase alpha error
- d) Increase beta error

Question 11:

A randomised control trial is being conducted in patients with direct inguinal hernia, to compare the outcomes of a new surgery 'A' and the gold standard surgery 'B'. The p value of this trial is found to be 0.04. What can we conclude from this?

- a) Type II error is small and we can accept the findings of the study
- b) The probability of false negative conclusion that operation A is better than B is 4%
- c) The power of the study to detect the difference between operation A and B is 96%
- d) The probability of a false positive conclusion that operation A is better than B is 4%

Question 12:

You are an investigator conducting a statistical test. You obtain the p value as 0.01 for your study. What does this mean?

- a) The probability of finding a significant difference is 1%
- b) There is a 1% probability that the difference is not by chance
- c) The difference is not significant 1% of times and significant 99% of times
- d) The power of the test used is 99%

Question 13:

In a well-designed cancer study, similar remission rates were obtained for a new treatment (33.2%) and the usual treatment (32.2%), with the p-value being 0.04. Which of the following statements is true regarding the study?

- a) Both the treatments are equally effective
- b) Neither of the treatments is effective
- c) The new treatment is more effective than the usual treatment
- d) The information given is not adequate to compare the efficacy of the treatments

Answer Key

Question No.	Correct Option
1	b
2	b
3	a
4	b
5	a
6	c
7	b
8	c
9	b
10	b
11	d
12	c

13	c
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Detailed Explanations

Solution to Question 1:

The hypothesis that an apparent effect is only due to chance is a null hypothesis. If this is rejected as false, we have an alternative hypothesis.

Solution to Question 2:

Significance level in hypothesis testing is called alpha level (α) or criterion.

Solution to Question 3:

p-value can be defined as the probability of declaring a significant difference when there is none in reality.

Solution to Question 4:

The power of a test is $1 - \beta$.

The p value is the probability that a type I error is being made. It is the chance that the presence of difference is concluded when actually there may be none. When the p value is less than α , the result is statistically significant.

Solution to Question 5:

This is an example of a type I error.

In this scenario, let us take the null hypothesis as the efficacy of two drugs being the same, which as the question states, is true in reality. The alternative hypothesis will then be that one drug is better than the other.

The questions states that p-value was found to be < 0.001 , which is statistically significant and so the null hypothesis is rejected. But we already know that the null hypothesis is true and we are forced to accept the alternative hypothesis even though it is false.

When an alternative hypothesis is accepted even when it is false and the null hypothesis is true, it is called as a type I error or an alpha error or a false positive error.

Solution to Question 6:

When the difference is significant, it means that the difference is unlikely by chance and p will be ≤ 0.05 .

Solution to Question 7:

By convention, if the p value ≤ 0.05 , the difference is said to be statistically significant. Thus, the null hypothesis is rejected and the study is accepted.

Solution to Question 8:

When the null hypothesis is true but is still rejected, it is a type I error and not a type II error. Type II errors are when an alternative hypothesis is rejected even though it is true.

Solution to Question 9:

Type II errors are when an alternative hypothesis is rejected when it is true.

In this scenario, the null hypothesis is that there is no difference and the alternative hypothesis is that there is a difference.

We failed to reject the null hypothesis that says there is no difference, and so we are making a type II error. In other words, we have rejected the alternative hypothesis that there is a difference when it is true.

Solution to Question 10:

The power of a study is its ability to avoid or decrease beta errors or type II errors.

The statistical power of a test is represented by $1 - \beta$.

Solution to Question 11:

Since the p -value of the study is found to be 0.04, we can infer that the probability of a false positive conclusion that the operation A is better than B is 4%.

In this scenario, the p value is 0.04 which makes it, by convention, statistically significant. We can reject the null hypothesis and accept the alternative hypothesis that operation A is better than

operation B.

Solution to Question 12:

If the p-value = 0.01, then the difference is not significant 1% of the time and significant 99% of the time.

In other words, there is a 99% probability that the difference is not by chance and there is a 1% probability that the difference is only by chance.

Solution to Question 13:

As the p-value is 0.04, we can conclude that the new treatment is more effective than the usual treatment.

Even though both treatments have similar rates of remission, a valid statistical test is required to determine the difference in their effectiveness. Statistically, p-value < 0.05 implies that the alternate hypothesis is true.

In this scenario, we can consider:

- Null hypothesis: Both the treatments are equally effective (in other terms, there is no difference between the effect of the two treatments).
- Alternative hypothesis: New treatment is more effective than the usual treatment.

As the p-value is 0.04, which is statistically significant (< 0.05), the null hypothesis is rejected and the alternative hypothesis is accepted. Hence the new treatment is more effective than the usual treatment.

Tests of Significance

Question 1:

Which of these tests is used in case of a nominal data?

- a) Student's t test
- b) Analysis of variance
- c) Chi-square
- d) Paired t test

Question 2:

Which of the following is true for non-parametric tests?

- a) ANOVA is an example of a non parametric test
- b) It is used for skewed distributions
- c) It involves the assumption that the data has a normal distribution
- d) It cannot be used for small sample sizes

Question 3:

Which of these terms is used for paired t tests?

- a) Independent samples t tests
- b) Two samples t tests
- c) Dependent samples t tests
- d) One sample t tests

Question 4:

The test of significance done for two or more proportions:

- a) Chi- square test
- b) Student's test
- c) Z test

d) ANOVA test

Question 5:

A researcher can use the chi-square test in all of the following situations, except:

- a) To test the hypothesis that drug A has the same potency as drug B**
- b) To calculate the degree of freedom of a 3 x 4 contingency table**
- c) To compare the mean corpuscular volume in normal subjects versus those suffering from hereditary spherocytosis**
- d) To compare the proportions of overweight boys and girls in a class**

Question 6:

Which of the following is not true about the Chi-square test?

- a) Tests the significance of difference between two proportions**
- b) Tells about the presence or absence of an association between two variables**
- c) Directly measures the strength of association**
- d) It is usually used for skewed data**

Question 7:

A Chi-square test would be most appropriate for testing which of the following hypotheses?

- a) The mean AIPGE score of Chennai students is greater than Mumbai students**
- b) A smaller proportion of people who were immunized against chicken pox subsequently develop zoster than those who were not immunized**
- c) The mean BP of indian, american and english hypertensive men taking ACE inhibitors is the same as that of indian, american and english hypertensive women on ACEIs**
- d) The mean cost of treating a patient with CAD by angioplasty is greater than the mean cost of providing medical treatment**

Question 8:

The test of choice to measure the difference in means between two groups in a study, with one group being a control is:

- a) Chi-square test
- b) Z test
- c) Unpaired T test
- d) Paired T test

Question 9:

A cardiologist wants to study the effect of the atorvastatin drug. She measures the initial cholesterol levels of 50 patients. After a month's treatment, she measures the cholesterol level again. Which of the following tests will she use to test the statistical significance of the change in the cholesterol level?

- a) Paired t test
- b) Unpaired t test
- c) ANOVA
- d) Chi-square test

Question 10:

In a particular trial, the association of lung cancer with smoking is found to be 40% in one sample and 60% in another. What is the best test to compare the results?

- a) Chi-square test
- b) Fischer test
- c) Paired t test
- d) Unpaired test

Question 11:

A pediatrician carried out a study to assess the difference in growth velocities in boys and girls. The average height of a group of 20 boys aged 10 years was 140 ± 13 cm and for 20 girls of the same age was 135 ± 7 cm. To test the statistical significance of the difference in height, which test would she use?

- a) Chi square test
- b) Paired t test
- c) Student t test
- d) Analysis of variance test

Question 12:

A clinical trial was done between two groups to measure their blood pressure after treatment. Which of the following tests would be appropriate to use?

- a) Paired t test
- b) Unpaired t test
- c) ANOVA
- d) Chi-square test

Question 13:

Analysis of variance test is used in which scenario among the following?

- a) Comparing proportions in more than 2 groups
- b) Comparing means in more than 2 groups
- c) Comparing means in same groups before and after intervention
- d) Comparing means in less than or equal to 2 groups

Question 14:

A study was done to assess nutritional status among young children in rural and urban areas. 30 children in rural and 20 children in urban areas were malnourished. Which of the following tests of significance is the most appropriate in this case?

- a) Chi Square test
- b) Paired T test
- c) Standard error of the mean
- d) ANOVA

Question 15:

A study was conducted to compare the incidence of postpartum depression in mothers as compared to the sex of the child. For assessment, Edinburg Depression scaled score (EDSS) was used. The study was divided into mothers with male babies as one group to be compared with mothers with female babies. Which of the following statistical test will be used to determine the association?

- a) Students t test
- b) Paired t test
- c) ANOVA
- d) Chi square test

Question 16:

Which of the following is the most appropriate test for testing the statistical significance of the difference in heights of school children?

- a) Student's t test
- b) Chi-square test
- c) Paired t test
- d) ANOVA

Question 17:

An investigator wants to study the association between maternal intake of iron supplements and the incidence of low birth weight. He collects relevant data from 100 pregnant women regarding the status of usage of iron supplements and the status of birth weight in their newborns. Which of the following tests will he use for his study?

- a) Paired t test
- b) Student's t test
- c) ANOVA
- d) Chi-square test

Question 18:

In a study, two groups of newborns are checked for differences in their birth weights based on whether mothers received supplements or not. Which of the following would be most appropriate test of significance?

- a) ANOVA
- b) Un-paired t test
- c) Chi-square test
- d) paired t test

Question 19:

A study was planned to find out the effect of iron supplementation during pregnancy on the birth weight of newborn children. Two groups, one with iron supplementation and the other without, were compared. Birth weights of newborns were recorded as means + SD. Which statistical test is appropriate for the comparison of birth weights between the two groups?

- a) Paired t test
- b) Unpaired t test
- c) Chi-square test
- d) ANOVA

Question 20:

While applying the Chi-square test to a contingency table of 4 rows and 4 columns, what would the degree of freedom be in this case?

- a) 1
- b) 4
- c) 8
- d) 9

Question 21:

In a Chi-square test for the following data, the degrees of freedom will be:

Progression to AIDS	A	B	AB	O
0 - 5 years	20	30	28	18
5 - 10 years	11	12	16	12
10 - 15 years	12	9	8	10

- a) 12
- b) 6
- c) 9
- d) 20

Answer Key

Question No.	Correct Option
1	c
2	b
3	c
4	a
5	c
6	c
7	b
8	c
9	a
10	a
11	c
12	b
13	b
14	a
15	d
16	d
17	d
18	b
19	b
20	d
21	b

Detailed Explanations

Solution to Question 1:

Chi-square test deals with nominal/ non-parametric data (qualitative).

Nominal data is one with a name and does not have a numerical value. Ex blood group, gender, religion

Student's t test, ANOVA, and paired t test are used for parametric data (quantitative).

Solution to Question 2:

Non-parametric tests are tests that do not assume that the data is normally distributed. Thus, they are used for skewed (asymmetric) distributions {skewed either to the lesser side (left-sided skewed) or towards the greater side (right-sided skewed)}.

The chi-square test is an example of a non-parametric test that is used to test hypotheses in relation to nominal scale data or categorical data.

ANOVA is an example of a parametric test.

Solution to Question 3:

Paired t tests are also called correlated pairs t test, paired samples t test, or dependent samples t test.

Solution to Question 4:

Chi-Square(χ^2) test is a test used for proportions and associations.

It is a non-parametric (non-numeric) test and it is used to test hypotheses in relation to nominal scale data or categorical data.

Solution to Question 5:

The Chi-square test is not used for comparing means for different groups as means can only be calculated for quantitative data.

The Chi-square test is used for qualitative data. It can be used:

- To test the null hypothesis
- To compare proportions
- To calculate the degree of freedom

Solution to Question 6:

A Chi-square test does not directly measure the strength of association. It merely tells about the presence or lack of association.

Solution to Question 7:

The Chi-square test is used to compare proportions or qualitative variables e.g., whether people developed zoster or not.

Options A and D: Two independent groups are being compared with quantitative variables (exam scores and cost, respectively). Thus, an unpaired t test will be used.

Option C: The blood pressure of 3 groups are being compared, thus making analysis of variance (ANOVA) the most appropriate method of testing significance.

Solution to Question 8:

The test of choice to measure the difference in means between two groups in a study, with one group being a control is unpaired T test.

Solution to Question 9:

In this scenario, the variable (cholesterol level) is quantitative. Moreover, the comparison is being done before and after the treatment (dependent). Hence, a paired t test will be used.

Solution to Question 10:

Chi-square tests are best used in comparing proportions and associations.

Solution to Question 11:

In the given scenario, two independent groups are being compared with quantitative variables. Thus, a student t test or unpaired t-test is used.

Solution to Question 12:

In the given scenario, two groups are being compared independent of each other. Hence, students t test/ unpaired t test should be used.

T tests are used if 2 groups are being compared. If the same group is compared before and after, paired T test is used. Otherwise, unpaired T test is used.

Solution to Question 13:

Analysis of variance or ANOVA is used when more than 2 groups are compared for a quantitative difference (means).

Solution to Question 14:

The most appropriate test of significance, in this case, is the chi-square test. It is used to compare qualitative data between 2 groups with independent measures.

Option B: Paired t-test compares paired data or pre-intervention and post-intervention data of the same group. This doesn't apply here as the groups are not paired.

Option C: The standard error of the mean is calculated for quantitative data, with a measure of central tendency as mean.

Option D: ANOVA is used to compare more than 2 groups.

Solution to Question 15:

The Chi-square test is used to compare qualitative data and compares 2 groups with independent measures. The variables used here are yes/no signifying the presence/absence of postpartum depression. Hence it is the most appropriate test here.

Option A: Student's t-test is for quantitative data. Here the data is qualitative, hence it can't be used.

Option B: Paired t-test compares paired data or pre-intervention and post-intervention data of the same group. This doesn't apply here as the groups are not paired.

Option C: ANOVA is used to compare more than 2 groups.

Solution to Question 16:

Analysis of variance or ANOVA is used when more than 2 variable groups are compared.

In the scenario of comparing heights of school children, we would divide the school children into classes or grades or sections or by schools themselves, etc. This would involve comparing more than 2 groups and thus ANOVA is used.

Remember if the question limits its comparison within 2 groups, for example, comparison between boys and girls in a class then we have to choose unpaired t test.

Solution to Question 17:

Here, two qualitative variables are being evaluated. First, whether the child belongs to the low birth weight category or not (and not the actual weight of the child, which is a quantitative value). Second, whether or not the mother took iron supplements in the gestational period. Hence, the chi-square test is used as the test the significance of the association.

Solution to Question 18:

The unpaired t-test is used to test the difference between two groups of parametric variables or means when the samples are independent of each other. Here, birth weights (quantitative variable) of the two groups of newborns are being compared, which are independent of each other.

Solution to Question 19:

In the given scenario, we aim to compare only the birth weights of the two groups, which is a quantitative data, and thus an unpaired t test is most appropriate.

Solution to Question 20:

In this scenario, $df = (4 - 1) (4 - 1) = 9$.

In a Chi-square test for a contingency table, the degrees of freedom $(df) = (r-1) (c-1)$, where r is the number of rows and c is the number of columns.

Solution to Question 21:

In this scenario, contingency table has 3 rows and 4 columns. Hence, the degree of freedom, $df = (3 - 1) (4 - 1) = 6$.

In a Chi-square test for a contingency table, the degrees of freedom $(df) = (r-1) (c-1)$, where r is the number of rows and c is the number of columns.

Correlational and Predictive Techniques

Question 1:

A study is conducted to find the relationship between the height and weight of school students. Which of the following would be useful in this study?

- a) Coefficient of variation
- b) Coefficient of correlation
- c) Coefficient of regression
- d) Coefficient of mean deviation

Question 2:

Which among the following is false about correlation?

- a) It indicates the strength of relationship between 2 variables
- b) Correlation coefficient of 1 indicates a linear relationship
- c) Correlation cannot measure risk
- d) Correlation implies a causal relationship

Question 3:

During a study conducted to assess the effect of smoking on life expectancy, the correlation coefficient was found to be 0.60. Calculate the coefficient of determination for this study.

- a) 0.6
- b) 0.36
- c) 0.4
- d) 1.2

Question 4:

What is the best way to study relationship between two variables?

- a) Forest plot
- b) Scattergram

- c) Funnel plot
- d) Frequency polygon

Question 5:

A psychiatry resident is conducting a study to assess the relationship between prefrontal blood flow under cognitive load and the severity of psychotic symptoms in schizophrenic patients. He calculates the correlation coefficient as -1.24 . You can therefore conclude that:

- a) Prefrontal blood flow is a good predictor of the severity of psychotic symptoms
- b) Prefrontal blood flow accounts for most of the variance in symptoms
- c) Schizophrenia causes low prefrontal blood flow
- d) The correlation coefficient has been calculated incorrectly

Question 6:

In a random sample of 5000 corporate workers, the correlation coefficient of work satisfaction and life expectancy was found to be $+0.7$ with a p-value of 0.01 . What is your interpretation?

- a) Work satisfaction does not improve life expectancy
- b) Strong statistically significant positive association between the variables
- c) 70% people who enjoy work will live longer
- d) 30% people who do not enjoy work will have a shorter life expectancy

Question 7:

A group of cardiologists found a correlation coefficient of 0.90 (p-value = 0.01) between the systolic BP and serum cholesterol levels of patients attending their clinics. Which of the following statements is an incorrect interpretation of this study?

- a) The magnitude of systolic BP and serum cholesterol levels are closely related
- b) A patient with a high systolic BP is likely to have high serum cholesterol
- c) A patient with a low systolic BP is likely to have low serum cholesterol
- d) 80% of the variation in BP can be explained by serum cholesterol and vice versa

Question 8:

Among the given options, what can be the likely value of the coefficient of correlation between infant mortality rate and economic status?

- a) +1
- b) -1
- c) +0.22
- d) -0.8

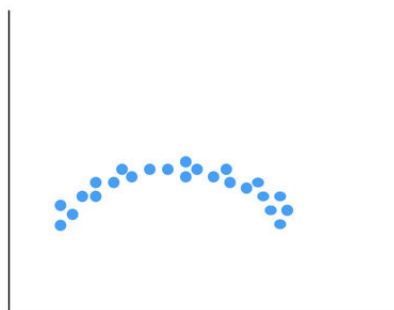
Question 9:

High salt intake in individuals is associated with high blood pressure. What type of correlation is this?

- a) Negative correlation
- b) Positive correlation
- c) Non-linear correlation
- d) Zero correlation

Question 10:

A scattergram was plotted as shown below to study the relationship between two variables. What is the correct interpretation?

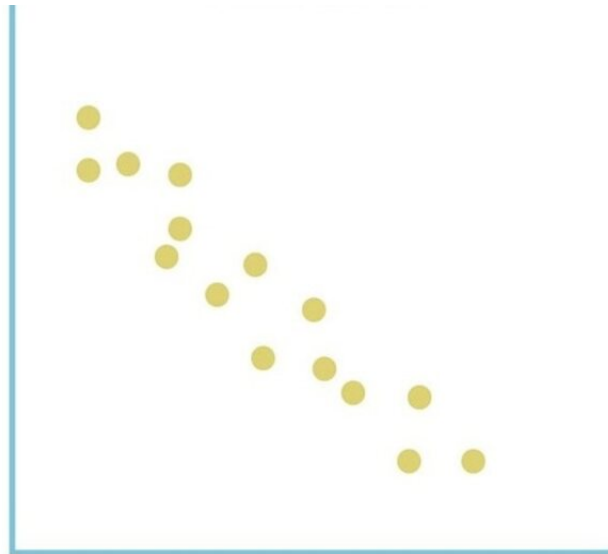


- a) There is correlation between both variables and the Pearson's coefficient is 1
- b) There is correlation between both variables and the Pearson's coefficient is -1

- c) There is non-linear correlation between both variables
- d) There is no association between both variables

Question 11:

What conclusion can be drawn from the following scattergram?



- a) There is a perfect negative correlation between the two variables
- b) There is a high positive correlation between the two variables
- c) There is a high negative correlation between the two variables
- d) There is a perfect positive correlation between the two variables

Question 12:

The value of one parameter can be estimated from another related parameter by:

- a) Standard deviation
- b) Correlation
- c) Scattergram
- d) Regression

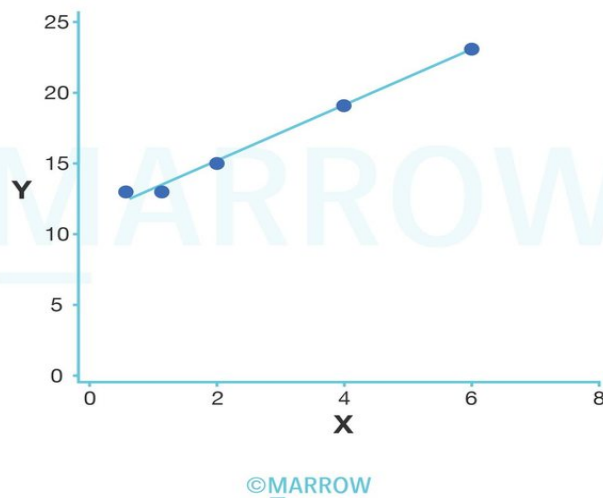
Question 13:

If correlation is given by the equation $y = a + bx$, what would be the nature of the graph?

- a) Straight line
- b) Parabola
- c) Hyperbola
- d) Sigmoid curve

Question 14:

Look at the image below and select the linear regression equation corresponding to the graph.



- a) $y=10-x$
- b) $x=10+y$
- c) $y=11+2x$
- d) $x=11+2y$

Question 15:

In a study, the total cholesterol was calculated as $a + b_1x_1 + b_2x_2 + b_3x_3$, where x_1 indicates calorie intake, x_2 indicates physical activity, and x_3 indicates body mass index. What is this an example of?

- a) Simple linear regression
- b) Simple curvilinear regression
- c) Multiple linear regression
- d) Logistic regression

Question 16:

An epidemiologist has determined five independent factors influencing the outcome of a particular disease. Which of the following statistical methods can he use to compare these factors?

- a) ANOVA
- b) Multiple linear regression
- c) T-test
- d) Logistic regression

Question 17:

What test is used to compare Kaplan-Meier survival curves?

- a) ANOVA
- b) Bland-Altman analysis
- c) Chi-square test
- d) Cox proportional hazards test

Question 18:

Which of the following is not used to compare survival curves?

- a) Mantel-Haenszel test
- b) Log-rank test
- c) Cox regression
- d) Fisher's exact test

Question 19:

What is the Spearman coefficient used to determine?

- a) Regression
- b) Correlation
- c) Comparison of variation

d) Standard deviation comparison

Question 20:

A researcher wants to compare a newly developed test for COVID with the gold standard test. Which of the following methods can he use?

- a) Regression analysis
- b) Bland-Altman Test
- c) Cox regression
- d) ANOVA

Answer Key

Question No.	Correct Option
1	b
2	d
3	b
4	b
5	d
6	b
7	a
8	d
9	b
10	c
11	c
12	d
13	a
14	c
15	c
16	d
17	d
18	d
19	b
20	b

Detailed Explanations

Solution to Question 1:

The coefficient of correlation can be used to determine the relationship between both variables in this scenario. It is used to express the strength and direction of the relationship between two variables.

It is represented by the letter 'r'. Its value ranges from -1 to +1.

Solution to Question 2:

A correlation between 2 variables only indicates the strength of the relationship between two variables, not causation.

A correlation coefficient of 1 indicates a linear relationship. It cannot measure the risk.

Solution to Question 3:

The coefficient of determination (CD) for this study is 0.36 (36%).

It is calculated by r^2 , where r is the correlation coefficient. In this study, the correlation coefficient (r) is 0.6.

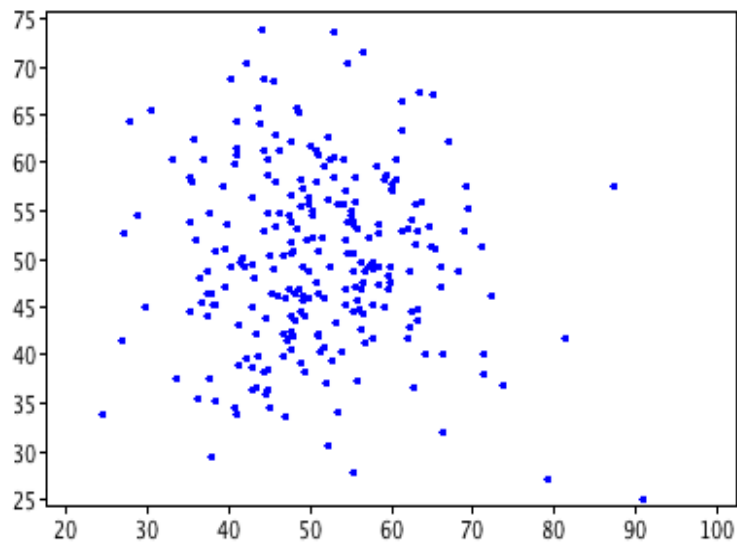
Hence $CD = (0.6)^2 = 0.36$

The coefficient of determination refers to the percentage variability in one variable due to a change in the other variable. In the above study, a coefficient of determination of 0.36 means that 36% of the variation in life expectancy is accounted for by smoking.

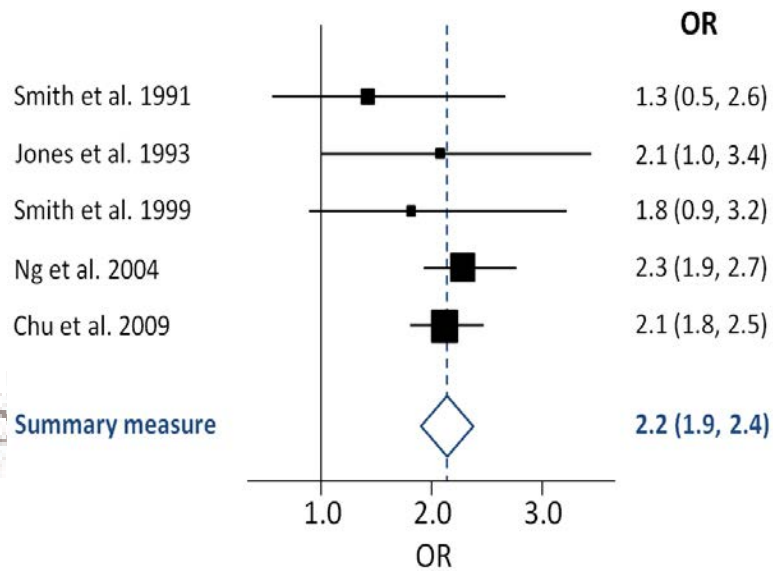
Solution to Question 4:

A scattergram/ scatter plot is the best way to study the relationship between 2 variables.

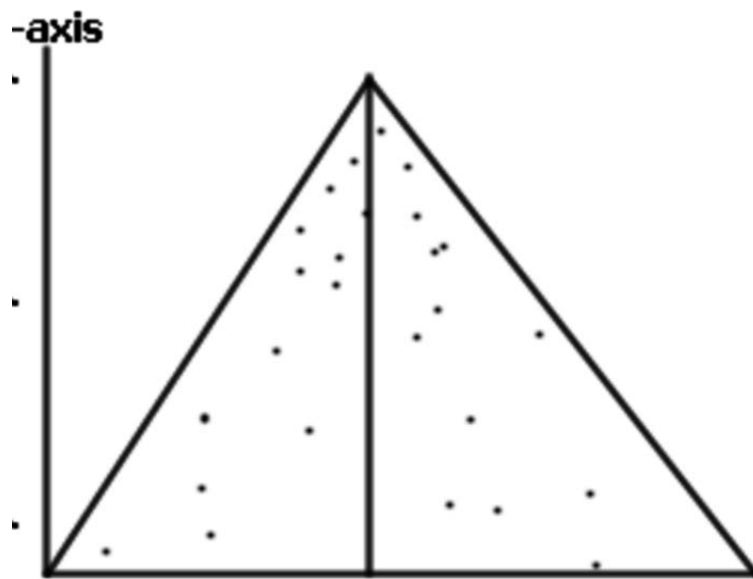
The following image shows a scatter plot.



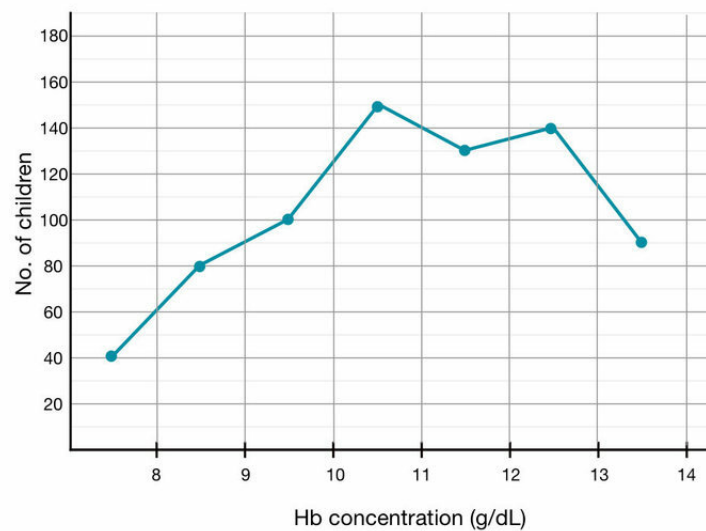
The following image shows a forest plot.



The following image shows a funnel plot.



The following image shows a frequency polygon



Solution to Question 5:

In this scenario, the correlation coefficient (r) of -1.24 is incorrect, as ' r ' can only lie between -1 to +1.

Solution to Question 6:

The above study is statistically significant (p -value = 0.01) and indicates a strong positive correlation (correlation coefficient = +0.7) between work satisfaction and life expectancy.

Correlation is a measure of the statistical association of the variables, not a causal relationship. Here, the coefficient of determination (r^2) = $0.7 \times 0.7 = 0.49$. This means that 49% of the variation in life expectancy can be explained by satisfaction at work.

Solution to Question 7:

Correlation is an indicator of the strength and direction of the relationship between 2 variables, not the magnitude.

The given study is statistically significant (p -value = 0.01) with a strong positive correlation ($r = +0.90$) between systolic BP and cholesterol levels. Thus, it is likely that patients with high systolic BP will have high cholesterol levels and those with low BP will have low cholesterol levels.

The coefficient of determination (r^2) here is $0.9 \times 0.9 = 0.81$. This indicates that around 80% of the variation in systolic BP among the patients can be explained by serum cholesterol levels.

Solution to Question 8:

Among the given options, the coefficient of correlation between infant mortality rate and economic status (r) is likely to be -0.8.

It is known that low economic status is associated with a high infant mortality rate. This indicates a negative correlation. However, the infant mortality rate is also influenced by other factors, indicating a moderate correlation (not equal to 1). This makes -0.8 the most likely value of correlation.

Solution to Question 9:

An increase in blood pressure with salt intake is an example of a positive correlation.

The association of decreased life expectancy with increased smoking is an example of negative correlation.

Solution to Question 10:

The given scattergram depicts a non-linear correlation between the two variables.

Correlation expresses the strength and direction of the relationship between two variables in terms of a correlation coefficient. There are two types of correlation coefficients:

Pearson coefficient	Spearman coefficient
Used for data with continuous intervals	Used for data with discrete or derived intervals

Pearson coefficient	Spearman coefficient
Used to find linear relationships (e.g. salt intake vs blood pressure)	Used to find linear or non-linear relationships (e.g. socioeconomic status vs parity)

Solution to Question 11:

The given scattergram shows a high negative correlation between the two variables.

Solution to Question 12:

The value of one parameter can be estimated from another related parameter by using regression techniques.

Solution to Question 13:

The equation, $y = a + bx$ indicates a straight line.

Solution to Question 14:

Among the given options, the equation $y = 11 + 2x$ corresponds to the linear regression equation of the form $y = \alpha + \beta x$.

When 2 variables are correlated, such that the dependent variable is represented on the y-axis and the independent variable on the x-axis, then the value of y can be obtained by using the linear regression equation $y = \alpha + \beta x$, where:

y - dependent variable

α - intercept

β - slope of the line

x - independent variable

In the equation $y = 11 + 2x$, the intercept (α) = 11, while the slope of the line (β) = 2.

Option A: $y = 10 - x$, represents an intercept of 10, and a slope of -1 indicating a negative correlation. However, the given image shows a positive correlation.

Options B and D: They do not fit the linear regression equation of the form $y = \alpha + \beta x$.

Solution to Question 15:

The given equation of $y = a + b_1x_1 + b_2x_2 + \dots + b_nx_n$ is an example of multiple linear regression.

In this equation:

$b_1, b_2, \dots, b_n$ represent different regression coefficients

$x_1, x_2, \dots, x_n$ represent the known variables

Solution to Question 16:

In this scenario, the outcome to be determined (dependent variable) is the occurrence of the disease, which is categorical (qualitative) in nature. Logistic regression can be used when a categorical dependent variable is being compared with either a categorical or numerical independent variable.

Linear regression is used when the outcome is numerical.

Dependent variable (outcome)			
Independent variable(risk factors, exposure)		Qualitative (categorical)	Quantitative (numerical)
Qualitative (categorical)	Logistic regression Chi-square	Linear regression T-test ANOVA	
Quantitative (numerical)	Logistic regression	Linear regression	

Solution to Question 17:

The Cox proportional hazards test is used to compare Kaplan-Meier survival curves.

Survival analysis is used for analyzing the expected duration of time until one event occurs, such as death. The following methods are used for this are:

- Life table analysis
- Kaplan-Meier analysis - it involves calculating the survival function (the probability that a person will survive past a certain time)

Life tables and survival functions are graphically represented in a Kaplan-Meier plot or curve.

Solution to Question 18:

Fisher's exact test is not used to compare survival curves. The following tests are used to compare the survival curves or Kaplan-Meier curves:

- Mantel-Haenszel test
- Log-rank test

- Cox regression or Cox proportional hazards analysis

Solution to Question 19:

Spearman coefficient is used to determine correlation. It is used to find linear or non-linear relationships for data with discrete ordered intervals.

Pearson correlation coefficient is used to find linear relationships for data with continuous intervals.

Solution to Question 20:

The Bland-Altman analysis is used for comparing a newly developed test with another test, which usually is the gold standard. This method involves the graphical construction of a Bland-Altman plot which is used to compare the measurements from both methods.

Facets of Clinical Research and Biostatistics

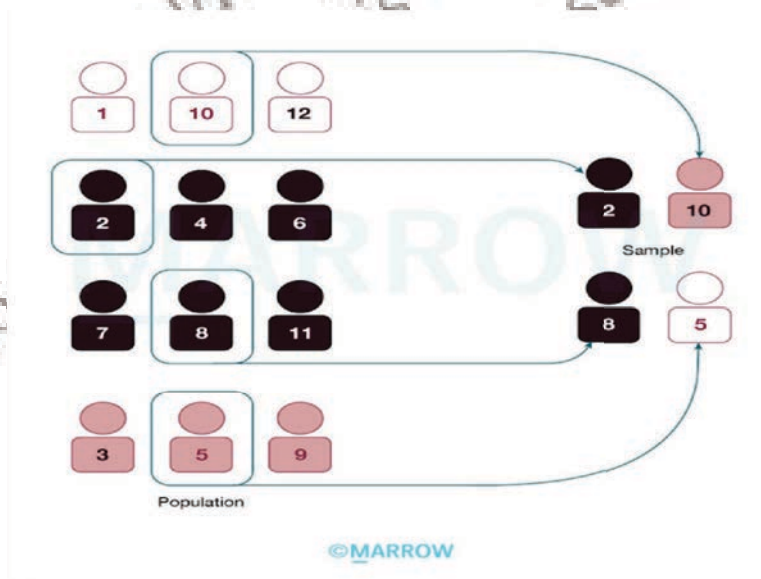
Question 1:

Which of the following is true about simple random sampling?

- a) It is the sampling method of choice for heterogenous population
- b) Members are selected without specifying any criteria for selection
- c) Every 5th or 10th member is picked at regular interval
- d) Population is first divided into strata then sampled randomly

Question 2:

A study on the prevalence of diabetes was conducted and the sampling method shown below was used. Identify the sampling method used.



- a) Simple random
- b) Cluster
- c) Systematic
- d) Stratified random

Question 3:

Which of the following is incorrect about cluster sampling?

- a) Clusters are created based on similarities
- b) It is a two stage sampling
- c) It is cheaper than most other methods
- d) It is a method used for rapid assessment

Question 4:

Which sampling method is used for assessing the immunization status of children under the immunization program?

- a) Systematic sampling
- b) Cluster sampling
- c) Group sampling
- d) Stratified random sampling

Question 5:

A research is being conducted to assess the nutritional status of the children in a particular district. For this purpose, 2 children are randomly selected from every school in the district. What is this type of sampling known as?

- a) Simple random
- b) Systematic random
- c) Stratified random
- d) Cluster

Question 6:

In which of the following sampling techniques is design effect used?

- a) Simple random
- b) Stratified random
- c) Cluster
- d) Systematic random

Question 7:

Simple random sampling is ideal for which of the following study groups?

- a) A study being conducted on all vaccinated people of an area
- b) A study on occupational exposure in all workers of the same factory
- c) A study being conducted on prevalence of acute gastroenteritis in all school children of an area
- d) A study on prevalence of hypertension in men aged 50-75 years in an area

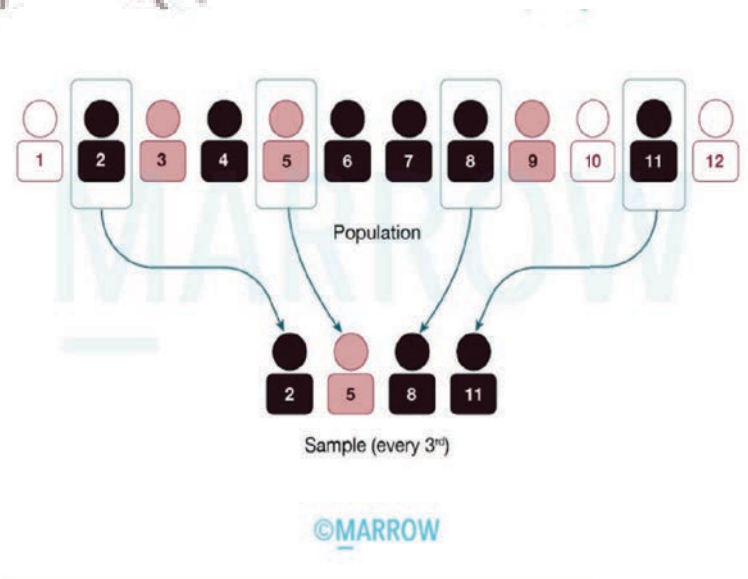
Question 8:

In a study, first schools were sampled, then sections and finally students. What is this type of sampling known as?

- a) Stratified sampling
- b) Snowball sampling
- c) Systematic sampling
- d) Multistage sampling

Question 9:

For an epidemiological study, sampling was done as shown below. What is this type of sampling known as?



- a) Cluster sampling

- b) Simple random sampling
- c) Stratified random sampling
- d) Systematic sampling

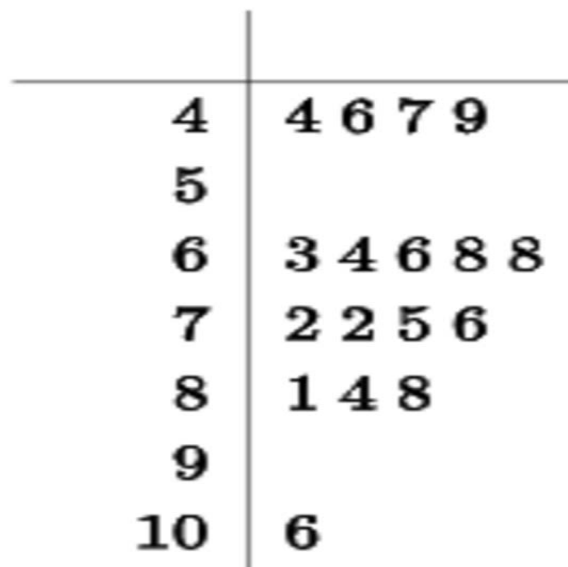
Question 10:

In a study, subjects were assigned according to the potential factor that will influence the outcome of the study. What is this type of sampling known as?

- a) Stratified
- b) Systematic
- c) Simple
- d) Cluster

Question 11:

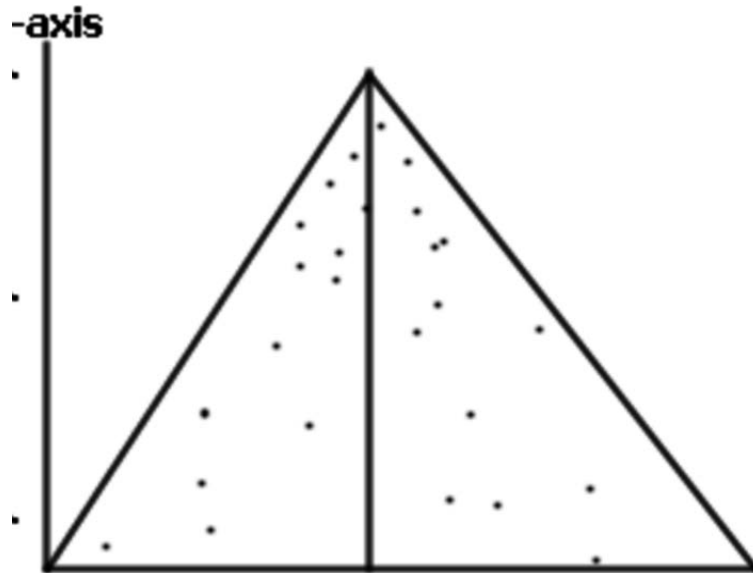
Identify the given statistical plot.



- a) Forest plot
- b) Funnel plot
- c) Stem-and-leaf plot
- d) Box and Whisker plot

Question 12:

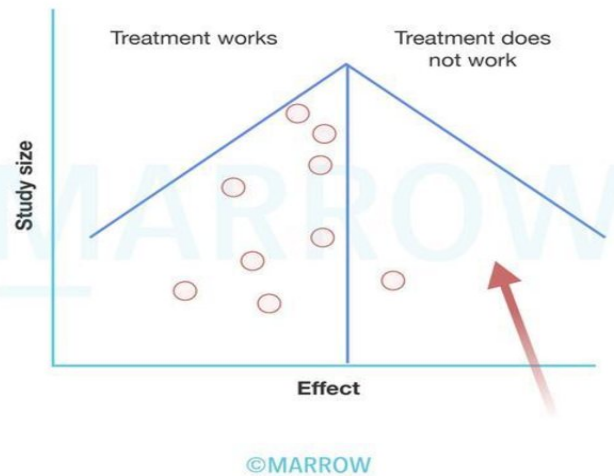
Identify the type of statistical plot shown below.



- a) Funnel plot
- b) Forest plot
- c) McNemar plot
- d) Spot map

Question 13:

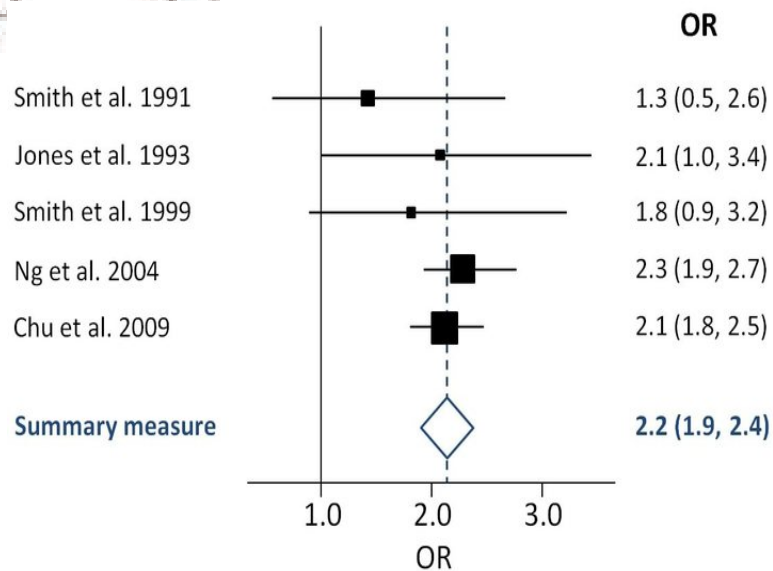
The plot from the data obtained from a study is shown below. Which of the following results in this type of plot?



- a) Sampling error
- b) Berksonian bias
- c) Large sample size of people
- d) Publication bias

Question 14:

Results of various studies carried out to study the role of smoking in breast cancer are plotted below. Identify the plot.

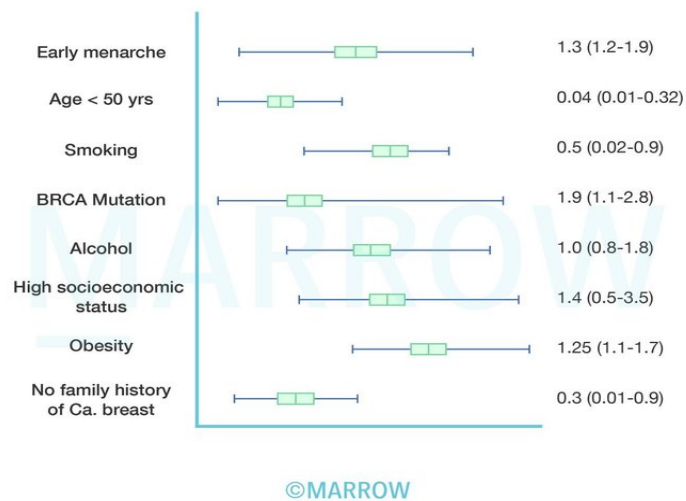


- a) McNemar plot

- b) Box-and-whisker plot
- c) Forest plot
- d) Fisher's exact plot

Question 15:

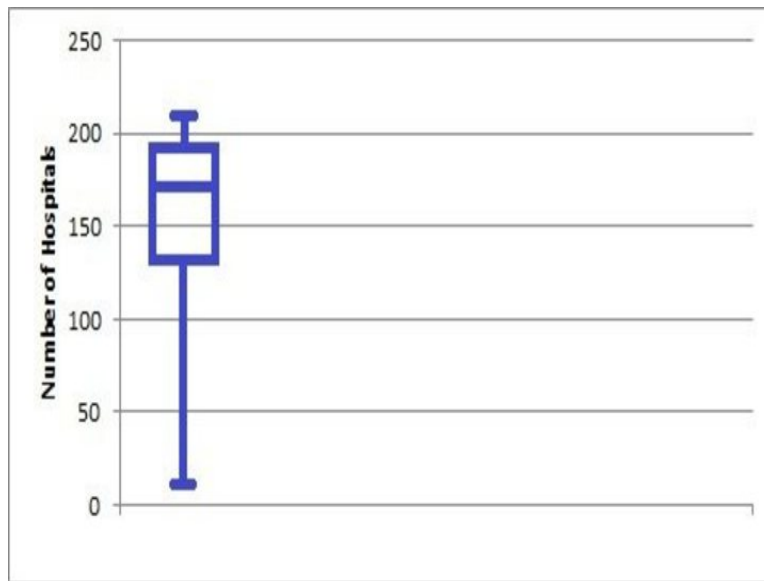
A study was done to assess the risk factors for breast cancer in 100 females from urban cities. A statistical group and analysis are shown below. The values indicate the odds ratio and the 95% confidence interval. How many risk factors may be ascertained from the statistical analysis of the research?



- a) 3
- b) 4
- c) 5
- d) 6

Question 16:

Identify the following statistical plot.



- a) Fisher's box plot
- b) Stem and leaf plot
- c) Box-and-whisker plot
- d) McNemar's plot

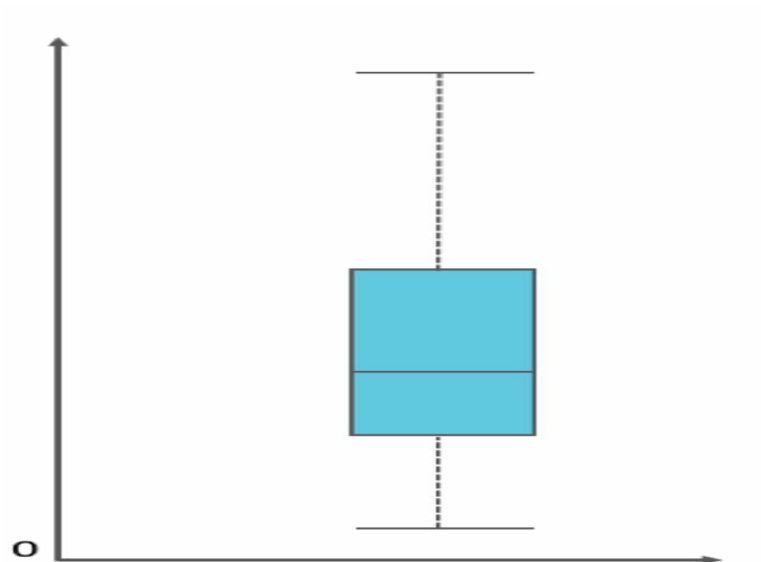
Question 17:

What is Dixon's Q test used for?

- a) Outliers
- b) Interquartile range
- c) Difference of proportions
- d) Medians

Question 18:

Regarding the given box-whisker plot, the true statement is:



- a) Mean > Median > Mode
- b) Mode > Median > Mean
- c) Mode = Mean = Median
- d) Peaked symmetrical distribution

Answer Key

Question No.	Correct Option
1	b
2	d
3	a
4	b
5	d
6	c
7	b
8	d
9	d
10	a
11	c
12	a
13	d
14	c

15	a
16	c
17	a
18	a

Detailed Explanations

Solution to Question 1:

In simple random sampling, samples are drawn in such a way that every element in a population has equal chances of being included. In other words, the members are selected without specifying any criteria for selection.

Solution to Question 2:

In this study, the population is separated into homogenous sub-groups and from these, random samples are drawn. This sampling method is called stratified random sampling.

Solution to Question 3:

In cluster sampling, the clusters are created randomly. On the contrary, in stratified random sampling, the strata or groups are created based on similarities.

Solution to Question 4:

Cluster sampling is the method of sampling done in immunization programs for assessing the immunization status of children.

Solution to Question 5:

In the cluster sampling technique, the sub-groups of the population/ clusters are naturally occurring. These clusters are randomly made rather than based on any particular characteristic. The population of children here is naturally divided into various schools without being based on any characteristics.

Solution to Question 6:

The design effect is used in cluster sampling to allow for design structure and correlation between clusters.

The design effect (DEFF) is used to decide the sample size needed for cluster sampling. This is done by calculating the difference in sampling variability to the variability expected with simple random sampling (SRS).

It can more simply be stated as the actual sample size divided by the effective sample size (what you would expect if you were using SRS). For example, let's say you were using cluster sampling. A DEFF of 2 means the variance is twice as large as you would expect with SRS. It also means that if you used cluster sampling, you'd have to use twice the sample size.

Solution to Question 7:

Simple random sampling is ideal for a homogeneous population as the sample would be completely representative of the population and it would eliminate the need for systematization or stratification. Here, workers of the same factory form a homogeneous population.

Option A: For a vaccinated population - cluster sampling

Options C and D: For a heterogeneous population - stratified sampling

Solution to Question 8:

Multistage sampling is the taking of samples in stages using smaller and smaller sampling units at each stage. It involves dividing the population into groups (or clusters). Then, one or more clusters are chosen at random and everyone within the chosen cluster is sampled.

Snowball sampling is a type of non-random sampling preferred in case of diseases with social stigma (eg., alcoholism, iv drug abuse etc).

Solution to Question 9:

Systematic sampling involves a systematic pattern to sampling, like choosing every 3rd person from an area or hospital, etc.

Solution to Question 10:

In stratified random sampling, the population is first divided into homogeneous groups or strata. From these strata, random samples are drawn.

For example: If a researcher wants to conduct a research study to determine if alcohol consumption is a risk factor for lung cancer. He must first stratify the population into smokers and non-smokers because smoking is a known risk factor and also smoking is associated with

alcohol consumption.

Solution to Question 11:

The given image shows a stem-and-leaf plot.

To construct the above stem-and leaf plot, the given data is first arranged in ascending order. E.g., 44, 46, 47, 49, 63, 64, 66, 68, 68, 72, 72, 75, 76, 81, 84, 88, 106

While constructing the plot, the leaf contains the last digit of the number and the stem contains all of the other digits. For example, for the number 106 the stem contains 10 and the leaf contains 6; similarly for the number 46 stem contains 4 and the leaf contains 6. Following this, the whole stem-and-leaf plot is given below.

Stem	Leaf
4	4 6 7 9
5	
6	3 4 6 8 8
7	2 2 5 6
8	1 4 8
9	
10	6

Solution to Question 12:

The image shown is of a symmetric funnel plot.

Solution to Question 13:

The image shown is that of an asymmetric funnel plot. This usually occurs in meta-analysis due to publication bias or other reporting bias.

Solution to Question 14:

The image shown is that of a forest plot used for plotting the results of a meta-analysis.

Solution to Question 15:

Among the factors shown in the above statistical group, the odds ratio of early menarche, BRCA mutation, and obesity show positive association and are statistically significant. Hence these 3 factors can be ascertained as risk factors.

High socioeconomic status is not a risk factor. Even though there is a positive association, it is not statistically significant. Any association that has a 95% confidence interval range that includes the value of 1.0 is considered to be non-significant. The value of 1.0 in the range means that the two factors may co-exist together but one is not causing the other to happen.

Odds ratio (OR) is the key parameter in case-control studies to assess the strength of association between the exposure and outcome. The interpretation of the odds ratio can be done as below:

- OR ≥ 1 indicates a positive association. If both upper and lower limits of the 95% confidence limits are also ≥ 1 , it suggests that the odds ratio is significant and the factor can be ascertained as a risk factor.
- OR = 1 indicates no association. Irrespective of the odds ratio, if the lower limit of 95% CI is ≤ 1 and the upper limit of 95% CI ≥ 1 , the OR is non-significant and cannot be ascertained either as a protective or a risk factor.
- OR ≤ 1 indicates a negative association. If both upper and lower limits of the 95% confidence limits are also ≤ 1 , it suggests that the odds ratio is significant and the factor can be ascertained as a protective factor.

Interpretation of the individual factors given in the question:

Factor	Odds ratio(95%confidence limits)	Interpretation
Early menarche	1.3(1.2-1.9)	Positive association, significant, risk factor
Age <50 years	0.04 (0.01-0.32)	Negative association, significant, protective factor
Smoking	0.5 (0.02-0.9)	Negative association, significant, protective factor
BRCA mutation	1.9 (1.1-2.8)	Positive association, significant, risk factor
Alcohol	1.0 (0.8-1.8)	No association, non-significant
High socioeconomic status	1.4 (0.5-3.5)	Positive association, non-significant
Obesity	1.24 (1.1-1.7)	Positive association, significant, risk factor
No family history of Ca. breast	0.3 (0.01-0.9)	Negative association, significant, protective factor

Solution to Question 16:

The image shown is that of a box-and-whisker plot.

McNemar's test is a statistical test used on paired nominal data. It is used as a transmission disequilibrium test for detecting linkage disequilibrium.

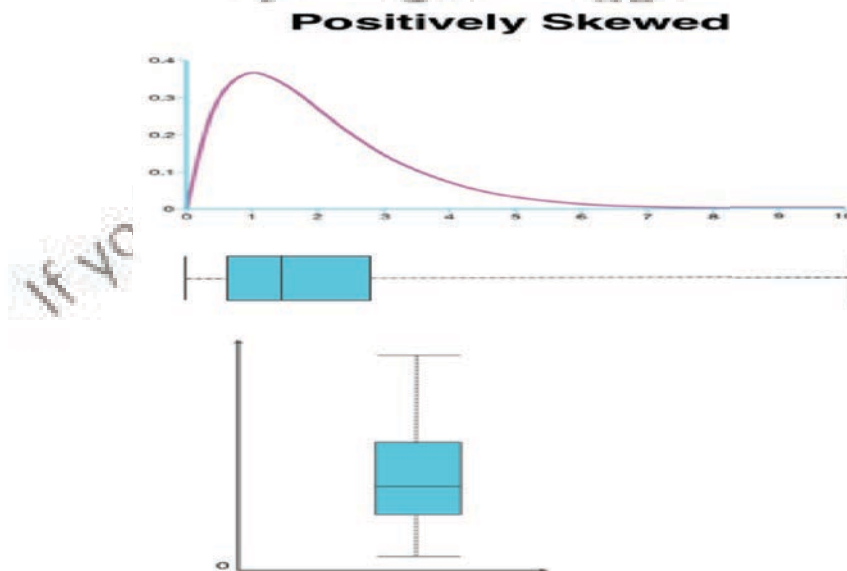
Solution to Question 17:

The Dixon's Q test is used for identification and rejection of outliers. It is usually used in normal distributions.

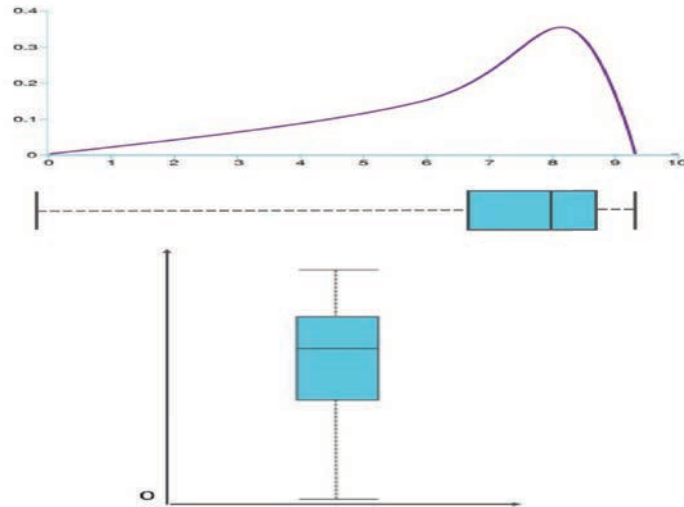
Solution to Question 18:

The given box-whisker plot represents a positively (or right) skewed distribution. In a positively skewed distribution, the mean $>$ median $>$ mode.

To further understand and correlate the fact, have a look at the following images:



Negatively Skewed



Sold by @Itachibot
If you purchased this from someone else,
you may have been scammed.

Mental Health

Question 1:

Which of the following is false about mental health?

- a) Mental disorders are universal, not specific to any group
- b) Mental health is the mere absence of mental illness
- c) Heredity can contribute to mental illnesses
- d) Substance abuse is a type of mental illness

Question 2:

Which of the following are sociopathological factors associated with mental health?

- a) 1 & 2 only
- b) 3, 4 & 5
- c) 1, 2 & 4
- d) 2, 3 & 5

Question 3:

The incidence of mental illness is higher in individuals who are exposed to causal factors during which period of life?

- a) Prenatal period
- b) Adolescence
- c) First 5 years of life
- d) Perinatal period

Question 4:

All of the following can be used for the secondary prevention of mental illness in patients except:

- a) Improving social environment
- b) Early diagnosis

- c) Family counseling
- d) Case work

Question 5:

Patients with which of the following conditions are considered to have a major mental illness?

- a) Obsessive compulsive disorder
- b) Neurosis
- c) Dementia
- d) Manic depressive psychosis

Question 6:

Which of the following is not a service offered to patients with mental health disorders?

- a) Rehabilitation
- b) After-care services
- c) Use of modern psychoactive drugs
- d) Employment services

Question 7:

A 32-year-old man is brought to the casualty with a suspected drug overdose. His sister tells you that he has recently started taking a 'superman drug' after hearing about it from his friends. What drug is she referring to?

- a) Cannabis
- b) Steroids
- c) Amphetamines
- d) Cocaine

Question 8:

What is currently the most widely used illicit substance in the world?

- a) Cocaine

- b) Amphetamines
- c) Heroin
- d) Cannabis

Question 9:

What is the most widely used legal addictive substance in India?

- a) Alcohol
- b) Tobacco
- c) Cannabis
- d) Benzodiazepines

Question 10:

Which of the following is not an objective of the National Mental Health Programme of 1982?

- a) Ensure equal human rights to mentally ill
- b) Promote community participation in mental health services
- c) Promote application of mental health knowledge
- d) Provide accessibility of mental health care

Question 11:

Which of the following is not a component of the District Mental Health Programme?

- a) Training
- b) Early detection
- c) Providing data for future planning
- d) Screening

Question 12:

Which of the following is not included in the Mental Healthcare Act of 2017?

- a) Advance directive rights of the mentally ill
- b) Decriminalizing suicide

- c) Establishing mobile mental health camps
- d) Procedure for admission, treatment and discharge of the mentally ill

Question 13:

According to the WHO Mental Health Gap Action Programme, which of the following is not included in the Mental Health Care Act, 2017?

- a) Screening of family members
- b) Mobilization of resources
- c) Social support
- d) Interpersonal therapy

Question 14:

The National Suicide Prevention Strategy aims to reduce suicide mortality by _____ in the country by 2030.

- a) 5%
- b) 10%
- c) 15%
- d) 20%

Question 15:

Which of the following is not a component of the REDS path in the National suicide prevention strategy?

- a) Reduce suicidal deaths by 2% in the next 5 years
- b) Enhance health services to provide suicide prevention strategies
- c) Develop community resilience, and societal support for suicide prevention
- d) Strengthen surveillance and evidence generation

Answer Key

Question No.	Correct Option
1	b
2	c
3	c
4	a
5	d
6	d
7	c
8	d
9	b
10	a
11	d
12	c
13	a
14	b
15	a

Detailed Explanations

Solution to Question 1:

Mental health is a state of well-being in which an individual realizes his or her own abilities and not merely the absence of mental illness. It is the ability to cope with stress, be productive, and contribute to the community.

Mental disorders are universal and are not specific to any particular group. They are associated with hereditary (family history of psychiatric disorders) and environmental factors. Substance abuse and alcohol addiction are considered types of mental illness.

Solution to Question 2:

Economic insecurity, anxiety, and frustration are the sociopathological factors associated with mental health.

The factors leading to mental health illnesses include:

- Organic - cerebral arteriosclerosis, neoplasms, endocrine, metabolic and chronic diseases
- Hereditary - schizophrenia
- Socio-pathological factors - worries, anxiety, frustration, emotional stress, tension, unhappy marriages, broken homes, poverty, industrialization, urbanization, changing family structure,

population mobility, economic insecurity, cruelty, rejection, neglect.

- Environmental - environmental toxins, substance abuse, radiation, nutritional factors, mineral deficiencies, infectious diseases during prenatal, perinatal, and postnatal periods (like rubella), road traffic accidents.

Solution to Question 3:

Exposure to causal factors during the initial 5 years of life leads to an increased incidence of mental illness.

This period is critical for lifelong mental health behavior and learning, even more crucial than adolescence. During early childhood, the growth and development of children are most vulnerable to external environmental stressors. A continuous, intimate, and warm relationship with the child's parents during infancy and childhood is very important.

Solution to Question 4:

Improving the social environment is a method of primary prevention of mental illness, not secondary prevention.

The levels of prevention of mental illness and measures are given below:

- Primary prevention - improving the social environment and the promotion of mental health awareness
- Secondary prevention - early diagnosis and family-based health services like casework and family counseling
- Tertiary prevention - measures to reduce the duration of mental illness as well family and community stresses to prevent further breakdown and disruption

Solution to Question 5:

Patients with manic depressive psychosis are considered to have a major mental illness.

Mental health disorders are classified as:

- Major (psychotic or not in touch with reality) - include schizophrenia, manic depressive psychosis, and paranoia
- Minor (non-psychotic, remain in touch with reality) - include neuroses (like OCD), personality and character disorders

Dementia occurs secondary to organic causes.

Solution to Question 6:

Employment services are not included under mental health services.

Mental health services include:

- Early diagnosis and treatment
- Group and individual psychotherapy
- Mental health education
- Use of modern psychoactive drugs
- Rehabilitation
- After-care services - help drug-dependent persons to adapt to regular life following treatment and rehabilitation

Solution to Question 7:

Amphetamines are also known as the superman drug.

This is because their effects include increased self-confidence and endurance. However, its usage leads to tolerance and dependence.

Solution to Question 8:

Cannabis is the most widely used illicit substance in the world.

Cannabis is obtained from *Cannabis sativa*, *C. indica*, and *C. americana*. It is colloquially called marijuana. Other forms of this drug are charas or hashish and bhang.

The image given below shows the cannabis plant.



Solution to Question 9:

Tobacco is the most widely used legal addictive substance in India, followed closely by alcohol as the second most widely consumed substance.

Solution to Question 10:

Ensuring equal rights to mentally ill people is not an objective of the National Mental Health Programme (NMHP). It was launched in 1982 with the following objectives:

- Ensure availability and accessibility of minimum mental health care
- Encourage the application of mental health knowledge in general care
- Promote community participation in mental health services development

The National Mental Health Programme was relaunched as The Mental Health Care Act, 2017 which includes provisions for equal rights for those with mental illness.

Solution to Question 11:

Screening is not a component of the District Mental Health Programme. It includes the following components:

- Training of all mental health workers at the nodal institute in the state
- Increasing public awareness by education to reduce stigma
- Early detection and treatment
- Providing data for future planning, improvement in service and research

Activities by the DMHP for positive mental health include:

- School mental health services
- College counseling services
- Workplace stress management
- Suicide prevention services

Solution to Question 12:

Establishing mobile mental health camps for the benefit of mentally ill patients is not included under the Mental Healthcare Act of 2017.

This act encourages the mobilization of resources to improve mental health care across the country as well as reduce stigma and improve social inclusion/support for those with mental illness. It also has provisions to improve outcomes and protect their rights.

The provisions of this act include:

- Rights of persons with mental illness to access health care, live without discrimination and enjoy confidentiality.
- Advance directives to allow mentally ill people to make decisions about their treatment
- Procedures for admission, treatment (behavioral and interpersonal therapy, medications), and discharge
- Decriminalizing suicide and prohibiting electroconvulsive therapy unless anesthesia is used

The act also has clear instructions for the central and state mental health authorities and other mental health establishments on their endeavors.

Solution to Question 13:

According to the WHO Mental Health Gap Action Programme (mhGAP), screening of family members is not included in the Mental Health Care Act, 2017.

The Mental Healthcare Act of 2017 has provisions that include:

- Rights of persons with mental illness to access health care, live without discrimination and enjoy confidentiality.
- Advance directives to allow mentally ill people to make decisions about their treatment
- Procedures for admission, treatment (behavioral and interpersonal therapy, medications), and discharge
- Decriminalizing suicide and prohibiting electroconvulsive therapy unless anesthesia is used

These are achieved through the mobilization of resources to required areas, and by providing social support.

Solution to Question 14:

The National Suicide Prevention Strategy (NSPS) aims to reduce suicide mortality by 10% in India by 2030. It is a multisectoral approach that will involve national and international government ministries, families, communities, and educational institutes to help prevent suicide.

India has a high burden of suicide. It is the leading cause of death among individuals between 15-29 years of age.

Epidemiology

- Young and middle-aged adults between 18-45 years
- Males more than females

- Daily wage earners and people who earn <1,00,000 per annum
- Individuals with family issues and illnesses

Hanging and poisoning were the most common methods of suicide. It accounts for >3/4th of all suicides.

Risk factors for suicide

- Psychiatric illnesses- Depression, BPD, schizophrenia, and anxiety disorders
- Alcohol and other substance abuse disorders
- Major physical illness
- Job or financial loss
- Relationship or social loss
- Domestic violence or abuse
- Lack of social support
- The stigma associated with help-seeking behavior of mental disorders
- Lack of social support and sense of isolation.

Solution to Question 15:

The National Suicide Prevention Strategy (NSPS) is a multisectoral approach that will involve national and international government ministries, families and communities, and educational institutes. It aims to reduce suicide mortality by 10% in India by 2030.

It recommends REDS path for suicide prevention.

- Reinforce leadership, partnership, and institutional capacity in the country
- Enhance the health services capacity to provide suicide prevention services
- Develop community resilience, societal support for suicide prevention and reduce stigma against suicidal behaviours
- Strengthen surveillance and evidence generation.

Objectives

- To establish effective surveillance mechanism for suicide prevention in next 3 years
- To establish psychiatric OPDs that provide suicide prevention services through District Mental Health Programme, within next 5 years
- To integrate mental well-being curriculum in all educational institutes within next 8 years.

Genetics and Health

Question 1:

A 42-year-old male prisoner is being evaluated for aggressive tendencies and antisocial behavior. Which of the following genotypes is associated with these symptoms?

- a) XXY
- b) XO
- c) XXX
- d) XYY

Question 2:

What is the risk of developing Down's syndrome in a baby born to a 45-year-old woman?

- a) 1 in 1000
- b) 1 in 50
- c) 1 in 3000
- d) 1 in 100

Question 3:

Which of the following conditions is not autosomal dominant?

- a) Neurofibromatosis
- b) Retinoblastoma
- c) Marfan's syndrome
- d) Ataxia telangiectasia

Question 4:

Which of the following is incorrect regarding autosomal recessive disorders?

- a) Both male and female offspring are affected
- b) Heterozygotes are not affected

- c) Most inborn errors of metabolism have this form of inheritance
- d) It affects more people at a later age than in childhood

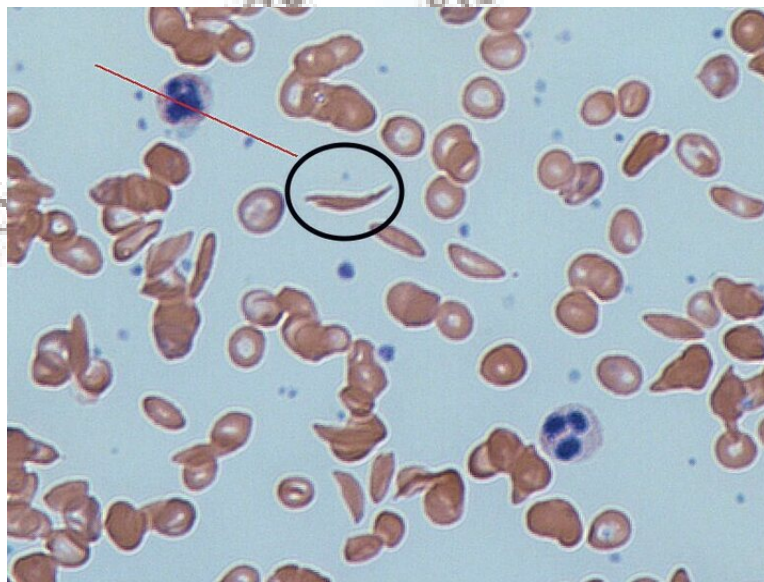
Question 5:

A 6-year-old boy with recurrent hemarthrosis was found to have deficiency of factor VIII. He was adopted and his family history is unknown. What is the inheritance pattern of this condition?

- a) X-linked recessive
- b) Autosomal dominant
- c) X-linked dominant
- d) Autosomal recessive

Question 6:

A pathologist is examining the peripheral smear of a patient with anemia. Which of the following is incorrect about the epidemiology of this disease?



- a) It is prevalent in areas with high frequency of malaria
- b) It is an autosomal dominant disorder
- c) Couples at risk can undergo prenatal testing and diagnosis
- d) It manifests during the first year of life

Question 7:

What was the primary goal of the Human Genome Project?

- a) To introduce a gene sequence into a cell
- b) To develop new diagnostic techniques for genetic disorders
- c) To determine the sequence of nucleotide base pairs in human DNA
- d) To study the precise genetic composition of specific populations

Question 8:

What is Hardy-Weinberg law related to?

- a) Population genetics
- b) Behavioural genetics
- c) Ecological genetics
- d) Developmental genetics

Question 9:

The Hardy-Weinberg law is applicable in populations with which of the following?

- a) Migration
- b) Small population
- c) Random mating
- d) Gene outflow

Question 10:

Match the following.

- a) 1-d, 2-c, 3-b, 4-a
- b) 1-c, 2-d, 3-a, 4-b
- c) 1-b, 2-a, 3-d, 4-c
- d) 1-a, 2-b, 3-c, 4-d

Question 11:

A young woman with a family history of beta-thalassemia is planning to marry her first cousin. Which of the following will you recommend for this woman?

- a) Retrospective genetic counseling
- b) Amniocentesis
- c) Positive eugenics
- d) Prospective genetic counseling

Question 12:

A 40-year-old pregnant woman is planned for amniocentesis. What is the gestational age at which it can be done at the earliest?

- a) 10th week of gestation
- b) 14th week of gestation
- c) 20th week of gestation
- d) 18th week of gestation

Question 13:

What is a Punnett square used for?

- a) Random sampling
- b) Finding genotype of offspring
- c) Population genetics
- d) Quantifying mutations

Question 14:

The risk of genetic diseases in consanguineous marriage between first cousins is?

- a) 1-2%
- b) 4-8%
- c) 8-10%
- d) 12-14%

Answer Key

Question No.	Correct Option
1	d
2	b
3	d
4	d
5	a
6	b
7	c
8	a
9	c
10	b
11	d
12	b
13	b
14	b

Detailed Explanations

Solution to Question 1:

XYY genotype is often associated with aggressive tendencies, antisocial, and often, criminal behavior. It is called Jacob's syndrome.

Option A: XXY genotype is associated with Klinefelter's syndrome.

Option B: XO genotype is associated with Turner's syndrome.

Option C: Triple X syndrome is associated with genotype XXX. It is also called super female.

Solution to Question 2:

The risk of developing Down's syndrome in a baby born to a 45-year-old woman is 1 in 50.

The risk of developing of Down's syndrome in a baby born to a 20-year-old woman is around 1 in 3000 and it increases with rising maternal age. It is an autosomal disorder caused by trisomy of chromosome 21.

Solution to Question 3:

Ataxia-telangiectasia or Louis-Bar syndrome is an autosomal recessive condition.

Autosomal dominant conditions include:

- Neurofibromatosis
- Marfan's syndrome
- Retinoblastoma
- Huntington's chorea
- Polycystic kidney disease

Solution to Question 4:

Autosomal recessive disorders usually manifest in childhood or earlier.

Heterozygotes are unaffected by the disease. When both parents are heterozygous, each offspring (both male and female) has a 1:4 chance of being affected. Examples of autosomal recessive disorders include several inborn errors of metabolism like phenylketonuria, cystic fibrosis, and galactosemia.

Solution to Question 5:

The given clinical scenario is suggestive of hemophilia A. It is an X-linked recessive disorder.

Hemophilia is of two types:

- Hemophilia A - due to mutations in the F8 gene leading to factor VIII deficiency
- Hemophilia B - due to mutations in the F9 gene leading to factor IX deficiency

Affected children often present with prolonged bleeding after procedures like circumcision or major bleeds like intracranial hemorrhages. When the child begins to walk, bleeding into the joint (hemarthrosis) may occur, causing painful swellings and reddening.

Laboratory investigations show elevated aPTT, with normal PT and platelet count. It is treated by the administration of factor VIII or factor IX concentrates.

Solution to Question 6:

The given peripheral smear shows sickling of RBCs, suggestive of sickle cell anemia. It is an autosomal recessive disorder.

The prevalence is high among Africans. This disorder usually has its onset during the first year of life. Couples at risk can opt for prenatal testing and diagnosis as well.

Solution to Question 7:

The primary goal of the Human Genome Project was to determine the sequence of nucleotide base pairs that make up human DNA and to map all genes from the human genome (20,500 genes).

The project was launched in 1990 and deemed complete in 2003.

Solution to Question 8:

The Hardy-Weinberg law is related to population genetics. Population genetics is the study of the precise genetic composition of a population and various factors determining the incidence of inherited traits in them.

Founder effect is a violation to this law. It is due to difference in expression of the genes in founding population. This is responsible for the difference in the distribution of disease in population.

Solution to Question 9:

The Hardy-Weinberg law is applicable for infinitely large, static populations with random mating.

Solution to Question 10:

Positive eugenics encourages the breeding of those with desirable phenotypes.

Negative eugenics discourages the breeding of those with undesirable phenotypes.

Euthenics is the environmental manipulation to enable genes to express themselves.

Euphenics is the science of making phenotypic improvements in humans after birth, generally to affect a problematic genetic condition.

Solution to Question 11:

This young woman must be advised to undergo prospective counseling.

It refers to the process explaining to heterozygous individuals the risk of having affected children if they marry another heterozygous individual. It is performed prior to conception.

Retrospective counseling is performed after the genetic disorder has already occurred in the first child.

Solution to Question 12:

Amniocentesis can be done as early as 14 weeks of gestation.

It refers to the examination of a sample of amniotic fluid to diagnose chromosomal disorders and some metabolic errors. It is indicated in the following cases:

- Pregnant women aged 35 years or above
- Family history of:
- Chromosomal abnormalities
- Metabolic errors
- Chromosomal translocation
- Sex-linked genetic disease

Solution to Question 13:

A Punnett square is a diagram used to predict the probability of a particular genotype in an offspring.

It is a standard tool for genetic counselors. The image given below shows a Punnett square.

	A	a
A	AA	Aa
a	Aa	aa

Solution to Question 14:

The risk of genetic diseases in consanguineous marriage between first cousins is 4-8%.

The risk of genetic diseases in non-consanguineous marriage is 2-4%.